SHOTGUN METAGENOMICS · STOOL DNA · 5.7 MILLION READ PAIRS, EVERY ONE ANALYSED

# Gut Health Test Metagenomic Report

What your gut bacteria can produce, who is living there, how healthy the community looks, and which published disease patterns it resembles — read straight from the DNA in your sample.




**+2.00**
HEALTHY RANGE
26 healthy · 3 disease markers
GMWI2 gut-health index

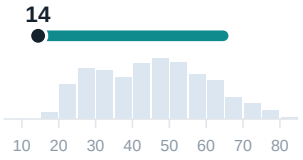
### WHAT STOOD OUT

- **General gut health** — healthy range (index +2.00).
- **Overgrown** — *Enterocloster GWHFCPF00000000* (detected, +116% vs typical carrier), *Blautia_A luti* (detected, +99% vs typical carrier). *Enterocloster GWHFCPF00000000* is an opportunist: normal at low levels, associated with inflammation when it expands.
- **Depleted** — *GGB74412 SGB5075*, *Faecalibacterium intestinale* and 3 more well below the reference carriers.
- **Vitamin K2** — low, 12th percentile.
- **p-Cresol** — above average, 84th percentile.
- **Indole-3-propionate** — above average in the favourable direction, 84th percentile.
- **Rheumatoid arthritis** — notably high resemblance, 95th percentile.

## Biological Age Estimate

section 8

**14**
ESTIMATED AGE (YRS)
90% interval 14–65




## Biofilm Score

section 11

| | |
|---|---|
| Biofilm-forming (bad) | 5/100 |
| Protective gut-lining (good) | 97/100 |

## Mycobiome

section 12

| | |
|---|---|
| Myco-Score (experimental) | 70/100 |

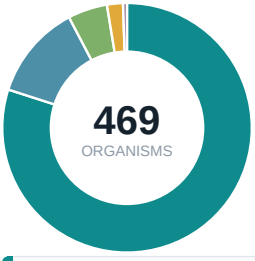
## Skin

section 13

| | |
|---|---|
| Gut-skin axis (beta) | 54/100 |

## Microbiome Diversity

sections 3 to 5, 15, 16




- Firmicutes 80%
- Bacteroidota 12%
- Proteobacteria 5%
- Actinobacteria 2%
- Tenericutes 1%

**3.72**
DIVERSITY
Shannon index · >99th pct

**0.77**
EVENNESS
0 to 1 · 97th pct

**0**
EXPECTED SPECIES MISSING
in most reference adults, absent here

**0 / 7**
DEPLETED / EXPANDED
none need qualification

**469**
ALL ORGANISMS DETECTED
275 genera, every catalogue pooled

**65**
RESOLVED TO STRAIN
read at single-base resolution

## Disease Signatures

sections 10, 20

| | |
|---|---|
| Rheumatoid arthritis | 95th |
| Alzheimer's disease | 88th |
| Coeliac disease | 77th |
| Type 1 diabetes | 64th |
| Alopecia areata | 47th |
| Atherosclerosis | 39th |
| Preclinical amyloid | 38th |
| IBS, mixed type | 34th |

Top 8 of 33 scored (33 in the library). Resemblance is not disease — section 10

## Microbial Functions

sections 9, 19

| | |
|---|---|
| Vitamin K2 ▼ | 12th |
| p-Cresol ▲ | 84th |
| Imidazole propionate ▲ | 83rd |
| Hydrogen sulfide ▲ | 81st |
| GABA ▼ | 20th |
| Fibre breakdown ▼ | 21st |
| Propionate ▲ | 78th |
| Unconjugated bile acids | 95th |

▲ worse ▼ better · **37 more** (10 notable) in section 9

## No disease-causing organisms found

Section 7

2 traces, unconfirmed · 2 species unresolved · 1 toxin-negative

## Possible solutions

**198** studied options for the **36** findings in this report

Section 14 →

# Table of contents

This report is interactive. Click a section for results at a glance, then a reading for its detailed explanation. Use ‘Back to overview’ to return.

**1 Your results at a glance**
every score on one page

**2 What stood out**
the findings worth your attention first

**3 Your gut community**
diversity, composition and community type

**4 Your microbial groups**
curated sets of species, each summed to one reading

**5 Organisms that need attention**
the ones flagged, one by one

**6 Every organism found**
the full inventory, searchable

**7 Pathogen screening**
what was searched for, and what was found

**8 Estimated biological age of biota**
how your community compares by age

**9 Your Microbial Functions**
what your microbes can make, break down and transform

**10 Resemblance to published disease patterns**
how your community compares with published patterns

**11 Biofilm-related potential**
organisms that build and degrade the mucus layer

**12 Your gut mycobiome**
the fungal side of the community

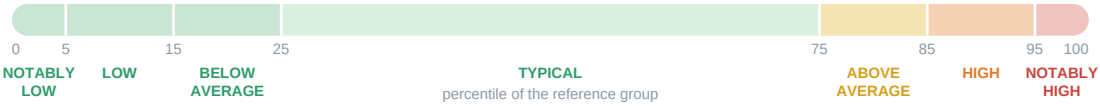
**13 Gut-skin axis**
one score for how your gut leans across published skin panels

**14 What you can do about it**
options with the evidence behind each one

**24 Your information & report context**
what was supplied, and what was assumed

## How to read this report

**One scale, everywhere.** Every reading — organism, group, metabolic function, disease pattern — is a **percentile against a reference group**, translated into the same seven words. The 60th percentile means 60% of the reference had a lower value than you; the reference’s middle half (25th to 75th) is **typical**.




**Colour is the judgement; the word is the position.** Where research links higher with worse outcomes (▲) the high end is red; where higher is favourable (▼) the bar is mirrored and high is green; where the evidence is mixed the bar is grey and no verdict is given. Disease-pattern resemblance always reads higher as less favourable. Everything here is *capacity* or *composition* read from DNA — never a measured amount of a compound in your body, and never a diagnosis.

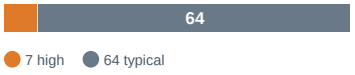
PART A — AT A GLANCE

# 2 What stood out

The findings that matter most, each with the graphic it comes from. Everything here is expanded in the sections named at the end of each line.




**General gut health: +2.00, healthy range** on an index trained on 8,069 labelled stool samples. Raised pattern scores below are therefore not just tracking disturbance. Section 3.



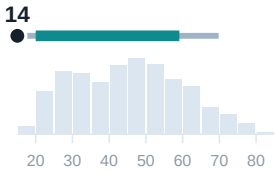

**7 expanded** (*Anaerostipes hadrus*, *Asaccharobacter celatus*); 64 typical. A further **58** expected species were not placed against the reference at all (*Acetobacter sp CAG 267*, *Alistipes obesi*, *Alistipes sp CAG 268*, +55), so nothing here says whether they are present or absent. Sections 5 and 16.



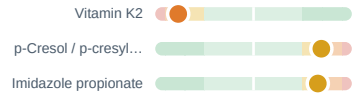

**Overgrown: *Enterocloster GWHFCPF00000000*** at a detected level of your community, +116% against the typical carrier, 96th percentile among reference adults who carry it — opportunist; formerly Clostridium bolteae, C. Also above range: *Blautia_A luti*, *Bacteroides coagulans* and 2 more. Section 5 lists every overgrown organism; section 16 has a page on each.



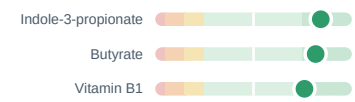

**2 uncommon in the reference**. Sections 5 and 16.



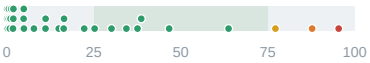

**Your gut community reads like an adolescent's — 14 years**. Built by an ensemble of a transformer neural network and a linear carriage model reading species composition, then moved by 1 disease pattern scoring high, 31 patterns in the typical range or better, a healthy general gut-health index (+2.00). Communities carrying fewer disease-associated patterns and a healthier index read younger; this is an estimate of the community, not of you. Section 8.



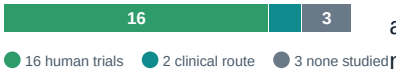

**7 of 45 metabolic functions** sit outside the typical band in the less favourable direction: **Vitamin K2 (menaquinone, MK-n)** (12th), **p-Cresol / p-cresyl sulfate** (84th), **Imidazole propionate (ImP)** (83rd), +4 more. Sections 9 and 19.




**Favourably placed: Indole-3-propionate (IPA)** (84th), **Butyrate (SCFA)** (82nd), **Vitamin B1 (thiamine)** (76th). Section 9.




Closest disease pattern: **Rheumatoid arthritis — gut pattern**, 95th percentile (notably high). Sections 10 and 20.




**21 evidence cards** for the readings that fired: 16 with human trial or guideline evidence for an exact product, diet or drug; 2 pointing to a conventional clinical route; 3 stating that nothing studied exists for that reading. Section 14, and under every reading in Part B.




**Two measurements of hydrogen-sulfide capacity disagree** (+2 more): the gene search puts it at the 81st percentile, the named species at the 16th. Not an error in either — they count different things, and each card explains the split and which to weight. Sections 4, 9, 15 and 19.

PART A — MICROBIOME DIVERSITY: WHAT'S THERE AND WHAT'S MISSING

# 3 Your gut community

Every reference catalogue available was searched, so an organism one catalogue cannot name is still found by another. These are **measured identifications**, not closest matches. Human DNA had already been removed by the sequencing provider.

| 469 | 3.72 | 0.77 | 6.19 |
|---|---|---|---|
| ORGANISMS DETECTED | DIVERSITY | EVENNESS | FIRMICUTES : BACTEROIDETES |
| 123 on the reference catalogue | Shannon index, reference-comparable | 0 to 1, reference-comparable | two dominant groups |




COMPOSITION BY PHYLUM

| | |
|---|---|
| Firmicutes | 79.8% |
| Bacteroidota | 12.9% |
| Pseudomonadota | 4.7% |
| Actinomycetota | 2.4% |
| Thermodesulfobacteriota | 0.1% |

## Most abundant organisms

| | | | |
|---|---|---|---|
| *Blautia A wexlerae* | 7.2% | *Ruminococcaceae bacterium AF68 02b2A* | 2.2% |
| *Faecalibacterium prausnitzii* | 6.2% | *Phocaeicola dorei* | 2.1% |
| *Lachnospira eligens* | 0.4% | *CAG-882 sp003486385* | 2.0% |
| *Anaerostipes hadrus* | 4.5% | *Bacteroides clarus* | 1.4% |
| *Ruminococcus bromii* | 3.4% | *PeH17 sp000435055* | 1.3% |
| *GGB6612 SGB9346* | 2.6% | *Ruminococcaceae bacterium AM53 2BH* | 1.2% |
| *Coprococcus eutactus* | 2.5% | *Bacteroides finegoldii* | 1.2% |
| *Eubacterium_R sp900539775* | 2.2% | *Mediterraneibacter lactaris* | 1.1% |

## General gut-health index




**+2.00**
HEALTHY RANGE
26 healthy · 3 disease markers

**GMWI2** is an independent, published index of general gut health trained on 8,069 stool metagenomes across 54 studies. Positive values lean healthy, negative lean disturbed. It is repeated beside every pattern score in the next section because **a community that is generally disturbed will resemble many disease patterns at once**; the index tells you whether that is what is happening here.

The general dysbiosis index is +2.00 (not dysbiotic). Profile scores are not being driven by broad gut disturbance.

### What is driving this index

the organisms present, weighted as the published model weights them

| Pulling the index down | |
|---|---|
| *Clostridium sp CAG 242* | -0.37 |
| *Clostridiaceae* (genus) | -0.02 |
| *Clostridiales Family XIII Incertae Sedis* (genus) | -0.00 |

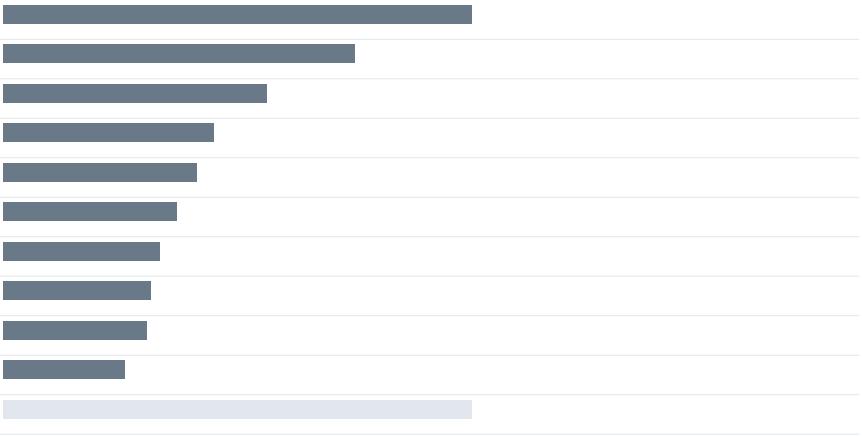
| Holding it up | |
|---|---|
| *Gemmiger formicilis* | +0.28 |
| *Clostridium sp CAG 167* | +0.24 |
| *Dorea formicigenerans* | +0.17 |
| *Eubacterium eligens* | +0.15 |
| *Ruminococcus torques* | +0.15 |
| *Ruminococcus bicirculans* | +0.14 |

The single largest downward pull is *Clostridium sp CAG 242* at -0.37. Each weight is the model's coefficient for that organism when present; the index is their sum. These are the organisms whose change would move the number.

## What your community is made of

genus shares of 331 organisms, as non-overlapping partitions

| | |
|---|---|
| *Blautia* | **10.4%** |
| *Faecalibacterium* | **7.8%** |
| *Lachnospira* | **5.8%** |
| *Anaerostipes* | **4.7%** |
| *Ruminococcus* | **4.3%** |
| *Bacteroides* | **3.8%** |
| *Ruminococcaceae* | **3.5%** |
| *Roseburia* | **3.3%** |
| *Coprococcus* | **3.2%** |
| *Phocaeicola* | **2.7%** |
| 226 other genera | 50.6% |



Shares of unified inventory (marker + whole-genome lanes) (organism_inventory.primary). These are partitions of one total, so they add to 100%: a genus is not counted twice and nothing is dropped. 5% of your community sits in genome bins that have been assembled and counted but not yet given a name — a real part of your community, not a gap in the measurement.

## Which research community type yours resembles

a resemblance, not a diagnosis

Your community is closest to the **Bacteroides-dominant** reference group, the group distinguished in that cohort by *Bacteroides*, *Alistipes*, *Ruminococcus*. It is 21% closer to that group than to the next, which is a clear rather than a borderline placement.

| REFERENCE GROUP | DISTANCE | RESEMBLANCE |
|---|---|---|
| **Bacteroides-dominant** | 0.486 | 57% |
| Prevotella-dominant | 0.586 | 43% |

Groups were fixed once on 3,027 reference samples from 22 studies, 2 groups chosen by held-out silhouette among those that survived study-by-study resampling. Your sample is compared with that frozen reference and never with the other samples analysed alongside it. The reference cohort predates several genus reclassifications, so your organisms were restated under its names to be comparable: *Phocaeicola* as *Bacteroides* (3%); *Mediterraneibacter* as *Ruminococcus* (2%); *Anaerobutyricum* as *Eubacterium* (1%). Same organisms, older names. The comparison uses the 63% of your community that the reference's genus list names, against 99% for a typical reference sample. The gap is newer unnamed genome bins that the older reference catalogue has no entry for; they are excluded from the comparison rather than forced into a neighbouring genus.

A community type is a resemblance to a published composition, not a diagnosis and not a probability of any disease. A *Bacteroides*-dominant result does not establish a high-protein or high-fat diet. Relative composition cannot establish how much microbial mass you carry; that needs an absolute-load assay.

## Diversity, in species you can picture

companion views; the index above is unchanged

**94.3**
EFFECTIVE SPECIES
how many equally-common species…

**44.9**
INVERSE SIMPSON
weighted toward the common organis…

**7.4%**
LARGEST SINGLE ORGANISM
share of the named community

**27.0%**
TOP FIVE TOGETHER
of 331 organisms counted

Counted over 331 named organisms from the organism_inventory.primary lane. Natural logarithms. These describe the same community as the index above and do not replace it; comparing them between samples needs the same catalogue and the same floor.

## Composition ratios

descriptive; no healthy cut-point is defined for any of these

| RATIO | VALUE | WHAT WENT INTO IT | READING |
|---|---|---|---|
| **Proteobacteria (Pseudomonadota) : Actinobacteriota** | **2.576** | Proteobacteria against Actinobacteriota (phylum level) | Descriptive. No established healthy cut-point in stool. |
| **Prevotella : Bacteroides** | **0.028** | Prevotella against Bacteroides (genus level) | Diet-response context only (E17). Genus membership follows the declared taxonomy: Segatella copri was Prevotella copri. |
| **Fusobacterium : Faecalibacterium** | below the floor | none found against Faecalibacterium (genus level) | The numerator was searched for and not detected, so the ratio is below what this depth can resolve rather than measured at zero. |

## Aerotolerance balance

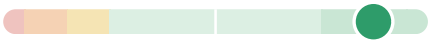
an ecological proxy, not a measurement of oxygen in your gut

Of the organisms whose oxygen requirement is established, **98.7%** of the abundance is strictly anaerobic and **1.3%** tolerates oxygen. A gut community is normally dominated by strict anaerobes. Oxygen requirements are established for **73%** of the abundance here; the rest is counted as unknown rather than assigned to either side.

Oxygen-tolerant organisms here: *GGB6565 SGB9274* (0.54%), *Sutterella wadsworthensis* (0.22%), *Streptococcus salivarius* (0.05%), *Campylobacter hominis* (0.04%), *Streptococcus thermophilus* (0.04%).

# 4 Your microbial groups at a glance

Curated sets of species that share a job or an origin, each summed to one reading and placed against 3,027 matched reference adults. ▼ higher is favourable, ▲ higher is adverse, no arrow: it depends on context. Under each reading are the organisms carrying it and the common members that were not found.

| GROUP | WHERE YOU SIT | READING |
|---|---|---|
| **FUNCTIONAL GROUPS** | | |
| **Butyrate producers ▼**<br>16.82% of reads · 14 of 19 species |  | 87th HIGH |
| ● *Faecalibacterium prausnitzii* **37%** · ● *Anaerostipes hadrus* **27%** · ● *Coprococcus eutactus* **15%** · ● *Roseburia faecis* **3%** +10 more not detected: *Eubacterium ventriosum* | | |
| **Sulfate-reducing bacteria ▲**<br>0.00% of reads · 1 of 4 species |  | 16th BELOW AVERAGE |
| **none detected** — usually present: *Bilophila wadsworthia* | | |
| **Mucin degraders ▲**<br>0.45% of reads · 4 of 8 species |  | 33rd TYPICAL |
| ● *Barnesiella intestinihominis* **51%** · ● *Mediterraneibacter torques* **23%** · ● *Bacteroides thetaiotaomicron* **20%** · ● *Akkermansia muciniphila* **5%** not detected: *Bacteroides caccae*, *Bacteroides fragilis* | | |
| **Methanogens (archaea) ▲**<br>0.00% of reads · 0 of 3 species |  | 35th NOT DETECTED · TYPICAL |
| none of this group's species were detected | | |
| **Polyphenol metabolisers (urolithin and equol) ▼**<br>0.45% of reads · 2 of 5 species |  | 43rd TYPICAL |
| ● *Adlercreutzia equolifaciens* **95%** · ● *Eggerthella lenta* **5%** not detected: *Gordonibacter pamelaeae* | | |
| **ORGANISMS FROM ELSEWHERE IN THE BODY** | | |
| **Oral-origin organisms found in stool ▲**<br>0.08% of reads · 3 of 33 species |  | 70th TYPICAL |
| ● *Streptococcus salivarius* **66%** · ● *Streptococcus parasanguinis* **19%** · ● *Rothia mucilaginosa* **15%** | | |
| **COMMONLY BENEFICIAL ORGANISMS** | | |
| **Resident bifidobacteria ▼**<br>0.03% of reads · 1 of 5 species |  | 10th LOW |
| ● *Bifidobacterium catenulatum* **100%** not detected: *Bifidobacterium longum*, *Bifidobacterium adolescentis* | | |
| **Health-associated core species ▼**<br>22.27% of reads · 20 of 25 species |  | 82nd ABOVE AVERAGE |
| ● *Faecalibacterium prausnitzii* **28%** · ● *Anaerostipes hadrus* **20%** · ● *Ruminococcus bromii* **15%** · ● *Coprococcus eutactus* **11%** +16 more not detected: *Bifidobacterium longum*, *Eubacterium ventriosum*, *Bifidobacterium adolescentis* | | |
| **Supplement and fermented-dairy species**<br>0.03% of reads · 1 of 14 species |  | 80th ABOVE AVERAGE |
| ● *Streptococcus thermophilus* **100%** | | |
| **OPPORTUNISTIC ORGANISMS** | | |
| **Opportunistic pathogens and pathobionts ▲**<br>0.00% of reads · 1 of 20 species |  | 3rd NOTABLY LOW |
| **none detected** — usually present: *Escherichia coli*, *Bilophila wadsworthia* | | |
| **CELL-SURFACE CHARACTERISTICS** | | |
| **Hexa-acylated LPS carriers ▲**<br>0.00% of reads · 1 of 17 species |  | 7th LOW |
| **none detected** — usually present: *Escherichia coli*, *Bilophila wadsworthia* | | |

# 5 Organisms that need attention: overgrown, missing, low

Your community, read in both directions. The full inventory of everything detected is the section before this; this page is the part that inventory cannot show — **what is expanded past its range**, **what is not there that usually is**, and what is present but low. Every organism here has its own page in section 16: what it does, why it matters, and what has been shown to move it.

| 2 | 0 | 5 | 3 |
|---|---|---|---|
| OVERGROWN | MISSING | DEPLETED | WORTH WATCHING |
| opportunists or very high | usually carried, not found | beneficial, well below range | expanded, not yet a concern |

Reference: 3,027 adult stool metagenomes from 22 studies. 3,027 matched reference adults 469 organisms detected in all; percentiles are against the adults who carry each organism. An organism with no percentile is one the reference set has no measurement for, and is flagged on presence and class alone.

## OVERGROWN — at a level that is a concern 2

Opportunists above the 90th percentile among reference adults who carry them, any organism above the 97th, and opportunists that few adults carry but that are several times above the level of the people who do. These are the readings to act on first.

| ORGANISM | CLASS | SHARE | YOUR LEVEL | READING |
|---|---|---|---|---|
| ***Enterocloster GWHFCPF00000000***<br>A group of bile-tolerant Lachnospiraceae, formerly filed under Clostridium<br>counted within *Clostridium sp AM22 11AC* · whole-genome reading 0.278% | Opportunist | no share of its own<br>carried by 24% · typical 0.128% | +116%<br>96th pct | OVERGROWN |
| ***Blautia_A luti***<br>One of the most abundant gut genera<br>counted within *Blautia A wexlerae* · whole-genome reading 0.315% | Conditional | no share of its own<br>carried by 13% · typical 0.159% | +99%<br>>99th pct | OVERGROWN |

## MISSING — commonly carried, not detected here 0

Species most reference adults carry that were not found at a depth where they should have been. The organisms that should be there.

*None in this sample.*

## DEPLETED — well below the reference 5

Beneficial organisms present but below the 10th percentile of people who carry them.

| ORGANISM | CLASS | SHARE | YOUR LEVEL | READING |
|---|---|---|---|---|
| ***GGB74412 SGB5075***<br>Pectin degraders producing acetate and the precursors of butyrate; associated with plant-rich diets and lower in some … | Beneficial | 0.025%<br>carried by 47% · typical 0.494% | –90%<br>2nd pct | LOW |
| ***Faecalibacterium intestinale***<br>The archetypal beneficial gut bacterium: one of the largest butyrate producers, with anti-inflammatory effects shown i…<br>counted within *Faecalibacterium prausnitzii* · whole-genome reading 0.244% | Beneficial | no share of its own<br>carried by 21% · typical 0.692% | –65%<br>10th pct | LOW |
| ***Wujia chipingensis***<br>Butyrate producers repeatedly associated with better quality-of-life measures and lower in depression cohorts; C. catu… | Beneficial | 0.086%<br>carried by 22% · typical 1.01% | –89%<br>4th pct | LOW |
| ***Faecalibacterium butyricigenerans***<br>The archetypal beneficial gut bacterium: one of the largest butyrate producers, with anti-inflammatory effects shown i… | Beneficial | 0.002%<br>carried by 31% · typical 0.664% | –89%<br><1st pct | LOW |
| ***Roseburia inulinivorans***<br>Flagellated butyrate producers that thrive on dietary fibre, especially beta-glucans and inulin | Beneficial | 0.004%<br>carried by 85% · typical 0.585% | –94%<br>9th pct | LOW |

## WORTH WATCHING — above the reference range, or uncommon to carry 3

Conditional organisms between the 90th and 97th percentile among carriers - normal residents that have expanded, a signal about diet and balance more than a problem in themselves - and opportunists that fewer than one in ten reference adults carry at all, whatever their level. A beneficial organism that is high is not listed: a supplement being taken is not an overgrowth.

| ORGANISM | CLASS | SHARE | YOUR LEVEL | READING |
|---|---|---|---|---|
| ***Bacteroides coagulans***<br>The dominant genus of a Western-diet gut<br>counted within *Bacteroides clarus* · whole-genome reading 0.042% | Conditional | no share of its own<br>carried by 1% · typical 0.011% | +279%<br>91st pct | HIGH |
| ***Dorea formicigenerans***<br>Gas producers (hydrogen, carbon dioxide) that break down mucus-associated sugars | Conditional | 0.582%<br>carried by 85% · typical 0.212% | +491%<br>96th pct | HIGH |
| ***Blautia_A MGYG000001338***<br>One of the most abundant gut genera<br>counted within *Blautia A wexlerae* · whole-genome reading 6.35% | Conditional | no share of its own<br>carried by 93% · typical 2.20% | +189%<br>90th pct | HIGH |

PART A — EVERY ORGANISM DETECTED

# 6.1 Your organisms, classified

**469 organisms** were found in this sample, pooling every reference catalogue available. Each is placed in one of four classes from the human literature, and — where a reference population exists for it — its level is compared with 3,027 matched reference adults. The bar is the community by share of reads: a dominant organism is a wide band, a trace one is invisible here and listed below. The grey tail is the 3% of sequence that matched nothing on record — shown so the parts add to the whole.

Detection pools **9 methods**, each with its own reference catalogue, so a species one catalogue has no genome for is still found when another can name it. Counted separately: **209** named species and **252** unnamed species-level clusters — organisms known only from assembled genomes, real and counted, whose names will change when they are formally described; **6** populations that the catalogues place in sibling species and that are listed once; **2** placed above species. Each organism has one share of the whole, on the scale of the marker catalogue that measures the composition; for **83** organisms that catalogue has no entry, so their shares (6.3% together, marked °) are estimated from the whole-genome search and drawn from the unclassified band; **138** populations the whole-genome search tells apart within a relative's share are listed with that relative and carry no share of their own; for **8** calls the whole-genome competition rejected, the marker share went to the relative the reads belong to (marked §). **159** organisms rest on a single detection method, said so beside their identifiers.




| Beneficial 36% | Conditional 34% | Opportunist 1% | Unknown 26% | Unclassified 3% |
|---|---|---|---|---|
| 71 organisms · 5 flagged | 135 organisms · 4 flagged | 9 organisms · 1 flagged | 254 organisms | sequence, not organisms |
| Higher levels are associated with better outcomes in human studies. Butyrate producers, mucus-layer keepers and the classic probiotic genera live here. | Cuts both ways: part of a healthy gut at typical levels, associated with problems when expanded or when the community around it is disturbed. Most of the gut is this. | Normal at low levels. Expansion is what the research associates with inflammation, infection or a disturbed community. Not a pathogen label — that is the pathogen screen's decision. | Not enough human evidence to say. Many are known only from genomes and have never been grown in a laboratory. | DNA that matched nothing in any catalogue. Not counted in the four classes; shown so the bar adds to 100%. |

**4 organisms at a level that is a concern** and **6 worth watching**. The most consequential is ***Enterocloster GWHFCPF00000000*** — opportunist, 0.000% of the community, high — 96th percentile among reference adults who carry it (+116% against the typical carrier). Flagged rows are tinted throughout this section.

Class is a property of the organism, from the literature; the flag is a property of this sample, from its percentile. **vs typical carrier** is how far the level sits from the median of reference adults who carry the organism at all (that median is the figure in brackets): +400% is five times their level, –90% a tenth of it, ×110 a hundred and ten times. Where fewer than ten reference adults carry an organism, no rank is stated. An organism with no reference was measured on a catalogue the reference population was not, so there is nothing to compare it against; the detection stands on its own.

## Furthest from the reference

Against the typical carrier in 3,027 matched reference adults. Left: the largest excesses among organisms at 0.1% or more. Right: the deepest shortfalls among organisms carried by at least half of the reference adults.

| MOST ABOVE THE REFERENCE | SHARE | VS TYPICAL | MOST BELOW THE REFERENCE | SHARE | VS TYPICAL |
|---|---|---|---|---|---|
| *Finegoldia magna* | 0.188% | **×154** >99th pct | *Roseburia inulinivorans* | 0.004% | **–94%** 9th pct |
| *Blautia faecicola* | 0.112% | **×24** 88th pct | *Romboutsia timonensis* | 0.048% | **–93%** <1st pct |
| *Anaerostipes hadrus* | 4.53% | **×23** 99th pct | *Alistipes shahii* | 0.076% | **–91%** 11th pct |
| *Coprococcus catus* | 0.534% | **×11** 99th pct | *Hominenteromicrobium mulieris* | 0.031% | **–87%** <1st pct |
| *Coprococcus eutactus* | 2.46% | **×11** 89th pct | *Lachnospira pectinoschiza* | 0.109% | **–87%** 22nd pct |
| *Asaccharobacter celatus* | 0.212% | **+790%** 96th pct | *Dysosmobacter segnis* | 0.073% | **–83%** 14th pct |
| *Bacteroides clarus* | 1.39% | **+658%** 86th pct | *Candidatus Cibionibacter quicibialis* | 0.186% | **–76%** 19th pct |
| *Lachnospira eligens_A* | 4.86% | **+587%** >99th pct | *Ruthenibacterium lactatiformans* | 0.025% | **–75%** 19th pct |

# 6.2 The 30 most abundant, one by one

Together these make up **61%** of the composition. Each has what it is, what the research says about it, its class, and whether the amount found is a concern.

| # | ORGANISM | WHAT IT IS | CLASS · SHARE | LEVEL |
|---|---|---|---|---|
| 1 | ***Blautia A wexlerae***<br>also Blautia_A wexlerae / wexlerae_B includes reads the marker catalogue read as Blautia massiliensis ex Durand et al 2017, Blautia A sp934394595<br>SGB 4837 · GTDB GCF_025148125.1 · mOTU 000158 · 8 methods | One of the most abundant gut genera, producing acetate and some members butyrate. Reported protective against visceral fat and some infections, but also expanded in obesity, type 2 diabetes and IBS cohorts. A common sign of a Western diet rather than a problem in itself. (genus-level) | Conditional<br>**7.21%**§ | reference catalogue reads this species differently |
| 2 | ***Faecalibacterium prausnitzii***<br>also Faecalibacterium duncaniae<br>SGB 15318 · GlobDB GCF_003324185 · mOTU 000004 · 9 methods | One of the most abundant butyrate producers in adult stool. Reduced relative abundance is one of the most replicated findings in Crohn's disease and ulcerative colitis, and lower levels are reported across several inflammatory and metabolic conditions. It is also simply variable between healthy people and across countries; a low level alone is not a diagnosis and no product restores the species itself. | Beneficial<br>**6.18%** | **+82%** vs typical carrier (5.40%)<br>80th percentile |
| 3 | ***Lachnospira eligens_A***<br>GlobDB GCF_009680455 · mOTU 000113 · UHGG 212 · 6 methods | Pectin degraders producing acetate and the precursors of butyrate; associated with plant-rich diets and lower in some asthma and IBD cohorts. (genus-level) | Beneficial<br>**4.86%**† | **+587%** vs typical carrier (0.704%)<br>>99th percentile‡ |
| 4 | ***Anaerostipes hadrus***<br>SGB 4540 · GlobDB GCF_030296915 · mOTU 000226 · 9 methods | Butyrate producer that also converts lactate and acetate to butyrate; part of the core butyrate-producing set of adult stool. | Beneficial<br>**4.53%** | **×23** vs typical carrier (0.476%)<br>99th percentile |
| 5 | ***Ruminococcus bromii***<br>also Ruminococcoides intestinale_A<br>SGB 4290 · GlobDB GCF_014288065 · mOTU 000081 · 8 methods | Keystone degrader of resistant starch: it breaks down starch granules other species cannot, feeding the wider community. Its baseline level predicted who responded to a resistant-starch feeding trial with a butyrate rise. | Beneficial<br>**3.40%** | **–34%** vs typical carrier (5.06%)<br>42nd percentile‡ |
| 6 | ***GGB6612 SGB9346***<br>also Scatacola_A faecigallinarum<br>SGB 9346 · GlobDB GCA_018713605 · mOTU 000927 · 7 methods | A member of the genus *Scatacola* for which no human evidence has been curated. Its presence is recorded; nothing here says it is helpful or harmful. | Unknown<br>**2.58%** | only 6 reference adults carry it:<br>too few to rank against |
| 7 | ***Coprococcus eutactus***<br>also Coprococcus aceti<br>SGB 5117 · GlobDB GCF_025567285 · mOTU 000279 · 9 methods | Butyrate producer; with Dialister, Coprococcus was depleted in depression cohorts in a large population study. | Beneficial<br>**2.46%** | **×11** vs typical carrier (0.549%)<br>89th percentile |
| 8 | **Eubacterium_R sp900539775**<br>not yet formally named · known from genomes<br>SGB 4316 · GlobDB GCA_900539775 · mOTU 005902 · 8 methods | A catch-all genus that has been progressively split up. Several members are butyrate producers and health-associated; others have little evidence. Read the species where named. (genus-level) | Conditional<br>**2.22%** | only 2 reference adults carry it:<br>too few to rank against |
| 9 | ***Ruminococcaceae bacterium AF68 02b2A***<br>also Ruminococcus_C sp000980705<br>SGB 4421 · GlobDB GCA_000980705 · mOTU 000903 · 8 methods | A member of the Oscillospiraceae (formerly Ruminococcaceae) not yet assigned to a genus; the family includes Faecalibacterium and many fibre degraders and is associated with leanness and diversity. (genus-level) | Conditional<br>**2.16%** | **+95%** vs typical carrier (1.10%)<br>75th percentile‡ |
| 10 | ***Phocaeicola dorei***<br>formerly Bacteroides dorei<br>SGB 1815 · GlobDB GCF_013009555 · mOTU 001486 · 9 methods | Common Bacteroides species; a rise preceded seroconversion in a Finnish type 1 diabetes birth cohort, while other cohorts associate it with lower atherosclerosis. Context-dependent. | Conditional<br>**2.06%** | **+165%** vs typical carrier (1.57%)<br>65th percentile |
| 11 | **CAG-882 sp003486385**<br>not yet formally named · known from genomes<br>SGB 5062 · GlobDB GCA_003486385 · mOTU 000322 · 8 methods | An organism of the genus *CAG-882* known only from genomes assembled out of stool samples. It has never been grown in a laboratory, so its physiology and any effect on health are not yet described. | Unknown<br>**2.00%** | only 7 reference adults carry it:<br>too few to rank against |
| 12 | ***Bacteroides clarus***<br>SGB 1832 · GlobDB GCF_900129655 · mOTU 002222 · 9 methods | Among the most abundant gut genera. Break down complex dietary fibre and host glycans and produce propionate; a normal, dominant part of a Western-diet gut. Several species are also opportunists outside the gut, and some strains carry an enterotoxin, so expansion in an unstable community is watched rather than welcomed. (genus-level) | Conditional<br>**1.39%** | **+658%** vs typical carrier (0.125%)<br>86th percentile |

| # | ORGANISM | WHAT IT IS | CLASS · SHARE | LEVEL |
|---|---|---|---|---|
| 13 | **PeH17 sp000435055**<br>not yet formally named · known from genomes<br>SGB 14306 · GTDB GCA_000435055.1 · mOTU 000199 · 7 methods | An organism of the genus *PeH17* known only from genomes assembled out of stool samples. It has never been grown in a laboratory, so its physiology and any effect on health are not yet described. | Unknown<br>**1.34%** | no reference |
| 14 | ***Ruminococcaceae bacterium AM53 2BH***<br>also CAG-115 sp003531585<br>SGB 4191 · GlobDB GCA_003531585 · mOTU 000306 · 9 methods | A member of the Oscillospiraceae (formerly Ruminococcaceae) not yet assigned to a genus; the family includes Faecalibacterium and many fibre degraders and is associated with leanness and diversity. (genus-level) | Conditional<br>**1.20%** | **+508%** vs typical carrier (0.196%)<br>89th percentile‡ |
| 15 | ***Bacteroides finegoldii***<br>also Bacteroides zhangwenhongii<br>SGB 1862 · GlobDB GCF_009193325 · mOTU 002746 · 8 methods | Among the most abundant gut genera. Break down complex dietary fibre and host glycans and produce propionate; a normal, dominant part of a Western-diet gut. Several species are also opportunists outside the gut, and some strains carry an enterotoxin, so expansion in an unstable community is watched rather than welcomed. (genus-level) | Conditional<br>**1.17%** | **+107%** vs typical carrier (0.256%)<br>63rd percentile |
| 16 | ***Mediterraneibacter lactaris***<br>formerly Ruminococcus lactaris<br>SGB 4557 · GlobDB GCF_025152405 · mOTU 001332 · 9 methods | Formerly Ruminococcus gnavus and R. torques: mucus degraders that are normal residents at low levels but among the most consistently expanded organisms in Crohn's disease, and producers of an inflammatory polysaccharide. Level matters more than presence. | Beneficial<br>**1.13%** | **+480%** vs typical carrier (0.463%)<br>95th percentile |
| 17 | **Limisoma sp000437795**<br>not yet formally named · known from genomes<br>SGB 2071 · GlobDB GCA_022009815 · mOTU 000178 · 8 methods | An organism of the genus *Limisoma* known only from genomes assembled out of stool samples. It has never been grown in a laboratory, so its physiology and any effect on health are not yet described. | Unknown<br>**1.04%** | only 7 reference adults carry it:<br>too few to rank against |
| 18 | ***Gemmiger formicilis***<br>SGB 15300 · GlobDB GCF_900167555 · mOTU 000050 · 9 methods | Butyrate producer related to Faecalibacterium, reported lower in IBD and obesity cohorts. Several members are known only from genomes and carry placeholder names. (genus-level) | Beneficial<br>**1.04%** | **+38%** vs typical carrier (0.696%)<br>62nd percentile‡ |
| 19 | **CAG-196 sp002102975**<br>not yet formally named · known from genomes<br>1 method: MP3 | An organism of the genus *CAG-196* known only from genomes assembled out of stool samples. It has never been grown in a laboratory, so its physiology and any effect on health are not yet described. | Unknown<br>**1.01%**◦ | no reference |
| 20 | ***Dorea longicatena***<br>also Dorea_A longicatena<br>SGB 4581 · GlobDB GCF_025150085 · mOTU 000608 · 8 methods | Common Lachnospiraceae member with mixed metabolic associations. | Conditional<br>**1.01%** | **+233%** vs typical carrier (0.582%)<br>89th percentile |
| 21 | **Faecalibacterium sp900539945**<br>not yet formally named · known from genomes<br>SGB 15346 · GlobDB GCA_025757845 · mOTU 000011 · 8 methods | The archetypal beneficial gut bacterium: one of the largest butyrate producers, with anti-inflammatory effects shown in cells and animals. Reduced in Crohn's disease more consistently than almost any other organism, and lower across many inflammatory and metabolic conditions. (genus-level) | Beneficial<br>**0.998%** | **+132%** vs typical carrier (0.477%)<br>94th percentile‡ |
| 22 | ***Blautia faecis***<br>also Blautia_A faecis<br>SGB 4820 · GlobDB GCF_020554845 · mOTU 000200 · 6 methods | One of the most abundant gut genera, producing acetate and some members butyrate. Reported protective against visceral fat and some infections, but also expanded in obesity, type 2 diabetes and IBS cohorts. A common sign of a Western diet rather than a problem in itself. (genus-level) | Conditional<br>**0.945%** | ≈ **typical** vs typical carrier (0.517%)<br>52nd percentile‡ |
| 23 | ***Roseburia amylophila***<br>also Roseburia_C amylophila<br>SGB 4938 · GlobDB GCF_025567465 · mOTU 000332 · 8 methods | Butyrate producers that thrive on dietary fibre. Lower levels reported in ulcerative colitis, type 2 diabetes and atherosclerosis cohorts. (genus-level) | Beneficial<br>**0.941%** | **+69%** vs typical carrier (0.572%)<br>80th percentile‡ |
| 24 | ***Fusicatenibacter saccharivorans***<br>also Fusicatenibacter sp. (SGB4874)<br>SGB 4874 · GTDB GCF_027667425.1 · mOTU 000133 · 8 methods | Butyrate producer reported lower in ulcerative colitis and depression cohorts. (genus-level) | Beneficial<br>**0.938%** | **+133%** vs typical carrier (0.820%)<br>74th percentile |
| 25 | **CAG-267 sp001917135**<br>not yet formally named · known from genomes<br>SGB 9333 · GlobDB GCA_001917135 · mOTU 002009 · 9 methods | An organism of the genus *CAG-267* known only from genomes assembled out of stool samples. It has never been grown in a laboratory, so its physiology and any effect on health are not yet described. | Unknown<br>**0.907%** | only 8 reference adults carry it:<br>too few to rank against |

| # | ORGANISM | WHAT IT IS | CLASS · SHARE | LEVEL |
|---|---|---|---|---|
| 26 | **Anaerotardibacter sp000436075**<br>not yet formally named · known from genomes<br>SGB 14322 · GlobDB GCA_000436075 · mOTU 000070 · 8 methods | An organism of the genus *Anaerotardibacter* known only from genomes assembled out of stool samples. It has never been grown in a laboratory, so its physiology and any effect on health are not yet described. | Unknown<br>**0.876%** | **+461%** vs typical carrier (0.163%)<br>95th percentile‡ |
| 27 | **Roseburia sp900550935**<br>not yet formally named · known from genomes<br>SGB 4948 · GlobDB GCA_900550935 · mOTU 001783 · 7 methods | Butyrate producers that thrive on dietary fibre. Lower levels reported in ulcerative colitis, type 2 diabetes and atherosclerosis cohorts. (genus-level) | Beneficial<br>**0.862%** | only 2 reference adults carry it:<br>too few to rank against |
| 28 | ***Roseburia sp AM59 24XD***<br>also Jutongia sp003524945<br>SGB 4654 · GlobDB GCF_003478445 · mOTU 002534 · 8 methods | Butyrate producers that thrive on dietary fibre. Lower levels reported in ulcerative colitis, type 2 diabetes and atherosclerosis cohorts. (genus-level) | Beneficial<br>**0.854%** | only 2 reference adults carry it:<br>too few to rank against |
| 29 | **Clostridium sp CAG 138**<br>not yet formally named · known from genomes<br>1 method: MP3 | A large and much-split genus. Most gut members are harmless fermenters; a few named species (C. perfringens, C. difficile) are pathogens and are handled by the pathogen screen. Unnamed members are read as neutral residents. (genus-level) | Conditional<br>**0.837%**◦ | no reference |
| 30 | ***Faecalibacillus intestinalis***<br>SGB 6754 · GlobDB GCF_003024685 · mOTU 000457 · 8 methods | Recently described Erysipelotrichaceae members; limited human evidence. (genus-level) | Conditional<br>**0.828%** | **+112%** vs typical carrier (0.418%)<br>86th percentile‡ |

# 6.3 Every organism detected

All 469 organisms, largest share first, colour-coded by class. **Level** is how far the organism sits from the typical carrier in 3,027 matched reference adults, with its percentile; **carried by** is the share of that group in which the organism is found at all. Rows tinted red are at a level that is a concern; amber, worth watching. **65** were also read to strain level, marked in the last column.

| ORGANISM | CLASS | SHARE | LEVEL | CARRIED BY | PERCENTILE | STRAIN |
|---|---|---|---|---|---|---|
| ***Blautia A wexlerae***<br>also Blautia_A wexlerae / wexlerae_B<br>includes reads the marker catalogue read as Blautia massiliensis ex Durand et al 2017, Blautia A sp934394595<br>SGB 4837 · GTDB GCF_025148125.1 · mOTU 000158 · 8 methods | Conditional | **7.21%**§ | reference catalogue reads this species differently | 87% | not comparable | 155 markers; nearest GCF_013300825.1 (1.431) |
| **Blautia_A MGYG000001338**<br>not yet formally named · known from genomes<br>GlobDB MGYG000001338 · UHGG 2045 · 4 methods | Conditional | within *Blautia* above | ▲ high<br>**+189%** vs typical carrier (2.20%) | 93% | 90th‡ | — |
| ***Blautia_A fusiformis***<br>GlobDB GCF_020687435 · mOTU 000153 · UHGG 371 · 5 methods | Conditional | within *Blautia* above | **–68%** vs typical carrier (0.666%) | 82% | 7th‡ | — |
| **Blautia_A GWHBMYI00000000**<br>not yet formally named · known from genomes<br>GlobDB GWHBMYI00000000 · 1 method: GlobDB sketch | Conditional | within *Blautia* above | **–11%** vs typical carrier (0.470%) | 36% | 39th‡ | — |
| ***Ruminococcus sp AF41 9***<br>also Blautia_A sp003477525<br>SGB 25497 · GlobDB GCA_003477525 · mOTU 006994 · 6 methods | Conditional | within *Blautia* above | only 6 reference adults carry it:<br>too few to rank against | 6% | not in reference set | 31 markers |
| ***Blautia_A luti***<br>GlobDB GCF_033096465 · mOTU 003244 · UHGG 431 · 4 methods | Conditional | within *Blautia* above | ▲ notably high<br>**+99%** vs typical carrier (0.159%) | 13% | >99th‡ | — |
| ***Blautia_A intestinihominis***<br>GlobDB GCF_040096655 · mOTU 002376 · UHGG 3260 · 6 methods | Conditional | within *Blautia* above | only 7 reference adults carry it:<br>too few to rank against | 7% | not in reference set | — |
| **Blautia_A sp900066355**<br>not yet formally named · known from genomes<br>SGB 4815 · GlobDB GCA_900066355 · mOTU 000678 · 4 methods | Conditional | within *Blautia* above | **+25%** vs typical carrier (0.241%) | 42% | 62nd‡ | — |

| ORGANISM | CLASS | SHARE | LEVEL | CARRIED BY | PERCENTILE | STRAIN |
|---|---|---|---|---|---|---|
| **Blautia_A sp900120195**<br>not yet formally named · known from genomes<br>mOTU 001556 · UHGG 413 · 2 methods: Kraken, mOTUs | Conditional | within *Blautia* above | no reference | — | not in reference set | — |
| **Blautia_A sp047027955**<br>not yet formally named · known from genomes<br>GlobDB GCA_047027955 · 2 methods: GTDB sketch, GlobDB sketch | Conditional | within *Blautia* above | **–54%** vs typical carrier (0.452%) | 47% | 19th‡ | — |
| ***Blautia massiliensis***<br>also Blautia_A complex (fusiformis, luti_A, sp005848555)<br>SGB 4826 · 1 method: Jun23 | Conditional | within *Blautia* above | no reference | — | not in reference set | — |
| **Blautia sp**<br>not yet formally named · known from genomes<br>1 method: MP3 | Conditional | within *Blautia* above | no reference | — | not in reference set | — |
| ***Blautia sp MCC283***<br>also Blautia_A sp900066335<br>SGB 4828 · UHGG 544 · 2 methods: Jun23, Kraken | Conditional | within *Blautia* above | no reference | — | not in reference set | — |
| **Blautia_A SPIREOTU_01410522**<br>not yet formally named · known from genomes<br>GlobDB SPIREOTU_01410522 · 1 method: GlobDB sketch | Conditional | within *Blautia* above | only 8 reference adults carry it: too few to rank against | 8% | not in reference set | — |
| ***Faecalibacterium prausnitzii***<br>also Faecalibacterium duncaniae<br>SGB 15318 · GlobDB GCF_003324185 · mOTU 000004 · 9 methods | Beneficial | **6.18%** | **+82%** vs typical carrier (5.40%) | 98% | 80th | ● 162 markers; nearest GCA_002549855.1 (0.844) |
| ***Faecalibacterium longum***<br>GTDB GCF_051201525.1 · mOTU 000007 · UHGG 3565 · 4 methods | Beneficial | within *Faecalibacterium* above | no reference | — | not in reference set | — |
| ***Faecalibacterium prausnitzii_I***<br>GlobDB GCF_003287495 · mOTU 000097 · UHGG 3235 · 6 methods | Beneficial | within *Faecalibacterium* above | **+34%** vs typical carrier (0.440%) | 43% | 72nd‡ | — |
| ***Faecalibacterium intestinale***<br>GlobDB GCF_040096575 · mOTU 000042 · UHGG 3636 · 5 methods | Beneficial | within *Faecalibacterium* above | ▼ low<br>**–65%** vs typical carrier (0.692%) | 21% | 10th‡ | — |
| ***Faecalibacterium taiwanense***<br>GlobDB GCF_036632915 · mOTU 000012 · UHGG 3233 · 5 methods | Beneficial | within *Faecalibacterium* above | **–60%** vs typical carrier (0.733%) | 60% | 13th‡ | — |
| **Faecalibacterium GCMETA_00000982**<br>not yet formally named · known from genomes<br>GlobDB GCMETA_00000982 · 1 method: GlobDB sketch | Beneficial | within *Faecalibacterium* above | **–26%** vs typical carrier (0.747%) | 41% | 42nd‡ | — |
| **Faecalibacterium HRGMV2_2685**<br>not yet formally named · known from genomes<br>1 method: Kraken panel | Beneficial | within *Faecalibacterium* above | no reference | — | not in reference set | — |
| ***Faecalibacterium tardum***<br>mOTU 000103 · 1 method: mOTUs | Beneficial | within *Faecalibacterium* above | no reference | — | not in reference set | — |
| **Faecalibacterium sp CAG 74**<br>not yet formally named · known from genomes<br>1 method: MP3 | Beneficial | within *Faecalibacterium* above | no reference | — | not in reference set | — |
| ***Faecalibacterium langellae***<br>1 method: Kraken panel | Beneficial | within *Faecalibacterium* above | no reference | — | not in reference set | — |
| ***Faecalibacterium prausnitzii_E***<br>UHGG 449 · 1 method: Kraken | Beneficial | within *Faecalibacterium* above | no reference | — | not in reference set | — |
| ***Faecalibacterium hattorii***<br>UHGG 173 · 1 method: Kraken | Beneficial | within *Faecalibacterium* above | no reference | — | not in reference set | — |
| ***Lachnospira eligens_A***<br>GlobDB GCF_009680455 · mOTU 000113 · UHGG 212 · 6 methods | Beneficial | **4.86%**† | **+587%** vs typical carrier (0.704%) | 71% | >99th‡ | — |
| ***Anaerostipes hadrus***<br>SGB 4540 · GlobDB GCF_030296915 · mOTU 000226 · 9 methods | Beneficial | **4.53%** | **×23** vs typical carrier (0.476%) | 87% | 99th | ● 196 markers; nearest GCA_003473625.1 (0.579) |

| ORGANISM | CLASS | SHARE | LEVEL | CARRIED BY | PERCENTILE | STRAIN |
|---|---|---|---|---|---|---|
| ***Anaerostipes amylophilus***<br>UHGG 150 · 1 method: Kraken | Beneficial | within *Anaerostipes* above | no reference | — | not in reference set | — |
| **Anaerostipes sp018918155**<br>not yet formally named · known from genomes<br>UHGG 1258 · 1 method: Kraken | Beneficial | within *Anaerostipes* above | no reference | — | not in reference set | — |
| ***Ruminococcus bromii***<br>also Ruminococcoides intestinale_A<br>SGB 4290 · GlobDB GCF_014288065 · mOTU 000081 · 8 methods | Beneficial | **3.40%** | **–34%** vs typical carrier (5.06%) | 12% | 42nd‡ | 198 markers; nearest GCA_003539535.1 (0.437) |
| **Ruminococcoides sp003438075**<br>not yet formally named · known from genomes<br>UHGG 3264 · 2 methods: Kraken, MP3 | Conditional | within *Ruminococcoides* above | **+563%** vs typical carrier (0.012%) | 11% | 78th | — |
| ***GGB6612 SGB9346***<br>also Scatacola_A faecigallinarum<br>SGB 9346 · GlobDB GCA_018713605 · mOTU 000927 · 7 methods | Unknown | **2.58%** | only 6 reference adults carry it: too few to rank against | 6% | not in reference set | 185 markers; nearest GCA_018713605.1 (0.161) |
| **Scatacola_A HRGMV2_3309**<br>not yet formally named · known from genomes<br>2 methods: Kraken panel, SingleM | Unknown | within *Scatacola* above | no reference | — | not in reference set | — |
| ***Coprococcus eutactus***<br>also Coprococcus aceti<br>SGB 5117 · GlobDB GCF_025567285 · mOTU 000279 · 9 methods | Beneficial | **2.46%** | **×11** vs typical carrier (0.549%) | 36% | 89th | 199 markers; nearest GCA_003487145.1 (0.572) |
| **Coprococcus_A GCMETA_00096139**<br>not yet formally named · known from genomes<br>GlobDB GCMETA_00096139 · 2 methods: GlobDB sketch, SingleM | Beneficial | within *Coprococcus* above | **+260%** vs typical carrier (0.164%) | 44% | 98th‡ | — |
| **Coprococcus MOTU40_000857**<br>not yet formally named · known from genomes<br>1 method: SingleM | Beneficial | within *Coprococcus* above | no reference | — | not in reference set | — |
| **Eubacterium_R sp900539775**<br>not yet formally named · known from genomes<br>SGB 4316 · GlobDB GCA_900539775 · mOTU 005902 · 8 methods | Conditional | **2.22%** | only 2 reference adults carry it: too few to rank against | 2% | not in reference set | 199 markers; nearest GCA_900539775.1 (0.019) |
| **Eubacterium sp CAG 38**<br>not yet formally named · known from genomes<br>1 method: MP3 | Conditional | within *Eubacterium* above | **+43%** vs typical carrier (0.198%) | 61% | 57th | — |
| ***Ruminococcaceae bacterium AF68 02b2A***<br>also Ruminococcus_C sp000980705<br>SGB 4421 · GlobDB GCA_000980705 · mOTU 000903 · 8 methods | Conditional | **2.16%** | **+95%** vs typical carrier (1.10%) | 16% | 75th‡ | 198 markers; nearest GCA_027680725.1 (0.577) |
| **Ruminococcus mOTUv4.0_109744**<br>not yet formally named · known from genomes<br>mOTU 109744 · 1 method: mOTUs | Conditional | within *Ruminococcus* above | no reference | — | not in reference set | — |
| ***Phocaeicola dorei***<br>formerly Bacteroides dorei<br>SGB 1815 · GlobDB GCF_013009555 · mOTU 001486 · 9 methods | Conditional | **2.06%** | **+165%** vs typical carrier (1.57%) | 57% | 65th | 70 markers; nearest GCA_000157075.2 (0.306) |
| **Phocaeicola MGYG000000243**<br>not yet formally named · known from genomes<br>GlobDB MGYG000000243 · UHGG 517 · 4 methods | Conditional | within *Phocaeicola* above | **–54%** vs typical carrier (1.13%) | 27% | 30th‡ | — |
| **CAG-882 sp003486385**<br>not yet formally named · known from genomes<br>SGB 5062 · GlobDB GCA_003486385 · mOTU 000322 · 8 methods | Unknown | **2.00%** | only 7 reference adults carry it: too few to rank against | 7% | not in reference set | 194 markers; nearest GCA_036718785.1 (0.673) |
| **CAG-882 sp900545175**<br>not yet formally named · known from genomes<br>UHGG 5079 · 1 method: Kraken | Unknown | within *CAG-882* above | no reference | — | not in reference set | — |
| ***Bacteroides clarus***<br>SGB 1832 · GlobDB GCF_900129655 · mOTU 002222 · 9 methods | Conditional | **1.39%** | **+658%** vs typical carrier (0.125%) | 13% | 86th | 184 markers; nearest GCA_023443365.1 (0.593) |

| ORGANISM | CLASS | SHARE | LEVEL | CARRIED BY | PERCENTILE | STRAIN |
|---|---|---|---|---|---|---|
| ***Bacteroides ovatus***<br>mOTU 000239 · UHGG 2098 · 3 methods | Conditional | within *Bacteroides* above | no reference | — | not in reference set | — |
| **Bacteroides GCMETA_00083129**<br>not yet formally named · known from genomes<br>GlobDB GCMETA_00083129 · 1 method: GlobDB sketch | Conditional | within *Bacteroides* above | **–59%** vs typical carrier (0.277%) | 46% | 33rd‡ | — |
| **Bacteroides GCMETA_00046043**<br>not yet formally named · known from genomes<br>GlobDB GCMETA_00046043 · 1 method: GlobDB sketch | Conditional | within *Bacteroides* above | **–71%** vs typical carrier (0.244%) | 26% | 12th‡ | — |
| ***Bacteroides gallinarum***<br>UHGG 2515 · 1 method: Kraken | Conditional | within *Bacteroides* above | no reference | — | not in reference set | — |
| ***Bacteroides coagulans***<br>1 method: MP3 | Conditional | within *Bacteroides* above | ▲ high<br>**+279%** vs typical carrier (0.011%) | 1% | 91st | — |
| **PeH17 sp000435055**<br>not yet formally named · known from genomes<br>SGB 14306 · GTDB GCA_000435055.1 · mOTU 000199 · 7 methods | Unknown | **1.34%** | no reference | — | not in reference set | ● 187 markers; nearest GCA_003512725.1 (0.617) |
| **PeH17 GCMETA_00600598**<br>not yet formally named · known from genomes<br>GlobDB GCMETA_00600598 · 2 methods: GlobDB sketch, SingleM | Unknown | within *PeH17* above | **+50%** vs typical carrier (0.854%) | 18% | 67th‡ | — |
| ***Ruminococcaceae bacterium AM53 2BH***<br>also CAG-115 sp003531585<br>SGB 4191 · GlobDB GCA_003531585 · mOTU 000306 · 9 methods | Conditional | **1.20%** | **+508%** vs typical carrier (0.196%) | 18% | 89th‡ | ● 181 markers; nearest GCA_000436675.1 (0.706) |
| **CAG-115 sp000432175**<br>not yet formally named · known from genomes<br>UHGG 2985 · 1 method: Kraken | Unknown | within *CAG-115* above | no reference | — | not in reference set | — |
| **CAG-115 sp945923775**<br>not yet formally named · known from genomes<br>1 method: SingleM | Unknown | within *CAG-115* above | no reference | — | not in reference set | — |
| ***Bacteroides finegoldii***<br>also Bacteroides zhangwenhongii<br>SGB 1862 · GlobDB GCF_009193325 · mOTU 002746 · 8 methods | Conditional | **1.17%** | **+107%** vs typical carrier (0.256%) | 27% | 63rd | ● 185 markers; nearest GCA_937983295.1 (0.597) |
| ***Mediterraneibacter lactaris***<br>formerly Ruminococcus lactaris<br>SGB 4557 · GlobDB GCF_025152405 · mOTU 001332 · 9 methods | Beneficial | **1.13%** | **+480%** vs typical carrier (0.463%) | 47% | 95th | ● 180 markers |
| **Mediterraneibacter GCMETA_00081181**<br>not yet formally named · known from genomes<br>GlobDB GCMETA_00081181 · 1 method: GlobDB sketch | Conditional | within *Mediterraneibacter* above | **–86%** vs typical carrier (0.766%) | 67% | 27th‡ | — |
| **Limisoma sp000437795**<br>not yet formally named · known from genomes<br>SGB 2071 · GlobDB GCA_022009815 · mOTU 000178 · 8 methods | Unknown | **1.04%** | only 7 reference adults carry it:<br>too few to rank against | 7% | not in reference set | ● 185 markers; nearest GCA_001701305.1 (0.450) |
| ***Gemmiger formicilis***<br>SGB 15300 · GlobDB GCF_900167555 · mOTU 000050 · 9 methods | Beneficial | **1.04%** | **+38%** vs typical carrier (0.696%) | 29% | 62nd‡ | ● 154 markers; nearest GCA_003525645.1 (0.516) |
| **Gemmiger MOTU40_023295**<br>not yet formally named · known from genomes<br>GlobDB MOTU40_023295 · 1 method: GlobDB sketch | Beneficial | within *Gemmiger* above | **+82%** vs typical carrier (0.522%) | 39% | 64th‡ | — |
| **Gemmiger sp900540775**<br>not yet formally named · known from genomes<br>1 method: Kraken panel | Beneficial | within *Gemmiger* above | no reference | — | not in reference set | — |
| **Gemmiger sp900540595**<br>not yet formally named · known from genomes<br>UHGG 5075 · 2 methods: Kraken, Kraken panel | Beneficial | within *Gemmiger* above | no reference | — | not in reference set | — |
| **CAG-196 sp002102975**<br>not yet formally named · known from genomes<br>1 method: MP3 | Unknown | **1.01%**◦ | no reference | — | not in reference set | — |

| ORGANISM | CLASS | SHARE | LEVEL | CARRIED BY | PERCENTILE | STRAIN |
|---|---|---|---|---|---|---|
| ***Dorea longicatena***<br>also Dorea_A longicatena<br>SGB 4581 · GlobDB GCF_025150085 · mOTU 000608 · 8 methods | Conditional | **1.01%** | **+233%** vs typical carrier (0.582%) | 85% | 89th | 125 markers; nearest GCF_025150085.1 (0.697) |
| **Dorea GWHBMXR00000000**<br>not yet formally named · known from genomes<br>GlobDB GWHBMXR00000000 · 2 methods: GlobDB sketch, SingleM | Conditional | within *Dorea* above | **+110%** vs typical carrier (0.244%) | 28% | 86th‡ | — |
| **Dorea_A MGYG000000038**<br>not yet formally named · known from genomes<br>UHGG 172 · 3 methods | Conditional | within *Dorea* above | no reference | — | not in reference set | — |
| ***Dorea_A amylophila***<br>GlobDB GCF_025567265 · mOTU 000388 · 4 methods | Conditional | within *Dorea* above | **–24%** vs typical carrier (0.452%) | 45% | 40th‡ | — |
| **Dorea_A sp019421265**<br>not yet formally named · known from genomes<br>UHGG 3241 · 1 method: Kraken | Conditional | within *Dorea* above | no reference | — | not in reference set | — |
| **Dorea_A GEMOTU_42092**<br>not yet formally named · known from genomes<br>1 method: SingleM | Conditional | within *Dorea* above | no reference | — | not in reference set | — |
| **Faecalibacterium sp900539945**<br>not yet formally named · known from genomes<br>SGB 15346 · GlobDB GCA_025757845 · mOTU 000011 · 8 methods | Beneficial | **0.998%** | **+132%** vs typical carrier (0.477%) | 53% | 94th‡ | 142 markers; nearest GCA_037068045.1 (0.450) |
| ***Blautia faecis***<br>also Blautia_A faecis<br>SGB 4820 · GlobDB GCF_020554845 · mOTU 000200 · 6 methods | Conditional | **0.945%** | **≈ typical** vs typical carrier (0.517%) | 82% | 52nd‡ | 165 markers; nearest GCA_017851495.1 (0.467) |
| ***Roseburia amylophila***<br>also Roseburia_C amylophila<br>SGB 4938 · GlobDB GCF_025567465 · mOTU 000332 · 8 methods | Beneficial | **0.941%** | **+69%** vs typical carrier (0.572%) | 15% | 80th‡ | 67 markers; nearest GCA_014865845.1 (0.513) |
| ***Fusicatenibacter saccharivorans***<br>also Fusicatenibacter sp. (SGB4874)<br>SGB 4874 · GTDB GCF_027667425.1 · mOTU 000133 · 8 methods | Beneficial | **0.938%** | **+133%** vs typical carrier (0.820%) | 92% | 74th | 156 markers |
| **Fusicatenibacter GCMETA_00500418**<br>not yet formally named · known from genomes<br>GlobDB GCMETA_00500418 · 2 methods: GlobDB sketch, SingleM | Beneficial | within *Fusicatenibacter* above | **≈ typical** vs typical carrier (1.02%) | 77% | 49th‡ | — |
| **CAG-267 sp001917135**<br>not yet formally named · known from genomes<br>SGB 9333 · GlobDB GCA_001917135 · mOTU 002009 · 9 methods | Unknown | **0.907%** | only 8 reference adults carry it: too few to rank against | 8% | not in reference set | 162 markers; nearest GCA_021768105.1 (0.593) |
| **CAG-267 MGYG000331255**<br>not yet formally named · known from genomes<br>1 method: SingleM | Unknown | within *CAG-267* above | no reference | — | not in reference set | — |
| **Anaerotardibacter sp000436075**<br>not yet formally named · known from genomes<br>SGB 14322 · GlobDB GCA_000436075 · mOTU 000070 · 8 methods | Unknown | **0.876%** | **+461%** vs typical carrier (0.163%) | 20% | 95th‡ | — |
| **Anaerotardibacter sp965147435**<br>not yet formally named · known from genomes<br>UHGG 5044 · 2 methods: Kraken, SingleM | Unknown | within *Anaerotardibacter* above | no reference | — | not in reference set | — |
| **Roseburia sp900550935**<br>not yet formally named · known from genomes<br>SGB 4948 · GlobDB GCA_900550935 · mOTU 001783 · 7 methods | Beneficial | **0.862%** | only 2 reference adults carry it: too few to rank against | 2% | not in reference set | 148 markers; nearest GCA_902461975.1 (0.258) |
| ***Roseburia sp AM59 24XD***<br>also Jutongia sp003524945<br>SGB 4654 · GlobDB GCF_003478445 · mOTU 002534 · 8 methods | Beneficial | **0.854%** | only 2 reference adults carry it: too few to rank against | 2% | not in reference set | 145 markers; nearest GCA_000432755.1 (0.529) |
| **Clostridium sp CAG 138**<br>not yet formally named · known from genomes<br>1 method: MP3 | Conditional | **0.837%**◦ | no reference | — | not in reference set | — |

| ORGANISM | CLASS | SHARE | LEVEL | CARRIED BY | PERCENTILE | STRAIN |
|---|---|---|---|---|---|---|
| ***Faecalibacillus intestinalis***<br>SGB 6754 · GlobDB GCF_003024685 · mOTU 000457 · 8 methods | Conditional | **0.828%** | **+112%** vs typical carrier (0.418%) | 50% | 86th‡ | 111 markers; nearest GCA_001916255.1 (0.157) |
| **Hominimerdicola MGYG000002321**<br>not yet formally named · known from genomes includes reads the marker catalogue read as Ruminococcus bicirculans ex Wegman et al 2014<br>GlobDB MGYG000002321 · UHGG 3288 · 4 methods | Unknown | **0.822%**§ | **–55%** vs typical carrier (1.98%) | 44% | 41st‡ | — |
| ***Alistipes putredinis***<br>SGB 2318 · GlobDB GCF_000154465 · mOTU 000454 · 9 methods | Conditional | **0.799%** | **–50%** vs typical carrier (3.30%) | 75% | 33rd | 142 markers; nearest GCF_000154465.1 (0.080) |
| **Alistipes sp047233565**<br>not yet formally named · known from genomes<br>GlobDB GCA_047233565 · 2 methods: GTDB sketch, GlobDB sketch | Conditional | within *Alistipes* above | **≈ typical** vs typical carrier (0.332%) | 23% | 52nd‡ | — |
| **Alistipes MGYG000004006**<br>not yet formally named · known from genomes<br>UHGG 5167 · 3 methods | Conditional | within *Alistipes* above | no reference | — | not in reference set | — |
| **Alistipes sp CAG 268**<br>not yet formally named · known from genomes<br>1 method: MP3 | Conditional | within *Alistipes* above | no reference | — | not in reference set | — |
| ***Alistipes obesi***<br>1 method: MP3 | Conditional | within *Alistipes* above | no reference | — | not in reference set | — |
| ***Bacteroides uniformis***<br>SGB 1836 · GlobDB GCF_025147485 · mOTU 000312 · 9 methods | Conditional | **0.779%** | **–58%** vs typical carrier (4.38%) | 92% | 32nd | 53 markers; nearest GCF_025147485.1 (0.381) |
| ***Ruminococcus bicirculans***<br>also Hominimerdicola aceti<br>SGB 4262 · GTDB GCF_025567495.1 · mOTU 000045 · 6 methods | Conditional | **0.759%**° | **≈ typical** vs typical carrier (0.411%) | 60% | 50th | 127 markers |
| ***Blautia luti***<br>also Blautia_A sp900066205<br>SGB 4829 · SGB 4832 · GlobDB GCF_013304735 · 7 methods | Conditional | **0.755%** | **+93%** vs typical carrier (0.245%) | 11% | 73rd‡ | 65 markers; nearest GCF_013304735.1 (0.370) |
| ***Allocoprococcus comes***<br>GlobDB GCF_025149785 · mOTU 000371 · UHGG 2019 · 6 methods | Unknown | **0.718%**† | **+80%** vs typical carrier (0.509%) | 70% | 80th‡ | — |
| ***Eubacterium siraeum***<br>also Ruminiclostridium_E siraeum<br>SGB 4198 · GlobDB GCA_025150425 · mOTU 000039 · 9 methods | Beneficial | **0.617%** | **+207%** vs typical carrier (0.396%) | 56% | 64th | 93 markers; nearest GCA_000210635.1 (0.615) |
| **Ruminiclostridium_E sp900539195**<br>not yet formally named · known from genomes<br>UHGG 5004 · 1 method: Kraken | Unknown | within *Ruminiclostridium* above | no reference | — | not in reference set | — |
| **Ruminiclostridium_E sp902490585**<br>not yet formally named · known from genomes<br>UHGG 5082 · 1 method: Kraken | Unknown | within *Ruminiclostridium* above | no reference | — | not in reference set | — |
| **Ruminiclostridium_E GCMETA_00067854**<br>not yet formally named · known from genomes<br>1 method: SingleM | Unknown | within *Ruminiclostridium* above | no reference | — | not in reference set | — |
| ***Roseburia faecis***<br>SGB 4925 · GlobDB GCF_045061065 · mOTU 000212 · 9 methods | Beneficial | **0.586%** | **–53%** vs typical carrier (1.47%) | 81% | 35th | 92 markers; nearest GCA_000433615.1 (0.388) |
| ***Dorea formicigenerans***<br>SGB 4575 · GTDB GCF_025150245.1 · mOTU 000271 · 8 methods | Conditional | **0.582%** | **▲ high**<br>**+491%** vs typical carrier (0.212%) | 85% | 96th | 112 markers; nearest GCF_025150245.1 (0.308) |
| ***Parabacteroides distasonis***<br>SGB 1934 · GlobDB GCF_000012845 · mOTU 000211 · 9 methods | Conditional | **0.570%** | **+57%** vs typical carrier (0.950%) | 85% | 64th | 113 markers; nearest GCF_000012845.1 (0.341) |
| **Parabacteroides MGYG000321710**<br>not yet formally named · known from genomes<br>GlobDB MGYG000321710 · 1 method: GlobDB sketch | Conditional | within *Parabacteroides* above | no reference | — | not in reference set | — |

| ORGANISM | CLASS | SHARE | LEVEL | CARRIED BY | PERCENTILE | STRAIN |
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| ***Parabacteroides merdae***<br>UHGG 184 · 1 method: Kraken | Conditional | within *Parabacteroides* above | no reference | — | not in reference set | — |
| ***Eubacterium ramulus***<br>also Eubacterium_I ramulus_A<br>SGB 4960 · GlobDB GCF_003122485 · mOTU 001873 · 8 methods | Conditional | **0.569%**† | **+287%** vs typical carrier (0.080%) | 68% | 87th | 101 markers; nearest GCF_003122485.1 (0.481) |
| ***Phocaeicola massiliensis***<br>formerly Bacteroides massiliensis<br>SGB 1812 · GTDB GCF_000382445.1 · mOTU 002531 · 5 methods | Conditional | **0.549%** | reference catalogue reads this species differently | 33% | not comparable | 85 markers; nearest GCA_000438395.1 (0.502) |
| ***Coprococcus catus***<br>also Coprococcus_A intestinihominis<br>SGB 4670 · GTDB GCF_040096555.1 · mOTU 000794 · 8 methods | Beneficial | **0.534%** | **×11** vs typical carrier (0.114%) | 69% | 99th | 112 markers; nearest GCA_024462395.1 (0.374) |
| ***Anaerobutyricum soehngenii***<br>SGB 4537 · GlobDB GCF_900209925 · mOTU 000333 · 7 methods | Beneficial | **0.527%** | **+51%** vs typical carrier (0.325%) | 55% | 71st‡ | 78 markers; nearest GCA_018379315.1 (1.056) |
| ***GGB9634 SGB15106***<br>also Vescimonas sanitatis<br>SGB 15106 · GlobDB GCF_049060325 · mOTU 000023 · 8 methods | Unknown | **0.526%** | ≈ **typical** vs typical carrier (0.492%) | 27% | 44th‡ | 65 markers; nearest GCF_049060325.1 (1.201) |
| **Vescimonas sp900545585**<br>not yet formally named · known from genomes<br>SGB 15093 · GlobDB GCA_900545585 · mOTU 000162 · 6 methods | Unknown | within *Vescimonas* above | **–81%** vs typical carrier (0.318%) | 15% | <1st‡ | — |
| **Vescimonas sp900547745**<br>not yet formally named · known from genomes<br>GlobDB GCA_900547745 · 2 methods: GTDB sketch, GlobDB sketch | Unknown | within *Vescimonas* above | only 4 reference adults carry it: too few to rank against | 4% | not in reference set | — |
| **Duodenibacillus sp900544255**<br>not yet formally named · known from genomes<br>SGB 9274 · GlobDB GCA_900544255 · mOTU 000506 · 8 methods | Conditional | **0.525%** | only 3 reference adults carry it: too few to rank against | 3% | not in reference set | 114 markers; nearest GCA_900544255.1 (0.670) |
| ***Agathobacter rectalis***<br>formerly Eubacterium rectale<br>SGB 4933 · GlobDB GCF_000020605 · mOTU 000006 · 9 methods | Beneficial | **0.518%** | **–50%** vs typical carrier (2.15%) | 88% | 37th | 87 markers; nearest GCF_000020605.1 (0.488) |
| **CAG-177 sp000431775**<br>not yet formally named · known from genomes<br>SGB 4364 · GTDB GCA_037368805.1 · mOTU 000522 · 6 methods | Unknown | **0.515%** | no reference | — | not in reference set | 73 markers; nearest GCA_021618165.1 (0.497) |
| **CAG-177 GWHGOKI00000000**<br>not yet formally named · known from genomes<br>GlobDB GWHGOKI00000000 · 2 methods: GlobDB sketch, SingleM | Unknown | within *CAG-177* above | only 1 reference adult carry it: too few to rank against | 1% | not in reference set | — |
| ***Roseburia hominis***<br>also Roseburia sp902781225<br>SGB 4936 · GlobDB GCF_000225345 · mOTU 000252 · 9 methods | Beneficial | **0.491%** | **+438%** vs typical carrier (0.229%) | 83% | 90th | 83 markers; nearest GCA_963654245.1 (1.681) |
| **CAG-313 sp003539625**<br>not yet formally named · known from genomes<br>SGB 6579 · GlobDB GCA_003539625 · mOTU 003189 · 8 methods | Unknown | **0.464%** | only 3 reference adults carry it: too few to rank against | 3% | not in reference set | 75 markers; nearest GCA_036793425.1 (0.558) |
| **CAG-313 GCMETA_00004955**<br>not yet formally named · known from genomes<br>1 method: SingleM | Unknown | within *CAG-313* above | no reference | — | not in reference set | — |
| **Angelakisella sp900547385**<br>not yet formally named · known from genomes<br>SGB 15356 · GlobDB GCA_900547385 · mOTU 000026 · 7 methods | Unknown | **0.463%** | **+85%** vs typical carrier (0.243%) | 18% | 78th‡ | 60 markers; nearest GCA_021640885.1 (0.469) |
| ***Adlercreutzia equolifaciens***<br>SGB 14797 · SGB 14798 · GTDB GCF_000478885.1 · 4 methods | Beneficial | **0.428%** | no reference | — | not in reference set | — |

| ORGANISM | CLASS | SHARE | LEVEL | CARRIED BY | PERCENTILE | STRAIN |
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| **Adlercreutzia MOTU40_022750**<br>not yet formally named · known from genomes<br>GlobDB MOTU40_022750 · 1 method: GlobDB sketch | Beneficial | within *Adlercreutzia* above | only 9 reference adults carry it: too few to rank against | 9% | not in reference set | — |
| ***Adlercreutzia rubneri***<br>GTDB GCF_009755265.1 · mOTU 000131 · UHGG 3487 · 3 methods | Beneficial | within *Adlercreutzia* above | no reference | — | not in reference set | — |
| **Adlercreutzia MOTU40_022751**<br>not yet formally named · known from genomes<br>GlobDB MOTU40_022751 · 1 method: GlobDB sketch | Beneficial | within *Adlercreutzia* above | only 8 reference adults carry it: too few to rank against | 8% | not in reference set | — |
| **Adlercreutzia MOTU40_004459**<br>not yet formally named · known from genomes<br>1 method: SingleM | Beneficial | within *Adlercreutzia* above | no reference | — | not in reference set | — |
| ***Lachnospira eligens***<br>formerly Eubacterium eligens<br>SGB 5082 · 4 methods | Beneficial | **0.407%**† | reference catalogue reads this species differently | 81% | not comparable | 189 markers; nearest GCA_051978475.1 (0.922) |
| **Lachnospira MOTU40_010844**<br>not yet formally named · known from genomes<br>GlobDB MOTU40_010844 · 1 method: GlobDB sketch | Beneficial | within *Lachnospira* above | only 7 reference adults carry it: too few to rank against | 7% | not in reference set | — |
| **Lachnospira sp003537285**<br>not yet formally named · known from genomes<br>UHGG 435 · 1 method: Kraken | Beneficial | within *Lachnospira* above | no reference | — | not in reference set | — |
| ***Clostridiaceae bacterium***<br>also Claveliimonas acetigenes<br>SGB 4629 · GlobDB GCF_025567385 · mOTU 027746 · 7 methods | Conditional | **0.402%** | **+144%** vs typical carrier (0.128%) | 22% | >99th‡ | 38 markers |
| ***Collinsella aerofaciens***<br>also Collinsella sp023656745<br>SGB 14535_group · SGB 14535 · 3 methods | Opportunist | **0.399%** | **+5%** vs typical carrier (0.841%) | 86% | 51st | — |
| **Collinsella sp003458415**<br>not yet formally named · known from genomes<br>GlobDB GCF_003458415 · 2 methods: GTDB sketch, GlobDB sketch | Conditional | within *Collinsella* above | no reference | — | not in reference set | — |
| **Collinsella MOTU40_005860**<br>not yet formally named · known from genomes<br>GlobDB MOTU40_005860 · 1 method: GlobDB sketch | Conditional | within *Collinsella* above | no reference | — | not in reference set | — |
| ***Collinsella aerofaciens H***<br>also Collinsella aerofaciens_F / aerofaciens_H<br>mOTU 000129 · 2 methods: mOTUs, SingleM | Conditional | within *Collinsella* above | no reference | — | not in reference set | — |
| **Collinsella MOTU40_045831**<br>not yet formally named · known from genomes<br>1 method: SingleM | Conditional | within *Collinsella* above | no reference | — | not in reference set | — |
| **Collinsella sp902464575**<br>not yet formally named · known from genomes<br>1 method: SingleM | Conditional | within *Collinsella* above | no reference | — | not in reference set | — |
| ***Anthropogastromicrobium aceti***<br>also Anthropogastromicrobium sp934292725<br>SGB 5090 · GlobDB GCF_025567095 · mOTU 000235 · 8 methods | Unknown | **0.390%** | **–19%** vs typical carrier (0.421%) | 70% | 34th‡ | 37 markers |
| **COE1 sp001916965**<br>not yet formally named · known from genomes<br>SGB 4882 · GlobDB GCA_001916965 · mOTU 000316 · 9 methods | Unknown | **0.386%** | **+284%** vs typical carrier (0.100%) | 27% | 78th‡ | 56 markers |
| **COE1 sp900753305**<br>not yet formally named · known from genomes<br>UHGG 2697 · 1 method: Kraken | Unknown | within *COE1* above | no reference | — | not in reference set | — |
| **Merdicola sp900552655**<br>not yet formally named · known from genomes<br>SGB 6321 · GlobDB GCA_900552655 · mOTU 004511 · 7 methods | Unknown | **0.381%** | **–11%** vs typical carrier (0.469%) | 11% | 36th‡ | 37 markers; nearest GCA_036300105.1 (0.264) |
| ***Faecalibacterium wellingii***<br>SGB 15340 · GlobDB GCF_023347535 · mOTU 000125 · 7 methods | Beneficial | **0.369%** | **–37%** vs typical carrier (0.537%) | 36% | 33rd‡ | 27 markers; nearest GCA_018381455.1 (0.711) |

| ORGANISM | CLASS | SHARE | LEVEL | CARRIED BY | PERCENTILE | STRAIN |
|---|---|---|---|---|---|---|
| **Firmicutes bacterium CAG 176**<br>not yet formally named · known from genomes<br>1 method: MP3 | Unknown | **0.366%**° | no reference | — | not in reference set | — |
| ***Agathobaculum butyriciproducens***<br>SGB 14993 · GlobDB GCF_003096535 · mOTU 000033 · 9 methods | Beneficial | **0.363%** | **–23%** vs typical carrier (0.214%) | 89% | 42nd | 57 markers |
| **Onthocola_B sp000437355**<br>not yet formally named · known from genomes<br>SGB 6502 · GlobDB GCA_000437355 · mOTU 001214 · 8 methods | Unknown | **0.359%** | no reference | — | not in reference set | 39 markers |
| **Clostridium sp CAG 413**<br>not yet formally named · known from genomes<br>1 method: MP3 | Conditional | **0.358%**° | **+29%** vs typical carrier (0.772%) | 5% | 55th | — |
| ***Jilunia laotingensis***<br>also Bacteroides laotingensis<br>SGB 1802 · GlobDB GCF_014385165 · mOTU 023556 · 7 methods | Conditional | **0.343%** | no reference | — | not in reference set | 48 markers |
| **HGM10766 sp900757295**<br>not yet formally named · known from genomes<br>SGB 6524 · GlobDB GCA_025992315 · mOTU 008608 · 7 methods | Unknown | **0.337%** | only 3 reference adults carry it:<br>too few to rank against | 3% | not in reference set | 36 markers |
| ***Ellagibacter isourolithinifaciens***<br>SGB 14816 · SGB 14816_group · GlobDB GCF_008831035 · 7 methods | Unknown | **0.327%** | only 7 reference adults carry it:<br>too few to rank against | 7% | not in reference set | — |
| **Ellagibacter sp900554945**<br>not yet formally named · known from genomes<br>UHGG 4012 · 1 method: Kraken | Unknown | within *Ellagibacter* above | no reference | — | not in reference set | — |
| **Ellagibacter MOTU40_010708**<br>not yet formally named · known from genomes<br>1 method: SingleM | Unknown | within *Ellagibacter* above | no reference | — | not in reference set | — |
| ***Clostridium fessum***<br>also Otoolea fessa<br>SGB 4705 · GlobDB GCF_003024715 · mOTU 000318 · 7 methods | Conditional | **0.319%** | **+50%** vs typical carrier (0.221%) | 72% | 72nd‡ | 45 markers |
| ***Anaerobutyricum hallii***<br>formerly Eubacterium hallii<br>SGB 4532 · GlobDB GCF_000173975 · mOTU 000622 · 8 methods | Beneficial | **0.313%** | **+253%** vs typical carrier (0.252%) | 86% | 80th | 39 markers |
| ***Alistipes finegoldii***<br>SGB 2301 · GlobDB GCF_000265365 · mOTU 000410 · 8 methods | Conditional | **0.306%** | **+224%** vs typical carrier (0.229%) | 71% | 72nd | 25 markers |
| ***Alistipes onderdonkii***<br>SGB 2303 · GTDB GCF_025145285.1 · mOTU 000215 · 6 methods | Conditional | **0.304%** | no reference | — | not in reference set | 21 markers |
| ***Oscillibacter sp ER4***<br>also Hominicoprocola sp934835795<br>SGB 15254 · 3 methods | Conditional | **0.299%** | no reference | — | not in reference set | 21 markers |
| ***Hominicoprocola fusiformis***<br>GTDB GCA_020687125.1 · mOTU 000021 · UHGG 4814 · 3 methods | Unknown | within *Hominicoprocola* above | no reference | — | not in reference set | — |
| **Hominicoprocola GCMETA_00000610**<br>not yet formally named · known from genomes<br>GlobDB GCMETA_00000610 · 2 methods: GlobDB sketch, SingleM | Unknown | within *Hominicoprocola* above | **–38%** vs typical carrier (0.593%) | 44% | 20th‡ | — |
| ***Blautia obeum***<br>formerly Ruminococcus obeum<br>SGB 4811 · GlobDB GCF_025147765 · mOTU 000471 · 7 methods | Conditional | **0.293%** | **+248%** vs typical carrier (0.173%) | 83% | 81st | — |
| ***Clostridium sp AM22 11AC***<br>also Enterocloster sp000431375<br>SGB 4749 · GTDB GCF_003473545.1 · mOTU 001218 · 7 methods | Conditional | **0.290%** | no reference | — | not in reference set | 27 markers |

| ORGANISM | CLASS | SHARE | LEVEL | CARRIED BY | PERCENTILE | STRAIN |
|---|---|---|---|---|---|---|
| **Enterocloster GWHFCPF00000000**<br>not yet formally named · known from genomes<br>GlobDB GWHFCPF00000000 · 1 method: GlobDB sketch | Opportunist | within *Enterocloster* above | ▲ high<br>**+116%** vs typical carrier (0.128%) | 24% | 96th‡ | — |
| **Enterocloster sp900541315**<br>not yet formally named · known from genomes<br>UHGG 1909 · 1 method: Kraken | Opportunist | within *Enterocloster* above | no reference | — | not in reference set | — |
| **F23-B02 sp001916715**<br>not yet formally named · known from genomes<br>SGB 15180 · GlobDB GCA_001916715 · mOTU 000128 · 8 methods | Unknown | **0.284%** | only 9 reference adults carry it:<br>too few to rank against | 9% | not in reference set | — |
| ***Bacteroides cellulosilyticus***<br>SGB 1844 · GlobDB GCF_028539415 · mOTU 000775 · 8 methods | Beneficial | **0.282%** | **+187%** vs typical carrier (0.305%) | 49% | 67th | 35 markers |
| ***Odoribacter laneus***<br>also Culturomica lanea<br>SGB 1792 · GlobDB GCF_000243215 · mOTU 014895 · 8 methods | Beneficial | **0.282%** | **+23%** vs typical carrier (0.154%) | 5% | 53rd | — |
| ***Alistipes SGB2313***<br>also Alistipes avistercoris<br>SGB 2313 · GlobDB GCA_019115075 · mOTU 001060 · 5 methods | Conditional | **0.281%** | only 2 reference adults carry it:<br>too few to rank against | 2% | not in reference set | — |
| ***GGB1420 SGB1957***<br>also Caccoplasma intestinavium<br>SGB 1957 · GlobDB GCA_018715885 · mOTU 002371 · 8 methods | Unknown | **0.279%** | **+90%** vs typical carrier (0.327%) | 10% | 64th | — |
| ***Mediterraneibacter faecis***<br>SGB 4563 · GlobDB GCA_001312505 · mOTU 000056 · 7 methods | Conditional | **0.277%** | **–64%** vs typical carrier (0.687%) | 77% | 18th‡ | — |
| **CAG-314 sp000437915**<br>not yet formally named · known from genomes<br>SGB 14114 · GlobDB GCA_000437915 · mOTU 002724 · 7 methods | Unknown | **0.263%** | **–36%** vs typical carrier (0.408%) | 30% | 37th‡ | — |
| ***Blautia intestinalis***<br>also Blautia_A intestinalis<br>SGB 4815 · UHGG 529 · 2 methods: Jan26, Kraken | Conditional | **0.261%** | no reference | — | not in reference set | — |
| **CAG-269 sp003525075**<br>not yet formally named · known from genomes<br>SGB 6340 · GlobDB GCA_003525075 · mOTU 000448 · 9 methods | Unknown | **0.255%** | **–55%** vs typical carrier (0.565%) | 23% | 13th‡ | 22 markers |
| **CAG-269 MOTU40_079257**<br>not yet formally named · known from genomes<br>GlobDB MOTU40_079257 · mOTU 079257 · 2 methods: GlobDB sketch, mOTUs | Unknown | within *CAG-269* above | no reference | — | not in reference set | — |
| **Lawsonibacter sp900066825**<br>not yet formally named · known from genomes<br>UHGG 229 · 1 method: Kraken | Conditional | **0.252%**◦ | no reference | — | not in reference set | — |
| **Ventricola sp900542445**<br>not yet formally named · known from genomes<br>SGB 15373 · GlobDB GCA_900542445 · mOTU 001250 · 7 methods | Unknown | **0.243%** | **+52%** vs typical carrier (0.163%) | 27% | 78th‡ | — |
| ***Brotocaccenecus cirricatena***<br>also Vescimonas cirricatena<br>SGB 15091 · GlobDB GCF_020687045 · mOTU 000100 · 6 methods | Unknown | **0.232%** | **–72%** vs typical carrier (0.957%) | 17% | 12th‡ | — |
| ***Barnesiella intestinihominis***<br>SGB 1965 · GlobDB GCF_000296465 · mOTU 000336 · 9 methods | Beneficial | **0.230%** | **–48%** vs typical carrier (0.844%) | 51% | 35th | — |
| **HGM13006 sp900756575**<br>not yet formally named · known from genomes<br>SGB 14969 · GlobDB GCA_900756575 · mOTU 000691 · 6 methods | Unknown | **0.227%** | only 8 reference adults carry it:<br>too few to rank against | 8% | not in reference set | — |

| ORGANISM | CLASS | SHARE | LEVEL | CARRIED BY | PERCENTILE | STRAIN |
|---|---|---|---|---|---|---|
| **UMGS995 sp900547465**<br>not yet formally named · known from genomes<br>SGB 6440 · GlobDB GCA_900547465 · mOTU 001559 · 7 methods | Unknown | **0.226%** | only 1 reference adult carry it: too few to rank against | 1% | not in reference set | — |
| ***Blautia stercoris***<br>SGB 4788 · GlobDB GCA_003481745 · mOTU 001030 · 8 methods | Conditional | **0.223%** | **–16%** vs typical carrier (0.334%) | 24% | 38th‡ | — |
| ***GGB47687 SGB2286***<br>also Tidjanibacter inops<br>SGB 2286 · GlobDB GCF_900104605 · mOTU 001635 · 9 methods | Unknown | **0.221%** | **–32%** vs typical carrier (0.288%) | 18% | 42nd | — |
| ***Sutterella wadsworthensis***<br>SGB 9283 · GlobDB GCF_003315195 · mOTU 000497 · 9 methods | Conditional | **0.214%** | only 9 reference adults carry it: too few to rank against | 9% | not in reference set | — |
| ***Asaccharobacter celatus***<br>1 method: MP3 | Unknown | **0.212%**◦ | **+790%** vs typical carrier (0.066%) | 63% | 96th | — |
| ***Blautia glucerasea***<br>also Blautia_A acetigignens<br>SGB 4816 · GlobDB GCF_025566865 · mOTU 001618 · 7 methods | Conditional | **0.212%** | **+66%** vs typical carrier (0.144%) | 12% | 67th‡ | — |
| **Acetatifactor sp900755865**<br>not yet formally named · known from genomes<br>SGB 4906 · GlobDB GCA_900755865 · mOTU 016271 · 6 methods | Unknown | **0.211%** | only 3 reference adults carry it: too few to rank against | 3% | not in reference set | — |
| **Acetatifactor SPIREOTU_02871567**<br>not yet formally named · known from genomes<br>GlobDB SPIREOTU_02871567 · 1 method: GlobDB sketch | Unknown | within *Acetatifactor* above | **–84%** vs typical carrier (0.506%) | 71% | 17th‡ | — |
| **Gemmiger sp900539695**<br>not yet formally named · known from genomes<br>SGB 15291 · GlobDB GCA_025758045 · mOTU 000198 · 8 methods | Beneficial | **0.209%** | **+45%** vs typical carrier (0.172%) | 30% | 80th‡ | — |
| ***Oliverpabstia intestinalis***<br>SGB 4868 · GlobDB GCF_009696065 · mOTU 000330 · 7 methods | Unknown | **0.208%** | **–59%** vs typical carrier (0.568%) | 62% | 16th‡ | — |
| ***Oliverpabstia tarda***<br>GlobDB GCA_019421285 · UHGG 4823 · 3 methods | Unknown | within *Oliverpabstia* above | **–57%** vs typical carrier (0.595%) | 36% | 14th‡ | — |
| **Apopatocola sp003522945**<br>not yet formally named · known from genomes<br>SGB 15244 · GlobDB GCA_003522945 · mOTU 000073 · 6 methods | Unknown | **0.203%** | **–8%** vs typical carrier (0.217%) | 11% | 46th‡ | — |
| **Apopatocola MGYG000330632**<br>not yet formally named · known from genomes<br>GlobDB MGYG000330632 · 2 methods: GlobDB sketch, SingleM | Unknown | within *Apopatocola* above | **+100%** vs typical carrier (0.113%) | 18% | 83rd‡ | — |
| **Eubacterium_F sp000433735**<br>not yet formally named · known from genomes<br>SGB 4643 · GlobDB GCA_000433735 · mOTU 001032 · 7 methods | Conditional | **0.201%** | **+26%** vs typical carrier (0.182%) | 23% | 61st‡ | — |
| **UBA11512 sp003522145**<br>not yet formally named · known from genomes<br>SGB 14193 · GlobDB GCA_003522145 · mOTU 003597 · 7 methods | Unknown | **0.200%** | only 1 reference adult carry it: too few to rank against | 1% | not in reference set | — |
| **CAG-433 sp000433675**<br>not yet formally named · known from genomes<br>includes reads the marker catalogue read as CAG-433 sp019423565<br>GlobDB GCA_000433675 · mOTU 000192 · UHGG 3098 · 5 methods | Unknown | **0.196%**§ | no reference | — | not in reference set | — |
| ***Finegoldia magna***<br>SGB 6686_group · GlobDB GCF_003182075 · mOTU 000404 · 9 methods | Unknown | **0.188%**† | **×154** vs typical carrier (0.003%) | 1% | >99th | — |

| ORGANISM | CLASS | SHARE | LEVEL | CARRIED BY | PERCENTILE | STRAIN |
|---|---|---|---|---|---|---|
| ***Candidatus Cibionibacter quicibialis***<br>also Gemmiger qucibialis<br>SGB 15286 · GlobDB GCA_004000625 · mOTU 000308 · 8 methods | Unknown | **0.186%** | **–76%** vs typical carrier (0.857%) | 83% | 19th‡ | — |
| **Clostridium sp CAG 167**<br>not yet formally named · known from genomes<br>1 method: MP3 | Conditional | **0.186%**° | **+218%** vs typical carrier (0.163%) | 26% | 75th | — |
| **Firm-07 sp900548735**<br>not yet formally named · known from genomes<br>SGB 14212 · GlobDB GCA_900548735 · mOTU 000899 · 6 methods | Unknown | **0.185%** | no reference | — | not in reference set | — |
| **UMGS1781 sp900553365**<br>not yet formally named · known from genomes<br>SGB 6290 · GlobDB GCA_900553365 · mOTU 004493 · 6 methods | Unknown | **0.182%** | no reference | — | not in reference set | — |
| **CAG-762 sp000432575**<br>not yet formally named · known from genomes<br>SGB 6460 · GlobDB GCA_000432575 · mOTU 010900 · 8 methods | Unknown | **0.181%** | no reference | — | not in reference set | — |
| ***Odoribacter splanchnicus***<br>SGB 1790 · GlobDB GCF_000190535 · mOTU 002115 · 8 methods | Beneficial | **0.177%** | **+7%** vs typical carrier (0.442%) | 74% | 52nd | — |
| **CAG-269 sp934130935**<br>not yet formally named · known from genomes<br>SGB 6343 · GlobDB GCA_934130935 · mOTU 008568 · 7 methods | Unknown | **0.169%** | no reference | — | not in reference set | — |
| **Lachnospira sp003451515**<br>not yet formally named · known from genomes<br>SGB 236902 · GTDB GCA_003464165.1 · mOTU 000505 · 4 methods | Beneficial | **0.164%** | no reference | — | not in reference set | — |
| ***Lachnospiraceae bacterium***<br>also Choladousia sp902363665<br>SGB 4782 · GlobDB GCA_902363665 · mOTU 000712 · 7 methods | Conditional | **0.163%** | **+24%** vs typical carrier (0.120%) | 33% | 70th‡ | — |
| **Choladousia sp902363135**<br>not yet formally named · known from genomes<br>GlobDB GCA_902363135 · mOTU 000840 · UHGG 279 · 4 methods | Unknown | within *Choladousia* above | **–43%** vs typical carrier (0.136%) | 60% | 7th‡ | — |
| ***Roseburia intestinalis***<br>SGB 4951 · GlobDB GCF_900537995 · mOTU 000297 · 8 methods | Beneficial | **0.162%** | **–10%** vs typical carrier (0.389%) | 73% | 48th | — |
| ***Levyella massiliensis***<br>also Murdochiella massiliensis_A<br>SGB 5300 · GlobDB GCF_000308275 · mOTU 002758 · 7 methods | Unknown | **0.158%** | no reference | — | not in reference set | — |
| ***Phascolarctobacterium_A succinatutens***<br>GlobDB GCF_000188175 · mOTU 000234 · UHGG 2079 · 5 methods | Beneficial | **0.154%**† | only 6 reference adults carry it: too few to rank against | 6% | not in reference set | — |
| ***Alistipes communis***<br>SGB 2290 · GlobDB GCF_006542665 · mOTU 001284 · 6 methods | Conditional | **0.145%** | **–34%** vs typical carrier (0.218%) | 50% | 20th‡ | — |
| ***GGB3653 SGB4964***<br>also Maccoyibacter intestinihominis<br>SGB 4964 · GlobDB GCF_040096355 · mOTU 000232 · 7 methods | Unknown | **0.142%** | **–27%** vs typical carrier (0.242%) | 68% | 29th‡ | — |
| ***Lachnospira hominis ex Liu et al 2021***<br>also Lachnospira hominis<br>SGB 5087 · GlobDB GCF_014287955 · mOTU 001554 · 7 methods | Beneficial | **0.129%** | **–47%** vs typical carrier (0.252%) | 34% | 24th‡ | — |
| ***Vescimonas coprocola***<br>SGB 15089 · GlobDB GCF_018408575 · mOTU 000080 · 9 methods | Unknown | **0.128%** | **+67%** vs typical carrier (0.153%) | 76% | 62nd | — |

| ORGANISM | CLASS | SHARE | LEVEL | CARRIED BY | PERCENTILE | STRAIN |
|---|---|---|---|---|---|---|
| ***Pseudoruminococcus massiliensis***<br>SGB 4280 · GlobDB GCF_900291955 · mOTU 000773 · 7 methods | Unknown | **0.122%** | **–95%** vs typical carrier (2.76%) | 13% | 8th‡ | — |
| ***Eubacterium_I ramulus***<br>GlobDB GCF_000469345 · mOTU 000772 · UHGG 2239 · 6 methods | Conditional | **0.122%**† | **–45%** vs typical carrier (0.258%) | 58% | 26th‡ | — |
| **CAG-433 sp900546245**<br>not yet formally named · known from genomes<br>SGB 6421 · GlobDB GCA_900546245 · UHGG 5810 · 6 methods | Unknown | **0.119%** | only 1 reference adult carry it: too few to rank against | 1% | not in reference set | — |
| ***Anaerotignum faecicola***<br>also Anaerotignum_A faecicola<br>SGB 5190 · GlobDB GCA_003865035 · mOTU 000096 · 6 methods | Conditional | **0.118%** | **–29%** vs typical carrier (0.180%) | 45% | 31st‡ | — |
| **Bilophila sp900550745**<br>not yet formally named · known from genomes<br>SGB 15451 · GlobDB GCA_900550745 · mOTU 005917 · 6 methods | Opportunist | **0.117%** | only 2 reference adults carry it: too few to rank against | 2% | not in reference set | — |
| ***Bilophila wadsworthia***<br>mOTU 000928 · 1 method: mOTUs | Opportunist | within *Bilophila* above | no reference | — | not in reference set | — |
| **HGM12814 sp900770165**<br>not yet formally named · known from genomes<br>UHGG 4696 · 1 method: Kraken | Unknown | **0.116%**◦ | no reference | — | not in reference set | — |
| **CAG-170 sp002404795**<br>not yet formally named · known from genomes<br>SGB 15236 · GlobDB MOTU40_000495 · GTDB GCA_002404795.1 · 7 methods | Unknown | **0.113%** | **–46%** vs typical carrier (0.237%) | 32% | 25th‡ | — |
| **CAG-170 sp902796245**<br>not yet formally named · known from genomes<br>SGB 15235 · GlobDB GCA_902796245 · mOTU 003975 · 5 methods | Unknown | within *CAG-170* above | no reference | — | not in reference set | — |
| **UBA11524 sp000437595**<br>not yet formally named · known from genomes<br>SGB 15368 · GlobDB MOTU40_000003 · GTDB GCA_000437595.1 · 8 methods | Unknown | **0.112%** | only 3 reference adults carry it: too few to rank against | 3% | not in reference set | — |
| ***Blautia faecicola***<br>also Oliverpabstia faecicola<br>SGB 4867 · GlobDB GCF_004123145 · mOTU 001286 · 7 methods | Conditional | **0.112%** | **×24** vs typical carrier (0.007%) | 43% | 88th | — |
| ***Lachnospira pectinoschiza***<br>also Lachnospira intestinalis<br>SGB 5089 · GlobDB GCF_040095895 · mOTU 000439 · 8 methods | Beneficial | **0.109%** | **–87%** vs typical carrier (0.727%) | 54% | 22nd | — |
| ***Clostridiaceae bacterium AF18 31LB***<br>also Hominiventricola aquisgranensis<br>SGB 4767 · GTDB GCF_040096225.1 · mOTU 000512 · 5 methods | Conditional | **0.108%** | no reference | — | not in reference set | — |
| **Hominiventricola GEMOTU_42051**<br>not yet formally named · known from genomes<br>GlobDB GEMOTU_42051 · 1 method: GlobDB sketch | Unknown | within *Hominiventricola* above | **≈ typical** vs typical carrier (0.134%) | 15% | 60th‡ | — |
| ***Hominilimicola fabiformis***<br>SGB 4269 · GlobDB GCF_020687385 · mOTU 000168 · 7 methods | Unknown | **0.107%** | **–56%** vs typical carrier (0.318%) | 65% | 12th‡ | — |
| ***Coprobacter fastidiosus***<br>SGB 1963 · GlobDB GCF_030296935 · mOTU 001330 · 7 methods | Conditional | **0.107%** | **+143%** vs typical carrier (0.101%) | 26% | 70th | — |
| ***Hominifimenecus microfluidus***<br>SGB 4777 · GlobDB GCF_020687485 · mOTU 005112 · 5 methods | Unknown | **0.105%** | **+12%** vs typical carrier (0.128%) | 19% | 68th‡ | — |
| ***Peptoniphilaceae bacterium AM52 5BH***<br>also Anaerococcus sp027669155<br>SGB 6678 · GlobDB MOTU40_011098 · GTDB GCA_027669155.1 · 6 methods | Unknown | **0.105%** | no reference | — | not in reference set | — |

| ORGANISM | CLASS | SHARE | LEVEL | CARRIED BY | PERCENTILE | STRAIN |
|---|---|---|---|---|---|---|
| **Anaerococcus MOTU40_011099**<br>not yet formally named · known from genomes<br>GlobDB MOTU40_011099 · 1 method: GlobDB sketch | Unknown | within *Anaerococcus* above | no reference | — | not in reference set | — |
| **Anaerococcus sp. (SGB6678)**<br>not yet formally named · known from genomes<br>SGB 6678 · 1 method: Jun23 | Unknown | within *Anaerococcus* above | no reference | — | not in reference set | — |
| **Anaerococcus MGYG000303866**<br>not yet formally named · known from genomes<br>GlobDB MGYG000303866 · 1 method: GlobDB sketch | Unknown | within *Anaerococcus* above | no reference | — | not in reference set | — |
| **Anaerococcus MOTU40_016832**<br>not yet formally named · known from genomes<br>GlobDB MOTU40_016832 · 1 method: GlobDB sketch | Unknown | within *Anaerococcus* above | no reference | — | not in reference set | — |
| ***Anaerococcus prevotii***<br>1 method: MP3 | Unknown | within *Anaerococcus* above | no reference | — | not in reference set | — |
| **Anaerococcus MOTU40_024141**<br>not yet formally named · known from genomes<br>1 method: SingleM | Unknown | within *Anaerococcus* above | no reference | — | not in reference set | — |
| ***Gallintestinimicrobium propionicum***<br>SGB 4993 · GlobDB GCF_025567525 · mOTU 000260 · 9 methods | Unknown | **0.103%** | **–73%** vs typical carrier (0.437%) | 36% | 3rd‡ | — |
| ***Mediterraneibacter torques***<br>formerly Ruminococcus torques<br>SGB 4608 · GTDB GCF_000153925.1 · mOTU 002160 · 7 methods | Opportunist | **0.103%** | **+68%** vs typical carrier (0.441%) | 78% | 64th | — |
| ***Dorea hominis***<br>also Dorea_D hominis<br>an earlier catalogue listed this population as Mediterraneibacter gnavus<br>SGB 4571 · GlobDB GCF_014287705 · mOTU 003001 · 7 methods | Conditional | **0.102%** | only 4 reference adults carry it:<br>too few to rank against | 4% | not in reference set | — |
| **F23-B02 sp002472405**<br>not yet formally named · known from genomes<br>SGB 53522 · GTDB GCA_002472405.1 · mOTU 000223 · 6 methods | Unknown | **0.096%** | no reference | — | not in reference set | — |
| **CAG-269 sp900554175**<br>not yet formally named · known from genomes<br>SGB 6331 · GlobDB GCA_900554175 · mOTU 011127 · 6 methods | Unknown | **0.094%** | only 2 reference adults carry it:<br>too few to rank against | 2% | not in reference set | — |
| ***Dorea sp AF36 15AT***<br>also Claveliimonas ammoniilytica<br>SGB 4552 · GlobDB GCF_025567415 · mOTU 000335 · 6 methods | Conditional | **0.092%** | ≈ **typical** vs typical carrier (0.118%) | 37% | 43rd‡ | — |
| ***Bacteroides thetaiotaomicron***<br>SGB 1861 · GTDB GCF_000011065.1 · mOTU 000293 · 7 methods | Conditional | **0.092%** | **–49%** vs typical carrier (0.544%) | 79% | 36th | — |
| ***Coprococcus comes***<br>also Allocoprococcus comes_A<br>SGB 4577 · 4 methods | Beneficial | **0.091%**† | reference catalogue reads this species differently | 79% | not comparable | 107 markers; nearest GCA_934329715.1 (1.933) |
| **JAZOXB01 sp040096185**<br>not yet formally named · known from genomes<br>SGB 15216 · GTDB GCF_040096185.1 · mOTU 000028 · 5 methods | Unknown | **0.089%** | no reference | — | not in reference set | — |
| **JAZOXB01 SPECIV4_04115**<br>not yet formally named · known from genomes<br>GlobDB SPECIV4_04115 · 1 method: GlobDB sketch | Unknown | within *JAZOXB01* above | **–41%** vs typical carrier (0.179%) | 66% | 24th‡ | — |
| ***Wujia chipingensis***<br>also Coprococcus_C hominis<br>SGB 5111 · GlobDB GCF_020687505 · mOTU 000368 · 6 methods | Beneficial | **0.086%** | ▼ low<br>**–89%** vs typical carrier (1.01%) | 22% | 4th‡ | — |
| **Tyzzerella sp**<br>not yet formally named · known from genomes<br>1 method: MP3 | Unknown | **0.086%**◦ | no reference | — | not in reference set | — |

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| ***Hallella colorans***<br>also Prevotella colorans<br>SGB 1502 · GTDB GCF_003096815.1 · mOTU 002387 · 6 methods | Conditional | **0.086%** | **+111%** vs typical carrier (0.006%) | 0% | 71st | — |
| **Prevotella sp CAG 279**<br>not yet formally named · known from genomes<br>1 method: MP3 | Conditional | within *Prevotella* above | **+163%** vs typical carrier (0.655%) | 17% | 73rd | — |
| **Prevotella MGYG000001040**<br>not yet formally named · known from genomes<br>UHGG 1679 · 2 methods: Kraken, Kraken panel | Conditional | within *Prevotella* above | no reference | — | not in reference set | — |
| **Prevotella MGYG000304160**<br>not yet formally named · known from genomes<br>GlobDB MGYG000304160 · 1 method: GlobDB sketch | Conditional | within *Prevotella* above | no reference | — | not in reference set | — |
| ***Prevotella buccalis***<br>SGB 1517_group · GTDB GCF_025151385.1 · 3 methods | Conditional | within *Prevotella* above | **+344%** vs typical carrier (0.011%) | 3% | 86th | — |
| **Scatousia sp000431315**<br>not yet formally named · known from genomes<br>1 method: Kraken panel | Unknown | **0.086%**° | no reference | — | not in reference set | — |
| **CAG-103 sp900543625**<br>not yet formally named · known from genomes<br>includes reads the marker catalogue read as CAG-103 sp905215475<br>GlobDB GCA_900543625 · mOTU 000031 · UHGG 1791 · 4 methods | Unknown | **0.084%**§ | **–75%** vs typical carrier (0.467%) | 28% | 4th‡ | — |
| **CAG-698 sp000431235**<br>not yet formally named · known from genomes<br>UHGG 5662 · 1 method: Kraken | Unknown | **0.083%**° | no reference | — | not in reference set | — |
| ***Roseburia zhanii***<br>also Roseburia_B zhanii<br>SGB 4963 · GlobDB GCF_014287635 · mOTU 007284 · 6 methods | Beneficial | **0.081%** | **–18%** vs typical carrier (0.143%) | 13% | 15th‡ | — |
| ***Hominiventricola filiformis***<br>SGB 4768 · GlobDB GCF_020686965 · mOTU 000726 · 6 methods | Unknown | **0.081%** | **–9%** vs typical carrier (0.123%) | 50% | 42nd‡ | — |
| ***Lachnospiraceae bacterium CLA AA H215***<br>also Pseudolachnospira sp. (SGB4777)<br>SGB 4777 · 1 method: Jun23 | Conditional | **0.080%**° | no reference | — | not in reference set | — |
| ***Alistipes shahii***<br>SGB 2295 · GTDB GCF_025145845.1 · mOTU 000521 · 7 methods | Conditional | **0.076%** | **–91%** vs typical carrier (0.140%) | 72% | 11th | — |
| **Evtepia sp022764045**<br>not yet formally named · known from genomes<br>SGB 58425 · mOTU 026614 · 2 methods: Jan26, mOTUs | Unknown | **0.075%** | no reference | — | not in reference set | — |
| ***Dysosmobacter segnis***<br>also Oscillibacter segnis<br>an earlier catalogue listed this population as Oscillibacter valericigenes<br>SGB 15053 · GTDB GCF_014297285.1 · mOTU 000049 · 6 methods | Conditional | **0.073%** | **–83%** vs typical carrier (0.545%) | 73% | 14th | — |
| ***Oscillibacter faecalis A***<br>also Oscillibacter faecalis / faecalis_A<br>GlobDB GCF_018408705 · mOTU 000476 · 3 methods | Conditional | within *Oscillibacter* above | **–61%** vs typical carrier (0.068%) | 39% | <1st‡ | — |
| **Oscillibacter sp947631165**<br>not yet formally named · known from genomes<br>GlobDB GCF_947631165 · 2 methods: GTDB sketch, GlobDB sketch | Conditional | within *Oscillibacter* above | no reference | — | not in reference set | — |
| **Oscillibacter sp CAG 241**<br>not yet formally named · known from genomes<br>1 method: MP3 | Conditional | within *Oscillibacter* above | **–74%** vs typical carrier (0.170%) | 61% | 28th | — |
| ***Peptoniphilus grossensis***<br>also Peptoniphilus_A hominis<br>SGB 6701 · GTDB GCF_040096015.1 · mOTU 000738 · 5 methods | Unknown | **0.073%** | no reference | — | not in reference set | — |

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| **Peptoniphilus_A HRGMV2_3624**<br>not yet formally named · known from genomes<br>GlobDB HRGMV2_3624 · 2 methods: GlobDB sketch, Kraken panel | Unknown | within *Peptoniphilus* above | no reference | — | not in reference set | — |
| ***Peptoniphilus_A lacydonensis***<br>mOTU 015314 · 1 method: mOTUs | Unknown | within *Peptoniphilus* above | no reference | — | not in reference set | — |
| **Anaerotruncus sp CAG 528**<br>not yet formally named · known from genomes<br>1 method: MP3 | Unknown | **0.072%**◦ | **–70%** vs typical carrier (0.660%) | 6% | 36th | — |
| **UBA10677 sp900543755**<br>not yet formally named · known from genomes<br>SGB 14020 · GlobDB GCA_900543755 · mOTU 007064 · 6 methods | Unknown | **0.072%** | only 8 reference adults carry it: too few to rank against | 8% | not in reference set | — |
| **Hominisplanchenecus mOTUv4.0_058984**<br>not yet formally named · known from genomes<br>mOTU 058984 · 1 method: mOTUs | Unknown | **0.072%**◦ | no reference | — | not in reference set | — |
| ***Simiaoa sunii***<br>also Acetatifactor intestinalis<br>SGB 4910 · GTDB GCF_009695995.1 · mOTU 000040 · 7 methods | Unknown | **0.071%** | no reference | — | not in reference set | — |
| **Firmicutes bacterium CAG 24053 14**<br>not yet formally named · known from genomes<br>1 method: MP3 | Unknown | **0.070%**◦ | no reference | — | not in reference set | — |
| ***Finegoldia magna_J***<br>UHGG 3862 · 1 method: Kraken | Unknown | **0.070%**† | no reference | — | not in reference set | — |
| ***Varibaculum cambriense***<br>also Varibaculum cambriense_A<br>SGB 15480 · GlobDB GCF_022133705 · mOTU 001326 · 8 methods | Unknown | **0.070%** | only 8 reference adults carry it: too few to rank against | 0% | not in reference set | — |
| ***Pilosibacter fragilis***<br>an earlier catalogue listed this population as Lacrimispora amygdalina<br>SGB 4716 · GlobDB GCF_032490495 · mOTU 000085 · 7 methods | Unknown | **0.070%** | **–45%** vs typical carrier (0.174%) | 38% | 10th‡ | — |
| ***Clostridium sp AF12 28***<br>also Pilosibacter sp900066055<br>SGB 4715 · GlobDB GCA_003480435 · mOTU 001377 · 5 methods | Conditional | within *Pilosibacter* above | only 9 reference adults carry it: too few to rank against | 9% | not in reference set | — |
| **CAG-269 sp900556695**<br>not yet formally named · known from genomes<br>SGB 6359 · GlobDB GCA_900556695 · mOTU 002982 · 6 methods | Unknown | **0.069%** | only 1 reference adult carry it: too few to rank against | 1% | not in reference set | — |
| **Negativibacillus sp000435195**<br>not yet formally named · known from genomes<br>SGB 14909 · GTDB GCA_000435195.1 · mOTU 000480 · 6 methods | Unknown | **0.069%** | no reference | — | not in reference set | — |
| **Negativibacillus HRGMV2_2750**<br>not yet formally named · known from genomes<br>GlobDB HRGMV2_2750 · 2 methods: GlobDB sketch, Kraken panel | Unknown | within *Negativibacillus* above | **–23%** vs typical carrier (0.104%) | 15% | 40th‡ | — |
| **MGYG000000876**<br>not yet formally named · known from genomes<br>UHGG 1493 · 1 method: Kraken | Beneficial | **0.068%**◦ | no reference | — | not in reference set | — |
| **GCMETA_00081199**<br>not yet formally named · known from genomes<br>GlobDB GCMETA_00081199 · 1 method: GlobDB sketch | Unknown | **0.067%**◦ | no reference | — | not in reference set | — |
| ***Blautia schinkii***<br>also Blautia_A schinkii_B<br>SGB 4825 · GlobDB GCF_013304825 · UHGG 5466 · 5 methods | Conditional | **0.066%** | only 8 reference adults carry it: too few to rank against | 8% | not in reference set | — |

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|---|---|---|---|---|---|---|
| **HGM13006 sp900757695**<br>not yet formally named · known from genomes<br>SGB 14972 · GlobDB GCA_959607825 · mOTU 005111 · 6 methods | Unknown | **0.066%** | only 4 reference adults carry it: too few to rank against | 4% | not in reference set | — |
| **SFEL01 sp004557245**<br>not yet formally named · known from genomes<br>SGB 14311 · GTDB GCA_004557245.1 · mOTU 000022 · 5 methods | Unknown | **0.064%** | no reference | — | not in reference set | — |
| **SFEL01 MGYG000004245**<br>not yet formally named · known from genomes<br>GlobDB MGYG000004245 · UHGG 5439 · 2 methods: GlobDB sketch, Kraken | Unknown | within *SFEL01* above | **–81%** vs typical carrier (0.443%) | 27% | 11th‡ | — |
| ***Finegoldia magna_K***<br>1 method: SingleM | Unknown | **0.064%**† | no reference | — | not in reference set | — |
| ***Roseburia lenta***<br>also Pararoseburia lenta<br>SGB 4957 · GlobDB GCF_014287435 · mOTU 002744 · 6 methods | Beneficial | **0.060%** | only 5 reference adults carry it: too few to rank against | 5% | not in reference set | — |
| ***Oscillibacter sp MSJ 31***<br>also Hominicoprocola sp900317525<br>SGB 15249 · GlobDB GCF_018918025 · mOTU 000079 · 8 methods | Conditional | **0.060%** | **–56%** vs typical carrier (0.208%) | 40% | 2nd‡ | — |
| **Brotonthovivens sp905193415**<br>not yet formally named · known from genomes<br>SGB 47850 · GlobDB GCA_905193415 · mOTU 003694 · 6 methods | Unknown | **0.057%** | only 1 reference adult carry it: too few to rank against | 1% | not in reference set | — |
| **SVMC01 sp958436045**<br>not yet formally named · known from genomes<br>SGB 14921 · GlobDB GCA_958436045 · mOTU 001646 · 6 methods | Unknown | **0.057%** | **+37%** vs typical carrier (0.051%) | 26% | 81st‡ | — |
| **Faecousia sp. (SGB105735)**<br>not yet formally named · known from genomes<br>SGB 105735 · 1 method: Jan26 | Unknown | **0.054%** | no reference | — | not in reference set | — |
| **Faecousia GCMETA_00037679**<br>not yet formally named · known from genomes<br>GlobDB GCMETA_00037679 · 1 method: GlobDB sketch | Unknown | within *Faecousia* above | only 1 reference adult carry it: too few to rank against | 1% | not in reference set | — |
| **MOTU40_046809 MOTU40_023428**<br>not yet formally named · known from genomes<br>GlobDB MOTU40_023428 · mOTU 023428 · 2 methods: GlobDB sketch, mOTUs | Unknown | **0.053%**◦ | only 1 reference adult carry it: too few to rank against | 1% | not in reference set | — |
| **UBA4285 sp900545225**<br>not yet formally named · known from genomes<br>SGB 5060 · GlobDB GCA_900545225 · mOTU 025819 · 6 methods | Unknown | **0.053%** | ≈ **typical** vs typical carrier (0.089%) | 19% | 53rd‡ | — |
| ***Bianquea renquensis***<br>UHGG 4274 · 1 method: Kraken | Unknown | **0.053%**◦ | no reference | — | not in reference set | — |
| **UBA3402 sp934418055**<br>not yet formally named · known from genomes<br>SGB 4708 · GlobDB GCA_934418055 · mOTU 004540 · 6 methods | Conditional | **0.052%** | only 2 reference adults carry it: too few to rank against | 2% | not in reference set | — |
| **UBA3402 MGYG000330431**<br>not yet formally named · known from genomes<br>GlobDB MGYG000330431 · 1 method: GlobDB sketch | Unknown | within *UBA3402* above | **+11%** vs typical carrier (0.057%) | 40% | 55th‡ | — |
| **UBA3402 sp003478355**<br>not yet formally named · known from genomes<br>mOTU 001254 · UHGG 561 · 2 methods: Kraken, mOTUs | Unknown | within *UBA3402* above | no reference | — | not in reference set | — |
| ***Finegoldia magna_F***<br>mOTU 001292 · 1 method: mOTUs | Unknown | **0.050%**† | no reference | — | not in reference set | — |
| ***Streptococcus salivarius***<br>also Streptococcus complex (sp001556435, salivarius_E)<br>SGB 8007 · SGB 8007_group · UHGG 318 · 5 methods | Conditional | **0.050%** | **+155%** vs typical carrier (0.073%) | 70% | 68th | — |

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|---|---|---|---|---|---|---|
| ***Streptococcus salivarius_D***<br>GlobDB GCF_002094975 · 2 methods: GTDB sketch, GlobDB sketch | Conditional | within *Streptococcus* above | **–60%** vs typical carrier (0.236%) | 25% | 36th‡ | — |
| **Streptococcus SPIREOTU_03006347**<br>not yet formally named · known from genomes<br>GlobDB SPIREOTU_03006347 · 1 method: GlobDB sketch | Conditional | within *Streptococcus* above | **–75%** vs typical carrier (0.275%) | 45% | 13th‡ | — |
| ***Streptococcus salivarius_E***<br>GlobDB GCF_016648815 · 2 methods: GTDB sketch, GlobDB sketch | Conditional | within *Streptococcus* above | **–88%** vs typical carrier (0.488%) | 12% | 8th‡ | — |
| ***Fusicatenibacter faecihominis***<br>also Limivivens faecihominis<br>SGB 4780 · GlobDB GCF_020686945 · mOTU 005955 · 6 methods | Beneficial | **0.050%** | **–39%** vs typical carrier (0.130%) | 47% | 28th‡ | — |
| **Limivivens sp900066135**<br>not yet formally named · known from genomes<br>UHGG 3232 · 2 methods: Kraken, Kraken panel | Unknown | within *Limivivens* above | no reference | — | not in reference set | — |
| **CAKTFW01 MOTU40_018619**<br>not yet formally named · known from genomes<br>mOTU 018619 · 1 method: mOTUs | Unknown | **0.048%**° | no reference | — | not in reference set | — |
| **Catenibacillus sp018369015**<br>not yet formally named · known from genomes<br>GlobDB GCA_018369015 · mOTU 113844 · 3 methods | Unknown | **0.048%**° | **–42%** vs typical carrier (0.030%) | 12% | 17th‡ | — |
| ***Pelethousia gallinarum***<br>mOTU 000484 · 1 method: mOTUs | Unknown | **0.048%**° | no reference | — | not in reference set | — |
| ***Romboutsia timonensis***<br>GlobDB GCF_900106845 · mOTU 001768 · 3 methods | Conditional | **0.048%**° | **–93%** vs typical carrier (0.140%) | 53% | <1st‡ | — |
| **Scatomorpha sp900545405**<br>not yet formally named · known from genomes<br>GlobDB GCA_900545405 · mOTU 000596 · 3 methods | Unknown | **0.048%**° | **–81%** vs typical carrier (0.096%) | 19% | <1st‡ | — |
| **UMGS902 sp003343845**<br>not yet formally named · known from genomes<br>GlobDB GCA_003343845 · mOTU 001327 · 3 methods | Unknown | **0.048%**° | only 2 reference adults carry it:<br>too few to rank against | 2% | not in reference set | — |
| **Laedolimicola sp934373375**<br>not yet formally named · known from genomes<br>SGB 4771 · GlobDB GCA_934373375 · 4 methods | Conditional | **0.047%** | only 2 reference adults carry it:<br>too few to rank against | 2% | not in reference set | — |
| ***Laedolimicola ammoniilytica***<br>GlobDB GCF_025567165 · mOTU 001713 · UHGG 197 · 5 methods | Unknown | within *Laedolimicola* above | ≈ **typical** vs typical carrier (0.114%) | 24% | 50th‡ | — |
| ***Laedolimicola intestinihominis***<br>mOTU 000872 · UHGG 1834 · 2 methods: Kraken, mOTUs | Unknown | within *Laedolimicola* above | no reference | — | not in reference set | — |
| ***Fenollaria massiliensis***<br>SGB 6716 · GlobDB GCF_000312505 · mOTU 000530 · 7 methods | Unknown | **0.046%** | no reference | — | not in reference set | — |
| **Fenollaria MOTU40_023700**<br>not yet formally named · known from genomes<br>GlobDB MOTU40_023700 · 1 method: GlobDB sketch | Unknown | within *Fenollaria* above | no reference | — | not in reference set | — |
| ***Flavonifractor plautii***<br>1 method: Kraken panel | Opportunist | **0.046%**° | no reference | — | not in reference set | — |
| ***Peptoniphilus lacrimalis***<br>also Peptoniphilus_A lacrimalis<br>SGB 6705 · GlobDB GCF_000378725 · mOTU 002759 · 7 methods | Unknown | **0.046%** | **×44** vs typical carrier (0.003%) | 1% | 96th | — |
| ***Ezakiella coagulans***<br>SGB 6728 · GlobDB GCF_003096635 · mOTU 000763 · 6 methods | Unknown | **0.045%** | no reference | — | not in reference set | — |
| ***Waltera acetigignens***<br>also Acetatifactor acetigignens<br>SGB 4914 · GTDB GCF_025567015.1 · mOTU 000051 · 6 methods | Unknown | **0.042%** | no reference | — | not in reference set | — |

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| ***Clostridium SGB48024***<br>also Brotaphodocola faecis<br>SGB 48024 · GlobDB GCF_040096075 · mOTU 005909 · 6 methods | Conditional | **0.042%** | **–34%** vs typical carrier (0.102%) | 13% | 15th‡ | — |
| **Brotaphodocola sp003478505**<br>not yet formally named · known from genomes<br>GlobDB GCF_003478505 · 2 methods: GTDB sketch, GlobDB sketch | Unknown | within *Brotaphodocola* above | **–64%** vs typical carrier (0.146%) | 16% | 6th‡ | — |
| ***Clostridium sp AF36 4***<br>also Eubacterium_F sp003491505<br>SGB 4644 · GTDB GCF_003477605.1 · mOTU 000291 · 6 methods | Conditional | **0.041%** | no reference | — | not in reference set | — |
| **Limivicinus sp003150355**<br>not yet formally named · known from genomes<br>SGB 15196 · GlobDB GCA_003150355 · mOTU 000276 · 6 methods | Unknown | **0.037%** | only 3 reference adults carry it: too few to rank against | 3% | not in reference set | — |
| ***Lachnospiraceae bacterium OM04 12BH***<br>also Suilimivivens aceti<br>SGB 4893 · GlobDB GCF_025567405 · mOTU 000905 · 6 methods | Conditional | **0.036%** | only 1 reference adult carry it: too few to rank against | 1% | not in reference set | — |
| **Suilimivivens HRGMV2_2207**<br>not yet formally named · known from genomes<br>1 method: Kraken panel | Unknown | within *Suilimivivens* above | no reference | — | not in reference set | — |
| **Suilimivivens sp000438155**<br>not yet formally named · known from genomes<br>GlobDB GCA_000438155 · mOTU 000156 · 3 methods | Unknown | within *Suilimivivens* above | **–47%** vs typical carrier (0.071%) | 20% | 20th‡ | — |
| **RUG11198 sp934149245**<br>not yet formally named · known from genomes<br>SGB 47358 · GlobDB GCA_934149245 · mOTU 008610 · 5 methods | Unknown | **0.036%** | only 2 reference adults carry it: too few to rank against | 2% | not in reference set | — |
| **Limivivens sp905214955**<br>not yet formally named · known from genomes<br>SGB 71759 · GlobDB GCA_905214955 · mOTU 011894 · 6 methods | Unknown | **0.036%** | **–5%** vs typical carrier (0.051%) | 34% | 38th‡ | — |
| ***Campylobacter hominis***<br>also Campylobacter_B hominis<br>SGB 19429 · GlobDB GCF_000017585 · mOTU 001293 · 6 methods | Unknown | **0.035%** | **+349%** vs typical carrier (0.007%) | 2% | 83rd | — |
| ***Clostridium SGB4714***<br>also Pilosibacter sp900066785<br>SGB 4714 · GlobDB GCA_003478035 · mOTU 000605 · 6 methods | Conditional | **0.035%** | **–19%** vs typical carrier (0.152%) | 25% | 40th‡ | — |
| ***Streptococcus thermophilus***<br>SGB 8002 · GlobDB GCF_010120595 · mOTU 000815 · 8 methods | Conditional | **0.035%** | **+136%** vs typical carrier (0.076%) | 36% | 67th | — |
| **Ventricola sp900542395**<br>not yet formally named · known from genomes<br>SGB 15374 · GlobDB GCA_900542395 · mOTU 000218 · 6 methods | Unknown | **0.034%** | **–54%** vs typical carrier (0.166%) | 30% | 17th‡ | — |
| ***Clostridiaceae unclassified SGB4769***<br>also Hominiventricola sp018223365<br>SGB 4769 · GlobDB GCA_018223365 · UHGG 706 · 5 methods | Conditional | **0.034%** | **–39%** vs typical carrier (0.147%) | 11% | <1st‡ | — |
| **CAG-81 MGYG000000233**<br>not yet formally named · known from genomes<br>UHGG 499 · 1 method: Kraken | Unknown | **0.033%**◦ | no reference | — | not in reference set | — |
| **Vescimonas sp900552475**<br>not yet formally named · known from genomes<br>SGB 15081 · GlobDB GCA_900552475 · mOTU 001788 · 7 methods | Unknown | **0.032%** | only 9 reference adults carry it: too few to rank against | 9% | not in reference set | — |
| **Butyribacter sp000436755**<br>not yet formally named · known from genomes<br>UHGG 2716 · 1 method: Kraken | Unknown | **0.032%**◦ | no reference | — | not in reference set | — |

| ORGANISM | CLASS | SHARE | LEVEL | CARRIED BY | PERCENTILE | STRAIN |
|---|---|---|---|---|---|---|
| ***Hominisplanchenecus faecis***<br>1 method: Kraken panel | Unknown | **0.031%**◦ | no reference | — | not in reference set | — |
| ***Hominenteromicrobium mulieris***<br>SGB 14861 · GlobDB GCF_020687165 · mOTU 000060 · 7 methods | Unknown | **0.031%** | **–87%** vs typical carrier (0.633%) | 71% | <1st‡ | — |
| ***Alistipes ihumii***<br>also Alistipes_A ihumii<br>SGB 2328 · GlobDB GCF_025144665 · mOTU 002372 · 7 methods | Conditional | **0.031%** | **–55%** vs typical carrier (0.107%) | 32% | 16th‡ | — |
| ***Peptoniphilus gorbachii***<br>also Peptoniphilus_A gorbachii<br>SGB 6699 · GlobDB GCF_016908115 · mOTU 000779 · 6 methods | Unknown | **0.031%** | no reference | — | not in reference set | — |
| **HGM13222 sp900757485**<br>not yet formally named · known from genomes<br>SGB 14177 · GlobDB GCA_900757485 · mOTU 010941 · 6 methods | Unknown | **0.030%** | **–33%** vs typical carrier (0.078%) | 17% | 41st‡ | — |
| **CAG-349 sp001940895**<br>not yet formally named · known from genomes<br>SGB 13975 · GlobDB GCA_001940895 · mOTU 002973 · 6 methods | Unknown | **0.030%** | no reference | — | not in reference set | — |
| **CAG-349 sp003539515**<br>not yet formally named · known from genomes<br>mOTU 000053 · 1 method: mOTUs | Unknown | within *CAG-349* above | no reference | — | not in reference set | — |
| **HGM16780 sp900538765**<br>not yet formally named · known from genomes<br>UHGG 3914 · 1 method: Kraken | Unknown | **0.029%**◦ | no reference | — | not in reference set | — |
| **CAG-793 sp900557475**<br>not yet formally named · known from genomes<br>SGB 6286 · GlobDB GCA_963558395 · mOTU 012681 · 6 methods | Unknown | **0.029%** | only 1 reference adult carry it: too few to rank against | 1% | not in reference set | — |
| **SPIREOTU_01076626**<br>not yet formally named · known from genomes<br>GlobDB SPIREOTU_01076626 · 1 method: GlobDB sketch | Unknown | **0.029%**◦ | no reference | — | not in reference set | — |
| ***GGB43675 SGB61016***<br>also Cryptoclostridium obscurum<br>SGB 61016 · GlobDB GCA_904420535 · mOTU 065587 · 5 methods | Unknown | **0.029%** | only 3 reference adults carry it: too few to rank against | 3% | not in reference set | — |
| ***Bifidobacterium catenulatum***<br>also Bifidobacterium hominis<br>SGB 17239 · GlobDB GCF_040095915 · mOTU 000400 · 4 methods | Beneficial | **0.028%** | no reference | — | not in reference set | — |
| **Limivicinus sp900547315**<br>not yet formally named · known from genomes<br>SGB 15198 · GlobDB GCA_900547315 · mOTU 000836 · 6 methods | Unknown | **0.026%** | only 5 reference adults carry it: too few to rank against | 5% | not in reference set | — |
| **Faecousia sp000435995**<br>not yet formally named · known from genomes<br>SGB 14924 · GlobDB GCA_900545515 · mOTU 005107 · 7 methods | Unknown | **0.026%** | **–53%** vs typical carrier (0.105%) | 31% | 6th‡ | — |
| ***Peptoniphilus rachelemmaiella***<br>also Aedoeadaptatus rachelemmaiella<br>SGB 109696 · GTDB GCF_943169845.1 · mOTU 002789 · 4 methods | Unknown | **0.026%** | no reference | — | not in reference set | — |
| **Aedoeadaptatus GCMETA_03560848**<br>not yet formally named · known from genomes<br>GlobDB GCMETA_03560848 · 1 method: GlobDB sketch | Unknown | within *Aedoeadaptatus* above | no reference | — | not in reference set | — |
| ***Aedoeadaptatus acetigenes***<br>mOTU 001151 · 1 method: mOTUs | Unknown | within *Aedoeadaptatus* above | no reference | — | not in reference set | — |
| ***Clostridiales bacterium S5 A14a***<br>also S5-A14a sp000758905<br>SGB 3977 · GlobDB GCA_000758905 · mOTU 002989 · 6 methods | Unknown | **0.025%** | no reference | — | not in reference set | — |

| ORGANISM | CLASS | SHARE | LEVEL | CARRIED BY | PERCENTILE | STRAIN |
|---|---|---|---|---|---|---|
| ***GGB74412 SGB5075***<br>also Lachnospira rogosae<br>SGB 5075 · GlobDB GCF_040096455 · mOTU 000761 · 6 methods | Beneficial | **0.025%** | ▼ notably low<br>**–90%** vs typical carrier (0.494%) | 47% | 2nd‡ | — |
| ***Ruthenibacterium lactatiformans***<br>SGB 15271 · GTDB GCF_000949455.1 · mOTU 002162 · 7 methods | Unknown | **0.025%** | **–75%** vs typical carrier (0.074%) | 82% | 19th | — |
| **Ruthenibacterium GCMETA_00001082**<br>not yet formally named · known from genomes<br>GlobDB GCMETA_00001082 · 1 method: GlobDB sketch | Unknown | within *Ruthenibacterium* above | **–38%** vs typical carrier (0.073%) | 89% | 24th‡ | — |
| **CAG-611 sp900552235**<br>not yet formally named · known from genomes<br>SGB 146335 · GlobDB GCA_900552235 · 3 methods | Unknown | **0.025%** | no reference | — | not in reference set | — |
| **CAG-611 sp000434175**<br>not yet formally named · known from genomes<br>SGB 6465 · mOTU 000229 · 2 methods: Jun23, mOTUs | Unknown | within *CAG-611* above | no reference | — | not in reference set | — |
| **Woodwardiibium sp900760275**<br>not yet formally named · known from genomes<br>SGB 14954 · GlobDB GCA_900760275 · UHGG 5250 · 5 methods | Unknown | **0.025%** | **–40%** vs typical carrier (0.085%) | 10% | <1st‡ | — |
| ***Eggerthella lenta***<br>SGB 14809 · GlobDB GCF_000024265 · mOTU 000459 · 8 methods | Opportunist | **0.024%** | **+27%** vs typical carrier (0.047%) | 47% | 56th | — |
| **Eggerthella sp CAG 209**<br>not yet formally named · known from genomes<br>1 method: MP3 | Opportunist | within *Eggerthella* above | no reference | — | not in reference set | — |
| ***Akkermansia muciniphila***<br>GTDB GCF_000020225.1 · mOTU 000414 · 2 methods: GTDB sketch, mOTUs | Beneficial | **0.024%**◦ | no reference | — | not in reference set | — |
| ***Allocoprobacillus halotolerans***<br>mOTU 106389 · 1 method: mOTUs | Unknown | **0.024%**◦ | no reference | — | not in reference set | — |
| **Avispirillum MOTU40_108702**<br>not yet formally named · known from genomes<br>GlobDB MOTU40_108702 · mOTU 108702 · 2 methods: GlobDB sketch, mOTUs | Unknown | **0.024%**◦ | no reference | — | not in reference set | — |
| **CAG-449 sp000432895**<br>not yet formally named · known from genomes<br>mOTU 010896 · 1 method: mOTUs | Unknown | **0.024%**◦ | no reference | — | not in reference set | — |
| **CAKTXU01 sp902492455**<br>not yet formally named · known from genomes<br>GlobDB GCA_902492455 · mOTU 009263 · 3 methods | Unknown | **0.024%**◦ | only 9 reference adults carry it:<br>too few to rank against | 9% | not in reference set | — |
| **CALWJF01 mOTUv4.0_023553**<br>not yet formally named · known from genomes<br>mOTU 023553 · 1 method: mOTUs | Unknown | **0.024%**◦ | no reference | — | not in reference set | — |
| **Caccenecus MOTU40_078079**<br>not yet formally named · known from genomes<br>GlobDB MOTU40_078079 · mOTU 078079 · 2 methods: GlobDB sketch, mOTUs | Unknown | **0.024%**◦ | no reference | — | not in reference set | — |
| ***Egerieenecus merdigallinarum***<br>GlobDB GCA_018711985 · mOTU 000544 · 3 methods | Unknown | **0.024%**◦ | **–65%** vs typical carrier (0.069%) | 29% | 3rd‡ | — |
| ***Faecivivens stercoripullorum***<br>mOTU 003761 · 1 method: mOTUs | Unknown | **0.024%**◦ | no reference | — | not in reference set | — |
| **JAFBIO01 sp021531255**<br>not yet formally named · known from genomes<br>mOTU 032367 · 1 method: mOTUs | Unknown | **0.024%**◦ | no reference | — | not in reference set | — |
| **Lawsonella sp018376445**<br>not yet formally named · known from genomes<br>SGB 3665 · GlobDB GCA_018376445 · mOTU 015136 · 4 methods | Unknown | **0.024%**◦ | no reference | — | not in reference set | — |
| ***Lawsonibacter asaccharolyticus***<br>SGB 15154 · GlobDB MOTU40_001350 · GTDB GCA_003112755.1 · 4 methods | Conditional | **0.024%**◦ | **–75%** vs typical carrier (0.110%) | 78% | <1st‡ | — |

| ORGANISM | CLASS | SHARE | LEVEL | CARRIED BY | PERCENTILE | STRAIN |
|---|---|---|---|---|---|---|
| **Limiplasma MOTU40_108708**<br>not yet formally named · known from genomes<br>mOTU 108708 · 1 method: mOTUs | Unknown | **0.024%**◦ | no reference | — | not in reference set | — |
| **Neoclostridium mOTUv4.0_075132**<br>not yet formally named · known from genomes<br>mOTU 075132 · 1 method: mOTUs | Unknown | **0.024%**◦ | no reference | — | not in reference set | — |
| ***Olegusella massiliensis***<br>mOTU 007163 · 1 method: mOTUs | Unknown | **0.024%**◦ | no reference | — | not in reference set | — |
| **Oxalobacter MOTU40_046776**<br>not yet formally named · known from genomes<br>GlobDB MOTU40_046776 · mOTU 046776 · 2 methods: GlobDB sketch, mOTUs | Unknown | **0.024%**◦ | no reference | — | not in reference set | — |
| ***Peptostreptococcus anaerobius***<br>mOTU 015132 · 1 method: mOTUs | Unknown | **0.024%**◦ | no reference | — | not in reference set | — |
| **Pseudoscillospira sp018369915**<br>not yet formally named · known from genomes<br>GlobDB GCA_020555865 · mOTU 009017 · 3 methods | Unknown | **0.024%**◦ | **–58%** vs typical carrier (0.055%) | 34% | 12th‡ | — |
| **QAKS01 sp003343685**<br>not yet formally named · known from genomes<br>mOTU 016529 · 1 method: mOTUs | Unknown | **0.024%**◦ | no reference | — | not in reference set | — |
| **QALR01 sp900553085**<br>not yet formally named · known from genomes<br>GlobDB GCA_934446365 · mOTU 008528 · 3 methods | Unknown | **0.024%**◦ | only 3 reference adults carry it: too few to rank against | 3% | not in reference set | — |
| **RUG11247 sp902767315**<br>not yet formally named · known from genomes<br>GlobDB GCA_902767315 · mOTU 004159 · 3 methods | Unknown | **0.024%**◦ | only 6 reference adults carry it: too few to rank against | 6% | not in reference set | — |
| **Scatosoma sp003150575**<br>not yet formally named · known from genomes<br>mOTU 002732 · 1 method: mOTUs | Unknown | **0.024%**◦ | no reference | — | not in reference set | — |
| **Spyradocola sp905193155**<br>not yet formally named · known from genomes<br>GlobDB GCA_905193155 · mOTU 000978 · 3 methods | Unknown | **0.024%**◦ | only 4 reference adults carry it: too few to rank against | 4% | not in reference set | — |
| **UBA3818 sp900557155**<br>not yet formally named · known from genomes<br>GlobDB GCA_934875375 · mOTU 007830 · 3 methods | Unknown | **0.024%**◦ | **–79%** vs typical carrier (0.086%) | 44% | <1st‡ | — |
| **UMGS1322 sp900550765**<br>not yet formally named · known from genomes<br>GlobDB GCA_958448705 · mOTU 005957 · 3 methods | Unknown | **0.024%**◦ | only 2 reference adults carry it: too few to rank against | 2% | not in reference set | — |
| **UMGS1449 sp934087515**<br>not yet formally named · known from genomes<br>mOTU 063387 · 1 method: mOTUs | Unknown | **0.024%**◦ | no reference | — | not in reference set | — |
| **UMGS856 sp900546265**<br>not yet formally named · known from genomes<br>GlobDB GCA_900546265 · mOTU 005110 · 3 methods | Unknown | **0.024%**◦ | **–87%** vs typical carrier (0.070%) | 11% | <1st‡ | — |
| ***Winkia anitrata***<br>mOTU 023794 · 1 method: mOTUs | Unknown | **0.024%**◦ | no reference | — | not in reference set | — |
| ***Yeguia hominis***<br>mOTU 034420 · 1 method: mOTUs | Unknown | **0.024%**◦ | no reference | — | not in reference set | — |
| ***Anaerococcus senegalensis***<br>SGB 6664 · GTDB GCF_000321005.1 · mOTU 001708 · 4 methods | Unknown | **0.024%** | no reference | — | not in reference set | — |
| ***Sanguibacteroides justesenii***<br>also Gabonibacter massiliensis<br>SGB 1776 · GlobDB GCF_001487125 · mOTU 028428 · 7 methods | Unknown | **0.023%** | **+18%** vs typical carrier (0.030%) | 6% | 54th | — |
| **SVKR01 sp938023775**<br>not yet formally named · known from genomes<br>includes reads the marker catalogue read as SVKR01 sp963649145<br>GlobDB GCA_938023775 · mOTU 003599 · 3 methods | Unknown | **0.022%**§ | no reference | — | not in reference set | — |

| ORGANISM | CLASS | SHARE | LEVEL | CARRIED BY | PERCENTILE | STRAIN |
|---|---|---|---|---|---|---|
| **UMGS882 sp900546385**<br>not yet formally named · known from genomes<br>SGB 14199 · GlobDB GCA_900546385 · mOTU 001568 · 6 methods | Unknown | **0.021%** | only 1 reference adult carry it: too few to rank against | 1% | not in reference set | — |
| **HGM11386 sp900761915**<br>not yet formally named · known from genomes<br>SGB 14050 · GlobDB GCA_900761915 · mOTU 007045 · 5 methods | Unknown | **0.021%** | only 4 reference adults carry it: too few to rank against | 4% | not in reference set | — |
| **Enterenecus sp900549885**<br>not yet formally named · known from genomes<br>GlobDB GCA_900549885 · 2 methods: GTDB sketch, GlobDB sketch | Unknown | **0.020%**◦ | **–73%** vs typical carrier (0.084%) | 33% | <1st‡ | — |
| **CAKSIN01 sp934500705**<br>not yet formally named · known from genomes<br>SGB 4143 · GlobDB GCA_934500705 · mOTU 004628 · 6 methods | Unknown | **0.020%** | no reference | — | not in reference set | — |
| **JAHZGC01 sp019421015**<br>not yet formally named · known from genomes<br>SGB 72479 · GlobDB GCA_019421015 · mOTU 004647 · 5 methods | Unknown | **0.020%** | **+37%** vs typical carrier (0.026%) | 13% | 69th‡ | — |
| ***Anaerotignum sp MSJ 24***<br>also Metalachnospira sp900553995<br>SGB 5180 · GlobDB GCF_018918145 · mOTU 001449 · 5 methods | Conditional | **0.019%** | only 4 reference adults carry it: too few to rank against | 4% | not in reference set | — |
| **Hominimerdicola sp900539095**<br>not yet formally named · known from genomes<br>UHGG 1864 · 1 method: Kraken | Conditional | **0.019%**◦ | no reference | — | not in reference set | — |
| **Faecivicinus sp900759005**<br>not yet formally named · known from genomes<br>SGB 15409 · GlobDB GCA_900759005 · mOTU 003404 · 6 methods | Unknown | **0.019%** | only 4 reference adults carry it: too few to rank against | 4% | not in reference set | — |
| **QAKL01 sp003343815**<br>not yet formally named · known from genomes<br>GlobDB GCA_003343815 · 2 methods: GTDB sketch, GlobDB sketch | Unknown | **0.018%**◦ | only 2 reference adults carry it: too few to rank against | 2% | not in reference set | — |
| ***Solibaculum mannosilyticum***<br>UHGG 350 · 1 method: Kraken | Unknown | **0.018%**◦ | no reference | — | not in reference set | — |
| **CAG-170 sp900545925**<br>not yet formally named · known from genomes<br>SGB 15233 · GlobDB GCA_900545925 · mOTU 000995 · 7 methods | Unknown | **0.018%** | **–71%** vs typical carrier (0.207%) | 12% | <1st‡ | — |
| **UBA1822 sp002314995**<br>not yet formally named · known from genomes<br>UHGG 6031 · 1 method: Kraken | Unknown | **0.018%**◦ | no reference | — | not in reference set | — |
| **UMGS911 sp003343545**<br>not yet formally named · known from genomes<br>SGB 14182 · GlobDB GCA_905205065 · UHGG 3700 · 5 methods | Unknown | **0.018%** | **–58%** vs typical carrier (0.076%) | 18% | 6th‡ | — |
| **MOTU40_086824**<br>not yet formally named · known from genomes<br>GlobDB MOTU40_086824 · 1 method: GlobDB sketch | Unknown | **0.018%**◦ | no reference | — | not in reference set | — |
| **UBA3818 sp904397845**<br>not yet formally named · known from genomes<br>UHGG 3744 · 1 method: Kraken | Unknown | **0.017%**◦ | no reference | — | not in reference set | — |
| ***Negativibacillus massiliensis***<br>SGB 14912 · GlobDB GCF_900148495 · mOTU 000248 · 7 methods | Unknown | **0.017%** | **–59%** vs typical carrier (0.088%) | 17% | 30th | — |
| ***Bacteroides xylanisolvens***<br>SGB 1867 · GTDB GCF_000210075.1 · UHGG 2057 · 6 methods | Conditional | **0.017%** | **–73%** vs typical carrier (0.193%) | 65% | 28th | — |
| ***Corynebacterium pyruviciproducens***<br>SGB 17029 · GTDB GCF_000411375.1 · mOTU 003635 · 5 methods | Unknown | **0.017%** | no reference | — | not in reference set | — |

| ORGANISM | CLASS | SHARE | LEVEL | CARRIED BY | PERCENTILE | STRAIN |
|---|---|---|---|---|---|---|
| **Corynebacterium GCMETA_00600295**<br>not yet formally named · known from genomes<br>GlobDB GCMETA_00600295 · 1 method: GlobDB sketch | Unknown | within *Corynebacterium* above | no reference | — | not in reference set | — |
| **Corynebacterium SPIREOTU_02163612**<br>not yet formally named · known from genomes<br>GlobDB SPIREOTU_02163612 · 1 method: GlobDB sketch | Unknown | within *Corynebacterium* above | no reference | — | not in reference set | — |
| ***Hoylesella timonensis***<br>also Prevotella timonensis<br>SGB 1507 · GlobDB GCF_000455445 · mOTU 000358 · 6 methods | Conditional | **0.017%** | **+204%** vs typical carrier (0.010%) | 3% | 70th | — |
| **CAG-170 sp900553545**<br>not yet formally named · known from genomes<br>SGB 15229 · GlobDB GCA_900553545 · 4 methods | Unknown | **0.015%** | **–64%** vs typical carrier (0.079%) | 33% | 3rd‡ | — |
| ***Streptococcus parasanguinis***<br>also Streptococcus parasanguinis_G<br>SGB 8071_group · SGB 8071 · GlobDB GCF_000164675 · 5 methods | Conditional | **0.015%** | **+41%** vs typical carrier (0.034%) | 57% | 58th | — |
| ***Porphyromonas sp HMSC065F10***<br>also Porphyromonas_A sp001808555<br>includes reads the marker catalogue read as Porphyromonas bennonis<br>SGB 1990_group · GlobDB GCF_001808555 · 5 methods | Unknown | **0.015%**§ | **+495%** vs typical carrier (0.006%) | 1% | 83rd | — |
| **Porphyromonas_A sp946222055**<br>not yet formally named · known from genomes<br>GlobDB GCA_946222055 · mOTU 005147 · 3 methods | Unknown | within *Porphyromonas* above | no reference | — | not in reference set | — |
| ***Porphyromonas uenonis***<br>GlobDB GCF_000482365 · mOTU 001491 · 3 methods | Unknown | within *Porphyromonas* above | no reference | — | not in reference set | — |
| **Dialister sp900541485**<br>not yet formally named · known from genomes<br>SGB 5832 · GlobDB GCA_965119665 · mOTU 000468 · 5 methods | Conditional | **0.013%** | only 3 reference adults carry it: too few to rank against | 3% | not in reference set | — |
| **Mobiluncus SPIREOTU_02229928**<br>not yet formally named · known from genomes<br>GlobDB SPIREOTU_02229928 · GTDB GCF_034180645.1 · 2 methods: GTDB sketch, GlobDB sketch | Unknown | **0.013%**° | no reference | — | not in reference set | — |
| **MOTU40_072568 GCMETA_00070180**<br>not yet formally named · known from genomes<br>GlobDB GCMETA_00070180 · 1 method: GlobDB sketch | Unknown | **0.013%**° | only 1 reference adult carry it: too few to rank against | 1% | not in reference set | — |
| **QANA01 MOTU40_062001**<br>not yet formally named · known from genomes<br>GlobDB MOTU40_062001 · 1 method: GlobDB sketch | Unknown | **0.013%**° | only 9 reference adults carry it: too few to rank against | 9% | not in reference set | — |
| ***Oscillospiraceae bacterium Marseille Q3528***<br>also JAGTTR01 sp018223385<br>SGB 4778 · 2 methods: Jun23, Jan26 | Conditional | **0.013%** | no reference | — | not in reference set | — |
| ***GGB42378 SGB59548***<br>also Ventrousia excrementavium<br>SGB 59548 · GlobDB GCA_018714845 · mOTU 007486 · 5 methods | Unknown | **0.012%** | only 4 reference adults carry it: too few to rank against | 4% | not in reference set | — |
| **Paludihabitans MOTU40_028425**<br>not yet formally named · known from genomes<br>GlobDB MOTU40_028425 · 1 method: GlobDB sketch | Unknown | **0.012%**° | only 7 reference adults carry it: too few to rank against | 7% | not in reference set | — |
| **RGIG7193 sp944381515**<br>not yet formally named · known from genomes<br>GlobDB GCA_944381515 · 2 methods: GTDB sketch, GlobDB sketch | Unknown | **0.012%**° | only 2 reference adults carry it: too few to rank against | 2% | not in reference set | — |
| **Pauljensenia sp000278725**<br>not yet formally named · known from genomes<br>GlobDB GCF_000278725 · 2 methods: GTDB sketch, GlobDB sketch | Unknown | **0.012%**° | **–59%** vs typical carrier (0.035%) | 12% | <1st‡ | — |

| ORGANISM | CLASS | SHARE | LEVEL | CARRIED BY | PERCENTILE | STRAIN |
|---|---|---|---|---|---|---|
| ***Rothia mucilaginosa***<br>GlobDB GCF_000175615 · 2 methods: GTDB sketch, GlobDB sketch | Conditional | **0.011%**° | no reference | — | not in reference set | — |
| **UMGS1766 sp959602365**<br>not yet formally named · known from genomes<br>SGB 15117 · GlobDB GCA_959602365 · mOTU 007132 · 5 methods | Unknown | **0.011%** | **–66%** vs typical carrier (0.070%) | 11% | 9th‡ | — |
| **UMGS1766 MOTU40_078088**<br>not yet formally named · known from genomes<br>GlobDB MOTU40_078088 · mOTU 078088 · 2 methods: GlobDB sketch, mOTUs | Unknown | within *UMGS1766* above | only 1 reference adult carry it: too few to rank against | 1% | not in reference set | — |
| ***Cloacibacillus porcorum***<br>GTDB GCF_001701045.1 · 1 method: GTDB sketch | Unknown | **0.011%**° | no reference | — | not in reference set | — |
| **Clostridium sp900540255**<br>not yet formally named · known from genomes<br>GlobDB GCF_039753335 · 2 methods: GTDB sketch, GlobDB sketch | Conditional | **0.011%**° | **–87%** vs typical carrier (0.107%) | 19% | <1st‡ | — |
| **RGIG4076 sp905198155**<br>not yet formally named · known from genomes<br>GlobDB GCA_905198155 · 2 methods: GTDB sketch, GlobDB sketch | Unknown | **0.011%**° | only 4 reference adults carry it: too few to rank against | 4% | not in reference set | — |
| ***Prevotella disiens***<br>SGB 1559 · GlobDB GCF_000613345 · mOTU 005950 · 6 methods | Conditional | **0.010%** | **+421%** vs typical carrier (0.007%) | 3% | 83rd | — |
| **CAG-302 sp934162325**<br>not yet formally named · known from genomes<br>SGB 6478 · GlobDB GCA_934162325 · 4 methods | Unknown | **0.010%** | only 3 reference adults carry it: too few to rank against | 3% | not in reference set | — |
| **Faecousia sp900544405**<br>not yet formally named · known from genomes<br>SGB 15012 · GlobDB GCA_900544405 · mOTU 000648 · 5 methods | Unknown | **0.009%** | only 3 reference adults carry it: too few to rank against | 3% | not in reference set | — |
| ***Hydrogenoanaerobacterium saccharovorans***<br>also Avimicrobium caecorum<br>SGB 15350 · GlobDB GCA_904394705 · mOTU 000421 · 5 methods | Unknown | **0.009%** | **–59%** vs typical carrier (0.063%) | 49% | 12th‡ | — |
| ***Bacteroides nordii***<br>SGB 1858 · GlobDB GCA_000613465 · mOTU 001891 · 6 methods | Conditional | **0.009%** | **–83%** vs typical carrier (0.167%) | 23% | <1st‡ | — |
| ***Lactobacillus crispatus***<br>GlobDB GCF_018987235 · 2 methods: GTDB sketch, GlobDB sketch | Beneficial | **0.008%**° | no reference | — | not in reference set | — |
| ***Protoclostridium gallicola***<br>GTDB GCA_904387375.1 · 1 method: GTDB sketch | Unknown | **0.008%**° | no reference | — | not in reference set | — |
| **QAKD01 sp003343965**<br>not yet formally named · known from genomes<br>GlobDB GCA_944383465 · 2 methods: GTDB sketch, GlobDB sketch | Unknown | **0.008%**° | **–73%** vs typical carrier (0.036%) | 19% | <1st‡ | — |
| ***Alistipes provencensis***<br>SGB 2298 · GlobDB GCF_900083545 · mOTU 002727 · 6 methods | Conditional | **0.007%** | only 1 reference adult carry it: too few to rank against | 1% | not in reference set | — |
| **Lientehia sp934351765**<br>not yet formally named · known from genomes<br>SGB 69322 · GlobDB GCF_052118925 · mOTU 012588 · 4 methods | Unknown | **0.007%** | only 7 reference adults carry it: too few to rank against | 7% | not in reference set | — |
| **UMGS1623 sp934647945**<br>not yet formally named · known from genomes<br>SGB 14932 · GlobDB GCA_934647945 · 4 methods | Unknown | **0.006%** | **–69%** vs typical carrier (0.060%) | 46% | <1st‡ | — |
| ***Phascolarctobacterium succinatutens***<br>also Phascolarctobacterium_A succinatutens_B<br>SGB 5763 · SGB 5763_group · mOTU 000008 · 5 methods | Beneficial | **0.006%**† | **–73%** vs typical carrier (1.47%) | 24% | 23rd | — |

| ORGANISM | CLASS | SHARE | LEVEL | CARRIED BY | PERCENTILE | STRAIN |
|---|---|---|---|---|---|---|
| ***Alistipes indistinctus***<br>also Alistipes_A indistinctus<br>SGB 2325 · GlobDB GCF_025144995 · mOTU 002384 · 6 methods | Conditional | **0.005%** | **–83%** vs typical carrier (0.058%) | 41% | 18th | — |
| **V9D3004 sp900760345**<br>not yet formally named · known from genomes<br>SGB 4638 · GTDB GCA_900760345.1 · mOTU 008552 · 3 methods | Unknown | **0.005%** | no reference | — | not in reference set | — |
| **V9D3004 BCRBG_63118**<br>not yet formally named · known from genomes<br>GlobDB BCRBG_63118 · 1 method: GlobDB sketch | Unknown | within *V9D3004* above | only 2 reference adults carry it: too few to rank against | 2% | not in reference set | — |
| ***Roseburia inulinivorans***<br>SGB 4940 · mOTU 000002 · UHGG 553 · 6 methods | Beneficial | **0.004%** | ▼ low<br>**–94%** vs typical carrier (0.585%) | 85% | 9th | — |
| **HGM11616 sp900761075**<br>not yet formally named · known from genomes<br>SGB 3813 · GlobDB GCA_900761075 · mOTU 009262 · 5 methods | Unknown | **0.004%** | **–36%** vs typical carrier (0.035%) | 19% | 16th‡ | — |
| **CAKUFI01 sp934647305**<br>not yet formally named · known from genomes<br>SGB 15407 · GlobDB GCA_934647305 · 4 methods | Unknown | **0.004%** | only 3 reference adults carry it: too few to rank against | 3% | not in reference set | — |
| ***Parabacteroides goldsteinii***<br>also Parabacteroides segnis<br>SGB 1941 · GTDB GCF_000969835.1 · UHGG 2291 · 5 methods | Conditional | **0.004%** | **–82%** vs typical carrier (0.088%) | 23% | 18th | — |
| **Flemingiibacterium sp900546315**<br>not yet formally named · known from genomes<br>SGB 14252 · GlobDB GCA_900546315 · mOTU 000565 · 5 methods | Unknown | **0.004%** | **–14%** vs typical carrier (0.026%) | 16% | 38th‡ | — |
| ***Fenollaria timonensis***<br>also Fenollaria sp900539225<br>SGB 6719 · GTDB GCA_900539225.1 · 3 methods | Unknown | **0.004%** | no reference | — | not in reference set | — |
| **UMGS1696 sp900554225**<br>not yet formally named · known from genomes<br>SGB 14255 · GlobDB GCA_900554225 · 4 methods | Unknown | **0.003%** | only 5 reference adults carry it: too few to rank against | 5% | not in reference set | — |
| **Firmicutes bacterium CAG 124**<br>not yet formally named · known from genomes<br>1 method: MP3 | Unknown | **0.003%**° | no reference | — | not in reference set | — |
| **RGIG1902 sp934636325**<br>not yet formally named · known from genomes<br>SGB 53819 · GlobDB GCA_934636325 · mOTU 018067 · 4 methods | Unknown | **0.003%** | no reference | — | not in reference set | — |
| **Prevotella MGYG000003691**<br>not yet formally named · known from genomes<br>includes reads the marker catalogue read as Hoylesella buccalis<br>1 method: Kraken panel | Conditional | **0.002%**§ | no reference | — | not in reference set | — |
| **CAG-238 sp934277655**<br>not yet formally named · known from genomes<br>SGB 80143 · GlobDB GCA_934277655 · mOTU 003774 · 5 methods | Unknown | **0.002%** | only 4 reference adults carry it: too few to rank against | 4% | not in reference set | — |
| ***Holdemania filiformis***<br>SGB 4046 · mOTU 001900 · 3 methods | Conditional | **0.002%** | no reference | — | not in reference set | — |
| ***Faecalibacterium butyricigenerans***<br>SGB 15315 · GlobDB GCF_023347275 · mOTU 000140 · 6 methods | Beneficial | **0.002%** | ▼ notably low<br>**–89%** vs typical carrier (0.664%) | 31% | <1st‡ | — |
| **Onthomonas sp900752305**<br>not yet formally named · known from genomes<br>SGB 15042 · GlobDB GCA_900752305 · mOTU 000589 · 5 methods | Unknown | **0.002%** | only 2 reference adults carry it: too few to rank against | 2% | not in reference set | — |
| **Blautia_A sp900763175**<br>not yet formally named · known from genomes<br>SGB 4817 · mOTU 011614 · 3 methods | Conditional | **0.002%** | no reference | — | not in reference set | — |

| ORGANISM | CLASS | SHARE | LEVEL | CARRIED BY | PERCENTILE | STRAIN |
|---|---|---|---|---|---|---|
| **Acetatifactor hominis**<br>not yet formally named · known from genomes<br>SGB 4909 · 2 methods: Jun23, Jan26 | Unknown | **0.002%** | no reference | — | not in reference set | — |
| **Neoanaerotignum_A sp944377425**<br>not yet formally named · known from genomes<br>SGB 105679 · GlobDB GCA_944377425 · mOTU 006348 · 4 methods | Unknown | **0.002%** | only 7 reference adults carry it:<br>too few to rank against | 7% | not in reference set | — |
| ***Mediterraneibacter butyricigenes***<br>also Mediterraneibacter_A butyricigenes<br>SGB 25493 · mOTU 004667 · UHGG 403 · 4 methods | Conditional | **0.001%** | no reference | — | not in reference set | — |
| **Firmicutes bacterium CAG 95**<br>not yet formally named · known from genomes<br>1 method: MP3 | Unknown | **0.001%**° | **–95%** vs typical carrier (0.052%) | 31% | 8th | — |
| ***Clostridiaceae bacterium Marseille Q3526***<br>also Otoolea sp000435655<br>SGB 4703 · mOTU 001497 · 4 methods | Conditional | **0.001%** | **–94%** vs typical carrier (0.069%) | 15% | 8th | — |
| ***Lentihominibacter faecis***<br>SGB 3957 · GlobDB GCF_014306095 · mOTU 000249 · 4 methods | Unknown | **0.000%** | **–66%** vs typical carrier (0.084%) | 45% | 9th‡ | — |
| ***Lentihominibacter butyricus***<br>GlobDB GCF_025567305 · 2 methods: GTDB sketch, GlobDB sketch | Unknown | within *Lentihominibacter* above | only 7 reference adults carry it:<br>too few to rank against | 7% | not in reference set | — |

**Share** is the organism's percentage of the whole community, on one scale. † the marker catalogue measured these populations as one species; its total is divided among them in proportion to the reads that mapped uniquely to each genome. ° the marker catalogue has no entry for this organism; the share is its whole-genome reading brought onto the same scale, and comes out of the unclassified band. § includes a share the marker catalogue had read under a relative; the whole-genome competition showed those reads belong to this organism, so the share came with them (the relative's name is given beneath). Where that makes the share differ fivefold from what the reference catalogue read under this name, the percentile is left out. *within*: a population the whole-genome search tells apart inside its genus, whose reads the marker catalogue counted under the relatives listed above it; it carries no share of its own. *not comparable*: the reference catalogue's reading of this species differs more than fivefold from the share shown - it draws the species' boundary or its markers differently - so its percentile is not a statement about the share and is left out. Where an organism has been reclassified, the current name is shown with the previous one beneath it; *also* gives its name in the Genome Taxonomy Database. A name made of a genus and a genome identifier marks an organism known only from assembled genomes.

## Organisms people ask about

The organisms most often named in gut-health writing, each looked up in your results. **20 of 34** were found in your sample. A blank here means this test searched and did not find it, which is not the same as the organism being absent from you — a low-abundance organism can sit below what one stool sample resolves.

| ORGANISM | IN YOUR SAMPLE | LEVEL | NOTE |
|---|---|---|---|
| *Akkermansia muciniphila* | found | **0.024%** | |
| *Bifidobacterium* | found | **0.028%** | genus total over 1 species |
| *Bifidobacterium adolescentis* | not found | — | searched, not detected |
| *Bifidobacterium longum* | not found | — | searched, not detected |
| *Blautia* | found | **0.223%** | genus total over 2 species |
| *Blautia obeum* | found | **0.293%** 90th pct | |
| *Blautia wexlerae* | found | **7.211%** 31th pct | listed here as *Blautia A wexlerae* |
| *Faecalibacterium* | found | **7.551%** | genus total over 15 species |
| *Faecalibacterium prausnitzii* | found | **6.182%** 81th pct | |
| *Lactobacillus* | found | **0.008%** | genus total over 1 species |
| *Roseburia* | found | **2.624%** | genus total over 6 species |
| *Ruminococcus* | not found | — | genus total over 1 species |
| *Ruminococcus bromii* | found | **3.396%** 93th pct | |
| *Agathobacter rectalis/Eubacterium rectale* | found | **0.518%** 52th pct | |
| *Alistipes finegoldii* | found | **0.306%** 87th pct | |
| *Alistipes putredinis* | found | **0.799%** 64th pct | |
| *Anaerobutyricum hallii/Eubacterium hallii* | found | **0.313%** 91th pct | |
| *Bacteroides* | found | **4.075%** | genus total over 13 species |

| ORGANISM | IN YOUR SAMPLE | LEVEL | NOTE |
|---|---|---|---|
| *Bacteroides fragilis* | not found | — | searched, not detected |
| *Bacteroides thetaiotaomicron* | **found** | **0.092%** 60th pct | |
| *Oxalobacter formigenes* | not found | — | searched, not detected |
| *Phocaeicola vulgatus* | not found | — | |
| *Prevotella* | **found** | **0.115%** | genus total over 8 species |
| *Prevotella/Segatella copri* | **found** | **0.115%** | genus total over 8 species |
| *Streptococcus thermophilus* | **found** | **0.035%** 92th pct | |
| *Enterococcus faecalis* | not found | — | searched, not detected |
| *Escherichia coli* | not found | — | searched, not detected |
| *Klebsiella pneumoniae* | not found | — | searched, not detected |
| *Ruminococcus/Mediterraneibacter gnavus* | not found | — | genus total over 1 species |
| *Bilophila wadsworthia* | **found** | **0.030%** | |
| *Desulfovibrio* | not found | — | searched, not detected |
| *Fusobacterium nucleatum* | not found | — | searched, not detected |
| *Proteobacteria/Pseudomonadota* | not found | — | |
| *Methanobrevibacter smithii* | not found | — | searched, not detected |

## What kind of organisms make up your community

**35.8%** beneficial · **34.2%** conditional · **26.4%** unknown · **0.7%** opportunist. Counted over 469 organisms in 239 genera. Organisms with no assigned class are shown as unclassified rather than dropped, and “unknown” describes the evidence, not the organism.

PART A — PATHOGENS: DISEASE-CAUSING ORGANISMS

# 7 Pathogens

A direct search of your sequencing reads for known disease-causing organisms — bacteria, protozoa, parasitic worms, microsporidia, fungi and viruses — against curated reference genomes. This is a **sequence** question, not a diagnosis: it reports what DNA is present, at what resolution, and how strong the evidence is. Section 18 takes every finding one at a time, and lists what could *not* be assessed in this sample.

| BACTERIAL PATHOGENS | PROTOZOA PATHOGENS | PARASITIC WORMS | MICROSPORIDIA PATHOGENS | FUNGAL PATHOGENS | VIRAL PATHOGENS |
|---|---|---|---|---|---|
| none | 2 | none | none | none | none |
| nameable | traces only | nameable | found | nameable | nameable |
| 6 normal or minor | 49 of 51 searched | 118 of 125 searched | all 15 searched | 6 normal or minor | 30 of 32 searched |

## Disease-causing organisms — 2 trace

Recognised causes of gut illness. A finding here is worth acting on, and worth showing to a doctor.

| ORGANISM | GROUP | WHAT THE SEQUENCE SHOWS | HOW MUCH | STRENGTH OF EVIDENCE |
|---|---|---|---|---|
| **Cryptosporidium canis** | Protozoa pathogens | trace, not enough to confirm | **small signal**<br>96 per million · 0.0096% of analysed DNA · 544 fragments in 1 regions | 0.04% of its genome covered · 100.0% identity |
| **Cyclospora cayetanensis (lineage A)** | Protozoa pathogens | trace, not enough to confirm | **faint signal**<br>9.4 per million · 0.0009% of analysed DNA · 53 fragments in 1 regions | 0.00% of its genome covered · 99.3% identity |

## Carried, but without the genes that cause disease — 1 named

For these organisms the disease is caused by a toxin or a pathotype, not by the species. Their DNA is here; the genes that would make them dangerous were looked for and not found. Carrying the harmless form is common, and for C. difficile it is associated with protection from infection rather than with disease.

| ORGANISM | GROUP | WHAT THE SEQUENCE SHOWS | HOW MUCH | STRENGTH OF EVIDENCE |
|---|---|---|---|---|
| **Clostridioides difficile** | Bacterial pathogens | enough sequence to name it | **faint signal**<br>4.2 per million · 0.0004% of analysed DNA · 24 fragments in 8 regions | 0.05% of its genome covered · 95.7% identity<br>disease genes not found |

## DNA that cannot be pinned to a species —

These organisms are the same genomic species as a relative that lives in most healthy guts, so alignment cannot tell them apart and a species name would be a guess. The evidence is shown in full, with the marker that would settle it. In stool, sequence like this usually comes from the harmless relative.

| ORGANISM | GROUP | WHAT THE SEQUENCE SHOWS | HOW MUCH | STRENGTH OF EVIDENCE |
|---|---|---|---|---|
| **Escherichia fergusonii** | Bacterial pathogens | shared sequence only — cannot tell apart<br>species not resolvable within the Escherichia coli / Shigella complex | **at the detection floor**<br>0.53 per million · well under a thousandth of a percent · 3 fragments in 3 regions | 0.01% of its genome covered · 100.0% identity |
| **Shigella/enteroinvasive Escherichia coli (combined group)** | Bacterial pathogens | shared sequence only — cannot tell apart<br>evidence aggregated from the complex; species not resolvable | —<br>no qualifying sequence to measure | 0.01% of its genome covered<br>disease genes not found |

## Sequence that belongs to relatives, not to these organisms —

The screen aligns reads to its own 485 reference genomes. Where a relative that is present has no reference of its own here, the sequence the two organisms share lands on the target and reads as a signal. These rows failed that check: the organism inventory, which maps the same reads competitively against every relative it found, did not find them, and the coverage is piled into the regions relatives share rather than spread across the genome. The relative that carries the reads is named on each row.

| ORGANISM | GROUP | WHAT THE SEQUENCE SHOWS | HOW MUCH | STRENGTH OF EVIDENCE |
|---|---|---|---|---|
| **Clostridium innocuum** | Bacterial pathogens | shared sequence only — cannot tell apart<br>shared sequence from relatives found in this sample; species not found by the organism inventory | **sequence from relatives**<br>4,501 fragments, shared with relatives · from Clostridium sp CAG 138 (0.837%), Clostridium sp CAG 413 (0.358%) · 4,501 fragments in 98 regions | 2.58% of its genome covered · 98.1% identity |

| ORGANISM | GROUP | WHAT THE SEQUENCE SHOWS | HOW MUCH | STRENGTH OF EVIDENCE |
|---|---|---|---|---|
| **Enterotoxigenic Bacteroides fragilis (ETBF)** | Bacterial pathogens | shared sequence only — cannot tell apart<br>shared sequence from relatives found in this sample; species not found by the organism inventory | **sequence from relatives**<br>3,328 fragments, shared with relatives · from Bacteroides clarus (1.39%), Bacteroides finegoldii (1.17%) · 3,328 fragments in 103 regions | 1.94% of its genome covered · 97.4% identity<br>disease genes not assessed |
| **Fusobacterium necrophorum** | Bacterial pathogens | shared sequence only — cannot tell apart<br>shared sequence from relatives found in this sample; species not found by the organism inventory | **sequence from relatives**<br>525 fragments, shared with relatives · from a relative that is present · 525 fragments in 6 regions | 0.86% of its genome covered · 100.0% identity |
| **Mediterraneibacter gnavus** | Bacterial pathogens | shared sequence only — cannot tell apart<br>shared sequence from relatives found in this sample; species not found by the organism inventory | **sequence from relatives**<br>19,156 fragments, shared with relatives · from Dorea hominis (0.102%), Mediterraneibacter lactaris (1.13%) · 19,156 fragments in 106 regions | 8.07% of its genome covered · 98.0% identity |

## Opportunists — harmless in most people, a problem in some — 2 named, 3 trace

These live in many healthy guts and cause disease mainly when something else has changed: recent antibiotics, a weakened immune system, surgery, or a large overgrowth. Presence alone is not illness.

| ORGANISM | GROUP | WHAT THE SEQUENCE SHOWS | HOW MUCH | STRENGTH OF EVIDENCE |
|---|---|---|---|---|
| **Burkholderia cepacia complex** | Bacterial pathogens | enough sequence to name it | **faint signal**<br>1.4 per million · 0.0001% of analysed DNA · 8 fragments in 5 regions | 0.01% of its genome covered · 96.0% identity |
| **Streptococcus gallolyticus group** | Bacterial pathogens | enough sequence to name it | **small signal**<br>23 per million · 0.0023% of analysed DNA · 131 fragments in 77 regions | 0.80% of its genome covered · 98.0% identity |
| **Apophysomyces, human-infecting members** | Fungal pathogens | trace, not enough to confirm | **at the detection floor**<br>0.35 per million · well under a thousandth of a percent · 2 fragments in 1 regions | 0.00% of its genome covered · 96.0% identity |
| **Enterococcus faecalis** | Bacterial pathogens | trace, not enough to confirm | **at the detection floor**<br>0.35 per million · well under a thousandth of a percent · 2 fragments in 2 regions | 0.01% of its genome covered · 97.4% identity |
| **Saksenaea, human-infecting members** | Fungal pathogens | trace, not enough to confirm | **small signal**<br>23 per million · 0.0023% of analysed DNA · 129 fragments in 1 regions | 0.00% of its genome covered · 100.0% identity |

## Common residents whose role is still unsettled — 2 named

Frequently found in healthy people. Research links some of them to disease, but whether they cause it is not established. Reported for completeness, not as a problem to fix.

| ORGANISM | GROUP | WHAT THE SEQUENCE SHOWS | HOW MUCH | STRENGTH OF EVIDENCE |
|---|---|---|---|---|
| **Campylobacter ureolyticus** | Bacterial pathogens | enough sequence to name it | **small signal**<br>14 per million · 0.0014% of analysed DNA · 81 fragments in 53 regions | 0.63% of its genome covered · 98.0% identity |
| **Peptostreptococcus anaerobius** | Bacterial pathogens | enough sequence to name it | **clear signal**<br>274 per million · 0.03% of analysed DNA · 1,552 fragments in 17 regions | 1.33% of its genome covered · 98.7% identity |

## Normal residents, food and environmental organisms — 1 named, 3 trace

Not pathogens. These are ordinary members of a healthy gut, or they arrive with food. They are listed because the screen searches for them by name, and finding them is expected rather than concerning.

| ORGANISM | GROUP | WHAT THE SEQUENCE SHOWS | HOW MUCH | STRENGTH OF EVIDENCE |
|---|---|---|---|---|
| **Geotrichum candidum** | Fungal pathogens | enough sequence to name it | **clear signal**<br>204 per million · 0.02% of analysed DNA · 1,156 fragments in 87 regions | 0.56% of its genome covered · 99.3% identity |

| ORGANISM | GROUP | WHAT THE SEQUENCE SHOWS | HOW MUCH | STRENGTH OF EVIDENCE |
|---|---|---|---|---|
| **Debaryomyces hansenii** | Fungal pathogens | trace, not enough to confirm | **at the detection floor** 0.18 per million · well under a thousandth of a percent · 1 fragments in 1 regions | 0.00% of its genome covered · 96.7% identity |
| **Malassezia globosa** | Fungal pathogens | trace, not enough to confirm | **at the detection floor** 0.18 per million · well under a thousandth of a percent · 1 fragments in 1 regions | 0.00% of its genome covered · 100.0% identity |
| **Malassezia restricta** | Fungal pathogens | trace, not enough to confirm | **at the detection floor** 0.53 per million · well under a thousandth of a percent · 3 fragments in 3 regions | 0.01% of its genome covered · 100.0% identity |

**How to read the two strengths. Enough sequence to name it** means distinct, organism-specific DNA passed every gate: the organism really is in the sample. **Trace** means matching DNA was found but too little of it to be sure of the species — it is neither a finding nor a clean negative, and at these amounts a close relative or a small amount of carry-over would look the same. **How much** is that organism's share of all the DNA analysed here; the bands describe the size of the signal in this sample and are not clinical thresholds, because no validated cut-off exists for stool sequencing.

**158 further targets** matched only sequence shared with relatives, so no organism can be named from it. Listed in section 18.

PART A — ESTIMATED BIOLOGICAL AGE OF BIOTA

# 8 Estimated biological age of biota

This page runs a **state-of-the-art machine learning ensemble** for microbiome age estimation: a linear carriage model over species presence and a **transformer neural network** applying multi-head attention across robust principal components, combined at fixed weights and trained on 4,070 healthy adults reconstructed from curatedMetagenomicData across 33 studies. **The interval, not the point, is the result.** It is the age this community reads as from its species patterns — a descriptive measure of the biota, not a health grade.

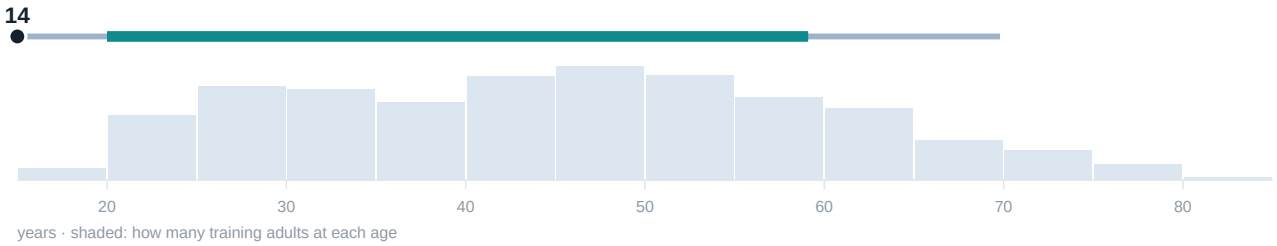
**14**
ESTIMATED AGE OF BIOTA (YEARS)

**14–65**
90% INTERVAL
the range the estimate sits in

**—**
DIFFERENCE FROM ACTUAL AGE
shown when actual age is on file

SUPPORTED

all gates passed · out-of-distribution check: in distribution · 81 of 123 detected species are in the model's namespace · mode: research




Dark bar: 80% interval. Light bar: 95% interval. Dot: point estimate. Red line: your actual age, when supplied. The shaded histogram is where the training adults sit; an estimate outside the well-populated range is less supported.

## What moved this estimate

Local contributions of species to *this* prediction, restricted to features that were stable across validation folds. They describe the arithmetic of the estimate, not age biology, causality or anything to act on.

| PUSHED THE ESTIMATE OLDER | | PUSHED THE ESTIMATE YOUNGER | |
|---|---|---|---|
| *Phascolarctobacterium succinatutens* | +1.0 y | *Campylobacter hominis* | -2.1 y |
| *Bacteroides ovatus* | +0.9 y | *Lawsonibacter asaccharolyticus* | -1.7 y |
| *Turicibacter sanguinis* | +0.9 y | *Prevotella buccalis* | -1.6 y |
| *Alistipes inops* | +0.7 y | *Prevotella timonensis* | -1.4 y |
| *Bacteroides cellulosilyticus* | +0.6 y | *Clostridium sp CAG 299* | -1.4 y |
| *Parasutterella excrementihominis* | +0.6 y | *Finegoldia magna* | -1.0 y |

## How the estimate was built

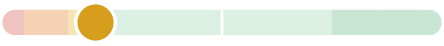
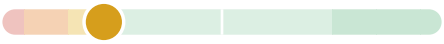
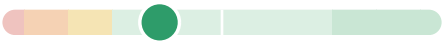
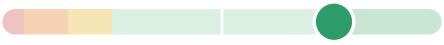
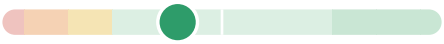
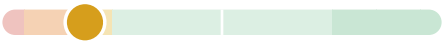
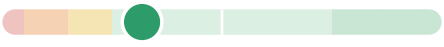
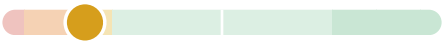
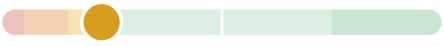
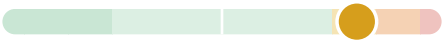
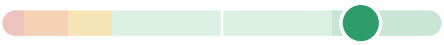
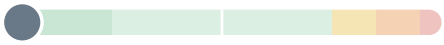
| ALGORITHM | ESTIMATE | WEIGHT |
|---|---|---|
| **Linear carriage model** ridge regression over standardised species presence/absence | 37.3 y | 80% |
| **Transformer** transformer-based robust PCA: multi-head attention over robust principal components, with a linear regression head | 48.5 y | 20% |
| **Model ensemble** | **39.5 y** | |
| **Microbiome health adjustment** | **-25.1 y** | |
| **Estimated age of biota** | **14 y** | |

PART A — WHAT YOUR BACTERIA CAN MAKE AND BREAK DOWN

# 9 Your Microbial Functions at a Glance

**What your microbes can make, break down and transform.**

Where your community's gene capacity for each function falls among 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort), grouped by what the function does and most notable first within each group. The bar is the same seven-band scale used throughout this report; the centre band is the middle half of the reference group. Every reading is *capacity* — the genes are present — not a measured amount of the compound.

| FUNCTION | WHERE YOU SIT | READING | |
|---|---|---|---|
| **FIBRE & DIETARY SUBSTRATES** | | | |
| **Dietary fibre and carbohydrate breakdown** ▼ |  | 21st | BELOW AVERAGE |
| Fewer carbohydrate-active enzyme families than most samples carry. | | | |
| **Cellulose** ▼ |  | 65th | TYPICAL |
| **General starch** ▼ |  | 23rd | BELOW AVERAGE |
| **Resistant starch** ▼ |  | 36th | TYPICAL |
| **Chitin** ▼ |  | 76th | ABOVE AVERAGE |
| **Pectin** ▼ |  | 40th | TYPICAL |
| **Inulin** ▼ |  | 55th | TYPICAL |
| **Fructooligosaccharides** ▼ |  | 55th | TYPICAL |
| **Galactooligosaccharides** ▼ |  | 19th | BELOW AVERAGE |
| **Xylooligosaccharides** ▼ |  | 34th | TYPICAL |
| **Isomaltooligosaccharides** ▼ |  | 32nd | TYPICAL |
| **Lactose** ▼ |  | 19th | BELOW AVERAGE |
| **Beta-glucans** ▼ |  | 58th | TYPICAL |
| **Arabinoxylan** ▼ |  | 23rd | BELOW AVERAGE |
| **Galactomannan and guar** ▼ |  | 40th | TYPICAL |
| **FERMENTATION & CROSS-FEEDING** | | | |
| **Hydrogen sulfide (H2S)** ▲ |  | 81st | ABOVE AVERAGE |
| Your gut carries more sulfide-producing capacity than most people's, largely from sulfate-reducing bacteria. | | | |
| **Propionate (SCFA)** ▲ |  | 78th | ABOVE AVERAGE |
| One genetic study reported that higher gut propionate production associates with increased Alzheimer's likelihood — the opposite direction to butyrate — and your capacity sits above most people's. | | | |
| **Butyrate (SCFA)** ▼ |  | 82nd | ABOVE AVERAGE |
| Your gut has more butyrate-producing capacity than most people, which is the favourable direction. | | | |
| **Fermentation routes and cross-feeding** |  | 79th | ABOVE AVERAGE |
| More fermentation-route capacity than most samples carry, across the acetate and lactate chemistry measured here. | | | |
| **Methane (CH4)** ▲ |  | 0 | NOT DETECTED |
| No reads matched this pathway in your sample. | | | |

| | | |
|---|---|---|
| Acetate production | 34th | TYPICAL |
| Acetate made from hydrogen and CO2 | 71st | TYPICAL |
| D-lactate production | 94th | HIGH |
| L-lactate production | 80th | ABOVE AVERAGE |
| Turning lactate into butyrate ▼ | 95th | NOTABLY HIGH |
| Turning lactate into propionate ▲ | 95th | NOTABLY HIGH |
| Succinate production ▲ | 26th | TYPICAL |
| Turning succinate into propionate ▲ | 14th | LOW |
| Turning propanediol into propionate ▲ | 59th | TYPICAL |
| Hydrogen production | 80th | ABOVE AVERAGE |
| Hydrogen used to make sulfide ▲ | 48th | TYPICAL |
| Turning taurine into sulfide ▲ | 95th | NOTABLY HIGH |

## VITAMINS & NUTRIENTS

**Vitamin K2 (menaquinone, MK-n) ▼** — 12th — LOW

Less bacterial menaquinone synthesis capacity than most people.

**Vitamin B1 (thiamine) ▼** — 76th — ABOVE AVERAGE

More of your bacteria carry a complete route to thiamine than in most samples.

**Folate (vitamin B9) ▼** — 29th — TYPICAL

Your community's folate synthesis capacity sits in the middle of the reference range.

**Vitamin B12 (cobalamin) ▼** — 38th — TYPICAL

Your community's B12 synthesis capacity sits in the middle of the reference range.

**Riboflavin (vitamin B2) ▼** — 60th — TYPICAL

Your community's riboflavin synthesis capacity sits in the middle of the reference range.

**Vitamin B6 (pyridoxal phosphate) ▼** — 42nd — TYPICAL

Your community's B6 synthesis capacity sits in the middle of the reference range.

**Vitamin B5 (pantothenate) ▼** — 58th — TYPICAL

Mid-range pantothenate synthesis capacity, which is the usual finding.

**Biotin (vitamin B7) ▼** — 52nd — TYPICAL

Your community's biotin synthesis capacity sits in the middle of the reference range.

**Vitamin B3 (niacin and NAD precursors) ▼** — 51st — TYPICAL

Your community's de-novo niacin capacity is mid-range, which is the usual finding.

## PROTEIN & NITROGEN METABOLISM

**p-Cresol / p-cresyl sulfate ▲** — 84th — ABOVE AVERAGE

More of your gut bacteria can convert tyrosine into p-cresol than in most people.

**Imidazole propionate (ImP) ▲** — 83rd — ABOVE AVERAGE

More of your gut bacteria carry the gene for making imidazole propionate than in most people.

**Indole-3-propionate (IPA) ▼** — 84th — ABOVE AVERAGE

Your gut has more indole-3-propionate capacity than most people, which is the favourable direction.

**Peptide uptake** — 28th — TYPICAL

A mid-range amount of peptide-uptake capacity, which is what most samples carry.

**Ammonia (from urea) ▲** — 70th — TYPICAL

Your urease capacity sits in the middle of the reference range.

**Indole / indoxyl sulfate ▼** — 33rd — TYPICAL

Tryptophan can be routed down several different paths and this test cannot tell which one dominates, so read this number together with indole-3-propionate below.

**Trimethylamine / TMAO ▲** — 65th — TYPICAL

TMAO has the strongest reported cognitive association of any metabolite in this panel, so being unremarkable here is genuinely worth knowing.

**Branched-chain amino acids (valine, leucine, isoleucine) ▲** — 37th — TYPICAL

Your community's branched-chain amino acid synthesis capacity sits in the middle of the reference range.

**Protein and nitrogen metabolism** — 45th — TYPICAL

Mid-range protein-fermentation capacity, which is the usual finding.

**Protein and peptide breakdown** — 59th — TYPICAL

**Amino-acid fermentation, including Stickland pairs** — 81st — ABOVE AVERAGE

**Aromatic precursor routes** — 44th — TYPICAL

## GUT-BRAIN & METABOLIC SIGNALLING

**gamma-aminobutyric acid (GABA) ▼** — 20th — BELOW AVERAGE

Less GABA-producing capacity than most people. One small study related low GABA production to a depression-associated brain signature; that is a single finding on an indirect measure and not a reason for concern.

**Histamine ▲** — 75th — ABOVE AVERAGE

Your gut carries more histamine-producing capacity than most people's.

**GABA breakdown** — 25th — TYPICAL

Mid-range GABA degradation capacity. This is shown beside your GABA production reading, which is a separate measurement over different genes; the two are never subtracted from one another.

**P9 protein (experimental)** — 30th — TYPICAL

A mid-range amount of read evidence on the tested sequence, read beside the family competition shown with it.

**Dopamine (from L-DOPA) and tyramine (from tyrosine)** — 33rd — TYPICAL

Your tyrosine decarboxylase capacity sits in the middle of the reference range.

**Tryptamine** — 49th — TYPICAL

Your tryptophan decarboxylase capacity sits in the middle of the reference range.

| | | |
|---|---|---|
| **GABA degradation** | 24th | BELOW AVERAGE |
| **4-aminobutyrate to butyrate** | 70th | TYPICAL |
| **Short-chain fatty acid signalling** ▼ | 95th | NOTABLY HIGH |

### PLANT-COMPOUND CONVERSION

| | | |
|---|---|---|
| **Glucosinolate to isothiocyanate conversion** | 13th | LOW |

Little or none of the characterised enzymes here. With only three reference sequences for a strain-specific reaction, a low value is weak evidence of absence rather than proof of it.

| | | |
|---|---|---|
| **Equol formation** | 95th | PRESENT |

Reads map to the published equol enzymes. That is the gene being present, not a measured equol response to soy, which is how producer status is actually defined.

| | | |
|---|---|---|
| **Urolithin A formation** | 0 | NOT DETECTED |

No reads matched this pathway in your sample.

### MUCUS, BILE & OTHER TRANSFORMATIONS

| | | |
|---|---|---|
| **Unconjugated bile acids** | 95th | NOTABLY HIGH |

Your gut has substantial bile-salt-deconjugating capacity. This is the gateway step that makes bile acids available for further bacterial processing, and it also affects fat absorption and cholesterol handling.

| | | |
|---|---|---|
| **Mucin glycans and lipid A** | 15th | LOW |

Less mucin-foraging and lipid A capacity than most samples carry.

| | | |
|---|---|---|
| **Secondary bile acids** ▲ | 71st | TYPICAL |

The organisms that perform this conversion are normally a very small fraction of the gut, so modest numbers here are expected and yours are unremarkable.

| | | |
|---|---|---|
| **Oxalate (degraded, not produced)** ▼ | 33rd | TYPICAL |

Your community's oxalate-degrading capacity sits in the middle of the reference range.

| | | |
|---|---|---|
| **Deconjugated glucuronides (oestrogens, drug metabolites, bilirubin)** ▲ | 37th | TYPICAL |

Your beta-glucuronidase capacity sits in the middle of the reference range.

| | | |
|---|---|---|
| **Mucin glycan foraging** | 22nd | BELOW AVERAGE |
| **Lipid A acylation and modification** | 18th | BELOW AVERAGE |
| **Hydroxysteroid dehydrogenase transformations** | 71st | TYPICAL |

### URATE & PURINES

| | | |
|---|---|---|
| **Urate degradation** | 77th | ABOVE AVERAGE |

More of the anaerobic urate-degrading route than most samples carry.

### POLYAMINES

| | | |
|---|---|---|
| **Spermidine formation** | 80th | ABOVE AVERAGE |

More spermidine-forming capacity than most samples carry, counted across both routes.

| | | |
|---|---|---|
| **Putrescine formation** | 61st | TYPICAL |

A mid-range amount of putrescine-forming capacity, similar to most samples in the reference set.

| | | |
|---|---|---|
| **Agmatine formation** | 43rd | TYPICAL |

A mid-range amount of arginine decarboxylase capacity, which is what most samples carry.

## GLUTATHIONE & ETHANOL

**Microbial ethanol formation** 76th ABOVE AVERAGE

More ethanol-forming gene capacity than most samples carry. Carrying adhE is common and is not the same as being a high-producing strain, which is what the fatty liver disease studies identified.

**Glutathione synthesis** 52nd TYPICAL

A mid-range amount of bacterial glutathione synthesis capacity, similar to most samples.

## TOXINS & VIRULENCE

**Colibactin and bacterial toxins (virulence factors, not metabolites)** ▲ 0 NOT DETECTED

No reads matched this pathway in your sample.

▲ research associates higher levels with worse outcomes · ▼ research associates higher levels with better outcomes · no arrow: the evidence is mixed, so the reading is shown in neutral grey. Section 19 has the findings, the references and the research on changing each pathway; every row above is a link to its card.

PART A — RESEMBLANCE TO PUBLISHED DISEASE PATTERNS

# 10 Resemblance to published disease patterns at a glance

For every pattern in the library: how closely this community resembles the group-level pattern published for people with that condition, as a percentile of the reference adults — the same scale as everything else. Resemblance is not a diagnosis and not a probability of disease. Every scored pattern has a full page, and the research on what has been tried for that condition, in Part B.

## The shape of the result

MIXED PICTURE

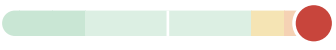
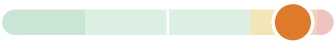
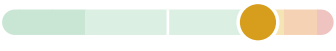
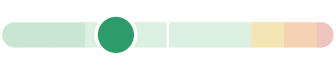
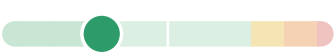
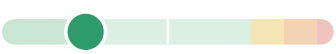
**3 of 33 patterns sit above the typical band; specificity is partial.**

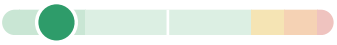
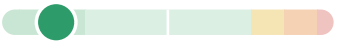
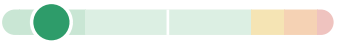
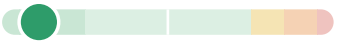
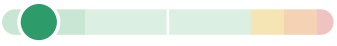
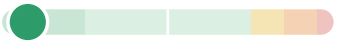
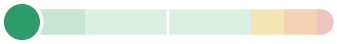
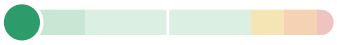
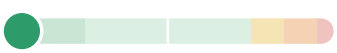
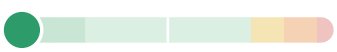
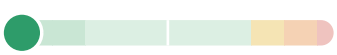
Several profiles score high and they share some but not all of their features. Read the ranked table with the specificity column: the profiles whose distinctive features drive the score matter more than those riding on shared butyrate-producer depletion or oral-taxa enrichment. GMWI2 general dysbiosis index: +2.00 (not dysbiotic).

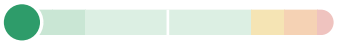
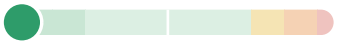
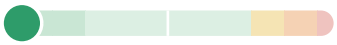
| 33 | 3 | 0.51 | 0 | +2.00 |
|---|---|---|---|---|
| PATTERNS SCORED | ABOVE TYPICAL | SPECIFICITY OF THOSE | NOT SCORED | GENERAL GUT-HEALTH INDEX |
| of 33 in the library | above 75th pct · 1 notably… | literature prior only | 0 declined · 0 unmeasurable | healthy range |

## Every pattern in the library, most resemblant first

**Maturity** is how far the underlying research has come (P0 preclinical → P5 clinical-grade); **evidence** lists which kinds of measurement the pattern uses. A pattern marked *declined* did not meet its own conditions for a meaningful score; *not measurable* means it needs an engine this pipeline does not yet run. Every scored pattern has a full page in Part B, and its row here is a link to it.

| PATTERN | MATURITY · EVIDENCE | RESEMBLANCE | READING |
|---|---|---|---|
| **Rheumatoid arthritis — gut pattern**<br>rheumatoid_arthritis competes with ad clinical, alopecia areata | P3 replicated<br>species · strain · genes |  | 95th NOTABLY HIGH |
| **Clinical Alzheimer's disease — stage-specific gut pattern**<br>alzheimers_disease competes with ra, celiac | P2 human cohort<br>species · pathways · genes · carriers |  | 88th HIGH |
| **Coeliac disease — gut pattern**<br>celiac_disease competes with ad clinical, alopecia areata | P2 human cohort<br>species · CAZymes · genes |  | 77th ABOVE AVERAGE |
| **Type 1 diabetes — gut pattern**<br>type_1_diabetes competes with ad clinical, alopecia areata | P2 human cohort<br>species · diversity · genes · carriers |  | 64th TYPICAL |
| **Alopecia areata — gut pattern**<br>alopecia_areata competes with ra, ad clinical | P1 one small study<br>species · pathways · genes |  | 47th TYPICAL |
| **Atherosclerotic cardiovascular disease — gut pattern**<br>atherosclerotic_cardiovascular_disease competes with ra, ad clinical | P3 replicated<br>species · genes · carriers |  | 39th TYPICAL |
| **Preclinical amyloid positivity — gut pattern**<br>alzheimers_disease competes with ra, celiac | P2 human cohort<br>species · genes |  | 38th TYPICAL |
| **IBS mixed type — gut pattern**<br>irritable_bowel_syndrome competes with ra, ad clinical | P1 one small study<br>species · genes |  | 34th TYPICAL |
| **IBS with constipation — gut pattern**<br>irritable_bowel_syndrome competes with ra, ad clinical | P1 one small study<br>species · genes |  | 30th TYPICAL |
| **Androgenetic alopecia — gut pattern (sex-stratified)**<br>androgenetic_alopecia competes with ra, ad clinical | P1 one small study<br>species · genes |  | 25th TYPICAL |

| PATTERN | MATURITY · EVIDENCE | RESEMBLANCE | READING |
|---|---|---|---|
| **Liver cirrhosis — gut pattern**<br>liver_cirrhosis competes with ra, ad clinical | P3 replicated<br>species · diversity · genes · carriers |  | 22nd BELOW AVERAGE |
| **Systemic lupus erythematosus — shared inflammatory gut pattern (not lupus-specific)**<br>systemic_lupus_erythematosus competes with ra, ad clinical | P1 one small study<br>species · strain |  | 16th BELOW AVERAGE |
| **Colorectal adenoma — gut pattern**<br>colorectal_adenoma competes with ra, ad clinical | P3 replicated<br>species · genes |  | 16th BELOW AVERAGE |
| **Colorectal cancer-associated community pattern**<br>colorectal_cancer · validation control competes with ra, ad clinical | P3 replicated<br>species · genes · carriers |  | 15th LOW |
| **Chronic kidney disease — gut pattern**<br>chronic_kidney_disease competes with ra, ad clinical | P2 human cohort<br>species · genes · carriers |  | 11th LOW |
| **Mild cognitive impairment — gut pattern**<br>alzheimers_disease competes with ra, celiac | P2 human cohort<br>species · pathways |  | 11th LOW |
| **Obesity — ecological pattern**<br>obesity competes with ra, ad clinical | P3 replicated<br>diversity · species · genes · carriers |  | 8th LOW |
| **IBS with diarrhoea — gut pattern**<br>irritable_bowel_syndrome competes with ra, ad clinical | P1 one small study<br>species · genes |  | 5th NOTABLY LOW |
| **Long COVID chronic-state community similarity**<br>post_acute_sequelae_of_covid19 competes with ra, ad clinical | P1 one small study<br>species · genes · diversity |  | 5th NOTABLY LOW |
| **Multiple sclerosis — gut pattern**<br>multiple_sclerosis competes with ad clinical, alopecia areata | P3 replicated<br>species · genes · carriers |  | 5th NOTABLY LOW |
| **Hypertension — gut pattern**<br>hypertension competes with ra, ad clinical | P2 human cohort<br>species · diversity · genes |  | 5th NOTABLY LOW |
| **Type 2 diabetes — gut pattern (metformin-gated)**<br>type_2_diabetes competes with ra, ad clinical | P3 replicated<br>species · genes · carriers |  | 2nd NOTABLY LOW |
| **ME/CFS-like gut microbiome similarity**<br>me_cfs competes with ra, ad clinical | P2 human cohort<br>species · genes · diversity |  | 2nd NOTABLY LOW |
| **Symptomatic dermographism (chronic inducible hives from skin stroking) — translated 16S pattern**<br>symptomatic_dermographism competes with ra, ad clinical | P1 one small study<br>species |  | 2nd NOTABLY LOW |
| **Irritable bowel syndrome — gut pattern**<br>irritable_bowel_syndrome competes with ra, ad clinical | P2 human cohort<br>species · diversity · genes |  | 1st NOTABLY LOW |
| **Chronic spontaneous urticaria (chronic hives) gut-microbiome similarity**<br>chronic_spontaneous_urticaria competes with ra, ad clinical | P2 human cohort<br>species · carriers |  | 1st NOTABLY LOW |
| **Ulcerative colitis — gut pattern**<br>ulcerative_colitis competes with ra, ad clinical | P3 replicated<br>species · diversity · genes · carriers |  | 1st NOTABLY LOW |

| PATTERN | MATURITY · EVIDENCE | RESEMBLANCE | READING |
|---|---|---|---|
| **Ankylosing spondylitis — gut pattern**<br>ankylosing_spondylitis competes with ad clinical, alopecia areata | P2 human cohort<br>species · genes · carriers |  | 1st NOTABLY LOW |
| **Parkinson's disease — gut pattern**<br>parkinsons_disease competes with ra, ad clinical | P3 replicated<br>species · genes · carriers |  | 1st NOTABLY LOW |
| **Inflammatory bowel disease (composite) — gut pattern**<br>inflammatory_bowel_disease competes with ra, ad clinical | P3 replicated<br>species · diversity · genes · carriers |  | <1st NOTABLY LOW |
| **Crohn's disease — gut pattern**<br>crohns_disease competes with ra, ad clinical | P3 replicated<br>species · diversity · genes · carriers |  | <1st NOTABLY LOW |
| **Metabolic dysfunction-associated steatotic liver disease — gut pattern**<br>masld competes with ra, ad clinical | P2 human cohort<br>species · diversity · genes · carriers |  | <1st NOTABLY LOW |
| **Major depressive disorder — gut pattern**<br>major_depressive_disorder competes with ra, ad clinical | P2 human cohort<br>species · genes |  | <1st NOTABLY LOW |

PART A — BIOFILM-RELATED POTENTIAL

# 11 Biofilm-related potential

Biofilms are organised communities of microbes held together by a shared matrix. They are a normal way for microbes to live, not a disease in themselves. Some support stable colonisation and a protective relationship with the intestinal lining; others help organisms persist, grow too close to tissue, or take part in inflammation. **This page reports concerning and protective readings separately, so that one never cancels the other.**

## Concerning biofilm-forming (bad)

**H-C · BIOFILM-ASSOCIATED COMMUNITY PATTERN**

**5** / 100 percentile

▼ lower reference range

Scope: limited to Biofilm-associated community pattern. Reference: 1,127 adults. Panel coverage: 0 of 2 features detected.
Reference interval 4–6 (200 resamples of the reference cohort).

### What drove this reading

| ORGANISM | LEVEL | RANK |
|---|---|---|
| *Mediterraneibacter gnavus* | 0% not detected | 22nd |
| *Escherichia coli and its close relatives* | 0% not detected | 10th |

## Protective gut lining (good)

**P-E · PROTECTIVE ECOLOGICAL SUPPORT (PARTIAL PANEL)**

**97** / 100 percentile

▲ upper reference range

Scope: limited to Protective ecological support (partial panel). Reference: 1,127 adults. Panel coverage: 3 of 3 features detected.
Reference interval 96–98 (200 resamples of the reference cohort).

### What contributed to this score

| ORGANISM | LEVEL | RANK |
|---|---|---|
| *Coprococcus species*<br>C. eutactus 5.8%, C. comes 1.8%, C. catus 1.3% | 8.90% | 98th |
| *Faecalibacterium prausnitzii* | 9.84% | 85th |
| *Blautia species*<br>B. obeum 0.6%, B. wexlerae 0.014% | 0.62% | 70th |

### What these numbers mean

**5** / 100 — **Biofilm-associated community pattern**
You sit in the lower part of the range: about 5 in every 100 reference adults measure lower than you. Higher means more of your community is made up of organisms that take part in biofilm growth.
Made up of ***Mediterraneibacter gnavus*** 0%, not detected (22nd); ***Escherichia coli and its close relatives*** 0%, not detected (10th) — none of these organisms was found in your sample, which is what places the reading low.

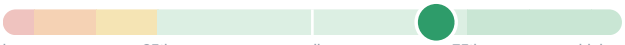
**97** / 100 — **Protective ecological support (partial panel)**
You sit in the upper part of the range: about 3 in every 100 reference adults measure higher than you. Higher means more of the community that keeps the gut lining intact and well fed.
Made up of ***Coprococcus species*** 8.9% (98th); ***Faecalibacterium prausnitzii*** 9.8% (85th); ***Blautia species*** 0.6% (70th).

A middle reading is the common one and is not a finding. These two numbers are never added, subtracted or averaged into a single biofilm score: a high concerning reading and a high protective reading can both be true at once, and collapsing them would hide exactly the case worth seeing. Both are read from which organisms are present and in what proportion, against 1,127 reference adults.

# 12 Your gut mycobiome

The fungi detected in your stool — and what the evidence says about them.

## Myco-Score (experimental)




**70**/100

Higher is healthier. From a neutral 50: +20 opportunistic fungi absent or very little. Policy weights, versioned (myco-score-v2); the evidence rules and every contribution are traced in the detail section.

| 0.0249% | 14 | 0% | 961 |
|---|---|---|---|
| FUNGAL DNA | FUNGI FOUND | OPPORTUNISTIC FUNGI | SPECIES SEARCHED |
| 1,410 of 5.7M read pairs | 3 confirmed, 1 likely, 10 trace | none found in your sample | 1,696 genomes, 3 methods |

Every one of the 5,660,139 quality-passed read pairs was searched against 961 fungal species (1,696 genomes, 321 genera) plus a single-copy-marker search and the targeted pathogen screen. Fungal signal is the share of all pairs confidently assigned to fungi after competition against near neighbours and bacterial decoys.

## What was searched for, and what was found

| GROUP | SPECIES SEARCHED | FOUND IN YOUR SAMPLE |
|---|---|---|
| **Candida and relatives** | 45 (408 genomes) | 1 · *Debaryomyces hansenii* (trace) 1 |
| **Saccharomyces and brewing/baking yeasts** | 23 (86 genomes) | none |
| **Malassezia and other skin yeasts** | 41 (68 genomes) | 2 · *Malassezia restricta* (trace) 3, *Malassezia globosa* (trace) 1 |
| **Aspergillus** | 137 (254 genomes) | none |
| **Penicillium** | 113 (223 genomes) | 2 · *Penicillium* 21, *Penicillium* (trace) 2 |
| **Mucor, Rhizopus and relatives** | 16 (31 genomes) | 1 · *Rhizomucor* (provisional) 45 |
| **Edible mushrooms** | 3 (3 genomes) | none |
| **Other moulds** | 73 (79 genomes) | none |
| **All other fungi** | 510 (544 genomes) | 8 · *Geotrichum candidum* 1053, *Geotrichum bryndzae*, *Leptographium* (trace) 276, *Fungi (placed above genus)* (trace) 5, *Schizothecium* (trace) 2, *Saksenaea, human-infecting members* (trace) 129, *Apophysomyces, human-infecting members* (trace) 2, *Sporothrix schenckii complex* (trace) 1 |

### Is this a normal amount of fungal DNA? context, not a target

In the largest survey of healthy adults (HMP, 147 volunteers), about **0.01%** of shotgun reads mapped to fungal genomes. Yours: **0.0249%** — within tenfold of the HMP figure. Fungi are a small fraction of stool DNA in everyone; there is no validated healthy range, and less is not better or worse. Genome size, cell walls and yesterday's meal all move this number.

| GENUS | HEALTHY ADULTS | USUALLY MEANS | YOU |
|---|---|---|---|
| ***Saccharomyces*** | in 97% of samples | bread, beer, wine, probiotics | not detected |
| ***Malassezia*** | in 88% of samples | skin yeast; stool carriage usually from skin | trace |
| ***Candida*** | in 81% of samples | common resident; C. albicans most often | not detected |
| ***Penicillium*** | commonly reported | mould on cheese and cured food; also inhaled from air | trace |
| ***Debaryomyces*** | commonly reported | salt-tolerant yeast of cheese rind and cured meat | trace |
| ***Cladosporium*** | commonly reported | one of the commonest airborne moulds; swallowed daily | not detected |
| ***Aspergillus*** | commonly reported | airborne mould; stool reads usually mean exposure, not colonisation | not detected |
| ***Pichia*** | commonly reported | fermentation yeast; competes with Candida in the gut | not detected |
| ***Kluyveromyces*** | commonly reported | dairy yeast of kefir and some cheeses | not detected |
| ***Cyberlindnera*** | commonly reported | fermentation yeast, often from plant food | not detected |
| ***Geotrichum*** | commonly reported | rind yeast of soft cheese; a normal transient | found |
| ***Rhodotorula*** | commonly reported | pink environmental yeast of damp surfaces and produce | not detected |

HMP prevalence is ITS2 amplicon on 317 stool samples from 147 healthy US adults; the shotgun read fraction is method-specific. Neither is a healthy range or a target: less fungal DNA is not better, and more is not worse. Source: Nash et al. 2017, Microbiome.

## Fungi detected

4 findings; each has a page in section 22

| FUNGUS | WHAT IT IS | DETECTION | FRAGMENTS | SHARE | STRAIN | EVIDENCE |
|---|---|---|---|---|---|---|
| ***Geotrichum candidum*** | food-associated | SUPPORTED | **1053** genome | **98%** | species only reference panel missing | — |
| ***Penicillium*** reported as Penicillium sp. B98-03 | environmental or food mould | SUPPORTED | **21** genome | **2%** | species only reference panel missing | — |
| ***Geotrichum bryndzae*** | no health evidence | SUPPORTED | **32** markers | — | not assessed | — |
| ***Rhizomucor*** | environmental or food mould | PROVISIONAL | **45** genome | — | not assessed | — |

**Traces, below the rule and not interpreted:** *Leptographium* 276 fragments (whole genome); *Fungi (placed above genus)* 5 fragments (whole genome); *Malassezia restricta* 3 fragments (whole genome); *Penicillium* 2 fragments (whole genome); *Schizothecium* 2 fragments (whole genome); *Saksenaea, human-infecting members* 129 fragments (pathogen screen); *Apophysomyces, human-infecting members* 2 fragments (pathogen screen); *Sporothrix schenckii complex* 1 fragment (pathogen screen); *Debaryomyces hansenii* 1 fragment (pathogen screen); *Malassezia globosa* 1 fragment (pathogen screen). One or two reads consistent with a species is a trace, not a detection.

## What kind of fungi

share of fungal DNA by organism class



food & drink yeasts 94% · environmental & food moulds 6% · no health evidence 0%

### Composition of the quantified fungal community

genome-coverage shares among supported species; not cell counts




Plus 336 fungal fragments placed only at genus level or below the support rule — counted in the signal, not inserted into the shares.

### Potentially beneficial evidence

*None activated under the current policy.* Unscored means unscored, not benign.

### Potential concerns

*None activated under the current policy.* Unscored means unscored, not benign.

## Bacteria that hold Candida in check

from your bacterial results; mouse and laboratory evidence

● *Bacteroides thetaiotaomicron* (0.09%, 36th pct) · — *Blautia producta* not detected · — *Lacticaseibacillus rhamnosus* not detected. Depleting these bacteria (antibiotics especially) is the best-described route to Candida expansion; their presence is context, not protection proven in you.

## What to do next

• Food- or probiotic-associated yeasts were found. Recording yeast probiotic and fermented-food intake and the time since the last dose separates a transient passenger from a persistent resident; a repeat sample after a break is the standard way to tell. Do not stop a beneficial therapy on a sequence result alone.

• This module reads stool DNA. It does not establish small-intestinal fungal overgrowth, mucosal invasion, fungemia, viability, growth phase or toxin production. No eradication target for all fungi follows from any result here, and no antifungal is selected by relative abundance.

Section 22 has each fungus in full: lanes, strain loci searched, evidence with its label, and the complete score trace.

PART A — SKIN-CONDITION RESEARCH PATTERNS

# 13 Gut-skin axis

## Gut-skin axis (beta)




**54**/100

Higher is the better direction. The weighted middle of 7 published panels, each weighted by how much of its own marker list this sample could measure. Carried mostly by Psoriatic disease, skin and joint pooled (59), Hidradenitis suppurativa (36), Chronic spontaneous urticaria (chronic hives) (44).

Researchers have compared the stool communities of people with several inflammatory skin diseases against healthy controls. This section asks, for each published list of organisms: **does this sample move in the direction that study reported?** There is no established, uniquely diagnostic stool signature for any of these conditions. A high reading is not a diagnosis and a low one does not rule anything out.

The scale is **pattern concordance** from low to high, drawn in neutral greys because it is agreement with a study's reported directions and not a health grade. It carries no per cent sign, because it is not a percentage or a probability. Where some organisms could not be measured, two separate figures appear: the index over the features that *were* measured, and the range the whole declared panel could occupy. The second is a coverage range, not a margin of error on the first.

| CONDITION AND COMPARISON | PANEL | INDEX | FEATURES | PANEL RANGE | EVIDENCE |
|---|---|---|---|---|---|
| **Plaque psoriasis** psoriasis vs controls | Plaque psoriasis — Deng 2026 single-marker SGB4348 | — (no usable features) | 0/1 single-marker result | 0–100 | tested in the same centre, not externally validated |
| **Plaque psoriasis** psoriasis vs healthy | Plaque psoriasis — Chang 2022 seven-taxon panel | **75** of 100 | 1/7 one feature only | 11–96 | one cohort, selected subset of features |
| **Plaque psoriasis** severe psoriasis vs healthy | Severe plaque psoriasis — Sá 2026 three-taxon panel | **41** of 100 | 1/3 one feature only | 14–80 | small cohort with heavy treatment confounding |
| **Psoriatic disease (skin and joint forms pooled)** pooled plaque psoriasis and psoriatic arthritis vs healthy | Psoriatic disease, skin and joint pooled — Xiao 2024 three species | **59** of 100 | 3/3 | 59–59 | pooled psoriasis and psoriatic arthritis contrast |
| **Psoriatic arthritis** treatment naive psoriatic arthritis vs healthy | Psoriatic arthritis — Liu 2024 nine gene families | — (none of the organisms could be measured) | 0/9 | 0–100 | one cohort, nominal significance only |
| **Psoriatic arthritis** psoriatic arthritis vs skin only psoriasis | Psoriatic arthritis versus skin-only psoriasis — KEGG 00072 capacity | — (none of the organisms could be measured) | 0/1 single-marker result | 0–100 | comparison between two disease subtypes |
| **Atopic dermatitis, adult** adult atopic dermatitis vs healthy | Adult atopic dermatitis (eczema) — Wang 2023 thirteen slots, translated from 16S | **41** of 100 | 2/13 | 6–91 | 16S-derived pattern applied to shotgun measurements |
| **Non-segmental vitiligo** advanced nonsegmental vitiligo vs healthy | Advanced non-segmental vitiligo — Luan 2023 eleven species | **24** of 100 | 2/11 | 4–86 | one cohort, selected subset of features |
| **Non-segmental vitiligo** active spreading vitiligo vs external controls | Actively spreading vitiligo — Ju 2025 four species, exploratory | — (no usable features) | 0/4 | 0–100 | cases and controls from different studies |
| **Hidradenitis suppurativa** hidradenitis suppurativa vs healthy | Hidradenitis suppurativa — Ogut 2022 single genus, translated from 16S | **36** of 100 | 1/1 single-marker result | 36–36 | one feature at nominal significance |

Panels for the same condition are shown side by side and never averaged into one condition score: they come from different cohorts, they disagree in places, and averaging would hide that. One organism appearing in several panels is one measurement, not several agreeing results.

# 14 What you can do about it

This report found **36** things worth doing something about. For **29** of them there is published research on changing them — **198** findings in all, each one named below with the exact food, compound, supplement, probiotic, herb, medication or procedure that was studied, where the finding comes from, and what was actually seen. Each problem is set out the same way: what was found, what might move it the right way, and what has been shown to move it the wrong way.

| WHAT WAS FOUND | WHERE IT IS EXPLAINED | OPTIONS ON RECORD |
|---|---|---|
| **1.** Geotrichum candidum was detected in the mycobiome › | MYCOBIOME section 22 | **13** |
| **2.** Penicillium sp. B98-03 was detected in the mycobiome › | MYCOBIOME section 22 | **13** |
| **3.** Enterocloster GWHFCPF00000000 is overgrown › | ORGANISM this reading, in full › | **3** · 1 to avoid |
| **4.** Blautia_A luti is overgrown › | ORGANISM this reading, in full › | **2** · 2 to avoid |
| **5.** Bacteroides coagulans needs qualification › | ORGANISM this reading, in full › | **7** · 9 to avoid |
| **6.** Dorea formicigenerans needs qualification › | ORGANISM this reading, in full › | **2** · 2 to avoid |
| **7.** GGB74412 SGB5075 is well below the usual level › | ORGANISM this reading, in full › | nothing on record |
| **8.** Faecalibacterium intestinale is well below the usual level › | ORGANISM this reading, in full › | **3** · 3 to avoid |
| **9.** Blautia_A MGYG000001338 needs qualification › | ORGANISM this reading, in full › | **2** · 2 to avoid |
| **10.** Faecalibacterium butyricigenerans is well below the usual level › | ORGANISM this reading, in full › | **3** · 3 to avoid |
| **11.** Roseburia inulinivorans is well below the usual level › | ORGANISM this reading, in full › | **7** · 3 to avoid |
| **12.** Wujia chipingensis is well below the usual level › | ORGANISM this reading, in full › | nothing on record |
| **13.** Hexa-acylated LPS carriers as a group is low › | MICROBIAL GROUP this reading, in full › | **9** · 41 to avoid |
| **14.** Resident bifidobacteria as a group is low › | MICROBIAL GROUP this reading, in full › | **2** · 2 to avoid |
| **15.** Sulfate-reducing bacteria as a group is below average › | MICROBIAL GROUP this reading, in full › | **1** · 1 to avoid |
| **16.** Supplement and fermented-dairy species as a group is above average › | MICROBIAL GROUP this reading, in full › | **1** · 1 to avoid |
| **17.** Urolithin A formation capacity is notably low › | METABOLITE PATHWAY this reading, in full › | **18** · 11 to avoid |
| **18.** Fermentation routes and cross-feeding capacity is above average › | METABOLITE PATHWAY this reading, in full › | **16** · 8 to avoid |
| **19.** GABA breakdown capacity is below average › | METABOLITE PATHWAY this reading, in full › | **4** · 2 to avoid |
| **20.** gamma-aminobutyric acid (GABA) capacity is below average › | METABOLITE PATHWAY this reading, in full › | **2** |
| **21.** Glucosinolate to isothiocyanate conversion capacity is low › | METABOLITE PATHWAY this reading, in full › | **38** · 40 to avoid |
| **22.** Histamine capacity is above average › | METABOLITE PATHWAY this reading, in full › | **2** · 2 to avoid |
| **23.** Hydrogen sulfide (H2S) capacity is above average › | METABOLITE PATHWAY this reading, in full › | **3** · 1 to avoid |

| WHAT WAS FOUND | WHERE IT IS EXPLAINED | OPTIONS ON RECORD |
|---|---|---|
| **24.** Imidazole propionate (ImP) capacity is above average › | METABOLITE PATHWAY this reading, in full › | **1** · 1 to avoid |
| **25.** Mucin glycans and lipid A capacity is low › | METABOLITE PATHWAY this reading, in full › | **5** · 2 to avoid |
| **26.** p-Cresol / p-cresyl sulfate capacity is above average › | METABOLITE PATHWAY this reading, in full › | **2** |
| **27.** Propionate (SCFA) capacity is above average › | METABOLITE PATHWAY this reading, in full › | **3** · 2 to avoid |
| **28.** Unconjugated bile acids capacity is high › | METABOLITE PATHWAY this reading, in full › | **4** |
| **29.** Urate degradation capacity is above average › | METABOLITE PATHWAY this reading, in full › | **29** · 15 to avoid |
| **30.** Dietary fibre and carbohydrate breakdown capacity is below average › | METABOLITE PATHWAY this reading, in full › | nothing on record |
| **31.** Equol formation capacity is high › | METABOLITE PATHWAY this reading, in full › | nothing on record |
| **32.** Microbial ethanol formation capacity is above average › | METABOLITE PATHWAY this reading, in full › | nothing on record |
| **33.** Spermidine formation capacity is above average › | METABOLITE PATHWAY this reading, in full › | nothing on record |
| **34.** Vitamin K2 (menaquinone, MK-n) capacity is low › | METABOLITE PATHWAY this reading, in full › | nothing on record |
| **35.** Rheumatoid arthritis pattern: notably high › | DISEASE PATTERN section 20 | **2** |
| **36.** Clinical Alzheimer's disease pattern: high › | DISEASE PATTERN section 20 | **1** |

‹ back to the list

MYCOBIOME

## 1. Geotrichum candidum was detected in the mycobiome

species

**WHAT MIGHT BRING IT DOWN** 10 of 13 on record

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **L. rhamnosus GR-1 plus L. reuteri RC-14** | medication - a prescriber's decision<br>prescriber decides | human trial<br>PMID 19187507 | Adjunct group had fewer positive yeast cultures and less symptomatic discharge.<br>Vaginal endpoint, not demonstrated gut fungal clearance; combination plus medication. |
| **Coconut-oil-rich diet**† | way of eating | animal study<br>taken by mouth in the study<br>PMID 27303684 | Lower intestinal colonization than comparison fat diets<br>Colonization, not invasive candidiasis; no established human gut treatment; whole oil differs from free fatty acids |
| **Nanoparticulate tartrate-modified ferrihydrite (Nano Fe(III)) versus soluble ferrous sulfate; T…**† | way of eating | animal study<br>taken by mouth in the study<br>PMID 25461615 | Both are host iron-delivery candidates with different microbial availability → Sulfate partly restored diversity after deficiency; Nano Fe(III) did not → Early iron-depletion imprint persisted; lower microbial availability did not equal microbiome restoration<br>Iron depletion at weaning; no gastrointestinal disease. Authors infer lower microbial availability from response patterns. No universal pathogen-inaccessibility claim; withholding iron also had ecological costs. |
| **HST013 — Microbiota-derived butyrate / host PPAR-γ pathway, mechanistic bridge**† | supplement or compound | animal study<br>taken by mouth in the study<br>PMID 28798125 | Mouse and epithelial mechanistic studies. Ecological cascade supported; product-specific transfer requires evidence.<br>Butyrate-producing community → epithelial PPAR-γ signaling `M` → colonocyte metabolic oxygen consumption and lower nitrate/oxygen availability `M` → restriction of Escherichia/Salmonella expansion `M`. Separate inflammatory nitrate study supports host nitrate → E. coli growth `M`. |
| **Monolaurin, local oral-cavity exposure**† | supplement or compound | animal study<br>taken by mouth in the study<br>PMID 30068882 | C. albicans: reduced fungal infection/biofilm burden in a mouse oral-candidiasis model<br>“Oral” here denotes mouth infection/local treatment, not demonstrated colon efficacy after swallowing |
| **S. cerevisiae CNCMI3856**† | supplement or compound | animal study<br>taken by mouth in the study<br>PMID 39243351 | Reduced mortality/weight loss, permeability and tissue injury; coaggregation and fecal yeast recovery observed.<br>Fecal recovery during treatment does not prove lasting human colonization or human Salmonella cure. |
| **Microbiota-derived SCFAs / B. thetaiotaomicron producer versus deficient mutant**† | fibre | animal study<br>reaches the gut unaided<br>PMID 42242208 | Fungal intracellular acidification and reduced hexose uptake → constrained C. albicans growth; community shifts strengthen resistance |
| **BPC-157 → reduced experimental cysteamine colitis injury**† | supplement or compound | animal study<br>route not stated<br>PMID 11595448 | Primary rat studies of induced colon lesions, including delayed treatment in chronic injury models.<br>Eligible as preclinical candidate. Chemical injury is not human IBD; human gut efficacy has not been established by these papers. |
| **Lacticaseibacillus rhamnosus Lcr35, native versus manufactured products**‡ | probiotic | laboratory culture<br>taken by mouth in the study<br>PMID 22542933 | Intestinal formulations improved acid/bile survival; vaginal formulation enhanced Candida inhibition and glycogen metabolism.<br>Exact manufacturing/formulation altered effect substantially; vaginal product cannot be assumed equivalent to oral product. |
| **Lcr35 plus sodium thiosulfate, vaginal-product formulation**‡ | medication - a prescriber's decision<br>prescriber decides | laboratory culture<br>taken by mouth in the study<br>PMID 33051479 | Formulation enhanced antifungal effect; sulfur-associated volatile S-methyl thioacetate linked to activity.<br>Sodium thiosulfate here is an excipient in an experimental formulation, not proof of oral sulfur supplementation benefit. |

### ALSO STUDIED ON THIS, DIRECTION NOT STATED 2

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Butyrate-supporting fiber network linked to host oxygen control**† | fibre | animal study<br>reaches the gut unaided<br>PMID 38838675 | Fiber-dependent butyrate generation → colonocyte metabolism/epithelial hypoxia → less oxygen-supported Candida expansion |
| **Capric acid; separately tested lauric acid**‡ | supplement or compound | laboratory culture<br>taken by mouth in the study<br>PMID 11600381 | Inactivated tested yeast cells in laboratory comparisons<br>Not proof that triglyceride oil, caprylic-acid capsules or monolaurin have the same result; oral exposure unresolved |

‹ back to the list

MYCOBIOME

## 2. Penicillium sp. B98-03 was detected in the mycobiome

species

**WHAT MIGHT BRING IT DOWN** 10 of 13 on record

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **L. rhamnosus GR-1 plus L. reuteri RC-14** | medication - a prescriber's decision<br>prescriber decides | human trial<br>PMID 19187507 | Adjunct group had fewer positive yeast cultures and less symptomatic discharge.<br>Vaginal endpoint, not demonstrated gut fungal clearance; combination plus medication. |
| **Coconut-oil-rich diet**† | way of eating | animal study<br>taken by mouth in the study<br>PMID 27303684 | Lower intestinal colonization than comparison fat diets<br>Colonization, not invasive candidiasis; no established human gut treatment; whole oil differs from free fatty acids |
| **Nanoparticulate tartrate-modified ferrihydrite (Nano Fe(III)) versus soluble ferrous sulfate; T…**† | way of eating | animal study<br>taken by mouth in the study<br>PMID 25461615 | Both are host iron-delivery candidates with different microbial availability → Sulfate partly restored diversity after deficiency; Nano Fe(III) did not → Early iron-depletion imprint persisted; lower microbial availability did not equal microbiome restoration<br>Iron depletion at weaning; no gastrointestinal disease. Authors infer lower microbial availability from response patterns. No universal pathogen-inaccessibility claim; withholding iron also had ecological costs. |
| **HST013 — Microbiota-derived butyrate / host PPAR-γ pathway, mechanistic bridge**† | supplement or compound | animal study<br>taken by mouth in the study<br>PMID 28798125 | Mouse and epithelial mechanistic studies. Ecological cascade supported; product-specific transfer requires evidence.<br>Butyrate-producing community → epithelial PPAR-γ signaling `M` → colonocyte metabolic oxygen consumption and lower nitrate/oxygen availability `M` → restriction of Escherichia/Salmonella expansion `M`. Separate inflammatory nitrate study supports host nitrate → E. coli growth `M`. |
| **Monolaurin, local oral-cavity exposure**† | supplement or compound | animal study<br>taken by mouth in the study<br>PMID 30068882 | C. albicans: reduced fungal infection/biofilm burden in a mouse oral-candidiasis model<br>“Oral” here denotes mouth infection/local treatment, not demonstrated colon efficacy after swallowing |
| **S. cerevisiae CNCMI3856**† | supplement or compound | animal study<br>taken by mouth in the study<br>PMID 39243351 | Reduced mortality/weight loss, permeability and tissue injury; coaggregation and fecal yeast recovery observed.<br>Fecal recovery during treatment does not prove lasting human colonization or human Salmonella cure. |
| **Microbiota-derived SCFAs / B. thetaiotaomicron producer versus deficient mutant**† | fibre | animal study<br>reaches the gut unaided<br>PMID 42242208 | Fungal intracellular acidification and reduced hexose uptake → constrained C. albicans growth; community shifts strengthen resistance |
| **BPC-157 → reduced experimental cysteamine colitis injury**† | supplement or compound | animal study<br>route not stated<br>PMID 11595448 | Primary rat studies of induced colon lesions, including delayed treatment in chronic injury models.<br>Eligible as preclinical candidate. Chemical injury is not human IBD; human gut efficacy has not been established by these papers. |
| **Lacticaseibacillus rhamnosus Lcr35, native versus manufactured products**‡ | probiotic | laboratory culture<br>taken by mouth in the study<br>PMID 22542933 | Intestinal formulations improved acid/bile survival; vaginal formulation enhanced Candida inhibition and glycogen metabolism.<br>Exact manufacturing/formulation altered effect substantially; vaginal product cannot be assumed equivalent to oral product. |
| **Lcr35 plus sodium thiosulfate, vaginal-product formulation**‡ | medication - a prescriber's decision<br>prescriber decides | laboratory culture<br>taken by mouth in the study<br>PMID 33051479 | Formulation enhanced antifungal effect; sulfur-associated volatile S-methyl thioacetate linked to activity.<br>Sodium thiosulfate here is an excipient in an experimental formulation, not proof of oral sulfur supplementation benefit. |

ALSO STUDIED ON THIS, DIRECTION NOT STATED 2

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Butyrate-supporting fiber network linked to host oxygen control**† | fibre | animal study<br>reaches the gut unaided<br>PMID 38838675 | Fiber-dependent butyrate generation → colonocyte metabolism/epithelial hypoxia → less oxygen-supported Candida expansion |
| **Capric acid; separately tested lauric acid**‡ | supplement or compound | laboratory culture<br>taken by mouth in the study<br>PMID 11600381 | Inactivated tested yeast cells in laboratory comparisons<br>Not proof that triglyceride oil, caprylic-acid capsules or monolaurin have the same result; oral exposure unresolved |

‹ back to the list · the reading this is about ›

ORGANISM

## 3. Enterocloster GWHFCPF00000000 is overgrown

0.28% of your community; 2× the typical carrier; 96th percentile

WHAT MIGHT BRING IT DOWN 3

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Dietary fibre (inulin, pectin)** | prebiotic fibre | human trial<br>source | Fibre shifts the Lachnospiraceae balance away from bile-tolerant members toward butyrate producers. |
| **Lower saturated-fat intake** | way of eating | human cohorts<br>source | Bile-acid load falls with dietary fat, and with it the bile-tolerant genera. |
| **Bifidobacterium longum strains**† | probiotic | animal study<br>taken by mouth in the study<br>source | — |

SHOWN TO PUSH IT THE OTHER WAY 1

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **High-fat, animal-based diet** | way of eating | human trial<br>source | — |

‹ back to the list · the reading this is about ›

ORGANISM

## 4. Blautia_A luti is overgrown

0.39% of your community; 99% above the typical carrier; 100th percentile

WHAT MIGHT BRING IT DOWN 2

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Increased dietary fibre diversity (vegetables, legumes over refined starch)** | way of eating | human trial<br>source | Relative abundance falls as fibre-fed competitors expand. |
| **Carnosic-acid-rich rosemary extract; lean and obese female Zucker rats**† | herb or botanical extract | animal study<br>reaches the gut unaided | M: caecal groups changed: Blautia coccoides and Bacteroides/Prevotella groups increased; the Lactobacillus/Leuconostoc/Pediococcus group decreased. [S14]<br>M/A: intestinal β-glucosidase activity decreased and faecal fibre excretion increased in both phenotypes. [S14] |

SHOWN TO PUSH IT THE OTHER WAY 2

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Refined carbohydrate intake** | way of eating | human cohorts<br>source | — |
| **Laminarin-derived β-1,3 oligosaccharides**‡ | prebiotic fibre | laboratory culture<br>reaches the gut unaided<br>PMID 33043794 | Increased growth |

ALSO STUDIED ON THIS, DIRECTION NOT STATED 1

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Laminarin**‡ | prebiotic fibre | laboratory culture<br>taken by mouth in the study<br>PMID 33043794 | B. uniformis JCM13288 releases β-glucan oligosaccharides → Blautia producta JCM1471 growth |

‹ back to the list · the reading this is about ›

ORGANISM

## 5. Bacteroides coagulans needs qualification

0.02% of your community; 4× the typical carrier; 91th percentile

WHAT MIGHT BRING IT DOWN 7

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Whole grains, legumes and resistant starch** | prebiotic fibre | human trial<br>source | Shifts the Bacteroides-to-Firmicutes balance toward fermenters. |
| **Curcumin in a 481-compound/47-gut-strain screen, followed by obese-mouse and donor-community-tr…**† | plant compound | animal study<br>taken by mouth in the study | M: isolate responses differed: E. lenta and B. thetaiotaomicron were inhibited, whereas Roseburia intestinalis and Eubacterium rectale could be stimulated under studied conditions. [S18]<br>M: mouse curcumin exposure shifted the community, including decreased Lactobacillus; some insulin-resistance effects transferred with donor communities. [S18] |
| **Carbohydrate fermentation plus antibiotic or motility disturbance**† | medication - a prescriber's decision<br>prescriber decides | animal study<br>taken by mouth in the study<br>PMID 25498344 | B. thetaiotaomicron succinate → C. difficile reduction to butyrate → pathogen growth advantage |
| **Host mucin glycans after antibiotics; not a supplement**† | medication - a prescriber's decision<br>prescriber decides | animal study<br>taken by mouth in the study<br>PMID 23995682 | B. thetaiotaomicron sialidase liberates sialic acid → C. difficile uptake → expansion |
| **Curcumin, isolated compound**† | plant compound | animal study<br>reaches the gut unaided<br>PMID 37244386 | E. lenta; collateral Bacteroides thetaiotaomicron<br>Direct exact-species laboratory observation; no human E. lenta depletion trial. The same study found context-dependent inhibition of other commensals and growth promotion of Roseburia intestinalis and Eubacterium rectale; selectivity is not assured |
| **0.5% inulin in nutrient-rich BHIS; six-species community**‡ | prebiotic fibre | laboratory culture<br>reaches the gut unaided<br>PMID 42429610 | Community competition shifts → R. gnavus relative enrichment; B. uniformis, B. thetaiotaomicron and A. muciniphila decline |
| **Chitosan / low-MW chitosan / chitooligosaccharides**‡¶ | supplement or compound | laboratory culture<br>route not stated<br>PMID 22528310 | Direct antimicrobial testing included Roseburia intestinalis, F. prausnitzii, B. thetaiotaomicron, B. vulgatus and other nonpathogenic anaerobes. |

**SHOWN TO PUSH IT THE OTHER WAY** 9

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Animal-protein, high-fat diet** | way of eating | human trial<br>source | — |
| **HST028 — NAC; microbiota-dependent absorption, AB024** | supplement or compound | human cohorts<br>PMID 40389439 | Rat perfusion and mechanistic microbial work plus a human predictive cohort. Drug-disposition pathway supported; antibiofilm efficacy not measured.<br>NAC exposure / derived cysteine → Bacteroides sphingolipid-related metabolism `M`; B. thetaiotaomicron/B. fragilis cysteine palmitoylation → altered NAC intestinal permeation `M`; human microbial features → absorption prediction `A`. |
| **Pea fiber**† | fibre | animal study<br>reaches the gut unaided<br>PMID 31539500 | Increased abundance |
| **Quercetin plus starch; laboratory pairing of B. thetaiotaomicron and E. ramulus**‡ | plant compound | laboratory culture<br>taken by mouth in the study | M: B. thetaiotaomicron liberated glucose/maltose from starch; E. ramulus did not utilise starch alone. [S20]<br>M: the released carbohydrates supported E. ramulus growth and quercetin degradation. [S20] |

**ALSO STUDIED ON THIS, DIRECTION NOT STATED** 14

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **N-acetylcysteine, exposure/biotransformation record** | supplement or compound | human cohorts<br>doi:10.1038/s41467-025-59916-7 | B. thetaiotaomicron/B. fragilis sphingolipid metabolism altered NAC-derived cysteine handling and intestinal drug permeability<br>Delivery evidence is community dependent. This is drug permeation, not a diagnostic measure of “leaky gut” or a demonstrated antibiofilm outcome |
| **Food-grade targeted fiber mixtures: pea/citrus/orange/tomato**† | fibre | animal study<br>reaches the gut unaided<br>PMID 34163075 | Gnotobiotic screens and human snack interventions altered targeted Bacteroides and expressed glycan-utilization functions. |
| **F. venenatum mycoprotein β-glucan**‡ | way of eating | laboratory culture<br>taken by mouth in the study<br>PMID 37253778 | B. cellulosilyticus WH2 or B. thetaiotaomicron VPI5482 releases glycans → B. longum NCIMB8809 / B. breve UCC2003 cross-feeding |
| **Same fungal β-glucan**‡ | way of eating | laboratory culture<br>taken by mouth in the study<br>PMID 37253778 | Cross-feeding |
| **Laminarin**‡ | prebiotic fibre | laboratory culture<br>taken by mouth in the study<br>PMID 33043794 | B. uniformis JCM13288 releases β-glucan oligosaccharides → Blautia producta JCM1471 growth |
| **Alginate; low F. prausnitzii / butyrate support**‡ | supplement or compound | laboratory culture<br>taken by mouth in the study<br>PMID 34053530 | Bacteroides breakdown left oligosaccharides usable by an F. prausnitzii isolate; F. prausnitzii rose in fecal fermentation. |

‹ back to the list · the reading this is about ›

ORGANISM

## 6. Dorea formicigenerans needs qualification

0.58% of your community; 6× the typical carrier; 96th percentile

**WHAT MIGHT BRING IT DOWN** 2

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Low-FODMAP diet (for gas symptoms)** | way of eating | human trial<br>source | Reduces fermentable substrate for gas producers; clinically supervised. |
| **TOTUM-070 proprietary plant-extract blend: olive leaf, artichoke leaf, chrysanthellum aerial pa…** | herb or botanical extract | human trial<br>PMID 41444426 | Dorea genus<br>Microbiome was a secondary endpoint, species not resolved; cannot assign blend effects to any individual ingredient. Multiple authors employed by or financially linked to manufacturer |

**SHOWN TO PUSH IT THE OTHER WAY** 2

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Fermentable fibre** | prebiotic fibre | human trial<br>source | — |
| **Oligonol, litchi-derived low-molecular-weight polyphenol preparation** | plant compound | human trial<br>PMID 35889878 | Dorea genus<br>Genus-level exploratory community endpoint; not exact D. formicigenerans, D. longicatena or CAG 317. No basis to label all Dorea pathogenic. Liver changes within one arm do not establish microbial causality |

‹ back to the list · the reading this is about ›

ORGANISM

## 7. GGB74412 SGB5075 is well below the usual level

0.02% of your community; 90% below the typical carrier; 2th percentile

No study on record says which way to bring it up. What has been studied on it is below, with the direction its source actually reported — which was none.

**ALSO STUDIED ON THIS, DIRECTION NOT STATED** 1

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Apple pectin / pectic oligosaccharides; F. prausnitzii / L. eligens**‡ | prebiotic fibre | laboratory culture<br>reaches the gut unaided<br>PMID 22101049 | Tested F. prausnitzii strains used apple pectin; F. prausnitzii and Eubacterium/Lachnospira eligens used galacturonide DP4/DP5. |

‹ back to the list · the reading this is about ›

ORGANISM

## 8. Faecalibacterium intestinale is well below the usual level

0.95% of your community; 65% below the typical carrier; 10th percentile

**WHAT MIGHT BRING IT UP** 3

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Standardized oral pomegranate extract** | herb or botanical extract | human trial<br>doi:10.3390/foods13010015 | Increased relative abundance of selected species; increases also included M. smithii and C. aerofaciens<br>Mixed ecological response; do not label every increased organism beneficial or infer lower methane |
| **Riboflavin/B2 in RIBOGUT; Randomized placebo-controlled healthy-adult trial; oral** | supplement or compound | human trial<br>PMID 35943883 | Vitamin exposure changed microbial activity → Pooled active groups showed a butyrate increase without Faecalibacterium abundance gain → Clinical benefit and a change in stool butyrate genes were not demonstrated<br>Existing fermenters, two studied exposure groups and short follow-up. Individual active arms had limited effects; pooled finding is not a guaranteed producer-expansion effect. |
| **Butyrate-producer preparations including B. pullicaecorum25-3T and six-isolate consortium**‡ | probiotic | laboratory culture<br>reaches the gut unaided<br>PMID 28904372 | F. prausnitzii preparation/consortium increased butyrate;25-3T/consortium improved barrier integrity.<br>Individual-strain and consortium effects differ; no commercial six-isolate equivalence asserted. |

**SHOWN TO PUSH IT THE OTHER WAY** 3

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **PHGG; lower Escherichia-Shigella / Klebsiella alongside support** | prebiotic fibre | human cohorts PMID 37541036 | In 41 human fecal inocula, PHGG fermentation enriched Bifidobacterium/Faecalibacterium and reduced Escherichia-Shigella, Klebsiella and Dorea; acetate/butyrate increased. |
| **Berberine-containing arms of PREMOTE; adults with newly diagnosed type 2 diabetes, after antibi…**‡ | medication - a prescriber's decision prescriber decides | laboratory culture taken by mouth in the study | M: treatment depleted several carbohydrate-fermenting taxa, including R. bromii, Roseburia, F. prausnitzii and some Bifidobacterium. [S07] M/A: circulating deoxycholic-acid species and FGF19 decreased; BA changes correlated with glycaemic improvement. Laboratory AF25-7 showed BA conversion and berberine growth sensitivity. [S07] |
| **Chitosan / low-MW chitosan / chitooligosaccharides**‡¶ | supplement or compound | laboratory culture route not stated PMID 22528310 | Direct antimicrobial testing included Roseburia intestinalis, F. prausnitzii, B. thetaiotaomicron, B. vulgatus and other nonpathogenic anaerobes. |

**ALSO STUDIED ON THIS, DIRECTION NOT STATED** 2

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Alginate; low F. prausnitzii / butyrate support**‡ | supplement or compound | laboratory culture taken by mouth in the study PMID 34053530 | Bacteroides breakdown left oligosaccharides usable by an F. prausnitzii isolate; F. prausnitzii rose in fecal fermentation. |
| **Apple pectin / pectic oligosaccharides; F. prausnitzii / L. eligens**‡ | prebiotic fibre | laboratory culture reaches the gut unaided PMID 22101049 | Tested F. prausnitzii strains used apple pectin; F. prausnitzii and Eubacterium/Lachnospira eligens used galacturonide DP4/DP5. |

‹ back to the list · the reading this is about ›

ORGANISM

## 9. Blautia_A MGYG000001338 needs qualification

6.3% of your community; 3× the typical carrier; 90th percentile

**WHAT MIGHT BRING IT DOWN** 2

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Increased dietary fibre diversity (vegetables, legumes over refined starch)** | way of eating | human trial source | Relative abundance falls as fibre-fed competitors expand. |
| **Carnosic-acid-rich rosemary extract; lean and obese female Zucker rats**† | herb or botanical extract | animal study reaches the gut unaided | M: caecal groups changed: Blautia coccoides and Bacteroides/Prevotella groups increased; the Lactobacillus/Leuconostoc/Pediococcus group decreased. [S14] M/A: intestinal β-glucosidase activity decreased and faecal fibre excretion increased in both phenotypes. [S14] |

**SHOWN TO PUSH IT THE OTHER WAY** 2

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Refined carbohydrate intake** | way of eating | human cohorts source | — |
| **Laminarin-derived β-1,3 oligosaccharides**‡ | prebiotic fibre | laboratory culture reaches the gut unaided PMID 33043794 | Increased growth |

**ALSO STUDIED ON THIS, DIRECTION NOT STATED** 1

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Laminarin**‡ | prebiotic fibre | laboratory culture taken by mouth in the study PMID 33043794 | B. uniformis JCM13288 releases β-glucan oligosaccharides → Blautia producta JCM1471 growth |

‹ back to the list · the reading this is about ›

ORGANISM

## 10. Faecalibacterium butyricigenerans is well below the usual level

0.00% of your community; 89% below the typical carrier; 0th percentile

### WHAT MIGHT BRING IT UP 3

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Standardized oral pomegranate extract** | herb or botanical extract | human trial<br>doi:10.3390/foods13010015 | Increased relative abundance of selected species; increases also included M. smithii and C. aerofaciens<br>Mixed ecological response; do not label every increased organism beneficial or infer lower methane |
| **Riboflavin/B2 in RIBOGUT; Randomized placebo-controlled healthy-adult trial; oral** | supplement or compound | human trial<br>PMID 35943883 | Vitamin exposure changed microbial activity → Pooled active groups showed a butyrate increase without Faecalibacterium abundance gain → Clinical benefit and a change in stool butyrate genes were not demonstrated<br>Existing fermenters, two studied exposure groups and short follow-up. Individual active arms had limited effects; pooled finding is not a guaranteed producer-expansion effect. |
| **Butyrate-producer preparations including B. pullicaecorum25-3T and six-isolate consortium**‡ | probiotic | laboratory culture<br>reaches the gut unaided<br>PMID 28904372 | F. prausnitzii preparation/consortium increased butyrate;25-3T/consortium improved barrier integrity.<br>Individual-strain and consortium effects differ; no commercial six-isolate equivalence asserted. |

### SHOWN TO PUSH IT THE OTHER WAY 3

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **PHGG; lower Escherichia-Shigella / Klebsiella alongside support** | prebiotic fibre | human cohorts<br>PMID 37541036 | In 41 human fecal inocula, PHGG fermentation enriched Bifidobacterium/Faecalibacterium and reduced Escherichia-Shigella, Klebsiella and Dorea; acetate/butyrate increased. |
| **Berberine-containing arms of PREMOTE; adults with newly diagnosed type 2 diabetes, after antibi…**‡ | medication - a prescriber's decision<br>prescriber decides | laboratory culture<br>taken by mouth in the study | M: treatment depleted several carbohydrate-fermenting taxa, including R. bromii, Roseburia, F. prausnitzii and some Bifidobacterium. [S07]<br>M/A: circulating deoxycholic-acid species and FGF19 decreased; BA changes correlated with glycaemic improvement. Laboratory AF25-7 showed BA conversion and berberine growth sensitivity. [S07] |
| **Chitosan / low-MW chitosan / chitooligosaccharides**‡¶ | supplement or compound | laboratory culture<br>route not stated<br>PMID 22528310 | Direct antimicrobial testing included Roseburia intestinalis, F. prausnitzii, B. thetaiotaomicron, B. vulgatus and other nonpathogenic anaerobes. |

### ALSO STUDIED ON THIS, DIRECTION NOT STATED 2

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Alginate; low F. prausnitzii / butyrate support**‡ | supplement or compound | laboratory culture<br>taken by mouth in the study<br>PMID 34053530 | Bacteroides breakdown left oligosaccharides usable by an F. prausnitzii isolate; F. prausnitzii rose in fecal fermentation. |
| **Apple pectin / pectic oligosaccharides; F. prausnitzii / L. eligens**‡ | prebiotic fibre | laboratory culture<br>reaches the gut unaided<br>PMID 22101049 | Tested F. prausnitzii strains used apple pectin; F. prausnitzii and Eubacterium/Lachnospira eligens used galacturonide DP4/DP5. |

‹ back to the list · the reading this is about ›

ORGANISM

## 11. Roseburia inulinivorans is well below the usual level

0.00% of your community; 94% below the typical carrier; 9th percentile

### WHAT MIGHT BRING IT UP 7

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Whole grains** | food | human trial<br>source | — |
| **Inulin and fructo-oligosaccharides** | prebiotic fibre | human trial<br>source | — |
| **Oat and barley beta-glucan** | prebiotic fibre | human trial<br>source | — |
| **Resistant starch** | prebiotic fibre | human trial<br>source | — |
| **Psyllium husk; stool water / ecological support** | prebiotic fibre | human trial<br>source | In constipated subjects, increased stool water accompanied changes in Lachnospira, Roseburia and Faecalibacterium. |
| **Therapeutic ferrous sulfate with or without vitamin E; Small double-blind randomized US infant/…** | supplement or compound | human trial<br>PMID 27548249 | Iron deficiency was corrected in both groups → Roseburia relative abundance increased with vitamin E; Escherichia fell over time overall → No persistent inflammatory-marker increase; clinical benefit from Roseburia change unproven<br>Documented iron deficiency; baseline diet measured; different setting from African fortification trials. No untreated-iron comparison establishing all effects of iron. An antioxidant add-on can have ecological signals without validating a general protective formula. |
| **Arabinoxylan + galactomannan, 1:1**‡ | prebiotic fibre | laboratory culture<br>reaches the gut unaided<br>doi:10.1038/s41467-026-73297-5 | Dominates fiber-enriched consortium |

### SHOWN TO PUSH IT THE OTHER WAY 3

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Low-fibre diet, antibiotics** | way of eating | human cohorts<br>source | — |
| **Curcumin in a 481-compound/47-gut-strain screen, followed by obese-mouse and donor-community-tr…**† | plant compound | animal study<br>taken by mouth in the study | M: isolate responses differed: E. lenta and B. thetaiotaomicron were inhibited, whereas Roseburia intestinalis and Eubacterium rectale could be stimulated under studied conditions. [S18]<br>M: mouse curcumin exposure shifted the community, including decreased Lactobacillus; some insulin-resistance effects transferred with donor communities. [S18] |
| **Chitosan / low-MW chitosan / chitooligosaccharides**‡¶ | supplement or compound | laboratory culture<br>route not stated<br>PMID 22528310 | Direct antimicrobial testing included Roseburia intestinalis, F. prausnitzii, B. thetaiotaomicron, B. vulgatus and other nonpathogenic anaerobes. |

### ALSO STUDIED ON THIS, DIRECTION NOT STATED 1

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Locust-bean-gum-derived mannan/galactomannan oligosaccharides**‡ | prebiotic fibre | laboratory culture<br>reaches the gut unaided<br>doi:10.3390/microorganisms10122496 | Direct use, butyrate production |

‹ back to the list · the reading this is about ›

ORGANISM

## 12. Wujia chipingensis is well below the usual level

0.09% of your community; 89% below the typical carrier; 4th percentile

**Nothing on record.** No study in any setting — trial, animal or laboratory — has been recorded here for how to bring it up. That is the state of the research, not a judgement that this does not matter, and not a reason to do nothing: the sections above explain the finding and a clinician can act on it.

‹ back to the list · the reading this is about ›

MICROBIAL GROUP

## 13. Hexa-acylated LPS carriers as a group is low

7th percentile; 1 of 17 members detected

### WHAT MIGHT BRING IT UP 9

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **NaFeEDTA-containing or ferrous-fumarate-containing multiple micronutrient powders; Two infant r…** | mineral or protein supplement | human trial<br>PMID 25143342 | Iron supplied with complementary food → Enterobacteria/pathogenic E. coli signals increased → Fecal calprotectin increased; diarrhea-treatment difference was not statistically significant<br>Weaning, maize porridge, high baseline pathogen carriage; different formulations/doses in separate trials. Cannot generalize the combined signal to all iron products, adults, or every baseline community. Serum iron status and stool ecology are different endpoints. |
| **Chondroitin sulfate**† | way of eating | animal study<br>taken by mouth in the study<br>PMID 23898195 | Desulfovibrio piger and hydrogen sulfide<br>Potential opposing ecological effect, not a clinical adverse-event finding. Exact species and metabolite measured; consumer effects remain unestablished |
| **Lactulose alone**† | prebiotic fibre | animal study<br>taken by mouth in the study<br>PMID 38512401 | Direct growth; ~10-fold higher mouse colonization |
| **HST003 — Microbially synthesized glutamine; nitrogen-recycling ecology**† | supplement or compound | animal study<br>taken by mouth in the study<br>PMID 32466801 | Human microbiome/metabolome association plus mouse intervention. Experimental disease cascade supported; oral supplement→Klebsiella bloom not tested.<br>Myeloma-associated nitrogen environment → enrichment of nitrogen-recycling organisms `A/M`; intestinal K. pneumoniae → de-novo glutamine production `M` → accelerated myeloma progression in mice `M`. Glutamine-deficient diet → slower progression `M`. |
| **Bifidobacterium longum subsp. longum JCM1217; strain-dependent carbohydrate transporters**† | probiotic | animal study<br>reaches the gut unaided<br>PMID 21270894 | Carbohydrate uptake → acetate → epithelial defense / less Shiga-toxin passage → improved survival after E. coli O157:H7 challenge.<br>Bacterial load, toxin abundance and toxin translocation are different targets. Transporter genotype and available carbohydrate modify the response; human protection untested. |
| **L-serine restriction + mucolytic community → AIEC access to host nutrients/epithelial niche**† | probiotic | animal study<br>reaches the gut unaided<br>source | Follow-up mouse work found A. muciniphila-mediated mucus degradation permitted AIEC to obtain host-derived nutrients during serine restriction; restriction could increase E. coli in that setting.<br>A documented direction reversal from community context. The right model is conditional rules, not permanent “serine bad” or “Akkermansia good” scores. |
| **Lactiplantibacillus plantarum CIRM653**† | probiotic | animal study<br>reaches the gut unaided<br>PMID 29022382 | Culture biofilm biomass fell, but treated mice maintained pathogen burden while controls declined; type3 pili increased.<br>Opposing gut outcome: an anti-biofilm culture hit cannot be assumed to decolonize gut Klebsiella. |
| **Fiber-containing versus fiber-free diet after microbiota depletion**‡ | fibre | laboratory culture<br>reaches the gut unaided<br>PMID 38512401 | Fiber suppresses colonization; fiber-free human recovery permits bloom |

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **XOS with tested lactobacilli**‡ | probiotic | laboratory culture<br>reaches the gut unaided<br>PMID 36569071 | No inhibitory enhancement reported |

## SHOWN TO PUSH IT THE OTHER WAY 41

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **HST007 — Oral L-glutamine in a short-bowel elemental diet**† | way of eating | animal study<br>taken by mouth in the study<br>PMID 12483001 | Rat small-bowel resection study with matched nutritional comparator. Microbial and host outcomes measured together; mediation not isolated.<br>Oral glutamine → greater mucosal growth `M`; oral glutamine → smaller increase in cecal bacterial counts and less extraintestinal bacterial translocation `M`; mucosal improvement as mediator of microbial reduction `I, authors' interpretation`. |
| **Lacticaseibacillus rhamnosus CNCM I-3690, named Lactobacillus rhamnosus in source**† | probiotic | animal study<br>taken by mouth in the study<br>PMID 30022049 | Bilophila wadsworthia<br>Exact strain and target; preclinical. Cannot transfer to LGG, arbitrary L. rhamnosus products, or human clearance |
| **Lacticaseibacillus rhamnosus HN001 / DR20**† | probiotic | animal study<br>taken by mouth in the study<br>PMID 12208607 | Protection against E. coli O157:H7 and, separately, Salmonella Typhimurium infection; immune-mediated effects rather than assumed direct eradication<br>HN001 is not LGG; exact commercial strain appears in Ritual identity map below |
| **HST004 — Microbially synthesized glutamine in myeloma-associated gut–lung signaling**† | supplement or compound | animal study<br>taken by mouth in the study<br>PMID 36077725 | Human association, host-cell work and a myeloma mouse model. Disease-model cascade supported; no general-population risk estimate.<br>Gut K. pneumoniae enrichment → glutamine production `M` → increased TNF-α secretion in lung fibroblast experiments `M`; disrupting bacterial glutamine synthesis → less lung inflammatory disease in myeloma mice `M`. Human gut enrichment/pneumonia relationship `A`. |

## ALSO STUDIED ON THIS, DIRECTION NOT STATED 8

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **HST002 — Glutamine / glutamate-family amino-acid availability**† | supplement or compound | animal study<br>taken by mouth in the study<br>PMID 31449551 | Nutrient-rescue cultures, ex-vivo host fluids/tissue material and mouse organ-fitness experiments. Not an oral-glutamine colonization trial.<br>Glutamate-family nutrient supply → rescue of growth in metabolically deficient K. pneumoniae `M`; bacterial metabolic capacity → organ-specific fitness including gut fitness `M`; oral glutamine in an intact human ecosystem → wild-type intestinal expansion `U`. |
| **HST020 — Human lactoferrin as an iron source for clinical K. pneumoniae isolates**‡ | mineral or protein supplement | laboratory culture<br>taken by mouth in the study<br>PMID 14521878 | Direct iron-utilization experiments; most tested isolates could use human lactoferrin as sole iron source. Microbial capability, not an intervention outcome.<br>Lactoferrin-bound iron → iron acquisition supporting tested K. pneumoniae isolates `M`; this capability → clinical intestinal expansion after bovine oral lactoferrin `U`. |
| **Inulin**‡ | prebiotic fibre | laboratory culture<br>reaches the gut unaided<br>PMID 38512401 | Not directly utilized in tested culture |
| **EGCG and myricetin-related compounds**‡ | plant compound | laboratory culture<br>reaches the gut unaided<br>PMID 29855532 | Curli-dependent E. coli biofilm effects<br>Not all flavonoids were equally active; laboratory strain mechanism, not all pathogenic strains |
| **Allium ursinum and A. oschaninii extracts**‡ | herb or botanical extract | laboratory culture<br>reaches the gut unaided<br>PMID 32120894 | Effects on tested K. pneumoniae/C. albicans biofilms<br>Neither plant is Allium sativum; no copying to every garlic product |
| **Artemisia essential oils from three separately named species**‡§ | essential oil | laboratory culture<br>needs an enteric-protected form<br>PMID 33588649 | E. coli attachment/biofilm-formation effects<br>Extract species-specific findings before target ranking; no generic wormwood product equivalence |

‹ back to the list · the reading this is about ›

MICROBIAL GROUP

## 14. Resident bifidobacteria as a group is low

10th percentile; 1 of 5 members detected

### WHAT MIGHT BRING IT UP 2

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Standard/functional kimchi; additional arm with heat-killed L. plantarum nF1** | food | human trial<br>PMID 35721806 | Within-group enzyme activity and symptoms decreased; functional kimchi increased B. adolescentis signal.<br>All groups received kimchi despite title calling study placebo-controlled; fecal enzymes were measured, not inferred from genes. |
| **Same oligosaccharides**‡ | prebiotic fibre | laboratory culture<br>reaches the gut unaided<br>PMID 33043794 | Increased growth |

### SHOWN TO PUSH IT THE OTHER WAY 2

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Acid-hydrolyzed agaro-oligosaccharides; R. gnavus / F. nucleatum suppression**† | prebiotic fibre | animal study<br>reaches the gut unaided<br>PMID 39570663 | Culture inhibition of R. gnavus and F. nucleatum; tested bifidobacteria/Lactobacillales were spared. B. longum dominated an R. gnavus co-culture under AOS exposure. |
| **Acid-hydrolyzed agaro-oligosaccharides**‡ | prebiotic fibre | laboratory culture<br>reaches the gut unaided<br>doi:10.1099/mic.0.001510 | R. gnavus inhibition plus spared B. longum → competitive B. longum advantage |

### ALSO STUDIED ON THIS, DIRECTION NOT STATED 6

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **F. venenatum mycoprotein β-glucan**‡ | way of eating | laboratory culture<br>taken by mouth in the study<br>PMID 37253778 | B. cellulosilyticus WH2 or B. thetaiotaomicron VPI5482 releases glycans → B. longum NCIMB8809 / B. breve UCC2003 cross-feeding |
| **Same fungal β-glucan**‡ | way of eating | laboratory culture<br>taken by mouth in the study<br>PMID 37253778 | Cross-feeding |
| **Arabinoxylan breakdown products**‡ | prebiotic fibre | laboratory culture<br>taken by mouth in the study<br>PMID 37938239 | Cross-feeding |
| **Same oligosaccharides**‡ | prebiotic fibre | laboratory culture<br>taken by mouth in the study<br>doi:10.3390/microorganisms10122496 | Acetate cross-feeding supports butyrate |
| **Locust-bean-gum-derived mannan/galactomannan oligosaccharides**‡ | prebiotic fibre | laboratory culture<br>reaches the gut unaided<br>doi:10.3390/microorganisms10122496 | B. adolescentis E194a/ATCC15703 acetate production → R. hominis A2-183/DSM16839 → butyrate |
| **Intact fungal β-glucan**‡¶ | supplement or compound | laboratory culture<br>route not stated<br>PMID 37253778 | Fails to use intact substrate |

‹ back to the list · the reading this is about ›

MICROBIAL GROUP

## 15. Sulfate-reducing bacteria as a group is below average

16th percentile; 1 of 4 members detected

WHAT MIGHT BRING IT UP 1

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Chondroitin sulfate**† | way of eating | animal study<br>taken by mouth in the study<br>PMID 23898195 | Desulfovibrio piger and hydrogen sulfide<br>Potential opposing ecological effect, not a clinical adverse-event finding. Exact species and metabolite measured; consumer effects remain unestablished |

SHOWN TO PUSH IT THE OTHER WAY 1

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Lacticaseibacillus rhamnosus CNCM I-3690, named Lactobacillus rhamnosus in source**† | probiotic | animal study<br>taken by mouth in the study<br>PMID 30022049 | Bilophila wadsworthia<br>Exact strain and target; preclinical. Cannot transfer to LGG, arbitrary L. rhamnosus products, or human clearance |

‹ back to the list · the reading this is about ›

MICROBIAL GROUP

## 16. Supplement and fermented-dairy species as a group is above average

80th percentile; 1 of 14 members detected

WHAT MIGHT BRING IT DOWN 1

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Limosilactobacillus reuteri DSM17938** | probiotic | human trial<br>PMID 28429333 | Mean breath methane decreased from20.8 to8.9ppm after4weeks; hydrogen did not significantly decrease.<br>Measured gas association, not randomized efficacy, eradication or lower metagenomic genes. |

SHOWN TO PUSH IT THE OTHER WAY 1

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Bifidobacterium bifidum PRL2010 + Bifidobacterium breve UCC2003; mucin-based coculture**‡ | probiotic | laboratory culture<br>taken by mouth in the study<br>doi:10.1186/s12866-014-0282-7 | growth of B. breve UCC2003 using products released by PRL2010. in vitro coculture and molecular analysis. positive cross-feeding. |

### ALSO STUDIED ON THIS, DIRECTION NOT STATED 8

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **F. venenatum mycoprotein β-glucan**‡ | way of eating | laboratory culture<br>taken by mouth in the study<br>PMID 37253778 | B. cellulosilyticus WH2 or B. thetaiotaomicron VPI5482 releases glycans → B. longum NCIMB8809 / B. breve UCC2003 cross-feeding |
| **Same fungal β-glucan**‡ | way of eating | laboratory culture<br>taken by mouth in the study<br>PMID 37253778 | Cross-feeding |
| **LNT and LNnT; B. breve selection**‡ | prebiotic fibre | laboratory culture<br>taken by mouth in the study<br>doi:10.1128/AEM.01843-13 | Across 24 B. breve strains, HMO consumption varied; multiple isolates used LNT/LNnT. |
| **Extracellular 2′FL cleavage by community members → enables B. breve utilization**‡ | probiotic | laboratory culture<br>taken by mouth in the study<br>PMID 37973815 | Infant microbiome cultivation/metagenomics found B. breve could use 2′FL in a consortium; metagenomics implicated extracellular fucosidases of coexisting members such as R. gnavus.<br>Marks beneficial cross-feeding and imperfect exclusivity. The implicated R. gnavus contribution is a metagenomic mechanistic inference, not proven adult clinical treatment response. |
| **3′SL and 6′SL; sialylated HMO selection**‡ | prebiotic fibre | laboratory culture<br>reaches the gut unaided<br>source | In that B. breve panel, growth on these structures was observed for KA179, not universally across B. breve. |
| **β-1,3 galactobiose and galactotriose**‡§ | enzyme | laboratory culture<br>needs an enteric-protected form<br>PMID 32385941 | bga-cluster-dependent use |

‹ back to the list · the reading this is about ›

METABOLITE PATHWAY

## 17. Urolithin A formation capacity is notably low

0th percentile against the reference cohort

**WHAT MIGHT BRING IT UP** 10 of 18 on record

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Standardized oral pomegranate extract** | herb or botanical extract | human trial<br>doi:10.3390/foods13010015 | Increased relative abundance of selected species; increases also included M. smithii and C. aerofaciens<br>Mixed ecological response; do not label every increased organism beneficial or infer lower methane |
| **Ferrous bisglycinate or ferrous sulfate versus placebo; Secondary microbiome analysis of a doub…** | mineral or protein supplement | human trial<br>PMID 37199608 | Oral iron exposure in predominantly iron-replete adults → Bisglycinate increased relative Enterobacteriaceae; sulfate increased an EPEC-associated qPCR gene signal → Clinical infection or disease from these changes was not demonstrated<br>Only 1% had iron-deficiency anemia at baseline. No diversity effect. Family/virulence-gene signals are not diagnosis; better absorption does not guarantee less ecological selection. |
| **Therapeutic ferrous sulfate with or without vitamin E; Small double-blind randomized US infant/…** | supplement or compound | human trial<br>PMID 27548249 | Iron deficiency was corrected in both groups → Roseburia relative abundance increased with vitamin E; Escherichia fell over time overall → No persistent inflammatory-marker increase; clinical benefit from Roseburia change unproven<br>Documented iron deficiency; baseline diet measured; different setting from African fortification trials. No untreated-iron comparison establishing all effects of iron. An antioxidant add-on can have ecological signals without validating a general protective formula. |
| **Arabinoxylan oligosaccharides (AXOS); bifidobacteria / butyrate producers** | probiotic | human trial<br>PMID 30827722 | Increased bifidobacteria and butyrate-producing species in adults with metabolic-syndrome features; lipid/glucose markers were not improved. |
| **A. soehngenii CH106 next-generation strain, 2025** | supplement or compound | human trial<br>PMID 40371708 | Glycemic variability and diastolic pressure improved; HbA1c difference emerged after 16 weeks including follow-up. Exact strain produces butyrate in culture from sugars or lactate plus acetate; no direct demonstrated human butyrate increase in this trial.<br>Geographic response heterogeneity, modest clinical effect sizes, multiple mechanisms plausible. Manufacturer/developer involvement. CH106 has 37 genome mutations relative to L2-7, including tetO inactivation, so they need separate nodes. Archives PRJNA1219376 and parental PRJEB22345. |
| **HST032 — Slippery-elm bark powder, study-specific preparation** | herb or botanical extract | human cohorts<br>PMID 29565634 | Anaerobic community-culture study. Taxa observed; metabolite/host cascade not demonstrated.<br>Herbal substrate → changed pooled human fecal-community composition `M`, including expansion of reported butyrate-associated taxa; taxa → greater SCFA output and epithelial protection `I, authors' prediction`; actual host protection `U`. |
| **Indole-3-carbinol(I3C) → AhR/gut-barrier signaling**† | way of eating | animal study<br>taken by mouth in the study<br>PMID 32366032 | Dietary mouse studies protected fromDSS colitis and altered intestinal microbiota/immune signaling; recent study found interplay with dietaryzinc. |
| **Direct urolithin A, separated here from the paper's synthetic UAS03 analogue; epithelial cells …**† | plant compound | animal study<br>taken by mouth in the study | M: urolithin A activated AhR/Nrf2-dependent responses. [S23]<br>M: epithelial tight-junction expression and barrier function improved; receptor-pathway loss experiments weakened the response. [S23] |
| **HST003 — Microbially synthesized glutamine; nitrogen-recycling ecology**† | supplement or compound | animal study<br>taken by mouth in the study<br>PMID 32466801 | Human microbiome/metabolome association plus mouse intervention. Experimental disease cascade supported; oral supplement→Klebsiella bloom not tested.<br>Myeloma-associated nitrogen environment → enrichment of nitrogen-recycling organisms `A/M`; intestinal K. pneumoniae → de-novo glutamine production `M` → accelerated myeloma progression in mice `M`. Glutamine-deficient diet → slower progression `M`. |

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **L-serine restriction + mucolytic community → AIEC access to host nutrients/epithelial niche**† | probiotic | animal study<br>reaches the gut unaided<br>source | Follow-up mouse work found A. muciniphila-mediated mucus degradation permitted AIEC to obtain host-derived nutrients during serine restriction; restriction could increase E. coli in that setting.<br>A documented direction reversal from community context. The right model is conditional rules, not permanent “serine bad” or “Akkermansia good” scores. |

## SHOWN TO PUSH IT THE OTHER WAY 11

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Limosilactobacillus reuteri DSM17938** | probiotic | human trial<br>PMID 28429333 | Mean breath methane decreased from20.8 to8.9ppm after4weeks; hydrogen did not significantly decrease.<br>Measured gas association, not randomized efficacy, eradication or lower metagenomic genes. |
| **HST008 — Oral L-glutamine; AR005** | supplement or compound | human trial<br>PMID 30108163 | Randomized human trial in selected postinfectious IBS-D with baseline hyperpermeability. Host outcome measured; ecological continuation not measured.<br>Oral glutamine → improved measured permeability and IBS-D symptoms `M`; improved barrier → less oxygen/nitrate-supported Enterobacteriaceae growth `I`, using the separate mechanism in HST013; actual Klebsiella change `U`. |
| **Calcium-D-glucarate → intestinal bacterial β-glucuronidase activity**† | way of eating | animal study<br>taken by mouth in the study<br>PMID 2346674 | Animal study; dietary calcium-glucarate reduced activity in bacterial flora from proximal intestine70% and colon54%. Rats, not a human stool trial. |
| **HST004 — Microbially synthesized glutamine in myeloma-associated gut–lung signaling**† | supplement or compound | animal study<br>taken by mouth in the study<br>PMID 36077725 | Human association, host-cell work and a myeloma mouse model. Disease-model cascade supported; no general-population risk estimate.<br>Gut K. pneumoniae enrichment → glutamine production `M` → increased TNF-α secretion in lung fibroblast experiments `M`; disrupting bacterial glutamine synthesis → less lung inflammatory disease in myeloma mice `M`. Human gut enrichment/pneumonia relationship `A`. |

### ALSO STUDIED ON THIS, DIRECTION NOT STATED 11

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **high indole branch; Pectin / fermentable dietary fiber, with exact fiber recorded** | prebiotic fibre | human cohorts<br>PMID 38918470 | Defined microbial communities, human fecal communities and colonized-animal experiments showed that fermentable fiber can redirect microbial tryptophan metabolism away from indole and toward indolelactate/indolepropionate through community metabolic interactions.<br>Candidate dietary/prebiotic strategy for shifting tryptophan metabolism. The effect depends on resident producers, substrate competition and fiber preparation. Do not call all indoles harmful, nor assume tryptophan supplementation alone produces IPA. |
| **Triphala fruit powder, Banyan Botanicals**‡ | herb or botanical extract | human tissue or stool, outside the body<br>taken by mouth in the study<br>PMID 29565634 | Restructured relative abundances; profile differed from control and licorice<br>Matched powder only; comparison used carbohydrate-poor control; not a human feeding trial |
| **HST014 — Dietary threonine; BX025**‡ | way of eating | laboratory culture<br>taken by mouth in the study<br>PMID 20335627 | Weanling-pig dietary experiment. Host material-supply response measured; microbial continuation unknown.<br>Threonine adequacy → preserved mucins, villus architecture and MUC2 expression `M`; both deficiency and excess → impaired aspects of mucosal integrity `M`; resulting changes in mucus-consuming taxa or infection `U`. |
| **Bifidobacterium adolescentis 22L or Bifidobacterium breve 12L or Bifidobacterium thermophilum J…**‡ | probiotic | laboratory culture<br>taken by mouth in the study<br>PMID 26441950 | PRL2010 growth from starch/xylan degradation products. in vitro pairwise coculture. positive with substrate/partner specificity. |
| **Lacticaseibacillus paracasei CNCM I-1572 / DG, drinkable vials**‡ | probiotic | laboratory culture<br>taken by mouth in the study<br>PMID 30116228 | Culturable, strain-PCR-confirmed DG recovered from every participant; recovery continued up to five days after stopping.<br>Recovery demonstrates viable transit, not indefinite engraftment or efficacy. |
| **Cellulose; cellulolytic guild support**‡ | fibre | laboratory culture<br>reaches the gut unaided<br>PMID 38484069 | Human-associated ruminococcal species encode functional cellulosomes; cultured R. champanellensis degrades microcrystalline cellulose. |

‹ back to the list · the reading this is about ›

METABOLITE PATHWAY

## 18. Fermentation routes and cross-feeding capacity is above average

79th percentile against the reference cohort

**WHAT MIGHT BRING IT DOWN** 10 of 16 on record

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **MicrobiomeX standardized citrus extract** | herb or botanical extract | human trial<br>PMID 37761122 | Butyrate fraction increased; calprotectin decrease was only a trend<br>Relative SCFA profile, not established absolute butyrate production or clinical remission |
| **Same general inulin/psyllium combination with longer follow-up** | prebiotic fibre | human trial<br>source | Delayed arrival/fermentation → altered timing, but not reduced 24-hour fermentation |
| **Heat-killed L. plantarum L-137** | probiotic | human trial<br>PMID 34559598 | Selected stool SCFA/community and ex-vivo T-cell changes reported without overt intervention-related adverse effects.<br>No randomized comparator for long-term effects; not proof of general immune/barrier improvement. |
| **Individual essential-oil constituents, particularly thymol and geraniol; mixed human faecal fer…** | essential oil | human cohorts | M: stronger tested exposures to thymol or geraniol suppressed total bacterial abundance and fermentation in mixed communities. [S11]<br>M: weaker thymol exposure had little effect; responses of selected commensals and fermentation products differed between compounds and between mixed-community and isolate experiments. [S11] |
| **PHGG; lower Escherichia-Shigella / Klebsiella alongside support** | prebiotic fibre | human cohorts<br>PMID 37541036 | In 41 human fecal inocula, PHGG fermentation enriched Bifidobacterium/Faecalibacterium and reduced Escherichia-Shigella, Klebsiella and Dorea; acetate/butyrate increased. |
| **Sucrosomial iron: ferric pyrophosphate in the studied phospholipid/sucrester formulation, compa…**† | way of eating | animal study<br>taken by mouth in the study<br>PMID 34697758 | Formulation-specific oral iron exposure → Higher Shannon diversity and lower Proteobacteria; selected SCFA-associated taxa increased → Actual SCFA output and clinical infection protection were not established<br>Iron-deficiency model; measured community before/after treatment. Reduced bacterial accessibility is a mechanistic possibility, not a measured universal exclusion of iron uptake. Sucrosomial and arbitrary liposomal products are not interchangeable. |
| **Tangeretin**† | way of eating | animal study<br>taken by mouth in the study<br>PMID 34182753 | Positive: reduced disease severity and restored junction markers; microbial and SCFA changes accompanied benefit.<br>Preventive dietary model, not a demonstrated rescue of an established human biofilm. |
| **Vitamin B12-restricted versus sufficient mouse diets; cobalamin form not resolved in this extra…**† | herb or botanical extract | animal study<br>taken by mouth in the study<br>PMID 35674742 | Deficiency impaired ileal mitochondrial metabolism → Reduced SCFA-associated activity and host oxygen utilization increased epithelial oxygen availability → Salmonella expansion increased in the deficient model<br>Baseline B12 deficiency, antibiotic exposure, oxygen-related host physiology. Not proof that giving extra B12 to a replete person suppresses pathogens; matched against NUT030. |
| **CB-a barrier / SCFA study, 2026**† | supplement or compound | animal study<br>taken by mouth in the study<br>PMID 42549889 | Measured fecal SCFA restoration, particularly butyrate ($p<0.05$), alongside reduced inflammatory markers and restoration of ZO-1/occludin, altered GPR41/43/109A and HDAC1/2 measures. This supports a barrier-and-metabolite mechanism to investigate.<br>Mouse study, strain isolated from chicken feces, no verified retail formulation. Correlated pathway measurements do not by themselves prove butyrate is the sole necessary mediator. No extrapolation to TO-A, CBM588, or unspecified C. butyricum supplements. |
| **Inulin or FOS in Western-style rat diets**† | prebiotic fibre | animal study<br>reaches the gut unaided<br>PMID 15016747 | Fermentation/lower pH with altered luminal environment → increased Salmonella Enteritidis colonization/translocation |

### SHOWN TO PUSH IT THE OTHER WAY 8

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Acetylated/butyrylated high-amylose maize starch, HAMSAB** | prebiotic fibre | human trial<br>PMID 39196205 | Ambulatory blood-pressure improvement reported<br>Chemically modified studied starch is not equivalent to generic resistant starch or acetate tablets |
| **HST032 — Slippery-elm bark powder, study-specific preparation** | herb or botanical extract | human cohorts<br>PMID 29565634 | Anaerobic community-culture study. Taxa observed; metabolite/host cascade not demonstrated.<br>Herbal substrate → changed pooled human fecal-community composition `M`, including expansion of reported butyrate-associated taxa; taxa → greater SCFA output and epithelial protection `I, authors' prediction`; actual host protection `U`. |
| **Latilactobacillus sakei LZ217** | probiotic | human cohorts<br>PMID 35053965 | Identified folate product as 5-methyltetrahydrofolate; simulated gastrointestinal-juice resistance and increased butyrate in fecal-slurry fermentation were reported.<br>The human material was cultured fecal slurry, not an intervention in people. Gastric-juice tolerance does not establish durable engraftment. No retail strain substitution. |
| **Apigenin**† | prebiotic fibre | animal study<br>taken by mouth in the study<br>PMID 36590200 | Positive: improved mucus/junction and inflammatory endpoints with microbiota/SCFA changes.<br>Do not label each changed taxon as a direct antimicrobial target without separate susceptibility evidence. |

### ALSO STUDIED ON THIS, DIRECTION NOT STATED 7

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **RS1 and RS5; structurally distinct expansion candidates**† | herb or botanical extract | animal study<br>reaches the gut unaided<br>PMID 41926473 | RS1/2/3/4 human-fecal fermentation comparisons and RS5 starch–fatty-acid studies demonstrate structure-dependent fermentability. |
| **Propionibacterium freudenreichii DSM 20271; co-fermented wheat bran with Levilactobacillus brev…**‡ | fibre | laboratory culture<br>taken by mouth in the study<br>PMID 31333632 | The exact P. freudenreichii strain produced B12; co-fermentation supported a fortified food ingredient while altering microbiological safety indicators.<br>Prepared food production, not in-vivo supplementation or microbiome engraftment. The B12 producer and co-culture strain have distinct roles. |
| **Same mucin-fed community**‡ | prebiotic fibre | laboratory culture<br>taken by mouth in the study<br>PMID 39385462 | Fermentation substrates → B. hydrogenotrophica DSM10507 acetogenesis and M. smithii DSM11975 methanogenesis enzyme expression |
| **Alginate; low F. prausnitzii / butyrate support**‡ | supplement or compound | laboratory culture<br>taken by mouth in the study<br>PMID 34053530 | Bacteroides breakdown left oligosaccharides usable by an F. prausnitzii isolate; F. prausnitzii rose in fecal fermentation. |
| **B. pullicaecorum25-3T**‡ | food | laboratory culture<br>reaches the gut unaided<br>PMID 25179909 | Acid-exposed cells remained active despite temporary noncultivability; cultivation recovered; SCFA output shifted toward acetate.<br>Delivery survival did not guarantee preserved butyrate output or engraftment. |
| **k■pos for Digestion; product identity**‡ | prebiotic fibre | laboratory culture<br>reaches the gut unaided<br>source | Manufacturer lists kpHMO plus effera human lactoferrin, produced by fermentation; individual HMO structures/ratios were not disclosed sufficiently for structure-specific equivalence. |

‹ back to the list · the reading this is about ›

METABOLITE PATHWAY

## 19. GABA breakdown capacity is below average

25th percentile against the reference cohort

### WHAT MIGHT BRING IT UP 4

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Mucin processed by B. thetaiotaomicron or glucanase**† | enzyme | animal study<br>taken by mouth in the study<br>PMID 33035488 | Complex glycan breakdown → accessible substrates → greater C. albicans growth |
| **A. muciniphila MucT with E. hallii L2-7 / A. soehngenii lineage**‡ | probiotic | laboratory culture<br>taken by mouth in the study<br>PMID 28928206 | Mucin-derived sugars / 1,2-propanediol feed partner → partner pseudovitamin B12 → Akkermansia succinate-to-propionate flux increases.<br>Pseudovitamin B12 is not assumed nutritionally equivalent to human-active B12. Host benefit and net mucus balance were not measured. Same study as PRX004. |
| **Lactiplantibacillus plantarum 299v, DSM 9843**‡ | probiotic | laboratory culture<br>reaches the gut unaided<br>doi:10.1093/jambio/lxac066 | The same primary study identified 299v among selected GABA-producing strains; production varied by strain and community context. Adding oligofructose-enriched inulin did not further stimulate GABA production under the conditions tested.<br>Eligible strain-specific candidate; preserve the negative synbiotic interaction result. Do not convert the existence of both probiotic and prebiotic into a claim that their combination raises GABA more. Gene presence alone does not establish net GABA output. |
| **Levilactobacillus brevis LB01, CGMCC 16921**‡ | probiotic | laboratory culture<br>reaches the gut unaided<br>PMID 36724279 | A study of commercially available strains found GABA production by LB01 in pure culture and human-fecal simulated-colon systems.<br>Exact-strain, laboratory/ex-vivo-supported GABA-production candidate. It does not establish lasting engraftment, human brain GABA increases, or treatment of anxiety, depression or insomnia. |

### SHOWN TO PUSH IT THE OTHER WAY 2

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Clostridium butyricum MIYAIRI588 in clindamycin/CDI mouse context**† | spore probiotic | animal study<br>taken by mouth in the study<br>PMID 34294848 | Community modulation → less gut succinate → reduced C. difficile proliferation; parallel IL-17A / plasma-B-cell / IgA responses protect tissue.<br>Do not label all IL-17 increases harmful: direction was protective in this infection model. Not every C. butyricum is MIYAIRI588. Succinate change is ecological, not direct strain consumption proven here. |
| **Carbohydrate fermentation plus antibiotic or motility disturbance**† | medication - a prescriber's decision<br>prescriber decides | animal study<br>taken by mouth in the study<br>PMID 25498344 | B. thetaiotaomicron succinate → C. difficile reduction to butyrate → pathogen growth advantage |

### ALSO STUDIED ON THIS, DIRECTION NOT STATED 2

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **R. bromii starch breakdown → indirect R. gnavus feeding**‡ | way of eating | laboratory culture<br>taken by mouth in the study<br>PMID 30455672 | Co-culture experiments: R. gnavus ATCC29149 did not use potato soluble/maize resistant starch alone but used products liberated by R. bromii L2-63.<br>Precisely why a monoculture “cannot use” result does not mean a food cannot feed that organism in a community. Net human harm is untested, and desired butyrate cross-feeding may coexist. |
| **Alginate; low F. prausnitzii / butyrate support**‡ | supplement or compound | laboratory culture<br>taken by mouth in the study<br>PMID 34053530 | Bacteroides breakdown left oligosaccharides usable by an F. prausnitzii isolate; F. prausnitzii rose in fecal fermentation. |

‹ back to the list · the reading this is about ›

METABOLITE PATHWAY

## 20. gamma-aminobutyric acid (GABA) capacity is below average

20th percentile against the reference cohort

### WHAT MIGHT BRING IT UP 2

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Lactiplantibacillus plantarum 299v, DSM 9843**‡ | probiotic | laboratory culture<br>reaches the gut unaided<br>doi:10.1093/jambio/lxac066 | The same primary study identified 299v among selected GABA-producing strains; production varied by strain and community context. Adding oligofructose-enriched inulin did not further stimulate GABA production under the conditions tested.<br>Eligible strain-specific candidate; preserve the negative synbiotic interaction result. Do not convert the existence of both probiotic and prebiotic into a claim that their combination raises GABA more. Gene presence alone does not establish net GABA output. |
| **Levilactobacillus brevis LB01, CGMCC 16921**‡ | probiotic | laboratory culture<br>reaches the gut unaided<br>PMID 36724279 | A study of commercially available strains found GABA production by LB01 in pure culture and human-fecal simulated-colon systems.<br>Exact-strain, laboratory/ex-vivo-supported GABA-production candidate. It does not establish lasting engraftment, human brain GABA increases, or treatment of anxiety, depression or insomnia. |

‹ back to the list · the reading this is about ›

METABOLITE PATHWAY

## 21. Glucosinolate to isothiocyanate conversion capacity is low

13th percentile against the reference cohort

**WHAT MIGHT BRING IT UP** 10 of 38 on record

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Triticum aestivum wheatgrass juice** | food | human trial PMID 11989836 | Improved clinical activity, including rectal bleeding, compared with placebo<br>Whole juice; no stool community or antimicrobial endpoint |
| **B. infantis NLS super strain** | way of eating | human trial PMID 23314670 | Some symptom improvements without established correction of intestinal permeability<br>Does not replace a gluten-free diet or establish safety of gluten exposure |
| **C. butyricum CBM588, Butirrisan studied formulation** | probiotic | human trial doi:10.3390/microorganisms13051139 | IBS-D symptom improvements with diet cointervention in open-label prospective study<br>Exact commercial strain available; cannot separate probiotic from diet/placebo/regression effects |
| **L. acidophilus NCFM** | probiotic | human trial source | Overall IBS severity improved similarly with probiotic and placebo; subgroup pain observations cannot be called universal efficacy<br>Retain alongside PB013 rather than hiding null |
| **Partially hydrolyzed guar gum (PHGG); low bifidobacteria / stool consistency** | probiotic | human trial PMID 31509971 | Increased fecal Bifidobacterium in adults with diarrhea tendency. |
| **Studied high-dose multistrain probiotic formulation** | probiotic | human trial PMID 35654766 | Greater short-term symptom improvement with usual treatment<br>Whole formulation, background psychiatric care and baseline context matter; no direct correction of a depression microbiome score was established |
| **NaFeEDTA-containing or ferrous-fumarate-containing multiple micronutrient powders; Two infant r…** | mineral or protein supplement | human trial PMID 25143342 | Iron supplied with complementary food → Enterobacteria/pathogenic E. coli signals increased → Fecal calprotectin increased; diarrhea-treatment difference was not statistically significant<br>Weaning, maize porridge, high baseline pathogen carriage; different formulations/doses in separate trials. Cannot generalize the combined signal to all iron products, adults, or every baseline community. Serum iron status and stool ecology are different endpoints. |
| **Riboflavin/B2 in RIBOGUT; Randomized placebo-controlled healthy-adult trial; oral** | supplement or compound | human trial PMID 35943883 | Vitamin exposure changed microbial activity → Pooled active groups showed a butyrate increase without Faecalibacterium abundance gain → Clinical benefit and a change in stool butyrate genes were not demonstrated<br>Existing fermenters, two studied exposure groups and short follow-up. Individual active arms had limited effects; pooled finding is not a guaranteed producer-expansion effect. |
| **Serum-derived bovine immunoglobulin/protein isolate (SBI)** | supplement or compound | human trial PMID 24833942 | Positive within-group: symptom days and severity improved in active groups.<br>Analysis emphasized change within groups, not established superiority over placebo; no significant quality-of-life change. |
| **Therapeutic ferrous sulfate with or without vitamin E; Small double-blind randomized US infant/…** | supplement or compound | human trial PMID 27548249 | Iron deficiency was corrected in both groups → Roseburia relative abundance increased with vitamin E; Escherichia fell over time overall → No persistent inflammatory-marker increase; clinical benefit from Roseburia change unproven<br>Documented iron deficiency; baseline diet measured; different setting from African fortification trials. No untreated-iron comparison establishing all effects of iron. An antioxidant add-on can have ecological signals without validating a general protective formula. |

## SHOWN TO PUSH IT THE OTHER WAY 40

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **GOS plus combined sodium iron EDTA and ferrous fumarate MNP; Double-blind infant randomized tri…** | prebiotic fibre | human trial PMID 28774885 | Iron repletion with a fermentable co-substrate → GOS mitigated depletion of bifidobacteria/lactobacilli and pathogen virulence-gene changes → Anemia improved while enterocyte-damage and morbidity signals were lower than with iron alone Weaning infants; exact low-dose iron mixture and GOS preparation. Same-trial endpoints support a combination, not proof that microbiota mediated every clinical result or that any prebiotic protects against any iron. |
| **Bifidobacterium animalis subsp. lactis CECT8145 / BPL1, live versus heat-killed** | probiotic | human trial PMID 30262813 | Anthropometric adiposity markers improved, especially with heat-killed cells and in women; Akkermansia increase suggested as a mechanism. Live and killed forms are distinct preparations; not proof of treating any specific stool disease score. |
| **Bacillus/Heyndrickxia coagulans SNZ1969** | spore probiotic | human trial PMID 34119240 | Transit improved; selected weekly bowel scores improved; Lactobacillales enrichment/Synergistales reduction associated with transit. Associations do not establish which microbial changes caused symptom change or permanent engraftment. |
| **Ascorbic acid in adults with suboptimal serum vitamin C; Secondary microbiome analysis of rando…** | supplement or compound | human trial PMID 40187667 | Serum/nutrient exposure changed → Desulfovibrio decreased; several immune and microbial endpoints changed → Microbiota changes correlated with mental-vitality endpoints Low baseline serum vitamin C and individual response heterogeneity. Correlations do not establish a microbial mechanism for mental improvement or neurotransmitter levels in the brain. |

## ALSO STUDIED ON THIS, DIRECTION NOT STATED 16

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **L. rhamnosus GR-1 plus L. reuteri RC-14** | probiotic | human trial PMID 19295645 | Tinidazole plus probiotics yielded 87.5% cure versus 50% with tinidazole plus placebo. Urogenital evidence only; effect is not attributed to a single strain. |
| **Exact B. TO-A human viable recovery, 2014** | spore probiotic | human trial doi:10.1248/bpb.b13-00641** | Fecal strain detection by culture plus strain-specific qPCR. All participants positive during administration; averages approximately 10^4.8 CFU/g by culture and 10^5.8 cells/g by qPCR. Half remained qPCR-positive three days after withdrawal; only one remained positive after one week. Different standalone product and study exposure; not a test of current AOR capsules. Manufacturer involvement. |
| **B. subtilis TO-A, historical B. mesentericusTO-A, FERM BP07462** | supplement or compound | human trial doi:10.1248/bpb.b13-00641 | All 24 volunteers had viable strain recovered during administration; mostly disappeared after withdrawal Standalone tested TO-A tablets, not proof of every component in AOR capsule |
| **effera recombinant human lactoferrin; product-family evidence** | mineral or protein supplement | human trial PMID 42178844 | Adult randomized study compared two effera arms with bovine lactoferrin; exploratory taxa/metabolite changes occurred, without a main diversity effect. |
| **Zinc carnosine, alone/with bovine colostrum → attenuated exercise-associated permeability** | mineral or protein supplement | human trial PMID 27357095 | Four-arm double-blind crossover in 8 volunteers; exercise challenge; parallel epithelial-cell experiments. Small, short study. Combination effects and heat/exercise context should not be transferred wholesale to chronic disease or zinc alone. |
| **Bifidobacterium longum 35624; paper calls it Bifidobacterium infantis 35624; encapsulated viabl…** | probiotic | human trial PMID 16863564 | abdominal pain and composite symptoms in women with IBS. human dose-ranging randomized placebo-controlled trial. positive at tested 10^8 CFU arm; other tested doses not significant. |

‹ back to the list · the reading this is about ›

METABOLITE PATHWAY

## 22. Histamine capacity is above average

75th percentile against the reference cohort

### WHAT MIGHT BRING IT DOWN 2

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Dietitian-guided limited low-FODMAP diet (restriction → reintroduction → personalisation)** | way of eating | human trial<br>McIntosh K et al. Gut 2017;66:1241 | three-week low-FODMAP versus high-FODMAP diet Studied in people with clinically confirmed IBS, ages 18–70, in Canada, n = 37.<br>Stool hdc genes are not a histamine measurement. In diagnosed IBS, a three-week randomized low-FODMAP diet lowered urinary histamine and improved symptoms; the causal Klebsiella-histamine work was in mice. |
| **L. reuteri ATCC PTA6475, hdc-positive; hdc mutants and H2-receptor blockade**† | supplement or compound | animal study<br>route not stated<br>PMID 22384111 | Histidine → histamine → H2 receptor / cAMP-PKA → lower TNF signaling and less experimental colitis.<br>Histamine can have different effects through other receptors/compartments; this is not evidence that histamine intolerance improves. H2-blocking medication is a mechanistic modifier, not an instruction to stop it. |

### SHOWN TO PUSH IT THE OTHER WAY 2

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Oral diamine oxidase(DAO) → dietary histamine handling**‡ | way of eating | laboratory culture<br>taken by mouth in the study<br>PMID 31807350 | Open-label human histamine-intolerance study reported symptom improvement during supplement intake. |
| **Lactiplantibacillus plantarum LP115 → host epithelialDAO response**‡ | probiotic | laboratory culture<br>reaches the gut unaided<br>doi:10.3390/biology14060734 | 2025 cell-model study foundLP115 stimulated epithelialDAO secretion in the presence of exogenous histamine. |

### ALSO STUDIED ON THIS, DIRECTION NOT STATED 1

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **L.plantarum WYP → histamine degradation**‡¶ | supplement or compound | laboratory culture<br>route not stated<br>PMID 42195967 | 2026 phenotype/genome study found 78.63% histamine degradation in vitro and demonstrated simulated gastric/intestinal tolerance. |

‹ back to the list · the reading this is about ›

METABOLITE PATHWAY

## 23. Hydrogen sulfide (H2S) capacity is above average

81th percentile against the reference cohort

### WHAT MIGHT BRING IT DOWN 3

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Bismuth subsalicylate** | medication - a prescriber's decision<br>prescriber decides | human cohorts<br>Suarez FL et al. Gastroenterology 1998;114:923 | 524 mg four times daily for 3–7 days Studied in people with clinically confirmed healthy adults, ages 18–60, in USA, n = 10.<br>Bismuth subsalicylate reduced ex-vivo fecal hydrogen-sulfide release in ten healthy adults (a surrogate, with salicylate and bismuth cautions), and a small sulfide-reducing diet study in UC gave mixed results. No disease endpoint has been shown to change. |
| **Garlic foods and commercial garlic preparations with different alliin/alliinase delivery; 13 he…**[‡§] | herb or botanical extract | laboratory culture<br>needs an enteric-protected form | M: preparation and meal context altered apparent allicin bioavailability, measured using breath allyl methyl sulphide. [S01]<br>M: an enteric label did not ensure equivalent conversion; some alliinase-inactivated foods still generated the breath metabolite through other garlic sulphur substrates. [S01] |
| **Porcine gastric mucin in a 15-species chemostat**[‡§] | prebiotic fibre | laboratory culture<br>needs an enteric-protected form<br>PMID 39385462 | Glycan/sulfate release and fermentation → D. piger DSM749 sulfate-reduction pathway expression |

### SHOWN TO PUSH IT THE OTHER WAY 1

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Chondroitin sulfate**[†] | way of eating | animal study<br>taken by mouth in the study<br>PMID 23898195 | Desulfovibrio piger and hydrogen sulfide<br>Potential opposing ecological effect, not a clinical adverse-event finding. Exact species and metabolite measured; consumer effects remain unestablished |

### ALSO STUDIED ON THIS, DIRECTION NOT STATED 1

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Dietary heme in a purified mouse diet; Mouse dietary experiment plus biochemical barrier measur…**[†] | way of eating | animal study<br>taken by mouth in the study<br>PMID 26216954 | Luminal cytotoxic products occurred with heme → Microbial sulfide and mucus degradation enabled epithelial exposure → Epithelial hyperproliferation/hyperplasia depended on the microbiota<br>Heme matrix, sulfide producers, mucus-degrading organisms and host mucus barrier. Does not establish identical harm for all heme foods, doses, or human supplements; Akkermansia cannot be assigned a fixed harmful polarity. |

‹ back to the list · the reading this is about ›

METABOLITE PATHWAY

## 24. Imidazole propionate (ImP) capacity is above average

83th percentile against the reference cohort

### WHAT MIGHT BRING IT DOWN 1

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Urolithin A (Mitopure)** | herb or botanical extract | human trial<br>Liu S et al. JAMA Netw Open 2022;5:e2144279 | 1,000 mg/day urolithin A for four months Studied in people with clinically confirmed older adults, ages 65–90, in USA, n = 66.<br>Whether someone converts ellagitannins to urolithins or daidzein to equol is determined by a substrate challenge with urine measurement. Direct urolithin A supplementation missed its primary endpoints and improved a secondary muscle measure in older adults; it bypasses the microbes rather than restoring them. |

**SHOWN TO PUSH IT THE OTHER WAY** 1

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Direct urolithin A, separated here from the paper's synthetic UAS03 analogue; epithelial cells …**† | plant compound | animal study<br>taken by mouth in the study | M: urolithin A activated AhR/Nrf2-dependent responses. [S23]<br>M: epithelial tight-junction expression and barrier function improved; receptor-pathway loss experiments weakened the response. [S23] |

**ALSO STUDIED ON THIS, DIRECTION NOT STATED** 1

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Ellagic acid; human intestinal Gordonibacter urolithinfaciens and G. pamelaeae isolates**‡ | plant compound | laboratory culture<br>reaches the gut unaided | M: ellagic acid was converted through urolithin M5 and M6. [S22]<br>M: urolithin C accumulated. [S22] |

‹ back to the list · the reading this is about ›

**METABOLITE PATHWAY**

## 25. Mucin glycans and lipid A capacity is low

15th percentile against the reference cohort

**WHAT MIGHT BRING IT UP** 5

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **HST032 — Slippery-elm bark powder, study-specific preparation** | herb or botanical extract | human cohorts<br>PMID 29565634 | Anaerobic community-culture study. Taxa observed; metabolite/host cascade not demonstrated.<br>Herbal substrate → changed pooled human fecal-community composition `M`, including expansion of reported butyrate-associated taxa; taxa → greater SCFA output and epithelial protection `I, authors' prediction`; actual host protection `U`. |
| **Apigenin**† | prebiotic fibre | animal study<br>taken by mouth in the study<br>PMID 36590200 | Positive: improved mucus/junction and inflammatory endpoints with microbiota/SCFA changes.<br>Do not label each changed taxon as a direct antimicrobial target without separate susceptibility evidence. |
| **Dietary fiber deprivation**† | fibre | animal study<br>reaches the gut unaided<br>doi:10.1016/j.cell.2016.10.043 | Increased microbial mucin foraging → thinner mucus barrier → greater Citrobacter rodentium susceptibility |
| **L-serine restriction + mucolytic community → AIEC access to host nutrients/epithelial niche**† | probiotic | animal study<br>reaches the gut unaided<br>source | Follow-up mouse work found A. muciniphila-mediated mucus degradation permitted AIEC to obtain host-derived nutrients during serine restriction; restriction could increase E. coli in that setting.<br>A documented direction reversal from community context. The right model is conditional rules, not permanent “serine bad” or “Akkermansia good” scores. |
| **A. muciniphila MucT with E. hallii L2-7 / A. soehngenii lineage**‡ | probiotic | laboratory culture<br>taken by mouth in the study<br>PMID 28928206 | Mucin-derived sugars / 1,2-propanediol feed partner → partner pseudovitamin B12 → Akkermansia succinate-to-propionate flux increases.<br>Pseudovitamin B12 is not assumed nutritionally equivalent to human-active B12. Host benefit and net mucus balance were not measured. Same study as PRX004. |

### SHOWN TO PUSH IT THE OTHER WAY 2

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **HST018 — Zinc gluconate in antibiotic/peptidoglycan injury**† | medication - a prescriber's decision<br>prescriber decides | animal study<br>taken by mouth in the study<br>PMID 42694231 | August 2026 mouse intervention after antibiotic disruption and peptidoglycan challenge; study-specific administration. Parallel host/community outcomes; not infection clearance.<br>Zinc gluconate → restoration of junction/mucin proteins, altered macrophage markers and less inflammatory signaling `M`; zinc gluconate → altered community structure/diversity `M`; microbiota mediation of host recovery `I`. |
| **0.5% inulin in nutrient-rich BHIS; six-species community**‡ | prebiotic fibre | laboratory culture<br>reaches the gut unaided<br>PMID 42429610 | Community competition shifts → R. gnavus relative enrichment; B. uniformis, B. thetaiotaomicron and A. muciniphila decline |

### ALSO STUDIED ON THIS, DIRECTION NOT STATED 7

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Urolithin A**‡ | plant compound | human tissue or stool, outside the body<br>taken by mouth in the study<br>PMID 42336837 | Positive mechanistic: strengthened barrier-associated immune/mucin responses; human biopsies responded ex vivo.<br>Human-tissue response is not a human oral efficacy trial. Adds a distinct mechanistic observation to BX036. |
| **Dietary L-threonine across deficient, adequate and excess intakes**‡ | way of eating | laboratory culture<br>taken by mouth in the study<br>PMID 20335627 | Nonmonotonic: adequate intake supported barrier/mucin measures; both deficiency and excess impaired aspects of the mucosa.<br>This is not a “more is better” supplement edge; nutritional adequacy cannot be measured from stool microbial genes. |
| **HST014 — Dietary threonine; BX025**‡ | way of eating | laboratory culture<br>taken by mouth in the study<br>PMID 20335627 | Weanling-pig dietary experiment. Host material-supply response measured; microbial continuation unknown.<br>Threonine adequacy → preserved mucins, villus architecture and MUC2 expression `M`; both deficiency and excess → impaired aspects of mucosal integrity `M`; resulting changes in mucus-consuming taxa or infection `U`. |
| **B. bifidum PRL2010 + B. breve UCC2003**‡ | prebiotic fibre | laboratory culture<br>taken by mouth in the study<br>doi:10.1186/s12866-014-0282-7 | Mucin-derived carbohydrate cross-feeding supported UCC2003 growth<br>Research pair; no generic claim that adding mucin is beneficial or safe |
| **B. bifidum PRL2010 with B. breve UCC2003**‡ | prebiotic fibre | laboratory culture<br>taken by mouth in the study<br>PMID 25420416 | PRL2010 degrades mucin or 3′-sialyllactose → liberated carbohydrates including sialic acid → UCC2003 growth.<br>Growth support does not establish benefit to the host or prove all liberated sugars are captured by desirable organisms; competing consumers were not tested in these experiments. |
| **Akkermansia muciniphila MucT / ATCC BAA-835 with A. caccae L1-92 or Faecalibacterium prausnitzi…**‡ | probiotic | laboratory culture<br>reaches the gut unaided<br>PMID 28928206 | Mucin degradation → released sugars and acetate → partner growth and butyrate.<br>Mucin utilization is not equivalent to damaging the mucus barrier, nor does this culture establish barrier repair. Intact host mucus secretion, diet and partner abundance are modifiers. |

‹ back to the list · the reading this is about ›

METABOLITE PATHWAY

## 26. p-Cresol / p-cresyl sulfate capacity is above average

84th percentile against the reference cohort

### WHAT MIGHT BRING IT DOWN 2

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **SYNERGY synbiotic (mixed prebiotic + nine-strain probiotic) in CKD (Rossi 2016)** | supplement or compound | human trial<br>Rossi M et al. Clin J Am Soc Nephrol 2016;11:223 (SYNERGY) | 15 g/day mixed prebiotic plus nine-strain probiotic escalating 45 → 90 billion CFU/day for six weeks (crossover) Studied in people with clinically confirmed CKD stage 4–5 (pre-dialysis), ages 18–85, in Australia, n = 37.<br>In pre-dialysis CKD, a synbiotic and resistant starch each lowered serum p-cresyl sulfate in small trials without hard outcomes. In someone with normal kidney function this reading has no supported intervention. |
| **Lactiplantibacillus plantarum P9** | probiotic | human trial<br>PMID 37019193 | Complete/spontaneous bowel movements improved; p-cresol, methylamine and TMA decreased; Ruminococcus_B gnavus relative abundance increased.<br>R. gnavus increase is a material collateral effect for an R. gnavus-targeted plan; taxa effects are not universally beneficial. |

‹ back to the list · the reading this is about ›

METABOLITE PATHWAY

## 27. Propionate (SCFA) capacity is above average

78th percentile against the reference cohort

### WHAT MIGHT BRING IT DOWN 3

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Ginsenoside compound K, retaining exact compound identity rather than generic ginseng**† | supplement or compound | animal study<br>route not stated<br>PMID 36361652 | In diabetic db/db mice, compound K altered the microbiota, lowered serum imidazole propionate and improved kidney-related inflammatory outcomes involving TLR4 signaling.<br>Animal-supported candidate for ImP-related dysmetabolism. Record measured serum ImP separately from host TLR4 effects. This is not evidence of direct UrdA inhibition, nor proof that any ginseng product has the same exposure or outcome in people. |
| **Propionate in an inflamed respiratory niche**† | supplement or compound | animal study<br>route not stated<br>PMID 34986344 | Salmonella anaerobic respiration permits propionate catabolism → carbon supply/colonization advantage |
| **Five-spore blend: B. indicusHU36, B. subtilisHU58, B. coagulansSC208, B. licheniformisSL307, B....**‡ | spore probiotic | human tissue or stool, outside the body<br>reaches the gut unaided<br>PMID 34600678 | Propionate increased in selected compartments; ammonium/lactate decreased; Akkermansia/Bifidobacteria increased, Lactobacillus decreased.<br>Mixture effect; paper's historical SC208 taxonomy retained; model does not establish same human effect. |

### SHOWN TO PUSH IT THE OTHER WAY 2

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **A. muciniphila MucT with E. hallii L2-7 / A. soehngenii lineage**‡ | probiotic | laboratory culture<br>taken by mouth in the study<br>PMID 28928206 | Mucin-derived sugars / 1,2-propanediol feed partner → partner pseudovitamin B12 → Akkermansia succinate-to-propionate flux increases.<br>Pseudovitamin B12 is not assumed nutritionally equivalent to human-active B12. Host benefit and net mucus balance were not measured. Same study as PRX004. |
| **Phosphate-cross-linked tapioca RS4**‡¶ | supplement or compound | laboratory culture<br>route not stated<br>PMID 32004499 | Enrichment linked to propionate shift |

ALSO STUDIED ON THIS, DIRECTION NOT STATED 1

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Veillonella atypica marathon-runner isolate, study-specific**† | lifestyle | animal study<br>taken by mouth in the study<br>PMID 31235964 | Host exercise lactate reaches gut → bacterial lactate-to-propionate metabolism → longer mouse treadmill running.<br>Study runner isolate is not assumed ATCC17744 or every commercial V. atypica. Human performance benefit and long-term safety were not shown by this mouse mechanism study. |

‹ back to the list · the reading this is about ›

METABOLITE PATHWAY

## 28. Unconjugated bile acids capacity is high

95th percentile against the reference cohort

WHAT MIGHT BRING IT DOWN 4

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Limosilactobacillus reuteri NCIMB 30242 (bile salt hydrolase-active)** | probiotic | human trial<br>Jones ML et al. Br J Nutr 2012;107:1505 | 2.9 × 10^9 CFU capsule twice daily for nine weeks Studied in people with clinically confirmed hypercholesterolaemia, ages 20–75, in Canada, n = 127.<br>Total stool BSH gene capacity is not the same as one BSH-active strain. L. reuteri NCIMB 30242 lowered LDL by 9–12% in two trials in people with measured high cholesterol; the reading here does not select or predict that. |
| **Lactobacillus reuteri NCIMB30242, BSH-active live capsules** | probiotic | human trial<br>PMID 22990854 | LDL cholesterol fell 11.64% versus placebo; plasma deconjugated bile acids increased and absorbed plant sterols decreased.<br>Bile-salt hydrolase can be desirable for this endpoint but is not universally good across bile-acid disorders. |
| **Berberine; high-fat-fed mice, intestinal-FXR-deficient comparison, and complementary microbial …**† | plant compound | animal study<br>taken by mouth in the study | M: berberine reduced BSH activity and increased tauro-conjugated BAs, especially taurocholic acid. [S06a], [S06b]<br>M: intestinal FXR signalling increased, with downstream hepatic lipid-handling changes. [S06a] |
| **Resveratrol; atherosclerosis-prone mice with microbiota-depletion and BA-receptor perturbation …**† | plant compound | animal study<br>reaches the gut unaided | M: resveratrol changed the microbiota, reduced microbial trimethylamine/TMAO formation, and increased BSH-related BA deconjugation/excretion. [S25]<br>M: ileal BA/FXR–FGF15 signalling decreased and hepatic BA synthesis increased. [S25] |

ALSO STUDIED ON THIS, DIRECTION NOT STATED 1

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Purified Lactobacillaceae bile-salt-hydrolase cocktails; L. gasseri ATCC33323 BSH mutants**† | probiotic | animal study<br>reaches the gut unaided<br>PMID 36914755 | Substrate-specific deconjugation → changed conventional and microbial-conjugated bile-acid pool → altered C. difficile germination/growth.<br>Purified enzymes, intact cells and a commercial probiotic are different preparations. BSH does not perform 7α-dehydroxylation; total BSH gene abundance cannot predict one net direction for every bile pool. |

‹ back to the list · the reading this is about ›

METABOLITE PATHWAY

## 29. Urate degradation capacity is above average

77th percentile against the reference cohort

**WHAT MIGHT BRING IT DOWN** 10 of 29 on record

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **B. longum subsp. infantis EVC001 in breastfed infants** | prebiotic fibre | human trial<br>source | HMO utilization → acetate/lactate and lower fecal pH → altered Enterobacteriaceae / endotoxin milieu.<br>The pH-mediated suppression edge was not isolated causally in the infant trial. Breast-milk/HMO context and baseline colonization matter; lower pH is not universally preferable in adults. |
| **HST008 — Oral L-glutamine; AR005** | supplement or compound | human trial<br>PMID 30108163 | Randomized human trial in selected postinfectious IBS-D with baseline hyperpermeability. Host outcome measured; ecological continuation not measured.<br>Oral glutamine → improved measured permeability and IBS-D symptoms `M`; improved barrier → less oxygen/nitrate-supported Enterobacteriaceae growth `I`, using the separate mechanism in HST013; actual Klebsiella change `U`. |
| **L. gasseri PA-3** | supplement or compound | human trial<br>PMID 35227163 | Reduced serum urate in a marginally elevated-urate population<br>Not an established treatment for every gout state or for urate stones |
| **Same general inulin/psyllium combination with longer follow-up** | prebiotic fibre | human trial<br>source | Delayed arrival/fermentation → altered timing, but not reduced 24-hour fermentation |
| **Lactiplantibacillus plantarum 299v; DSM 9843; viable preparation** | probiotic | human trial<br>PMID 15702859 | viable gastrointestinal transit/fecal recovery with and without acid inhibition. human placebo-controlled double-blind study. positive survival through GI transit. |
| **Individual essential-oil constituents, particularly thymol and geraniol; mixed human faecal fer…** | essential oil | human cohorts | M: stronger tested exposures to thymol or geraniol suppressed total bacterial abundance and fermentation in mixed communities. [S11]<br>M: weaker thymol exposure had little effect; responses of selected commensals and fermentation products differed between compounds and between mixed-community and isolate experiments. [S11] |
| **PHGG preparation, human fecal fermentation** | prebiotic fibre | human cohorts<br>PMID 37541036 | Mixed-community response → acetate/butyrate increase and lower Klebsiella / Escherichia-Shigella signals |
| **PHGG; lower Escherichia-Shigella / Klebsiella alongside support** | prebiotic fibre | human cohorts<br>PMID 37541036 | In 41 human fecal inocula, PHGG fermentation enriched Bifidobacterium/Faecalibacterium and reduced Escherichia-Shigella, Klebsiella and Dorea; acetate/butyrate increased. |
| **Dietary glucosinolates; B. thetaiotaomicron biochemical work and mono-associated mice**† | way of eating | animal study<br>taken by mouth in the study | M: the examined commensal converted glucosinolates into isothiocyanates through a characterised bacterial pathway. [S34]<br>M: loss of pathway activity reduced isothiocyanate production in mono-associated mice, supporting a microbial contribution in vivo. [S34] |
| **Lacticaseibacillus rhamnosus HN001 / DR20**† | probiotic | animal study<br>taken by mouth in the study<br>PMID 12208607 | Protection against E. coli O157:H7 and, separately, Salmonella Typhimurium infection; immune-mediated effects rather than assumed direct eradication<br>HN001 is not LGG; exact commercial strain appears in Ritual identity map below |

## SHOWN TO PUSH IT THE OTHER WAY 15

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **A. soehngenii CH106 next-generation strain, 2025** | supplement or compound | human trial<br>PMID 40371708 | Glycemic variability and diastolic pressure improved; HbA1c difference emerged after 16 weeks including follow-up. Exact strain produces butyrate in culture from sugars or lactate plus acetate; no direct demonstrated human butyrate increase in this trial.<br>Geographic response heterogeneity, modest clinical effect sizes, multiple mechanisms plausible. Manufacturer/developer involvement. CH106 has 37 genome mutations relative to L2-7, including tetO inactivation, so they need separate nodes. Archives PRJNA1219376 and parental PRJEB22345. |
| **Studied oral liposomal iron versus intravenous iron gluconate; Open-label randomized CKD trial,…** | mineral or protein supplement | human trial<br>PMID 25395392 | Hemoglobin improved through both routes → IV treatment more rapidly replenished stores and maintained hemoglobin after withdrawal → Different adverse-event frequencies were measured, but bacterial exposure/engraftment was not<br>Non-dialysis CKD with documented iron-deficiency anemia. Clinical absorption and tolerability evidence; no microbiome endpoint. It does not prove that luminal organisms cannot access liposomal iron. |
| **Latilactobacillus sakei LZ217** | probiotic | human cohorts<br>PMID 35053965 | Identified folate product as 5-methyltetrahydrofolate; simulated gastrointestinal-juice resistance and increased butyrate in fecal-slurry fermentation were reported.<br>The human material was cultured fecal slurry, not an intervention in people. Gastric-juice tolerance does not establish durable engraftment. No retail strain substitution. |
| **Dietary L-serine → competitive advantage for AIEC LF82 in the inflamed gut**† | way of eating | animal study<br>taken by mouth in the study<br>PMID 31686025 | Mechanistic bacterial and mouse work: inflammation changed metabolic programs; removing serine blunted blooms in the tested setting.<br>Keep strain, inflammatory context, and nutrient separate. This does not establish that glutamine feeds R. gnavus or justify general amino-acid/protein restriction. |

ALSO STUDIED ON THIS, DIRECTION NOT STATED 17

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **high indole branch; Pectin / fermentable dietary fiber, with exact fiber recorded** | prebiotic fibre | human cohorts<br>PMID 38918470 | Defined microbial communities, human fecal communities and colonized-animal experiments showed that fermentable fiber can redirect microbial tryptophan metabolism away from indole and toward indolelactate/indolepropionate through community metabolic interactions.<br>Candidate dietary/prebiotic strategy for shifting tryptophan metabolism. The effect depends on resident producers, substrate competition and fiber preparation. Do not call all indoles harmful, nor assume tryptophan supplementation alone produces IPA. |
| **Veillonella atypica marathon-runner isolate, study-specific**† | lifestyle | animal study<br>taken by mouth in the study<br>PMID 31235964 | Host exercise lactate reaches gut → bacterial lactate-to-propionate metabolism → longer mouse treadmill running.<br>Study runner isolate is not assumed ATCC17744 or every commercial V. atypica. Human performance benefit and long-term safety were not shown by this mouse mechanism study. |
| **B. longum 105-A / JCM31944 and B. infantis DSM20088 ALDH constructs**† | probiotic | animal study<br>reaches the gut unaided<br>PMID 34675385 | Aromatic lactate dehydrogenase → tryptophan/phenylalanine/tyrosine-derived lactates → AhR/HCA3 signaling and altered human immune-cell responses.<br>The tested ALDH locus does not establish equal production by every commercial Bifidobacterium. A receptor effect does not by itself establish a disease benefit. |
| **RS1 and RS5; structurally distinct expansion candidates**† | herb or botanical extract | animal study<br>reaches the gut unaided<br>PMID 41926473 | RS1/2/3/4 human-fecal fermentation comparisons and RS5 starch–fatty-acid studies demonstrate structure-dependent fermentability. |
| **LGG-derived p40 protein**† | supplement or compound | animal study<br>route not stated<br>source | Epithelial survival/barrier support through ADAM17/HB-EGF/EGFR pathway<br>Research protein; do not assign its active dose to retail LGG CFU |
| **Propionibacterium freudenreichii DSM 20271; co-fermented wheat bran with Levilactobacillus brev…**‡ | fibre | laboratory culture<br>taken by mouth in the study<br>PMID 31333632 | The exact P. freudenreichii strain produced B12; co-fermentation supported a fortified food ingredient while altering microbiological safety indicators.<br>Prepared food production, not in-vivo supplementation or microbiome engraftment. The B12 producer and co-culture strain have distinct roles. |

‹ back to the list · the reading this is about ›

METABOLITE PATHWAY

## 30. Dietary fibre and carbohydrate breakdown capacity is below average

21th percentile against the reference cohort

**Nothing on record.** No study in any setting — trial, animal or laboratory — has been recorded here for how to bring it up. That is the state of the research, not a judgement that this does not matter, and not a reason to do nothing: the sections above explain the finding and a clinician can act on it.

‹ back to the list · the reading this is about ›

METABOLITE PATHWAY

## 31. Equol formation capacity is high

95th percentile against the reference cohort

**Nothing on record.** No study in any setting — trial, animal or laboratory — has been recorded here for how to bring it down. That is the state of the research, not a judgement that this does not matter, and not a reason to do nothing: the sections above explain the finding and a clinician can act on it.

‹ back to the list · the reading this is about ›

METABOLITE PATHWAY

## 32. Microbial ethanol formation capacity is above average

76th percentile against the reference cohort

**Nothing on record.** No study in any setting — trial, animal or laboratory — has been recorded here for how to bring it down. That is the state of the research, not a judgement that this does not matter, and not a reason to do nothing: the sections above explain the finding and a clinician can act on it.

‹ back to the list · the reading this is about ›

METABOLITE PATHWAY

## 33. Spermidine formation capacity is above average

80th percentile against the reference cohort

**Nothing on record.** No study in any setting — trial, animal or laboratory — has been recorded here for how to bring it down. That is the state of the research, not a judgement that this does not matter, and not a reason to do nothing: the sections above explain the finding and a clinician can act on it.

‹ back to the list · the reading this is about ›

METABOLITE PATHWAY

## 34. Vitamin K2 (menaquinone, MK-n) capacity is low

12th percentile against the reference cohort

**Nothing on record.** No study in any setting — trial, animal or laboratory — has been recorded here for how to bring it up. That is the state of the research, not a judgement that this does not matter, and not a reason to do nothing: the sections above explain the finding and a clinician can act on it.

Gene capacity for menaquinone synthesis does not measure coagulation or vitamin K sufficiency. People on warfarin should keep vitamin K intake stable under their prescriber.

‹ back to the list

DISEASE PATTERN

## 35. Rheumatoid arthritis pattern: notably high

A community-level resemblance to a published pattern, not a diagnosis

### WHAT HAS BEEN TRIED IN THIS CONDITION 2

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **L. acidophilus + L. casei + B. bifidum capsule (Zamani 2016, rheumatoid arthritis)** | probiotic | human trial<br>Zamani B et al. Int J Rheum Dis 2016;19:869 | 2 × 10^9 CFU each of three organisms daily for eight weeks Studied in people with clinically confirmed rheumatoid arthritis, ages 20–80, female, in Iran, n = 46.<br>Rheumatoid arthritis is diagnosed by rheumatology with serology and examination. Two small probiotic trials were positive on disease activity, one was null; many RA patients are immunosuppressed, which changes live-microbe safety. |
| **Lacticaseibacillus casei 01 (Alipour 2014, rheumatoid arthritis)** | probiotic | human trial<br>Alipour B et al. Int J Rheum Dis 2014;17:519 | 10^8 CFU/day L. casei 01 for eight weeks Studied in people with clinically confirmed rheumatoid arthritis, ages 20–80, female, in Iran, n = 46.<br>Rheumatoid arthritis is diagnosed by rheumatology with serology and examination. Two small probiotic trials were positive on disease activity, one was null; many RA patients are immunosuppressed, which changes live-microbe safety. |

‹ back to the list

DISEASE PATTERN

### 36. Clinical Alzheimer's disease pattern: high

A community-level resemblance to a published pattern, not a diagnosis

**WHAT HAS BEEN TRIED IN THIS CONDITION** 1

| WHAT | KIND | WHERE THIS COMES FROM | WHAT WAS SEEN |
|---|---|---|---|
| **Probiotic milk with Lactobacillus acidophilus, L. casei, Bifidobacterium bifidum and L. ferment…** | probiotic | human trial<br>Akbari E et al. Front Aging Neurosci 2016;8:256 | 200 mL/day probiotic milk (2 × 10^9 CFU/g of each organism) for 12 weeks Studied in people with clinically confirmed Alzheimer disease, ages 60–95, in Iran, n = 60.<br>One 60-person trial reported MMSE gains; a later trial in severe disease was null and pooled analyses are not robust. FMT in Alzheimer disease is animal and case-level only. |

† Shown in animals, not yet in people. ‡ Shown in laboratory culture rather than in a living body: it tells you the organism is susceptible to the substance, not that eating it will do this in you. § Would need an enteric-coated or delayed-release preparation to reach the colon intact; the plain form is absorbed or destroyed before it arrives. ¶ The source does not say how it was delivered.

Every row is a description of published research, not a recommendation, and not a claim that it is safe or right for you. Nothing here is a prescription; anything marked as a prescriber's decision is one. Talk to a qualified clinician before acting on any of it.

## Personalised synbiotic options: what the model predicts

Your own community is rebuilt as a metabolic model from the organisms found in your sample and run four ways for each fibre — as it is, with a five-species probiotic consortium added, with the fibre added, and with both — so the question “is the combination worth more than either part alone?” can be asked directly, and the question “who would this fibre actually feed in *my* gut?” can be answered organism by organism. These are **experimental predictions from a model**, not measurements, and not a guarantee of how you would respond.

| COMPONENT | STATE | DETAIL |
|---|---|---|
| **Simulation engine** | ready | MICOM 0.37.0; community growth held at 0.99 of its maximum while each output is bounded |
| **Metabolic model library** | ready | AGORA2 species models, checksum verified |
| **Numerical solver** | exact | every value is the optimum of a linear programme; each arm's growth maximum was recomputed by independent worker processes and agreed to every digit (0 programmes solved, 982 served from cache) |

**The consortium added in the model is *Akkermansia muciniphila*, *Clostridium beijerinckii*, *Clostridium butyricum*, *Bifidobacterium longum* and *Anaerobutyricum hallii*.** These are species added to a metabolic model following the published protocol — not a named product, not a dose, and not a recommendation to take any of them.

**50 of the 313 organisms found in your sample have a metabolic model in the library**, covering 52% of the bacterial abundance they represent. The model library predates some renames, so two of your organisms were matched *M. lactaris* under its earlier name *R. lactaris*; *A. rectalis* under its earlier name *E. rectale*; *L. massiliensis* under its earlier name *M. massiliensis*. They are the same organism, not an extra one. In this diet definition the model's total growth is capped by **alagly** supply, so growth is identical in every arm and only what the community *makes* is compared.

### Butyrate potential under each option mmol/gDW/h in the model; ranked by the combined option

| FIBRE | AS YOU ARE | + CONSORTIUM | + FIBRE | + BOTH | INTERACTION | VERDICT |
|---|---|---|---|---|---|---|
| **Inulin**<br>Chicory root, alliums, artichoke | 219.4 | 219.0 | 418.6 | **440.1** | +21.9 | pair gains |
| **Maltodextrin**<br>Defined maltodextrin preparations | 219.4 | 219.0 | 417.9 | **437.0** | +19.6 | pair gains |
| **Pectin**<br>Apple, citrus peel, vegetables | 219.4 | 219.0 | 355.0 | **375.2** | +20.6 | pair gains |
| **Starch**<br>Cooked and cooled starchy foods | 219.4 | 219.0 | 344.0 | **343.6** | -0.0 | fibre alone suffices |
| **Arabinoxylan***<br>Wheat and rye bran, psyllium | 219.4 | 219.0 | 229.8 | **300.8** | +71.4 | pair gains |
| **Cellulose***<br>Hemp seed, intact plant material | 219.4 | 219.0 | 219.4 | **219.0** | 0.0 | no gain |

**Interaction** is the combined option minus both single options plus the baseline (F11 – F10 – F01 + F00): above zero, the pair does more than the sum of its parts in the model; below zero, less. A fibre that adds nothing on its own is still shown, and a verdict of “fibre alone suffices” means the consortium adds little on top of it. * Arabinoxylan: An

arabinoxylan exchange standing in for psyllium husk. Psyllium is a gel-forming mucilage whose physical behaviour the model does not represent. Cellulose: A cellulose exchange standing in for hemp seed. The model sees purified cellulose, not the whole seed's protein, fat and other fibre.

## Who each fibre would feed in your community

**share of the fibre each organism can take up when the community makes the most butyrate it can**

| FIBRE | BENEFICIAL OR ADDED ORGANISMS | ORGANISMS THIS REPORT FLAGGED | OTHER ORGANISMS |
|---|---|---|---|
| **Inulin** 12 of your organisms can use it | *F. prausnitzii E* ≤100%; *B. longum* ≤89%; *A. hallii* ≤37%; *B. cellulosilyticus* ≤6%; *F. saccharivorans* ≤6%; +4 more | none in the model | *D. longicatena* ≤22%; *B. uniformis* ≤8%; *C. aerofaciens H* ≤3% |
| **Maltodextrin** 18 of your organisms can use it | *F. prausnitzii E* ≤27% **required**; *B. longum* ≤22%; *C. butyricum* ≤22%; *R. bromii* ≤15%; *F. saccharivorans* ≤4%; +5 more | *D. formicigenerans* ≤3% (p96) | *D. longicatena* ≤4%; *A. putredinis* ≤4%; *B. luti* ≤3%; *P. distasonis* ≤3%; *E. ramulus* ≤3%; +2 more |
| **Pectin** 3 of your organisms can use it | *F. prausnitzii E* ≤61% **required**; *C. beijerinckii* ≤46%; *C. butyricum* ≤46% | none in the model | none in the model |
| **Starch** 9 of your organisms can use it | *F. prausnitzii E* ≤50%; *C. beijerinckii* ≤39%; *C. butyricum* ≤39%; *R. bromii* ≤22%; *C. eutactus* ≤20%; +3 more | none in the model | *P. distasonis* ≤5% |
| **Arabinoxylan** 3 of your organisms can use it | *B. longum* ≤33% **required**; *B. cellulosilyticus* ≤2% **required**; *R. intestinalis* ≤1% | none in the model | none in the model |
| **Cellulose** | no organism in your modelled community carries a transporter for it | — | — |

"≤ 40%" means that in some butyrate-optimal state of your community this organism takes up to 40% of the fibre supplied; **required** means no such state exists without it. A flagged organism listed here has the transporter and the community state permits the uptake — a specific, testable hypothesis about your sample, which is why it is shown rather than assumed either way.

*D. formicigenerans* can use only Maltodextrin of the fibres tested. This concerns direct uptake only: an organism can still gain from products another organism releases, which the model does allow and which the detail section shows.

## Who would make the butyrate

**each organism's share of the community's potential; a range is what it must and what it can contribute**

| SCENARIO | REQUIRED PRODUCERS |
|---|---|
| **As you are** 219.4 mmol/gDW/h | no organism is required (optional: *F. prausnitzii E*, *A. hadrus*, *C. eutactus*, *C. sp. AF12_28*, +13 more) |
| **Inulin + consortium** 440.1 mmol/gDW/h | *F. prausnitzii E* 9%–19%; *C. butyricum* 5%–15% (added); *C. beijerinckii* 5%–15% (added); *A. hallii* 5%–15% (added); *A. hadrus* 4%–14% (optional: *C. eutactus*, *C. sp. AF12_28*, *R. bacterium AM53 2BH*, *G. formicilis*, +8 more) |

## How much of the goal set these options cover

| | | |
|---|---|---|
| **Goal coverage (C)** | **83%** |  |
| | | share of modelled goals with a favourable edge |
| **Pair-supported coverage (P)** | **67%** |  |
| | | share where the pair, not either part alone, carries it |

Over 6 modelled goals, each weighted equally. A candidate whose model did not solve is left out of the set rather than counted against it. These are percentages of a modelled goal set — not a response rate, and not a share of people helped.

**On high-fibre diet, no fibre supplement changes the prediction.** Every candidate reaches the same output as the consortium alone, because that diet already supplies what the supplement would have added — and it reaches 109% of the best combined result on the baseline diet without any supplement at all. If your own diet is already high in fibre, the model puts the benefit in the organisms rather than in the fibre. This is a range across predeclared dietary assumptions, not a confidence interval and not a probability of response.

**The method is validated against published predictions.** Run against the released predictions of the study it follows, this pipeline reproduces every value to 1e-08, including the result that a fibre alone can outrank a fibre-plus-probiotic combination, and that ranking by total output and by output per unit of growth picks different winners. That replay and your solved model are different things, and the data file labels them differently.

## What the published synbiotic trials actually show

| 4 of 17 | 4 of 17 | 9 of 17 | 8 of 17 |
|---|---|---|---|
| demonstrated synergy | no synergy demonstrated | undetermined | missing a component-only arm |

Four of these are negative results: a strain, a substrate and the two together, with the combination no better than the strain alone. Eight lack a component-only arm and cannot show synergy in either direction. Neither fact is an absence of data.

Units are mmol per gram of community dry weight per hour in the model: a production potential, not a stool concentration, an absorbed amount or a dose. Fibre caps in the model are exchange limits, not grams to eat. The consortium is a species-level perturbation following the published protocol, not a named product. Organisms the model library lacks are counted as unmapped, never substituted. Feeding of a flagged organism is a specific model edge — it carries the transporter and the community state permits the uptake — not a rule that fibre feeds pathogens. In ISAPP terms every row is a *candidate combination*; a positive contrast in the model is evidence about the model, not a demonstrated host benefit.

## Measurements not in this edition

**Your sample could not answer these.** Plant-compound conversion (a declared negative control matched for Urolithin C to urolithin A (9-dehydroxylation); the step is withheld because the panel cannot currently tell the two enzymes apart in this sample); Beta-glucuronidase substrate classes (The structural class of a beta-glucuronidase is defined by the length of a loop at one of two active-site positions, measured in a curated alignment guided by solved structures. Reproducing that from a pairwise alignment to the E. coli enzyme works for enzymes close to it and fails for the divergent gut ones: in testing, the Bacteroides fragilis enzyme came out as Loop 1 where the published structure is mini-Loop 1, and the Bacteroides uniformis enzyme came out as No Loop where it is the published example of Loop 2. Those are the gut organisms the breakdown would exist to describe, so no break); Options for the organisms flagged here (10 organism(s) are flagged in this sample and the intervention registry holds no studied option that names any of them); Change over time (only one sample is on file for this participant, so nothing can be compared with it yet); Return toward a pre-event community (a recovery estimate needs a recorded perturbation and at least 3 eligible samples before it); External laboratory results (None are needed: every reading in this report is produced from your sequencing data. A supplied result would be shown beside the finding it relates to, not used to replace it.).

PART B

# In detail

Every reading from Part A, one at a time and in the same order: what it is, exactly where you sit, what that means, the research behind the reading — and, under each one, **what the human research says about changing it**: the exact products, diets, drugs or clinical routes that have been studied, in whom, with the evidence for and against side by side and the safety rules that apply. Where nothing has been shown to work, the card says so. Where fecal microbiota transplantation is the only route with evidence, the card says that and gives its regulatory status.

IN THIS PART


- 15 · Your microbial groups in detail, and every species detected
- 16 · Organisms that need attention, one by one
- 18 · Pathogens, one at a time, and what was not assessed
- 19 · Your metabolic functions in detail (45 pathways)
- 20 · Disease-pattern resemblance, every scored pattern
- 23 · Gut-skin axis, panel by panel
- 24 · Your information, and what this report assumed
- 25 · What none of this can tell you · 26 · Sequencing quality and reference coverage
- 27 · Measured accuracy · 28 · Technical record

PART B — WHO IS THERE

# 15 Your microbial groups in detail

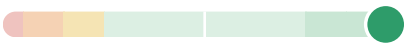
Curated sets of species that share a job or an origin — the fibre fermenters, the mouth bacteria that should not be in stool in quantity, the organisms that make methane or sulfide — each summed to one reading and placed against 3,027 matched reference adults. The arrow says which direction the research reads as favourable; no arrow means it depends on context. Under each group are the species that made up your reading.

# Functional groups

‹ at a glance

## Butyrate producers ▼

16.82% of classified reads · **+135%** vs typical · 14 of 19 species detected · 2 in the literature not nameable here



**95th** HIGH

The organisms that make butyrate, the short-chain fatty acid your colon cells burn for fuel. Depletion of this group is the most common finding in gut disease research and the least specific: it accompanies inflammatory bowel disease, recent antibiotics, low-fibre eating and general ill health alike.

Carriage of the pathway is not production. Butyrate output depends on fermentable fibre reaching the colon, and a person with a rich butyrate- producer community on a low-fibre diet produces little of it.

### What is driving this reading 14 of 19 species detected; the reading is their sum

| ORGANISM | CLASS | SHARE OF GROUP | IN YOUR SAMPLE | YOUR LEVEL |
|---|---|---|---|---|
| ***Faecalibacterium prausnitzii*** | ● Beneficial | 37% | 6.18% | **+82%** vs typical carrier<br>**80th** percentile<br>carried by 98% of adults |
| ***Anaerostipes hadrus*** | ● Beneficial | 27% | 4.53% | **×23** vs typical carrier<br>**99th** percentile<br>carried by 87% of adults |
| ***Coprococcus eutactus*** | ● Beneficial | 15% | 2.46% | **×11** vs typical carrier<br>**89th** percentile<br>carried by 36% of adults |
| ***Roseburia faecis*** | ● Beneficial | 3% | 0.586% | **–53%** vs typical carrier<br>**35th** percentile<br>carried by 81% of adults |
| ***Coprococcus catus*** | ● Beneficial | 3% | 0.534% | **×11** vs typical carrier<br>**99th** percentile<br>carried by 69% of adults |
| ***Agathobacter rectalis*** | ● Beneficial | 3% | 0.518% | **–50%** vs typical carrier<br>**37th** percentile<br>carried by 88% of adults |
| ***Roseburia hominis*** | ● Beneficial | 3% | 0.491% | **+438%** vs typical carrier<br>**90th** percentile<br>carried by 83% of adults |
| ***Lachnospira eligens*** | ● Beneficial | 2% | 0.407% | **97th** percentile<br>carried by 81% of adults |
| ***Agathobaculum butyriciproducens*** | ● Beneficial | 2% | 0.363% | **–23%** vs typical carrier<br>**42nd** percentile<br>carried by 89% of adults |
| ***Anaerobutyricum hallii*** | ● Beneficial | 2% | 0.313% | **+253%** vs typical carrier<br>**80th** percentile<br>carried by 86% of adults |
| ***Odoribacter splanchnicus*** | ● Beneficial | 1% | 0.177% | **+7%** vs typical carrier<br>**52nd** percentile<br>carried by 74% of adults |
| ***Roseburia intestinalis*** | ● Beneficial | 1% | 0.162% | **–10%** vs typical carrier<br>**48th** percentile<br>carried by 73% of adults |
| ***Coprococcus comes*** | ● Beneficial | 1% | 0.091% | **94th** percentile<br>carried by 79% of adults |
| ***Roseburia inulinivorans***<br>full page → | ● Beneficial | 0% | 0.004% | **–94%** vs typical carrier<br>**9th** percentile<br>carried by 85% of adults |

**Why.** This reading is carried by *Faecalibacterium prausnitzii*, which alone is **37%** of the group's reads and sits at the **80th** percentile among adults who carry it. *Anaerostipes hadrus* contributes another **27%**. Changing this reading means changing them.

**Not detected (5 of 19):** *Eubacterium ventriosum* (52%), *Butyrivibrio crossotus* (18%), *Butyricicoccus pullicaecorum* (3%), *Anaerostipes caccae* (2%), *Subdoligranulum variabile* (0%). Percentages are how many reference adults carry each.

**Two measurements agree**

This agrees with the butyrate production capacity reading (82nd percentile), which measures the same function by counting the pathway's genes across the whole community instead of these species.

## What your reading means high

This group is 16.82% of your classified reads, counted over 14 of the 19 species this group is defined by. That places you at the 95th percentile among reference adults who have the group. A group reading is the sum of its species, so it moves when they do.

## How can I improve this?

Higher is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Faecalibacterium prausnitzii* (6.18% of your community); *Anaerostipes hadrus* (4.53% of your community); *Coprococcus eutactus* (2.46% of your community); *Roseburia faecis* (0.59% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for each species is on its own page, and the members not detected are listed above: for a low reading those absentees are the gap, and they are the specific thing any change would have to supply.

## The research behind it

The organisms that make butyrate, the short-chain fatty acid your colon cells burn for fuel. Depletion of this group is the most common finding in gut disease research and the least specific: it accompanies inflammatory bowel disease, recent antibiotics, low-fibre eating and general ill health alike.

Carriage of the pathway is not production. Butyrate output depends on fermentable fibre reaching the colon, and a person with a rich butyrate- producer community on a low-fibre diet produces little of it.

Odoribacter splanchnicus is included as a characterised butyrate producer but is also elevated in the Alzheimer's continuum panel. Group membership is about metabolic capacity, not about whether an organism is desirable.

‹ at a glance

### Sulfate-reducing bacteria ▲

0.00% of classified reads · 1 of 4 species detected · 2 in the literature not nameable here



16th BELOW AVERAGE

Bacteria that breathe sulfate instead of oxygen, releasing hydrogen sulfide as they do. They are normal members of most healthy communities. Sustained high sulfide interferes with how colon cells use butyrate for fuel, and sulfate-reducer expansion is reported in ulcerative colitis.

A small group in a low-abundance niche, so this reading is noisier than the larger groups and more sensitive to sequencing depth.

## What is driving this reading 0 of 4 species detected; the reading is their sum

**Why.** This reading is low because **1** species most reference adults carry were not detected: *Bilophila wadsworthia* (61%).

**Not detected (4 of 4):** *Bilophila wadsworthia* (61%), *Desulfovibrio piger* (20%), *Desulfovibrio fairfieldensis* (2%), *Desulfovibrio legallii* (0%). Percentages are how many reference adults carry each.

**Two measurements of the same function disagree**

**Read this beside the hydrogen-sulfide capacity reading.** The sulfate-reducing species reading sits at the 16th percentile; the hydrogen-sulfide capacity reading is at the 81st. The two are measured differently — genes in the whole community against the abundance of the species characterised for the job — and they can separate when the genes are carried by organisms outside the curated list, which the gene search counts and the species list does not; the characterised species are present below the profiler's limit of detection, so their genes are counted while the organisms are not. Of the two, the gene reading is the better guide here, because this capacity does not depend on any one set of species. Hydrogen sulfide has two independent sources: dissimilatory sulfate reduction by Desulfovibrio and relatives, and cysteine degradation, which is widespread in ordinary gut bacteria. The gene panel sees both; the group sees only the sulfate reducers, which are a minority of stool even when sulfide capacity is high.

## What your reading means below average

This group is 0.00% of your classified reads, counted over 1 of the 4 species this group is defined by. That places you at the 16th percentile among reference adults who have the group. A group reading is the sum of its species, so it moves when they do.

## How can I improve this?

Lower is the more favourable direction for this reading.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for each species is on its own page, and the members not detected are listed above: for a low reading those absentees are the gap, and they are the specific thing any change would have to supply.

## The research behind it

Bacteria that breathe sulfate instead of oxygen, releasing hydrogen sulfide as they do. They are normal members of most healthy communities. Sustained high sulfide interferes with how colon cells use butyrate for fuel, and sulfate-reducer expansion is reported in ulcerative colitis.

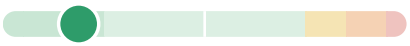
A small group in a low-abundance niche, so this reading is noisier than the larger groups and more sensitive to sequencing depth.

Bilophila wadsworthia rose sharply on an animal-based diet in the David et al. dietary intervention, making this group unusually diet-responsive over days rather than months.

‹ at a glance

## Mucin degraders ▲

0.45% of classified reads · **–66%** vs typical · 4 of 8 species detected



**19th** BELOW AVERAGE

Bacteria that feed on the mucus layer lining your gut. This is normal and part of how the mucus layer renews itself. When dietary fibre is scarce, these organisms turn to mucin as their main carbon source and the barrier thins — an effect demonstrated cleanly in gnotobiotic mice.

Akkermansia muciniphila carries the strongest and most contradictory literature of any species here. It is depleted in obesity and metabolic syndrome and improved insulin sensitivity in a human trial when given as a pasteurised supplement; it is elevated in Parkinson's disease and multiple sclerosis. Whether it is desirable depends on the question being asked, which is why this group is not scored as favourable.

### What is driving this reading

4 of 8 species detected; the reading is their sum

| ORGANISM | CLASS | SHARE OF GROUP | IN YOUR SAMPLE | YOUR LEVEL |
|---|---|---|---|---|
| ***Barnesiella intestinihominis*** | ● Beneficial | 51% | 0.230% | **–48%** vs typical carrier<br>**35th** percentile<br>carried by 51% of adults |
| ***Mediterraneibacter torques*** | ● Opportunist | 23% | 0.103% | **+68%** vs typical carrier<br>**64th** percentile<br>carried by 78% of adults |
| ***Bacteroides thetaiotaomicron*** | ● Conditional | 20% | 0.092% | **–49%** vs typical carrier<br>**36th** percentile<br>carried by 79% of adults |
| ***Akkermansia muciniphila*** | ● Beneficial | 5% | 0.024% | **24th** percentile<br>carried by 51% of adults |

**Why.** This reading is low because **2** species most reference adults carry were not detected: *Bacteroides caccae* (61%), *Bacteroides fragilis* (52%). What is present, led by *Barnesiella intestinihominis*, is not enough to make up the sum.

**Not detected (4 of 8):** *Bacteroides caccae* (61%), *Bacteroides fragilis* (52%), *Ruminococcus gnavus* (47%), *Bifidobacterium bifidum* (22%). Percentages are how many reference adults carry each.

### What your reading means

below average

This group is 0.45% of your classified reads, counted over 4 of the 8 species this group is defined by. That places you at the 19th percentile among reference adults who have the group. A group reading is the sum of its species, so it moves when they do.

### How can I improve this?

Lower is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Barnesiella intestinihominis* (0.23% of your community); *Mediterraneibacter torques* (0.10% of your community); *Bacteroides thetaiotaomicron* (0.09% of your community); *Akkermansia muciniphila* (0.02% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for each species is on its own page, and the members not detected are listed above: for a low reading those absentees are the gap, and they are the specific thing any change would have to supply.

### The research behind it

Bacteria that feed on the mucus layer lining your gut. This is normal and part of how the mucus layer renews itself. When dietary fibre is scarce, these organisms turn to mucin as their main carbon source and the barrier thins — an effect demonstrated cleanly in gnotobiotic mice.

Akkermansia muciniphila carries the strongest and most contradictory literature of any species here. It is depleted in obesity and metabolic syndrome and improved insulin sensitivity in a human trial when given as a pasteurised supplement; it is elevated in Parkinson's disease and multiple sclerosis. Whether it is desirable depends on the question being asked, which is why this group is not scored as favourable.

Ruminococcus gnavus is a mucin degrader and is enriched in inflammatory bowel disease, where its glucorhamnan polysaccharide induces TNF-alpha. Same functional group, very different clinical company.

## What the research says about changing this

**Mucin-degrading species (Akkermansia, R. gnavus) — a context-dependent group** LANE C HUMAN TRIAL · LIMITED

Fired by: mucin degraders relative abundance: typical

Akkermansia is linked to metabolic health in some cohorts and enriched in MS and Parkinson cohorts in others; R. gnavus blooms in Crohn's flares. The only trials are of one exact pasteurised Akkermansia product with mixed primary endpoints. Group abundance is not a target.

**Studied:** Pasteurised Akkermansia muciniphila MucT (ATCC BAA-835) (live biotherapeutic) · **in:** people with clinically confirmed overweight/obesity with insulin resistance, ages 18–70, in Belgium, n = 32 · **studied protocol (a source fact, not an instruction):** 10^10 pasteurised cells/day for three months

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: single study. Strongest design: randomized trial.

- In a 32-completer exploratory pilot, pasteurised A. muciniphila MucT (10^10 cells/day for three months) improved insulin sensitivity and lowered insulinaemia and cholesterol in overweight/obese insulin-resistant adults. [Depommier C et al. Nat Med 2019;25:1096] · lane C · randomized double-blind placebo-controlled exploratory pilot · n = 40 · benefit
- **Against or limiting —** A 2025 randomized trial of live A. muciniphila AKK-WST01 in obese type 2 diabetes found no overall weight or HbA1c benefit, only exploratory subgroup signals. [Zhang et al. 2025 — AKK-WST01 in obese type 2 diabetes] · lane X · randomized placebo-controlled · null
- **Against or limiting —** A 2026 pasteurised-Akkermansia trial missed its primary insulin-sensitivity endpoint. [Suenaert et al. 2026 — pasteurised A. muciniphila, primary insulin-sensitivity endpoint missed] · lane X · randomized placebo-controlled · null

**Applicability:** The study required a clinical diagnosis of insulin resistance. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “mucus-eating bacteria damaging your gut” · “boost Akkermansia”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

## Polyphenol metabolisers (urolithin and equol) ▼

0.45% of classified reads · **–65%** vs typical · 2 of 5 species detected · 4 in the literature not nameable here



**26th** TYPICAL

The bacteria that turn compounds in pomegranate, berries, nuts and soy into the forms your body can actually use. Whether you benefit from these foods depends partly on whether you carry the converters, and a substantial share of people do not.

Eggerthella lenta is a versatile metaboliser: it also inactivates the heart drug digoxin via its cgr operon, and it is enriched in several disease cohorts (rheumatoid arthritis, Parkinson's). It is counted here for its urolithin activity, not as a health marker.

### What is driving this reading 2 of 5 species detected; the reading is their sum

| ORGANISM | CLASS | SHARE OF GROUP | IN YOUR SAMPLE | YOUR LEVEL |
|---|---|---|---|---|
| ***Adlercreutzia equolifaciens*** | ● Beneficial | **95%** | 0.428% | **18th** percentile<br>carried by 64% of adults |
| ***Eggerthella lenta*** | ● Opportunist | **5%** | 0.024% | **+27%** vs typical carrier<br>**56th** percentile<br>carried by 47% of adults |

**Why.** Led by *Adlercreutzia equolifaciens* at **95%** of the group, the 18th percentile among carriers. The group as a whole sits within the reference range.

**Not detected (3 of 5):** *Gordonibacter pamelaeae* (62%), *Slackia isoflavoniconvertens* (37%), *Slackia exigua* (0%). Percentages are how many reference adults carry each.

**Two measurements of the same function disagree**

**Read this beside the urolithin and equol capacity reading.** The polyphenol-metabolising species reading sits at the 43rd percentile; the urolithin and equol capacity reading is at the 83rd. The two are measured differently — genes in the whole community against the abundance of the species characterised for the job — and they can separate when the genes are carried by organisms outside the curated list, which the gene search counts and the species list does not; the gene family is shared with enzymes that do a different job, so some matched fragments are not this pathway. Of the two, the species reading is the better guide here. Urolithin and equol production is strain-level: two isolates of the same species differ, and only a handful of characterised strains do it. The curated species are the better guide; a high gene reading without them is most likely a related enzyme.

### What your reading means typical

This group is 0.45% of your classified reads, counted over 2 of the 5 species this group is defined by. That places you at the 26th percentile among reference adults who have the group. A group reading is the sum of its species, so it moves when they do.

### How can I improve this?

Higher is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Adlercreutzia equolifaciens* (0.43% of your community); *Eggerthella lenta* (0.02% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for each species is on its own page, and the members not detected are listed above: for a low reading those absentees are the gap, and they are the specific thing any change would have to supply.

### The research behind it

The bacteria that turn compounds in pomegranate, berries, nuts and soy into the forms your body can actually use. Whether you benefit from these foods depends partly on whether you carry the converters, and a substantial share of people do not.

Eggerthella lenta is a versatile metaboliser: it also inactivates the heart drug digoxin via its cgr operon, and it is enriched in several disease cohorts (rheumatoid arthritis, Parkinson's). It is counted here for its urolithin activity, not as a health marker.

Presence of the converters does not guarantee the conversion happens at a useful rate — the metabotype is determined by the whole community and by diet, and the gene-level detail (the specific dehydroxylases) is below the resolution of species-level profiling.

‹ at a glance

### Methanogens (archaea) ▲

0.00% of classified reads · 0 of 3 species detected · 1 in the literature not nameable here



35th NOT DETECTED · TYPICAL

The methane-producing archaea in your gut. Roughly a third to a half of adults carry them in quantity and the rest carry almost none — this is a normal split, not a spectrum of health. Methane slows intestinal transit, and high carriage is consistently associated with constipation.

Bimodal by nature. A near-zero reading is the normal state for a large fraction of healthy adults and means the hydrogen goes elsewhere — typically to acetogens or sulfate reducers instead.

## What is driving this reading 0 of 3 species detected; the reading is their sum

**Why.** None of this group's species were detected in your sample.

**Not detected (3 of 3):** *Methanobrevibacter smithii* (30%), *Methanosphaera stadtmanae* (4%), *Methanobrevibacter woesei* (0%). Percentages are how many reference adults carry each.

**Two measurements of the same function disagree**

**Read this beside the methane capacity (mcrA) reading.** The methanogen abundance reading sits at the 35th percentile; the methane capacity (mcrA) reading is at the <1st. The two are measured differently — genes in the whole community against the abundance of the species characterised for the job — and they can separate when the characterised species are present below the profiler's limit of detection, so their genes are counted while the organisms are not; the organisms are present but fewer of the terminal-step genes were found than their genomes would predict. Neither reading is preferred: a split this size means one of the two engines is at its limit for this function. mcrA is exclusive to methanogenic archaea, so these two should track each other tightly. A split means one of the two engines is at its limit: methanogens are archaea, they are a small fraction of stool DNA, and the bacterial profiler under-reports them.

## What your reading means not detected · typical

This group is 0.00% of your classified reads, counted over 0 of the 3 species this group is defined by. That places you at the 35th percentile among reference adults who have the group. A group reading is the sum of its species, so it moves when they do.

## How can I improve this?

Lower is the more favourable direction for this reading.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for each species is on its own page, and the members not detected are listed above: for a low reading those absentees are the gap, and they are the specific thing any change would have to supply.

## The research behind it

The methane-producing archaea in your gut. Roughly a third to a half of adults carry them in quantity and the rest carry almost none — this is a normal split, not a spectrum of health. Methane slows intestinal transit, and high carriage is consistently associated with constipation.

Bimodal by nature. A near-zero reading is the normal state for a large fraction of healthy adults and means the hydrogen goes elsewhere — typically to acetogens or sulfate reducers instead.

Carriage is partly heritable and increases with age and with slow transit, so cause and effect run both ways with constipation.

# Organisms from elsewhere in the body

‹ at a glance

**Oral-origin organisms found in stool ▲**

0.08% of classified reads · **+51%** vs typical · 3 of 33 species detected



58th TYPICAL

Bacteria that normally live in your mouth, measured in stool. Ordinarily stomach acid and bile stop them; when they arrive in quantity it usually means one of those barriers is reduced. Acid-suppressing medication is the most common everyday cause by far.

Proton pump inhibitors raise this group substantially and are extremely common. Before reading anything else into a high value, check whether the person takes one.

## What is driving this reading 3 of 33 species detected; the reading is their sum

| ORGANISM | CLASS | SHARE OF GROUP | IN YOUR SAMPLE | YOUR LEVEL |
|---|---|---|---|---|
| ***Streptococcus salivarius*** | ● Conditional | 66% | 0.050% | **+155%** vs typical carrier<br>**68th** percentile<br>carried by 70% of adults |
| ***Streptococcus parasanguinis*** | ● Conditional | 19% | 0.015% | **+41%** vs typical carrier<br>**58th** percentile<br>carried by 57% of adults |
| ***Rothia mucilaginosa*** | ● Conditional | 15% | 0.011% | **39th** percentile<br>carried by 23% of adults |

**Why.** Led by *Streptococcus salivarius* at **66%** of the group, the 68th percentile among carriers. The group as a whole sits within the reference range.

**Not detected (30 of 33):** *Haemophilus parainfluenzae* (41%), *Veillonella parvula* (39%), *Veillonella atypica* (30%), *Veillonella dispar* (29%), *Streptococcus mitis* (22%), *Actinomyces odontolyticus* (21%), *Gemella sanguinis* (19%), *Streptococcus oralis* (16%), *Streptococcus australis* (15%), *Streptococcus gordonii* (12%), *Streptococcus vestibularis* (12%), *Streptococcus sanguinis* (11%), *Streptococcus anginosus group* (9%), *Gemella haemolysans* (6%), *Veillonella rogosae* (6%), *Actinomyces graevenitzii* (5%), *Parvimonas micra* (4%), *Solobacterium moorei* (3%), *Atopobium parvulum* (3%), *Gemella morbillorum* (3%), *Atopobium rimae* (2%), *Streptococcus cristatus* (2%), *Rothia dentocariosa* (2%), *Peptostreptococcus stomatis* (2%), *Fusobacterium nucleatum* (1%), *Actinomyces naeslundii* (1%), *Fusobacterium periodonticum* (1%), *Porphyromonas gingivalis* (0%), *Rothia aeria* (0%), *Porphyromonas endodontalis* (0%). Percentages are how many reference adults carry each.

## What your reading means typical

This group is 0.08% of your classified reads, counted over 3 of the 33 species this group is defined by. That places you at the 58th percentile among reference adults who have the group. A group reading is the sum of its species, so it moves when they do.

## How can I improve this?

Lower is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Streptococcus salivarius* (0.05% of your community); *Streptococcus parasanguinis* (0.01% of your community); *Rothia mucilaginosa* (0.01% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for each species is on its own page, and the members not detected are listed above: for a low reading those absentees are the gap, and they are the specific thing any change would have to supply.

## The research behind it

Bacteria that normally live in your mouth, measured in stool. Ordinarily stomach acid and bile stop them; when they arrive in quantity it usually means one of those barriers is reduced. Acid-suppressing medication is the most common everyday cause by far.

Proton pump inhibitors raise this group substantially and are extremely common. Before reading anything else into a high value, check whether the person takes one.

Periodontal disease increases the oral load being swallowed, which raises this group without any change in gut barrier function.

# Commonly beneficial organisms

‹ at a glance

**Health-associated core species** ▼

22.27% of classified reads · **+64%** vs typical · 20 of 25 species detected



**91st** HIGH

The organisms that healthy people consistently have more of. Most ferment fibre into short-chain fatty acids, most are oxygen-intolerant, and most are lost first when the gut is disturbed. This reading tracks the "core" of a well-functioning adult community.

"Health-associated" is a statistical description of who carries more, not a mechanism, and the set was derived from mostly Western industrialised cohorts. Some members (Akkermansia in particular) reverse direction in specific diseases, which is why disease profiles score species individually while this group provides the overview.

## What is driving this reading

20 of 25 species detected; the reading is their sum

| ORGANISM | CLASS | SHARE OF GROUP | IN YOUR SAMPLE | YOUR LEVEL |
|---|---|---|---|---|
| ***Faecalibacterium prausnitzii*** | ● Beneficial | 28% | 6.18% | **+82%** vs typical carrier<br>**80th** percentile<br>carried by 98% of adults |
| ***Anaerostipes hadrus*** | ● Beneficial | 20% | 4.53% | **×23** vs typical carrier<br>**99th** percentile<br>carried by 87% of adults |
| ***Ruminococcus bromii*** | ● Beneficial | 15% | 3.40% | **–34%** vs typical carrier<br>**42nd** percentile<br>carried by 12% of adults |
| ***Coprococcus eutactus*** | ● Beneficial | 11% | 2.46% | **×11** vs typical carrier<br>**89th** percentile<br>carried by 36% of adults |
| ***Mediterraneibacter lactaris*** | ● Beneficial | 5% | 1.13% | **+480%** vs typical carrier<br>**95th** percentile<br>carried by 47% of adults |
| ***Alistipes putredinis*** | ● Conditional | 4% | 0.799% | **–50%** vs typical carrier<br>**33rd** percentile<br>carried by 75% of adults |
| ***Eubacterium siraeum*** | ● Beneficial | 3% | 0.617% | **+207%** vs typical carrier<br>**64th** percentile<br>carried by 56% of adults |
| ***Coprococcus catus*** | ● Beneficial | 2% | 0.534% | **×11** vs typical carrier<br>**99th** percentile<br>carried by 69% of adults |
| ***Agathobacter rectalis*** | ● Beneficial | 2% | 0.518% | **–50%** vs typical carrier<br>**37th** percentile<br>carried by 88% of adults |
| ***Roseburia hominis*** | ● Beneficial | 2% | 0.491% | **+438%** vs typical carrier<br>**90th** percentile<br>carried by 83% of adults |
| ***Lachnospira eligens*** | ● Beneficial | 2% | 0.407% | **97th** percentile<br>carried by 81% of adults |
| ***Anaerobutyricum hallii*** | ● Beneficial | 1% | 0.313% | **+253%** vs typical carrier<br>**80th** percentile<br>carried by 86% of adults |
| ***Bacteroides cellulosilyticus*** | ● Beneficial | 1% | 0.282% | **+187%** vs typical carrier<br>**67th** percentile<br>carried by 49% of adults |
| ***Odoribacter splanchnicus*** | ● Beneficial | 1% | 0.177% | **+7%** vs typical carrier<br>**52nd** percentile<br>carried by 74% of adults |
| ***Roseburia intestinalis*** | ● Beneficial | 1% | 0.162% | **–10%** vs typical carrier<br>**48th** percentile<br>carried by 73% of adults |
| ***Coprococcus comes*** | ● Beneficial | 0% | 0.091% | **94th** percentile<br>carried by 79% of adults |
| ***Alistipes shahii*** | ● Conditional | 0% | 0.076% | **–91%** vs typical carrier<br>**11th** percentile<br>carried by 72% of adults |

| | | | | |
|---|---|---|---|---|
| ***Dysosmobacter segnis*** | Conditional | 0% | 0.073% | –**83%** vs typical carrier<br>**14th** percentile<br>carried by 73% of adults |
| ***Akkermansia muciniphila*** | Beneficial | 0% | 0.024% | **24th** percentile<br>carried by 51% of adults |
| ***Roseburia inulinivorans***<br>full page → | Beneficial | 0% | 0.004% | –**94%** vs typical carrier<br>**9th** percentile<br>carried by 85% of adults |

**Why.** This reading is carried by *Faecalibacterium prausnitzii*, which alone is **28%** of the group's reads and sits at the **80th** percentile among adults who carry it. *Anaerostipes hadrus* contributes another **20%**. Changing this reading means changing them.

**Not detected (5 of 25):** *Bifidobacterium longum* (67%), *Eubacterium ventriosum* (52%), *Bifidobacterium adolescentis* (51%), *Butyrivibrio crossotus* (18%), *Lachnospiraceae bacterium 2 1 46FAA* (1%). Percentages are how many reference adults carry each.

## What your reading means high

This group is 22.27% of your classified reads, counted over 20 of the 25 species this group is defined by. That places you at the 91st percentile among reference adults who have the group. A group reading is the sum of its species, so it moves when they do.

## How can I improve this?

Higher is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Faecalibacterium prausnitzii* (6.18% of your community); *Anaerostipes hadrus* (4.53% of your community); *Ruminococcus bromii* (3.40% of your community); *Coprococcus eutactus* (2.46% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for each species is on its own page, and the members not detected are listed above: for a low reading those absentees are the gap, and they are the specific thing any change would have to supply.

## The research behind it

The organisms that healthy people consistently have more of. Most ferment fibre into short-chain fatty acids, most are oxygen-intolerant, and most are lost first when the gut is disturbed. This reading tracks the "core" of a well-functioning adult community.

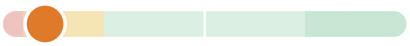
"Health-associated" is a statistical description of who carries more, not a mechanism, and the set was derived from mostly Western industrialised cohorts. Some members (Akkermansia in particular) reverse direction in specific diseases, which is why disease profiles score species individually while this group provides the overview.

This group overlaps heavily with the butyrate-producer group by design: fibre fermenters are health-associated because fibre fermentation is what a healthy colon does.

‹ at a glance

**Resident bifidobacteria** ▼

0.03% of classified reads · 1 of 5 species detected



10th LOW

The bifidobacteria that actually live in an adult colon, rather than the ones passing through from food or a supplement. They are consistently more abundant in healthy adults, and they feed the butyrate producers that line your gut wall. More of these is generally the direction you want.

A low reading is common and is not a diagnosis. Bifidobacterial abundance falls with age, with low-fibre eating and after antibiotics, and a single stool sample cannot separate those causes.

## What is driving this reading 1 of 5 species detected; the reading is their sum

| ORGANISM | CLASS | SHARE OF GROUP | IN YOUR SAMPLE | YOUR LEVEL |
|---|---|---|---|---|
| ***Bifidobacterium catenulatum*** | ● Beneficial | 100% | 0.028% | **45th** percentile carried by 10% of adults |

**Why.** This reading is low because **2** species most reference adults carry were not detected: *Bifidobacterium longum* (67%), *Bifidobacterium adolescentis* (51%). What is present, led by *Bifidobacterium catenulatum*, is not enough to make up the sum.

**Not detected (4 of 5):** *Bifidobacterium longum* (67%), *Bifidobacterium adolescentis* (51%), *Bifidobacterium pseudocatenulatum* (36%), *Bifidobacterium bifidum* (22%). Percentages are how many reference adults carry each.

## What your reading means low

This group is 0.03% of your classified reads, counted over 1 of the 5 species this group is defined by. That places you at the 10th percentile among reference adults who have the group. A group reading is the sum of its species, so it moves when they do.

## How can I improve this?

Higher is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Bifidobacterium catenulatum* (0.03% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for each species is on its own page, and the members not detected are listed above: for a low reading those absentees are the gap, and they are the specific thing any change would have to supply.

## The research behind it

The bifidobacteria that actually live in an adult colon, rather than the ones passing through from food or a supplement. They are consistently more abundant in healthy adults, and they feed the butyrate producers that line your gut wall. More of these is generally the direction you want.

A low reading is common and is not a diagnosis. Bifidobacterial abundance falls with age, with low-fibre eating and after antibiotics, and a single stool sample cannot separate those causes.

Bifidobacterium animalis and Bifidobacterium breve are deliberately not counted here. Both are predominantly supplement and fermented-dairy organisms in adults, so they are grouped with the other transient probiotic species where a high reading is read as intake rather than as colonisation.

‹ at a glance

### Supplement and fermented-dairy species

0.03% of classified reads · **+45%** vs typical · 1 of 14 species detected · 2 in the literature not nameable here



56th TYPICAL

The bacteria found in yoghurt, kefir and probiotic capsules. These are mostly transient in the adult gut, so a high reading usually means you are currently eating or taking them rather than that they have settled in — and it can also mean the opposite of good news, because the same organisms rise when stomach acid is low or when bacteria overgrow the small intestine. A high or low reading is not a health verdict on its own, which is why this scale is drawn in neutral grey.

A high Lactobacillus reading in an adult is most often diet or a supplement passing through; colonisation is uncommon. It is also raised in small-intestinal bacterial overgrowth and in people with low stomach acid (proton pump inhibitor use), where oral and upper-gut organisms reach the stool in larger numbers.

## What is driving this reading

1 of 14 species detected; the reading is their sum

| ORGANISM | CLASS | SHARE OF GROUP | IN YOUR SAMPLE | YOUR LEVEL |
|---|---|---|---|---|
| ***Streptococcus thermophilus*** | ● Conditional | **100%** | 0.035% | **+136%** vs typical carrier<br>**67th** percentile<br>carried by 36% of adults |

**Why.** Led by *Streptococcus thermophilus* at **100%** of the group, the 67th percentile among carriers. The group as a whole sits within the reference range.

**Not detected (13 of 14):** *Lactobacillus ruminis* (13%), *Bifidobacterium dentium* (12%), *Bifidobacterium animalis* (6%), *Lactobacillus delbrueckii* (5%), *Bifidobacterium breve* (5%), *Lactobacillus salivarius* (5%), *Lactobacillus fermentum* (3%), *Lactobacillus acidophilus* (3%), *Lactobacillus rhamnosus* (2%), *Lactobacillus plantarum* (2%), *Lactobacillus gasseri* (1%), *Lactobacillus reuteri* (1%), *Lactobacillus johnsonii* (0%). Percentages are how many reference adults carry each.

> **Two measurements agree**
>
> This agrees with the bile-salt deconjugation capacity reading (95th percentile), which measures the same function by counting the pathway's genes across the whole community instead of these species.

## What your reading means

typical

This group is 0.03% of your classified reads, counted over 1 of the 14 species this group is defined by. That places you at the 56th percentile among reference adults who have the group. A group reading is the sum of its species, so it moves when they do.

## How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

This reading follows the organisms carrying it, and the ones found in you are *Streptococcus thermophilus* (0.03% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for each species is on its own page, and the members not detected are listed above: for a low reading those absentees are the gap, and they are the specific thing any change would have to supply.

## The research behind it

The bacteria found in yoghurt, kefir and probiotic capsules. These are mostly transient in the adult gut, so a high reading usually means you are currently eating or taking them rather than that they have settled in — and it can also mean the opposite of good news, because the same organisms rise when stomach acid is low or when bacteria overgrow the small intestine. A high or low reading is not a health verdict on its own, which is why this scale is drawn in neutral grey.

A high Lactobacillus reading in an adult is most often diet or a supplement passing through; colonisation is uncommon. It is also raised in small-intestinal bacterial overgrowth and in people with low stomach acid (proton pump inhibitor use), where oral and upper-gut organisms reach the stool in larger numbers.

Bifidobacterium dentium is an oral species and is counted here for completeness; it is not a probiotic and is also a member of the oral-origin group.

## What the research says about changing this

**Probiotic-type species — stool abundance reflects transit as much as colonisation** LANE D HUMAN · NON-RANDOMISED

Fired by: probiotics relative abundance: somewhat high

Lactobacilli and bifidobacteria in stool largely reflect what passed through (fermented foods, supplements) rather than residents. Colonisation is person- and strain-specific, probiotic effects are strain- and indication-specific, and after antibiotics a probiotic slowed native recovery.

- An 11-strain probiotic colonised the gut mucosa in a person-, strain- and region-specific way; stool abundance of probiotic species reflects transit as much as colonisation. [Zmora N et al. Cell 2018;174:1388] · lane D · human intervention with endoscopic sampling · mixed
- **Against or limiting —** After antibiotics, a multistrain probiotic delayed recovery of the native microbiome relative to doing nothing, while autologous FMT restored it fastest. [Suez J et al. Cell 2018;174:1406] · lane X · human intervention after antibiotics · harm
- **Against or limiting —** AGA 2020 recommends probiotics for Crohn's disease, ulcerative colitis and IBS only in the context of a clinical trial, and finds the evidence formulation- and indication-specific. [Su GL et al. AGA Clinical Practice Guideline on probiotics. Gastroenterology 2020;159:697] · lane X · guideline

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “you need a probiotic” · “probiotic deficiency”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

# Opportunistic organisms

‹ at a glance

**Opportunistic pathogens and pathobionts ▲**

0.00% of classified reads · 1 of 20 species detected · 2 in the literature not nameable here



3rd NOTABLY LOW

Bacteria that most healthy adults carry at trace levels or not at all, and which take over when the normal community is disrupted — after antibiotics, during illness, or in a low-diversity gut. A high reading is a general sign of disturbance, not a diagnosis of any specific infection.

Escherichia coli is the largest term in this sum and is itself a mix of harmless commensal strains and pathogenic ones; species-level profiling does not separate them. Read a high E. coli-driven total as "Enterobacteriaceae expansion", which is meaningful on its own.

### What is driving this reading 0 of 20 species detected; the reading is their sum

**Why.** This reading is low because **2** species most reference adults carry were not detected: *Escherichia coli* (62%), *Bilophila wadsworthia* (61%).

**Not detected (20 of 20):** *Escherichia coli* (62%), *Bilophila wadsworthia* (61%), *Ruminococcus gnavus* (47%), *Haemophilus parainfluenzae* (41%), *Klebsiella pneumoniae* (21%), *Klebsiella variicola* (18%), *Streptococcus anginosus group* (9%), *Enterobacter cloacae complex* (7%), *Fusobacterium mortiferum* (5%), *Enterococcus faecalis* (5%), *Enterococcus faecium* (5%), *Clostridium perfringens* (4%), *Citrobacter freundii* (4%), *Fusobacterium varium* (3%), *Klebsiella oxytoca* (3%), *Fusobacterium nucleatum* (1%), *Pseudomonas aeruginosa group* (1%), *Proteus mirabilis* (1%), *Clostridioides difficile* (0%), *Staphylococcus aureus* (0%). Percentages are how many reference adults carry each.

### What your reading means notably low

This group is 0.00% of your classified reads, counted over 1 of the 20 species this group is defined by. That places you at the 3rd percentile among reference adults who have the group. A group reading is the sum of its species, so it moves when they do.

### How can I improve this?

Lower is the more favourable direction for this reading.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for each species is on its own page, and the members not detected are listed above: for a low reading those absentees are the gap, and they are the specific thing any change would have to supply.

### The research behind it

Bacteria that most healthy adults carry at trace levels or not at all, and which take over when the normal community is disrupted — after antibiotics, during illness, or in a low-diversity gut. A high reading is a general sign of disturbance, not a diagnosis of any specific infection.

Escherichia coli is the largest term in this sum and is itself a mix of harmless commensal strains and pathogenic ones; species-level profiling does not separate them. Read a high E. coli-driven total as "Enterobacteriaceae expansion", which is meaningful on its own.

Detection of Clostridioides difficile in a stool sample is not the same as C. difficile infection. Asymptomatic carriage runs around 5-15% of healthy adults and is higher in infants.

# Cell-surface characteristics

‹ at a glance

### Hexa-acylated LPS carriers ▲

0.00% of classified reads · 1 of 17 species detected · 1 in the literature not nameable here



7th LOW

Bacteria whose cell-surface lipopolysaccharide strongly activates the innate immune receptor TLR4. Not all Gram-negative bacteria do this — the Bacteroides that dominate most healthy guts carry a weakly-activating variant, which is why counting them all together would be misleading.

Carriage of hexa-acylated LPS is normal. Enterobacteriaceae are ordinary members of a healthy gut community at low abundance; the concern in the literature is expansion, not presence.

## What is driving this reading 0 of 17 species detected; the reading is their sum

**Why.** This reading is low because **2** species most reference adults carry were not detected: *Escherichia coli* (62%), *Bilophila wadsworthia* (61%).

**Not detected (17 of 17):** *Escherichia coli* (62%), *Bilophila wadsworthia* (61%), *Klebsiella pneumoniae* (21%), *Desulfovibrio piger* (20%), *Klebsiella variicola* (18%), *Enterobacter cloacae complex* (7%), *Citrobacter freundii* (4%), *Fusobacterium varium* (3%), *Klebsiella aerogenes* (3%), *Klebsiella oxytoca* (3%), *Fusobacterium nucleatum* (1%), *Citrobacter koseri* (1%), *Escherichia fergusonii* (1%), *Morganella morganii* (1%), *Proteus mirabilis* (1%), *Escherichia albertii* (0%), *Salmonella enterica* (0%). Percentages are how many reference adults carry each.

## What your reading means low

This group is 0.00% of your classified reads, counted over 1 of the 17 species this group is defined by. That places you at the 7th percentile among reference adults who have the group. A group reading is the sum of its species, so it moves when they do.

## How can I improve this?

Lower is the more favourable direction for this reading.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for each species is on its own page, and the members not detected are listed above: for a low reading those absentees are the gap, and they are the specific thing any change would have to supply.

## The research behind it

Bacteria whose cell-surface lipopolysaccharide strongly activates the innate immune receptor TLR4. Not all Gram-negative bacteria do this — the Bacteroides that dominate most healthy guts carry a weakly-activating variant, which is why counting them all together would be misleading.

Carriage of hexa-acylated LPS is normal. Enterobacteriaceae are ordinary members of a healthy gut community at low abundance; the concern in the literature is expansion, not presence.

This is not a blood endotoxin measurement. Whether LPS reaches the circulation depends on gut barrier integrity, which is not measured here and for which no validated stool test exists.

# 16 Organisms that need attention, one by one

One page per organism flagged in Part A. Each says what the organism is and does in a gut, why it is flagged in your sample, what has been shown to move it in the direction you would want — with the strength of that evidence stated on every line — and what competes with it. Laboratory and animal findings are listed and labelled as such; they are leads, not recommendations. Nothing here is a prescription.

| Human trial | Human cohorts | Animal model | Laboratory culture |
|---|---|---|---|
| given to people; the organism measured before and after | associations in people; nothing was given | a mouse or rat gut, not a human one | a culture dish; no gut at all |

## OVERGROWN — at a level that is a concern

‹ at a glance
what you can do about it ›

***Enterocloster GWHFCPF00000000*** ● **Opportunist** · genus-level knowledge — OVERGROWN

### What it is and what it does

A group of bile-tolerant Lachnospiraceae, formerly filed under Clostridium. They are ordinary low-level residents that thrive on bile acids and are expanded in autism-spectrum, IBD and post-antibiotic cohorts. E. bolteae in particular has been isolated from bloodstream and abdominal infections.

### Why it is flagged in your sample

| | |
|---|---|
| YOUR LEVEL | **no share of its own** — its reads are counted within *Clostridium sp AM22 11AC* in the composition; the whole-genome lane reads it at **0.278%** of its own scale |
| POSITION | **96th** percentile among reference adults who carry it, on that lane — **+116%** against their typical level of 0.128% |
| HOW COMMON | carried by **24%** of reference adults |
| CLASS | Opportunist — Normal at low levels. |
| DETECTED BY | GlobDB whole-genome sketch |
| IDENTIFIERS | GlobDB GWHFCPF00000000 |
| DETECTION | **supported** — competitive confirmation: 2,456 uniquely mapping fragments over 37.9% of the genome at 99.2% identity, spread evenly (evenness 0.72), with 1 relatives competing |

**What the level means.** Bile tolerance means a high-fat diet feeds them; expansion is a common signal of a fat-rich, fibre-poor pattern and of a recently disturbed community.

### What has been shown to lower it 3 items, strongest evidence first

| WHAT | KIND | EVIDENCE | NOTE |
|---|---|---|---|
| **Dietary fibre (inulin, pectin)** | prebiotic fibre | human trial | Fibre shifts the Lachnospiraceae balance away from bile-tolerant members toward butyrate producers. So D et al. Am J Clin Nutr 2018;107:965 (meta-analysis) |
| **Lower saturated-fat intake** | diet pattern | **human cohorts** | Bile-acid load falls with dietary fat, and with it the bile-tolerant genera. David LA et al. Nature 2014;505:559 |
| **Bifidobacterium longum strains** | probiotic | animal model | Wang Y et al. Front Microbiol 2020 (Clostridium cluster XIVa modulation) |

**Competes with:** *Faecalibacterium prausnitzii*, *Roseburia intestinalis*. Organisms that occupy the same niche or make compounds that inhibit it; raising them is an indirect way of lowering this one.

Known to push it the other way: High-fat, animal-based diet (human trial).

‹ at a glance
what you can do about it ›

### *Blautia_A luti* ● Conditional · genus-level knowledge

OVERGROWN

#### What it is and what it does

One of the most abundant gut genera. Blautia species produce acetate and, in some cases, butyrate, and B. producta is a documented coloniser that clears vancomycin-resistant Enterococcus. The genus is reported protective against visceral fat in Japanese cohorts, yet expanded in obesity, type 2 diabetes and IBS in Western ones. It is a marker of a Western, starch-rich diet more than of any single disease.

#### Why it is flagged in your sample

| | |
|---|---|
| YOUR LEVEL | **no share of its own** — its reads are counted within *Blautia A wexlerae* in the composition; the whole-genome lane reads it at **0.315%** of its own scale |
| POSITION | **>99th** percentile among reference adults who carry it, on that lane — **+99%** against their typical level of 0.159% |
| HOW COMMON | carried by **13%** of reference adults |
| CLASS | Conditional — Cuts both ways: part of a healthy gut at typical levels, associated with problems when expanded or when the community around it is disturbed. |
| DETECTED BY | GTDB whole-genome sketch, GlobDB whole-genome sketch, Kraken2/UHGG, mOTUs 4 |
| IDENTIFIERS | GCF_033096465, mOTU 003244, UHGG species 431; GTDB R232 Blautia_A luti |
| DETECTION | **supported** — competitive confirmation: 33,256 uniquely mapping fragments over 65.6% of the genome at 99.1% identity, spread evenly (evenness 0.72), with 25 relatives competing |

**What the level means.** High Blautia is common and mostly benign; it becomes a concern when it expands at the expense of Faecalibacterium and Roseburia, which signals a low-fibre, high-starch pattern.

#### What has been shown to lower it 1 item, strongest evidence first

| WHAT | KIND | EVIDENCE | NOTE |
|---|---|---|---|
| **Increased dietary fibre diversity (vegetables, legumes over refined starch)** | diet pattern | human trial | Relative abundance falls as fibre-fed competitors expand. Vandeputte D et al. Gut 2017 |

**Competes with:** *Faecalibacterium prausnitzii*, *Roseburia intestinalis*. Organisms that occupy the same niche or make compounds that inhibit it; raising them is an indirect way of lowering this one.

Known to push it the other way: Refined carbohydrate intake (human cohorts).

## DEPLETED — well below the reference

‹ at a glance
what you can do about it ›

### *GGB74412 SGB5075* ● Beneficial · genus-level knowledge

LOW

#### What it is and what it does

Pectin degraders producing acetate and the precursors of butyrate; associated with plant-rich diets and lower in some asthma and IBD cohorts.

#### Why it is flagged in your sample

| | |
|---|---|
| YOUR LEVEL | **0.025%** of the community |
| POSITION | **2nd** percentile among reference adults who carry it — –**90%** against their typical level of 0.494% |
| HOW COMMON | carried by **47%** of reference adults |
| CLASS | Beneficial — Higher levels are associated with better outcomes in human studies. |
| DETECTED BY | GTDB whole-genome sketch, GlobDB whole-genome sketch, MetaPhlAn 4.2 (Jan26), Kraken2/UHGG, mOTUs 4, rescue |
| IDENTIFIERS | SGB5075, GCF_040096455, mOTU 000761, UHGG species 245; GTDB R232 Lachnospira rogosae |
| DETECTION | **supported** — seen by 4 independent methods: genome_sketch, marker, read_classification, universal_marker |

**What the level means.** No intervention evidence is on record for this organism; the flag rests on its percentile alone and is listed to watch, not to act on.

#### What has been shown to raise it 0 items, strongest evidence first

Nothing in the literature we hold has been shown to raise this organism specifically. The general levers — fibre diversity, less processed food, avoiding unnecessary antibiotics — shift the whole community and are the honest fallback.

‹ at a glance
what you can do about it ›

## *Faecalibacterium intestinale* ● Beneficial · genus-level knowledge

LOW

### What it is and what it does

The archetypal beneficial gut bacterium: one of the largest butyrate producers, with anti-inflammatory effects shown in cells and animals. Reduced in Crohn's disease more consistently than almost any other organism, and lower across many inflammatory and metabolic conditions.

### Why it is flagged in your sample

| | |
|---|---|
| YOUR LEVEL | **no share of its own** — its reads are counted within *Faecalibacterium prausnitzii* in the composition; the whole-genome lane reads it at **0.244%** of its own scale |
| POSITION | **10th** percentile among reference adults who carry it, on that lane — –**65%** against their typical level of 0.692% |
| HOW COMMON | carried by **21%** of reference adults |
| CLASS | Beneficial — Higher levels are associated with better outcomes in human studies. |
| DETECTED BY | GTDB whole-genome sketch, GlobDB whole-genome sketch, Kraken2/UHGG, mOTUs 4, rescue |
| IDENTIFIERS | GCF_040096575, mOTU 000042, UHGG species 3636, panel BackhedF_2015_ERR525896_bin.27; GTDB R232 Faecalibacterium intestinale |
| DETECTION | **supported** — competitive confirmation: 7,205 uniquely mapping fragments over 59.0% of the genome at 98.8% identity, spread evenly (evenness 0.67), with 19 relatives competing |

**What the level means.** No intervention evidence is on record for this organism; the flag rests on its percentile alone and is listed to watch, not to act on.

### What has been shown to raise it 0 items, strongest evidence first

Nothing in the literature we hold has been shown to raise this organism specifically. The general levers — fibre diversity, less processed food, avoiding unnecessary antibiotics — shift the whole community and are the honest fallback.

‹ at a glance
what you can do about it ›

## *Wujia chipingensis* ● Beneficial · genus-level knowledge

LOW

### What it is and what it does

Butyrate producers repeatedly associated with better quality-of-life measures and lower in depression cohorts; C. catus and C. eutactus especially.

### Why it is flagged in your sample

| | |
|---|---|
| YOUR LEVEL | **0.086%** of the community |
| POSITION | **4th** percentile among reference adults who carry it — –**89%** against their typical level of 1.01% |
| HOW COMMON | carried by **22%** of reference adults |
| CLASS | Beneficial — Higher levels are associated with better outcomes in human studies. |
| DETECTED BY | MetaPhlAn 4 (Jun23), GTDB whole-genome sketch, GlobDB whole-genome sketch, MetaPhlAn 4.2 (Jan26), Kraken2/UHGG, mOTUs 4 |
| IDENTIFIERS | SGB5111, GCF_020687505, mOTU 000368, UHGG species 439; GTDB R232 Coprococcus_C hominis |
| DETECTION | **supported** — seen by 4 independent methods: genome_sketch, marker, read_classification, universal_marker |

**What the level means.** No intervention evidence is on record for this organism; the flag rests on its percentile alone and is listed to watch, not to act on.

### What has been shown to raise it 0 items, strongest evidence first

Nothing in the literature we hold has been shown to raise this organism specifically. The general levers — fibre diversity, less processed food, avoiding unnecessary antibiotics — shift the whole community and are the honest fallback.

‹ at a glance
what you can do about it ›

## *Faecalibacterium butyricigenerans* ● Beneficial · genus-level knowledge

LOW

### What it is and what it does

The archetypal beneficial gut bacterium: one of the largest butyrate producers, with anti-inflammatory effects shown in cells and animals. Reduced in Crohn's disease more consistently than almost any other organism, and lower across many inflammatory and metabolic conditions.

### Why it is flagged in your sample

| | |
|---|---|
| YOUR LEVEL | **0.002%** of the community |
| POSITION | **<1st** percentile among reference adults who carry it — –**89%** against their typical level of 0.664% |
| HOW COMMON | carried by **31%** of reference adults |
| CLASS | Beneficial — Higher levels are associated with better outcomes in human studies. |
| DETECTED BY | MetaPhlAn 4 (Jun23), GlobDB whole-genome sketch, MetaPhlAn 4.2 (Jan26), Kraken2/UHGG, mOTUs 4, rescue |
| IDENTIFIERS | SGB15315, GCF_023347275, mOTU 000140, UHGG species 2969; GTDB R232 Faecalibacterium butyricigenerans |
| DETECTION | **supported** — seen by 4 independent methods: genome_sketch, marker, read_classification, universal_marker |
| REFERENCE STRAINS | 1 in BacDive: culture collection DSM 103434; grows at 37 °C. Traits of reference strains, not measured in your sample. |

**What the level means.** No intervention evidence is on record for this organism; the flag rests on its percentile alone and is listed to watch, not to act on.

### What has been shown to raise it 0 items, strongest evidence first

Nothing in the literature we hold has been shown to raise this organism specifically. The general levers — fibre diversity, less processed food, avoiding unnecessary antibiotics — shift the whole community and are the honest fallback.

‹ at a glance
what you can do about it ›

## *Roseburia inulinivorans* ● Beneficial · genus-level knowledge

LOW

### What it is and what it does

Flagellated butyrate producers that thrive on dietary fibre, especially beta-glucans and inulin. Lower levels are reported in ulcerative colitis, type 2 diabetes and atherosclerosis cohorts. R. intestinalis has direct anti-inflammatory effects on the gut lining in cell and mouse work.

### Why it is flagged in your sample

| | |
|---|---|
| YOUR LEVEL | **0.004%** of the community |
| POSITION | **9th** percentile among reference adults who carry it — –**94%** against their typical level of 0.585% |
| HOW COMMON | carried by **85%** of reference adults |
| CLASS | Beneficial — Higher levels are associated with better outcomes in human studies. |
| DETECTED BY | MetaPhlAn 4 (Jun23), MetaPhlAn 4.2 (Jan26), Kraken2/UHGG, mOTUs 4, rescue, MetaPhlAn 3 |
| IDENTIFIERS | scoring Roseburia_inulinivorans, SGB4940, mOTU 000002, UHGG species 553, GCF_000174195; GTDB R232 Roseburia inulinivorans |
| DETECTION | **supported** — seen by 3 independent methods: marker, read_classification, universal_marker |
| REFERENCE STRAINS | 3 in BacDive: culture collection DSM 108070, DSM 16841; anaerobe; grows at 37 °C; isolated from human, feces (stool). Traits of reference strains, not measured in your sample. |

**What the level means.** Missing or low Roseburia is a fibre-intake signal as much as a health one: these are among the first organisms to fall on a low-fibre diet and the first to recover on a high-fibre one.

### What has been shown to raise it 4 items, strongest evidence first

| WHAT | KIND | EVIDENCE | NOTE |
|---|---|---|---|
| **Oat and barley beta-glucan** | prebiotic fibre | **human trial** | Martinez I et al. ISME J 2013;7:269 |
| **Inulin and fructo-oligosaccharides** | prebiotic fibre | **human trial** | Vandeputte D et al. Gut 2017 |
| **Whole grains** | food | **human trial** | Vanegas SM et al. Am J Clin Nutr 2017;105:635 |
| **Resistant starch** | prebiotic fibre | **human trial** | Walker AW et al. ISME J 2011 |

Known to push it the other way: Low-fibre diet, antibiotics (human cohorts).

## WORTH WATCHING — above the reference range, or uncommon to carry

‹ at a glance
what you can do about it ›

### *Bacteroides coagulans* ● Conditional · genus-level knowledge

HIGH

### What it is and what it does

The dominant genus of a Western-diet gut. Bacteroides species carry vast enzyme repertoires for breaking down complex plant fibres and host glycans and produce propionate and acetate. B. uniformis, B. ovatus and B. xylanisolvens are among the most versatile fibre degraders known. A high Bacteroides share is normal in Europe and North America and unusual in fibre-rich agrarian populations, where Prevotella dominates instead.

### Why it is flagged in your sample

| | |
|---|---|
| YOUR LEVEL | **no share of its own** — its reads are counted within *Bacteroides clarus* in the composition; the whole-genome lane reads it at **0.042%** of its own scale |
| POSITION | **91st** percentile among reference adults who carry it, on that lane — **+279%** against their typical level of 0.011% |
| HOW COMMON | carried by **1%** of reference adults |
| CLASS | Conditional — Cuts both ways: part of a healthy gut at typical levels, associated with problems when expanded or when the community around it is disturbed. |
| DETECTED BY | MetaPhlAn 3 |
| IDENTIFIERS | scoring Bacteroides_coagulans |
| DETECTION | **provisional** — one method (scoring) at 0.02; presence rests on competitive confirmation; confirmation pending: no reference genome identifier for this organism |
| REFERENCE STRAINS | 1 in BacDive: culture collection DSM 20705; anaerobe; grows at 37 °C; isolated from gastrointestinal tract, urogenital tract. Traits of reference strains, not measured in your sample. |

**What the level means.** Expansion is a dietary signal (animal protein and fat) more than a health one. It matters when it comes with depletion of butyrate producers.

### What has been shown to lower it 1 item, strongest evidence first

| WHAT | KIND | EVIDENCE | NOTE |
|---|---|---|---|
| **Whole grains, legumes and resistant starch** | prebiotic fibre | human trial | Shifts the Bacteroides-to-Firmicutes balance toward fermenters. Walker AW et al. ISME J 2011; David LA et al. Nature 2014 |

**Competes with:** *Prevotella copri*, *Faecalibacterium prausnitzii*. Organisms that occupy the same niche or make compounds that inhibit it; raising them is an indirect way of lowering this one.

Known to push it the other way: Animal-protein, high-fat diet (human trial).

‹ at a glance
what you can do about it ›

## *Dorea formicigenerans* ● Conditional · genus-level knowledge

HIGH

### What it is and what it does

Gas producers (hydrogen, carbon dioxide) that break down mucus-associated sugars. Expanded in IBS and some multiple sclerosis cohorts, lower in others; D. longicatena is common in healthy adults. Contributes to bloating when abundant.

### Why it is flagged in your sample

| | |
|---|---|
| YOUR LEVEL | **0.582%** of the community |
| POSITION | **96th** percentile among reference adults who carry it — **+491%** against their typical level of 0.212% |
| HOW COMMON | carried by **85%** of reference adults |
| CLASS | Conditional — Cuts both ways: part of a healthy gut at typical levels, associated with problems when expanded or when the community around it is disturbed. |
| DETECTED BY | MetaPhlAn 4 (Jun23), GTDB whole-genome sketch, MetaPhlAn 4.2 (Jan26), Kraken2/UHGG, mOTUs 4, rescue, MetaPhlAn 3, SingleM |
| IDENTIFIERS | scoring Dorea_formicigenerans, SGB4575, mOTU 000271, UHGG species 377, GCF_025150245.1; GTDB R232 Dorea formicigenerans |
| DETECTION | **supported** — seen by 5 independent methods: genome_sketch, marker, marker_window, read_classification, universal_marker |
| REFERENCE STRAINS | 4 in BacDive: culture collection DSM 105840, DSM 118841, DSM 3992; anaerobe; grows at 37 °C; isolated from feces (stool), suidae (pig,swine). Traits of reference strains, not measured in your sample. |
| STRAIN PLACEMENT | nearest reference genome GCF_025150245.1.fna, distance 0.307834; polymorphic sites at 0.22%; a consensus can hide co-resident strains. A placement, not proof of an identical strain. |

**What the level means.** High Dorea with bloating is a plausible link; missing Dorea is usually unremarkable.

### What has been shown to lower it 1 item, strongest evidence first

| WHAT | KIND | EVIDENCE | NOTE |
|---|---|---|---|
| **Low-FODMAP diet (for gas symptoms)** | diet pattern | human trial | Reduces fermentable substrate for gas producers; clinically supervised. Halmos EP et al. Gut 2015;64:93 |

Known to push it the other way: Fermentable fibre (human trial).

‹ at a glance
what you can do about it ›

## *Blautia_A MGYG000001338* ● Conditional · genus-level knowledge

HIGH

### What it is and what it does

One of the most abundant gut genera. Blautia species produce acetate and, in some cases, butyrate, and B. producta is a documented coloniser that clears vancomycin-resistant Enterococcus. The genus is reported protective against visceral fat in Japanese cohorts, yet expanded in obesity, type 2 diabetes and IBS in Western ones. It is a marker of a Western, starch-rich diet more than of any single disease.

### Why it is flagged in your sample

| | |
|---|---|
| YOUR LEVEL | **no share of its own** — its reads are counted within *Blautia A wexlerae* in the composition; the whole-genome lane reads it at **6.35%** of its own scale |
| POSITION | **90th** percentile among reference adults who carry it, on that lane — **+189%** against their typical level of 2.20% |
| HOW COMMON | carried by **93%** of reference adults |
| CLASS | Conditional — Cuts both ways: part of a healthy gut at typical levels, associated with problems when expanded or when the community around it is disturbed. |
| DETECTED BY | GlobDB whole-genome sketch, Kraken2/UHGG, rescue, SingleM |
| IDENTIFIERS | GlobDB MGYG000001338, UHGG species 2045 |
| DETECTION | **supported** — competitive confirmation: 57,937 uniquely mapping fragments over 81.7% of the genome at 99.2% identity, spread evenly (evenness 0.82), with 25 relatives competing |

**What the level means.** High Blautia is common and mostly benign; it becomes a concern when it expands at the expense of Faecalibacterium and Roseburia, which signals a low-fibre, high-starch pattern.

### What has been shown to lower it 1 item, strongest evidence first

| WHAT | KIND | EVIDENCE | NOTE |
|---|---|---|---|
| **Increased dietary fibre diversity (vegetables, legumes over refined starch)** | diet pattern | human trial | Relative abundance falls as fibre-fed competitors expand. Vandeputte D et al. Gut 2017 |

**Competes with:** *Faecalibacterium prausnitzii*, *Roseburia intestinalis*. Organisms that occupy the same niche or make compounds that inhibit it; raising them is an indirect way of lowering this one.

Known to push it the other way: Refined carbohydrate intake (human cohorts).

## PART B — PATHOGENS, ONE AT A TIME

# 17 Which strain, not just which species

Two people can carry the same species and a different **strain** of it, and the difference often matters more than the species does: whether an *E. coli* is a harmless resident or carries a toxin operon is a strain-level question. This section reports, for each organism with enough coverage, the fingerprint of the population actually living in this sample — read from the marker genes, at single-base resolution.

| ORGANISMS RESOLVED | CALLABLE BASES | DETECTED, UNRESOLVED |
|---|---|---|
| **69** | **5.9M** | **63** |
| population fingerprinted | compared at base level | too little coverage |

## What a fingerprint can and cannot tell you

**It is this sample's own population.** Not a match to a named laboratory strain. Naming one would require comparing the whole genome against that isolate, which marker genes cannot do.

**It is the dominant population.** Where two populations of one species are present, this reports the majority one. A less abundant second population can be hidden by it entirely.

**It enables comparison.** Because the fingerprint is read at single-base resolution, two samples can be compared organism by organism — which is how donor and recipient populations are told apart in the matching report.

**It does not attribute genes.** Resolving an organism's population does not show which organism carries a separately detected toxin or resistance gene. That needs physical linkage.

## Every organism resolved to population level

Ordered by how much sequence was callable, which is how much of the population could actually be characterised. **Markers** is how many marker genes resolved; **breadth** is how much of each was covered; **depth** is the average read depth over them.

| ORGANISM | SGB | MARKERS | CALLABLE BASES | BREADTH | DEPTH |
|---|---|---|---|---|---|
| *Lachnospira eligens* | SGB5082 | 189 | 184,506 | 98.9% | 20.21× |
| *Bacteroides finegoldii* | SGB1862 | 185 | 179,431 | 95.1% | 4.64× |
| *Coprococcus eutactus* | SGB5117 | 199 | 177,963 | 98.2% | 9.88× |
| *Anaerostipes hadrus* | SGB4540 | 196 | 165,842 | 98.8% | 17.46× |
| *Ruminococcus bromii* | SGB4290 | 198 | 160,005 | 98.4% | 12.77× |
| GGB3267 SGB4316 | SGB4316 | 199 | 149,432 | 97.4% | 7.85× |
| *Bacteroides clarus* | SGB1832 | 184 | 147,933 | 95.8% | 5.05× |
| *Ruminococcus lactaris* | SGB4557 | 180 | 145,879 | 94.7% | 4.2× |
| GGB51269 SGB5062 | SGB5062 | 194 | 144,014 | 96.2% | 6.96× |
| *Blautia faecis* | SGB4820 | 165 | 139,454 | 91.7% | 3.44× |
| *Ruminococcus SGB4421* | SGB4421 | 198 | 137,598 | 97.0% | 8.6× |
| GGB6601 SGB9333 | SGB9333 | 162 | 137,297 | 92.8% | 3.66× |
| GGB1495 SGB2071 | SGB2071 | 185 | 137,262 | 93.8% | 3.98× |
| *Parabacteroides distasonis* | SGB1934 | 113 | 135,890 | 91.7% | 2.7× |
| GGB13283 SGB4948 | SGB4948 | 148 | 132,153 | 91.6% | 3.53× |
| GGB9342 SGB14306 | SGB14306 | 187 | 130,657 | 94.7% | 4.79× |
| *Blautia wexlerae* | SGB4837 | 155 | 130,595 | 99.2% | 30.3× |
| GGB3175 SGB4191 | SGB4191 | 181 | 129,132 | 94.1% | 4.78× |

| ORGANISM | SGB | MARKERS | CALLABLE BASES | BREADTH | DEPTH |
|---|---|---|---|---|---|
| *Fusicatenibacter saccharivorans* | SGB4874 | 156 | 128,671 | 92.2% | 3.44× |
| *Faecalibacterium prausnitzii* | SGB15318 | 162 | 120,048 | 96.3% | 9.07× |
| *Alistipes putredinis* | SGB2318 | 142 | 118,206 | 92.4% | 3.14× |
| GGB6612 SGB9346 | SGB9346 | 185 | 117,779 | 97.7% | 9.47× |
| *Gemmiger formicilis* | SGB15300 | 154 | 115,358 | 91.7% | 3.77× |
| *Dorea longicatena* | SGB4581 | 125 | 111,495 | 91.6% | 2.94× |
| *Roseburia sp AM59 24XD* | SGB4654 | 145 | 106,380 | 91.7% | 3.45× |
| *Faecalibacterium SGB15346* | SGB15346 | 142 | 106,379 | 92.3% | 3.62× |
| *Jilunia laotingensis* | SGB1802 | 48 | 102,373 | 84.1% | 1.82× |
| *Faecalibacterium prausnitzii* | SGB15332 | 137 | 101,016 | 94.7% | 6.64× |
| *Dorea formicigenerans* | SGB4575 | 112 | 98,574 | 88.3% | 2.46× |
| *Ruminococcus bicirculans* | SGB4262 | 127 | 96,011 | 91.5% | 3.32× |
| *Coprococcus catus* | SGB4670 | 112 | 92,064 | 88.4% | 2.73× |
| *Eubacterium siraeum* | SGB4198 | 93 | 89,784 | 88.1% | 2.48× |
| *Coprococcus comes* | SGB4577 | 107 | 89,751 | 91.8% | 3.14× |
| *Faecalibacterium prausnitzii* | SGB15317 | 117 | 88,801 | 89.2% | 2.75× |
| *Faecalibacillus intestinalis* | SGB6754 | 111 | 86,916 | 92.2% | 3.37× |
| *Eubacterium ramulus* | SGB4960 | 101 | 86,868 | 90.0% | 2.54× |
| GGB6565 SGB9274 | SGB9274 | 114 | 81,834 | 88.4% | 2.71× |
| *Roseburia faecis* | SGB4925 | 92 | 77,647 | 89.1% | 2.65× |
| *Eubacterium rectale* | SGB4933 | 87 | 77,456 | 89.1% | 2.58× |
| *Anaerobutyricum soehngenii* | SGB4537 | 78 | 72,278 | 88.8% | 2.26× |
| *Phocaeicola massiliensis* | SGB1812 | 85 | 68,676 | 88.6% | 2.41× |
| *Phocaeicola dorei* | SGB1815 | 70 | 64,627 | 97.2% | 7.96× |
| *Roseburia hominis* | SGB4936 | 83 | 63,038 | 87.2% | 2.3× |
| GGB4750 SGB6579 | SGB6579 | 75 | 61,980 | 87.5% | 2.26× |
| *Bacteroides uniformis* | SGB1836 | 53 | 61,428 | 91.5% | 3.14× |
| *Roseburia sp AF02 12* | SGB4938 | 67 | 55,692 | 91.5% | 3.43× |
| GGB3303 SGB4364 | SGB4364 | 73 | 50,984 | 88.2% | 2.5× |
| GGB9635 SGB15106 | SGB15106 | 65 | 50,365 | 87.3% | 2.38× |
| GGB9747 SGB15356 | SGB15356 | 60 | 46,872 | 88.1% | 2.26× |
| *Agathobaculum butyriciproducens* | SGB14993 | 57 | 44,232 | 87.5% | 2.27× |
| *Bacteroides cellulosilyticus* | SGB1844 | 35 | 42,086 | 85.0% | 1.94× |
| GGB3612 SGB4882 | SGB4882 | 56 | 40,436 | 88.7% | 2.3× |
| *Clostridium fessum* | SGB4705 | 45 | 38,592 | 85.5% | 2.03× |
| *Anaerobutyricum hallii* | SGB4532 | 39 | 33,819 | 86.6% | 1.98× |
| *Clostridiaceae bacterium* | SGB4629 | 38 | 33,388 | 84.6% | 1.95× |

| ORGANISM | SGB | MARKERS | CALLABLE BASES | BREADTH | DEPTH |
|---|---|---|---|---|---|
| *Ruminococcus sp AF41 9* | SGB25497 | 31 | 32,629 | 84.9% | 1.9× |
| *Blautia luti* | SGB4832 | 29 | 32,414 | 85.6% | 1.84× |
| GGB4699 SGB6502 | SGB6502 | 39 | 31,948 | 89.3% | 2.15× |
| GGB4711 SGB6524 | SGB6524 | 36 | 31,922 | 89.7% | 2.18× |
| GGB4571 SGB6321 | SGB6321 | 37 | 27,401 | 85.3% | 2.16× |
| *Faecalibacterium prausnitzii* | SGB15316 | 36 | 26,410 | 87.0% | 2.38× |
| *Clostridiales bacterium KLE1615* | SGB5090 | 37 | 25,979 | 86.4% | 2.21× |
| *Faecalibacterium sp HTFF* | SGB15340 | 27 | 21,867 | 85.8% | 1.97× |
| *Dorea longicatena* | SGB4582 | 25 | 20,856 | 85.8% | 2.02× |
| *Clostridium sp AM22 11AC* | SGB4749 | 27 | 19,980 | 84.2% | 1.93× |
| *Alistipes finegoldii* | SGB2301 | 25 | 18,674 | 87.5% | 2.04× |
| GGB4585 SGB6340 | SGB6340 | 22 | 17,114 | 89.5% | 1.91× |
| *Alistipes onderdonkii* | SGB2303 | 21 | 16,125 | 84.9% | 2.16× |
| *Oscillibacter sp ER4* | SGB15254 | 21 | 13,767 | 86.0% | 2.32× |

**63 further organism(s)** were detected but carried too few resolvable marker genes to characterise a population. They are present; their strain is simply unresolved at this sequencing depth, which is a limit of the data rather than a finding about the organism.

This is the dominant population's consensus across marker genes. A second, less abundant population of the same species can be concealed by it, so 'one population' here means 'one detectable dominant population', not proof of a single strain.

Marker genes are a conserved subset of the genome. Agreement across them supports a related population; it does not establish whole-genome identity to a named isolate, and it cannot attribute an unlinked toxin or resistance gene to this organism.

Method: metaphlan 4.1.1 marker consensus (sample2markers) · database mpa_vJun23_CHOCOPhlAnSGB_202403.

## Placed among the reference genomes of its species

For every organism with enough reconstructed marker sequence, its consensus markers were aligned with the marker sequences of member genomes of the same species (GTDB R232) and a tree was inferred. The **nearest reference** is the genome the sample's population sits closest to in that tree and the **distance** is the tree's branch length between them — comparable within one tree, not a genome-wide identity. A placement is not proof of an identical resident strain: it says which known genome is closest, not that they are the same. **Polymorphic sites** in the consensus are the mixture evidence — a high rate suggests more than one population of the species is present, and the consensus then describes neither exactly. No pathogenicity or resistance is inferred from a neighbour.

| PLACED | UNRESOLVED | DEFERRED |
|---|---|---|
| 46 | 80 | 0 |
| of 61 eligible organisms | too few markers or references | left for the next run |

| ORGANISM | NEAREST REFERENCE | DISTANCE | MARKERS | POLYMORPHIC | GENOMES |
|---|---|---|---|---|---|
| *Blautia A wexlerae* SGB4837 | GCF_013300825.1 | 1.431 | 126 | 0.36% | 6 |
| *Faecalibacterium prausnitzii* SGB15318 | GCA_002549855.1 | 0.844 | 173 | 0.76% | 6 |
| *Anaerostipes hadrus* SGB4540 | GCA_003473625.1 | 0.579 | 178 | 0.21% | 6 |
| *Ruminococcus bromii* SGB4290 | GCA_003539535.1 | 0.437 | 195 | 0.10% | 6 |
| GGB6612 SGB9346 SGB9346 | GCA_018713605.1 | 0.161 | 190 | 0.07% | 6 |

| ORGANISM | NEAREST REFERENCE | DISTANCE | MARKERS | POLYMORPHIC | GENOMES |
|---|---|---|---|---|---|
| *Coprococcus eutactus* SGB5117 | GCA_003487145.1 | 0.572 | 180 | 0.07% | 6 |
| GGB3267 SGB4316 SGB4316 | GCA_900539775.1 | 0.019 | 200 | 0.06% | 6 |
| *Ruminococcaceae bacterium AF68 02b2A* SGB4421 | GCA_027680725.1 | 0.577 | 200 | 0.06% | 6 |
| *Phocaeicola dorei* SGB1815 | GCA_000157075.2 | 0.306 | 59 | 0.22% | 6 |
| GGB115571 SGB5062 SGB5062 | GCA_036718785.1 | 0.673 | 197 | 0.05% | 6 |
| *Bacteroides clarus* SGB1832 | GCA_023443365.1 | 0.593 | 191 | 0.07% | 6 |
| GGB9342 SGB14306 SGB14306 | GCA_003512725.1 | 0.617 | 186 | 0.06% | 6 |
| *Ruminococcaceae bacterium AM53 2BH* SGB4191 | GCA_000436675.1 | 0.706 | 187 | 0.07% | 6 |
| *Bacteroides finegoldii* SGB1862 | GCA_937983295.1 | 0.597 | 156 | 0.04% | 6 |
| GGB1495 SGB2071 SGB2071 | GCA_001701305.1 | 0.450 | 189 | 0.05% | 6 |
| *Gemmiger formicilis* SGB15300 | GCA_003525645.1 | 0.516 | 157 | 0.84% | 6 |
| *Dorea longicatena* SGB4581 | GCF_025150085.1 | 0.697 | 127 | 0.07% | 6 |
| *Faecalibacterium SGB15346* SGB15346 | GCA_037068045.1 | 0.450 | 147 | 0.27% | 6 |
| *Blautia faecis* SGB4820 | GCA_017851495.1 | 0.467 | 151 | 1.11% | 6 |
| *Roseburia amylophila* SGB4938 | GCA_014865845.1 | 0.513 | 172 | 0.04% | 6 |
| GGB6601 SGB9333 SGB9333 | GCA_021768105.1 | 0.593 | 166 | 0.05% | 6 |
| GGB13283 SGB4948 SGB4948 | GCA_902461975.1 | 0.258 | 24 | 0.07% | 6 |
| *Roseburia sp AM59 24XD* SGB4654 | GCA_000432755.1 | 0.529 | 177 | 0.06% | 6 |
| *Faecalibacillus intestinalis* SGB6754 | GCA_001916255.1 | 0.157 | 77 | 0.27% | 6 |
| *Alistipes putredinis* SGB2318 | GCF_000154465.1 | 0.080 | 167 | 0.07% | 6 |
| *Bacteroides uniformis* SGB1836 | GCF_025147485.1 | 0.381 | 98 | 0.12% | 6 |
| *Blautia luti* SGB4829 | GCF_013304735.1 | 0.370 | 65 | 0.06% | 5 |
| *Eubacterium siraeum* SGB4198 | GCA_000210635.1 | 0.615 | 82 | 0.04% | 6 |
| *Roseburia faecis* SGB4925 | GCA_000433615.1 | 0.388 | 92 | 0.12% | 6 |
| *Dorea formicigenerans* SGB4575 | GCF_025150245.1 | 0.308 | 109 | 0.22% | 6 |

| ORGANISM | NEAREST REFERENCE | DISTANCE | MARKERS | POLYMORPHIC | GENOMES |
|---|---|---|---|---|---|
| *Parabacteroides distasonis* SGB1934 | GCF_000012845.1 | 0.341 | 72 | 0.04% | 6 |
| *Eubacterium ramulus* SGB4960 | GCF_003122485.1 | 0.481 | 84 | 0.04% | 6 |
| *Phocaeicola massiliensis* SGB1812 | GCA_000438395.1 | 0.502 | 83 | 0.05% | 6 |
| *Coprococcus catus* SGB4670 | GCA_024462395.1 | 0.374 | 109 | 0.05% | 6 |
| *Anaerobutyricum soehngenii* SGB4537 | GCA_018379315.1 | 1.056 | 93 | 0.04% | 6 |
| GGB9634 SGB15106 SGB15106 | GCF_049060325.1 | 1.201 | 56 | 0.28% | 6 |
| GGB6565 SGB9274 SGB9274 | GCA_900544255.1 | 0.670 | 63 | 0.06% | 6 |
| *Agathobacter rectalis* SGB4933 | GCF_000020605.1 | 0.488 | 80 | 0.06% | 6 |
| GGB3303 SGB4364 SGB4364 | GCA_021618165.1 | 0.497 | 65 | 0.06% | 6 |
| *Roseburia hominis* SGB4936 | GCA_963654245.1 | 1.681 | 75 | 0.05% | 6 |
| GGB4750 SGB6579 SGB6579 | GCA_036793425.1 | 0.558 | 78 | 0.06% | 6 |
| GGB9747 SGB15356 SGB15356 | GCA_021640885.1 | 0.469 | 52 | 0.04% | 6 |
| *Lachnospira eligens* SGB5082 | GCA_051978475.1 | 0.922 | 162 | 0.12% | 6 |
| GGB140897 SGB6321 SGB6321 | GCA_036300105.1 | 0.264 | 34 | 0.04% | 6 |
| *Faecalibacterium wellingii* SGB15340 | GCA_018381455.1 | 0.711 | 31 | 0.30% | 6 |
| *Coprococcus comes* SGB4577 | GCA_934329715.1 | 1.933 | 149 | 0.06% | 6 |

**Eligible but unresolved**, with the reason:

• *Mediterraneibacter lactaris* (SGB4557): only 0 reference genomes could be fetched (0 members in GTDB R232); a comparative tree needs at least three

• *Fusicatenibacter saccharivorans* (SGB4874): only 0 reference genomes could be fetched (0 members in GTDB R232); a comparative tree needs at least three

• *Adlercreutzia equolifaciens* (SGB14797): the sample's consensus did not pass StrainPhlAn's marker filters (samples kept after filtering

• *Clostridiaceae bacterium* (SGB4629): the sample's consensus did not pass StrainPhlAn's marker filters (samples kept after filtering

• *Anthropogastromicrobium aceti* (SGB5090): the sample's consensus did not pass StrainPhlAn's marker filters (samples kept after filtering

• *GGB3612 SGB4882* (SGB4882): the sample's consensus did not pass StrainPhlAn's marker filters (samples kept after filtering

• *Agathobaculum butyriciproducens* (SGB14993): the sample's consensus did not pass StrainPhlAn's marker filters (samples kept after filtering

• *GGB4699 SGB6502* (SGB6502): the sample's consensus did not pass StrainPhlAn's marker filters (samples kept after filtering

• *Jilunia laotingensis* (SGB1802): the sample's consensus did not pass StrainPhlAn's marker filters (samples kept after filtering

• *GGB4711 SGB6524* (SGB6524): the sample's consensus did not pass StrainPhlAn's marker filters (samples kept after filtering

• *Clostridium fessum* (SGB4705): the sample's consensus did not pass StrainPhlAn's marker filters (samples kept after filtering

• *Anaerobutyricum hallii* (SGB4532): the sample's consensus did not pass StrainPhlAn's marker filters (samples kept after filtering

• *Clostridium sp AM22 11AC* (SGB4749): the sample's consensus did not pass StrainPhlAn's marker filters (samples kept after filtering

• *Bacteroides cellulosilyticus* (SGB1844): the sample's consensus did not pass StrainPhlAn's marker filters (samples kept after filtering

• *Odoribacter laneus* (SGB1792): the sample's consensus did not pass StrainPhlAn's marker filters (samples kept after filtering

65 further marker clades had fewer than 20 reconstructed markers and were not placed; they are present as organisms where the inventory lists them, and unresolved only at strain level.

Method: StrainPhlAn 4 on the mpa_vJan26_CHOCOPhlAnSGB_202605 markers, reference genomes from the organism's GTDB R232 species cluster (representative first, at most 6), PhyloPhlAn fast mode. Strain results never add to species counts.

## What was asked for, and what came back

A screen is only meaningful alongside its own coverage. Every resolution target registered for this sample is listed below with the state it reached. **An unrun, failed or unavailable assay is never reported as a negative result** — the distinction between "looked and found nothing" and "never looked" is preserved throughout.

| STATE | TARGETS | WHAT IT MEANS |
|---|---|---|
| **Completed** | 5 | the assay ran and produced a result |
| **Incompatible with DNA input** | 8 | an RNA genome cannot be assayed from DNA sequencing |
| **Reference or tool unavailable** | 15 | the tool or reference is not installed |
| **Registered, awaiting reference curation** | 16 | registered, with reference curation outstanding |

**44 targets registered**, of which **16** remain outstanding. The outstanding targets are listed in results.json under `resolution_census`, each with the reason it has not yet been determined.

Organism-specific typing: 9 schemes registered (4 with their tool installed). A scheme without its tool reports an unavailable capability for every sample; it never reports a negative type, and a species name is never allowed to stand in for a type.

Strain-comparison tools: 8 of 8 installed. These tools estimate different objects: a marker phylogeny, a genome-wide population comparison and a reference-mixture deconvolution are not three measurements of one quantity. Agreement between them is not multiplied into extra certainty, and discordance is preserved rather than averaged away.

## 18 Pathogens in detail

Each finding on its own, by group, with the evidence behind it kept as separate facts rather than one blended score: how much distinct sequence supported it, how much of the reference that sequence covered, what resolution it justifies, and what would confirm it. Groups with nothing found still appear, because a negative from a search that was capable of finding something is itself a result.

### Bacterial pathogens

5 supported, 1 candidate, out of 136 targets searched in this group.

‹ at a glance

#### Burkholderia cepacia complex

ENOUGH SEQUENCE TO NAME IT

**WHAT IT IS**

A group of closely related Gram-negative bacteria found in soil, water, plant roots and occasionally in contaminated liquids such as mouthwashes and disinfectants. They are best known as plant-associated organisms.

**WHY IT MATTERS**

For most people these bacteria are harmless. Their serious reputation comes almost entirely from cystic fibrosis and chronic granulomatous disease, where they can establish transmissible and difficult-to-treat lung infection. They are intrinsically resistant to many antibiotics. They are not established causes of gastroenteritis, and detecting them in stool is not evidence of lung infection or of any infection at all.

**HOW MUCH IS HERE**

**Faint signal** in this sample: 1.4 per million, 0.0001% of analysed DNA, from 8 distinct DNA fragments across 5 separate regions of its genome. A small fraction of the sample's dna. For this organism, it is normally arrives with food or from the environment. Coverage evenness 0.43 (1.0 is sequence spread across the whole genome as depth would predict; near 0 is reads piled into the regions relatives share). The aligned fragments cover **0.01%** of this organism's reference genome. A genuinely present organism is covered broadly; a small percentage means the reads sit on a slice of the genome that relatives may share, which is why breadth is reported next to the count.

**WHAT WOULD MAKE THE AMOUNT CONCERNING**

No validated stool threshold exists and sequencing cannot establish one. This finding acquires significance only alongside cystic fibrosis, chronic granulomatous disease, or an unexplained respiratory or bloodstream illness being investigated by a clinician, in which case sputum or blood culture is the appropriate test. In everyone else, a stool signal without symptoms should be treated as incidental.

**WHAT IS KNOWN TO ACT AGAINST IT**

Non-prescription agents with published activity against this organism. The tier after each name is how strong the evidence is, and it matters: a culture result is not a demonstration that swallowing the substance changes anything in a person. None of this is a prescription or a treatment plan.

**Allicin (crushed raw garlic, aqueous garlic extract)** (botanical) — laboratory culture only — no human evidence

Pure allicin and aqueous garlic extract inhibited and killed 38 clinical Burkholderia cepacia complex isolates in culture, including three Burkholderia cenocepacia strains, with allicin minimum inhibitory concentrations of 8-62 micrograms per millilitre. The mechanism is chemical modification of essential bacterial cysteine residues. This is culture work only; there are no human data. *Dose studied:* not established in humans; in vitro minimum inhibitory concentrations of 8-62 micrograms per millilitre of allicin. *Reaching the gut:* Allicin forms when raw garlic is crushed and does survive into the upper gut, but it reacts rapidly with glutathione and other thiols in the gut wall and blood, so very little intact allicin gets far. The same paper's authors and later work note this poor bioavailability explicitly. Colonic delivery in active form should therefore be treated as doubtful, not assumed. Aged or cooked garlic contains little or no allicin. *Caution:* Garlic thins the blood and can interact with warfarin, clopidogrel and other anticoagulants and antiplatelet drugs; stop it before planned surgery. It commonly causes heartburn, reflux and wind, and can worsen symptoms in irritable bowel syndrome because it is high in fructans. It does not treat cystic fibrosis lung infection and must never replace prescribed antibiotics. Garlic Revisited: Antimicrobial Activity of Allicin-Containing Garlic Extracts against Burkholderia cepacia Complex

**Manuka honey** (food) — laboratory culture only — no human evidence

Manuka honey inhibited all 55 Burkholderia clinical isolates from cystic fibrosis patients at 7% or less by weight, with a mean minimum inhibitory concentration of 4.6%, making Burkholderia somewhat more honey-susceptible than Pseudomonas aeruginosa in the same study. It also acted synergistically or additively with tobramycin and colistin. Culture work only; no human outcome data. *Dose studied:* not established in humans; in vitro minimum inhibitory concentration around 4.6% honey by weight. *Reaching the gut:* Poorly. Honey sugars are absorbed in the small intestine and the active methylglyoxal is diluted and chemically consumed en route, so colonic concentrations fall well short of the several-per-cent used in the laboratory. State this as a limitation of the agent. *Caution:* High in sugar, so unsuitable if diabetes is poorly controlled. Never give honey to infants under twelve months because of infant botulism risk. This study tested honey in a test tube against airway isolates; it is not evidence that eating honey affects the lungs or the gut, and it is no substitute for specialist cystic fibrosis care. A demonstration of the susceptibility of clinical isolates obtained from cystic fibrosis patients to manuka honey

organism sequence not infection diagnosis

| RESOLVED TO | ORGANISM CLASS | REFERENCE ROUTE |
|---|---|---|
| species complex | opportunist, context dependent | genome supported |
| **DISTINCT FRAGMENTS** | **INFORMATIVE REGIONS** | **INFORMATIVE BASES** |
| 8 | 5 | 1,041 |
| **SHARED-SEQUENCE FRAGMENTS** | **MEDIAN IDENTITY** | **INFORMATIVE BREADTH** |
| 63 (not counted as support) | 96.0% | 0.01% |

**What would confirm this:** clinician selected stool NAAT or culture, specialist referral.

**Caveat:** no extraction or library controls for this sample.

below identity gate · partial query alignment · controls unavailable · species resolution limited · not in organism inventory

‹ at a glance

## Campylobacter ureolyticus

ENOUGH SEQUENCE TO NAME IT

### HOW MUCH IS HERE

**Small signal** in this sample: 14 per million, 0.0014% of analysed DNA, from 81 distinct DNA fragments across 53 separate regions of its genome. A minor component of the sample's dna. For this organism, it is carried by many healthy people, so presence alone is unremarkable. Coverage evenness 0.47 (1.0 is sequence spread across the whole genome as depth would predict; near 0 is reads piled into the regions relatives share). The aligned fragments cover **0.63%** of this organism's reference genome. A genuinely present organism is covered broadly; a small percentage means the reads sit on a slice of the genome that relatives may share, which is why breadth is reported next to the count.

organism sequence not infection diagnosis

| RESOLVED TO | ORGANISM CLASS | REFERENCE ROUTE |
|---|---|---|
| species level | role in gut disease uncertain | genome supported |
| **DISTINCT FRAGMENTS** | **INFORMATIVE REGIONS** | **INFORMATIVE BASES** |
| 81 | 53 | 11,556 |
| **SHARED-SEQUENCE FRAGMENTS** | **MEDIAN IDENTITY** | **INFORMATIVE BREADTH** |
| 14 (not counted as support) | 98.0% | 0.66% |

**What would confirm this:** clinician selected stool NAAT or culture.

**Caveat:** no extraction or library controls for this sample.

below identity gate · partial query alignment · duplicate family collapsed · controls unavailable · not in organism inventory

‹ at a glance

## Clostridioides difficile

ENOUGH SEQUENCE TO NAME IT

### WHAT IT IS

A spore-forming bacterium that lives in the large bowel. Its spores survive for a long time on surfaces and resist alcohol hand gel. Only some strains carry the genes for the toxins that cause disease; strains without those genes do not cause the illness.

### WHY IT MATTERS

When antibiotics wipe out the normal gut bacteria, toxin-producing C. difficile can multiply and release toxins that inflame and damage the colon. This causes watery diarrhoea, abdominal pain and fever, and in severe cases pseudomembranous colitis, toxic megacolon, bowel perforation and death. It is a leading cause of healthcare-associated infection. Suspected C. difficile infection needs medical assessment and prescription treatment such as vancomycin or fidaxomicin, and recurrent cases may need faecal microbiota transplantation. Supplements are not a substitute for that, and delaying treatment in someone who is unwell is dangerous. Drugs that slow the bowel should be avoided.

### HOW MUCH IS HERE

**Faint signal** in this sample: 4.2 per million, 0.0004% of analysed DNA, from 24 distinct DNA fragments across 8 separate regions of its genome. A small fraction of the sample's dna. For this organism, it is carried by many healthy people, so presence alone is unremarkable. Coverage evenness 0.30 (1.0 is sequence spread across the whole genome as depth would predict; near 0 is reads piled into the regions relatives share). The aligned fragments cover **0.05%** of this organism's reference genome. A genuinely present organism is covered broadly; a small percentage means the reads sit on a slice of the genome that relatives may share, which is why breadth is reported next to the count.

### IS CARRYING THIS NORMAL

Carried without symptoms by 0–17.5% of healthy adults; the toxin-producing form by about 1.5% (95% CI 0.7–2.6%). Toxin-negative strains are associated with protection from C. difficile infection, not with disease. Infant carriage reaches 75%. Source: Meta-analysis of 51 studies, 39,447 participants (doi:10.1186/s13099-024-00674-0); non-toxigenic protection (doi:10.3389/fmicb.2018.01700)

#### THE GENES THAT DECIDE WHETHER IT MATTERS

The defining disease genes **were looked for and not found**. Without them this organism is not the disease-causing form, so this is carriage evidence rather than a disease finding.

#### WHAT WOULD MAKE THE AMOUNT CONCERNING

Sequencing on its own cannot establish that C. difficile is causing illness, and this is the single most important caveat for this organism. Diagnosis requires diarrhoea plus laboratory evidence of toxin, usually a toxin immunoassay interpreted alongside a glutamate dehydrogenase or nucleic acid test. Two things make a sequencing hit more meaningful: detection of the toxin genes tcdA and tcdB, and better still the binary toxin genes cdtA and cdtB or the tcdC deletion associated with more virulent strains, and a clear clinical picture of new diarrhoea after antibiotics. A detection in someone with formed stools and no symptoms should be treated as carriage, and treating asymptomatic carriage is explicitly not recommended.

#### WHAT IS KNOWN TO ACT AGAINST IT

Non-prescription agents with published activity against this organism. The tier after each name is how strong the evidence is, and it matters: a culture result is not a demonstration that swallowing the substance changes anything in a person. None of this is a prescription or a treatment plan.

**Multi-strain probiotics (preventive, during antibiotics)** (probiotic) — randomised human trial

A 2025 Cochrane review of 38 randomised trials in 13,179 participants found that probiotics taken alongside antibiotics may halve the incidence of C. difficile-associated diarrhoea, from 3.2 per cent to 1.6 per cent, with a number needed to treat of about 65. The evidence was graded low certainty, and this is prevention during antibiotic courses, not treatment of established infection. *Dose studied:* Varies by trial; typically 10^9 to 10^10 colony-forming units daily, started within about two days of the antibiotic and continued for its duration. *Reaching the gut:* The strains used in these trials, mainly Lactobacillus, Bifidobacterium and Saccharomyces species, are selected for acid and bile tolerance and are recoverable live from stool during supplementation. Bacterial strains can be killed by the concurrent antibiotic, which is one argument for the yeast-based options. *Caution:* Preventive only; this does not treat an established infection, and the certainty of the evidence is low. Live probiotics should not be given to severely immunosuppressed people, those with central venous catheters, or critically ill patients, in whom bloodstream infection from the probiotic organism has been reported. Probiotics for the prevention of Clostridioides difficile-associated diarrhea in adults and children

**Saccharomyces boulardii** (probiotic) — randomised human trial

In a randomised controlled trial of 120 patients with mild to moderate C. difficile infection, adding S. boulardii to vancomycin gave a higher global cure rate (96.6 per cent versus 85.3 per cent) and a lower recurrence rate (1.7 per cent versus 13.1 per cent). It was tested as an addition to antibiotic treatment, not instead of it. *Dose studied:* Two 250 mg capsules (500 mg) every 12 hours alongside vancomycin. *Reaching the gut:* S. boulardii is a yeast, not a bacterium, so it is unaffected by antibacterial drugs and can be given at the same time as vancomycin. It tolerates gastric acid and bile and is recovered live in stool during dosing, reaching steady-state colonic concentrations within about three days and clearing within days of stopping. *Caution:* An addition to treatment, not a replacement for it. S. boulardii must be avoided in critically ill patients, in people who are severely immunosuppressed, and in anyone with a central venous catheter, because fungaemia has occurred in these groups, sometimes from airborne contamination when capsules are opened at the bedside. Avoid in anyone with a yeast allergy. Efficacy and safety of Saccharomyces boulardii as adjunct therapy with Vancomycin in treating Clostridioides difficile infection: A randomized controlled trial

**Berberine** (botanical) — animal study only

Blocked relapse of C. difficile infection in mice after standard vancomycin treatment, with a follow-up study describing the mechanisms involved. Mouse evidence for prevention of relapse; there is no controlled human trial in C. difficile infection. *Dose studied:* not established for humans in C. difficile infection. *Reaching the gut:* Berberine's oral bioavailability is under one per cent, so nearly all of a dose stays in the gut lumen and reaches the colon, which is precisely where C. difficile grows and releases toxin. Of the botanicals listed here it has the clearest case for reaching the target site. *Caution:* Animal evidence only, and no substitute for prescription treatment. Berberine inhibits CYP3A4, CYP2D6 and P-glycoprotein, so it raises levels of many prescription drugs, which matters particularly in the older, multi-medicated patients who get C. difficile. Avoid in pregnancy, when breastfeeding, and in newborns. Berberine blocks the relapse of Clostridium difficile infection in C57BL/6 mice after standard vancomycin treatment

**Carvacrol** (botanical) — laboratory culture only — no human evidence

Reduced C. difficile sporulation and spore outgrowth in culture, and in a separate study reduced toxin production and the toxicity of culture supernatant to cells. Both findings are from culture-dish experiments; there is no human data. *Dose studied:* not established. *Reaching the gut:* Carvacrol is absorbed and conjugated in the upper small intestine, so a plain oregano oil capsule delivers little to the colon. Reaching C. difficile in the large bowel would need a delayed-release or encapsulated formulation, and none has been tested in people for this purpose. *Caution:* Culture-dish evidence only, and it does not treat infection. Broad antibacterial botanicals are a questionable idea in C. difficile precisely because the illness is caused by loss of normal gut bacteria, and carvacrol suppresses commensals too. Undiluted oregano oil burns mucous membranes. Avoid in pregnancy. Carvacrol reduces Clostridium difficile sporulation and spore outgrowth in vitro

**Epigallocatechin gallate (EGCG, green tea catechin)** (botanical) — laboratory culture only — no human evidence

Reduced virulence measures of the hypervirulent C. difficile PCR ribotype 027 strain in culture, including effects on toxin activity. The endpoint was virulence rather than killing the organism, and the work was entirely in culture. *Dose studied:* not established. *Reaching the gut:* EGCG is very poorly absorbed, so most of an oral dose does reach the colon, which suits this target. Against that, it is unstable at small-intestinal pH and is extensively broken down by gut bacteria, so a fair proportion of what arrives is metabolites rather than intact EGCG. *Caution:* Culture-dish evidence only, and no substitute for prescription treatment. Concentrated green tea extracts have caused liver injury, and several regulators limit EGCG per daily dose; taking them on an empty stomach increases the risk. Catechins reduce non-haem iron absorption. Inhibitory Effect of Epigallocatechin Gallate on the Virulence of Clostridium difficile PCR Ribotype 027

**trans-Cinnamaldehyde** (botanical) — laboratory culture only — no human evidence

Reduced C. difficile toxin production and the cytotoxicity of its culture supernatant to cells, alongside carvacrol, at concentrations below those that inhibited growth. This is an anti-toxin effect shown in culture only, with no animal or human data. *Dose studied:* not established. *Reaching the gut:* Cinnamaldehyde is chemically reactive and is largely absorbed and oxidised to cinnamic acid in the upper gut, so ordinary cinnamon oil or cinnamon powder is unlikely to deliver active compound to the colon. Encapsulated forms are used in animal feed for this reason but have not been tested against C. difficile in people. *Caution:* Culture-dish evidence only. Cassia cinnamon contains coumarin, which is toxic to the liver in regular large amounts. Cinnamon oil is a strong mucosal irritant when undiluted, and cinnamon may add to the effect of blood-glucose-lowering medicines. Carvacrol and trans-cinnamaldehyde reduce Clostridium difficile toxin production and cytotoxicity in vitro

organism sequence not infection diagnosis

| RESOLVED TO | ORGANISM CLASS | REFERENCE ROUTE |
|---|---|---|
| species level | depends on toxin or pathotype | genome supported |
| **DISTINCT FRAGMENTS** | **INFORMATIVE REGIONS** | **INFORMATIVE BASES** |
| 24 | 8 | 2,156 |
| **SHARED-SEQUENCE FRAGMENTS** | **MEDIAN IDENTITY** | **INFORMATIVE BREADTH** |
| 335 (not counted as support) | 95.7% | 0.05% |

**What would confirm this:** clinical toxin immunoassay, clinical stool NAAT, clinical stool culture.

**Caveat:** no extraction or library controls for this sample.

below identity gate · partial query alignment · controls unavailable · defining determinant not detected · not in organism inventory

‹ at a glance

## Peptostreptococcus anaerobius

ENOUGH SEQUENCE TO NAME IT

**WHAT IT IS**

An anaerobic Gram-positive bacterium found in the normal flora of the mouth, the gut and the female genital tract. It is a familiar component of polymicrobial anaerobic infections when it escapes those sites.

**WHY IT MATTERS**

In its usual niches it is an ordinary commensal, and it appears in polymicrobial abscesses, dental and soft-tissue infections when tissue is breached. It has attracted attention because it is more abundant in the stool and colonic tissue of people with colorectal cancer than in controls, and in mice it increased colonic tumour numbers when given repeatedly by gavage at high dose; a surface protein has been described that binds an integrin overexpressed on colorectal cancer cells and activates proliferative signalling. The framing matters. Those mouse experiments used 10^8 organisms daily for ten weeks in animals genetically primed to develop tumours, which bears no relation to ordinary human carriage, and the human abundance difference is measured in people who already have cancer. A stool detection is not a cancer finding, not a risk score and not an indication for investigation on its own. Follow bowel-screening invitations as normal, and take bowel symptoms to a doctor on their own merits.

**HOW MUCH IS HERE**

**Clear signal** in this sample: 274 per million, 0.03% of analysed DNA, from 1,552 distinct DNA fragments across 17 separate regions of its genome. Well represented in the sample's dna. For this organism, it is carried by many healthy people, so presence alone is unremarkable. In the community composition (section 6) this organism is **0.024%** of the whole, measured competitively against every relative found in this sample. That is the amount; the count above is this screen's own alignment, which credits shared sequence to the target it knows. Coverage evenness 0.07 (1.0 is sequence spread across the whole genome as depth would predict; near 0 is reads piled into the regions relatives share). The aligned fragments cover **1.33%** of this organism's reference genome. A genuinely present organism is covered broadly; a small percentage means the reads sit on a slice of the genome that relatives may share, which is why breadth is reported next to the count.

**WHAT WOULD MAKE THE AMOUNT CONCERNING**

No abundance threshold has been validated and sequencing cannot establish one, nor can it separate a colonic population from swallowed oral bacteria. Where this species is used in research it forms part of a multi-marker stool panel alongside Fusobacterium nucleatum, Parvimonas micra and a faecal immunochemical test, evaluated at population level in screening studies; that is a very different thing from a single species reading in one person's sample, and the panel context is exactly what a lone result lacks. There is no quantity that makes this finding actionable in an individual.

**WHAT TO DO ABOUT IT**

There is no reason to try to remove this organism and attempting to would be unwise. It is a normal member of the anaerobic flora of the mouth, gut and genital tract, and its presence in stool is an expected background finding rather than evidence of infection or of cancer. No human trial has shown that any supplement, botanical or probiotic reduces it, and none has shown any benefit from reducing it. The animal tumour data involved daily high-dose gavage into genetically susceptible mice and provide no basis for acting on an incidental human carriage result. Presenting a protocol here would manufacture alarm from an ordinary finding and encourage indefinite use of non-selective botanical antimicrobials, which suppress the beneficial anaerobes that maintain colonisation resistance and the colonic mucus barrier, at real cost and no demonstrated benefit. If you want to act on the ecology sensibly, look after your gums and teeth and eat enough fermentable fibre to support butyrate-producing anaerobes. Take bowel symptoms and screening invitations seriously in their own right.

organism sequence not infection diagnosis

| RESOLVED TO | ORGANISM CLASS | REFERENCE ROUTE |
|---|---|---|
| species level | role in gut disease uncertain | genome supported |
| **DISTINCT FRAGMENTS** | **INFORMATIVE REGIONS** | **INFORMATIVE BASES** |
| 1,552 | 17 | 27,799 |
| **SHARED-SEQUENCE FRAGMENTS** | **MEDIAN IDENTITY** | **INFORMATIVE BREADTH** |
| 385 (not counted as support) | 98.7% | 1.36% |

**What would confirm this:** clinician selected stool NAAT or culture.

**Caveat:** no extraction or library controls for this sample.

below identity gate · duplicate family collapsed · partial query alignment · controls unavailable

‹ at a glance

## Streptococcus gallolyticus group

ENOUGH SEQUENCE TO NAME IT

**WHAT IT IS**

A group of Gram-positive bacteria, formerly called the Streptococcus bovis group, that live in the human and animal intestine. It includes Streptococcus gallolyticus subspecies gallolyticus and Streptococcus infantarius.

**WHY IT MATTERS**

Please read this section with a clinician in mind. In the gut these are ordinary flora. But bloodstream infection or endocarditis caused by Streptococcus gallolyticus subspecies gallolyticus has a long-established and strong association with colorectal tumours, and it is standard practice to arrange a colonoscopy in that situation. In one comparative study, colorectal neoplasia was found in 49.8% of patients with Streptococcus gallolyticus bacteraemia, and in 91.4% of those cases the tumour was occult, meaning it had caused no symptoms and would otherwise have been missed. That is why the recommendation exists. It is equally important to be clear about what it does not cover: that recommendation applies to a positive blood culture with a clinical illness, not to a DNA sequence in stool. A stool detection is not a cancer finding and is not by itself a reason for colonoscopy. If you have had a positive blood culture with this organism, or if you have endocarditis, unexplained fever, anaemia, rectal bleeding, a change in bowel habit or unexplained weight loss, see a doctor about it, and say that this organism was detected. Follow national bowel-screening invitations regardless of this result.

**HOW MUCH IS HERE**

**Small signal** in this sample: 23 per million, 0.0023% of analysed DNA, from 131 distinct DNA fragments across 77 separate regions of its genome. A minor component of the sample's dna. For this organism, it is carried by many healthy people, so presence alone is unremarkable. Coverage evenness 0.46 (1.0 is sequence spread across the whole genome as depth would predict; near 0 is reads piled into the regions relatives share). The aligned fragments cover **0.80%** of this organism's reference genome. A genuinely present organism is covered broadly; a small percentage means the reads sit on a slice of the genome that relatives may share, which is why breadth is reported next to the count.

**WHAT WOULD MAKE THE AMOUNT CONCERNING**

No stool abundance threshold has been validated, and sequencing cannot establish one; there is no quantity of this organism in stool that constitutes evidence of a tumour or of invasive infection. What changes management is a positive blood culture with a clinical syndrome, or the presence of bowel symptoms or anaemia that merit investigation in their own right. Note also that the cancer signal is concentrated in one subspecies, Streptococcus gallolyticus subspecies gallolyticus, and short stool reads usually cannot resolve subspecies, so the finding is reported at group level and is correspondingly less specific.

**WHAT TO DO ABOUT IT**

There is no reason to try to remove this organism and attempting to would be unwise, and it would also be a distraction from the only action that matters here. These are normal gut flora, no supplement or probiotic has been shown in humans to reduce them, and there is no evidence whatsoever that reducing faecal carriage alters the risk of colorectal neoplasia or of endocarditis. Presenting a supplement list against this finding risks the serious harm of substituting something ineffective for the appropriate response. The appropriate response is clinical: discuss the finding with a doctor, particularly if you have had a positive blood culture with this organism, endocarditis, unexplained fever, iron-deficiency anaemia, rectal bleeding, a change in bowel habit or unexplained weight loss, and take up national bowel-screening invitations when they arrive. Those steps have demonstrated value; suppressing a commensal does not.

organism sequence not infection diagnosis

| RESOLVED TO | ORGANISM CLASS | REFERENCE ROUTE |
|---|---|---|
| group level | opportunist, context dependent | genome supported |
| **DISTINCT FRAGMENTS** | **INFORMATIVE REGIONS** | **INFORMATIVE BASES** |
| 131 | 77 | 18,169 |
| **SHARED-SEQUENCE FRAGMENTS** | **MEDIAN IDENTITY** | **INFORMATIVE BREADTH** |
| 205 (not counted as support) | 98.0% | 0.84% |

**What would confirm this:** clinician selected stool NAAT or culture, blood culture, specialist referral.

**Caveat:** no extraction or library controls for this sample.

partial query alignment · below identity gate · duplicate family collapsed · controls unavailable · species resolution limited · not in organism inventory

‹ at a glance

## Enterococcus faecalis

TRACE, NOT ENOUGH TO CONFIRM

**WHAT IT IS**

A Gram-positive bacterium and one of the most consistently present members of the normal human gut community. It colonises the intestine within days of birth and stays for life. Some strains are used as probiotics and in food fermentation.

**WHY IT MATTERS**

As a gut resident it contributes to normal gut function, including regulating intestinal pH and producing vitamins. It becomes a problem only when it gets somewhere it should not be: it is a common cause of urinary tract infection, and of line-associated bloodstream infection, endocarditis and intra-abdominal infection in people who are hospitalised, immunosuppressed, or have had abdominal or cardiac surgery. The genuinely reportable event is not the organism itself but acquired vancomycin resistance, vanA or vanB, if that is linked to the organism, because it narrows treatment options and can spread. Detection of the organism in stool is expected, not a finding.

**HOW MUCH IS HERE**

**At the detection floor** in this sample: 0.35 per million, well under a thousandth of a percent, from 2 distinct DNA fragments across 2 separate regions of its genome. Barely above what this depth can see. For this organism, it is carried by many healthy people, so presence alone is unremarkable. The aligned fragments cover **0.01%** of this organism's reference genome. A genuinely present organism is covered broadly; a small percentage means the reads sit on a slice of the genome that relatives may share, which is why breadth is reported next to the count.

**IS CARRYING THIS NORMAL**

A lifelong member of the normal gut community in the great majority of healthy adults. Source: OpenBiota pathogen catalogue, resident lane

**WHAT WOULD MAKE THE AMOUNT CONCERNING**

No meaningful threshold exists for an organism this ubiquitous, and one stool metagenome cannot establish absolute abundance. A very high enterococcal share alongside a collapse in overall diversity is a recognised pattern after broad-spectrum antibiotics or in critical illness, and in hospitalised patients it can precede bloodstream infection, but that is a hospital observation and not something a stool sequence in a well person establishes. Invasive infection is diagnosed by blood or urine culture with clinical assessment. The one result here that genuinely changes management is a linked acquired vancomycin-resistance gene, which is reported separately and is a matter for a clinician.

**WHAT TO DO ABOUT IT**

This is a dominant, lifelong member of the normal gut community, present in the great majority of healthy people at substantial numbers, and there is no reason to try to remove it. Attempting to would be unwise on two counts. First, it cannot realistically be achieved: enterococci are intrinsically tolerant of bile, acid and a wide range of antimicrobials, which is precisely why they persist. Second, the attempt does harm. Antibiotic pressure aimed at enterococci is the best-documented way of selecting for vancomycin-resistant strains, and broad-spectrum botanical antimicrobials are non-selective: they suppress the beneficial anaerobes that provide colonisation resistance, and the reliable consequence of losing those anaerobes is that enterococci and Enterobacteriaceae expand, not shrink. There is no human trial showing that any supplement usefully reduces Enterococcus faecalis in a healthy gut, and no evidence that doing so would benefit anyone. If invasive enterococcal infection is suspected, that is a clinical matter requiring blood or urine culture and prescribed treatment.

organism sequence not infection diagnosis

| RESOLVED TO | ORGANISM CLASS | REFERENCE ROUTE |
|---|---|---|
| species level | opportunist, context dependent | genome supported |
| **DISTINCT FRAGMENTS** | **INFORMATIVE REGIONS** | **INFORMATIVE BASES** |
| 2 | 2 | 302 |
| **SHARED-SEQUENCE FRAGMENTS** | **MEDIAN IDENTITY** | **INFORMATIVE BREADTH** |
| 210 (not counted as support) | 97.4% | 0.01% |

**What would confirm this:** clinician selected stool NAAT or culture.

below identity gate · partial query alignment

## Protozoa pathogens

0 supported, 2 candidate, out of 49 targets searched in this group.

‹ at a glance

### Cryptosporidium canis

TRACE, NOT ENOUGH TO CONFIRM

**WHAT IT IS**

A Cryptosporidium species whose natural host is the dog. It occasionally infects people, usually children or immunocompromised adults with close contact with dogs, and it produces the same kind of intestinal infection as the human-adapted species.

**WHY IT MATTERS**

Human infections are uncommon but real, and the illness is the familiar watery diarrhoea with cramping and weight loss. In immunocompetent people it is generally self-limiting. In immunocompromised people, as with the other species, it can become prolonged and severe. A detection is also useful information about the household: it points to a dog as the likely source, and the animal may benefit from veterinary attention. This should be discussed with a doctor.

**HOW MUCH IS HERE**

**Small signal** in this sample: 96 per million, 0.0096% of analysed DNA, from 544 distinct DNA fragments across 1 separate regions of its genome. A minor component of the sample's dna. For this organism, it is not a normal resident; healthy stool does not usually contain it. The aligned fragments cover **0.04%** of this organism's reference genome. A genuinely present organism is covered broadly; a small percentage means the reads sit on a slice of the genome that relatives may share, which is why breadth is reported next to the count.

**WHAT WOULD MAKE THE AMOUNT CONCERNING**

Quantity is not the question; presence is. Sequencing cannot show whether the oocysts are viable, whether the infection is active, or whether it explains any symptoms. Immune status and current symptoms are what matter.

**WHAT IS KNOWN TO ACT AGAINST IT**

Non-prescription agents with published activity against this organism. The tier after each name is how strong the evidence is, and it matters: a culture result is not a demonstration that swallowing the substance changes anything in a person. None of this is a prescription or a treatment plan.

**Bovine colostrum immunoglobulin concentrate** (nutrient) — human study, not randomised

Reduced stool weight and frequency in AIDS patients with Cryptosporidium parvum diarrhoea in an open non-randomised study, with the powder form working and capsules not. This was a different Cryptosporidium species in the same genus, stated as such, and later placebo-controlled work has been inconsistent. *Dose studied:* bovine immunoglobulin concentrate powder for 21 consecutive days. *Reaching the gut:* Intended to act in the gut lumen; partly degraded by stomach acid and pepsin, which is why the powder formulation and acid suppression have been used. *Caution:* Milk-derived; unsuitable with cows' milk protein allergy. Retail colostrum is not the studied product. Does not replace medical assessment. Treatment of severe diarrhea caused by Cryptosporidium parvum with oral bovine immunoglobulin concentrate in patients with AIDS

**Curcumin** (botanical) — animal study only

Eliminated Cryptosporidium parvum oocyst shedding with no recurrence in immunosuppressed mice and reduced intestinal damage. Again this is a different species within the same genus, stated as such, and mouse data only. *Dose studied:* oral curcumin daily for 11 days in mice; no established human dose. *Reaching the gut:* Poorly absorbed, so most of an oral dose remains in the intestinal lumen throughout its transit. *Caution:* Bleeding risk with anticoagulants, gallstone pain, rare liver injury with concentrated products, reduced iron absorption. No human evidence. Comparative efficacy of curcumin and paromomycin against Cryptosporidium parvum infection in a BALB/c model

**Zinc** (nutrient) — animal study only

Cut oocyst shedding by 98.3 per cent in mice infected with Cryptosporidium parvum, a closely related species in the same genus, stated as such. No human trial has shown zinc eliminates any Cryptosporidium species. *Dose studied:* commercial zinc preparation daily for 8 days in mice; WHO childhood diarrhoea dose is 10-20 mg elemental zinc daily for 10-14 days. *Reaching the gut:* Absorbed in the small intestine at the site of infection, with the remainder passing into the colon. *Caution:* Prolonged high-dose zinc causes copper deficiency with anaemia and nerve damage. Blocks quinolone and tetracycline absorption. Nausea on an empty stomach is common. The Effects of Commercially Available Syzygium aromaticum, Anethum graveolens, Lactobacillus acidophilus LB, and Zinc as Alternatives Therapy in Experimental Mice Challenged with Cryptosporidium parvum

organism sequence not infection diagnosis

| RESOLVED TO | ORGANISM CLASS | REFERENCE ROUTE |
|---|---|---|
| species level | established gut pathogen | genome supported |
| **DISTINCT FRAGMENTS** | **INFORMATIVE REGIONS** | **INFORMATIVE BASES** |
| 544 | 1 | 3,483 |
| **SHARED-SEQUENCE FRAGMENTS** | **MEDIAN IDENTITY** | **INFORMATIVE BREADTH** |
| 1 (not counted as support) | 100.0% | 0.04% |

**What would confirm this:** clinical stool NAAT, clinical stool antigen, modified acid fast stain.

below identity gate · duplicate family collapsed · partial query alignment

‹ at a glance

### Cyclospora cayetanensis (lineage A)

TRACE, NOT ENOUGH TO CONFIRM

**WHAT IT IS**

A single-celled intestinal parasite spread by eating fresh produce or water contaminated with human faeces. Outbreaks in Europe and North America are repeatedly traced to imported soft fruit, salad leaves, coriander and mangetout. Unusually, the oocysts need days to weeks in the environment before becoming infectious, which is why it does not spread directly from person to person.

**WHY IT MATTERS**

It causes a distinctive and often prolonged illness: watery diarrhoea that comes and goes in waves, profound fatigue, loss of appetite and weight loss, which can drag on for six weeks or more if untreated. It is one of the more miserable travellers' and foodborne infections. Importantly, it responds very well to a specific prescription antibiotic, co-trimoxazole, usually within a few days, so there is a real cost to not being treated. A confirmed detection should be taken to a doctor.

**HOW MUCH IS HERE**

**Faint signal** in this sample: 9.4 per million, 0.0009% of analysed DNA, from 53 distinct DNA fragments across 1 separate regions of its genome. A small fraction of the sample's dna. For this organism, it is not a normal resident; healthy stool does not usually contain it. The aligned fragments cover **0.00%** of this organism's reference genome. A genuinely present organism is covered broadly; a small percentage means the reads sit on a slice of the genome that relatives may share, which is why breadth is reported next to the count.

**WHAT WOULD MAKE THE AMOUNT CONCERNING**

This is not an overgrowth question. Any detection is significant because it indicates real exposure to an organism that does not belong there. Sequencing cannot say whether the parasite is still active or whether it explains current symptoms, but the relapsing pattern of the illness makes symptom history more informative than quantity.

**WHAT TO DO ABOUT IT**

Nothing is listed here, and that is a deliberate statement about the evidence rather than an oversight. Cyclospora cannot be maintained in laboratory culture, which is the practical reason almost no natural products have ever been tested against it, and there is no credible published evidence that any supplement, botanical, probiotic or food reduces Cyclospora infection in humans or animals. Any product marketed for it is making a claim with nothing behind it. This matters more than usual because there is an effective treatment: co-trimoxazole, taken as a prescription course, usually resolves the illness in a few days, with ciprofloxacin as an alternative for those with sulfonamide allergy. Please see a doctor rather than treating this yourself. Supportive measures such as fluid and electrolyte replacement are sensible alongside proper treatment, not instead of it.

organism sequence not infection diagnosis

| RESOLVED TO | ORGANISM CLASS | REFERENCE ROUTE |
|---|---|---|
| species level | established gut pathogen | genome supported |
| **DISTINCT FRAGMENTS** | **INFORMATIVE REGIONS** | **INFORMATIVE BASES** |
| 53 | 1 | 770 |
| **MEDIAN IDENTITY** | **INFORMATIVE BREADTH** | |
| 99.3% | 0.00% | |

**What would confirm this:** clinical stool NAAT, modified acid fast stain, clinical stool ova and parasite microscopy.

below identity gate · partial query alignment · duplicate family collapsed

## Parasitic worms

Nothing found. 118 of 125 targets in this group were searched and none returned supporting or candidate sequence. 6 matched only sequence shared with relatives, which cannot name an organism.

## Microsporidia pathogens

Nothing found. all 15 targets in this group were searched and none returned supporting or candidate sequence.

## Fungal pathogens

1 supported, 5 candidate, out of 92 targets searched in this group.

‹ at a glance

### Geotrichum candidum

ENOUGH SEQUENCE TO NAME IT

**WHAT IT IS**

The white, velvety mould that forms the rind of Camembert, Brie, Saint-Nectaire and many other surface-ripened cheeses, and which is also used in some fermented milks. It is deliberately added to cheese and is present in soil, silage and raw milk.

**WHY IT MATTERS**

In the overwhelming majority of cases it does not matter at all: it is a food organism eaten on purpose by millions of people. It has been reported very rarely as an opportunist in patients with leukaemia and profound neutropenia, which is true of a great many environmental fungi and does not make it a threat to anyone with a working immune system.

**HOW MUCH IS HERE**

**Clear signal** in this sample: 204 per million, 0.02% of analysed DNA, from 1,156 distinct DNA fragments across 87 separate regions of its genome. Well represented in the sample's dna. For this organism, it is normally arrives with food or from the environment. The aligned fragments cover **0.56%** of this organism's reference genome. A genuinely present organism is covered broadly; a small percentage means the reads sit on a slice of the genome that relatives may share, which is why breadth is reported next to the count.

**WHAT WOULD MAKE THE AMOUNT CONCERNING**

No meaningful threshold exists. Abundance tracks cheese and dairy consumption in the preceding days. A sequencing result cannot separate a mould eaten on a cheese rind from anything else, and there is no published level above which this becomes a clinical concern in an immunocompetent person.

**WHAT TO DO ABOUT IT**

This is a normal dietary finding and no treatment is appropriate. Geotrichum candidum is the cheese-rind mould; it is put into food on purpose and passes through the gut. Attempting to eliminate it would mean giving up surface-ripened cheeses for no benefit, and taking antifungal supplements would disturb the gut bacterial community while achieving nothing worthwhile. There is no evidence that carrying it does any harm to a person with normal immunity.

organism sequence not infection diagnosis

| RESOLVED TO | ORGANISM CLASS | REFERENCE ROUTE |
|---|---|---|
| species level | background, food or environmental | genome supported |
| **DISTINCT FRAGMENTS** | **INFORMATIVE REGIONS** | **INFORMATIVE BASES** |
| 1,156 | 87 | 132,970 |
| **SHARED-SEQUENCE FRAGMENTS** | **MEDIAN IDENTITY** | **INFORMATIVE BREADTH** |
| 1 (not counted as support) | 99.3% | 0.58% |

**What would confirm this:** clinical culture with species identification, specialist referral.

**Caveat:** no extraction or library controls for this sample.

below identity gate · partial query alignment · duplicate family collapsed · controls unavailable

‹ at a glance

## Apophysomyces, human-infecting members

TRACE, NOT ENOUGH TO CONFIRM

### HOW MUCH IS HERE

**At the detection floor** in this sample: 0.35 per million, well under a thousandth of a percent, from 2 distinct DNA fragments across 1 separate regions of its genome. Barely above what this depth can see. For this organism, it is normally arrives with food or from the environment. The aligned fragments cover **0.00%** of this organism's reference genome. A genuinely present organism is covered broadly; a small percentage means the reads sit on a slice of the genome that relatives may share, which is why breadth is reported next to the count.

organism sequence not infection diagnosis

| RESOLVED TO | ORGANISM CLASS | REFERENCE ROUTE |
|---|---|---|
| genus level | opportunist, context dependent | genome supported |
| **DISTINCT FRAGMENTS** | **INFORMATIVE REGIONS** | **INFORMATIVE BASES** |
| 2 | 1 | 119 |
| **SHARED-SEQUENCE FRAGMENTS** | **MEDIAN IDENTITY** | **INFORMATIVE BREADTH** |
| 67 (not counted as support) | 96.0% | 0.00% |

**What would confirm this:** tissue biopsy or imaging, clinical culture with species identification, specialist referral.

below identity gate · partial query alignment · species resolution limited

‹ at a glance

## Debaryomyces hansenii

TRACE, NOT ENOUGH TO CONFIRM

### WHAT IT IS

A salt-tolerant yeast that is one of the most common fungi in cheese, cured meats, salted fish, brines and pickles. Its ability to grow at high salt concentrations is exactly why it dominates those foods, and it is widely used as a ripening culture.

### WHY IT MATTERS

In healthy people it does not matter. It is among the most frequently eaten yeasts in a Western diet. Some research has linked Debaryomyces to impaired healing of intestinal wounds in Crohn's disease, but this is early experimental work in mice and tissue, not a demonstrated cause of disease in people, and it is not a basis for treating anybody.

### HOW MUCH IS HERE

**At the detection floor** in this sample: 0.18 per million, well under a thousandth of a percent, from 1 distinct DNA fragments across 1 separate regions of its genome. Barely above what this depth can see. For this organism, it is normally arrives with food or from the environment. The aligned fragments cover **0.00%** of this organism's reference genome. A genuinely present organism is covered broadly; a small percentage means the reads sit on a slice of the genome that relatives may share, which is why breadth is reported next to the count.

### WHAT WOULD MAKE THE AMOUNT CONCERNING

No threshold has been established and none is meaningful. Read counts follow what has been eaten. Sequencing gives no way to tell yeast that came off a slice of cheese from anything with clinical relevance.

### WHAT TO DO ABOUT IT

A normal dietary finding requiring no action. Debaryomyces hansenii comes from cheese, cured meat, salted fish and pickles, and detecting it says that you eat those foods. Trying to remove it is neither necessary nor advisable: it would mean an arbitrary dietary restriction with no health benefit, and antifungal supplements taken for it would suppress beneficial bacteria for nothing. The tentative research link to Crohn's disease healing is preliminary animal and laboratory work and does not justify treating a healthy person, or indeed a person with Crohn's disease, on the basis of a stool sequencing result.

organism sequence not infection diagnosis

| RESOLVED TO | ORGANISM CLASS | REFERENCE ROUTE |
|---|---|---|
| species level | background, food or environmental | genome supported |
| **DISTINCT FRAGMENTS** | **INFORMATIVE REGIONS** | **INFORMATIVE BASES** |
| 1 | 1 | 151 |
| **MEDIAN IDENTITY** | **INFORMATIVE BREADTH** | |
| 96.7% | 0.00% | |

**What would confirm this:** clinical culture with species identification, specialist referral.

below identity gate

‹ at a glance

## Malassezia globosa

TRACE, NOT ENOUGH TO CONFIRM

HOW MUCH IS HERE

**At the detection floor** in this sample: 0.18 per million, well under a thousandth of a percent, from 1 distinct DNA fragments across 1 separate regions of its genome. Barely above what this depth can see. For this organism, it is normally arrives with food or from the environment. The aligned fragments cover **0.00%** of this organism's reference genome. A genuinely present organism is covered broadly; a small percentage means the reads sit on a slice of the genome that relatives may share, which is why breadth is reported next to the count.

organism sequence not infection diagnosis

| RESOLVED TO | ORGANISM CLASS | REFERENCE ROUTE |
|---|---|---|
| species level | background, food or environmental | genome supported |
| DISTINCT FRAGMENTS | INFORMATIVE REGIONS | INFORMATIVE BASES |
| 1 | 1 | 151 |
| MEDIAN IDENTITY | INFORMATIVE BREADTH | |
| 100.0% | 0.00% | |

**What would confirm this:** clinical culture with species identification, dermatology referral with skin sampling.

‹ at a glance

## Malassezia restricta

TRACE, NOT ENOUGH TO CONFIRM

HOW MUCH IS HERE

**At the detection floor** in this sample: 0.53 per million, well under a thousandth of a percent, from 3 distinct DNA fragments across 3 separate regions of its genome. Barely above what this depth can see. For this organism, it is normally arrives with food or from the environment. The aligned fragments cover **0.01%** of this organism's reference genome. A genuinely present organism is covered broadly; a small percentage means the reads sit on a slice of the genome that relatives may share, which is why breadth is reported next to the count.

organism sequence not infection diagnosis

| RESOLVED TO | ORGANISM CLASS | REFERENCE ROUTE |
|---|---|---|
| species level | background, food or environmental | genome supported |
| DISTINCT FRAGMENTS | INFORMATIVE REGIONS | INFORMATIVE BASES |
| 3 | 3 | 417 |
| MEDIAN IDENTITY | INFORMATIVE BREADTH | |
| 100.0% | 0.01% | |

**What would confirm this:** clinical culture with species identification, dermatology referral with skin sampling.

‹ at a glance

## Saksenaea, human-infecting members

TRACE, NOT ENOUGH TO CONFIRM

HOW MUCH IS HERE

**Small signal** in this sample: 23 per million, 0.0023% of analysed DNA, from 129 distinct DNA fragments across 1 separate regions of its genome. A minor component of the sample's dna. For this organism, it is normally arrives with food or from the environment. The aligned fragments cover **0.00%** of this organism's reference genome. A genuinely present organism is covered broadly; a small percentage means the reads sit on a slice of the genome that relatives may share, which is why breadth is reported next to the count.

organism sequence not infection diagnosis

| RESOLVED TO | ORGANISM CLASS | REFERENCE ROUTE |
|---|---|---|
| genus level | opportunist, context dependent | genome supported |
| DISTINCT FRAGMENTS | INFORMATIVE REGIONS | INFORMATIVE BASES |
| 129 | 1 | 306 |
| SHARED-SEQUENCE FRAGMENTS | MEDIAN IDENTITY | INFORMATIVE BREADTH |
| 2 (not counted as support) | 100.0% | 0.00% |

**What would confirm this:** tissue biopsy or imaging, clinical culture with species identification, specialist referral.

partial query alignment · duplicate family collapsed · species resolution limited

## Viral pathogens

Nothing found. 30 of 32 targets in this group were searched and none returned supporting or candidate sequence. A further 32 are RNA viruses, which carry no DNA for a DNA test to find. 1 matched only sequence shared with relatives, which cannot name an organism.

## Matched only shared sequence

These targets picked up reads, but every read fell in sequence they share with close relatives. That is not evidence for the named organism and it is not evidence against it. They are listed so the signal is not silently discarded.

| ORGANISM | GROUP | WHAT THE SEQUENCE SHOWS | HOW MUCH | STRENGTH OF EVIDENCE |
|---|---|---|---|---|
| **Achromobacter xylosoxidans** | Bacterial pathogens | shared sequence only — cannot tell apart | **at the detection floor** 0.88 per million · well under a thousandth of a percent · 5 fragments in 3 regions | 0.01% of its genome covered · 98.6% identity |
| **Acinetobacter baumannii–calcoaceticus complex** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Acremonium / Sarocladium, clinically relevant members** | Fungal pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Aeromonas caviae** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Aeromonas dhakensis** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Aeromonas hydrophila** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Aeromonas jandaei** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Aeromonas trota** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Aeromonas veronii** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Aliarcobacter cryaerophilus** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Aliarcobacter skirrowii** | Bacterial pathogens | shared sequence only — cannot tell apart | **at the detection floor** 0.18 per million · well under a thousandth of a percent · 1 fragments in 1 regions | 0.01% of its genome covered · 96.0% identity |
| **Aspergillus nidulans** | Fungal pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Aspergillus terreus** | Fungal pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Bacillus cereus sensu lato** | Bacterial pathogens | shared sequence only — cannot tell apart | **faint signal** 4.1 per million · 0.0004% of analysed DNA · 23 fragments in 13 regions | 0.04% of its genome covered · 96.6% identity |
| **Bacteriophage background (including viral sequence clusters)** | Viral pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Basidiobolus, verified relevant relatives** | Fungal pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Blastocystis spp.** | Protozoa pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Brachyspira aalborgi** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Brachyspira pilosicoli** | Bacterial pathogens | shared sequence only — cannot tell apart | **faint signal** 5.1 per million · 0.0005% of analysed DNA · 29 fragments in 2 regions | 0.06% of its genome covered · 100.0% identity |
| **Campylobacter coli** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Campylobacter concisus** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |

| ORGANISM | GROUP | WHAT THE SEQUENCE SHOWS | HOW MUCH | STRENGTH OF EVIDENCE |
|---|---|---|---|---|
| **Campylobacter curvus** | Bacterial pathogens | shared sequence only — cannot tell apart | **at the detection floor**<br>0.18 per million · well under a thousandth of a percent · 1 fragments in 1 regions | 0.01% of its genome covered · 95.4% identity |
| **Campylobacter fetus** | Bacterial pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Campylobacter hyointestinalis** | Bacterial pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Campylobacter jejuni** | Bacterial pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Campylobacter lari** | Bacterial pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Campylobacter showae** | Bacterial pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Campylobacter upsaliensis** | Bacterial pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Candida albicans** | Fungal pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Candida auris (Candidozyma auris)** | Fungal pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Candida dubliniensis** | Fungal pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Candida krusei (Pichia kudriavzevii)** | Fungal pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Candida tropicalis** | Fungal pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Citrobacter freundii complex** | Bacterial pathogens | shared sequence only — cannot tell apart | **at the detection floor**<br>0.18 per million · well under a thousandth of a percent · 1 fragments in 1 regions | 0.00% of its genome covered · 96.7% identity |
| **Citrobacter koseri** | Bacterial pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Cladophialophora, clinically relevant species** | Fungal pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Clostridium baratii (neurotoxigenic)** | Bacterial pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Clostridium botulinum** | Bacterial pathogens | shared sequence only — cannot tell apart | **at the detection floor**<br>0.71 per million · well under a thousandth of a percent · 4 fragments in 3 regions | 0.01% of its genome covered · 95.3% identity |
| **Clostridium butyricum (neurotoxigenic)** | Bacterial pathogens | shared sequence only — cannot tell apart | **at the detection floor**<br>0.35 per million · well under a thousandth of a percent · 2 fragments in 2 regions | 0.01% of its genome covered · 97.7% identity |
| **Clostridium innocuum** | Bacterial pathogens | shared sequence only — cannot tell apart<br>shared sequence from relatives found in this sample; species not found by the organism inventory | **sequence from relatives**<br>4,501 fragments, shared with relatives · from Clostridium sp CAG 138 (0.837%), Clostridium sp CAG 413 (0.358%) · 4,501 fragments in 98 regions | 2.58% of its genome covered · 98.1% identity |
| **Clostridium perfringens** | Bacterial pathogens | shared sequence only — cannot tell apart | **at the detection floor**<br>0.71 per million · well under a thousandth of a percent · 4 fragments in 3 regions | 0.02% of its genome covered · 95.9% identity |
| **Clostridium septicum** | Bacterial pathogens | shared sequence only — cannot tell apart | **at the detection floor**<br>0.53 per million · well under a thousandth of a percent · 3 fragments in 3 regions | 0.01% of its genome covered · 98.7% identity |
| **Clostridium tertium** | Bacterial pathogens | shared sequence only — cannot tell apart | **small signal**<br>20 per million · 0.0020% of analysed DNA · 112 fragments in 3 regions | 0.09% of its genome covered · 98.0% identity |
| **Cronobacter malonaticus** | Bacterial pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |

| ORGANISM | GROUP | WHAT THE SEQUENCE SHOWS | HOW MUCH | STRENGTH OF EVIDENCE |
|---|---|---|---|---|
| **Cronobacter sakazakii** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Cronobacter turicensis** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Cryptococcus neoformans species complex** | Fungal pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Cryptosporidium muris** | Protozoa pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Curated clinically relevant nontuberculous mycobacteria** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Diffusely adherent Escherichia coli (DAEC)** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Eimeria spp.** | Protozoa pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Entamoeba dispar** | Protozoa pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Entamoeba histolytica** | Protozoa pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Entamoeba moshkovskii** | Protozoa pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Entamoeba nuttalli** | Protozoa pathogens | shared sequence only — cannot tell apart | **small signal** 15 per million · 0.0015% of analysed DNA · 84 fragments in 1 regions | 0.01% of its genome covered · 96.0% identity |
| **Enterobacter asburiae** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Enterobacter cloacae** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Enterobacter hormaechei** | Bacterial pathogens | shared sequence only — cannot tell apart | **at the detection floor** 0.35 per million · well under a thousandth of a percent · 2 fragments in 2 regions | 0.01% of its genome covered · 96.0% identity |
| **Enterobacter kobei** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Enterobacter ludwigii** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Enterobacter roggenkampii** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Enterobius vermicularis** | Parasitic worms | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Enterococcus casseliflavus** | Bacterial pathogens | shared sequence only — cannot tell apart | **at the detection floor** 0.35 per million · well under a thousandth of a percent · 2 fragments in 2 regions | 0.01% of its genome covered · 95.9% identity |
| **Enterococcus faecium** | Bacterial pathogens | shared sequence only — cannot tell apart | **small signal** 24 per million · 0.0024% of analysed DNA · 134 fragments in 96 regions | 0.67% of its genome covered · 98.0% identity |
| **Enterococcus gallinarum** | Bacterial pathogens | shared sequence only — cannot tell apart | **small signal** 11 per million · 0.0011% of analysed DNA · 61 fragments in 6 regions | 0.17% of its genome covered · 99.3% identity |
| **Enteroinvasive Escherichia coli (EIEC)** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Enterotoxigenic Bacteroides fragilis (ETBF)** | Bacterial pathogens | shared sequence only — cannot tell apart shared sequence from relatives found in this sample; species not found by the organism inventory | **sequence from relatives** 3,328 fragments, shared with relatives · from Bacteroides clarus (1.39%), Bacteroides finegoldii (1.17%) · 3,328 fragments in 103 regions | 1.94% of its genome covered · 97.4% identity disease genes not assessed |

| ORGANISM | GROUP | WHAT THE SEQUENCE SHOWS | HOW MUCH | STRENGTH OF EVIDENCE |
|---|---|---|---|---|
| **Epidermophyton, human-infecting species (Epidermophyton floccosum)** | Fungal pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Escherichia albertii** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Escherichia coli (commensal reference)** | Bacterial pathogens | shared sequence only — cannot tell apart | **faint signal** 2.8 per million · 0.0003% of analysed DNA · 16 fragments in 16 regions | 0.05% of its genome covered · 100.0% identity |
| **Escherichia fergusonii** | Bacterial pathogens | shared sequence only — cannot tell apart species not resolvable within the Escherichia coli / Shigella complex | **at the detection floor** 0.53 per million · well under a thousandth of a percent · 3 fragments in 3 regions | 0.01% of its genome covered · 100.0% identity |
| **Fonsecaea, clinically relevant species** | Fungal pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Fusarium, clinically relevant species complexes** | Fungal pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Fusobacterium necrophorum** | Bacterial pathogens | shared sequence only — cannot tell apart shared sequence from relatives found in this sample; species not found by the organism inventory | **sequence from relatives** 525 fragments, shared with relatives · from a relative that is present · 525 fragments in 6 regions | 0.86% of its genome covered · 100.0% identity |
| **Fusobacterium nucleatum** | Bacterial pathogens | shared sequence only — cannot tell apart | **at the detection floor** 0.71 per million · well under a thousandth of a percent · 4 fragments in 4 regions | 0.03% of its genome covered · 96.9% identity |
| **Giardia duodenalis** | Protozoa pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Hafnia alvei** | Bacterial pathogens | shared sequence only — cannot tell apart | **at the detection floor** 0.71 per million · well under a thousandth of a percent · 4 fragments in 4 regions | 0.01% of its genome covered · 99.3% identity |
| **Hafnia paralvei** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Helicobacter bizzozeronii** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Helicobacter cinaedi** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Helicobacter fennelliae** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Helicobacter heilmannii sensu stricto** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Helicobacter pylori** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Helicobacter salomonis** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Helicobacter suis** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Hymenolepis diminuta** | Parasitic worms | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Klebsiella aerogenes** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Klebsiella grimontii** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Klebsiella michiganensis** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Klebsiella oxytoca** | Bacterial pathogens | shared sequence only — cannot tell apart | **at the detection floor** 0.53 per million · well under a thousandth of a percent · 3 fragments in 3 regions | 0.01% of its genome covered · 99.3% identity |

| ORGANISM | GROUP | WHAT THE SEQUENCE SHOWS | HOW MUCH | STRENGTH OF EVIDENCE |
|---|---|---|---|---|
| **Klebsiella pneumoniae** | Bacterial pathogens | shared sequence only — cannot tell apart | **at the detection floor**<br>0.71 per million · well under a thousandth of a percent · 4 fragments in 4 regions | 0.01% of its genome covered · 100.0% identity |
| **Klebsiella quasipneumoniae** | Bacterial pathogens | shared sequence only — cannot tell apart | **at the detection floor**<br>0.18 per million · well under a thousandth of a percent · 1 fragments in 1 regions | 0.00% of its genome covered · 95.4% identity |
| **Klebsiella variicola** | Bacterial pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Laribacter hongkongensis** | Bacterial pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Leishmania spp.** | Protozoa pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Listeria monocytogenes** | Bacterial pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Lomentospora prolificans** | Fungal pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Magnusiomyces capitatus** | Fungal pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Malassezia furfur** | Fungal pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Malassezia pachydermatis** | Fungal pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Mediterraneibacter gnavus** | Bacterial pathogens | shared sequence only — cannot tell apart<br>shared sequence from relatives found in this sample; species not found by the organism inventory | **sequence from relatives**<br>19,156 fragments, shared with relatives · from Dorea hominis (0.102%), Mediterraneibacter lactaris (1.13%) · 19,156 fragments in 106 regions | 8.07% of its genome covered · 98.0% identity |
| **Morganella morganii** | Bacterial pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Mycobacterium tuberculosis complex** | Bacterial pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Necator americanus** | Parasitic worms | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Parvimonas micra** | Bacterial pathogens | shared sequence only — cannot tell apart | **small signal**<br>10 per million · 0.0010% of analysed DNA · 59 fragments in 13 regions | 0.31% of its genome covered · 97.9% identity |
| **Pentatrichomonas hominis** | Protozoa pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Plesiomonas shigelloides** | Bacterial pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Proteus mirabilis** | Bacterial pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Proteus penneri** | Bacterial pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Proteus vulgaris** | Bacterial pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Prototheca, clinically documented species** | Fungal pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Providencia alcalifaciens** | Bacterial pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Providencia rettgeri** | Bacterial pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Providencia stuartii** | Bacterial pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Pseudomonas aeruginosa** | Bacterial pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |

| ORGANISM | GROUP | WHAT THE SEQUENCE SHOWS | HOW MUCH | STRENGTH OF EVIDENCE |
|---|---|---|---|---|
| **Pseudoterranova spp.** | Parasitic worms | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Rhodotorula glutinis** | Fungal pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Rhodotorula minuta (Cystobasidium minutum)** | Fungal pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Salmonella bongori** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Salmonella enterica serovar Paratyphi A** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Salmonella enterica serovar Paratyphi B** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Salmonella enterica serovar Paratyphi C** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Salmonella enterica serovar Typhi** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Salmonella enterica subsp. arizonae** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Salmonella enterica subsp. diarizonae** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Salmonella enterica subsp. enterica** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Salmonella enterica subsp. houtenae** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Salmonella enterica subsp. indica** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Salmonella enterica subsp. salamae** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Scedosporium apiospermum / boydii complex** | Fungal pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Serratia marcescens** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Shigella boydii** | Bacterial pathogens | shared sequence only — cannot tell apart | **at the detection floor** 0.35 per million · well under a thousandth of a percent · 2 fragments in 1 regions | 0.01% of its genome covered · 100.0% identity |
| **Shigella dysenteriae** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Shigella flexneri** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Shigella sonnei** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |
| **Shigella/enteroinvasive Escherichia coli (combined group)** | Bacterial pathogens | shared sequence only — cannot tell apart evidence aggregated from the complex; species not resolvable | — no qualifying sequence to measure | 0.01% of its genome covered disease genes not found |
| **Sporothrix schenckii complex (pathogenic clade)** | Fungal pathogens | shared sequence only — cannot tell apart | **at the detection floor** 0.18 per million · well under a thousandth of a percent · 1 fragments in 1 regions | 0.00% of its genome covered · 97.1% identity |
| **Staphylococcus aureus** | Bacterial pathogens | shared sequence only — cannot tell apart | **faint signal** 3.0 per million · 0.0003% of analysed DNA · 17 fragments in 7 regions | 0.05% of its genome covered · 98.0% identity |
| **Stenotrophomonas maltophilia** | Bacterial pathogens | shared sequence only — cannot tell apart | — no qualifying sequence to measure | 0.00% of its genome covered |

| ORGANISM | GROUP | WHAT THE SEQUENCE SHOWS | HOW MUCH | STRENGTH OF EVIDENCE |
|---|---|---|---|---|
| **Streptococcus agalactiae** | Bacterial pathogens | shared sequence only — cannot tell apart | **faint signal**<br>2.7 per million · 0.0003% of analysed DNA · 15 fragments in 10 regions | 0.10% of its genome covered · 98.6% identity |
| **Streptococcus anginosus group** | Bacterial pathogens | shared sequence only — cannot tell apart | **faint signal**<br>9.5 per million · 0.0010% of analysed DNA · 54 fragments in 32 regions | 0.36% of its genome covered · 98.0% identity |
| **Strongyloides ratti** | Parasitic worms | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Syncephalastrum, human-infecting members** | Fungal pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Taenia solium** | Parasitic worms | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Trichophyton, human-infecting species** | Fungal pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Trichosporon / Cutaneotrichosporon, curated human-infecting members** | Fungal pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Trichosporon asahii** | Fungal pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Trichosporon inkin** | Fungal pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Tropheryma whipplei** | Bacterial pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Vibrio alginolyticus** | Bacterial pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Vibrio cholerae** | Bacterial pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Vibrio fluvialis** | Bacterial pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Vibrio furnissii** | Bacterial pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Vibrio mimicus** | Bacterial pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Vibrio parahaemolyticus** | Bacterial pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Vibrio vulnificus** | Bacterial pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Yersinia enterocolitica** | Bacterial pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |
| **Yersinia pseudotuberculosis** | Bacterial pathogens | shared sequence only — cannot tell apart | —<br>no qualifying sequence to measure | 0.00% of its genome covered |

## Toxin, virulence and resistance genes

Whether a gene is present is asked separately from which organism carries it. A resistance gene found in stool belongs to *some* organism in the community, and unless the read or contig physically links it to a named genome, that organism is unknown. Gene presence is also not a prediction of treatment failure.

| GENE | KIND | WHAT WAS FOUND | CARRIER ORGANISM | SUPPORT |
|---|---|---|---|---|
| **Aminoglycoside phosphotransferases (aph)** | amr<br>aminoglycoside | present, complete | carrier not established | 20 fragments |
| **Erm rRNA methyltransferases (erm)** | amr<br>macrolide lincosamide | present, complete | carrier not established | 66 fragments |
| **Lincosamide nucleotidyltransferases (lnu)** | amr<br>macrolide lincosamide | present, complete | carrier not established | 13 fragments |
| **Mef macrolide efflux (mef)** | amr<br>macrolide lincosamide | present, complete | carrier not established | 111 fragments |
| **Tetracycline ribosomal protection proteins (tet RPP)** | amr<br>tetracycline | present, complete | carrier not established | 72 fragments |

| GENE | KIND | WHAT WAS FOUND | CARRIER ORGANISM | SUPPORT |
|---|---|---|---|---|
| **Trimethoprim-resistant dihydrofolate reductases (dfr)** | amr<br>sulfonamide trimethoprim | present, complete | carrier not established | 29 fragments |

### What this screen did and did not cover

Every number below is counted from the run itself, not from a fixed marketing total. Organisms, determinants and subtypes are counted separately and never summed into a single figure, because they are different kinds of question.

| | | |
|---|---|---|
| **Targets in the catalogue** | 486 | every organism this screen carries a definition for |
| **Searched, answered at its own rank** | 409 | own reference sequence, aligned, and separable from its relatives |
| **Searched, answers for a group** | 30 | whole genome shared with a catalogue relative; the toxin and virulence gene screen below is what separates these |
| **Searched, masked by the host** | 1 | sequence also present in the human reference genome, so a matching read cannot be assigned to the organism rather than to you |
| **RNA genome, outside a DNA test** | 32 | no DNA exists in the sample to find; needs a stool RNA panel |
| **Reference not available yet** | 14 | no public genome or marker sequence good enough to search; named below |
| **Determinants searched** | 79 | toxin, virulence and resistance genes, searched independently of species |
| **Determinants not searched** | 14 | no usable reference for the gene family |

**The 14 targets with no reference yet.** *Acanthocephalus rauschi*, *Additional curated human DNA viruses (systemic watch)*, *Bertiella mucronata*, *Common food-animal genome competitors*, *Episthmium spp.*, *Fish ciliate competitors (including fish Balantidium)*, *Free-living and plant nematode competitors*, *Human filarial parasites*, *Inermicapsifer madagascariensis*, *Iodamoeba buetschlii*, *Neighboring fluke and tapeworm competitors*, *Other common dietary yeasts (curated group)*, *Other human stool-associated parvoviruses*, *Reclassified opportunistic cryptococcal yeasts (former Cryptococcus spp.)*. Each is in the catalogue and each will be searched as soon as a reference exists for it. Until then this report says nothing either way about them.

**The 31 targets that answer for a group.** *Animal Cryptosporidium competitors* (searched as *Cryptosporidium parvum*), *Animal Cystoisospora competitors* (searched as *Cystoisospora suis*), *Animal Giardia competitors* (searched as *Giardia duodenalis*), *Anncaliia connori* (searched as *Anncaliia algerae*), *Anncaliia vesicularum* (searched as *Anncaliia algerae*), *Aspergillus, clinically relevant cryptic species* (searched as *Aspergillus flavus*), *Bacteroides fragilis group* (searched as *Enterotoxigenic Bacteroides fragilis (ETBF)*), *Blastomyces, relevant relatives* (searched as *Blastomyces gilchristii*), *Cryptosporidium mink genotype* (searched as *Cryptosporidium parvum*), *Cyclospora ashfordi (lineage B)* (searched as *Cyclospora cayetanensis (lineage A)*), *Cyclospora cayetanensis sensu lato* (searched as *Cyclospora cayetanensis (lineage A)*), *Cyclospora henanensis (lineage C)* (searched as *Cyclospora cayetanensis (lineage A)*), *Echinostoma fujianensis* (searched as *Echinostoma echinatum*), *Echinostoma ilocanum* (searched as *Echinostoma echinatum*), *Enteroaggregative Escherichia coli (EAEC)* (searched as *Diffusely adherent Escherichia coli (DAEC)*), *Enterobacter cloacae complex* (searched as *Enterobacter cloacae*), *Enteropathogenic Escherichia coli (EPEC)* (searched as *Diffusely adherent Escherichia coli (DAEC)*), *Enterotoxigenic Escherichia coli (ETEC)* (searched as *Diffusely adherent Escherichia coli (DAEC)*), *Hybrid diarrheagenic Escherichia coli pathotypes* (searched as *Diffusely adherent Escherichia coli (DAEC)*), *Klebsiella oxytoca complex* (searched as *Klebsiella oxytoca*), *Klebsiella pneumoniae complex* (searched as *Klebsiella pneumoniae*), *Microsporidium ceylonensis* (searched as *Microsporidium africanum*), *Nakaseomyces glabratus group, unresolved (Candida glabrata / nivariensis / bracarensis)* (searched as *Candida dubliniensis*), *Nonhuman Sarcocystis competitors* (searched as *Sarcocystis sigmoideus*), *Pichia, curated dietary species* (searched as *Candida krusei (Pichia kudriavzevii)*), *Salmonella enterica, nontyphoidal serovars* (searched as *Salmonella enterica subsp. enterica*), *Shiga toxin-producing Escherichia coli (STEC)* (searched as *Diffusely adherent Escherichia coli (DAEC)*), *Shigella/enteroinvasive Escherichia coli (combined group)* (searched as *Diffusely adherent Escherichia coli (DAEC)*), *Strongyloides fuelleborni subsp. kellyi* (searched as *Strongyloides fuelleborni subsp. fuelleborni*), *Trachipleistophora anthropophthera* (searched as *Trachipleistophora hominis*), *Trichinella recognized genotypes (unnamed)* (searched as *Trichinella papuae*). These were searched. Their genomes are the same as a relative's in this catalogue, so a read cannot say which of the pair it came from; where the difference between them is a gene rather than a genome — the diarrhoea-causing *E. coli* types, toxin-producing *Bacteroides fragilis* — the gene screen above answers it.

**Limits that apply to every finding above.** This is a research screen on a stool sample, not a diagnostic test. Sequence presence is not infection, and it is not a measure of how much organism is there: read counts depend on how much of your sample was that organism's DNA, not on its abundance in you. A negative here does not rule out an organism that sheds intermittently, sits in tissue rather than the gut lumen, or has an RNA genome that a DNA library cannot see. Nothing here should change treatment without a clinician and, where listed, a confirmatory test.

## Antibiotic-resistance genes, by class

Resistance genes present in the community, grouped by the antibiotics they act against. **A gene is not a phenotype.** Carrying one of these does not establish that an antibiotic would fail for you, and not carrying one does not establish that it would work. Gut bacteria carry resistance genes routinely, most of them in harmless residents.

Read from the existing determinant screen and grouped; no call is recomputed and no phenotype is inferred.

| ANTIBIOTIC CLASS | GENE FAMILIES | EVIDENCE | HOW IT WORKS | READS |
|---|---|---|---|---|
| **Macrolides, lincosamides and streptogramin B** | **3** mef, erm, lnu | 3 acquired genes | ribosomal methylation (erm), efflux (mef) or enzymatic inactivation (lnu) | 190 |
| **Sulfonamides and trimethoprim** | **1** dfr | 1 acquired gene | alternative dihydropteroate synthase or dihydrofolate reductase | 29 |
| **Aminoglycosides** | **4** aac, aph, ant, rmt | 1 acquired gene · 3 unresolved | enzymatic modification (aac, aph, ant) or 16S methylation (armA) | 48 |

| ANTIBIOTIC CLASS | GENE FAMILIES | EVIDENCE | HOW IT WORKS | READS |
|---|---|---|---|---|
| **Tetracyclines** | **3** tet_efflux, tet_ribosomal_protection, tet_enzymatic | 1 acquired gene · 2 unresolved | ribosomal protection, efflux or enzymatic inactivation | 189 |
| **Oxazolidinones and phenicols** | **2** cfr, cat | 2 unresolved | 23S rRNA methylation (cfr) or enzymatic inactivation (cat) | 30 |

**13 distinct gene families** across 5 of 10 classes. Families are counted once even when they appear in two classes, and a drug is never counted again as its own class. All 13 detected determinants are unlinked: the reads show the gene is present in the community, not which organism carries it.

PART B — WHAT YOUR BACTERIA CAN MAKE AND BREAK DOWN

# 19 Your Microbial Functions in Detail

**What your microbes can make, break down and transform.**

One card per pathway, grouped as on the previous page and most notable first within each group: what the compound is, where you sit, what that means, and the research behind it.

## Fibre & Dietary Substrates

‹ at a glance
what you can do about it ›

## Dietary fibre and carbohydrate breakdown ▼

BELOW AVERAGE

The breadth of dietary carbohydrates your gut bacteria carry enzymes for, across starch, pectin, xylan and galactoside chemistry.

| | |
|---|---|
| YOUR RESULT | **549.3** copies per 100 genomes |
| PERCENTILE | **21st** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 578.7 – 811.5 (median 702.6) |
| MADE FROM | Dietary fibre, resistant starch and other carbohydrates reaching the colon |
| DNA FRAGMENTS MATCHED | 55,596 |




### What is driving this 19,053 matched fragments, by the organism each came closest to

**53% of the signal comes from organisms found in you:** *Lachnospira eligens* (5.27% of your community); *Bacteroides clarus* (1.39% of your community); *Faecalibacterium prausnitzii* (6.18% of your community). A further 30% matched *Bacteroides reticulotermitis*, whose genus is present but whose species was not resolved. The remaining 16% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Lachnospira eligens*** | 6% (1210 fragments) | yes · 5.265% | ● Beneficial |
| ***Bacteroides reticulotermitis*** | 6% (1204 fragments) | genus present; closest reference | — |
| ***Brachybacterium nesterenkovii*** | 3% (650 fragments) | not found; closest reference | — |
| ***Bacteroides clarus*** | 2% (464 fragments) | yes · 1.387% | ● Conditional |
| ***Faecalibacterium prausnitzii*** | 2% (459 fragments) | yes · 6.182% | ● Beneficial |

The organisms carrying this gene in your sample are *Lachnospira eligens*, *Bacteroides clarus*, *Faecalibacterium prausnitzii*; *Lachnospira eligens* alone accounts for 6% of the signal. Changing this reading means changing them.

### The organisms behind this, and what is known about them

| FOUND IN YOU | WHAT THE LITERATURE RECORDS |
|---|---|
| ***Faecalibacterium prausnitzii***<br>6.18% of your community · 2% of this signal | One of the most abundant butyrate producers in adult stool. Reduced relative abundance is one of the most replicated findings in Crohn's disease and ulcerative colitis, and lower levels are reported across several inflammatory and metabolic conditions. It is also simply variable between healthy people and across countries; a low level alone is not a diagnosis and no product restores the species itself.<br>■ when lower: Reduced F. prausnitzii is associated with Crohn's disease and with post-operative recurrence; the isolate had anti-inflammatory effects in vitro and in mouse colitis. *[Sokol H et al. PNAS 2008;105:16731. PMID 18936492]*<br>■ when lower: A 2017 review consolidates lower F. prausnitzii across IBD, IBS, obesity, coeliac disease and colorectal cancer cohorts, with phylogroup-dependent patterns. *[Lopez-Siles M et al. ISME J 2017;11:841. PMID 28072420]*<br>Association only — no study on record shows that changing this organism changes an outcome. |

### What your reading means below average · somewhat lower than most people

Fewer carbohydrate-active enzyme families than most samples carry. That can follow a narrow diet, and it is capacity rather than intake: adding a fibre nothing here can open would not help.

### How can I improve this?

Higher is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Lachnospira eligens* (5.27% of your community); *Bacteroides clarus* (1.39% of your community); *Faecalibacterium prausnitzii* (6.18% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it Research associates higher levels with better outcomes · Early evidence

Fourteen individual fibre and starch readings are built from these enzymes in the functions section. This panel is their shared evidence and a breadth summary; the specificity judgement happens per substrate.

A carbohydrate-active enzyme family usually acts on a class of molecules rather than on one, so a family alone answers a broader question than the one a reader asks. GH13 supports general amylolysis but not resistant starch in particular; a

beta-galactosidase is carried by every lactose-using organism and does not establish galactooligosaccharide use. The per-substrate readings therefore state whether the evidence is substrate-specific or only class-level, and a gene shared between two substrates appears on both cards and is counted once in any total. What none of this measures is how much fibre you eat: it is the capacity to use what arrives.

One of the genes behind this figure (amyA, lacZ, xynA) is matching distant relatives rather than close ones, so treat the number as an upper bound. The pathway is still detected — other genes in it match well.

Reference: Drula E. et al., "The carbohydrate-active enzyme database: functions and literature", Nucleic Acids Res 50:D571-D577 (2022), doi:10.1093/nar/gkab1045; Zheng J. et al., "dbCAN3", Nucleic Acids Res 51:W115-W121 (2023), doi:10.1093/nar/gkad328.

## Fermentation & Cross-feeding

‹ at a glance
what you can do about it ›

## Hydrogen sulfide (H2S) ▲

ABOVE AVERAGE

Hydrogen sulfide, a gas made by bacteria that breathe sulfate or break down sulfur-containing amino acids.

| | |
|---|---|
| YOUR RESULT | **96.5** copies per 100 genomes |
| PERCENTILE | 81st of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 61.8 – 88.0 (median 75.3) |
| MADE FROM | Dietary sulfate (preservatives, some drinking water), red meat, eggs, cruciferous vegetables |
| DNA FRAGMENTS MATCHED | 2,701 |




### What is driving this 2,564 matched fragments, by the organism each came closest to

**41% of the signal comes from organisms found in you:** *Blautia luti* (0.76% of your community); *Ruminococcus bromii* (3.40% of your community); *Blautia obeum* (0.29% of your community). A further 59% matched *Blautia wexlerae*, *Blautia producta*, whose genus is present but whose species was not resolved.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Blautia wexlerae*** | 42% (1090 fragments) | genus present; closest reference | — |
| ***Blautia luti*** | 16% (402 fragments) | yes · 0.755% | Conditional |
| ***Ruminococcus bromii*** | 12% (302 fragments) | yes · 3.396% | Beneficial |
| ***Blautia producta*** | 8% (197 fragments) | genus present; closest reference | — |
| ***Blautia obeum*** | 8% (193 fragments) | yes · 0.293% | Conditional |

The organisms carrying this gene in your sample are *Blautia luti*, *Ruminococcus bromii*, *Blautia obeum*. Changing this reading means changing them.

### The organisms behind this, and what is known about them

| FOUND IN YOU | WHAT THE LITERATURE RECORDS |
|---|---|
| ***Ruminococcus bromii***<br>3.40% of your community · 12% of this signal | Keystone degrader of resistant starch: it breaks down starch granules other species cannot, feeding the wider community. Its baseline level predicted who responded to a resistant-starch feeding trial with a butyrate rise.<br>■ : R. bromii is a keystone species for resistant starch degradation in the human colon. *[Ze X et al. ISME J 2012;6:1535. PMID 22343308]*<br>■ : Faecal butyrate rose with resistant potato starch only in baseline responders, and R. bromii predicted response. *[Venkataraman A et al. Microbiome 2016;4:33. PMID 27357127]*<br>Association only — no study on record shows that changing this organism changes an outcome. |
| ***Blautia obeum***<br>0.29% of your community · 8% of this signal | Abundant Lachnospiraceae member; Blautia is higher in some obesity cohorts and lower in others. No fixed reading.<br>■ : Review of Blautia as a widespread genus with mixed disease associations. *[Liu X et al. Gut Microbes 2021;13:1875796. PMID 33525961]*<br>Association only — no study on record shows that changing this organism changes an outcome. |

### What your reading means above average · somewhat higher than most people

Your gut carries more sulfide-producing capacity than most people's, largely from sulfate-reducing bacteria. Sustained high sulfide interferes with how colon cells use butyrate, and higher levels have been reported in ulcerative colitis. Dietary sulfate and sulfur-rich protein are the raw material.

**Why this colour.** Hydrogen sulfide stops colon cells burning butyrate and is raised in ulcerative colitis; a trace is normal signalling, a high reading is not.

**Two measurements of the same function disagree**

**Read this beside the sulfate-reducing species reading.** The hydrogen-sulfide capacity reading sits at the 81st percentile; the sulfate-reducing species reading is at the 16th. The two are measured differently — genes in the whole community against the abundance of the species characterised for the job — and they can separate when the genes are carried by organisms outside the curated list, which the gene search counts and the species list does not; the characterised species are present below the profiler's limit of detection, so their genes are counted while the organisms are not. Of the two, the gene reading is the better guide here, because this capacity does not depend on any one set of species. Hydrogen sulfide has two independent sources: dissimilatory sulfate reduction by Desulfovibrio and relatives, and cysteine degradation, which is widespread in ordinary gut bacteria. The gene panel sees both; the group sees only the sulfate reducers, which are a minority of stool even when sulfide capacity is high.

### How can I improve this?

Lower is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Blautia luti* (0.76% of your community); *Ruminococcus bromii* (3.40% of your community); *Blautia obeum* (0.29% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

**Research associates higher levels with worse outcomes · Reasonably supported**

Hydrogen sulfide impairs the ability of colon cells to burn butyrate for energy, and higher sulfide has been reported in ulcerative colitis. It is also a normal signalling molecule at lower concentrations, so this is a direction to notice rather than a hazard reading.

Roediger and colleagues showed that sulfide blocks butyrate oxidation in isolated colonocytes, which is the mechanism most often invoked for the ulcerative colitis association; Pitcher and Cummings measured higher faecal sulfide in active UC. Sulfate-reducing bacteria are also enriched in colorectal cancer cohorts, and hydrogen sulfide is genotoxic at colonic concentrations. Against that, sulfide at physiological levels is a signalling molecule with documented anti-inflammatory effects, and Desulfovibrio is a normal member of most healthy adult communities. The literature supports "sustained excess is adverse", not "sulfide is bad".

Reference: Pitcher MC, Cummings JH, Gut 39:1-4 (1996); Roediger WE et al., Gastroenterology 104:802-809 (1993); Carbonero F et al., Nat Rev Gastroenterol Hepatol 9:504-518 (2012)

### What the research says about changing this

**Higher hydrogen-sulfide (dsrA) capacity — surrogate evidence only** LANE D HUMAN · NON-RANDOMISED

Fired by: h2s gene capacity: somewhat high

Bismuth subsalicylate reduced ex-vivo fecal hydrogen-sulfide release in ten healthy adults (a surrogate, with salicylate and bismuth cautions), and a small sulfide-reducing diet study in UC gave mixed results. No disease endpoint has been shown to change.

**Studied:** Bismuth subsalicylate (otc drug) · **in:** people with clinically confirmed healthy adults, ages 18–60, in USA, n = 10 · **studied protocol (a source fact, not an instruction):** 524 mg four times daily for 3–7 days

Evidence lane D: non-randomized human, pilot, case-series or observational evidence. Replication: single study. Strongest design: nonrandomized study.

- Bismuth subsalicylate (524 mg four times daily for 3–7 days) reduced ex-vivo fecal hydrogen-sulfide release in ten healthy adults — a surrogate with no disease endpoint and salicylate/bismuth safety gates. [Suarez FL et al. Gastroenterology 1998;114:923] · lane D · single-arm in ten healthy adults · n = 10 · benefit

- **Against or limiting —** A small open-label eight-week sulfide-reducing diet in UC produced both responses and worsening — no basis for generic sulfur restriction. [4-SURE diet open-label study in ulcerative colitis (2022)] · lane X · open-label single-arm · n = 28 · mixed

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “sulfur toxicity” · “avoid sulfur foods”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance
what you can do about it ›

## Propionate (SCFA) ▲

ABOVE AVERAGE

Propionate, a short-chain fatty acid produced alongside butyrate but with a different reported relationship to cognition.

| | |
|---|---|
| YOUR RESULT | **26.4** copies per 100 genomes |
| PERCENTILE | 78th of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 16.4 – 25.3 (median 19.9) |
| MADE FROM | Dietary fibre, lactate, and the sugars fucose and rhamnose |
| DNA FRAGMENTS MATCHED | 5,120 |




### What is driving this 1,188 matched fragments, by the organism each came closest to

**21% of the signal comes from organisms found in you:** *Coprococcus eutactus* (2.46% of your community). A further 64% matched *Blautia caecimuris*, *Peptoniphilus stercorisuis*, *Clostridium symbiosum*, whose genus is present but whose species was not resolved. The remaining 15% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Blautia caecimuris*** | 31% (371 fragments) | genus present; closest reference | — |
| ***Coprococcus eutactus*** | 14% (161 fragments) | yes · 2.462% | ● Beneficial |
| ***Muricomes intestini*** | 9% (110 fragments) | not found; closest reference | — |
| ***Peptoniphilus stercorisuis*** | 5% (59 fragments) | genus present; closest reference | — |
| ***Clostridium symbiosum*** | 5% (56 fragments) | genus present; closest reference | — |

The organisms carrying this gene in your sample are *Coprococcus eutactus*. Changing this reading means changing them.

### The organisms behind this, and what is known about them

| FOUND IN YOU | WHAT THE LITERATURE RECORDS |
|---|---|
| ***Coprococcus eutactus***<br>2.46% of your community · 14% of this signal | Butyrate producer; with Dialister, Coprococcus was depleted in depression cohorts in a large population study.<br>■ when lower: Coprococcus and Dialister were consistently depleted in depression after correcting for antidepressants in the Flemish Gut Flora Project and LifeLines DEEP. *[Valles-Colomer M et al. Nat Microbiol 2019;4:623. PMID 30718848]*<br>Association only — no study on record shows that changing this organism changes an outcome. |

### What your reading means above average · somewhat higher than most people

One genetic study reported that higher gut propionate production associates with increased Alzheimer's likelihood — the opposite direction to butyrate — and your capacity sits above most people's. That was a single study using an indirect method, and propionate also has documented protective effects on mitochondria, so this is worth noting rather than acting on.

### How can I improve this?

Lower is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Coprococcus eutactus* (2.46% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it Research associates higher levels with worse outcomes · Early evidence

Propionate behaves differently from butyrate. A genetic (Mendelian randomisation) study found that higher gut propionate production was associated with an increased likelihood of Alzheimer's disease, while higher butyrate was associated with improved insulin resistance.

Sanna et al. used Mendelian randomisation across gut microbiome, short-chain fatty acid and disease datasets and reported that increased propionate production or absorption associated with an increased likelihood of Alzheimer's disease. This is a single study using an indirect method, so the finding is suggestive rather than established. Propionate also has documented protective effects on mitochondrial structure, so the overall picture is not settled.

Reference: Sanna S. et al., Nat Genet 51:600-605 (2019), doi:10.1038/s41588-019-0350-x; Cheng Y. et al., Biochem Pharmacol 169:113607 (2019), doi:10.1016/j.bcp.2019.08.015

‹ at a glance

## Butyrate (SCFA) ▼

ABOVE AVERAGE

Butyrate, a short-chain fatty acid and the main energy source for the cells lining the colon.

| | |
|---|---|
| YOUR RESULT | **32.1** copies per 100 genomes |
| PERCENTILE | **82nd** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 21.9 – 30.7 (median 26.8) |
| MADE FROM | Dietary fibre fermented by gut bacteria |
| DNA FRAGMENTS MATCHED | 1,217 |




### What is driving this 1,172 matched fragments, by the organism each came closest to

**83% of the signal comes from organisms found in you:** *Anaerostipes hadrus* (4.53% of your community); *Faecalibacterium intestinale*; *Roseburia hominis* (0.49% of your community); *Faecalibacterium langellae*. A further 17% matched *Faecalibacterium hominis*, whose genus is present but whose species was not resolved.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Anaerostipes hadrus*** | **16%** (186 fragments) | **yes** · 4.531% | ● Beneficial |
| ***Faecalibacterium intestinale*** | **8%** (92 fragments) | **yes** | ● Beneficial |
| ***Faecalibacterium hominis*** | **7%** (83 fragments) | genus present; closest reference | — |
| ***Roseburia hominis*** | **5%** (62 fragments) | **yes** · 0.492% | ● Beneficial |
| ***Faecalibacterium langellae*** | **5%** (54 fragments) | **yes** | ● Beneficial |

The organisms carrying this gene in your sample are *Anaerostipes hadrus*, *Faecalibacterium intestinale*, *Roseburia hominis*; *Anaerostipes hadrus* alone accounts for 16% of the signal. Changing this reading means changing them.

### The organisms behind this, and what is known about them

| FOUND IN YOU | WHAT THE LITERATURE RECORDS |
|---|---|
| ***Anaerostipes hadrus***<br>4.53% of your community · 16% of this signal | Butyrate producer that also converts lactate and acetate to butyrate; part of the core butyrate-producing set of adult stool.<br>■ : Anaerostipes is among the principal butyrate producers using the butyryl-CoA:acetate CoA-transferase route. *[Louis P, Flint HJ. Environ Microbiol 2017;19:29. PMID 27928878]*<br>Association only — no study on record shows that changing this organism changes an outcome. |
| ***Roseburia hominis***<br>0.49% of your community · 5% of this signal | Butyrate producer; reduced in ulcerative colitis cohorts alongside F. prausnitzii.<br>■ when lower: R. hominis and F. prausnitzii were both reduced in UC and inversely correlated with disease activity. *[Machiels K et al. Gut 2014;63:1275. PMID 24021287]*<br>Association only — no study on record shows that changing this organism changes an outcome. |

### What your reading means above average · somewhat higher than most people

Your gut has more butyrate-producing capacity than most people, which is the favourable direction. Butyrate feeds your colon lining, dampens inflammation, and in animal work restores memory function and improves neuronal mitochondrial function. This capacity is fibre-dependent — the bacteria are there, but they need fermentable fibre to work with.

**Two measurements agree**

This agrees with the butyrate-producing species reading (87th percentile), which measures the same function by counting the organisms characterised for the job instead of the genes.

38% of this gene family's matched fragments landed on reference proteins from organisms that carry the enzyme fold without the pathway; those are counted as background, not as capacity, and are excluded from the figure above.

### How can I improve this?

Higher is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Anaerostipes hadrus* (4.53% of your community); *Faecalibacterium intestinale*; *Roseburia hominis* (0.49% of your community); *Faecalibacterium langellae*. Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it Research associates higher levels with better outcomes · Reasonably supported

Butyrate is the most consistently beneficial gut metabolite in this panel. It strengthens the gut barrier, reduces inflammation, and improves mitochondrial function in neurons. People with Alzheimer's disease tend to have fewer butyrate-producing bacteria in their stool.

A specific molecular route from butyrate to tau pathology has now been described: butyrate drives acetylation of GSK3-beta at lysine 15, which regulates its inhibitory phosphorylation at serine 9 and in turn changes how much tau gets phosphorylated. In the same work, transplanting gut bacteria from Alzheimer's model mice into healthy mice produced dysbiosis, tau phosphorylation and cognitive impairment — and giving those mice Lactobacillus and Bifidobacterium reversed it. Separately, butyrate improves neuronal mitochondrial function, restored memory function in the APP/PS1 mouse model, and ameliorates diabetes-related cognitive impairment by restoring Parkin-mediated mitophagy. In humans, stool from people with Alzheimer's shows reduced short-chain-fatty-acid producers, and carriers of the APOE E4 risk allele specifically show lower abundance of butyrate-producing bacteria alongside enrichment of pro-inflammatory genera.

Reference: Zhang Y. et al., Mol Psychiatry 28:4421-4437 (2023), doi:10.1038/s41380-023-02216-7; Sepulveda-Rivera V. et al., Sci Rep 15:28505 (2025), doi:10.1038/s41598-025-13262-2; Cho J.H. et al., Autophagy 20:1505-1522 (2024), doi:10.1080/15548627.2023.2274205

‹ at a glance
what you can do about it ›

## Fermentation routes and cross-feeding

ABOVE AVERAGE

The specific routes your bacteria use to ferment carbohydrate into acids and gases.

| | |
|---|---|
| YOUR RESULT | **151.2** copies per 100 genomes |
| PERCENTILE | **79th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 130.9 – 147.7 (median 140.3) |
| MADE FROM | Pyruvate and other products of carbohydrate fermentation |
| DNA FRAGMENTS MATCHED | 18,226 |




### What is driving this 5,697 matched fragments, by the organism each came closest to

**44% of the signal comes from organisms found in you:** *Anaerostipes hadrus* (4.53% of your community); *Lachnospira eligens* (5.27% of your community). A further 24% matched *Streptococcus ictaluri*, whose genus is present but whose species was not resolved. The remaining 33% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Anaerostipes hadrus*** | **8%** (427 fragments) | **yes** · 4.531% | ● Beneficial |
| ***Streptococcus ictaluri*** | **6%** (360 fragments) | genus present; closest reference | — |
| ***Nosocomiicoccus ampullae*** | **4%** (255 fragments) | not found; closest reference | — |
| ***Ignavibacterium album*** | **4%** (233 fragments) | not found; closest reference | — |
| ***Lachnospira eligens*** | **4%** (226 fragments) | **yes** · 5.265% | ● Beneficial |

The organisms carrying this gene in your sample are *Anaerostipes hadrus*, *Lachnospira eligens*; *Anaerostipes hadrus* alone accounts for 8% of the signal. Changing this reading means changing them.

### The organisms behind this, and what is known about them

| FOUND IN YOU | WHAT THE LITERATURE RECORDS |
|---|---|
| ***Anaerostipes hadrus***<br>4.53% of your community · 8% of this signal | Butyrate producer that also converts lactate and acetate to butyrate; part of the core butyrate-producing set of adult stool.<br>■ : Anaerostipes is among the principal butyrate producers using the butyryl-CoA:acetate CoA-transferase route. *[Louis P, Flint HJ. Environ Microbiol 2017;19:29. PMID 27928878]*<br>Association only — no study on record shows that changing this organism changes an outcome. |

### What your reading means above average · somewhat higher than most people

More fermentation-route capacity than most samples carry, across the acetate and lactate chemistry measured here. The individual route readings show which specific chemistries are complete.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

This reading follows the organisms carrying it, and the ones found in you are *Anaerostipes hadrus* (4.53% of your community); *Lachnospira eligens* (5.27% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it The research picture is mixed · Early evidence

The existing butyrate, propionate, methane and sulfide readings say how much capacity is present. These say by which chemistry.

Propionate is reached by three unrelated routes - succinate, acrylate and propanediol - and a single propionate number cannot distinguish them, which matters because they use different substrates. Acetate is the case where a gene count misleads most easily: an AMP-forming acetyl-CoA synthetase runs towards acetate assimilation rather than formation, so finding it is evidence the community can consume acetate, not make it, and it is excluded here. Hydrogen is shown as production and consumption separately because a healthy community makes a great deal of it and netting the two would hide both.

One of the genes behind this figure (pta) is matching distant relatives rather than close ones, so treat the number as an upper bound. The pathway is still detected — other genes in it match well.

Reference: Reichardt N. et al., "Phylogenetic distribution of three pathways for propionate production within the human gut microbiota", ISME J 8:1323-1335 (2014), doi:10.1038/ismej.2014.14; Louis P., Flint H.J., Environ Microbiol 19:29-41 (2017), doi:10.1111/1462-2920.13589.

‹ at a glance

## Methane (CH4) ▲

NOT DETECTED

Methane, made by archaea (not bacteria) that consume the hydrogen other gut microbes produce.

| | |
|---|---|
| YOUR RESULT | **0** copies per 100 genomes |
| PERCENTILE | 0 of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 0 – 1.4 (median 0) |
| MADE FROM | Hydrogen and carbon dioxide from fibre fermentation |
| DNA FRAGMENTS MATCHED | 0 |




**Why this colour.** Methane slows transit and high methanogen carriage is one of the firmest associations with constipation-predominant IBS.

**Two measurements of the same function disagree**

**Read this beside the methanogen abundance reading.** The methane capacity (mcrA) reading sits at the <1st percentile; the methanogen abundance reading is at the 35th. The two are measured differently — genes in the whole community against the abundance of the species characterised for the job — and they can separate when the characterised species are present below the profiler's limit of detection, so their genes are counted while the organisms are not; the organisms are present but fewer of the terminal-step genes were found than their genomes would predict. Neither reading is preferred: a split this size means one of the two engines is at its limit for this function. mcrA is exclusive to methanogenic archaea, so these two should track each other tightly. A split means one of the two engines is at its limit: methanogens are archaea, they are a small fraction of stool DNA, and the bacterial profiler under-reports them.

### How can I improve this?

Lower is the more favourable direction for this reading.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

**Research associates higher levels with worse outcomes · Reasonably supported**

Methane slows intestinal transit. High methanogen carriage is strongly associated with constipation and with the constipation subtype of IBS, and is a normal variant in a large minority of healthy people.

Pimentel and colleagues infused methane into dog ileum and slowed transit by about 60%, which is the clearest causal evidence for the association. In humans, breath-methane-positive patients are overwhelmingly constipation-predominant, and Kim et al. showed that Methanobrevibacter smithii abundance in stool tracks breath methane. Methanogen carriage is also partly heritable and rises with age. Methane production is not itself a disease: it shifts transit, and whether that matters depends on where a person's transit already sits.

Reference: Pimentel M et al., Am J Physiol Gastrointest Liver Physiol 290:G1089 (2006); Kim G et al., Dig Dis Sci 57:3213-3218 (2012); Triantafyllou K et al., J Neurogastroenterol Motil 20:31-40 (2014)

## Vitamins & Nutrients

‹ at a glance
what you can do about it ›

## Vitamin K2 (menaquinone, MK-n) ▾

LOW

The capacity of your gut bacteria to make menaquinone (vitamin K2), which they use for their own respiration.

| | |
|---|---|
| YOUR RESULT | **11.7** copies per 100 genomes |
| PERCENTILE | 12th of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 15.9 – 30.4 (median 22.6) |
| MADE FROM | Built de novo by bacteria from chorismate |
| DNA FRAGMENTS MATCHED | 301 |




### What is driving this 485 matched fragments, by the organism each came closest to

**89% of the signal comes from organisms found in you:** *Phocaeicola vulgatus*; *Parabacteroides distasonis* (0.57% of your community); *Bacteroides thetaiotaomicron* (0.09% of your community); *Bacteroides xylanisolvens* (0.02% of your community). A further 11% matched *Bacteroides acidifaciens*, whose genus is present but whose species was not resolved.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Phocaeicola vulgatus*** | 22% (105 fragments) | yes | Conditional |
| ***Parabacteroides distasonis*** | 6% (31 fragments) | yes · 0.570% | Conditional |
| ***Bacteroides thetaiotaomicron*** | 5% (23 fragments) | yes · 0.092% | Conditional |
| ***Bacteroides acidifaciens*** | 4% (22 fragments) | genus present; closest reference | — |
| ***Bacteroides xylanisolvens*** | 4% (21 fragments) | yes · 0.017% | Conditional |

The organisms carrying this gene in your sample are *Phocaeicola vulgatus*, *Parabacteroides distasonis*, *Bacteroides thetaiotaomicron*; *Phocaeicola vulgatus* alone accounts for 22% of the signal. Changing this reading means changing them.

### The organisms behind this, and what is known about them

| FOUND IN YOU | WHAT THE LITERATURE RECORDS |
|---|---|
| ***Parabacteroides distasonis***<br>0.57% of your community · 6% of this signal | Common commensal; lower in some IBD/obesity cohorts and enriched in others; mouse work suggests metabolic benefit via succinate and bile acids. Not directional in humans.<br>■ : P. distasonis alleviated obesity and metabolic dysfunction in mice via succinate and secondary bile acids; human data are associative. *[Wang K et al. Cell Rep 2019;26:222. PMID 30605678]*<br>Association only — no study on record shows that changing this organism changes an outcome. |
| ***Bacteroides thetaiotaomicron***<br>0.09% of your community · 5% of this signal | Model glycan-degrading commensal with a large polysaccharide-utilisation repertoire; common and without a fixed directional reading.<br>■ : Bacteroides as glycan-degrading commensals. *[Wexler HM. Clin Microbiol Rev 2007;20:593. PMID 17934076]*<br>Association only — no study on record shows that changing this organism changes an outcome. |
| ***Bacteroides xylanisolvens***<br>0.02% of your community · 4% of this signal | Xylan-degrading Bacteroides; a common commensal.<br>■ : Bacteroides as diet-responsive commensals. *[Wexler HM. Clin Microbiol Rev 2007;20:593. PMID 17934076]*<br>Association only — no study on record shows that changing this organism changes an outcome. |

### What your reading means low · lower than most people

Less bacterial menaquinone synthesis capacity than most people. This has no established bearing on your vitamin K status, which comes from diet.

**Why this colour.** Long-chain menaquinones from gut bacteria contribute to vitamin K status.

### How can I improve this?

Higher is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Phocaeicola vulgatus*; *Parabacteroides distasonis* (0.57% of your community); *Bacteroides thetaiotaomicron* (0.09% of your community); *Bacteroides xylanisolvens* (0.02% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it Research associates higher levels with better outcomes · Early evidence

Gut bacteria are major menaquinone producers, but the long-chain forms they make are poorly absorbed from the colon and are made for bacterial electron transport rather than for the host.

Karl et al. profiled menaquinone forms in human stool and showed that long-chain MK-10 to MK-13 dominate, matching Bacteroides production, and that faecal menaquinone concentrations did not track markers of host vitamin K status. Ramotar et al. estimated colonic absorption of bacterial menaquinone as low. The interesting biology here is microbiological rather than nutritional: which quinone route a community uses (classical versus futalosine) separates Bacteroidota from Proteobacteria and tracks how the community respires. Claims that gut bacteria are a meaningful source of your vitamin K are not supported by the absorption data.

One of the genes behind this figure (menA, mqnE) is matching distant relatives rather than close ones, so treat the number as an upper bound. The pathway is still detected — other genes in it match well.

Reference: Karl J.P. et al., Am J Clin Nutr 106:1052-1061 (2017), doi:10.3945/ajcn.117.155424; Fenn K. et al., Nat Microbiol 2:16241 (2017)

## What the research says about changing this

**Menaquinone (vitamin K2) synthesis genes — not a vitamin K status** LANE A HUMAN EVIDENCE · STRONG

Fired by: k2 gene capacity: low

Gene capacity for menaquinone synthesis does not measure coagulation or vitamin K sufficiency. People on warfarin should keep vitamin K intake stable under their prescriber.

- Gene capacity for menaquinone synthesis does not measure coagulation status or sufficiency; warfarin users need stable intake managed by their prescriber. [NIH ODS — Vitamin K fact sheet] · lane A · reference

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “vitamin K deficiency” · “take K2”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

## Vitamin B1 (thiamine) ▼

ABOVE AVERAGE

Your gut bacteria's capacity to build thiamine from scratch, rather than only take in thiamine that is already there.

| | |
|---|---|
| YOUR RESULT | **230.5** copies per 100 genomes |
| PERCENTILE | **76th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 190.1 – 229.3 (median 213.7) |
| MADE FROM | Built de novo by bacteria from a thiazole ring and a pyrimidine ring, joined by thiamine phosphate synthase |
| DNA FRAGMENTS MATCHED | 14,552 |




### What is driving this

6,494 matched fragments, by the organism each came closest to

**14% of the signal comes from organisms found in you:** *Blautia faecicola* (0.11% of your community). A further 71% matched *Blautia wexlerae*, *Anaerostipes caccae*, *Blautia intestinihominis*, whose genus is present but whose species was not resolved. The remaining 15% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Blautia wexlerae*** | **6%** (423 fragments) | genus present; closest reference | — |
| ***Anaerostipes caccae*** | **4%** (247 fragments) | genus present; closest reference | — |
| ***Blautia intestinihominis*** | **3%** (209 fragments) | genus present; closest reference | — |
| ***Jingyaoa shaoxingensis*** | **3%** (191 fragments) | not found; closest reference | — |
| ***Blautia faecicola*** | **3%** (171 fragments) | **yes** · 0.111% | ● Conditional |

The organisms carrying this gene in your sample are *Blautia faecicola*. Changing this reading means changing them.

### What your reading means

above average · somewhat higher than most people

More of your bacteria carry a complete route to thiamine than in most samples. The colon absorbs little of what they make, so the likeliest consequence is within the community, where thiamine-dependent species have a nearer source.

### How can I improve this?

Higher is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Blautia faecicola* (0.11% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

Research associates higher levels with better outcomes · Early evidence

Most gut species cannot make thiamine and depend on the ones that can. This counts how much of your community carries the complete route.

Magnusdottir and colleagues reconstructed B-vitamin synthesis across gut genomes and found thiamine synthesis in a minority of them, with most species depending on what others make or on what arrives in food. That makes thiamine one of the clearer cases of cross-feeding in the gut, and it is why the two rings and their coupling step are counted separately: a community can carry the branches without the synthase that joins them. What this cannot tell you is your thiamine status. Dietary thiamine is absorbed in the small intestine, upstream of where most of this synthesis happens, and deficiency is a clinical diagnosis rather than a stool one.

Reference: Magnusdottir S., Ravcheev D., de Crecy-Lagard V., Thiele I., "Systematic genome assessment of B-vitamin biosynthesis suggests co-operation among gut microbes", Front Genet 6:148 (2015), doi:10.3389/fgene.2015.00148.

‹ at a glance

## Folate (vitamin B9) ▼

TYPICAL

The capacity of your gut bacteria to synthesise folate, which colonocytes can absorb to a limited extent.

| | |
|---|---|
| YOUR RESULT | **102.1** copies per 100 genomes |
| PERCENTILE | **29th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 99.1 – 138.6 (median 119.4) |
| MADE FROM | Built de novo by bacteria from GTP and para-aminobenzoate |
| DNA FRAGMENTS MATCHED | 2,025 |




### What is driving this

1,758 matched fragments, by the organism each came closest to

**28% of the signal comes from organisms found in you:** *Phocaeicola vulgatus*; *Bacteroides clarus* (1.39% of your community). A further 33% matched *Clostridium beijerinckii*, whose genus is present but whose species was not resolved. The remaining 39% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Clostridium beijerinckii*** | **15%** (257 fragments) | genus present; closest reference | — |
| ***Flammeovirga aprica*** | **12%** (206 fragments) | not found; closest reference | — |
| ***Phocaeicola vulgatus*** | **8%** (144 fragments) | **yes** | ● Conditional |
| ***Butyrivibrio fibrisolvens*** | **5%** (92 fragments) | not found; closest reference | — |
| ***Bacteroides clarus*** | **4%** (69 fragments) | **yes** · 1.387% | ● Conditional |

The organisms carrying this gene in your sample are *Phocaeicola vulgatus*, *Bacteroides clarus*. Changing this reading means changing them.

### What your reading means

typical · in line with most people

Your community's folate synthesis capacity sits in the middle of the reference range. Some colonic folate is absorbed, but diet supplies most of it, and this reading does not substitute for a blood measurement of folate status.

### How can I improve this?

Higher is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Phocaeicola vulgatus*; *Bacteroides clarus* (1.39% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

Research associates higher levels with better outcomes · Early evidence

Bifidobacterium and several other gut genera synthesise folate, and unlike B12 some of it is absorbed in the colon. The contribution to human folate status is modest compared with diet.

Rossi et al. and Pompei et al. showed that Bifidobacterium strains raise folate concentrations in the gut and in host plasma in rat models, and human colonocytes do express a proton-coupled folate transporter capable of absorbing bacterial folate. Magnusdottir et al.'s genome reconstruction found folate biosynthesis in a larger share of gut genomes than cobalamin. The realistic scale of the contribution is the honest caveat: in a fortified food supply, dietary folate dominates, and no trial has shown that manipulating bacterial folate synthesis changes clinically relevant folate status in humans.

One of the genes behind this figure (folP) is matching distant relatives rather than close ones, so treat the number as an upper bound. The pathway is still detected — other genes in it match well.

Reference: Rossi M. et al., Nutrients 3:118-134 (2011), doi:10.3390/nu3010118; Magnusdottir S. et al., Front Genet 6:148 (2015)

## What the research says about changing this

**Folate synthesis genes — not a folate status** LANE A HUMAN EVIDENCE · STRONG

Fired by: folate gene capacity: typical

Folate status is assessed by diet, medications, pregnancy and blood tests; microbial genes do not indicate deficiency, and high folic-acid intake can mask B12 deficiency.

- Folate status is assessed by diet, medication, pregnancy and blood tests; high folic-acid intake can mask B12 deficiency. No folate dose follows from microbial genes. [NIH ODS — Folate fact sheet] · lane A · reference

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “folate deficient” · “take folic acid”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

## Vitamin B12 (cobalamin) ▾

TYPICAL

The capacity of your gut bacteria to build vitamin B12 from scratch, and separately their capacity to compete for it.

| | |
|---|---|
| YOUR RESULT | **83.4** copies per 100 genomes |
| PERCENTILE | **38th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 75.2 – 101.4 (median 90.8) |
| MADE FROM | Built de novo by bacteria from small precursors; not from diet |
| DNA FRAGMENTS MATCHED | 2,574 |




### What is driving this 2,568 matched fragments, by the organism each came closest to

**69% of the signal comes from organisms found in you:** *Faecalibacterium prausnitzii* (6.18% of your community); *Faecalibacterium langellae*; *Phocaeicola dorei* (2.06% of your community); *Brotocaccenecus cirricatena* (0.23% of your community). A further 31% matched *Blautia wexlerae*, whose genus is present but whose species was not resolved.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Faecalibacterium prausnitzii*** | **15%** (395 fragments) | **yes** · 6.182% | Beneficial |
| ***Blautia wexlerae*** | **13%** (341 fragments) | genus present; closest reference | — |
| ***Faecalibacterium langellae*** | **8%** (197 fragments) | **yes** | Beneficial |
| ***Phocaeicola dorei*** | **3%** (87 fragments) | **yes** · 2.056% | Conditional |
| ***Brotocaccenecus cirricatena*** | **3%** (68 fragments) | **yes** · 0.232% | Unknown |

The organisms carrying this gene in your sample are *Faecalibacterium prausnitzii*, *Faecalibacterium langellae*, *Phocaeicola dorei*; *Faecalibacterium prausnitzii* alone accounts for 15% of the signal. Changing this reading means changing them.

### The organisms behind this, and what is known about them

| FOUND IN YOU | WHAT THE LITERATURE RECORDS |
|---|---|
| ***Faecalibacterium prausnitzii***<br>6.18% of your community · 15% of this signal | One of the most abundant butyrate producers in adult stool. Reduced relative abundance is one of the most replicated findings in Crohn's disease and ulcerative colitis, and lower levels are reported across several inflammatory and metabolic conditions. It is also simply variable between healthy people and across countries; a low level alone is not a diagnosis and no product restores the species itself.<br>■ when lower: Reduced F. prausnitzii is associated with Crohn's disease and with post-operative recurrence; the isolate had anti-inflammatory effects in vitro and in mouse colitis. *[Sokol H et al. PNAS 2008;105:16731. PMID 18936492]*<br>■ when lower: A 2017 review consolidates lower F. prausnitzii across IBD, IBS, obesity, coeliac disease and colorectal cancer cohorts, with phylogroup-dependent patterns. *[Lopez-Siles M et al. ISME J 2017;11:841. PMID 28072420]*<br>Association only — no study on record shows that changing this organism changes an outcome. |

### What your reading means typical · in line with most people

Your community's B12 synthesis capacity sits in the middle of the reference range. The result says nothing about your B12 status: the bacteria that make it live in the colon, past where B12 is absorbed, and a blood test is the only way to know.

**Why this colour.** Bacterial B12 made in the colon is poorly absorbed, so the benefit is small, but there is no harm in more of it.

### How can I improve this?

Higher is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Faecalibacterium prausnitzii* (6.18% of your community); *Faecalibacterium langellae*; *Phocaeicola dorei* (2.06% of your community); *Brotocaccenecus cirricatena* (0.23% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it Research associates higher levels with better outcomes · Early evidence

Gut bacteria are prolific B12 producers, but they make it in the colon, below the point where humans absorb it, and most of it is consumed by other bacteria. Higher synthesis capacity is a feature of the community, not a contribution to your B12 status.

Degnan et al. established that most human gut bacteria either synthesise corrinoids or scavenge them, and that corrinoid competition is a genuine structuring force in the community. Magnusdottir et al. reconstructed B-vitamin biosynthesis across 256 gut genomes and found only about 40% could make cobalamin de novo. The absorption argument is settled anatomy:

intrinsic-factor-mediated uptake occurs in the terminal ileum, while the bulk of bacterial synthesis happens distal to it. Where bacterial B12 does matter is between bacteria — corrinoid availability shapes which species can grow.

Reference: Degnan P.H. et al., Cell Metab 20:769-778 (2014), doi:10.1016/j.cmet.2014.10.002; Magnusdottir S. et al., Front Genet 6:148 (2015), doi:10.3389/fgene.2015.00148

## What the research says about changing this

**Vitamin B12 synthesis genes — colonic synthesis is not your B12 status** LANE A HUMAN EVIDENCE · STRONG

Fired by: b12 gene capacity: typical

Bacteria in the colon make B12 that is largely not absorbed there; this reading does not indicate deficiency or sufficiency. B12 status is a blood test, and diet, age, metformin and acid suppression matter more.

**Studied:** Serum vitamin B12 (and methylmalonic acid when appropriate) to assess status (clinical route) · **in:** ages 0–99 · **studied protocol (a source fact, not an instruction):** serum B12 ± methylmalonic acid

Evidence lane A: current authoritative guideline or regulatory label for an independently confirmed indication. Replication: single study. Strongest design: reference fact sheet. Guideline position: strong for.

- Suspected B12 deficiency is confirmed with serum B12 and, when appropriate, methylmalonic acid; supplementation follows diet, malabsorption and clinical context. Colonic microbial B12 synthesis is not host bioavailability. [NIH Office of Dietary Supplements — Vitamin B12 fact sheet for health professionals] · lane A · reference · benefit

**Conventional route:** Serum B12 (± methylmalonic acid) if deficiency is a concern.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “B12 deficient” · “your bacteria make enough B12”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

## Riboflavin (vitamin B2) ▾

TYPICAL

The capacity of your gut bacteria to make riboflavin, which they also use as an electron shuttle between themselves.

| | |
|---|---|
| YOUR RESULT | **200.1** copies per 100 genomes |
| PERCENTILE | **60th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 165.9 – 214.4 (median 190.8) |
| MADE FROM | Built de novo by bacteria from GTP and ribulose-5-phosphate |
| DNA FRAGMENTS MATCHED | 2,479 |




### What is driving this 4,007 matched fragments, by the organism each came closest to

**44% of the signal comes from organisms found in you:** *Phocaeicola vulgatus*; *Bacteroides thetaiotaomicron* (0.09% of your community); *Agathobacter rectalis* (0.52% of your community). A further 10% matched *Clostridium novyi*, whose genus is present but whose species was not resolved. The remaining 46% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Actinobacillus pleuropneumoniae*** | **15%** (596 fragments) | not found; closest reference | — |
| ***Phocaeicola vulgatus*** | **5%** (195 fragments) | **yes** | ● Conditional |
| ***Bacteroides thetaiotaomicron*** | **5%** (193 fragments) | **yes** · 0.092% | ● Conditional |
| ***Agathobacter rectalis*** | **5%** (187 fragments) | **yes** · 0.518% | ● Beneficial |
| ***Clostridium novyi*** | **3%** (136 fragments) | genus present; closest reference | — |

The organisms carrying this gene in your sample are *Phocaeicola vulgatus*, *Bacteroides thetaiotaomicron*, *Agathobacter rectalis*. Changing this reading means changing them.

### The organisms behind this, and what is known about them

| FOUND IN YOU | WHAT THE LITERATURE RECORDS |
|---|---|
| ***Bacteroides thetaiotaomicron***<br>0.09% of your community · 5% of this signal | Model glycan-degrading commensal with a large polysaccharide-utilisation repertoire; common and without a fixed directional reading.<br>■ : Bacteroides as glycan-degrading commensals. *[Wexler HM. Clin Microbiol Rev 2007;20:593. PMID 17934076]*<br>Association only — no study on record shows that changing this organism changes an outcome. |

### What your reading means typical · in line with most people

Your community's riboflavin synthesis capacity sits in the middle of the reference range. Riboflavin is one of the B vitamins the colon can absorb in small amounts, but dietary intake is the dominant source and this reading is a feature of your bacteria rather than a measure of your vitamin status.

### How can I improve this?

Higher is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Phocaeicola vulgatus*; *Bacteroides thetaiotaomicron* (0.09% of your community); *Agathobacter rectalis* (0.52% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it Research associates higher levels with better outcomes · Early evidence

Riboflavin is one of the few bacterial vitamins with a demonstrated effect on the community itself: Faecalibacterium prausnitzii uses it to transfer electrons, and riboflavin supplementation increases F. prausnitzii abundance in human trials.

Khan et al. showed that F. prausnitzii uses riboflavin as an extracellular electron shuttle, which explains how a strict anaerobe tolerates the oxygen gradient at the mucosal surface. Steinert et al. then supplemented healthy adults with riboflavin and observed an increase in F. prausnitzii and in butyrate, one of the cleaner prebiotic-mechanism demonstrations in the field. Separately, riboflavin biosynthesis intermediates are the antigens that MR1 presents to MAIT cells, making this pathway a direct link between bacterial metabolism and mucosal immunity. What the evidence does not support is treating this number as a nutritional riboflavin status.

Reference: Khan M.T. et al., ISME J 6:1578-1585 (2012), doi:10.1038/ismej.2012.5; Steinert R.E. et al., Gut Microbes 11:1300-1309 (2020), doi:10.1080/19490976.2020.1748208

## What the research says about changing this

**Riboflavin synthesis genes — the riboflavin-supplement trials measured the microbiome, not you**

LANE C HUMAN TRIAL · LIMITED

Fired by: riboflavin gene capacity: typical

Riboflavin supplementation was tested as a way to feed F. prausnitzii: two weeks in healthy adults did not significantly raise it at either dose, and three weeks in Crohn's disease did not alter the metagenome. This gene reading is not a riboflavin status.

**Studied:** Riboflavin (vitamin B2) supplement (vitamin supplement) · **in:** people with clinically confirmed healthy adults, ages 18–65, in Netherlands, n = 105 · **studied protocol (a source fact, not an instruction):** 50 or 100 mg/day riboflavin for two weeks

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: single study. Strongest design: randomized trial.

- Two weeks of riboflavin (50 or 100 mg/day) in 105 healthy adults did not significantly increase F. prausnitzii at either dose; pooled groups showed higher butyrate — a surrogate result only. [RIBOGUT trial (2022) — riboflavin 50 or 100 mg/day in healthy adults] · lane C · randomized double-blind placebo-controlled · n = 105 · mixed

- **Against or limiting —** Three weeks of riboflavin 100 mg/day in Crohn's disease did not materially alter metagenomic diversity, taxonomy or pathways. [RISE-UP (2020) — riboflavin 100 mg/day in Crohn's disease] · lane X · single-arm · n = 70 · null

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “riboflavin deficiency”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

## Vitamin B6 (pyridoxal phosphate) ▾

TYPICAL

Whether your bacteria carry either of the two known routes to vitamin B6.

| | |
|---|---|
| YOUR RESULT | **59.8** copies per 100 genomes |
| PERCENTILE | **42nd** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 54.3 – 69.6 (median 62.2) |
| MADE FROM | Built de novo by one of two unrelated routes, DXP-dependent or DXP-independent |
| DNA FRAGMENTS MATCHED | 2,652 |




### What is driving this 1,603 matched fragments, by the organism each came closest to

**84% of the signal comes from organisms found in you:** *Bacteroides thetaiotaomicron* (0.09% of your community); *Ruminococcus bromii* (3.40% of your community); *Blautia stercoris* (0.22% of your community); *Agathobaculum butyriciproducens* (0.36% of your community). The remaining 16% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Bacteroides thetaiotaomicron*** | **18%** (296 fragments) | **yes** · 0.092% | Conditional |
| ***Ruminococcus bromii*** | **8%** (132 fragments) | **yes** · 3.396% | Beneficial |
| ***[Lactobacillus] rogosae*** | **7%** (107 fragments) | not found; closest reference | — |
| ***Blautia stercoris*** | **4%** (65 fragments) | **yes** · 0.223% | Conditional |
| ***Agathobaculum butyriciproducens*** | **3%** (53 fragments) | **yes** · 0.363% | Beneficial |

The organisms carrying this gene in your sample are *Bacteroides thetaiotaomicron*, *Ruminococcus bromii*, *Blautia stercoris*; *Bacteroides thetaiotaomicron* alone accounts for 18% of the signal. Changing this reading means changing them.

### The organisms behind this, and what is known about them

| FOUND IN YOU | WHAT THE LITERATURE RECORDS |
|---|---|
| ***Bacteroides thetaiotaomicron***<br>0.09% of your community · 18% of this signal | Model glycan-degrading commensal with a large polysaccharide-utilisation repertoire; common and without a fixed directional reading.<br>■ : Bacteroides as glycan-degrading commensals. *[Wexler HM. Clin Microbiol Rev 2007;20:593. PMID 17934076]*<br>Association only — no study on record shows that changing this organism changes an outcome. |
| ***Ruminococcus bromii***<br>3.40% of your community · 8% of this signal | Keystone degrader of resistant starch: it breaks down starch granules other species cannot, feeding the wider community. Its baseline level predicted who responded to a resistant-starch feeding trial with a butyrate rise.<br>■ : R. bromii is a keystone species for resistant starch degradation in the human colon. *[Ze X et al. ISME J 2012;6:1535. PMID 22343308]*<br>■ : Faecal butyrate rose with resistant potato starch only in baseline responders, and R. bromii predicted response. *[Venkataraman A et al. Microbiome 2016;4:33. PMID 27357127]*<br>Association only — no study on record shows that changing this organism changes an outcome. |

### What your reading means typical · in line with most people

Your community's B6 synthesis capacity sits in the middle of the reference range. Pyridoxal phosphate is the cofactor for most amino-acid chemistry in a cell, so the capacity to make it is common without being universal, and a middling value is the ordinary finding.

### How can I improve this?

Higher is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Bacteroides thetaiotaomicron* (0.09% of your community); *Ruminococcus bromii* (3.40% of your community); *Blautia stercoris* (0.22% of your community); *Agathobaculum butyriciproducens* (0.36% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it Research associates higher levels with better outcomes · Early evidence

Two unrelated chemistries reach the same vitamin. Requiring both would call almost every community unable to make B6; accepting one isolated subunit would call almost all of them able to.

The DXP-independent route, a PdxS synthase working with its PdxT glutaminase, is the one most bacteria use; the DXP-dependent route through PdxA and PdxJ is found in Escherichia coli and its relatives. They are alternatives, so both are measured and the partner subunits are shown separately. Pyridoxal phosphate is the cofactor for most amino-acid chemistry in the cell, which is why the capacity is common without being universal.

Reference: Rodionov D.A. et al., Front Microbiol 10:1316 (2019), doi:10.3389/fmicb.2019.01316; Fitzpatrick T.B. et al., Biochem J 407:1-13 (2007), doi:10.1042/BJ20070765.

‹ at a glance

## Vitamin B5 (pantothenate) ▼

TYPICAL

Whether your bacteria can build pantothenate from both of its branches, rather than only use the vitamin once it exists.

| | |
|---|---|
| YOUR RESULT | **160.9** copies per 100 genomes |
| PERCENTILE | **58th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 131.7 – 175.6 (median 154.5) |
| MADE FROM | Built de novo from pantoate and beta-alanine, joined by pantothenate synthetase |
| DNA FRAGMENTS MATCHED | 3,694 |




### What is driving this 3,109 matched fragments, by the organism each came closest to

**84% of the signal comes from organisms found in you:** *Hominisplanchenecus faecis*; *Anaerostipes amylophilus*; *Phocaeicola vulgatus*; *Roseburia lenta* (0.06% of your community). A further 16% matched *Blautia wexlerae*, whose genus is present but whose species was not resolved.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Hominisplanchenecus faecis*** | **10%** (312 fragments) | **yes** | Unknown |
| ***Anaerostipes amylophilus*** | **8%** (238 fragments) | **yes** | Beneficial |
| ***Blautia wexlerae*** | **5%** (161 fragments) | genus present; closest reference | — |
| ***Phocaeicola vulgatus*** | **5%** (149 fragments) | **yes** | Conditional |
| ***Roseburia lenta*** | **4%** (127 fragments) | **yes** · 0.060% | Beneficial |

The organisms carrying this gene in your sample are *Hominisplanchenecus faecis*, *Anaerostipes amylophilus*, *Phocaeicola vulgatus*; *Hominisplanchenecus faecis* alone accounts for 10% of the signal. Changing this reading means changing them.

### What your reading means typical · in line with most people

Mid-range pantothenate synthesis capacity, which is the usual finding. About half of gut species can build it from both branches and the rest take up what is already available, so a middling value reflects the ordinary division of labour.

### How can I improve this?

Higher is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Hominisplanchenecus faecis*; *Anaerostipes amylophilus*; *Phocaeicola vulgatus*; *Roseburia lenta* (0.06% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it Research associates higher levels with better outcomes · Early evidence

Pantothenate is the precursor of coenzyme A, so the enzymes that consume it are universal and the enzymes that make it are not. Only the making is counted here.

Every living cell needs coenzyme A, so counting the consuming step would turn every organism into a producer; coaA is deliberately absent from this panel for that reason. Magnusdottir and colleagues found pantothenate synthesis in roughly half the gut genomes they reconstructed, with the beta-alanine branch the more commonly missing of the two, which is why that branch is measured on its own as well as in the total.

Reference: Magnusdottir S., Ravcheev D., de Crecy-Lagard V., Thiele I., Front Genet 6:148 (2015), doi:10.3389/fgene.2015.00148.

‹ at a glance

## Biotin (vitamin B7) ▾

TYPICAL

The capacity of your gut bacteria to synthesise biotin, and the complementary fact of how much of the community depends on scavenging it.

| | |
|---|---|
| YOUR RESULT | **64.1** copies per 100 genomes |
| PERCENTILE | **52nd** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 50.5 – 77.8 (median 63.0) |
| MADE FROM | Built de novo by bacteria from pimeloyl-ACP and alanine |
| DNA FRAGMENTS MATCHED | 1,025 |




### What is driving this 1,673 matched fragments, by the organism each came closest to

**40% of the signal comes from organisms found in you:** *Anaerostipes hadrus* (4.53% of your community); *Agathobacter rectalis* (0.52% of your community). A further 49% matched *Blautia intestinihominis*, *Blautia wexlerae*, whose genus is present but whose species was not resolved. The remaining 11% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Blautia intestinihominis*** | **15%** (248 fragments) | genus present; closest reference | — |
| ***Anaerostipes hadrus*** | **12%** (192 fragments) | **yes** · 4.531% | ● Beneficial |
| ***Blautia wexlerae*** | **8%** (141 fragments) | genus present; closest reference | — |
| ***Agathobacter rectalis*** | **8%** (127 fragments) | **yes** · 0.518% | ● Beneficial |
| ***Glaesserella parasuis*** | **5%** (85 fragments) | not found; closest reference | — |

The organisms carrying this gene in your sample are *Anaerostipes hadrus*, *Agathobacter rectalis*. Changing this reading means changing them.

### The organisms behind this, and what is known about them

| FOUND IN YOU | WHAT THE LITERATURE RECORDS |
|---|---|
| ***Anaerostipes hadrus***<br>4.53% of your community · 12% of this signal | Butyrate producer that also converts lactate and acetate to butyrate; part of the core butyrate-producing set of adult stool.<br>■ : Anaerostipes is among the principal butyrate producers using the butyryl-CoA:acetate CoA-transferase route. *[Louis P, Flint HJ. Environ Microbiol 2017;19:29. PMID 27928878]*<br>Association only — no study on record shows that changing this organism changes an outcome. |

### What your reading means typical · in line with most people

Your community's biotin synthesis capacity sits in the middle of the reference range. Biotin is one of the vitamins the colon does absorb, so bacterial production contributes to supply, but dietary intake is by far the larger source for most people and a typical reading needs no action.

### How can I improve this?

Higher is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Anaerostipes hadrus* (4.53% of your community); *Agathobacter rectalis* (0.52% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it Research associates higher levels with better outcomes · Preliminary

Biotin is the one vitamin with a demonstrated mechanistic link to hair growth, though only in germ-free mice under experimental deprivation. In humans with normal biotin status there is no evidence that bacterial biotin synthesis affects hair.

Hayashi et al. fed germ-free mice a biotin-deficient diet, observed alopecia accompanied by Lactobacillus murinus overgrowth, and reversed both with biotin supplementation — establishing that the gut community's biotin economy can affect hair in an extreme experimental setting. That is the entire mechanistic basis for the widely repeated claim linking gut bacteria to hair loss. Human biotin deficiency is rare and produces a distinctive picture (rash, neurological signs) unlike either common form of alopecia, and randomised evidence for biotin supplementation in hair loss with normal biotin status is absent. Magnusdottir et al. found biotin synthesis capacity in a minority of gut genomes, with most species dependent on scavenging.

One of the genes behind this figure (bioF) is matching distant relatives rather than close ones, so treat the number as an upper bound. The pathway is still detected — other genes in it match well.

Reference: Hayashi A. et al., Cell Rep 20:1513-1524 (2017), doi:10.1016/j.celrep.2017.07.057; Magnusdottir S. et al., Front Genet 6:148 (2015)

## What the research says about changing this

**Biotin synthesis genes — no hair or supplement claim follows** LANE A HUMAN EVIDENCE · STRONG

Fired by: biotin gene capacity: typical

Biotin deficiency is rare and this reading does not indicate it. High-dose biotin supplements interfere with laboratory tests including troponin.

- Biotin deficiency is rare; high-dose supplements interfere with laboratory assays including troponin (FDA). No hair-loss or alopecia biotin suggestion follows from microbial genes. [NIH ODS — Biotin fact sheet] · lane A · reference
- **Against or limiting —** FDA warns that high-dose biotin can cause falsely low troponin and other assay results. [FDA — biotin interference with troponin lab tests] · lane X · safety communication · harm

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “biotin for hair” · “biotin deficiency”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

## Vitamin B3 (niacin and NAD precursors) ▾

TYPICAL

Whether your bacteria can build the niacin ring from aspartate, as distinct from recycling nicotinamide that already exists.

| | |
|---|---|
| YOUR RESULT | **192.9** copies per 100 genomes |
| PERCENTILE | **51st** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 175.7 – 218.3 (median 191.9) |
| MADE FROM | Built de novo from aspartate through quinolinate, or salvaged from nicotinamide |
| DNA FRAGMENTS MATCHED | 6,956 |




### What is driving this 6,037 matched fragments, by the organism each came closest to

A further 10% matched *Coprococcus intestinihominis*, whose genus is present but whose species was not resolved. The remaining 90% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Ligilactobacillus ruminis*** | **16%** (962 fragments) | not found; closest reference | — |
| ***Treponema ruminis*** | **9%** (521 fragments) | not found; closest reference | — |
| ***Fusobacterium mortiferum*** | **7%** (412 fragments) | not found; closest reference | — |
| ***Tannerella forsythia*** | **4%** (250 fragments) | not found; closest reference | — |
| ***Coprococcus intestinihominis*** | **4%** (249 fragments) | genus present; closest reference | — |

None of the closest references is itself in your organism list: the fragments most likely come from close relatives that carry a similar gene. The genus is named where it was found.

### What your reading means typical · in line with most people

Your community's de-novo niacin capacity is mid-range, which is the usual finding. Most gut bacteria salvage nicotinamide rather than building the ring from aspartate, and this reading deliberately does not count that salvage.

### How can I improve this?

Higher is the more favourable direction for this reading.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it Research associates higher levels with better outcomes · Early evidence

The de-novo route and the salvage route both end at NAD. This measures the first and deliberately excludes the second.

A gene count that pooled the two routes would report almost every community as a producer, because salvage is close to universal. Rodionov and colleagues mapped both across gut genomes and found the de-novo route much less common, which is why the salvage enzymes are held out as a decoy set here rather than counted as production. Niacin is also one of the vitamins where host and bacterial supply are hard to separate: your own cells make NAD from tryptophan, and colonic absorption of bacterial niacin is limited.

Reference: Rodionov D.A. et al., "Micronutrient requirements and sharing capabilities of the human gut microbiome", Front Microbiol 10:1316 (2019), doi:10.3389/fmicb.2019.01316.

## Protein & Nitrogen Metabolism

‹ at a glance
what you can do about it ›

## p-Cresol / p-cresyl sulfate ▲

ABOVE AVERAGE

p-Cresol, which the host converts to p-cresyl sulfate, a compound that accumulates when kidney function declines.

| | |
|---|---|
| YOUR RESULT | **11.3** copies per 100 genomes |
| PERCENTILE | **84th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 4.9 – 9.5 (median 6.6) |
| MADE FROM | Dietary tyrosine and phenylalanine |
| DNA FRAGMENTS MATCHED | 492 |




### What is driving this

917 matched fragments, by the organism each came closest to

**1% of the signal comes from organisms found in you:** *Gallintestinimicrobium propionicum* (0.10% of your community). A further 99% matched *Blautia wexlerae*, *Acetatifactor muris*, *Collinsella acetigenes*, whose genus is present but whose species was not resolved.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Blautia wexlerae*** | **90%** (823 fragments) | genus present; closest reference | — |
| ***Acetatifactor muris*** | **7%** (62 fragments) | genus present; closest reference | — |
| ***Gallintestinimicrobium propionicum*** | **1%** (8 fragments) | **yes** · 0.103% | ● Unknown |
| ***Collinsella acetigenes*** | **1%** (6 fragments) | genus present; closest reference | — |
| ***Dorea ammoniilytica*** | **1%** (6 fragments) | genus present; closest reference | — |

The organisms carrying this gene in your sample are *Gallintestinimicrobium propionicum*. Changing this reading means changing them.

### What your reading means

above average · somewhat higher than most people

More of your gut bacteria can convert tyrosine into p-cresol than in most people. p-Cresol interferes with mitochondrial energy metabolism and raises superoxide production, and is enriched in people with diabetes and obesity. It is also cleared by the kidneys, so it matters more if kidney function is reduced. No study has linked it directly to Alzheimer's in humans.

### How can I improve this?

Lower is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Gallintestinimicrobium propionicum* (0.10% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

Research associates higher levels with worse outcomes · Early evidence

p-Cresol interferes with mitochondrial energy metabolism and increases production of damaging superoxide. It is enriched in people with diabetes and obesity, and in animals it disrupts brain signalling.

p-Cresol modifies mitochondrial oxidative metabolism and increases superoxide production. In mice, it induces autism-like social behaviour by reducing the function of central dopamine neurons, and it alters NMDA receptor subunit expression in the hippocampus and nucleus accumbens. No direct study links p-cresol to Alzheimer's disease in humans, so the connection rests on shared mechanisms rather than on evidence.

Reference: Bermudez-Martin P. et al., Microbiome 9:157 (2021), doi:10.1186/s40168-021-01103-z; Fromentin S. et al., Nat Med 28:303-314 (2022), doi:10.1038/s41591-022-01688-4

## What the research says about changing this

**Higher p-cresol (hpdB) capacity — evidence exists only in established kidney disease** LANE C HUMAN TRIAL · LIMITED

Fired by: pcresol gene capacity: somewhat high

In pre-dialysis CKD, a synbiotic and resistant starch each lowered serum p-cresyl sulfate in small trials without hard outcomes. In someone with normal kidney function this reading has no supported intervention.

**Studied:** SYNERGY synbiotic (mixed prebiotic + nine-strain probiotic) in CKD (Rossi 2016) (multi ingredient product) · **in:** people with clinically confirmed CKD stage 4–5 (pre-dialysis), ages 18–85, in Australia, n = 37 · **studied protocol (a source fact, not an instruction):** 15 g/day mixed prebiotic plus nine-strain probiotic escalating 45 → 90 billion CFU/day for six weeks (crossover)

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: single study. Strongest design: systematic review.

- In 37 pre-dialysis CKD patients, six weeks of the SYNERGY synbiotic reduced serum p-cresyl sulfate but not indoxyl sulfate — surrogate evidence that applies only with established CKD and a measured serum PCS. [Rossi M et al. Clin J Am Soc Nephrol 2016;11:223 (SYNERGY)] · lane C · randomized double-blind placebo-controlled crossover · n = 37 · mixed
- Resistant-starch studies in CKD show mixed surrogate results. [review — resistant starch in CKD (2024)] · lane D · review · mixed
- **Against or limiting —** A 16-week resistant-starch CKD trial lowered p-cresyl sulfate but not other biomarkers or vascular function, and reduced several alpha-diversity metrics. [Headley SA et al. 2024 — resistant starch in CKD] · lane X · randomized controlled · mixed

**Applicability:** The study required a clinical diagnosis of chronic kidney disease. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “uraemic toxins” · “kidney damage”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance
what you can do about it ›

## Imidazole propionate (ImP) ▲

ABOVE AVERAGE

Imidazole propionate (ImP), a small molecule made by gut bacteria from the amino acid histidine.

| | |
|---|---|
| YOUR RESULT | **53.5** copies per 100 genomes |
| PERCENTILE | 83rd of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 26.2 – 47.8 (median 38.0) |
| MADE FROM | Dietary histidine, via the intermediate urocanate |
| DNA FRAGMENTS MATCHED | 1,749 |




### What is driving this 3,040 matched fragments, by the organism each came closest to

**86% of the signal comes from organisms found in you:** *Faecalibacterium langellae*; *Faecalibacterium intestinale*; *Faecalibacterium duncaniae* (6.18% of your community); *Faecalibacterium tardum*. A further 14% matched *Faecalibacterium gallinarum*, whose genus is present but whose species was not resolved.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Faecalibacterium langellae*** | 13% (387 fragments) | yes | Beneficial |
| ***Faecalibacterium intestinale*** | 12% (372 fragments) | yes | Beneficial |
| ***Faecalibacterium gallinarum*** | 6% (184 fragments) | genus present; closest reference | — |
| ***Faecalibacterium duncaniae*** | 6% (169 fragments) | yes · 6.182% | Beneficial |
| ***Faecalibacterium tardum*** | 5% (165 fragments) | yes | Beneficial |

The organisms carrying this gene in your sample are *Faecalibacterium langellae*, *Faecalibacterium intestinale*, *Faecalibacterium duncaniae*; *Faecalibacterium langellae* alone accounts for 13% of the signal. Changing this reading means changing them.

### What your reading means above average · somewhat higher than most people

More of your gut bacteria carry the gene for making imidazole propionate than in most people. In studies, higher blood ImP tracks with insulin resistance and type 2 diabetes. Because carrying the gene is only weakly related to how much ImP actually ends up in blood, this is a reason to pay attention to metabolic health generally rather than a finding in itself.

**Two measurements of the same function disagree**

**Read this beside the polyphenol-metabolising species reading.** The urolithin and equol capacity reading sits at the 83rd percentile; the polyphenol-metabolising species reading is at the 43rd. The two are measured differently — genes in the whole community against the abundance of the species characterised for the job — and they can separate when the genes are carried by organisms outside the curated list, which the gene search counts and the species list does not; the gene family is shared with enzymes that do a different job, so some matched fragments are not this pathway. Of the two, the species reading is the better guide here. Urolithin and equol production is strain-level: two isolates of the same species differ, and only a handful of characterised strains do it. The curated species are the better guide; a high gene reading without them is most likely a related enzyme.

### How can I improve this?

Lower is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Faecalibacterium langellae*; *Faecalibacterium intestinale*; *Faecalibacterium duncaniae* (6.18% of your community); *Faecalibacterium tardum*. Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it Research associates higher levels with worse outcomes · Reasonably supported

ImP interferes with insulin signalling in host cells. Higher blood ImP is found in people with type 2 diabetes and prediabetes, and insulin resistance is itself an established risk factor for cognitive decline.

The link to cognition is indirect but mechanistically coherent: ImP activates p38-gamma and mTORC1, degrading insulin receptor substrate signalling. Impaired brain insulin signalling is a recognised feature of Alzheimer's disease. No study has yet shown that ImP itself causes cognitive decline in humans. Note also that in the source study of 294 stool samples, carrying the urdA gene was only weakly related to actual blood ImP levels. One overlap is worth flagging: Eggerthella, a documented urdA carrier, is among the pro-inflammatory genera found enriched in carriers of the APOE E4 Alzheimer's risk allele — so this pathway and that risk factor share an organism, though no causal link between them has been shown.

A stricter version of this measurement, which additionally requires the matched gene to carry the correct active-site amino acid, gives 8.8 copies per 100 genomes. The true value lies between the two.

Reference: Koh A. et al., Cell 175:947-961 (2018), doi:10.1016/j.cell.2018.09.055; Molinaro A. et al., Nat Commun 11:5881 (2020), doi:10.1038/s41467-020-19589-w; Sepulveda-Rivera V. et al., Sci Rep 15:28505 (2025), doi:10.1038/s41598-025-13262-2

## What the research says about changing this

**Polyphenol-conversion capacity (urolithin/equol) — status is defined by a challenge, not by genes** LANE C HUMAN TRIAL · LIMITED

Fired by: urda gene capacity: somewhat high

Whether someone converts ellagitannins to urolithins or daidzein to equol is determined by a substrate challenge with urine measurement. Direct urolithin A supplementation missed its primary endpoints and improved a secondary muscle measure in older adults; it bypasses the microbes rather than restoring them.

**Studied:** Urolithin A (Mitopure) (botanical supplement) · **in:** people with clinically confirmed older adults, ages 65–90, in USA, n = 66 · **studied protocol (a source fact, not an instruction):** 1,000 mg/day urolithin A for four months

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: single study. Strongest design: randomized trial.

- Direct urolithin A (1,000 mg/day for four months) in adults 65–90 improved a secondary muscle-endurance measure but missed its primary walking and ATP endpoints; it bypasses rather than restores microbes. [Liu S et al. JAMA Netw Open 2022;5:e2144279] · lane C · randomized double-blind placebo-controlled · n = 66 · mixed
- Equol-producer status is defined by a soy challenge (2 × 250 mL soy milk/day for three days) and urinary S-equol:daidzein, not by the presence of a species. [Setchell KDR, Cole SJ. J Nutr 2006;136:2188] · lane D · substrate challenge

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “non-producer” · “take urolithin”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

## Indole-3-propionate (IPA) ▾

ABOVE AVERAGE

Indole-3-propionate (IPA), one of the more strongly neuroprotective gut metabolites described so far.

| | |
|---|---|
| YOUR RESULT | **8.9** copies per 100 genomes |
| PERCENTILE | **84th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 3.1 – 6.3 (median 4.6) |
| MADE FROM | Dietary tryptophan, via a reductive pathway in a narrow group of bacteria |
| DNA FRAGMENTS MATCHED | 196 |




### What is driving this 327 matched fragments, by the organism each came closest to

**23% of the signal comes from organisms found in you:** *Blautia luti* (0.76% of your community); *Blautia obeum* (0.29% of your community); *Peptoniphilus lacrimalis* (0.05% of your community). A further 77% matched *Blautia wexlerae*, *Anaerotignum neopropionicum*, whose genus is present but whose species was not resolved.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Blautia wexlerae*** | **71%** (233 fragments) | genus present; closest reference | — |
| ***Blautia luti*** | **16%** (52 fragments) | **yes** · 0.755% | ● Conditional |
| ***Blautia obeum*** | **4%** (12 fragments) | **yes** · 0.293% | ● Conditional |
| ***Anaerotignum neopropionicum*** | **2%** (8 fragments) | genus present; closest reference | — |
| ***Peptoniphilus lacrimalis*** | **2%** (6 fragments) | **yes** · 0.046% | ● Unknown |

The organisms carrying this gene in your sample are *Blautia luti*, *Blautia obeum*, *Peptoniphilus lacrimalis*. Changing this reading means changing them.

### The organisms behind this, and what is known about them

| FOUND IN YOU | WHAT THE LITERATURE RECORDS |
|---|---|
| ***Blautia obeum***<br>0.29% of your community · 4% of this signal | Abundant Lachnospiraceae member; Blautia is higher in some obesity cohorts and lower in others. No fixed reading.<br>■ : Review of Blautia as a widespread genus with mixed disease associations. *[Liu X et al. Gut Microbes 2021;13:1875796. PMID 33525961]*<br>Association only — no study on record shows that changing this organism changes an outcome. |

### What your reading means above average · somewhat higher than most people

Your gut has more indole-3-propionate capacity than most people, which is the favourable direction. IPA crosses into the brain, where it neutralises hydroxyl radicals, reduces DNA damage, calms inflammatory activity in astrocytes and microglia, and interferes with the formation of amyloid-beta fibrils. It is reduced in people with obesity and diabetes, so having plenty of capacity is a good sign.

### How can I improve this?

Higher is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Blautia luti* (0.76% of your community); *Blautia obeum* (0.29% of your community); *Peptoniphilus lacrimalis* (0.05% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it Research associates higher levels with better outcomes · Reasonably supported

IPA crosses into the brain, where it neutralises hydroxyl radicals, reduces DNA damage, calms inflammatory activity in astrocytes and microglia, and interferes with the formation of amyloid-beta fibrils. It is reduced in people with obesity and diabetes.

After absorption from the intestine, IPA transits to the brain where it neutralises hydroxyl radicals, decreases DNA damage, alleviates neuroinflammation by inhibiting astrocytes and microglia, and hinders the formation of amyloid-beta fibrils. Clinical studies show significantly reduced indole metabolites, including IPA, in people who are obese or diabetic, tracking with gut dysbiosis.

Reference: Pappolla M.A. et al., Neurobiol Dis 156:105403 (2021), doi:10.1016/j.nbd.2021.105403; Dodd D. et al., Nature 551:648-652 (2017), doi:10.1038/nature24661

## What the research says about changing this

**Tryptophan-metabolite capacity (indole, IPA, tryptamine) — no supported targeted intervention** NONE NO SUPPORTED INTERVENTION

Fired by: indole gene capacity: typical; ipa gene capacity: somewhat high; tryptamine gene capacity: typical

Microbial tryptophan metabolism is an active research area with associations to gut barrier and immune signalling, but no human trial shows that changing these gene readings changes a health outcome. Tryptophan intake is never adjusted from stool DNA.

**No supported targeted intervention** — no human study on record shows that changing this reading changes a health outcome. That is a result, not a gap in the analysis.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “serotonin level” · “tryptophan deficiency”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

## Peptide uptake

TYPICAL

How much capacity your gut bacteria have to import peptides whole, rather than breaking them down outside the cell first.

| | |
|---|---|
| YOUR RESULT | **108.7** copies per 100 genomes |
| PERCENTILE | **28th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 106.4 – 137.1 (median 123.0) |
| MADE FROM | peptides from dietary and host protein |
| DNA FRAGMENTS MATCHED | 14,203 |




### What is driving this

4,184 matched fragments, by the organism each came closest to

**41% of the signal comes from organisms found in you:** *Faecalibacterium prausnitzii* (6.18% of your community). A further 53% matched *Blautia wexlerae*, *Bacteroides reticulotermitis*, *Clostridium scindens*, whose genus is present but whose species was not resolved. The remaining 5% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Faecalibacterium prausnitzii*** | **23%** (950 fragments) | **yes** · 6.182% | ● Beneficial |
| ***Blautia wexlerae*** | **18%** (755 fragments) | genus present; closest reference | — |
| ***Bacteroides reticulotermitis*** | **8%** (335 fragments) | genus present; closest reference | — |
| ***Clostridium scindens*** | **4%** (149 fragments) | genus present; closest reference | — |
| ***Robinsoniella peoriensis*** | **3%** (127 fragments) | not found; closest reference | — |

The organisms carrying this gene in your sample are *Faecalibacterium prausnitzii*; *Faecalibacterium prausnitzii* alone accounts for 23% of the signal. Changing this reading means changing them.

### The organisms behind this, and what is known about them

| FOUND IN YOU | WHAT THE LITERATURE RECORDS |
|---|---|
| ***Faecalibacterium prausnitzii***<br>6.18% of your community · 23% of this signal | One of the most abundant butyrate producers in adult stool. Reduced relative abundance is one of the most replicated findings in Crohn's disease and ulcerative colitis, and lower levels are reported across several inflammatory and metabolic conditions. It is also simply variable between healthy people and across countries; a low level alone is not a diagnosis and no product restores the species itself.<br>■ when lower: Reduced F. prausnitzii is associated with Crohn's disease and with post-operative recurrence; the isolate had anti-inflammatory effects in vitro and in mouse colitis. *[Sokol H et al. PNAS 2008;105:16731. PMID 18936492]*<br>■ when lower: A 2017 review consolidates lower F. prausnitzii across IBD, IBS, obesity, coeliac disease and colorectal cancer cohorts, with phylogroup-dependent patterns. *[Lopez-Siles M et al. ISME J 2017;11:841. PMID 28072420]*<br>Association only — no study on record shows that changing this organism changes an outcome. |

### What your reading means

typical · in line with most people

A mid-range amount of peptide-uptake capacity, which is what most samples carry. Read it beside the protein-fermentation reading rather than on its own, since the same peptides can go either way.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

This reading follows the organisms carrying it, and the ones found in you are *Faecalibacterium prausnitzii* (6.18% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

The research picture is mixed · Early evidence

Taking peptides in is a different capability from cutting proteins up. This is the uptake side, and it is a gene count rather than a rate.

Peptide transport is how most gut bacteria actually acquire amino acids: importing a short peptide costs less than making each residue, and the transporters are among the most highly expressed genes in gut isolates. What this cannot tell you is how much dietary protein reaches your colon. That is set by how much you eat and how well your small intestine digests it, and a transporter count in stool speaks to neither. Nor does high uptake capacity mean more protein fermentation: the same peptides can be built into bacterial protein instead, which is the opposite outcome.

Reference: Detmers F.J., Lanfermeijer F.C., Poolman B., "Peptides and ATP binding cassette peptide transporters", Res Microbiol 152:245-258 (2001), doi:10.1016/s0923-2508(01)01196-2.

‹ at a glance

## Ammonia (from urea) ▲

TYPICAL

A bacterial enzyme that splits urea into ammonia, raising the pH of the gut lumen.

| | |
|---|---|
| YOUR RESULT | **16.3** copies per 100 genomes |
| PERCENTILE | **70th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 8.2 – 17.6 (median 11.7) |
| MADE FROM | Urea, which diffuses into the gut from the bloodstream |
| DNA FRAGMENTS MATCHED | 575 |




### What is driving this 897 matched fragments, by the organism each came closest to

**8% of the signal comes from organisms found in you:** *Blautia luti* (0.76% of your community). A further 88% matched *Blautia difficilis*, *Clostridium yunnanense*, whose genus is present but whose species was not resolved. The remaining 4% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Blautia difficilis*** | **75%** (673 fragments) | genus present; closest reference | — |
| ***Blautia luti*** | **7%** (66 fragments) | **yes** · 0.755% | ● Conditional |
| ***Anaerovorax odorimutans*** | **2%** (17 fragments) | not found; closest reference | — |
| ***Clostridium yunnanense*** | **2%** (14 fragments) | genus present; closest reference | — |
| ***Ethanoligenens harbinense*** | **1%** (12 fragments) | not found; closest reference | — |

The organisms carrying this gene in your sample are *Blautia luti*. Changing this reading means changing them.

### What your reading means typical · in line with most people

Your urease capacity sits in the middle of the reference range. Urea hydrolysis is a normal gut activity, part of how nitrogen is recycled between host and bacteria, and a typical level has no significance in someone with healthy liver and kidneys.

**Why this colour.** Bacterial urease makes ammonia, the mechanism behind hepatic encephalopathy and the trait that lets Proteus form kidney stones and H. pylori survive acid.

### How can I improve this?

Lower is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Blautia luti* (0.76% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it Research associates higher levels with worse outcomes · Reasonably supported

Ammonia from bacterial urease is the mechanism behind hepatic encephalopathy in liver disease and rises with kidney failure. In healthy people, urease capacity is common and carries no known implication.

Wong et al. showed that end-stage renal disease expands urease- and uricase-containing families while contracting short-chain fatty acid producers, which is the clearest human evidence for the pH mechanism. Ni et al. found that a urease-positive community was sufficient to drive dysbiosis and worsen colitis in mice, and that replacing it with a urease-negative E. coli reversed the effect. In liver disease, the whole therapeutic strategy for encephalopathy — lactulose to acidify the colon, rifaximin to reduce the ammonia producers — is aimed at this enzyme's product. None of that makes urease capacity meaningful in a person with normal liver and kidney function.

Reference: Wong J. et al., Am J Nephrol 39:230-237 (2014); Ni J. et al., Sci Transl Med 9:eaah6888 (2017), doi:10.1126/scitranslmed.aah6888

‹ at a glance

## Indole / indoxyl sulfate ▾

TYPICAL

Indole, which the host converts to indoxyl sulfate — a compound that builds up when kidney function is impaired.

| | |
|---|---|
| YOUR RESULT | **10.0** copies per 100 genomes |
| PERCENTILE | **33rd** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 8.3 – 20.8 (median 13.3) |
| MADE FROM | Dietary tryptophan |
| DNA FRAGMENTS MATCHED | 237 |




### What is driving this 422 matched fragments, by the organism each came closest to

**50% of the signal comes from organisms found in you:** *Alistipes finegoldii* (0.31% of your community); *Agathobaculum butyriciproducens* (0.36% of your community); *Jilunia laotingensis* (0.34% of your community). A further 50% matched *Bacteroides eggerthii*, *Bacteroides intestinalis*, whose genus is present but whose species was not resolved.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Bacteroides eggerthii*** | **13%** (53 fragments) | genus present; closest reference | — |
| ***Alistipes finegoldii*** | **6%** (27 fragments) | **yes** · 0.307% | ● Conditional |
| ***Agathobaculum butyriciproducens*** | **6%** (26 fragments) | **yes** · 0.363% | ● Beneficial |
| ***Jilunia laotingensis*** | **6%** (26 fragments) | **yes** · 0.343% | ● Conditional |
| ***Bacteroides intestinalis*** | **6%** (25 fragments) | genus present; closest reference | — |

The organisms carrying this gene in your sample are *Alistipes finegoldii*, *Agathobaculum butyriciproducens*, *Jilunia laotingensis*. Changing this reading means changing them.

### The organisms behind this, and what is known about them

| FOUND IN YOU | WHAT THE LITERATURE RECORDS |
|---|---|
| ***Alistipes finegoldii***<br>0.31% of your community · 6% of this signal | Alistipes species with mixed associations; protective in mouse colitis, higher in some human disease cohorts.<br>■ : Review of Alistipes across cohorts. *[Parker BJ et al. Front Immunol 2020;11:906. PMID 32582143]*<br>Association only — no study on record shows that changing this organism changes an outcome. |

### What your reading means typical · in line with most people

Tryptophan can be routed down several different paths and this test cannot tell which one dominates, so read this number together with indole-3-propionate below. Your capacity for the indole route itself is unremarkable.

**Why this colour.** Indole from tryptophan tightens the gut barrier and signals through the AhR receptor; the exception is advanced kidney disease, where its metabolite indoxyl sulfate accumulates.

### How can I improve this?

Higher is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Alistipes finegoldii* (0.31% of your community); *Agathobaculum butyriciproducens* (0.36% of your community); *Jilunia laotingensis* (0.34% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it Research associates higher levels with better outcomes · Early evidence

Tryptophan can be routed down several paths, and they do not all point the same way. Indole itself has antioxidant properties, but its host-derived product indoxyl sulfate is a uraemic toxin. Read this panel alongside indole-3-propionate, which is the clearly beneficial branch.

Microbial tryptophan metabolites influence neuroinflammation and the gut-brain axis, and people with obesity or diabetes show reduced levels of the beneficial indole derivatives. Whether high tryptophanase capacity is good or bad depends on which downstream products dominate, which this test cannot determine. Indoxyl sulfate accumulation is primarily a concern in reduced kidney function.

Reference: Roager H.M., Licht T.R., Nat Commun 9:3294 (2018), doi:10.1038/s41467-018-05470-4

## What the research says about changing this

**Tryptophan-metabolite capacity (indole, IPA, tryptamine) — no supported targeted intervention** NONE NO SUPPORTED INTERVENTION

Fired by: indole gene capacity: typical; ipa gene capacity: somewhat high; tryptamine gene capacity: typical

Microbial tryptophan metabolism is an active research area with associations to gut barrier and immune signalling, but no human trial shows that changing these gene readings changes a health outcome. Tryptophan intake is never adjusted from stool DNA.

**No supported targeted intervention** — no human study on record shows that changing this reading changes a health outcome. That is a result, not a gap in the analysis.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “serotonin level” · “tryptophan deficiency”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

## Trimethylamine / TMAO ▲

TYPICAL

Trimethylamine (TMA), which the liver converts into trimethylamine N-oxide (TMAO).

| | |
|---|---|
| YOUR RESULT | **3.3** copies per 100 genomes |
| PERCENTILE | **65th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 1.6 – 3.9 (median 2.4) |
| MADE FROM | Dietary choline (eggs, liver, soy) and L-carnitine (red meat) |
| DNA FRAGMENTS MATCHED | 140 |




### What is driving this 246 matched fragments, by the organism each came closest to

A further 50% matched *Clostridium scindens*, *Streptococcus canis*, whose genus is present but whose species was not resolved. The remaining 50% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Clostridium scindens*** | **37%** (91 fragments) | genus present; closest reference | — |
| ***Sporomusa termitida*** | **29%** (72 fragments) | not found; closest reference | — |
| ***Sporomusa paucivorans*** | **8%** (20 fragments) | not found; closest reference | — |
| ***Streptococcus canis*** | **5%** (13 fragments) | genus present; closest reference | — |
| ***Terrisporobacter mayombei*** | **4%** (10 fragments) | not found; closest reference | — |

None of the closest references is itself in your organism list: the fragments most likely come from close relatives that carry a similar gene. The genus is named where it was found.

### What your reading means typical · in line with most people

TMAO has the strongest reported cognitive association of any metabolite in this panel, so being unremarkable here is genuinely worth knowing. Dietary choline and L-carnitine are the raw material if this ever becomes relevant.

### How can I improve this?

Lower is the more favourable direction for this reading.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it Research associates higher levels with worse outcomes · Reasonably supported

TMAO is the most consistently reported gut metabolite association in Alzheimer's disease research. It is elevated in the cerebrospinal fluid of people with Alzheimer's, and dietary TMAO accelerates brain ageing and cognitive decline in mice.

Vogt et al. found TMAO elevated in the cerebrospinal fluid of people with Alzheimer's disease and mild cognitive impairment, correlating with biomarkers of tau pathology and neuronal damage. TMAO has also been linked to amyloid-beta accumulation, tau aggregation and synaptic damage through mitochondrial impairment. Separately, higher circulating TMAO is associated with cardiovascular disease and newly diagnosed diabetes, both of which raise dementia risk.

Reference: Vogt N.M. et al., Alzheimers Res Ther 10:124 (2018), doi:10.1186/s13195-018-0451-2; Buawangpong N. et al., J Cell Physiol 237:1661-1685 (2022), doi:10.1002/jcp.30706

‹ at a glance

## Branched-chain amino acids (valine, leucine, isoleucine) ▲

TYPICAL

The capacity of your gut bacteria to synthesise the branched-chain amino acids valine, leucine and isoleucine.

| | |
|---|---|
| YOUR RESULT | **180.5** copies per 100 genomes |
| PERCENTILE | **37th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 173.6 – 199.0 (median 188.4) |
| MADE FROM | Pyruvate and other central carbon intermediates |
| DNA FRAGMENTS MATCHED | 6,681 |




### What is driving this 7,552 matched fragments, by the organism each came closest to

**100% of the signal comes from organisms found in you:** *Lachnospira eligens* (5.27% of your community); *Anaerobutyricum hallii* (0.31% of your community); *Roseburia faecis* (0.59% of your community); *Mediterraneibacter butyricigenes* (0.00% of your community).

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Lachnospira eligens*** | **5%** (357 fragments) | **yes** · 5.265% | Beneficial |
| ***Anaerobutyricum hallii*** | **5%** (348 fragments) | **yes** · 0.313% | Beneficial |
| ***Roseburia faecis*** | **4%** (298 fragments) | **yes** · 0.586% | Beneficial |
| ***Mediterraneibacter butyricigenes*** | **3%** (225 fragments) | **yes** · 0.001% | Conditional |
| ***Faecalibacillus faecis*** | **3%** (209 fragments) | **yes** | Conditional |

The organisms carrying this gene in your sample are *Lachnospira eligens*, *Anaerobutyricum hallii*, *Roseburia faecis*; *Lachnospira eligens* alone accounts for 5% of the signal. Changing this reading means changing them.

### The organisms behind this, and what is known about them

| FOUND IN YOU | WHAT THE LITERATURE RECORDS |
|---|---|
| ***Roseburia faecis***<br>0.59% of your community · 4% of this signal | Butyrate-producing Roseburia species commonly detected in adult stool.<br>■ when lower: Roseburia spp. reviewed as butyrate producers depleted in disease cohorts. *[Tamanai-Shacoori Z et al. Future Microbiol 2017;12:157. PMID 28139139]*<br>Association only — no study on record shows that changing this organism changes an outcome. |

### What your reading means typical · in line with most people

Your community's branched-chain amino acid synthesis capacity sits in the middle of the reference range. The raised capacity reported in insulin resistance was a group-level pattern; a typical reading is the expected state and carries no message about metabolic health on its own.

### How can I improve this?

Lower is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Lachnospira eligens* (5.27% of your community); *Anaerobutyricum hallii* (0.31% of your community); *Roseburia faecis* (0.59% of your community); *Mediterraneibacter butyricigenes* (0.00% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it Research associates higher levels with worse outcomes · Reasonably supported

Higher microbial branched-chain amino acid biosynthesis potential was the strongest functional signature of insulin resistance in a large European cohort, driven mainly by Prevotella copri and Bacteroides vulgatus.

Pedersen et al. profiled 277 non-diabetic Danish adults with serum metabolomics and shotgun metagenomics and found that insulin-resistant individuals had higher serum BCAAs and that the gut community's BCAA biosynthesis potential was the strongest associated functional module. They then showed that colonising mice with Prevotella copri — a main driver species — raised serum BCAAs and induced insulin resistance, which turns a correlation into a plausible causal chain. The honest limits: the driver species are diet-dependent and geographically variable, and serum BCAA elevation in insulin resistance has a substantial host-metabolism component that this measurement does not touch.

Reference: Pedersen H.K. et al., Nature 535:376-381 (2016), doi:10.1038/nature18646

‹ at a glance

## Protein and nitrogen metabolism

TYPICAL

How much of your community is set up to ferment protein rather than carbohydrate.

| | |
|---|---|
| YOUR RESULT | **298.5** copies per 100 genomes |
| PERCENTILE | **45th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 285.3 – 318.6 (median 302.0) |
| MADE FROM | Dietary and host protein reaching the colon |
| DNA FRAGMENTS MATCHED | 9,999 |




### What is driving this 9,823 matched fragments, by the organism each came closest to

**55% of the signal comes from organisms found in you:** *Dorea formicigenerans* (0.58% of your community); *Lachnospira pectinoschiza* (0.11% of your community); *Bacteroides thetaiotaomicron* (0.09% of your community). A further 31% matched *Agathobacter ruminis*, whose genus is present but whose species was not resolved. The remaining 14% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Agathobacter ruminis*** | **4%** (380 fragments) | genus present; closest reference | — |
| ***Dorea formicigenerans*** | **3%** (253 fragments) | **yes** · 0.582% | Conditional |
| ***Lachnospira pectinoschiza*** | **2%** (221 fragments) | **yes** · 0.109% | Beneficial |
| ***Bacteroides thetaiotaomicron*** | **2%** (203 fragments) | **yes** · 0.092% | Conditional |
| ***Butyrivibrio hungatei*** | **2%** (165 fragments) | not found; closest reference | — |

The organisms carrying this gene in your sample are *Dorea formicigenerans*, *Lachnospira pectinoschiza*, *Bacteroides thetaiotaomicron*. Changing this reading means changing them.

### The organisms behind this, and what is known about them

| FOUND IN YOU | WHAT THE LITERATURE RECORDS |
|---|---|
| ***Dorea formicigenerans***<br>0.58% of your community · 3% of this signal | Gas-producing Lachnospiraceae member; higher in some IBS and obesity cohorts.<br>■ : Dorea among taxa enriched in IBS in a large cohort study with functional readouts. *[Tap J et al. Gastroenterology 2017;152:111. PMID 27725146]*<br>Association only — no study on record shows that changing this organism changes an outcome. |
| ***Bacteroides thetaiotaomicron***<br>0.09% of your community · 2% of this signal | Model glycan-degrading commensal with a large polysaccharide-utilisation repertoire; common and without a fixed directional reading.<br>■ : Bacteroides as glycan-degrading commensals. *[Wexler HM. Clin Microbiol Rev 2007;20:593. PMID 17934076]*<br>Association only — no study on record shows that changing this organism changes an outcome. |

### What your reading means typical · in line with most people

Mid-range protein-fermentation capacity, which is the usual finding. This chemistry rises when fermentable fibre runs short, so it is worth reading beside the carbohydrate breadth reading rather than on its own.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

This reading follows the organisms carrying it, and the ones found in you are *Dorea formicigenerans* (0.58% of your community); *Lachnospira pectinoschiza* (0.11% of your community); *Bacteroides thetaiotaomicron* (0.09% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it The research picture is mixed · Early evidence

Protein fermentation produces branched-chain fatty acids, ammonia and phenolic compounds, which is a different chemistry from fibre fermentation and generally a less favourable one.

Generic peptidase abundance is not a measurement of dietary protein digestion: every organism carries peptidases for its own protein turnover, so a raw count tracks how many organisms are present. The informative markers are the ones specific to fermenting amino acids for energy - Stickland reductases, branched-chain dehydrogenases - and those are what this counts. Protein breakdown, branched-chain amino acid synthesis and branched-chain amino acid fermentation are three different things, and this report keeps three separate readings for them.

One of the genes behind this figure (pepT, ilvE) is matching distant relatives rather than close ones, so treat the number as an upper bound. The pathway is still detected — other genes in it match well.

Reference: Diether N.E., Willing B.P., "Microbial fermentation of dietary protein: an important factor in diet-microbe-host interaction", Microorganisms 7:19 (2019), doi:10.3390/microorganisms7010019.

## Gut-Brain & Metabolic Signalling

‹ at a glance
what you can do about it ›

## gamma-aminobutyric acid (GABA) ▼

BELOW AVERAGE

GABA, the main inhibitory neurotransmitter in the brain, which many gut bacteria also make from glutamate.

| | |
|---|---|
| YOUR RESULT | **10.5** copies per 100 genomes |
| PERCENTILE | 20th of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 12.4 – 26.6 (median 19.9) |
| MADE FROM | Dietary and microbial glutamate |
| DNA FRAGMENTS MATCHED | 365 |




### What is driving this

459 matched fragments, by the organism each came closest to

**100% of the signal comes from organisms found in you:** *Bacteroides clarus* (1.39% of your community); *Phocaeicola vulgatus*; *Phocaeicola massiliensis* (0.55% of your community); *Parabacteroides distasonis* (0.57% of your community).

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Bacteroides clarus*** | 27% (125 fragments) | yes · 1.387% | Conditional |
| ***Phocaeicola vulgatus*** | 9% (40 fragments) | yes | Conditional |
| ***Phocaeicola massiliensis*** | 6% (28 fragments) | yes · 0.549% | Conditional |
| ***Parabacteroides distasonis*** | 6% (27 fragments) | yes · 0.570% | Conditional |
| ***Phocaeicola dorei*** | 5% (21 fragments) | yes · 2.056% | Conditional |

The organisms carrying this gene in your sample are *Bacteroides clarus*, *Phocaeicola vulgatus*, *Phocaeicola massiliensis*; *Bacteroides clarus* alone accounts for 27% of the signal. Changing this reading means changing them.

### The organisms behind this, and what is known about them

| FOUND IN YOU | WHAT THE LITERATURE RECORDS |
|---|---|
| ***Parabacteroides distasonis***<br>0.57% of your community · 6% of this signal | Common commensal; lower in some IBD/obesity cohorts and enriched in others; mouse work suggests metabolic benefit via succinate and bile acids. Not directional in humans.<br>■ : P. distasonis alleviated obesity and metabolic dysfunction in mice via succinate and secondary bile acids; human data are associative. *[Wang K et al. Cell Rep 2019;26:222. PMID 30605678]*<br>Association only — no study on record shows that changing this organism changes an outcome. |

### What your reading means

below average · somewhat lower than most people

Less GABA-producing capacity than most people. One small study related low GABA production to a depression-associated brain signature; that is a single finding on an indirect measure and not a reason for concern.

**Why this colour.** Bacterial GABA is read as a gut-brain benefit in the evidence that exists, which is early.

### How can I improve this?

Higher is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Bacteroides clarus* (1.39% of your community); *Phocaeicola vulgatus*; *Phocaeicola massiliensis* (0.55% of your community); *Parabacteroides distasonis* (0.57% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

Research associates higher levels with better outcomes · Preliminary

Certain Bacteroides and Bifidobacterium species produce GABA in the gut, and one study found that GABA-producing capacity related inversely to a brain signature of depression. Whether gut GABA reaches the brain in meaningful amounts is unresolved, so this is reported as an interesting capacity, not a mood measurement.

Strandwitz et al. isolated a GABA-dependent Bacteroides from human stool, showed that Bacteroides species are major gut GABA producers, and found that relative Bacteroides abundance related inversely to depression- associated brain signatures on functional MRI in a small cohort. That is a correlation in a small sample on an indirect outcome. The mechanistic gap is real: GABA crosses the blood-brain barrier poorly, and the routes proposed for gut-derived GABA run through the vagus nerve and enteric nervous system rather than the circulation. Treat this as a genuine community feature with an unresolved connection to anything you feel.

Reference: Strandwitz P. et al., Nat Microbiol 4:396-403 (2019), doi:10.1038/s41564-018-0307-3; Otaru N. et al., Front Microbiol 12:656895 (2021)

## What the research says about changing this

**Neurotransmitter-related capacity (GABA, tyrosine decarboxylase) — a medication fact, not a mood reading** LANE D NO SUPPORTED INTERVENTION

Fired by: dopamine gene capacity: typical; gaba gene capacity: somewhat low

Bacterial GABA and tyrosine-decarboxylase genes do not measure brain neurotransmitters. The one clinically relevant fact is that tyrosine decarboxylase abundance correlated with levodopa dose requirement in Parkinson patients — information for a prescriber, not an action.

**No supported targeted intervention** — no human study on record shows that changing this reading changes a health outcome. That is a result, not a gap in the analysis.

- Bacterial tyrosine decarboxylase abundance correlated with levodopa dose requirement in Parkinson patients — a medication-context fact for the prescriber; there is no approved microbiome inhibitor or eradication protocol. [van Kessel SP et al. Nat Commun 2019;10:310] · lane D · human association + in vitro/rat

**Conventional route:** If taking levodopa: information card to the prescriber; no dose change from this report.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “your dopamine is low” · “GABA deficiency” · “mood bacteria”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance
what you can do about it ›

## Histamine ▲

ABOVE AVERAGE

Histamine, made by some gut bacteria from the amino acid histidine.

| | |
|---|---|
| YOUR RESULT | **0.5** copies per 100 genomes |
| PERCENTILE | **75th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 0 – 0.5 (median 0.2) |
| MADE FROM | Dietary histidine; also present directly in aged and fermented foods |
| DNA FRAGMENTS MATCHED | 1,025 |




### What is driving this 17 matched fragments, by the organism each came closest to

**94% of the signal comes from organisms found in you:** *Jilunia laotingensis* (0.34% of your community); *Bacteroides nordii* (0.01% of your community). A further 6% matched *Bifidobacterium magnum*, whose genus is present but whose species was not resolved.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Jilunia laotingensis*** | **76%** (13 fragments) | **yes** · 0.343% | ● Conditional |
| ***Bacteroides nordii*** | **18%** (3 fragments) | **yes** · 0.009% | ● Conditional |
| ***Bifidobacterium magnum*** | **6%** (1 fragments) | genus present; closest reference | — |

The organisms carrying this gene in your sample are *Jilunia laotingensis*, *Bacteroides nordii*; *Jilunia laotingensis* alone accounts for 76% of the signal. Changing this reading means changing them.

### What your reading means above average · too few reads for a stable reading; Poisson noise dominates at this count

Your gut carries more histamine-producing capacity than most people's. The table above names the organisms carrying the gene in your sample, and that is what decides whether it matters: histamine from Klebsiella, Morganella and their relatives has been linked to visceral pain in an IBS subgroup and to histamine-intolerance symptoms, while histamine from Lactobacillus reuteri is anti-inflammatory. When the producers found are not the anti-inflammatory exception, a high reading is worth taking seriously alongside any symptoms of flushing, headache, hives or gut pain after fermented or aged foods.

**Why this colour.** High bacterial histamine capacity is the reading people worry about: histamine intolerance, IBS pain and mast-cell symptoms. Coloured as adverse unless the producers found are the anti-inflammatory exception.

### How can I improve this?

Lower is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Jilunia laotingensis* (0.34% of your community); *Bacteroides nordii* (0.01% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it Research associates higher levels with worse outcomes · Early evidence

A subset of gut bacteria can produce histamine. One well-conducted study linked Klebsiella aerogenes histamine production to visceral pain in a subgroup of IBS patients, while Lactobacillus reuteri histamine has the opposite, anti-inflammatory effect. Direction depends on the producer.

De Palma et al. identified Klebsiella aerogenes as a histamine super-producer in a subset of IBS patients, showed it drove mast cell accumulation and visceral hypersensitivity when transferred to mice, and found that a low-fermentable-carbohydrate diet reduced both. That is a strong, mechanism-backed finding in a defined subgroup. Against it, Thomas et al. showed Lactobacillus reuteri histamine signalling through the H2 receptor suppresses TNF production — the same molecule from a different organism acting anti-inflammatorily. There is no evidence that total bacterial histidine decarboxylase capacity, which is what this panel measures, predicts anything on its own.

Based on 1025 matching DNA fragments, few enough that the exact figure carries real uncertainty; the broad position is more reliable than the precise number.

Reference: De Palma G. et al., Sci Transl Med 14:eabj1895 (2022), doi:10.1126/scitranslmed.abj1895; Thomas C.M. et al., PLoS One 7:e31951 (2012), doi:10.1371/journal.pone.0031951

## What the research says about changing this

**Higher histidine-decarboxylase capacity — the only trial is a diet trial in IBS** LANE C HUMAN TRIAL · LIMITED

Fired by: histamine gene capacity: somewhat high

Stool hdc genes are not a histamine measurement. In diagnosed IBS, a three-week randomized low-FODMAP diet lowered urinary histamine and improved symptoms; the causal Klebsiella-histamine work was in mice.

**Studied:** Dietitian-guided limited low-FODMAP diet (restriction → reintroduction → personalisation) (diet) · **in:** people with clinically confirmed IBS, ages 18–70, in Canada, n = 37 · **studied protocol (a source fact, not an instruction):** three-week low-FODMAP versus high-FODMAP diet

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: single study. Strongest design: randomized trial.

- A three-week randomized low-FODMAP diet in diagnosed IBS improved symptoms and lowered urinary histamine — a diet effect in IBS, not a treatment for stool histamine genes. [McIntosh K et al. Gut 2017;66:1241] · lane C · randomized controlled · n = 40 · benefit
- A high-urinary-histamine IBS subgroup and a K. aerogenes hdc variant were identified, with the causal treatment work done in mice. [De Palma G et al. Sci Transl Med 2022;14:eabj1895] · lane E · human subgroup + gnotobiotic mouse experiments

**Applicability:** The study required a clinical diagnosis of ibs. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “histamine intolerance” · “low-histamine diet for everyone”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance
what you can do about it ›

## GABA breakdown

TYPICAL

Your gut bacteria's capacity to break GABA down, shown beside their capacity to make it.

| | |
|---|---|
| YOUR RESULT | **8.2** copies per 100 genomes |
| PERCENTILE | **25th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 8.2 – 12.3 (median 9.5) |
| MADE FROM | GABA made by other bacteria or arriving in fermented food |
| DNA FRAGMENTS MATCHED | 561 |




### What is driving this

149 matched fragments, by the organism each came closest to

**No organism behind this reading was found in your sample.** The 149 matched fragments came closest to *Mycobacterium ostraviense*, *Mycobacterium aquaticum*, *Mycobacterium yunnanensis*, which the taxonomic profile did not detect in you. A read resembling an organism's copy of a gene is not that organism being present — some other member of your community carries a similar sequence, or the reference set lacks whichever one does. Treat this as a trace signal with no identified source rather than as evidence about those species.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Mycobacterium ostraviense*** | **24%** (36 fragments) | not found; closest reference | — |
| ***Mycobacterium aquaticum*** | **19%** (28 fragments) | not found; closest reference | — |
| ***Mycobacterium yunnanensis*** | **4%** (6 fragments) | not found; closest reference | — |
| ***Brenneria roseae*** | **3%** (5 fragments) | not found; closest reference | — |
| ***Streptomyces lavendulae*** | **3%** (5 fragments) | not found; closest reference | — |

None of the closest references is itself in your organism list: the fragments most likely come from close relatives that carry a similar gene. The genus is named where it was found.

### What your reading means

typical · detected below the confirmed identity threshold; read as indicative only

Mid-range GABA degradation capacity. This is shown beside your GABA production reading, which is a separate measurement over different genes; the two are never subtracted from one another.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

The research picture is mixed · Early evidence

The existing GABA panel measures production. This measures degradation. They are separate readings and are never subtracted from each other.

A community that both makes and breaks down GABA is not a community with no GABA, which is why these are reported side by side rather than netted. Neither number is a GABA concentration, and neither says anything about GABA reaching your brain: the blood-brain barrier is not permeable to it, so any gut-brain effect runs through other routes such as the vagus nerve or the enteric nervous system.

This is a provisional result rather than a confirmed one. Read it as a possible signal.

Reference: Strandwitz P. et al., "GABA-modulating bacteria of the human gut microbiota", Nat Microbiol 4:396-403 (2019), doi:10.1038/s41564-018-0307-3.

‹ at a glance

## P9 protein (experimental)

TYPICAL

Whether the reads carry the exact Akkermansia protein that was tested in the GLP-1 experiments, as opposed to something in the same family.

| | |
|---|---|
| YOUR RESULT | **0.053** copies per 100 genomes |
| PERCENTILE | **30th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 0 – 0.9 (median 0.2) |
| MADE FROM | not a metabolite; this is a bacterial protein-coding gene |
| DNA FRAGMENTS MATCHED | 2 |




### What is driving this 4 matched fragments, by the organism each came closest to

**100% of the signal comes from organisms found in you:** *Akkermansia muciniphila*.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Akkermansia muciniphila*** | **100%** (4 fragments) | **yes** | Beneficial |

The organisms carrying this gene in your sample are *Akkermansia muciniphila*; *Akkermansia muciniphila* alone accounts for 100% of the signal. Changing this reading means changing them.

### The organisms behind this, and what is known about them

| FOUND IN YOU | WHAT THE LITERATURE RECORDS |
|---|---|
| ***Akkermansia muciniphila*** | Mucin-degrading Verrucomicrobium. Higher levels are associated with metabolic health in several cohorts and a pasteurised strain (MucT) has human trial evidence, yet the same species is enriched in multiple sclerosis and Parkinson's disease cohorts. Endogenous abundance does not predict response to any Akkermansia product.<br>■ when higher: Pasteurised A. muciniphila MucT improved insulin sensitivity and metabolic markers in an exploratory 3-month trial of overweight/obese insulin-resistant volunteers. *[Depommier C et al. Nat Med 2019;25:1096. PMID 31263284]*<br>■ when higher: Akkermansia is enriched in multiple sclerosis patients across cohorts (iMSMS). *[iMSMS Consortium. Cell 2022;185:3467. PMID 36113426]* |

### What your reading means typical · too few reads for a stable reading; Poisson noise dominates at this count

A mid-range amount of read evidence on the tested sequence, read beside the family competition shown with it. Neither number is a hormone level, and the evidence for an effect is preclinical.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

This reading follows the organisms carrying it, and the ones found in you are *Akkermansia muciniphila*. Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it The research picture is mixed · Preliminary

A single protein with preclinical evidence. Finding its gene is a long way from a hormone response, and this is not comparable to a GLP-1 medication.

The protein was purified and tested in cell and mouse work, where it increased GLP-1 secretion through a host receptor. That is experimental evidence about a protein, not about a person: what this screen can say is whether reads map to the tested sequence at high identity. Detection is deliberately strict. One reference sequence means a diverged version in another strain will not reach the identity floor, and is reported as unresolved rather than counted, because transferring a functional annotation onto a divergent homolog is exactly the error the specification forbids here. Neither secretion nor any GLP-1 level is measured by these reads.

Based on 2 matching DNA fragments, few enough that the exact figure carries real uncertainty; the broad position is more reliable than the precise number.

Reference: Yoon H.S. et al., "Akkermansia muciniphila secretes a protein P9 that improves glucose homeostasis", Nat Microbiol 6:563-573 (2021), doi:10.1038/s41564-021-00880-5.

‹ at a glance

## Dopamine (from L-DOPA) and tyramine (from tyrosine)

TYPICAL

A bacterial enzyme that decarboxylates tyrosine to tyramine, and L-DOPA to dopamine, in the gut.

| | |
|---|---|
| YOUR RESULT | **0.8** copies per 100 genomes |
| PERCENTILE | **33rd** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 0.6 – 2.2 (median 1.1) |
| MADE FROM | Dietary tyrosine; and levodopa, if taken as medication |
| DNA FRAGMENTS MATCHED | 86 |




### What is driving this 23 matched fragments, by the organism each came closest to

**14% of the signal comes from organisms found in you:** *Eggerthella lenta* (0.02% of your community). A further 36% matched *Enterocloster lavalensis*, whose genus is present but whose species was not resolved. The remaining 50% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Enterocloster lavalensis*** | **22%** (5 fragments) | genus present; closest reference | — |
| ***Syntrophomonas wolfei*** | **13%** (3 fragments) | not found; closest reference | — |
| ***Sporomusa silvacetica*** | **9%** (2 fragments) | not found; closest reference | — |
| ***Eggerthella lenta*** | **9%** (2 fragments) | **yes** · 0.024% | ● Opportunist |
| ***Slackia heliotrinireducens*** | **9%** (2 fragments) | not found; closest reference | — |

The organisms carrying this gene in your sample are *Eggerthella lenta*. Changing this reading means changing them.

### The organisms behind this, and what is known about them

| FOUND IN YOU | WHAT THE LITERATURE RECORDS |
|---|---|
| ***Eggerthella lenta***<br>0.02% of your community · 9% of this signal | Actinobacterium notable for drug metabolism (inactivates digoxin via cgr genes; decarboxylates levodopa-derived dopamine). Higher in some IBD/RA cohorts. Relevant to medication history, not a target.<br>■ : E. lenta strains carrying the cgr operon inactivate digoxin; arginine suppresses this in mice. *[Haiser HJ et al. Science 2013;341:295. PMID 23869020]*<br>■ : E. lenta dehydroxylates dopamine produced from levodopa by Enterococcus faecalis tyrosine decarboxylase. *[Maini Rekdal V et al. Science 2019;364:eaau6323. PMID 31196984]*<br>Association only — no study on record shows that changing this organism changes an outcome. |

### What your reading means typical · too few reads for a stable reading; Poisson noise dominates at this count

Your tyrosine decarboxylase capacity sits in the middle of the reference range. This enzyme makes tyramine from dietary tyrosine and, in people taking levodopa, breaks the drug down before absorption; at a typical level neither effect stands out from the population.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

This reading follows the organisms carrying it, and the ones found in you are *Eggerthella lenta* (0.02% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it The research picture is mixed · Reasonably supported

Gut bacterial tyrosine decarboxylase converts levodopa into dopamine in the intestine, where it cannot reach the brain. Higher carriage is associated with needing a higher levodopa dose in Parkinson's disease. In people not taking levodopa, this mostly reflects tyramine-producing capacity.

van Kessel et al. showed that Enterococcus faecalis tyrDC decarboxylates levodopa in the gut, that carbidopa (the standard peripheral inhibitor) does not block the bacterial enzyme, and that tyrDC abundance in stool correlated with the levodopa dose Parkinson's patients required. Maini Rekdal et al. independently identified the same enzyme and the single active-site residue that determines whether it accepts L-DOPA, and found a small molecule that inhibits it. This is one of the best-characterised drug-microbiome interactions known. It says nothing about brain dopamine: dopamine made in the colon does not cross the

blood-brain barrier, which is the entire reason levodopa is given rather than dopamine itself.

This is a provisional result rather than a confirmed one — matches average 73% identity (at least 75% required), so the reads are matching distant relatives rather than the gene itself. Read it as a possible signal.

Reference: van Kessel S.P. et al., Nat Commun 10:310 (2019), doi:10.1038/s41467-019-08294-y; Maini Rekdal V. et al., Science 364:eaau6323 (2019), doi:10.1126/science.aau6323

## What the research says about changing this

**Neurotransmitter-related capacity (GABA, tyrosine decarboxylase) — a medication fact, not a mood reading** LANE D NO SUPPORTED INTERVENTION

Fired by: dopamine gene capacity: typical; gaba gene capacity: somewhat low

Bacterial GABA and tyrosine-decarboxylase genes do not measure brain neurotransmitters. The one clinically relevant fact is that tyrosine decarboxylase abundance correlated with levodopa dose requirement in Parkinson patients — information for a prescriber, not an action.

**No supported targeted intervention** — no human study on record shows that changing this reading changes a health outcome. That is a result, not a gap in the analysis.

- Bacterial tyrosine decarboxylase abundance correlated with levodopa dose requirement in Parkinson patients — a medication-context fact for the prescriber; there is no approved microbiome inhibitor or eradication protocol. [van Kessel SP et al. Nat Commun 2019;10:310] · lane D · human association + in vitro/rat

**Conventional route:** If taking levodopa: information card to the prescriber; no dose change from this report.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “your dopamine is low” · “GABA deficiency” · “mood bacteria”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

## Tryptamine

TYPICAL

Tryptamine, a molecule made by a small number of gut bacteria from tryptophan, which speeds intestinal transit through a serotonin receptor.

| | |
|---|---|
| YOUR RESULT | **0.1** copies per 100 genomes |
| PERCENTILE | **49th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 0 – 0.5 (median 0.1) |
| MADE FROM | Dietary tryptophan (protein) |
| DNA FRAGMENTS MATCHED | 3 |




### What is driving this

4 matched fragments, by the organism each came closest to

**No organism behind this reading was found in your sample.** The 4 matched fragments came closest to *Mycobacterium noviomagense*, which the taxonomic profile did not detect in you. A read resembling an organism's copy of a gene is not that organism being present — some other member of your community carries a similar sequence, or the reference set lacks whichever one does. Treat this as a trace signal with no identified source rather than as evidence about those species.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Mycobacterium noviomagense*** | **100%** (4 fragments) | not found; closest reference | — |

None of the closest references is itself in your organism list: the fragments most likely come from close relatives that carry a similar gene. The genus is named where it was found.

### What your reading means

typical · too few reads for a stable reading; Poisson noise dominates at this count

Your tryptophan decarboxylase capacity sits in the middle of the reference range. Tryptamine is made from dietary tryptophan by a small number of gut species and acts on serotonin receptors in the gut wall; a typical level is what most people carry and has no established consequence.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

The research picture is mixed · Early evidence

Tryptamine activates the intestinal 5-HT4 receptor and accelerates colonic transit in mice. The enzyme is carried by very few gut species, so most people have little or none.

Williams et al. found bacterial tryptophan decarboxylase in only two of more than a thousand sequenced human gut bacterial genomes — Clostridium sporogenes and Ruminococcus gnavus — while detecting the gene in about 10% of a human metagenomic cohort. Bhattarai et al. then showed that colonising germ-free mice with a tryptamine-producing strain accelerated colonic transit through the epithelial 5-HT4 receptor. This is a clean, small mechanism. What it does not support is any claim about mood: tryptamine made in the colon acts locally on gut receptors, and the separate literature on microbial effects on host serotonin runs through enterochromaffin cells, not through bacterial synthesis.

This is a provisional result rather than a confirmed one — matches average 73% identity (at least 75% required), so the reads are matching distant relatives rather than the gene itself. Read it as a possible signal.

Reference: Williams B.B. et al., Cell Host Microbe 16:495-503 (2014), doi:10.1016/j.chom.2014.09.001; Bhattarai Y. et al., Cell Host Microbe 23:775-785 (2018), doi:10.1016/j.chom.2018.05.004

## What the research says about changing this

**Tryptophan-metabolite capacity (indole, IPA, tryptamine) — no supported targeted intervention** NONE NO SUPPORTED INTERVENTION

Fired by: indole gene capacity: typical; ipa gene capacity: somewhat high; tryptamine gene capacity: typical

Microbial tryptophan metabolism is an active research area with associations to gut barrier and immune signalling, but no human trial shows that changing these gene readings changes a health outcome. Tryptophan intake is never adjusted from stool DNA.

**No supported targeted intervention** — no human study on record shows that changing this reading changes a health outcome. That is a result, not a gap in the analysis.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “serotonin level” · “tryptophan deficiency”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

## Plant-Compound Conversion

‹ at a glance
what you can do about it ›

## Glucosinolate to isothiocyanate conversion

LOW

Whether the characterised bacterial enzymes for converting glucosinolates to isothiocyanates are present in your sample.

| | |
|---|---|
| YOUR RESULT | **2.4** copies per 100 genomes |
| PERCENTILE | **13th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 3.7 – 11.2 (median 6.2) |
| MADE FROM | Glucosinolates from broccoli, rocket and other cruciferous vegetables |
| DNA FRAGMENTS MATCHED | 36 |




### What is driving this 128 matched fragments, by the organism each came closest to

**100% of the signal comes from organisms found in you:** *Bacteroides thetaiotaomicron* (0.09% of your community).

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Bacteroides thetaiotaomicron*** | **100%** (128 fragments) | **yes** · 0.092% | Conditional |

The organisms carrying this gene in your sample are *Bacteroides thetaiotaomicron*; *Bacteroides thetaiotaomicron* alone accounts for 100% of the signal. Changing this reading means changing them.

### The organisms behind this, and what is known about them

| FOUND IN YOU | WHAT THE LITERATURE RECORDS |
|---|---|
| ***Bacteroides thetaiotaomicron***<br>0.09% of your community · 100% of this signal | Model glycan-degrading commensal with a large polysaccharide-utilisation repertoire; common and without a fixed directional reading.<br>■ : Bacteroides as glycan-degrading commensals. *[Wexler HM. Clin Microbiol Rev 2007;20:593. PMID 17934076]*<br>Association only — no study on record shows that changing this organism changes an outcome. |

### What your reading means low · lower than most people

Little or none of the characterised enzymes here. With only three reference sequences for a strain-specific reaction, a low value is weak evidence of absence rather than proof of it.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

This reading follows the organisms carrying it, and the ones found in you are *Bacteroides thetaiotaomicron* (0.09% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it The research picture is mixed · Early evidence

Measured against the B. thetaiotaomicron locus itself at a high identity floor, because this route is described in one organism rather than carried by an enzyme family.

Three reference sequences is the right size for a reaction characterised in a single organism, and the high identity floor is what keeps a distant homolog from matching; the consequence is that this under-calls rather than over-calls. This bacterial route is not plant myrosinase, which is an enzyme of the vegetable and is destroyed by cooking, and it is not the only bacterial route described. A gene finding is also not a conversion: whether you actually produce isothiocyanates is settled by a substrate challenge with metabolomics, which can be imported beside this.

Reference: Liou C.S. et al., "A metabolic pathway for activation of dietary glucosinolates by a human gut symbiont", Cell 180:717-728 (2020), doi:10.1016/j.cell.2020.01.023.

‹ at a glance
what you can do about it ›

## Equol formation

PRESENT

Whether your gut bacteria carry the enzymes that turn the soy isoflavone daidzein into equol.

| | |
|---|---|
| YOUR RESULT | **0.1** copies per 100 genomes |
| PERCENTILE | **95th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 0 – 0 (median 0) |
| MADE FROM | daidzein from soy foods |
| DNA FRAGMENTS MATCHED | 12 |




### What is driving this

12 matched fragments, by the organism each came closest to

**No organism behind this reading was found in your sample.** The 12 matched fragments came closest to *Lactococcus garvieae*, which the taxonomic profile did not detect in you. A read resembling an organism's copy of a gene is not that organism being present — some other member of your community carries a similar sequence, or the reference set lacks whichever one does. Treat this as a trace signal with no identified source rather than as evidence about those species.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Lactococcus garvieae*** | **100%** (12 fragments) | not found; closest reference | — |

None of the closest references is itself in your organism list: the fragments most likely come from close relatives that carry a similar gene. The genus is named where it was found.

### What your reading means

present · more than half the reference cohort carries none of this, so any amount ranks above the middle; that is a statement about how uncommon this is and not about how much of it you have

Reads map to the published equol enzymes. That is the gene being present, not a measured equol response to soy, which is how producer status is actually defined.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

The research picture is mixed · Preliminary

Roughly a third of people carry equol-producing bacteria. This measures the published enzymes, and is not a measurement of equol.

The three reductases were purified and characterised, which is what makes a specific detector possible; the specification is explicit that aliases such as eqlA/B/C need strain-specific mapping and that a general reductase domain is not evidence of this chemistry. One reference sequence per step means low sensitivity by design, and a diverged enzyme is reported as unresolved rather than transferred onto this annotation. Equol producer status in the literature is defined by measuring equol in urine after a soy challenge, which DNA cannot substitute for.

Based on 12 matching DNA fragments, few enough that the exact figure carries real uncertainty; the broad position is more reliable than the precise number.

Reference: Schroder C. et al., "Identification and expression of genes involved in the conversion of daidzein and genistein by the equol-forming bacterium Slackia isoflavoniconvertens", Appl Environ Microbiol 79:3494-3502 (2013), doi:10.1128/AEM.03693-12.

‹ at a glance
what you can do about it ›

### Urolithin A formation

NOT DETECTED

Whether your gut bacteria carry the enzyme that makes urolithin A from the urolithin C produced earlier in the pathway.

| | |
|---|---|
| YOUR RESULT | **0** copies per 100 genomes |
| PERCENTILE | 0 of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 0 – 0.1 (median 0) |
| MADE FROM | ellagitannins in pomegranate, walnuts, raspberries and strawberries |
| DNA FRAGMENTS MATCHED | 0 |




#### What is driving this

2 matched fragments, by the organism each came closest to

A further 100% matched *Enterocloster bolteae*, whose genus is present but whose species was not resolved.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Enterocloster bolteae*** | **100%** (2 fragments) | genus present; closest reference | — |

None of the closest references is itself in your organism list: the fragments most likely come from close relatives that carry a similar gene. The genus is named where it was found.

#### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

#### The research behind it

The research picture is mixed · Preliminary

Only some people's microbiomes make urolithin A. This measures the published enzyme for the final step; it is not a urolithin measurement.

The enzyme responsible was identified in 2025 as a dehydroxylase encoded by an operon in gut Enterocloster species, and this panel holds the three proteins from that record rather than a gene-symbol search, which returns an unrelated fungal protein. Detection is strict: one reference sequence per subunit means a diverged version in another strain will not reach the identity floor, and is reported as unresolved rather than counted. Carrying the gene is also not the same as being a urolithin producer, which the specification is explicit about: that is established by measuring urolithin after an ellagitannin challenge, not from DNA.

Reference: Pidgeon R., Mitchell S., Shamash M., et al., "Diet-derived urolithin A is produced by a dehydroxylase encoded by human gut Enterocloster species", GenBank PQ855390.1 (2025).

## Mucus, Bile & Other Transformations

‹ at a glance
what you can do about it ›

## Unconjugated bile acids

NOTABLY HIGH

Deconjugated (unconjugated) bile acids — the first step your gut bacteria take in reworking bile.

| | |
|---|---|
| YOUR RESULT | **123.0** copies per 100 genomes |
| PERCENTILE | **95th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 65.7 – 95.4 (median 80.7) |
| MADE FROM | Liver-derived conjugated bile acids |
| DNA FRAGMENTS MATCHED | 2,023 |




### What is driving this

3,546 matched fragments, by the organism each came closest to

**68% of the signal comes from organisms found in you:** *Simiaoa sunii* (0.07% of your community); *Faecalibacterium duncaniae* (6.18% of your community); *Lachnospira eligens* (5.27% of your community). A further 32% matched *Blautia wexlerae*, *Eubacterium ventriosum*, whose genus is present but whose species was not resolved.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Simiaoa sunii*** | **18%** (645 fragments) | **yes** · 0.071% | Unknown |
| ***Blautia wexlerae*** | **9%** (314 fragments) | genus present; closest reference | — |
| ***Faecalibacterium duncaniae*** | **6%** (210 fragments) | **yes** · 6.182% | Beneficial |
| ***Eubacterium ventriosum*** | **5%** (173 fragments) | genus present; closest reference | — |
| ***Lachnospira eligens*** | **5%** (172 fragments) | **yes** · 5.265% | Beneficial |

The organisms carrying this gene in your sample are *Simiaoa sunii*, *Faecalibacterium duncaniae*, *Lachnospira eligens*; *Simiaoa sunii* alone accounts for 18% of the signal. Changing this reading means changing them.

### What your reading means

notably high · well above most people

Your gut has substantial bile-salt-deconjugating capacity. This is the gateway step that makes bile acids available for further bacterial processing, and it also affects fat absorption and cholesterol handling. Nearly everyone has plenty of it, so a high reading is not unusual.

**Two measurements agree**

This agrees with the probiotic-genus species reading (80th percentile), which measures the same function by counting the organisms characterised for the job instead of the genes.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

This reading follows the organisms carrying it, and the ones found in you are *Simiaoa sunii* (0.07% of your community); *Faecalibacterium duncaniae* (6.18% of your community); *Lachnospira eligens* (5.27% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

The research picture is mixed · Early evidence

This is an enabling step rather than an outcome. Deconjugation is what makes bile acids available for the secondary bile acid pathway, and it also affects fat absorption and cholesterol handling. Nearly everyone carries abundant capacity for it.

No direct association with cognition has been reported for bile salt hydrolase activity itself. It is reported here because it sits immediately upstream of the secondary bile acid pathway, and because it acts as an internal consistency check: substantial secondary bile acid capacity without deconjugation capacity would not make biological sense.

Reference: Jones B.V. et al., Proc Natl Acad Sci USA 105:13580-13585 (2008), doi:10.1073/pnas.0804437105

## What the research says about changing this

**Bile-salt hydrolase — the one strain with LDL trials, and why total capacity is not it** LANE C HUMAN TRIAL · LIMITED

Fired by: bsh gene capacity: notably high

Total stool BSH gene capacity is not the same as one BSH-active strain. L. reuteri NCIMB 30242 lowered LDL by 9–12% in two trials in people with measured high cholesterol; the reading here does not select or predict that.

**Studied:** Limosilactobacillus reuteri NCIMB 30242 (bile salt hydrolase-active) (probiotic) · **in:** people with clinically confirmed hypercholesterolaemia, ages 20–75, in Canada, n = 127 · **studied protocol (a source fact, not an instruction):** 2.9 × 10^9 CFU capsule twice daily for nine weeks

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: conflicting. Strongest design: randomized trial.

- In hypercholesterolaemic adults, a yogurt with bile-salt-hydrolase-active L. reuteri NCIMB 30242 lowered LDL by about 9% over six weeks — an exact strain/product result for that phenotype only. [Jones ML et al. Br J Nutr 2012;107:1505] · lane C · randomized double-blind placebo-controlled · n = 114 · benefit
- Capsules of the same strain (2.9 × 10^9 CFU twice daily, nine weeks) lowered LDL by about 12% in a second trial. Total stool BSH gene capacity does not predict this and does not justify another strain. [Jones ML et al. Eur J Clin Nutr 2012;66:1234] · lane C · randomized double-blind placebo-controlled · n = 127 · benefit
- Safety analysis of the L. reuteri NCIMB 30242 trials found no safety signal. [Jones ML et al. — safety of L. reuteri NCIMB 30242 (2012)] · lane C · safety analysis of RCT · null

**Applicability:** The study required a clinical diagnosis of hypercholesterolaemia. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “lowers your cholesterol” · “BSH deficiency”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance
what you can do about it ›

## Mucin glycans and lipid A

LOW

Capacity to forage the mucus layer's sugars, and to build the lipid A core of endotoxin.

| | |
|---|---|
| YOUR RESULT | **136.7** copies per 100 genomes |
| PERCENTILE | **15th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 164.2 – 262.9 (median 202.8) |
| MADE FROM | Host mucin glycans; bacterial outer-membrane biosynthesis |
| DNA FRAGMENTS MATCHED | 8,063 |




### What is driving this 4,943 matched fragments, by the organism each came closest to

**100% of the signal comes from organisms found in you:** *Phocaeicola vulgatus*; *Blautia massiliensis*; *Phocaeicola massiliensis* (0.55% of your community); *Bacteroides uniformis* (0.78% of your community).

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Phocaeicola vulgatus*** | **8%** (380 fragments) | **yes** | Conditional |
| ***Blautia massiliensis*** | **8%** (370 fragments) | **yes** | Conditional |
| ***Phocaeicola massiliensis*** | **5%** (247 fragments) | **yes** · 0.549% | Conditional |
| ***Bacteroides uniformis*** | **4%** (201 fragments) | **yes** · 0.779% | Conditional |
| ***Phocaeicola dorei*** | **4%** (172 fragments) | **yes** · 2.056% | Conditional |

The organisms carrying this gene in your sample are *Phocaeicola vulgatus*, *Blautia massiliensis*, *Phocaeicola massiliensis*; *Phocaeicola vulgatus* alone accounts for 8% of the signal. Changing this reading means changing them.

### The organisms behind this, and what is known about them

| FOUND IN YOU | WHAT THE LITERATURE RECORDS |
|---|---|
| ***Bacteroides uniformis***<br>0.78% of your community · 4% of this signal | Among the most prevalent Bacteroides in Western adults; level varies with diet and has no established directional meaning on its own.<br>■ : Bacteroides species are dominant, diet-responsive members of Western adult stool with both commensal and opportunistic behaviours. *[Wexler HM. Clin Microbiol Rev 2007;20:593. PMID 17934076]*<br>Association only — no study on record shows that changing this organism changes an outcome. |

### What your reading means low · lower than most people

Less mucin-foraging and lipid A capacity than most samples carry. Mucin foragers are ordinary residents, so a lower reading here is a description of your community rather than a sign of a healthier barrier.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

This reading follows the organisms carrying it, and the ones found in you are *Phocaeicola vulgatus*; *Blautia massiliensis*; *Phocaeicola massiliensis* (0.55% of your community); *Bacteroides uniformis* (0.78% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it The research picture is mixed · Early evidence

Mucin foraging is a normal part of how a gut community makes its living. These genes do not resolve an endotoxin structure or measure barrier damage.

Mucin-foraging organisms are ordinary residents and the mucus layer is continuously renewed, so carrying these enzymes is physiological rather than damaging; the question of barrier integrity needs a measurement this test does not make. For lipid A, the immune effect depends on how many acyl chains the finished molecule carries, and one modification gene does not resolve that structure. Total Gram-negative abundance resolves it even less. Where the structure cannot be determined the gene evidence is shown with its uncertainty rather than converted into an endotoxin figure.

One of the genes behind this figure (lpxA) is matching distant relatives rather than close ones, so treat the number as an upper bound. The pathway is still detected — other genes in it match well.

Reference: Tailford L.E. et al., "Mucin glycan foraging in the human gut microbiome", Front Genet 6:81 (2015), doi:10.3389/fgene.2015.00081.

‹ at a glance

## Secondary bile acids ▲

TYPICAL

Secondary bile acids, chiefly deoxycholic acid (DCA) and lithocholic acid (LCA), made by bacteria reworking the bile acids your liver produces.

| | |
|---|---|
| YOUR RESULT | **3.5** copies per 100 genomes |
| PERCENTILE | **71st** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 2.0 – 3.6 (median 2.5) |
| MADE FROM | Liver-derived primary bile acids, after bacterial deconjugation |
| DNA FRAGMENTS MATCHED | 339 |




### What is driving this

358 matched fragments, by the organism each came closest to

A further 73% matched *Blautia producta*, *Clostridium scindens*, whose genus is present but whose species was not resolved. The remaining 27% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Blautia producta*** | **62%** (223 fragments) | genus present; closest reference | — |
| ***Peptacetobacter hiranonis*** | **12%** (43 fragments) | not found; closest reference | — |
| ***Terrisporobacter glycolicus*** | **10%** (36 fragments) | not found; closest reference | — |
| ***Clostridium scindens*** | **9%** (31 fragments) | genus present; closest reference | — |
| ***Terrisporobacter mayombei*** | **4%** (14 fragments) | not found; closest reference | — |

None of the closest references is itself in your organism list: the fragments most likely come from close relatives that carry a similar gene. The genus is named where it was found.

### What your reading means

typical · detected below the confirmed identity threshold; read as indicative only

The organisms that perform this conversion are normally a very small fraction of the gut, so modest numbers here are expected and yours are unremarkable.

### How can I improve this?

Lower is the more favourable direction for this reading.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

Research associates higher levels with worse outcomes · Reasonably supported

A large study of people with Alzheimer's disease found lower levels of the liver's own primary bile acid and higher levels of the bacterially produced secondary bile acid DCA, with the ratio between them tracking cognitive decline.

MahmoudianDehkordi et al. reported reduced serum cholic acid and elevated DCA and its glycine- and taurine-conjugated forms in people with Alzheimer's disease compared with cognitively normal older adults. Bile acids and their receptors are present in the brain, and DCA has been shown to interact with nicastrin, a subunit of the gamma-secretase enzyme that generates amyloid-beta. Not all secondary bile acids behave the same way: tauroursodeoxycholic acid (TUDCA) inhibits amyloid-beta accumulation and improves mitochondrial function.

This is a provisional result rather than a confirmed one. Read it as a possible signal.

Reference: MahmoudianDehkordi S. et al., Alzheimers Dement 15:76-92 (2019), doi:10.1016/j.jalz.2018.07.217; Luan H. et al., Adv Biotechnol 1:7 (2023)

‹ at a glance

## Oxalate (degraded, not produced) ▼

TYPICAL

The capacity of your gut bacteria to break down dietary oxalate before your body absorbs it.

| | |
|---|---|
| YOUR RESULT | **0.4** copies per 100 genomes |
| PERCENTILE | **33rd** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 0.3 – 2.1 (median 0.6) |
| MADE FROM | Dietary oxalate (spinach, rhubarb, nuts, tea, chocolate) |
| DNA FRAGMENTS MATCHED | 9 |




### What is driving this

16 matched fragments, by the organism each came closest to

**44% of the signal comes from organisms found in you:** *Bifidobacterium hominis* (0.03% of your community). A further 33% matched *Lactobacillus acidophilus*, *Bifidobacterium magnum*, whose genus is present but whose species was not resolved. The remaining 22% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Bifidobacterium hominis*** | **25%** (4 fragments) | **yes** · 0.028% | Beneficial |
| ***Lactobacillus acidophilus*** | **12%** (2 fragments) | genus present; closest reference | — |
| ***Paraburkholderia haematera*** | **6%** (1 fragments) | not found; closest reference | — |
| ***Bifidobacterium magnum*** | **6%** (1 fragments) | genus present; closest reference | — |
| ***Paraburkholderia pallida*** | **6%** (1 fragments) | not found; closest reference | — |

The organisms carrying this gene in your sample are *Bifidobacterium hominis*; *Bifidobacterium hominis* alone accounts for 25% of the signal. Changing this reading means changing them.

### What your reading means

typical · too few reads for a stable reading; Poisson noise dominates at this count

Your community's oxalate-degrading capacity sits in the middle of the reference range. Oxalobacter formigenes and related organisms break down dietary oxalate before it can be absorbed; a typical level means the usual degree of protection against oxalate load, and says nothing about kidney stone risk by itself.

### How can I improve this?

Higher is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Bifidobacterium hominis* (0.03% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

Research associates higher levels with better outcomes · Reasonably supported

Oxalate absorbed from the gut is excreted by the kidney, where it can form calcium oxalate stones. Bacteria that degrade oxalate in the gut reduce how much reaches the kidney, and higher degradation capacity is the favourable direction.

Kaufman et al. found Oxalobacter formigenes colonisation associated with a roughly 70% lower odds of recurrent calcium oxalate stones in a case-control study, and Sidhu et al. showed the mechanism in cystic fibrosis patients, who lose the organism to repeated antibiotics and develop hyperoxaluria. Against a simple reading: attempts to treat hyperoxaluria by administering O. formigenes have given mixed results in trials, and oxalate degradation capacity is distributed across many species beyond Oxalobacter, so carriage of one organism is not the whole picture. The direction of the association is consistent; the size of the effect from any one person's community is not established.

Based on 9 matching DNA fragments, few enough that the exact figure carries real uncertainty; the broad position is more reliable than the precise number.

Reference: Kaufman D.W. et al., J Am Soc Nephrol 19:1197-1203 (2008), doi:10.1681/ASN.2007101058; Sidhu H. et al., Lancet 352:1026-1029 (1998)

‹ at a glance

## Deconjugated glucuronides (oestrogens, drug metabolites, bilirubin) ▲

TYPICAL

A bacterial enzyme that reverses one of the liver's main routes for getting rid of drugs, hormones and toxins.

| | |
|---|---|
| YOUR RESULT | **83.1** copies per 100 genomes |
| PERCENTILE | **37th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 75.0 – 114.6 (median 91.9) |
| MADE FROM | Glucuronide conjugates made by your own liver |
| DNA FRAGMENTS MATCHED | 2,529 |




### What is driving this

4,838 matched fragments, by the organism each came closest to

**71% of the signal comes from organisms found in you:** *Faecalibacterium prausnitzii* (6.18% of your community); *Roseburia hominis* (0.49% of your community); *Lachnospira eligens* (5.27% of your community). A further 29% matched *Bacteroides reticulotermitis*, *Faecalibacterium hominis*, whose genus is present but whose species was not resolved.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Faecalibacterium prausnitzii*** | **27%** (1323 fragments) | **yes** · 6.182% | ● Beneficial |
| ***Bacteroides reticulotermitis*** | **14%** (680 fragments) | genus present; closest reference | — |
| ***Roseburia hominis*** | **7%** (347 fragments) | **yes** · 0.492% | ● Beneficial |
| ***Lachnospira eligens*** | **6%** (314 fragments) | **yes** · 5.265% | ● Beneficial |
| ***Faecalibacterium hominis*** | **3%** (146 fragments) | genus present; closest reference | — |

The organisms carrying this gene in your sample are *Faecalibacterium prausnitzii*, *Roseburia hominis*, *Lachnospira eligens*; *Faecalibacterium prausnitzii* alone accounts for 27% of the signal. Changing this reading means changing them.

### The organisms behind this, and what is known about them

| FOUND IN YOU | WHAT THE LITERATURE RECORDS |
|---|---|
| ***Faecalibacterium prausnitzii***<br>6.18% of your community · 27% of this signal | One of the most abundant butyrate producers in adult stool. Reduced relative abundance is one of the most replicated findings in Crohn's disease and ulcerative colitis, and lower levels are reported across several inflammatory and metabolic conditions. It is also simply variable between healthy people and across countries; a low level alone is not a diagnosis and no product restores the species itself.<br>■ when lower: Reduced F. prausnitzii is associated with Crohn's disease and with post-operative recurrence; the isolate had anti-inflammatory effects in vitro and in mouse colitis. *[Sokol H et al. PNAS 2008;105:16731. PMID 18936492]*<br>■ when lower: A 2017 review consolidates lower F. prausnitzii across IBD, IBS, obesity, coeliac disease and colorectal cancer cohorts, with phylogroup-dependent patterns. *[Lopez-Siles M et al. ISME J 2017;11:841. PMID 28072420]*<br>Association only — no study on record shows that changing this organism changes an outcome. |
| ***Roseburia hominis***<br>0.49% of your community · 7% of this signal | Butyrate producer; reduced in ulcerative colitis cohorts alongside F. prausnitzii.<br>■ when lower: R. hominis and F. prausnitzii were both reduced in UC and inversely correlated with disease activity. *[Machiels K et al. Gut 2014;63:1275. PMID 24021287]*<br>Association only — no study on record shows that changing this organism changes an outcome. |

### What your reading means

typical · in line with most people

Your beta-glucuronidase capacity sits in the middle of the reference range. Some is universal — every gut community carries it — and it is part of how the gut recycles the liver's tagged compounds, including bile and hormones.

**Why this colour.** Reactivates toxins, hormones and drugs the liver had tagged for excretion; raised in colorectal-cancer cohorts and the cause of irinotecan gut toxicity.

### How can I improve this?

Lower is the more favourable direction for this reading.

This reading follows the organisms carrying it, and the ones found in you are *Faecalibacterium prausnitzii* (6.18% of your community); *Roseburia hominis* (0.49% of your community); *Lachnospira eligens* (5.27% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

Research associates higher levels with worse outcomes · Reasonably supported

Your liver tags compounds with glucuronic acid to excrete them. Gut bacterial beta-glucuronidase removes that tag, sending the compound back into circulation. This is well established for the chemotherapy drug irinotecan and is the proposed route for

oestrogen recycling.

The best-evidenced case is irinotecan: bacterial GUS reactivates the inactive SN-38G metabolite in the gut, producing the dose-limiting diarrhoea, and Wallace et al. showed that a selective bacterial GUS inhibitor prevented that toxicity in mice without harming the bacteria. Pollet et al. mapped GUS diversity across the human gut and showed that loop structure, not abundance, governs which substrates an enzyme processes. The oestrogen story rests mostly on in vitro deconjugation and on correlational studies; it is plausible and popular, and it is not established that between-person GUS variation changes circulating hormone levels in a way that matters.

Reference: Wallace B.D. et al., Science 330:831-835 (2010), doi:10.1126/science.1191175; Pollet R.M. et al., Structure 25:967-977 (2017), doi:10.1016/j.str.2017.05.003

## Urate & Purines

‹ at a glance
what you can do about it ›

## Urate degradation

ABOVE AVERAGE

Your gut bacteria's capacity to break urate down without oxygen, using the route described in the 2023 work on bacterial purine handling.

| | |
|---|---|
| YOUR RESULT | **180.3** copies per 100 genomes |
| PERCENTILE | **77th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 132.8 – 176.4 (median 149.9) |
| MADE FROM | urate arriving from host purine turnover and from purine-rich food |
| DNA FRAGMENTS MATCHED | 5,709 |




### What is driving this 7,674 matched fragments, by the organism each came closest to

**34% of the signal comes from organisms found in you:** *Faecalibacterium prausnitzii* (6.18% of your community). A further 42% matched *Blautia wexlerae*, *Anaerostipes rhamnosivorans*, whose genus is present but whose species was not resolved. The remaining 24% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Faecalibacterium prausnitzii*** | **17%** (1272 fragments) | **yes** · 6.182% | ● Beneficial |
| ***Blautia wexlerae*** | **12%** (920 fragments) | genus present; closest reference | — |
| ***Anaerostipes rhamnosivorans*** | **8%** (648 fragments) | genus present; closest reference | — |
| ***Sporomusa ovata*** | **7%** (506 fragments) | not found; closest reference | — |
| ***Longibaculum muris*** | **5%** (380 fragments) | not found; closest reference | — |

The organisms carrying this gene in your sample are *Faecalibacterium prausnitzii*; *Faecalibacterium prausnitzii* alone accounts for 17% of the signal. Changing this reading means changing them.

### The organisms behind this, and what is known about them

| FOUND IN YOU | WHAT THE LITERATURE RECORDS |
|---|---|
| ***Faecalibacterium prausnitzii***<br>6.18% of your community · 17% of this signal | One of the most abundant butyrate producers in adult stool. Reduced relative abundance is one of the most replicated findings in Crohn's disease and ulcerative colitis, and lower levels are reported across several inflammatory and metabolic conditions. It is also simply variable between healthy people and across countries; a low level alone is not a diagnosis and no product restores the species itself.<br>■ when lower: Reduced F. prausnitzii is associated with Crohn's disease and with post-operative recurrence; the isolate had anti-inflammatory effects in vitro and in mouse colitis. *[Sokol H et al. PNAS 2008;105:16731. PMID 18936492]*<br>■ when lower: A 2017 review consolidates lower F. prausnitzii across IBD, IBS, obesity, coeliac disease and colorectal cancer cohorts, with phylogroup-dependent patterns. *[Lopez-Siles M et al. ISME J 2017;11:841. PMID 28072420]*<br>Association only — no study on record shows that changing this organism changes an outcome. |

### What your reading means above average · somewhat higher than most people

More of the anaerobic urate-degrading route than most samples carry. The mouse work associates this route with lower host urate, but nothing here measures your urate, and the human evidence is association rather than effect.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

This reading follows the organisms carrying it, and the ones found in you are *Faecalibacterium prausnitzii* (6.18% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it The research picture is mixed · Preliminary

Gut bacteria can degrade urate anaerobically. This is a count of the genes for that route. It is not a serum urate measurement and does not diagnose or rule out gout.

Two 2023 studies showed that gut bacteria degrade urate anaerobically and that this contributes to whole-body purine handling in mice, with human cohort association. The route runs through xanthine and purine fermentation to short-chain fatty acids, and

it is not the uricase reaction: uricase requires molecular oxygen and most gut anaerobes do not use it. What is measured here is gene capacity in stool. Serum urate is set mostly by renal excretion and host production, and no gene count here can substitute for a blood test or explain an individual's urate level.

One of the genes behind this figure (xdhC) is matching distant relatives rather than close ones, so treat the number as an upper bound. The pathway is still detected — other genes in it match well.

Reference: Liu Y. et al., "Gut microbial DNA and purine degradation", Cell 186:3400-3413 (2023), doi:10.1016/j.cell.2023.06.010; Kasahara K. et al., "Gut bacterial metabolism contributes to host global purine homeostasis", Cell Host Microbe 31:1038-1053 (2023), doi:10.1016/j.chom.2023.05.011.

## Polyamines

‹ at a glance
what you can do about it ›

## Spermidine formation

ABOVE AVERAGE

Capacity to make spermidine, counted over both of the routes bacteria actually use rather than only the one in the textbooks.

| | |
|---|---|
| YOUR RESULT | **118.1** copies per 100 genomes |
| PERCENTILE | **80th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 90.3 – 113.3 (median 99.9) |
| MADE FROM | putrescine, with either S-adenosylmethionine or aspartate semialdehyde |
| DNA FRAGMENTS MATCHED | 3,458 |




### What is driving this

3,757 matched fragments, by the organism each came closest to

**85% of the signal comes from organisms found in you:** *Lachnospira eligens* (5.27% of your community); *Lachnospira hominis* (0.13% of your community); *Faecalibacterium duncaniae* (6.18% of your community); *Faecalibacterium intestinale*. The remaining 15% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Lachnospira eligens*** | **7%** (252 fragments) | **yes** · 5.265% | Beneficial |
| ***Lachnospira hominis*** | **5%** (196 fragments) | **yes** · 0.129% | Beneficial |
| ***Faecalibacterium duncaniae*** | **5%** (188 fragments) | **yes** · 6.182% | Beneficial |
| ***Faecalibacterium intestinale*** | **4%** (133 fragments) | **yes** | Beneficial |
| ***Lacrimispora defluvii*** | **4%** (131 fragments) | not found; closest reference | — |

The organisms carrying this gene in your sample are *Lachnospira eligens*, *Lachnospira hominis*, *Faecalibacterium duncaniae*; *Lachnospira eligens* alone accounts for 7% of the signal. Changing this reading means changing them.

### What your reading means

above average · somewhat higher than most people

More spermidine-forming capacity than most samples carry, counted across both routes. This is gene capacity, not a spermidine measurement, and the longevity literature on spermidine is mostly from model organisms.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

This reading follows the organisms carrying it, and the ones found in you are *Lachnospira eligens* (5.27% of your community); *Lachnospira hominis* (0.13% of your community); *Faecalibacterium duncaniae* (6.18% of your community); *Faecalibacterium intestinale*. Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

The research picture is mixed · Early evidence

Spermidine has been studied for autophagy and healthy ageing. This reading is the genetic capacity of your bacteria to produce it, not a spermidine level.

Most gut Bacteroidetes make spermidine through carboxyspermidine rather than through the aminopropyltransferase route, and a 2023 study described a further variant running through carboxyaminopropylagmatine. A pipeline that looks only for spermidine synthase therefore reports an absence where the capacity is present, which is why both routes are counted here. Spermidine's association with autophagy and longevity comes largely from model organisms and from observational human diet data; supplement trials in humans are small and mixed, and none of that work measured bacterial gene capacity of the kind reported here.

Reference: Hanfrey C.C. et al., "Alternative spermidine biosynthetic route is critical for growth of Campylobacter jejuni and is the dominant polyamine pathway in human gut microbiota", J Biol Chem 286:43301-43312 (2011), doi:10.1074/jbc.M111.307835.

‹ at a glance

## Putrescine formation

TYPICAL

How much capacity your gut bacteria have to make putrescine, the smallest of the polyamines.

| | |
|---|---|
| YOUR RESULT | **206.2** copies per 100 genomes |
| PERCENTILE | **61st** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 182.2 – 220.3 (median 194.4) |
| MADE FROM | arginine and ornithine from dietary and host protein |
| DNA FRAGMENTS MATCHED | 4,727 |




### What is driving this 5,349 matched fragments, by the organism each came closest to

**57% of the signal comes from organisms found in you:** *Blautia faecicola* (0.11% of your community). The remaining 43% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Blautia faecicola*** | **34%** (1826 fragments) | **yes** · 0.111% | ● Conditional |
| ***Glutamicibacter creatinolyticus*** | **10%** (538 fragments) | not found; closest reference | — |
| ***Escherichia coli*** | **7%** (362 fragments) | not found; closest reference | — |
| ***Pseudobutyrivibrio ruminis*** | **6%** (302 fragments) | not found; closest reference | — |
| ***Pseudobutyrivibrio xylanivorans*** | **4%** (200 fragments) | not found; closest reference | — |

The organisms carrying this gene in your sample are *Blautia faecicola*; *Blautia faecicola* alone accounts for 34% of the signal. Changing this reading means changing them.

### What your reading means typical · in line with most people

A mid-range amount of putrescine-forming capacity, similar to most samples in the reference set. Polyamine capacity is read alongside the spermidine and agmatine readings rather than interpreted as a good or bad number by itself.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

This reading follows the organisms carrying it, and the ones found in you are *Blautia faecicola* (0.11% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it The research picture is mixed · Early evidence

Polyamines are needed for the constant renewal of the gut lining, and are also elevated in some disease states. A higher or lower reading is not good or bad on its own.

Putrescine, spermidine and spermine are required for cell proliferation, and the gut epithelium renews itself every few days, so a supply matters. The same chemistry is implicated in tumour biology, and bacterial polyamine production has been associated with colorectal cancer tissue in observational work. Stable-isotope and multi-omics work in 2025 traced polyamine flux in gut communities and found production and consumption happening simultaneously in the same sample, which is why this reading is presented as capacity rather than as an amount, and why its direction is context-dependent rather than favourable or adverse.

One of the genes behind this figure (speC) is matching distant relatives rather than close ones, so treat the number as an upper bound. The pathway is still detected — other genes in it match well.

Reference: Nakamura A. et al., "Symbiotic polyamine metabolism regulates epithelial proliferation and macrophage differentiation in the colon", Nat Commun 12:2105 (2021), doi:10.1038/s41467-021-22212-1.

‹ at a glance

## Agmatine formation

TYPICAL

Capacity to turn arginine into agmatine, which feeds the polyamine routes and has its own activity in the nervous system.

| | |
|---|---|
| YOUR RESULT | **32.5** copies per 100 genomes |
| PERCENTILE | **43rd** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 26.9 – 43.5 (median 34.7) |
| MADE FROM | arginine from dietary and host protein |
| DNA FRAGMENTS MATCHED | 1,062 |




### What is driving this 1,331 matched fragments, by the organism each came closest to

**28% of the signal comes from organisms found in you:** *Phocaeicola vulgatus*. A further 53% matched *Anaerostipes rhamnosivorans*, *Bacteroides difficilis*, whose genus is present but whose species was not resolved. The remaining 18% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Anaerostipes rhamnosivorans*** | **23%** (304 fragments) | genus present; closest reference | — |
| ***Phocaeicola vulgatus*** | **14%** (187 fragments) | **yes** | ● Conditional |
| ***Bacillus mycoides*** | **5%** (62 fragments) | not found; closest reference | — |
| ***Burkholderia arboris*** | **4%** (60 fragments) | not found; closest reference | — |
| ***Bacteroides difficilis*** | **4%** (50 fragments) | genus present; closest reference | — |

The organisms carrying this gene in your sample are *Phocaeicola vulgatus*. Changing this reading means changing them.

### What your reading means typical · in line with most people

A mid-range amount of arginine decarboxylase capacity, which is what most samples carry. Read it beside the putrescine and spermidine readings, since agmatine is the entry point to both of those routes.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

This reading follows the organisms carrying it, and the ones found in you are *Phocaeicola vulgatus*. Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it The research picture is mixed · Early evidence

Agmatine is a polyamine precursor and a neuroactive molecule. This counts the gene for making it; it is not a measurement of agmatine.

Arginine decarboxylase is also part of how several gut bacteria survive stomach acid, consuming a proton with each decarboxylation, so a high reading can reflect acid resistance rather than a deliberate investment in polyamine production. Agmatine itself binds imidazoline and alpha-2 adrenergic receptors and has been studied in animal models of pain and mood, but the doses used in those studies are pharmacological and bear no relation to what gut bacteria produce. Nothing here measures agmatine reaching your circulation or your brain.

Reference: Piletz J.E. et al., "Agmatine: clinical applications after 100 years in translation", Drug Discov Today 18:880-893 (2013), doi:10.1016/j.drudis.2013.05.017.

## Glutathione & Ethanol

‹ at a glance
what you can do about it ›

## Microbial ethanol formation

ABOVE AVERAGE

Capacity of your gut bacteria to produce ethanol, and the separate ethanolamine route that produces acetaldehyde.

| | |
|---|---|
| YOUR RESULT | **69.8** copies per 100 genomes |
| PERCENTILE | **76th** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 53.6 – 69.0 (median 60.4) |
| MADE FROM | dietary sugars, and ethanolamine from cell membrane turnover |
| DNA FRAGMENTS MATCHED | 3,560 |




### What is driving this 4,702 matched fragments, by the organism each came closest to

**40% of the signal comes from organisms found in you:** *Lachnospira eligens* (5.27% of your community); *Ruminococcus bromii* (3.40% of your community). A further 52% matched *Blautia wexlerae*, *Ruminococcus callidus*, whose genus is present but whose species was not resolved. The remaining 8% matched reference sequences from organisms not detected in you, which is a resemblance rather than a presence.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Blautia wexlerae*** | **16%** (754 fragments) | genus present; closest reference | — |
| ***Lachnospira eligens*** | **10%** (482 fragments) | **yes** · 5.265% | ● Beneficial |
| ***Ruminococcus bromii*** | **6%** (262 fragments) | **yes** · 3.396% | ● Beneficial |
| ***Ruminococcus callidus*** | **4%** (211 fragments) | genus present; closest reference | — |
| ***[Ruminococcus] lactaris*** | **3%** (145 fragments) | not found; closest reference | — |

The organisms carrying this gene in your sample are *Lachnospira eligens*, *Ruminococcus bromii*. Changing this reading means changing them.

### The organisms behind this, and what is known about them

| FOUND IN YOU | WHAT THE LITERATURE RECORDS |
|---|---|
| ***Ruminococcus bromii***<br>3.40% of your community · 6% of this signal | Keystone degrader of resistant starch: it breaks down starch granules other species cannot, feeding the wider community. Its baseline level predicted who responded to a resistant-starch feeding trial with a butyrate rise.<br>■ : R. bromii is a keystone species for resistant starch degradation in the human colon. *[Ze X et al. ISME J 2012;6:1535. PMID 22343308]*<br>■ : Faecal butyrate rose with resistant potato starch only in baseline responders, and R. bromii predicted response. *[Venkataraman A et al. Microbiome 2016;4:33. PMID 27357127]*<br>Association only — no study on record shows that changing this organism changes an outcome. |

### What your reading means above average · somewhat higher than most people

More ethanol-forming gene capacity than most samples carry. Carrying adhE is common and is not the same as being a high-producing strain, which is what the fatty liver disease studies identified.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

This reading follows the organisms carrying it, and the ones found in you are *Lachnospira eligens* (5.27% of your community); *Ruminococcus bromii* (3.40% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it The research picture is mixed · Preliminary

Some gut bacteria make ethanol. High-producing strains have been linked to fatty liver disease in specific studies. This is a gene count and does not diagnose anything.

The clearest evidence concerns particular high-alcohol-producing Klebsiella pneumoniae strains found in people with non-alcoholic fatty liver disease, where colonising mice with those strains produced liver changes. That is a strain-level finding: carrying adhE is ordinary and most bacteria that carry it are not high producers, so this reading cannot identify auto-brewery syndrome or fatty liver. Acetaldehyde, the intermediate, is the genotoxic part of this chemistry. A separate reading for acetaldehyde disposal is deliberately not given, because the aldehyde dehydrogenase family is not substrate-specific enough

for a count over it to mean acetaldehyde rather than any other aldehyde.

Reference: Yuan J. et al., "Fatty liver disease caused by high-alcohol-producing Klebsiella pneumoniae", Cell Metab 30:675-688 (2019), doi:10.1016/j.cmet.2019.08.018.

‹ at a glance

## Glutathione synthesis

TYPICAL

Your gut bacteria's capacity to make glutathione, the main small-molecule antioxidant inside cells.

| | |
|---|---|
| YOUR RESULT | **5.2** copies per 100 genomes |
| PERCENTILE | **52nd** of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 2.8 – 8.9 (median 4.9) |
| MADE FROM | glutamate, cysteine and glycine |
| DNA FRAGMENTS MATCHED | 412 |




### What is driving this

220 matched fragments, by the organism each came closest to

**56% of the signal comes from organisms found in you:** *Ellagibacter isourolithinifaciens* (0.33% of your community); *Adlercreutzia rubneri*; *Anaerobutyricum hallii* (0.31% of your community). A further 44% matched *Finegoldia dalianensis*, *Clostridium thermobutyricum*, whose genus is present but whose species was not resolved.

| ORGANISM | SHARE OF SIGNAL | IN YOUR SAMPLE | CLASS |
|---|---|---|---|
| ***Finegoldia dalianensis*** | **12%** (26 fragments) | genus present; closest reference | — |
| ***Ellagibacter isourolithinifaciens*** | **8%** (18 fragments) | **yes** · 0.327% | Unknown |
| ***Adlercreutzia rubneri*** | **8%** (17 fragments) | **yes** | Beneficial |
| ***Anaerobutyricum hallii*** | **6%** (12 fragments) | **yes** · 0.313% | Beneficial |
| ***Clostridium thermobutyricum*** | **5%** (11 fragments) | genus present; closest reference | — |

The organisms carrying this gene in your sample are *Ellagibacter isourolithinifaciens*, *Adlercreutzia rubneri*, *Anaerobutyricum hallii*. Changing this reading means changing them.

### What your reading means

typical · in line with most people

A mid-range amount of bacterial glutathione synthesis capacity, similar to most samples. Many gut anaerobes import glutathione rather than making it, so the synthesis reading is only part of how a community handles it.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

This reading follows the organisms carrying it, and the ones found in you are *Ellagibacter isourolithinifaciens* (0.33% of your community); *Adlercreutzia rubneri*; *Anaerobutyricum hallii* (0.31% of your community). Anything that changes this reading has to change them.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

The research picture is mixed · Preliminary

Some gut bacteria synthesise glutathione. This counts those genes. It is not a measurement of glutathione in you, and it is not a marker of your own antioxidant status.

Bacterial glutathione synthesis is real but patchy: many gut anaerobes lack the route entirely and import what they need, and the fused gshF enzyme found in some Gram-positives does both steps at once, so a search for the two-enzyme route alone will miss it. A 2026 study described glutathione handling in Eubacterium rectale specifically. What this cannot establish is any effect on your own glutathione: host glutathione is made by your own cells from cysteine supply, and there is no established route by which bacterial glutathione in the colon becomes host glutathione in the liver.

One of the genes behind this figure (gshB, gshF) is matching distant relatives rather than close ones, so treat the number as an upper bound. The pathway is still detected — other genes in it match well.

Reference: Smirnova G.V., Oktyabrsky O.N., "Glutathione in bacteria", Biochemistry (Moscow) 70:1199-1211 (2005), doi:10.1007/s10541-005-0248-3.

## Toxins & Virulence

‹ at a glance

## Colibactin and bacterial toxins (virulence factors, not metabolites) ▲

NOT DETECTED

Three specific bacterial genes whose products damage or transform colon cells: a Fusobacterium adhesin, a Bacteroides fragilis toxin, and the colibactin biosynthesis machinery of certain Escherichia coli.

| | |
|---|---|
| YOUR RESULT | **0** copies per 100 genomes |
| PERCENTILE | 0 of 100 adult stool samples (MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort) |
| REFERENCE MIDDLE HALF | 0 – 0 (median 0) |
| MADE FROM | Not a metabolite — these are toxins and biosynthetic enzymes |
| DNA FRAGMENTS MATCHED | 0 |




### How can I improve this?

Lower is the more favourable direction for this reading.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

Research associates higher levels with worse outcomes · Reasonably supported

These are the mechanistically specific colorectal cancer virulence factors, with genuine causal evidence in animal models and, for colibactin, a distinctive mutational signature in human tumours. They are also carried by many healthy people, and this panel can only report detected or not detected.

Colibactin has the strongest evidence: Pleguezuelos-Manzano et al. exposed human intestinal organoids to pks+ E. coli and produced a mutational signature they then found in human colorectal tumours — a direct molecular fingerprint linking the bacterium to the mutations. Rubinstein et al. showed FadA binds E-cadherin and activates beta-catenin signalling, and Wu et al. showed enterotoxigenic B. fragilis drives IL-17-dependent colon tumorigenesis in mice. Against acting on any of this: pks+ E. coli is carried by around a fifth of healthy adults, ETBF by a substantial minority, and the overwhelming majority never develop colorectal cancer. Carriage is a risk factor at population scale, not a prediction for an individual.

Reference: Pleguezuelos-Manzano C. et al., Nature 580:269-273 (2020), doi:10.1038/s41586-020-2080-8; Rubinstein M.R. et al., Cell Host Microbe 14:195-206 (2013); Wu S. et al., Nat Med 15:1016-1022 (2009)

‹ the functions overview

# Substrates, routes and transformations, one at a time

What your microbes carry the genes to do, grouped by the kind of chemistry. Each row says how specific the evidence is, because a gene family that acts on a whole class of molecules is not evidence about one member of it — and each says the one conclusion it does not support.

**Evidence level: research.** These are research-grade readings. The biology behind them is published and the genes are real, but none of them has the weight of a clinical measurement, and none should be acted on as though it had.

### GABA: two counts, deliberately not combined

| READING | VALUE | UNIT |
|---|---|---|
| Genes for making GABA | **10.5** | copies per 100 bacterial genomes |
| Genes for breaking GABA down | — no supported reading | copies per 100 bacterial genomes |

Two separate measurements, shown side by side. Their difference is not a net flux and neither is their ratio: both describe how many genes are present, not how much GABA is made or broken down.

### Three protein readings that are not the same reading

| READING | WHAT IT MEANS |
|---|---|
| **Protein breakdown** | cutting proteins and peptides up (proteolysis) |
| **BCAA synthesis** | making branched-chain amino acids (the existing bcaa panel) |
| **BCAA fermentation** | fermenting them into branched-chain fatty acids (bcfa) |

These are three different things and this report keeps three separate readings for them. A community can be high in all three.

### Where your own body finishes the job

| MICROBES MAKE | YOUR ENZYME | WHICH BECOMES |
|---|---|---|
| **phenylacetate** | hepatic glutamine N-acyltransferase | phenylacetylglutamine (PAGln) |
| **trimethylamine (TMA)** | hepatic FMO3 | trimethylamine N-oxide (TMAO) |

Microbes make phenylacetate; your liver conjugates it to PAGln. The circulating metabolite associated with cardiovascular outcomes is the host product, and this measures only the microbial precursor.

### What the difficult panels are told apart from

Some reactions are carried by enzymes that look very like commoner ones. For those, the panel is built around the look-alike it has to reject: if a read matches the decoy, the panel has failed rather than found something. These are the decoys.

| READING | THE LOOK-ALIKE | WHY IT IS HARD |
|---|---|---|
| **urolithin 9 dehydroxylation** | *urdA* | UrdA acts on urocanate and makes imidazole propionate. It is close enough to be mistaken for a urolithin dehydroxylase, so a panel that matches it is failing rather than finding. Residue discrimination must use aligned reference numbering over covered diagnostic positions. |
| **equol daidzein conversion** | *generic_reductase* | An unassigned oxidoreductase domain matches many things. The aliases eqlA, eqlB and eqlC need source- and strain-specific mapping before they can be read as these reactions. |

## Each reading in full

One card per reading, all in the same four parts: what is driving it, what it means, how it might be moved, and the research behind it. Each card names the genes the reading is measured from and where each of those sits in the reference cohort, so the number above can be taken apart rather than taken on trust.

‹ at a glance

**Cellulose** · **headline for this chemistry is the carbohydrates panel**



**65th** of the reference cohort

### What is driving this

| ORGANISM CARRYING IT | SHARE OF THIS READING | FRAGMENTS |
|---|---|---|
| *Ruminococcus callidus ATCC 27760* | **20%** | 323 |
| *Ruminococcus champanellensis (strain DSM 18848 / JCM 17042 / KCTC 15320 / 18P13)* | **15%** | 239 |
| *Coprococcus aceti* | **11%** | 178 |
| *Lachnospira eligens (strain ATCC 27750 / DSM 3376 / VPI C15-48 / C15-B4) (Eubacterium eligens)* | **11%** | 174 |
| *Sellimonas catena* | **8%** | 125 |
| *[Eubacterium] siraeum DSM 15702* | **7%** | 117 |

and 7 more organisms with smaller shares.

| THE GENE BEHIND IT | HOW MUCH IS IN YOU | WHERE THAT SITS |
|---|---|---|
| **Endo-beta-1,4-glucanase** celA | 86.0 copies per 100 genomes | 41st of 100 people |
| **Endoglucanase** celB | 48.9 copies per 100 genomes | 84th of 100 people |

### What your reading means

Your capacity for cellulose sits in the middle of the reference range, at the 65th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

GH5, GH9

**What it does not mean.** A beta-glucanase is not a cellulase. Cellulose is crystalline beta-1,4-glucan and needs processive enzymes and a binding module; an enzyme that clears the soluble mixed-linkage glucan of oats does not touch it.

‹ at a glance

## General starch · headline for this chemistry is the carbohydrates panel



**23rd** of the reference cohort

### What is driving this

| ORGANISM CARRYING IT | SHARE OF THIS READING | FRAGMENTS |
|---|---|---|
| *Amedibacillus dolichus* | **23%** | 2,097 |
| *Blautia wexlerae* | **13%** | 1,207 |
| *Bacteroides thetaiotaomicron (strain ATCC 29148 / DSM 2079 / JCM 5827 / CCUG 10774 / NCTC 10582 / VPI-5482 / E50)* | **6%** | 533 |
| *Klebsiella pneumoniae* | **6%** | 527 |
| *Ruminococcus bromii* | **5%** | 477 |
| *Lachnospira eligens (strain ATCC 27750 / DSM 3376 / VPI C15-48 / C15-B4) (Eubacterium eligens)* | **4%** | 391 |

and 29 more organisms with smaller shares.

| THE GENE BEHIND IT | HOW MUCH IS IN YOU | WHERE THAT SITS |
|---|---|---|
| **Alpha-glucosidase** aglA | 141.7 copies per 100 genomes | 35th of 100 people |
| **Alpha-amylase** amyA | 120.6 copies per 100 genomes | 29th of 100 people |
| **Glucoamylase** gluA | 39.5 copies per 100 genomes | 25th of 100 people |
| **Oligo-1,6-glucosidase** malL | 91.1 copies per 100 genomes | 46th of 100 people |
| **Debranching** pulA | 101.9 copies per 100 genomes | 43rd of 100 people |

### What your reading means

Your capacity for general starch sits lower than most people's, at the 23rd of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

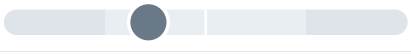
### The research behind it

GH13, GH15, GH31

**What it does not mean.** Amylolytic capacity is the broad comparator, not evidence about any particular starch. It says the community can digest starch it can reach.

‹ at a glance

## Resistant starch · headline for this chemistry is the carbohydrates panel



**36th** of the reference cohort

### What is driving this

| ORGANISM CARRYING IT | SHARE OF THIS READING | FRAGMENTS |
|---|---|---|
| *Blautia wexlerae* | **21%** | 945 |
| *Klebsiella pneumoniae* | **12%** | 527 |
| *Ruminococcus bromii* | **10%** | 477 |
| *Lachnospira eligens (strain ATCC 27750 / DSM 3376 / VPI C15-48 / C15-B4) (Eubacterium eligens)* | **9%** | 391 |
| *Blautia obeum A2-162* | **8%** | 343 |
| *Agathobacter ruminis* | **5%** | 211 |

and 15 more organisms with smaller shares.

| THE GENE BEHIND IT | HOW MUCH IS IN YOU | WHERE THAT SITS |
|---|---|---|
| **Alpha-amylase** amyA | 120.6 copies per 100 genomes | 29th of 100 people |
| **Oligo-1,6-glucosidase** malL | 91.1 copies per 100 genomes | 46th of 100 people |
| **Debranching** pulA | 101.9 copies per 100 genomes | 43rd of 100 people |

### What your reading means

Your capacity for resistant starch sits in the middle of the reference range, at the 36th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

GH13

**What it does not mean.** GH13 alone supports general amylolysis, not resistant-starch utilisation. Reaching granular or retrograded starch needs the binding modules and the surface machinery that a general amylase does not carry, and the physical type of the starch (RS1 to RS5) is a property of the food rather than of you.

‹ at a glance

## Chitin · headline for this chemistry is the carbohydrates panel



**76th** of the reference cohort

### What is driving this

| ORGANISM CARRYING IT | SHARE OF THIS READING | FRAGMENTS |
|---|---|---|
| *Streptomyces melanosporofaciens* | **26%** | 824 |
| *Nonomuraea thailandensis* | **23%** | 742 |
| *Streptomyces rugosispiralis* | **16%** | 502 |
| *Streptomyces yatensis* | **14%** | 458 |
| *Streptosporangium saharense* | **12%** | 398 |
| *Laedolimicola ammoniilytica* | **6%** | 183 |

and 2 more organisms with smaller shares.

| THE GENE BEHIND IT | HOW MUCH IS IN YOU | WHERE THAT SITS |
|---|---|---|
| **Chitinase** chiA | 133.4 copies per 100 genomes | 76th of 100 people |

### What your reading means

Your capacity for chitin sits higher than most people's, at the 76th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

GH18

**What it does not mean.** GH18 alone does not demonstrate an intact chitin-utilisation pathway. Many GH18 proteins act on peptidoglycan or on host glycans, and releasing chito-oligosaccharides without the downstream hexosaminidase leaves them unused.

‹ at a glance

## Pectin · headline for this chemistry is the carbohydrates panel



**40th** of the reference cohort

### What is driving this

| ORGANISM CARRYING IT | SHARE OF THIS READING | FRAGMENTS |
|---|---|---|
| *Bacteroides clarus YIT 12056* | **14%** | 326 |
| *Bacteroides reticulotermitis JCM 10512* | **11%** | 262 |
| *Lachnospira eligens (strain ATCC 27750 / DSM 3376 / VPI C15-48 / C15-B4) (Eubacterium eligens)* | **9%** | 227 |
| *Cupriavidus metallidurans (strain ATCC 43123 / DSM 2839 / NBRC 102507 / CH34) (Ralstonia metallidurans)* | **7%** | 165 |
| *Butyrivibrio proteoclasticus (strain ATCC 51982 / DSM 14932 / B316) (Clostridium proteoclasticum)* | **6%** | 146 |
| *Phocaeicola vulgatus (strain ATCC 8482 / DSM 1447 / JCM 5826 / CCUG 4940 / NBRC 14291 / NCTC 11154) (Bacteroides vulgatus)* | **6%** | 145 |

and 17 more organisms with smaller shares.

| THE GENE BEHIND IT | HOW MUCH IS IN YOU | WHERE THAT SITS |
|---|---|---|
| **Pectate lyase** pelA | 76.8 copies per 100 genomes | 47th of 100 people |
| **Polygalacturonase** pgl | 103.3 copies per 100 genomes | 39th of 100 people |
| **Rhamnogalacturonase** rgl | 11.2 copies per 100 genomes | 67th of 100 people |

### What your reading means

Your capacity for pectin sits in the middle of the reference range, at the 40th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

GH28, PL1

**What it does not mean.** A polygalacturonase acts on the homogalacturonan backbone. The branched rhamnogalacturonan regions of real pectin need their own enzymes, so backbone capacity is not whole-pectin capacity.

‹ at a glance

## Inulin · headline for this chemistry is the carbohydrates panel



**55th** of the reference cohort

### What is driving this

| ORGANISM CARRYING IT | SHARE OF THIS READING | FRAGMENTS |
|---|---|---|
| *Bifidobacterium amazonense* | **18%** | 222 |
| *Collinsella aerofaciens (strain ATCC 25986 / DSM 3979 / JCM 10188 / KCTC 3647 / NCTC 11838 / VPI 1003)* | **12%** | 141 |
| *Blautia faecis* | **10%** | 123 |
| *Clostridium carnis* | **9%** | 114 |
| *Phocaeicola vulgatus (strain ATCC 8482 / DSM 1447 / JCM 5826 / CCUG 4940 / NBRC 14291 / NCTC 11154) (Bacteroides vulgatus)* | **7%** | 86 |
| *Blautia segnis* | **7%** | 85 |

and 10 more organisms with smaller shares.

| THE GENE BEHIND IT | HOW MUCH IS IN YOU | WHERE THAT SITS |
|---|---|---|
| **Levanase** levB | 15.2 copies per 100 genomes | 21st of 100 people |
| **Fructan** sacA | 93.2 copies per 100 genomes | 70th of 100 people |

### What your reading means

Your capacity for inulin sits in the middle of the reference range, at the 55th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

GH32

**What it does not mean.** GH32 alone does not separate inulin from fructooligosaccharides. Long-chain inulin has to be cut outside the cell before anything can take it up, so a GH32 that handles short fructans is not evidence of inulin utilisation.

‹ at a glance

## Fructooligosaccharides · headline for this chemistry is the carbohydrates panel



**55th** of the reference cohort

### What is driving this

| ORGANISM CARRYING IT | SHARE OF THIS READING | FRAGMENTS |
|---|---|---|
| *Bifidobacterium amazonense* | **18%** | 222 |
| *Collinsella aerofaciens (strain ATCC 25986 / DSM 3979 / JCM 10188 / KCTC 3647 / NCTC 11838 / VPI 1003)* | **12%** | 141 |
| *Blautia faecis* | **10%** | 123 |
| *Clostridium carnis* | **9%** | 114 |
| *Phocaeicola vulgatus (strain ATCC 8482 / DSM 1447 / JCM 5826 / CCUG 4940 / NBRC 14291 / NCTC 11154) (Bacteroides vulgatus)* | **7%** | 86 |
| *Blautia segnis* | **7%** | 85 |

and 10 more organisms with smaller shares.

| THE GENE BEHIND IT | HOW MUCH IS IN YOU | WHERE THAT SITS |
|---|---|---|
| **Levanase** levB | 15.2 copies per 100 genomes | 21st of 100 people |
| **Fructan** sacA | 93.2 copies per 100 genomes | 70th of 100 people |

### What your reading means

Your capacity for fructooligosaccharides sits in the middle of the reference range, at the 55th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

GH32

**What it does not mean.** Sharing GH32 with inulin is expected and is not double evidence. Short-chain fructans are taken up whole and cut inside, which is a different route from the one inulin needs.

‹ at a glance

## Galactooligosaccharides · headline for this chemistry is the carbohydrates panel



**19th** of the reference cohort

### What is driving this

| ORGANISM CARRYING IT | SHARE OF THIS READING | FRAGMENTS |
|---|---|---|
| *Brachybacterium nesterenkovii* | **18%** | 515 |
| *Bacteroides reticulotermitis JCM 10512* | **12%** | 337 |
| *Blautia wexlerae* | **10%** | 289 |
| *Alicyclobacillus acidocaldarius subsp. acidocaldarius (strain ATCC 27009 / DSM 446 / BCRC 14685 / JCM 5260 / KCTC 1825 / NBRC 15652 / NCIMB 11725 / NRRL B-14509 / 104-IA) (Bacillus acidocaldarius)* | **8%** | 231 |
| *Blautia aquisgranensis* | **8%** | 221 |
| *Ktedonospora formicarum* | **8%** | 219 |

and 10 more organisms with smaller shares.

| THE GENE BEHIND IT | HOW MUCH IS IN YOU | WHERE THAT SITS |
|---|---|---|
| **Beta-galactosidase** lacZ | 177.6 copies per 100 genomes | 19th of 100 people |
| **Beta-mannosidase** manA | 27.3 copies per 100 genomes | 21st of 100 people |

### What your reading means

Your capacity for galactooligosaccharides sits lower than most people's, at the 19th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

GH2

**What it does not mean.** GH2 and GH42 alone do not establish specific GOS utilisation. Every lactose-using organism carries a beta-galactosidase; taking up an oligosaccharide of three to eight sugars needs a transporter that lactose does not.

‹ at a glance

## Xylooligosaccharides · headline for this chemistry is the carbohydrates panel



**34th** of the reference cohort

### What is driving this

| ORGANISM CARRYING IT | SHARE OF THIS READING | FRAGMENTS |
|---|---|---|
| *Phocaeicola salanitronis (strain DSM 18170 / JCM 13657 / CCUG 60908 / BL78) (Bacteroides salanitronis)* | **34%** | 347 |
| *Proteiniphilum saccharofermentans* | **16%** | 162 |
| *Butyrivibrio fibrisolvens* | **11%** | 111 |
| *Hydrogenispora ethanolica* | **10%** | 104 |
| *Xylanibacter ruminicola (Prevotella ruminicola)* | **8%** | 84 |
| *Zunongwangia atlantica 22II14-10F7* | **7%** | 71 |

and 2 more organisms with smaller shares.

| THE GENE BEHIND IT | HOW MUCH IS IN YOU | WHERE THAT SITS |
|---|---|---|
| **Xylan 1,4-beta-xylosidase** xylB | 85.3 copies per 100 genomes | 34th of 100 people |

### What your reading means

Your capacity for xylooligosaccharides sits in the middle of the reference range, at the 34th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

GH43

**What it does not mean.** Using xylooligosaccharides does not mean using arabinoxylan. The polymer has to be cut by an endo-xylanase first, and an organism that lives on the released oligosaccharides may carry no xylanase at all.

‹ at a glance

## Isomaltooligosaccharides · headline for this chemistry is the carbohydrates panel



**32nd** of the reference cohort

### What is driving this

| ORGANISM CARRYING IT | SHARE OF THIS READING | FRAGMENTS |
|---|---|---|
| *Amedibacillus dolichus* | **26%** | 2,097 |
| *Blautia wexlerae* | **15%** | 1,207 |
| *Klebsiella pneumoniae* | **6%** | 527 |
| *Ruminococcus bromii* | **6%** | 477 |
| *Lachnospira eligens (strain ATCC 27750 / DSM 3376 / VPI C15-48 / C15-B4) (Eubacterium eligens)* | **5%** | 391 |
| *Blautia obeum A2-162* | **4%** | 343 |

and 22 more organisms with smaller shares.

| THE GENE BEHIND IT | HOW MUCH IS IN YOU | WHERE THAT SITS |
|---|---|---|
| **Alpha-glucosidase** aglA | 141.7 copies per 100 genomes | 35th of 100 people |
| **Alpha-amylase** amyA | 120.6 copies per 100 genomes | 29th of 100 people |
| **Oligo-1,6-glucosidase** malL | 91.1 copies per 100 genomes | 46th of 100 people |
| **Debranching** pulA | 101.9 copies per 100 genomes | 43rd of 100 people |

### What your reading means

Your capacity for isomaltooligosaccharides sits in the middle of the reference range, at the 32nd of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

GH13, GH31

**What it does not mean.** An alpha-glucosidase is not specific to the alpha-1,6 linkages that define IMO. Commercial IMO preparations also differ widely in how much of them actually resists digestion, which is a property of the product.

‹ at a glance

## Lactose · headline for this chemistry is the carbohydrates panel



**19th** of the reference cohort

### What is driving this

| ORGANISM CARRYING IT | SHARE OF THIS READING | FRAGMENTS |
|---|---|---|
| *Brachybacterium nesterenkovii* | **18%** | 515 |
| *Bacteroides reticulotermitis JCM 10512* | **12%** | 337 |
| *Blautia wexlerae* | **10%** | 289 |
| *Alicyclobacillus acidocaldarius subsp. acidocaldarius (strain ATCC 27009 / DSM 446 / BCRC 14685 / JCM 5260 / KCTC 1825 / NBRC 15652 / NCIMB 11725 / NRRL B-14509 / 104-IA) (Bacillus acidocaldarius)* | **8%** | 231 |
| *Blautia aquisgranensis* | **8%** | 221 |
| *Ktedonospora formicarum* | **8%** | 219 |

and 10 more organisms with smaller shares.

| THE GENE BEHIND IT | HOW MUCH IS IN YOU | WHERE THAT SITS |
|---|---|---|
| **Beta-galactosidase** lacZ | 177.6 copies per 100 genomes | 19th of 100 people |
| **Beta-mannosidase** manA | 27.3 copies per 100 genomes | 21st of 100 people |

### What your reading means

Your capacity for lactose sits lower than most people's, at the 19th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

GH2

**What it does not mean.** This is the community's capacity to use lactose that reaches the colon. It is not a lactase-persistence genotype, not a breath test, and not an explanation of dairy symptoms, which depend on your own small-intestinal lactase.

‹ at a glance

## Beta-glucans · headline for this chemistry is the carbohydrates panel



**58th** of the reference cohort

### What is driving this

| ORGANISM CARRYING IT | SHARE OF THIS READING | FRAGMENTS |
|---|---|---|
| *Escherichia coli (strain K12)* | **29%** | 591 |
| *Bacillus subtilis (strain 168)* | **28%** | 561 |
| *Clostridium longisporum* | **15%** | 293 |
| *Kordia algicida OT-1* | **7%** | 151 |
| *Ewingella americana (strain ATCC 33852 / DSM 4580 / CCUG 14506 / JCM 5911 / LMG 7869 / NCTC 12157 / CDC 1468-78)* | **7%** | 133 |
| *Robinsoniella peoriensis* | **4%** | 89 |

and 10 more organisms with smaller shares.

| THE GENE BEHIND IT | HOW MUCH IS IN YOU | WHERE THAT SITS |
|---|---|---|
| **Beta-glucosidase** bglA | 129.1 copies per 100 genomes | 63rd of 100 people |
| **Beta-1,3-1,4-glucanase** licB | 6.6 copies per 100 genomes | 29th of 100 people |

### What your reading means

Your capacity for beta-glucans sits in the middle of the reference range, at the 58th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

## The research behind it

GH16, GH3

**What it does not mean.** Cereal and fungal beta-glucans are different molecules with different linkages, and the enzymes are not interchangeable. An oat beta-glucan finding says nothing about a mushroom one.

‹ at a glance

## Arabinoxylan · headline for this chemistry is the carbohydrates panel



**23rd** of the reference cohort

## What is driving this

| ORGANISM CARRYING IT | SHARE OF THIS READING | FRAGMENTS |
|---|---|---|
| *Phocaeicola salanitronis (strain DSM 18170 / JCM 13657 / CCUG 60908 / BL78) (Bacteroides salanitronis)* | **14%** | 347 |
| *Faecalibacterium prausnitzii* | **10%** | 248 |
| *Bacteroides graminisolvens DSM 19988 = JCM 15093* | **7%** | 178 |
| *Proteiniphilum saccharofermentans* | **7%** | 162 |
| *Butyrivibrio fibrisolvens* | **6%** | 159 |
| *Bacteroides reticulotermitis JCM 10512* | **5%** | 134 |

and 17 more organisms with smaller shares.

| THE GENE BEHIND IT | HOW MUCH IS IN YOU | WHERE THAT SITS |
|---|---|---|
| **Alpha-N-arabinofuranosidase** abfA | 51.0 copies per 100 genomes | 45th of 100 people |
| **Xylan 1,4-beta-xylosidase** xylB | 85.3 copies per 100 genomes | 34th of 100 people |
| **Endo-1,4-beta-xylanase** xynA | 147.8 copies per 100 genomes | 19th of 100 people |

## What your reading means

Your capacity for arabinoxylan sits lower than most people's, at the 23rd of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

## How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

## The research behind it

GH10, GH43, GH51

**What it does not mean.** A xylanase opens the backbone; the arabinose side chains and their ferulate cross-links are what make cereal arabinoxylan hard to use, and they need their own enzymes.

‹ at a glance

## Galactomannan and guar · headline for this chemistry is the carbohydrates panel



**40th** of the reference cohort

## What is driving this

| ORGANISM CARRYING IT | SHARE OF THIS READING | FRAGMENTS |
|---|---|---|
| *Bacteroides uniformis* | **18%** | 49 |
| *Clostridium oryzae* | **18%** | 49 |
| *Inconstantimicrobium mannanitabidum* | **16%** | 44 |
| *Lachnoclostridium phytofermentans (strain ATCC 700394 / DSM 18823 / ISDg) (Clostridium phytofermentans)* | **14%** | 39 |
| *Eubacterium ruminantium* | **11%** | 29 |
| *Jilunia laotingensis* | **8%** | 23 |

and 2 more organisms with smaller shares.

| THE GENE BEHIND IT | HOW MUCH IS IN YOU | WHERE THAT SITS |
|---|---|---|
| **Mannan endo-1,4-beta-mannosidase** manB | 34.9 copies per 100 genomes | 40th of 100 people |

### What your reading means

Your capacity for galactomannan and guar sits in the middle of the reference range, at the 40th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

GH26

**What it does not mean.** The mannan backbone and its galactose side chains need different enzymes, and partially hydrolysed guar behaves differently from the intact gum, which is a property of the preparation rather than of the community.

‹ at a glance

## Acetate production · headline for this chemistry is the fermentation panel



**34th** of the reference cohort

### What is driving this

| ORGANISM CARRYING IT | SHARE OF THIS READING | FRAGMENTS |
|---|---|---|
| *Streptococcus ictaluri 707-05* | **30%** | 301 |
| *Ignavibacterium album (strain DSM 19864 / JCM 16511 / NBRC 101810 / Mat9-16)* | **17%** | 171 |
| *Odoribacter laneus YIT 12061* | **14%** | 145 |
| *Malonomonas rubra DSM 5091* | **10%** | 99 |
| *Seohaeicola zhoushanensis* | **9%** | 87 |
| *Salipiger pallidus* | **7%** | 71 |

and 2 more organisms with smaller shares.

| THE GENE BEHIND IT | HOW MUCH IS IN YOU | WHERE THAT SITS |
|---|---|---|
| **Phosphate acetyltransferase** pta | 66.3 copies per 100 genomes | 34th of 100 people |

### What your reading means

Your capacity for acetate production sits in the middle of the reference range, at the 34th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

pta

**What it does not mean.** Phosphotransacetylase is the committed step. Acetate kinase completes the route and is measured by the butyrate panel as a decoy, so it corroborates here rather than being required: duplicating it as a target would take reads from that panel rather than add a reading.

‹ at a glance

## Acetate made from hydrogen and CO2 · headline for this chemistry is the fermentation panel



**71st** of the reference cohort

### What is driving this

| ORGANISM CARRYING IT | SHARE OF THIS READING | FRAGMENTS |
|---|---|---|
| *Blautia wexlerae* | **31%** | 434 |
| *Blautia faecicola* | **14%** | 195 |
| *Blautia argi* | **13%** | 189 |
| *Blautia pseudococcoides* | **13%** | 185 |
| *Clostridium autoethanogenum* | **6%** | 78 |
| *Hominisplanchenecus murintestinalis* | **5%** | 71 |

and 10 more organisms with smaller shares.

| THE GENE BEHIND IT | HOW MUCH IS IN YOU | WHERE THAT SITS |
|---|---|---|
| **CO-methylating acetyl-CoA synthase** acsB | 21.1 copies per 100 genomes | 85th of 100 people |
| **Carbon monoxide dehydrogenase** cooS | 67.4 copies per 100 genomes | 65th of 100 people |

### What your reading means

Your capacity for acetate made from hydrogen and co2 sits in the middle of the reference range, at the 71st of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

acsB, cooS

**What it does not mean.** Reconstructed separately from Pta-AckA because it is a different chemistry with a different substrate: it consumes hydrogen rather than sugar.

‹ at a glance

## D-lactate production · headline for this chemistry is the fermentation panel



**94th** of the reference cohort

### What is driving this

| ORGANISM CARRYING IT | SHARE OF THIS READING | FRAGMENTS |
|---|---|---|
| *Nosocomiicoccus ampullae* | **26%** | 320 |
| *Anaerostipes hadrus* | **25%** | 302 |
| *Nosocomiicoccus massiliensis* | **12%** | 151 |
| *Lachnospira eligens (strain ATCC 27750 / DSM 3376 / VPI C15-48 / C15-B4) (Eubacterium eligens)* | **11%** | 137 |
| *Blautia luti* | **10%** | 122 |
| *Gemmiger formicilis* | **7%** | 81 |

and 2 more organisms with smaller shares.

| THE GENE BEHIND IT | HOW MUCH IS IN YOU | WHERE THAT SITS |
|---|---|---|
| **D-lactate dehydrogenase** ldhA | 84.9 copies per 100 genomes | 94th of 100 people |

### What your reading means

Your capacity for d-lactate production sits higher than most people's, at the 94th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

ldhA

**What it does not mean.** Stereoisomer reported where the enzyme resolves it.

‹ at a glance

## L-lactate production · headline for this chemistry is the fermentation panel



**80th** of the reference cohort

### What is driving this

| ORGANISM CARRYING IT | SHARE OF THIS READING | FRAGMENTS |
|---|---|---|
| *Clostridium tetani (strain Massachusetts / E88)* | **40%** | 231 |
| *Blautia wexlerae* | **16%** | 93 |
| *Mediterraneibacter torques* | **14%** | 81 |
| *Anaerostipes hadrus* | **9%** | 53 |
| *Lachnospira eligens (strain ATCC 27750 / DSM 3376 / VPI C15-48 / C15-B4) (Eubacterium eligens)* | **6%** | 36 |
| *[Ruminococcus] lactaris ATCC 29176* | **5%** | 30 |

and 2 more organisms with smaller shares.

| THE GENE BEHIND IT | HOW MUCH IS IN YOU | WHERE THAT SITS |
|---|---|---|
| **L-lactate dehydrogenase** ldh | 62.8 copies per 100 genomes | 80th of 100 people |

### What your reading means

Your capacity for l-lactate production sits higher than most people's, at the 80th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

ldh

**What it does not mean.** Stereoisomer reported where the enzyme resolves it.

‹ at a glance

## Turning lactate into butyrate · headline for this chemistry is the butyrate panel



**95th** of the reference cohort

### What is driving this

| ORGANISM CARRYING IT | SHARE OF THIS READING | FRAGMENTS |
|---|---|---|
| *Anaerostipes hadrus* | **21%** | 112 |
| *Rikenella microfusus* | **14%** | 74 |
| *Faecalibacterium hominis (ex Afrizal et al. 2022)* | **10%** | 53 |
| *Faecalibacterium intestinale* | **9%** | 48 |
| *Christensenella minuta* | **7%** | 40 |
| *Faecalibacterium langellae* | **6%** | 33 |

and 10 more organisms with smaller shares.

| THE GENE BEHIND IT | HOW MUCH IS IN YOU | WHERE THAT SITS |
|---|---|---|
| **Butyryl-CoA:acetate CoA-transferase** but | 32.1 copies per 100 genomes | 82nd of 100 people |
| **Butyryl-CoA:acetate CoA-transferase) — out of scope (counted as background** but | 8.6 copies per 100 genomes | no reference range |

### What your reading means

Your capacity for turning lactate into butyrate sits higher than most people's, at the 95th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

lctA, lctB, lctC, but

**What it does not mean.** A cross-feeding sink for lactate, not a subtraction from the lactate a community can make. Both are capacities and neither measures an amount.

‹ at a glance

## Turning lactate into propionate · headline for this chemistry is the propionate panel



**95th** of the reference cohort

### What is driving this

| ORGANISM CARRYING IT | SHARE OF THIS READING | FRAGMENTS |
|---|---|---|
| *Lachnospira eligens (strain ATCC 27750 / DSM 3376 / VPI C15-48 / C15-B4) (Eubacterium eligens)* | **20%** | 313 |
| *Syntrophaceticus schinkii* | **16%** | 243 |
| *Pseudoflavonifractor phocaeensis* | **14%** | 219 |
| *Anaerostipes hadrus* | **13%** | 196 |
| *Allocoprococcus comes* | **8%** | 130 |
| *Brotocaccenecus cirricatena* | **8%** | 119 |

and 4 more organisms with smaller shares.

| THE GENE BEHIND IT | HOW MUCH IS IN YOU | WHERE THAT SITS |
|---|---|---|
| **Lactoyl-CoA dehydratase** lcdA | 19.8 copies per 100 genomes | 88th of 100 people |
| **Lactoyl-CoA dehydratase subunit** lcdB | 164.3 copies per 100 genomes | 95th of 100 people |

### What your reading means

Your capacity for turning lactate into propionate sits higher than most people's, at the 95th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

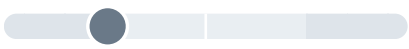
### The research behind it

lcdA, lcdB

**What it does not mean.** The acrylate route, distinct from the succinate and propanediol routes.

‹ at a glance

## Succinate production · headline for this chemistry is the propionate panel



**26th** of the reference cohort

### What is driving this

| ORGANISM CARRYING IT | SHARE OF THIS READING | FRAGMENTS |
|---|---|---|
| *Parabacteroides distasonis (strain ATCC 8503 / DSM 20701 / CIP 104284 / JCM 5825 / NCTC 11152)* | **38%** | 109 |
| *Bacteroides fragilis (strain YCH46)* | **12%** | 35 |
| *Bacteroides thetaiotaomicron (strain ATCC 29148 / DSM 2079 / JCM 5827 / CCUG 10774 / NCTC 10582 / VPI-5482 / E50)* | **12%** | 35 |

| ORGANISM CARRYING IT | SHARE OF THIS READING | FRAGMENTS |
|---|---|---|
| *Roseicitreum antarcticum* | **11%** | 30 |
| *Pelagibius litoralis* | **10%** | 28 |
| *Rhizosaccharibacter radicis* | **6%** | 16 |

and 2 more organisms with smaller shares.

| THE GENE BEHIND IT | HOW MUCH IS IN YOU | WHERE THAT SITS |
|---|---|---|
| **Malate dehydrogenase** mdh | 34.9 copies per 100 genomes | 26th of 100 people |

### What your reading means

Your capacity for succinate production sits in the middle of the reference range, at the 26th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

mdh

**What it does not mean.** A fermentative intermediate that other organisms consume.

‹ at a glance

## Turning succinate into propionate · headline for this chemistry is the propionate panel



**14th** of the reference cohort

### What is driving this

| ORGANISM CARRYING IT | SHARE OF THIS READING | FRAGMENTS |
|---|---|---|
| *Geofilum rubicundum JCM 15548* | **59%** | 629 |
| *Bacteroides stercoris* | **19%** | 202 |
| *Bacteroides graminisolvens DSM 19988 = JCM 15093* | **4%** | 40 |
| *Bacteroides reticulotermitis JCM 10512* | **3%** | 37 |
| *Porphyromonas crevioricanis* | **3%** | 37 |
| *Oceanobacillus luteolus* | **2%** | 23 |

and 10 more organisms with smaller shares.

| THE GENE BEHIND IT | HOW MUCH IS IN YOU | WHERE THAT SITS |
|---|---|---|
| **Methylmalonyl-CoA decarboxylase** mmdA | 61.4 copies per 100 genomes | 13th of 100 people |
| **Propionyl-CoA:succinate CoA transferase** scpA | 13.6 copies per 100 genomes | 20th of 100 people |

### What your reading means

Your capacity for turning succinate into propionate sits lower than most people's, at the 14th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

mmdA, scpA

**What it does not mean.** The succinate route to propionate, which is why succinate rarely accumulates in a community that carries it.

‹ at a glance

## Turning propanediol into propionate · headline for this chemistry is the propionate panel



**59th** of the reference cohort

### What is driving this

| ORGANISM CARRYING IT | SHARE OF THIS READING | FRAGMENTS |
|---|---|---|
| *Blautia caecimuris* | **58%** | 204 |
| *Muricomes intestini* | **16%** | 56 |
| *Anaerobutyricum hallii* | **6%** | 21 |
| *Faecalicatena orotica* | **5%** | 17 |
| *Cuneatibacter caecimuris* | **5%** | 16 |
| *Intestinimonas butyriciproducens* | **3%** | 12 |

and 5 more organisms with smaller shares.

| THE GENE BEHIND IT | HOW MUCH IS IN YOU | WHERE THAT SITS |
|---|---|---|
| **Propanediol dehydratase large subunit** pduC | 0.8 copies per 100 genomes | 14th of 100 people |
| **Propionaldehyde dehydrogenase** pduP | 15.6 copies per 100 genomes | 74th of 100 people |

### What your reading means

Your capacity for turning propanediol into propionate sits in the middle of the reference range, at the 59th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

pduC, pduP

**What it does not mean.** The deoxy-sugar route; a different chemistry from succinate or acrylate.

‹ at a glance

## Hydrogen production · headline for this chemistry is the fermentation panel



**80th** of the reference cohort

### What is driving this

| ORGANISM CARRYING IT | SHARE OF THIS READING | FRAGMENTS |
|---|---|---|
| *Blautia hydrogenotrophica (strain DSM 10507 / JCM 14656 / S5a33) (Ruminococcus hydrogenotrophicus)* | **29%** | 821 |
| *Blautia wexlerae* | **26%** | 718 |
| *Roseburia intestinalis L1-82* | **16%** | 463 |
| *Faecalibacterium prausnitzii* | **9%** | 242 |
| *Saccharicrinis fermentans DSM 9555 = JCM 21142* | **5%** | 150 |
| *Ruminococcus bromii* | **5%** | 147 |

and 2 more organisms with smaller shares.

| THE GENE BEHIND IT | HOW MUCH IS IN YOU | WHERE THAT SITS |
|---|---|---|
| **[FeFe] hydrogenase** hydA | 167.0 copies per 100 genomes | 80th of 100 people |

### What your reading means

Your capacity for hydrogen production sits higher than most people's, at the 80th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

## The research behind it

hydA

**What it does not mean.** Direction-resolved where the HydDB class allows it. A healthy community produces a great deal of hydrogen; there is no adverse direction here.

‹ at a glance

### Hydrogen used to make sulfide · headline for this chemistry is the h2s panel



**48th** of the reference cohort

## What is driving this

| ORGANISM CARRYING IT | SHARE OF THIS READING | FRAGMENTS |
|---|---|---|
| *Uncultured sulfate-reducing bacterium* | **57%** | 4 |
| *Desulfovibrio simplex* | **29%** | 2 |
| *Uncultured Desulfovibrio sp* | **14%** | 1 |

| THE GENE BEHIND IT | HOW MUCH IS IN YOU | WHERE THAT SITS |
|---|---|---|
| **Dissimilatory sulfite reductase, alpha subunit** dsrA | 0.6 copies per 100 genomes | 48th of 100 people |

## What your reading means

Your capacity for hydrogen used to make sulfide sits in the middle of the reference range, at the 48th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

## How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

## The research behind it

dsrA

**What it does not mean.** One of several sulfur routes, kept apart from the taurine, isethionate, sulfoacetate and cysteine routes because their endpoints differ. dsrA is measured by the hydrogen-sulfide panel, which owns it; this route reads that measurement rather than duplicating the target.

‹ at a glance

### Turning taurine into sulfide · headline for this chemistry is the h2s panel



**95th** of the reference cohort

## What is driving this

| ORGANISM CARRYING IT | SHARE OF THIS READING | FRAGMENTS |
|---|---|---|
| *Oleidesulfovibrio alaskensis (strain ATCC BAA-1058 / DSM 17464 / G20) (Desulfovibrio alaskensis)* | **17%** | 123 |
| *Tropicibacter naphthalenivorans* | **17%** | 120 |
| *Ruminococcus bromii* | **16%** | 116 |
| *Achromobacter pestifer* | **14%** | 100 |
| *Clostridium chromiireducens* | **9%** | 63 |
| *Bilophila wadsworthia (strain 3 1 6)* | **8%** | 58 |

and 6 more organisms with smaller shares.

| THE GENE BEHIND IT | HOW MUCH IS IN YOU | WHERE THAT SITS |
|---|---|---|
| **Taurine--pyruvate aminotransferase** tpa | 25.3 copies per 100 genomes | 95th of 100 people |
| **Isethionate sulfite-lyase, IslA** isl | 7.5 copies per 100 genomes | 94th of 100 people |

## What your reading means

Your capacity for turning taurine into sulfide sits higher than most people's, at the 95th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

tpa, isl

**What it does not mean.** Bilophila abundance is not this pathway, and this pathway is not every sulfur pathway.

‹ at a glance

## Protein and peptide breakdown · headline for this chemistry is the nitrogen panel



**59th** of the reference cohort

### What is driving this

| ORGANISM CARRYING IT | SHARE OF THIS READING | FRAGMENTS |
|---|---|---|
| *Phocaeicola dorei CL02T12C06* | **14%** | 47 |
| *Enterocloster clostridioformis* | **10%** | 35 |
| *Desulfosporosinus orientis (strain ATCC 19365 / DSM 765 / NCIMB 8382 / VKM B-1628 / Singapore I) (Desulfotomaculum orientis)* | **9%** | 31 |
| *Alistipes putredinis DSM 17216* | **8%** | 29 |
| *Bacteroides clarus YIT 12056* | **8%** | 27 |
| *Mammaliicoccus sciuri (Staphylococcus sciuri)* | **8%** | 27 |

and 10 more organisms with smaller shares.

| THE GENE BEHIND IT | HOW MUCH IS IN YOU | WHERE THAT SITS |
|---|---|---|
| **Tripeptide aminopeptidase** pepT | 78.0 copies per 100 genomes | 63rd of 100 people |
| **Periplasmic dipeptide transport** dppA | 6.7 copies per 100 genomes | 49th of 100 people |

### What your reading means

Your capacity for protein and peptide breakdown sits in the middle of the reference range, at the 59th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

pepT, dppA

**What it does not mean.** Every organism carries peptidases for its own protein turnover. Counting them measures how many organisms are present, not how much of your dinner reaches the colon undigested.

‹ at a glance

## Amino-acid fermentation, including Stickland pairs · headline for this chemistry is the nitrogen panel



**81st** of the reference cohort

### What is driving this

| ORGANISM CARRYING IT | SHARE OF THIS READING | FRAGMENTS |
|---|---|---|
| *Romboutsia weinsteinii* | **23%** | 42 |
| *Clostridium scindens (strain ATCC 35704 / DSM 5676 / VPI 13733 / 19)* | **19%** | 35 |
| *Faecalicatena orotica* | **9%** | 17 |
| *Allocoprococcus comes* | **6%** | 11 |

| ORGANISM CARRYING IT | SHARE OF THIS READING | FRAGMENTS |
|---|---|---|
| *Dorea acetigenes* | **6%** | 10 |
| *Hespellia stercorisuis DSM 15480* | **5%** | 9 |

and 16 more organisms with smaller shares.

| THE GENE BEHIND IT | HOW MUCH IS IN YOU | WHERE THAT SITS |
|---|---|---|
| **Glycine reductase selenoprotein A** grdA | 7.9 copies per 100 genomes | 80th of 100 people |
| **D-proline reductase** prdA | 15.4 copies per 100 genomes | 78th of 100 people |
| **(R)-phenyllactate dehydrogenase** fldH | 0.9 copies per 100 genomes | 55th of 100 people |

### What your reading means

Your capacity for amino-acid fermentation, including stickland pairs sits higher than most people's, at the 81st of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

grdA, prdA, fldH

**What it does not mean.** Fermenting an amino acid for energy is the opposite of making it. The two capacities can coexist in a community and say different things.

‹ at a glance

## Aromatic precursor routes · headline for this chemistry is the nitrogen panel



**44th** of the reference cohort

### What is driving this

| ORGANISM CARRYING IT | SHARE OF THIS READING | FRAGMENTS |
|---|---|---|
| *Blautia wexlerae* | **36%** | 430 |
| *Blautia schinkii* | **20%** | 233 |
| *Vreelandella neptunia* | **8%** | 96 |
| *Pseudomonas alloputida* | **8%** | 90 |
| *Dryocola clanedunensis* | **6%** | 66 |
| *Phocaeicola sartorii* | **5%** | 64 |

and 18 more organisms with smaller shares.

| THE GENE BEHIND IT | HOW MUCH IS IN YOU | WHERE THAT SITS |
|---|---|---|
| **4-hydroxyphenylacetate decarboxylase, large subunit** hpdB | 11.3 copies per 100 genomes | 84th of 100 people |
| **Phenylacetate-CoA ligase** padI | 50.3 copies per 100 genomes | 37th of 100 people |
| **Indolepyruvate ferredoxin oxidoreductase** porA | 0.4 copies per 100 genomes | 73rd of 100 people |

### What your reading means

Your capacity for aromatic precursor routes sits in the middle of the reference range, at the 44th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

hpdB, padI, porA

**What it does not mean.** These are microbial precursors. The host performs a further step on several of them, and the blood metabolite is the host's product.

‹ at a glance

## GABA degradation · headline for this chemistry is the gababreak panel



**24th** of the reference cohort

### What is driving this

| ORGANISM CARRYING IT | SHARE OF THIS READING | FRAGMENTS |
|---|---|---|
| *Mycobacterium ostraviense* | **21%** | 36 |
| *Pseudolactococcus insecticola* | **18%** | 31 |
| *Mycobacterium aquaticum* | **18%** | 30 |
| *Aquirufa nivalisilvae* | **8%** | 13 |
| *Brevibacterium yomogidense* | **4%** | 6 |
| *Mycobacterium yunnanensis* | **4%** | 6 |

and 10 more organisms with smaller shares.

| THE GENE BEHIND IT | HOW MUCH IS IN YOU | WHERE THAT SITS |
|---|---|---|
| **4-aminobutyrate aminotransferase, degradation** gabT | 5.3 copies per 100 genomes | 37th of 100 people |
| **Succinate-semialdehyde dehydrogenase** gabD | 8.2 copies per 100 genomes | 25th of 100 people |

### What your reading means

Your capacity for gaba degradation sits lower than most people's, at the 24th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

gabT, gabD

**What it does not mean.** a net GABA balance. Degradation and synthesis are separate measurements and subtracting one percentile from another is not a flux

‹ at a glance

## 4-aminobutyrate to butyrate · headline for this chemistry is the gababreak panel



**70th** of the reference cohort

### What is driving this

| ORGANISM CARRYING IT | SHARE OF THIS READING | FRAGMENTS |
|---|---|---|
| *Eubacterium ramulus* | **18%** | 45 |
| *Pseudolactococcus insecticola* | **12%** | 31 |
| *Dorea amylophila* | **11%** | 27 |
| *Hyphomonas jannaschiana VP2* | **10%** | 24 |
| *Anaerotruncus massiliensis (ex Togo et al. 2026)* | **9%** | 22 |
| *Methylobacterium aquaticum* | **8%** | 20 |

and 10 more organisms with smaller shares.

| THE GENE BEHIND IT | HOW MUCH IS IN YOU | WHERE THAT SITS |
|---|---|---|
| **4-aminobutyrate aminotransferase, degradation** gabT | 5.3 copies per 100 genomes | 37th of 100 people |
| **4-hydroxybutyrate dehydrogenase** abfH | 18.5 copies per 100 genomes | 81st of 100 people |

## What your reading means

Your capacity for 4-aminobutyrate to butyrate sits in the middle of the reference range, at the 70th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

## How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

## The research behind it

gabT, abfH

**What it does not mean.** that GABA is the main butyrate source here; the fibre routes are usually much larger and are measured separately

‹ at a glance

## Mucin glycan foraging · headline for this chemistry is the mucinlps panel



**22nd** of the reference cohort

## What is driving this

| ORGANISM CARRYING IT | SHARE OF THIS READING | FRAGMENTS |
|---|---|---|
| *Bacteroides stercoris* | **16%** | 374 |
| *Bacteroides uniformis (strain ATCC 8492 / DSM 6597 / CCUG 4942 / CIP 103695 / JCM 5828 / KCTC 5204 / NCTC 13054 / VPI 0061)* | **12%** | 296 |
| *Blautia massiliensis (ex Durand et al. 2017)* | **9%** | 208 |
| *Phocaeicola coprophilus DSM 18228 = JCM 13818* | **8%** | 198 |
| *Bacteroides clarus YIT 12056* | **6%** | 141 |
| *Hoylesella enoeca* | **5%** | 131 |

and 14 more organisms with smaller shares.

| THE GENE BEHIND IT | HOW MUCH IS IN YOU | WHERE THAT SITS |
|---|---|---|
| **Exo-alpha-sialidase** nanH | 29.5 copies per 100 genomes | 16th of 100 people |
| **Alpha-L-fucosidase** fucA | 93.3 copies per 100 genomes | 17th of 100 people |
| **Sulfatase** sulf | 152.9 copies per 100 genomes | 26th of 100 people |

## What your reading means

Your capacity for mucin glycan foraging sits lower than most people's, at the 22nd of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

## How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

## The research behind it

nanH, fucA, sulf

**What it does not mean.** damage to your mucus barrier. Mucin foraging is physiological and these organisms are normal residents; genetic capacity is not an image of mucosal erosion

‹ at a glance

## Lipid A acylation and modification · headline for this chemistry is the mucinlps panel



**18th** of the reference cohort

## What is driving this

| ORGANISM CARRYING IT | SHARE OF THIS READING | FRAGMENTS |
|---|---|---|
| *Dysgonomonas hofstadii* | **18%** | 86 |

| ORGANISM CARRYING IT | SHARE OF THIS READING | FRAGMENTS |
|---|---|---|
| *Bacteroides oleiciplenus YIT 12058* | **13%** | 61 |
| *Bacteroides stercoris* | **9%** | 45 |
| *Candidatus Kryptonium thompsonii* | **7%** | 33 |
| *Indolivaga macrotermitis* | **7%** | 32 |
| *Parabacteroides gordonii MS-1 = DSM 23371* | **5%** | 23 |

and 18 more organisms with smaller shares.

| THE GENE BEHIND IT | HOW MUCH IS IN YOU | WHERE THAT SITS |
|---|---|---|
| **UDP-GlcNAc acyltransferase** lpxA | 13.9 copies per 100 genomes | 17th of 100 people |
| **UDP-3-O-acyl-GlcNAc deacetylase** lpxC | 14.5 copies per 100 genomes | 23rd of 100 people |
| **Lauroyl acyltransferase** lpxL | 18.6 copies per 100 genomes | 22nd of 100 people |

### What your reading means

Your capacity for lipid a acylation and modification sits lower than most people's, at the 18th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

lpxA, lpxC, lpxL

**What it does not mean.** a TLR4 stimulus or an endotoxin concentration. The number of acyl chains decides the immune effect, one modification gene does not resolve the structure, and total Gram-negative abundance resolves nothing at all

‹ at a glance

## Hydroxysteroid dehydrogenase transformations · headline for this chemistry is the mucinlps panel



**71st** of the reference cohort

### What is driving this

| ORGANISM CARRYING IT | SHARE OF THIS READING | FRAGMENTS |
|---|---|---|
| *Blautia caecimuris* | **72%** | 171 |
| *Hespellia stercorisuis DSM 15480* | **6%** | 14 |
| *Nocardioides lianchengensis* | **5%** | 12 |
| *Bacteroides caecimuris* | **5%** | 11 |
| *Sporanaerobacter acetigenes DSM 13106* | **4%** | 10 |
| *Clostridium collagenovorans DSM 3089* | **3%** | 7 |

and 2 more organisms with smaller shares.

| THE GENE BEHIND IT | HOW MUCH IS IN YOU | WHERE THAT SITS |
|---|---|---|
| **7-alpha-hydroxysteroid dehydrogenase** hdhA | 29.7 copies per 100 genomes | 71st of 100 people |

### What your reading means

Your capacity for hydroxysteroid dehydrogenase transformations sits in the middle of the reference range, at the 71st of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

hdhA

**What it does not mean.** one linear sequence of bile-acid chemistry. Some HSDHs act on conjugated substrates, so these reactions form a graph rather than a chain

‹ at a glance

## Short-chain fatty acid signalling · headline for this chemistry is the butyrate panel



**95th** of the reference cohort

### What is driving this

| ORGANISM CARRYING IT | SHARE OF THIS READING | FRAGMENTS |
|---|---|---|
| *Streptococcus ictaluri 707-05* | **12%** | 301 |
| *Ignavibacterium album (strain DSM 19864 / JCM 16511 / NBRC 101810 / Mat9-16)* | **7%** | 171 |
| *Odoribacter laneus YIT 12061* | **6%** | 145 |
| *Blautia difficilis* | **5%** | 137 |
| *Faecalibacterium langellae* | **5%** | 127 |
| *Anaerostipes hadrus* | **4%** | 112 |

and 37 more organisms with smaller shares.

| THE GENE BEHIND IT | HOW MUCH IS IN YOU | WHERE THAT SITS |
|---|---|---|
| **Butyryl-CoA:acetate CoA-transferase** but | 32.1 copies per 100 genomes | 82nd of 100 people |
| **Butyryl-CoA:acetate CoA-transferase) — out of scope (counted as background** but | 8.6 copies per 100 genomes | no reference range |
| **Butyrate kinase** buk | 0.9 copies per 100 genomes | 75th of 100 people |
| **Butyrate kinase) — out of scope (counted as background** buk | 29.7 copies per 100 genomes | no reference range |
| **Phosphate acetyltransferase** pta | 66.3 copies per 100 genomes | 34th of 100 people |
| **Acetate kinase** ackA | 109.6 copies per 100 genomes | no reference range |

### What your reading means

Your capacity for short-chain fatty acid signalling sits higher than most people's, at the 95th of 99 reference samples. This counts the genes for the step, which is a capacity and not a measurement of how much is made.

### How can I improve this?

Neither direction is established as better for this reading, so there is no target to move it towards.

No study on record shows that a diet, a supplement or a probiotic moves this particular reading. What is established for the organisms behind it is in the research below. Treating an unproven lever as a proven one is the failure this report is built to avoid, so the gap is printed rather than filled.

### The research behind it

but, buk, pta

**What it does not mean.** a GLP-1 level. The receptors that read these signals are yours, not your microbes', and the step from a fatty acid in the lumen to a hormone in your blood is host physiology this test does not see

### The fermentation steps your community can run

Each line is one step, from what goes in to what comes out. A **supported** step is one this community carries the genes for. A step marked *literature only* is consistent with what is known but is not established by these genes, and is shown so the gap is visible rather than filled in. None of these is a flow: they are things the community could do, with no rate attached, and a step that uses a compound is not evidence the compound is there.

| FROM | TO | ROUTE | EVIDENCE |
|---|---|---|---|
| substrate | **acetate** | acetate.pta_ack | genes present |
| hydrogen | **acetate** | acetate.wood_ljungdahl | genes present |
| carbon dioxide | **acetate** | acetate.wood_ljungdahl | genes present |

| FROM | TO | ROUTE | EVIDENCE |
|---|---|---|---|
| substrate | **D-lactate** | lactate.d_formation | genes present |
| substrate | **L-lactate** | lactate.l_formation | genes present |
| D-lactate | **butyrate** | lactate.utilisation_butyrate | genes present |
| L-lactate | **butyrate** | lactate.utilisation_butyrate | genes present |
| D-lactate | **propionate** | lactate.utilisation_propionate | genes present |
| L-lactate | **propionate** | lactate.utilisation_propionate | genes present |
| substrate | **succinate** | succinate.formation | genes present |
| succinate | **propionate** | succinate.to_propionate | genes present |
| fucose | **propionate** | propionate.propanediol | genes present |
| rhamnose | **propionate** | propionate.propanediol | genes present |
| substrate | **hydrogen** | hydrogen.production_fefe | genes present |
| hydrogen | **hydrogen sulfide** | hydrogen.consumption_sulfate | genes present |
| sulfate | **hydrogen sulfide** | hydrogen.consumption_sulfate | genes present |
| taurine | **hydrogen sulfide** | sulfur.taurine | genes present |
| hydrogen | **methane** | hydrogen.consumption_methanogenesis | literature only |

Supported biochemical opportunities, not flows. No edge carries a rate. An edge does not assume its substrate is present: a lactate sink is capacity to use lactate, not evidence that there is lactate to use. Production and consumption of the same metabolite are separate measurements; neither is subtracted from the other. Filling a node does not mean an added organism would establish there.

### Why these fibre readings cannot be added together

**9** of the **24** genes behind these fibre readings serve more than one fibre. The same stretch of DNA that helps digest starch also helps digest resistant starch, so each reading counts it and a total over the readings counts it several times.

| GENE | COUNTS TOWARD |
|---|---|
| ***aglA*** | imo, starch general |
| ***amyA*** | imo, resistant starch, starch general |
| ***lacZ*** | gos, lactose |
| ***levB*** | fos, inulin |
| ***malL*** | imo, resistant starch, starch general |
| ***manA*** | galactomannan, gos, lactose |
| ***pulA*** | imo, resistant starch, starch general |
| ***sacA*** | fos, inulin |
| ***xylB*** | arabinoxylan, xos |

Adding the cards together would count every shared gene once per card. Any total over these substrates must be taken with `aggregate_fragments`, which counts each gene once.

## Medicines your gut bacteria are known to alter

Each of these transformations has been demonstrated with purified enzymes or defined strains. They are here as mechanism, not as a result about you: none of them is a measurement of what is happening to a medicine you take, and nothing in this section is a reason to change a dose or stop a drug. That is a conversation with the person who prescribed it.

| MEDICINE | WHAT THE BACTERIA DO | IN THIS REPORT | WHAT IT DOES NOT MEAN |
|---|---|---|---|
| **Digoxin**<br>Eggerthella lenta, and only some strains of it | Cgr2 reduces the lactone ring of digoxin to dihydrodigoxin, which is inactive. Strains of E. lenta differ in whether they carry a functional cgr operon, and the reaction is inhibited by arginine rather than increased by it. | not measured | Nothing here estimates a digoxin level or supports a dose change. Carrying E. lenta is not the same as carrying a reducing strain, and this report does not measure the cgr operon: too few characterised sequences exist for a read-based search to be sensitive. |
| **Levodopa (L-DOPA)**<br>Enterococcus faecalis, then Eggerthella lenta | E. faecalis TyrDC decarboxylates levodopa to dopamine in the small intestine, before the drug reaches the brain. E. lenta Dadh then dehydroxylates that dopamine to m-tyramine. The first step is not blocked by carbidopa, the inhibitor given alongside levodopa for the human enzyme. | tyrDC detected | Carrying these genes does not mean your levodopa is being consumed, and no dose change follows from this. Both genes are measured in this report because they have other roles; their presence is common. |

| MEDICINE | WHAT THE BACTERIA DO | IN THIS REPORT | WHAT IT DOES NOT MEAN |
|---|---|---|---|
| **Irinotecan (via its metabolite SN-38)** many gut bacteria, with large differences between enzymes | The liver inactivates SN-38 by attaching glucuronic acid. Bacterial beta-glucuronidases remove it again in the gut, regenerating the active drug in the intestinal lining, which is a proposed mechanism of the diarrhoea that limits this chemotherapy. | gus detected | Your total beta-glucuronidase reading does not estimate this. The enzymes that perform the reaction efficiently are one structural class among several, and this report cannot resolve which class yours belong to. |
| **Sulfasalazine** a broad range of gut bacteria | Sulfasalazine is a prodrug: it passes the small intestine intact and gut bacterial azoreductases cleave its azo bond to release the active 5-aminosalicylate in the colon, which is how the drug is designed to work. | not measured | This is the intended mechanism of the drug rather than a problem, and no reading here estimates how much is released. Azoreductase is not measured in this report: the protein name covers an enzyme family too broad for a count over it to mean this reaction. |

## Neurotransmitters, and what can actually be measured

This is the part of the field where the distance between what a stool sample can show and what people want to conclude is widest, so each of these says where its evidence stops. Three of the five have no specific test behind them at all, and say so rather than borrowing a number from a related molecule.

| MOLECULE | WHAT THE EVIDENCE SHOWS | IN THIS REPORT | WHERE IT STOPS |
|---|---|---|---|
| **Dopamine-related microbial metabolism** | Enterococcus faecalis tyrosine decarboxylase decarboxylates levodopa to dopamine in the small intestine, and Eggerthella lenta dopamine dehydroxylase then removes a hydroxyl to give m-tyramine. Both enzymes were purified and both reactions were demonstrated; both genes are measured in this report. | tyrDC detected | Not brain dopamine, and not a medication instruction. These reactions happen in the gut lumen on a drug or on dietary tyrosine; dopamine made there does not cross into the brain, and nothing here indicates whether your levodopa, if you take any, is being consumed. |
| **Serotonin-related microbial metabolism** | A 2025 result describes a Limosilactobacillus mucosae and Ligilactobacillus ruminis consortium whose effect depends on 5-hydroxytryptophan being available, with a traceable candidate decarboxylase. The chemistry is consortium-level and 5-HTP-dependent, and the construct's substrate specificity is recorded as unresolved in the source itself. | no specific test exists<br>There is no validated serotonin-specific gene panel. The available decarboxylase evidence is for tryptamine, which is a different molecule, and turning a tryptamine enzyme into a serotonin detector because a homolog can be retrieved is precisely the error this card exists to avoid. The tryptamine reading is linked instead, labelled as tryptamine. | This is not a serotonin measurement and not an inference about serotonin in your blood or brain. Most of the body's serotonin is made by your own enterochromaffin cells, not by bacteria; the two pools do not exchange across the blood-brain barrier. |
| **Acetylcholine** | Some lactobacilli have been reported to produce acetylcholine in culture, and choline is the shared precursor of both acetylcholine and the trimethylamine route this report does measure. The enzymes responsible in bacteria are not characterised to the level a specific detector needs. | no specific test exists<br>No validated substrate-specific panel exists. Acetyltransferases are one of the largest and least substrate-specific enzyme families in bacteria, so a count over them would measure the family and be reported as acetylcholine, which is worse than reporting nothing. | A generic acetyltransferase count is not an acetylcholine assay. Nothing here measures acetylcholine, and nothing here bears on cognition or neuromuscular function. |
| **Norepinephrine** | The microbial contribution here is handling rather than synthesis: bacterial beta-glucuronidase can deconjugate catecholamine glucuronides in the gut lumen, freeing the parent compound. Host adrenergic signalling is host physiology and is a separate matter from anything in a stool sample. | no specific test exists<br>Bacterial norepinephrine synthesis is not established, so there is nothing specific to search for. The deconjugation capacity is measured by the existing beta-glucuronidase reading, which is linked here and is not substrate-specific to catecholamines. | No bacterial abundance converts to a norepinephrine level. Microbial deconjugation and your own adrenergic response are different things, and this card keeps them apart rather than joining them. |
| **Histamine detail** | Bacterial histidine decarboxylase converts histidine to histamine, and it is measured in this report whatever organism carries it. A specific Klebsiella aerogenes strain producing histamine was linked to visceral hypersensitivity in a mouse model and to a patient subgroup, which is a strain-level finding rather than a species-level one. | hdcA detected | Potential production is not a diagnosis of histamine intolerance. That condition is defined clinically and is not established from stool DNA, and a high reading here is not a reason to start a low-histamine diet. |

**Why there is no breakdown of your beta-glucuronidase by substrate.** The structural class of a beta-glucuronidase is defined by the length of a loop at one of two active-site positions, measured in a curated alignment guided by solved structures. Reproducing that from a pairwise alignment to the E. coli enzyme works for enzymes close to it and fails for the divergent gut ones: in testing, the Bacteroides fragilis enzyme came out as Loop 1 where the published structure is mini-Loop 1, and the Bacteroides uniformis enzyme came out as No Loop where it is the published example of Loop 2. Those are the gut organisms the breakdown would exist to describe, so no breakdown is shown. Your total beta-glucuronidase reading is unaffected and is reported as usual.

The test that withheld it: Bacteroides fragilis should be mL1, came out as L1; Bacteroides uniformis should be L2, came out as NL; Parabacteroides merdae should be mL2, came out as NL. The curated loop alignment published as Data S3 of the source, or a profile model built from the solved structures, in place of pairwise alignment to a single anchor. The per-reference counts this needs are already recorded, so the breakdown appears as soon as the assignment agrees with the published examples.

## Synbiotic scenarios in detail: every arm, every output, every organism

The full accounting behind the summary in the actions section. For each fibre: what the community could release of four fermentation products in each of the four arms, then which organisms would take up the fibre and which would make the butyrate. Each is a range over every state in which the community makes at least 99% of its butyrate maximum: the least an organism contributes in any such state, and the most.

### Inulin + consortium — pair gains · model cap 6.14 mmol/gDW/h on EX_inulin_m

| OUTPUT (mmol/gDW/h) | AS YOU ARE | + CONSORTIUM | + FIBRE | + BOTH |
|---|---|---|---|---|
| **butyrate** | 219.4 | 219.0 | 418.6 | 440.1 |
| **propionate** | 218.5 | 314.2 | 219.6 | 326.7 |
| **acetate** | 504.6 | 505.8 | 887.8 | 918.1 |
| **lactate** | 336.4 | 337.2 | 404.3 | 530.7 |

Each output is its own exchange bounded separately; they are never summed into one short-chain-fatty-acid number. Community growth 0.0539 per hour in every arm (diet-capped).

**Organisms that would take up the fibre (+ both)**

| ORGANISM | MUST | CAN | STATUS IN THIS REPORT |
|---|---|---|---|
| *F. prausnitzii E* | none | 100% | beneficial |
| *B. longum* | none | 89% | added by the consortium |
| *A. hallii* | none | 37% | added by the consortium; beneficial |
| *D. longicatena* | none | 22% | conditional |
| *B. uniformis* | none | 8% | conditional |
| *B. cellulosilyticus* | none | 6% | beneficial |
| *F. saccharivorans* | none | 6% | beneficial |
| *A. soehngenii* | none | 4% | beneficial |
| *R. faecis* | none | 4% | beneficial |
| *R. intestinalis* | none | 4% | beneficial |
| *A. rectalis* | none | 3% | beneficial |
| *C. aerofaciens H* | none | 3% | conditional |

**Organisms that would make the butyrate (+ both)**

| ORGANISM | MUST | CAN | STATUS IN THIS REPORT |
|---|---|---|---|
| *F. prausnitzii E* **required** | 9% | 19% | beneficial |
| *C. butyricum* **required** | 5% | 15% | added by the consortium |
| *C. beijerinckii* **required** | 5% | 15% | added by the consortium |
| *A. hallii* **required** | 5% | 15% | added by the consortium; beneficial |
| *A. hadrus* **required** | 4% | 14% | beneficial |
| *C. eutactus* | none | 6% | beneficial |
| *C. sp. AF12_28* | none | 5% | conditional |
| *R. bacterium AM53 2BH* | none | 4% | conditional |
| *G. formicilis* | none | 3% | beneficial |
| *R. faecis* | none | 2% | beneficial |
| *C. catus* | none | 2% | beneficial |
| *A. soehngenii* | none | 2% | beneficial |
| *A. rectalis* | none | 2% | beneficial |
| *E. ramulus* | none | 2% | conditional |

+3 more with smaller possible shares

**Organisms that make the butyrate as you are**

| ORGANISM | MUST | CAN | STATUS IN THIS REPORT |
|---|---|---|---|
| *F. prausnitzii E* | none | 56% | beneficial |
| *A. hadrus* | none | 41% | beneficial |

| ORGANISM | MUST | CAN | STATUS IN THIS REPORT |
|---|---|---|---|
| *C. eutactus* | none | 17% | beneficial |
| *C. sp. AF12_28* | none | 15% | conditional |
| *R. bacterium AM53 2BH* | none | 11% | conditional |
| *G. formicilis* | none | 9% | beneficial |
| *R. faecis* | none | 5% | beneficial |
| *C. catus* | none | 5% | beneficial |
| *A. soehngenii* | none | 5% | beneficial |
| *A. rectalis* | none | 5% | beneficial |
| *E. ramulus* | none | 5% | conditional |
| *R. hominis* | none | 4% | beneficial |
| *F. magna K* | none | 3% | unknown |
| *A. hallii* | none | 2% | added by the consortium; beneficial |

+3 more with smaller possible shares

## Maltodextrin + consortium — pair gains · model cap 30.28 mmol/gDW/h on EX_dextrin_m

| OUTPUT (mmol/gDW/h) | AS YOU ARE | + CONSORTIUM | + FIBRE | + BOTH |
|---|---|---|---|---|
| **butyrate** | 219.4 | 219.0 | 417.9 | 437.0 |
| **propionate** | 218.5 | 314.2 | 222.6 | 329.4 |
| **acetate** | 504.6 | 505.8 | 897.2 | 926.8 |
| **lactate** | 336.4 | 337.2 | 401.9 | 530.7 |

Each output is its own exchange bounded separately; they are never summed into one short-chain-fatty-acid number. Community growth 0.0539 per hour in every arm (diet-capped).

### Organisms that would take up the fibre (+ both)

| ORGANISM | MUST | CAN | STATUS IN THIS REPORT |
|---|---|---|---|
| *F. prausnitzii E* **required** | 3% | 27% | beneficial |
| *B. longum* | none | 22% | added by the consortium |
| *C. butyricum* | none | 22% | added by the consortium |
| *R. bromii* | none | 15% | beneficial |
| *D. longicatena* | none | 4% | conditional |
| *F. saccharivorans* | none | 4% | beneficial |
| *A. putredinis* | none | 4% | conditional |
| *B. luti* | none | 3% | conditional |
| *R. faecis* | none | 3% | beneficial |
| *D. formicigenerans* | none | 3% | flagged high (p96) |
| *P. distasonis* | none | 3% | conditional |
| *E. ramulus* | none | 3% | conditional |
| *C. bacterium Marseille Q3526* | none | 2% | conditional |
| *R. hominis* | none | 2% | beneficial |

+4 more with smaller possible shares

### Organisms that would make the butyrate (+ both)

| ORGANISM | MUST | CAN | STATUS IN THIS REPORT |
|---|---|---|---|
| *F. prausnitzii E* **required** | 10% | 19% | beneficial |
| *C. beijerinckii* **required** | 6% | 15% | added by the consortium |
| *C. butyricum* **required** | 6% | 15% | added by the consortium |
| *A. hadrus* **required** | 5% | 14% | beneficial |
| *A. hallii* **required** | 5% | 13% | added by the consortium; beneficial |
| *C. eutactus* | none | 6% | beneficial |
| *C. sp. AF12_28* | none | 5% | conditional |
| *R. bacterium AM53 2BH* | none | 4% | conditional |
| *G. formicilis* | none | 3% | beneficial |

| ORGANISM | MUST | CAN | STATUS IN THIS REPORT |
|---|---|---|---|
| *R. faecis* | none | 2% | beneficial |
| *C. catus* | none | 2% | beneficial |
| *A. soehngenii* | none | 2% | beneficial |
| *A. rectalis* | none | 2% | beneficial |
| *E. ramulus* | none | 2% | conditional |
| +3 more with smaller possible shares | | | |

**Organisms that make the butyrate as you are**

| ORGANISM | MUST | CAN | STATUS IN THIS REPORT |
|---|---|---|---|
| *F. prausnitzii E* | none | 56% | beneficial |
| *A. hadrus* | none | 41% | beneficial |
| *C. eutactus* | none | 17% | beneficial |
| *C. sp. AF12_28* | none | 15% | conditional |
| *R. bacterium AM53 2BH* | none | 11% | conditional |
| *G. formicilis* | none | 9% | beneficial |
| *R. faecis* | none | 5% | beneficial |
| *C. catus* | none | 5% | beneficial |
| *A. soehngenii* | none | 5% | beneficial |
| *A. rectalis* | none | 5% | beneficial |
| *E. ramulus* | none | 5% | conditional |
| *R. hominis* | none | 4% | beneficial |
| *F. magna K* | none | 3% | unknown |
| *A. hallii* | none | 2% | added by the consortium; beneficial |
| +3 more with smaller possible shares | | | |

## Pectin + consortium — pair gains · model cap 0.4 mmol/gDW/h on EX_pect_m

| OUTPUT (mmol/gDW/h) | AS YOU ARE | + CONSORTIUM | + FIBRE | + BOTH |
|---|---|---|---|---|
| **butyrate** | 219.4 | 219.0 | 355.0 | 375.2 |
| **propionate** | 218.5 | 314.2 | 218.5 | 326.7 |
| **acetate** | 504.6 | 505.8 | 746.4 | 781.0 |
| **lactate** | 336.4 | 337.2 | 401.9 | 520.7 |

Each output is its own exchange bounded separately; they are never summed into one short-chain-fatty-acid number. Community growth 0.0539 per hour in every arm (diet-capped).

**Organisms that would take up the fibre (+ both)**

| ORGANISM | MUST | CAN | STATUS IN THIS REPORT |
|---|---|---|---|
| *F. prausnitzii E* **required** | 6% | 61% | beneficial |
| *C. beijerinckii* | none | 46% | added by the consortium |
| *C. butyricum* | none | 46% | added by the consortium |

**Organisms that would make the butyrate (+ both)**

| ORGANISM | MUST | CAN | STATUS IN THIS REPORT |
|---|---|---|---|
| *F. prausnitzii E* | none | 22% | beneficial |
| *C. beijerinckii* | none | 17% | added by the consortium |
| *C. butyricum* | none | 17% | added by the consortium |
| *A. hadrus* | none | 16% | beneficial |
| *A. hallii* | none | 16% | added by the consortium; beneficial |
| *C. eutactus* | none | 7% | beneficial |
| *C. sp. AF12_28* | none | 6% | conditional |
| *R. bacterium AM53 2BH* | none | 4% | conditional |
| *G. formicilis* | none | 4% | beneficial |
| *R. faecis* | none | 2% | beneficial |

| ORGANISM | MUST | CAN | STATUS IN THIS REPORT |
|---|---|---|---|
| *C. catus* | none | 2% | beneficial |
| *A. soehngenii* | none | 2% | beneficial |
| *A. rectalis* | none | 2% | beneficial |
| *E. ramulus* | none | 2% | conditional |
| +5 more with smaller possible shares | | | |

**Organisms that make the butyrate as you are**

| ORGANISM | MUST | CAN | STATUS IN THIS REPORT |
|---|---|---|---|
| *F. prausnitzii E* | none | 56% | beneficial |
| *A. hadrus* | none | 41% | beneficial |
| *C. eutactus* | none | 17% | beneficial |
| *C. sp. AF12_28* | none | 15% | conditional |
| *R. bacterium AM53 2BH* | none | 11% | conditional |
| *G. formicilis* | none | 9% | beneficial |
| *R. faecis* | none | 5% | beneficial |
| *C. catus* | none | 5% | beneficial |
| *A. soehngenii* | none | 5% | beneficial |
| *A. rectalis* | none | 5% | beneficial |
| *E. ramulus* | none | 5% | conditional |
| *R. hominis* | none | 4% | beneficial |
| *F. magna K* | none | 3% | unknown |
| *A. hallii* | none | 2% | added by the consortium; beneficial |
| +3 more with smaller possible shares | | | |

## Starch + consortium — fibre alone suffices · model cap 16.65 mmol/gDW/h on EX_strch1_m

| OUTPUT (mmol/gDW/h) | AS YOU ARE | + CONSORTIUM | + FIBRE | + BOTH |
|---|---|---|---|---|
| **butyrate** | 219.4 | 219.0 | 344.0 | 343.6 |
| **propionate** | 218.5 | 314.2 | 218.5 | 326.7 |
| **acetate** | 504.6 | 505.8 | 749.7 | 749.2 |
| **lactate** | 336.4 | 337.2 | 401.9 | 499.5 |

Each output is its own exchange bounded separately; they are never summed into one short-chain-fatty-acid number. Community growth 0.0539 per hour in every arm (diet-capped).

**Organisms that would take up the fibre (+ both)**

| ORGANISM | MUST | CAN | STATUS IN THIS REPORT |
|---|---|---|---|
| *F. prausnitzii E* | none | 50% | beneficial |
| *C. beijerinckii* | none | 39% | added by the consortium |
| *C. butyricum* | none | 39% | added by the consortium |
| *R. bromii* | none | 22% | beneficial |
| *C. eutactus* | none | 20% | beneficial |
| *R. faecis* | none | 5% | beneficial |
| *P. distasonis* | none | 5% | conditional |
| *A. rectalis* | none | 4% | beneficial |
| *R. intestinalis* | none | 1% | beneficial |

**Organisms that would make the butyrate (+ both)**

| ORGANISM | MUST | CAN | STATUS IN THIS REPORT |
|---|---|---|---|
| *F. prausnitzii E* | none | 24% | beneficial |
| *C. beijerinckii* | none | 19% | added by the consortium |
| *C. butyricum* | none | 19% | added by the consortium |
| *A. hadrus* | none | 18% | beneficial |
| *A. hallii* | none | 17% | added by the consortium; beneficial |

| ORGANISM | MUST | CAN | STATUS IN THIS REPORT |
|---|---|---|---|
| *C. eutactus* | none | 7% | beneficial |
| *C. sp. AF12_28* | none | 7% | conditional |
| *R. bacterium AM53 2BH* | none | 5% | conditional |
| *G. formicilis* | none | 4% | beneficial |
| *R. faecis* | none | 2% | beneficial |
| *C. catus* | none | 2% | beneficial |
| *A. soehngenii* | none | 2% | beneficial |
| *A. rectalis* | none | 2% | beneficial |
| *E. ramulus* | none | 2% | conditional |
| +5 more with smaller possible shares | | | |

**Organisms that make the butyrate as you are**

| ORGANISM | MUST | CAN | STATUS IN THIS REPORT |
|---|---|---|---|
| *F. prausnitzii E* | none | 56% | beneficial |
| *A. hadrus* | none | 41% | beneficial |
| *C. eutactus* | none | 17% | beneficial |
| *C. sp. AF12_28* | none | 15% | conditional |
| *R. bacterium AM53 2BH* | none | 11% | conditional |
| *G. formicilis* | none | 9% | beneficial |
| *R. faecis* | none | 5% | beneficial |
| *C. catus* | none | 5% | beneficial |
| *A. soehngenii* | none | 5% | beneficial |
| *A. rectalis* | none | 5% | beneficial |
| *E. ramulus* | none | 5% | conditional |
| *R. hominis* | none | 4% | beneficial |
| *F. magna K* | none | 3% | unknown |
| *A. hallii* | none | 2% | added by the consortium; beneficial |
| +3 more with smaller possible shares | | | |

## Arabinoxylan + consortium — pair gains · model cap 4.09 mmol/gDW/h on EX_arabinoxyl_m

| OUTPUT (mmol/gDW/h) | AS YOU ARE | + CONSORTIUM | + FIBRE | + BOTH |
|---|---|---|---|---|
| **butyrate** | 219.4 | 219.0 | 229.8 | 300.8 |
| **propionate** | 218.5 | 314.2 | 218.6 | 326.7 |
| **acetate** | 504.6 | 505.8 | 525.6 | 654.7 |
| **lactate** | 336.4 | 337.2 | 350.4 | 436.5 |

Each output is its own exchange bounded separately; they are never summed into one short-chain-fatty-acid number. Community growth 0.0539 per hour in every arm (diet-capped).

**Organisms that would take up the fibre (+ both)**

| ORGANISM | MUST | CAN | STATUS IN THIS REPORT |
|---|---|---|---|
| *B. longum* **required** | 32% | 33% | added by the consortium |
| *B. cellulosilyticus* **required** | <1% | 2% | beneficial |
| *R. intestinalis* | none | 1% | beneficial |

**Organisms that would make the butyrate (+ both)**

| ORGANISM | MUST | CAN | STATUS IN THIS REPORT |
|---|---|---|---|
| *F. prausnitzii E* | none | 28% | beneficial |
| *C. beijerinckii* | none | 22% | added by the consortium |
| *C. butyricum* | none | 22% | added by the consortium |
| *A. hallii* | none | 21% | added by the consortium; beneficial |
| *A. hadrus* | none | 20% | beneficial |
| *C. eutactus* | none | 8% | beneficial |

| ORGANISM | MUST | CAN | STATUS IN THIS REPORT |
|---|---|---|---|
| *C. sp. AF12_28* | none | 8% | conditional |
| *R. bacterium AM53 2BH* | none | 5% | conditional |
| *G. formicilis* | none | 5% | beneficial |
| *R. faecis* | none | 3% | beneficial |
| *C. catus* | none | 2% | beneficial |
| *A. soehngenii* | none | 2% | beneficial |
| *A. rectalis* | none | 2% | beneficial |
| *E. ramulus* | none | 2% | conditional |
| +5 more with smaller possible shares | | | |

**Organisms that make the butyrate as you are**

| ORGANISM | MUST | CAN | STATUS IN THIS REPORT |
|---|---|---|---|
| *F. prausnitzii E* | none | 56% | beneficial |
| *A. hadrus* | none | 41% | beneficial |
| *C. eutactus* | none | 17% | beneficial |
| *C. sp. AF12_28* | none | 15% | conditional |
| *R. bacterium AM53 2BH* | none | 11% | conditional |
| *G. formicilis* | none | 9% | beneficial |
| *R. faecis* | none | 5% | beneficial |
| *C. catus* | none | 5% | beneficial |
| *A. soehngenii* | none | 5% | beneficial |
| *A. rectalis* | none | 5% | beneficial |
| *E. ramulus* | none | 5% | conditional |
| *R. hominis* | none | 4% | beneficial |
| *F. magna K* | none | 3% | unknown |
| *A. hallii* | none | 2% | added by the consortium; beneficial |
| +3 more with smaller possible shares | | | |

## Cellulose + consortium — no gain · model cap 0.37 mmol/gDW/h on EX_cellul_m

| OUTPUT (mmol/gDW/h) | AS YOU ARE | + CONSORTIUM | + FIBRE | + BOTH |
|---|---|---|---|---|
| **butyrate** | 219.4 | 219.0 | 219.4 | 219.0 |
| **propionate** | 218.5 | 314.2 | 218.5 | 314.2 |
| **acetate** | 504.6 | 505.8 | 504.6 | 505.8 |
| **lactate** | 336.4 | 337.2 | 336.4 | 337.2 |

Each output is its own exchange bounded separately; they are never summed into one short-chain-fatty-acid number. Community growth 0.0539 per hour in every arm (diet-capped).

**Organisms that would make the butyrate (+ both)**

| ORGANISM | MUST | CAN | STATUS IN THIS REPORT |
|---|---|---|---|
| *F. prausnitzii E* | none | 38% | beneficial |
| *C. beijerinckii* | none | 30% | added by the consortium |
| *C. butyricum* | none | 30% | added by the consortium |
| *A. hadrus* | none | 28% | beneficial |
| *A. hallii* | none | 27% | added by the consortium; beneficial |
| *C. eutactus* | none | 11% | beneficial |
| *C. sp. AF12_28* | none | 11% | conditional |
| *R. bacterium AM53 2BH* | none | 7% | conditional |
| *G. formicilis* | none | 6% | beneficial |
| *R. faecis* | none | 4% | beneficial |
| *C. catus* | none | 3% | beneficial |
| *A. soehngenii* | none | 3% | beneficial |
| *A. rectalis* | none | 3% | beneficial |

| ORGANISM | MUST | CAN | STATUS IN THIS REPORT |
|---|---|---|---|
| *E. ramulus* | none | 3% | conditional |
| +5 more with smaller possible shares | | | |

**Organisms that make the butyrate as you are**

| ORGANISM | MUST | CAN | STATUS IN THIS REPORT |
|---|---|---|---|
| *F. prausnitzii E* | none | 56% | beneficial |
| *A. hadrus* | none | 41% | beneficial |
| *C. eutactus* | none | 17% | beneficial |
| *C. sp. AF12_28* | none | 15% | conditional |
| *R. bacterium AM53 2BH* | none | 11% | conditional |
| *G. formicilis* | none | 9% | beneficial |
| *R. faecis* | none | 5% | beneficial |
| *C. catus* | none | 5% | beneficial |
| *A. soehngenii* | none | 5% | beneficial |
| *A. rectalis* | none | 5% | beneficial |
| *E. ramulus* | none | 5% | conditional |
| *R. hominis* | none | 4% | beneficial |
| *F. magna K* | none | 3% | unknown |
| *A. hallii* | none | 2% | added by the consortium; beneficial |
| +3 more with smaller possible shares | | | |

## How many organisms hold each function

**fewest confirmed carriers first**

Two communities can make the same amount of something with very different security. If one organism is doing all of it, the function travels with that organism; if four share it, losing any one changes little. The count below is an *effective* one: two organisms splitting a job evenly count as two, but two where one does 99% of the work count as barely more than one.

| FUNCTION | EFFECTIVE CARRIERS | NAMED | TOP SHARE | SIGNAL CONFIRMED |
|---|---|---|---|---|
| gamma-aminobutyric acid (GABA) | 1.0 | 1 | 100% | 6% † |
| Histamine | 1.0 | 1 | 100% | 67% |
| p-Cresol / p-cresyl sulfate | 1.0 | 1 | 100% | 1% † |
| Ammonia (from urea) | 1.0 | 1 | 100% | 8% † |
| Peptide uptake | 1.3 | 2 | 92% | 32% † |
| Indole-3-propionate (IPA) | 1.6 | 2 | 80% | 20% † |
| Microbial ethanol formation | 1.8 | 2 | 72% | 17% † |
| Deconjugated glucuronides (oestrogens, drug metabolites, bilirubin) | 1.8 | 2 | 71% | 16% † |
| Vitamin B3 (niacin and NAD precursors) | 1.9 | 2 | 62% | 5% † |
| Riboflavin (vitamin B2) | 2.0 | 2 | 53% | 4% † |
| Vitamin K2 (menaquinone, MK-n) | 2.0 | 2 | 53% | 10% † |
| Mucin glycans and lipid A | 2.0 | 2 | 53% | 12% † |
| Folate (vitamin B9) | 2.0 | 2 | 52% | 3% † |
| Protein and nitrogen metabolism | 2.0 | 2 | 50% | 18% † |
| Putrescine formation | 2.1 | 6 | 82% | 43% † |
| Fermentation routes and cross-feeding | 2.6 | 3 | 60% | 23% † |
| Propionate (SCFA) | 2.8 | 4 | 64% | 25% † |
| Unconjugated bile acids | 2.9 | 4 | 64% | 53% |

| FUNCTION | EFFECTIVE CARRIERS | NAMED | TOP SHARE | SIGNAL CONFIRMED |
|---|---|---|---|---|
| Hydrogen sulfide (H2S) | 2.9 | 3 | 45% | 37% † |
| Glutathione synthesis | 3.0 | 3 | 35% | 26% † |
| Vitamin B6 (pyridoxal phosphate) | 3.6 | 4 | 44% | 32% † |
| Vitamin B1 (thiamine) | 3.6 | 4 | 34% | 18% † |
| Dietary fibre and carbohydrate breakdown | 3.7 | 4 | 43% | 10% † |
| Indole / indoxyl sulfate | 4.0 | 4 | 30% | 43% † |
| Biotin (vitamin B7) | 4.1 | 5 | 49% | 25% † |
| Butyrate (SCFA) | 4.2 | 5 | 45% | 70% |
| Urate degradation | 4.2 | 5 | 40% | 15% † |
| Imidazole propionate (ImP) | 4.3 | 5 | 33% | 71% |
| Branched-chain amino acids (valine, leucine, isoleucine) | 4.9 | 5 | 25% | 40% † |
| Vitamin B5 (pantothenate) | 5.6 | 8 | 34% | 52% |
| Vitamin B12 (cobalamin) | 5.7 | 8 | 38% | 54% |
| Spermidine formation | 7.4 | 8 | 23% | 43% † |

The bar runs to 4 effective carriers, past which more organisms stop changing the answer. “Named” counts only organisms found in your own inventory as well as on the gene: a reference organism whose copy of a gene matched a read, but which was not detected in you, is context rather than a carrier. **Signal confirmed** is how much of the function's signal those named carriers account for. † On 26 rows it is under half, meaning most of the signal sits on organisms this assay could not place in you — there read the carrier count as the fewest it could be, not as the number there are. This is redundancy among the carriers this assay resolved — not a guarantee of ecological resilience.

PART B — DISEASE-PATTERN RESEMBLANCE

# 20 Resemblance to published disease patterns, in detail

Researchers have described how the gut communities of people with certain conditions differ, on average, from those of healthy people. This section asks, for each of the patterns in the library: **how closely does this sample resemble that published group-level pattern?** The answer is a percentile of a matched reference group, on the same scale as every other reading in this report. It is not a diagnosis and not a probability of disease.

## What we knew about you when scoring

Several patterns depend on facts the sequencing cannot see — medication, how long you have been ill, recent antibiotics. Those supplied are listed; those not supplied are listed with what they would have changed. Where a pattern needs one of them, the ordinary case was assumed (no medication, no recent antibiotics) and the pattern was scored on that stated assumption rather than withheld; section 0 lists every assumption.

| FACTOR | STATUS | EFFECT ON SCORING |
|---|---|---|
| Sequencing run / flowcell | recorded: 23VJTHLT4:8 | used where a pattern declares it |
| Metformin | not supplied · assumed absent | produces much of the published type 2 diabetes signature; assumed absent when no medication list is supplied (participant mode abstains) |
| Proton-pump inhibitor | not supplied · assumed absent | large documented shift; recorded and lowers technical reliability |
| Antibiotics in the last 6 months | not supplied | dominates the community for months; profiles abstain within 90 days |
| Gluten-free diet | not supplied | large effect in prevalent celiac disease; required for the celiac profile |
| Country | not supplied | geography often exceeds the disease effect; reference matched on it |
| Age | not supplied | large across the lifespan; mandatory for age-specific profiles |
| Sex | not supplied | androgenetic alopecia signals are sex-specific; mandatory there |
| Stool form | not supplied | correlates with diversity; recorded |
| Hospitalisation / ICU | not supplied | drives E. coli, Klebsiella and resistance genes; recorded |
| Kidney function | not supplied | decides whether uremic-toxin capacity matters; context for CKD |
| Other medications | not supplied | recorded |

What was recorded, what was missing, and which profiles refused to score because of it. A missing confounder does not lower a score; it widens what the score could mean.

## How the pages that follow are built

Each pattern is made of one or more **study modules** — one per independent published cohort, so a finding replicated by three groups counts three times and a finding from one group once. Modules are scored separately so you can see which line of evidence is driving the result, then fused with equal weight per independent study. Below each result is a table of every contributing feature with its measured value, its position in the comparison group, the direction the research reports, and its weight, so the score can be recomputed by hand.

Every scored pattern follows, most resemblant first, each with the human research on what has been tried for that condition. Patterns the engine declined to score are listed with the reason at the end.

‹ at a glance
what you can do about it ›

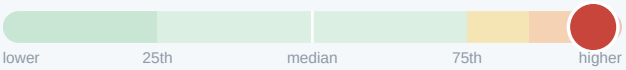
# Rheumatoid arthritis — gut pattern

NOTABLY HIGH

Pattern version **1.0.0** (ra) · maturity **P3** (independent human replication with coherent task and assay) · research beta · mixed_us_chinese_cohorts

The rheumatoid arthritis finding is strain-level, which is a problem for a species-level pipeline. Scher et al. found Prevotella copri expanded in new-onset untreated RA, and later work showed immune reactivity to specific P. copri peptides in a subset of patients. Since P. copri is also high in healthy people on high-fibre and non-Western diets, species abundance alone cannot carry the finding. Three later results constrain this profile further, and are the reason the species feature is an anchor rather than a scored direction. First, replication is poor. Several cohorts, including a 2024 preclinical-RA study, could not reproduce group-wise Prevotellaceae or P. copri differences, and attempts to reproduce the anti-P. copri antibody finding returned only modest or null results with reactivity varying by strain. Second, the direction is not even consistent within the species. In a comprehensively phenotyped anti-CCP-positive at-risk cohort (2024), three Prevotellaceae strains were enriched and five were depleted in those progressing towards arthritis. A single species-level abundance cannot represent a signal whose components move in opposite directions. Third, and decisively for the clade question: Manghi et al. resolved the Segatella (formerly Prevotella) copri complex into 13 species-level clades and tested them against 1,635 disease cases and 1,854 controls across 22 studies, including a rheumatoid arthritis cohort. None of the complex's species was associated with any health condition. So there is no established RA clade to score against, even though the clades themselves are now measurable (see `copri_complex` below). What remains is this: P. copri expansion is the finding. Where the organism is absent, the profile does not assert the pattern at all — the accessory taxa from Scher's cohort cannot carry it alone. Zhang et al. added a Chinese shotgun cohort with oral and gut samples, finding Haemophilus depleted and Lactobacillus salivarius enriched, partly normalised by treatment.

**COMBINED RESEMBLANCE**
percentile of the matched reference group




**95th**
likely range 31–100

**Read this percentile narrowly.** *Prevotella copri* — the organism this profile's central finding is about — was looked for and not detected, so the module carrying that finding is not scored. The percentile above therefore rests on one study module (Zhang 2015 Chinese RA shotgun (gut)), and reflects accessory taxa rather than the organism the condition is associated with.

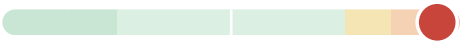
## What is driving it

**Scher 2013 new-onset untreated RA (16S, US)**
species · association only · 16S study · study scher_2013 · **Prevotella copri not detected, so this pattern is not asserted** · 3/3 features · 100% of combined




— ANCHOR ABSENT

**Zhang 2015 Chinese RA shotgun (gut)**
species · study zhang_2015 · 6/6 features · 100% of combined



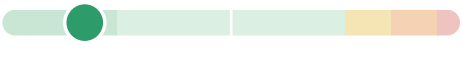
95th NOTABLY HIGH

**RA-associated P. copri clade**
strain · study scher_2013 · 0/1 features · not measurable — a validated reference panel for this specific strain-level target. Strain resolution itself is available and reported separately: the marker lane fingerprints the dominant population of every organism with enough coverage. What is missing here is not the engine but a published, discriminating marker set for this named target




— NOT COMPUTABLE

**Metabolite gene capacity**
genes · genomic capacity · study functional · 1/1 features · not in combined



18th BELOW AVERAGE

## The Prevotella copri complex, clade by clade

No member of the Segatella (Prevotella) copri complex was detected. All 13 clades were assessed, including the 9 found in humans, so this is a measured absence across the whole complex rather than an unresolved species-level reading.

No clade of this complex has an established association with any disease, including rheumatoid arthritis. Manghi et al. tested all 13 clades against 1,635 disease cases and 1,854 controls across 22 studies and 11 conditions, a set that included a rheumatoid arthritis cohort, and found none of the complex's species associated with any health condition. Earlier rheumatoid-arthritis reports named the species as a whole, not a clade, and those reports have replicated poorly: in an anti-CCP-positive at-risk cohort three Prevotellaceae strains were enriched and five depleted in the people who progressed. So this section reports what is present and makes no disease claim from it.

Tett 2019 Cell Host Microbe doi:10.1016/j.chom.2019.08.018; Manghi 2023 Cell Host Microbe doi:10.1016/j.chom.2023.09.013; Hitch 2022 Syst Appl Microbiol doi:10.1016/j.syapm.2022.126354.

## How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

| | | |
|---|---|---|
| **Technical reliability** | 1.00 | sequencing depth, host fraction, reference match |
| **Independent evidence coverage** | 1.00 | share of the pattern's independent evidence clusters this sample could measure |
| **Evidence maturity** | P3 | independent human replication with coherent task and assay |
| **Assay transportability** | native | whether the studies measured what this sample measured |
| **Population match** | 0.60 | how well the reference group matches you |
| **Specificity** | unknown | does this pattern separate its condition from general disturbance? measured where labelled data exist |
| **Direction conflict** | material | studies disagreeing on direction for one or more features |

**Competing patterns** scoring nearly as high: ad clinical (88th), alopecia areata (47th), acvd (39th). Margin over the next pattern: 8 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

## Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| **Scher 2013 new-onset untreated RA (16S, US)** · species · association only · 16S study · study scher_2013 · **Prevotella copri not detected, so this pattern is not asserted** | | | | | | | | | |
| *Prevotella copri* | not detected | 25 | ▲ | 0.50 | 0.60 | 0.20 | 0.50 | -0.23 | |
| *Bacteroides vulgatus* | not detected | 3 | ▼ | 0.50 | 0.60 | 0.30 | 0.60 | +1.00 | |
| *Bacteroides uniformis* | 1.860% | 50 | ▼ | 0.40 | 0.60 | 0.30 | 0.60 | -0.00 | |
| **Zhang 2015 Chinese RA shotgun (gut)** · species · study zhang_2015 | | | | | | | | | |
| *Haemophilus parainfluenzae* | not detected | 29 | ▼ | 0.80 | 0.80 | 0.60 | 0.80 | +0.18 | |
| *Lactobacillus salivarius* | not detected | 48 | ▲ | 0.80 | 0.80 | 0.50 | 0.80 | -0.02 | |
| *Bifidobacterium dentium* | not detected | 44 | ▲ | 0.50 | 0.60 | 0.40 | 0.60 | -0.05 | |
| *Ruminococcus lactaris* | 2.685% | 99 | ▲ | 0.50 | 0.60 | 0.50 | 0.60 | +0.80 | |
| *Clostridium asparagiforme* | not detected | 41 | ▲ | 0.50 | 0.60 | 0.50 | 0.60 | -0.08 | |
| *Klebsiella pneumoniae* | not detected | 40 | ▼ | 0.40 | 0.50 | 0.40 | 0.50 | +0.09 | |
| **RA-associated P. copri clade** · strain · study scher_2013 | | | | | | | | | |
| Prevotella copri RA clade | missing | — | ▲ | 1.00 | 1.00 | 1.00 | 1.00 | — | |
| **Metabolite gene capacity** · genes · genomic capacity · study functional | | | | | | | | | |
| butyrate | 32.1/100 | 82 | ▼ | 0.80 | 0.70 | 0.20 | 0.70 | -0.26 | |

FINE PRINT

**Confidence grade insufficient.** Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal; +1 more, all in results.json.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

**Measured discrimination: not available.** No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Rheumatoid arthritis is diagnosed clinically with serology (RF, anti-CCP) and imaging. This profile does not diagnose or predict it.

The central RA finding is a Prevotella copri strain this pipeline cannot resolve. The species is common in healthy people. The strain module is reported as not computable, and the profile score rests mostly on the Chinese cohort.

P. copri is the anchor of the Scher module. Where it is absent, that module reports `anchor_absent` and takes no part in the score: a sample with no Prevotella cannot resemble a Prevotella-expansion signature, however its Bacteroides species happen to fall. In that case the profile score reflects the Chinese cohort alone and should be read as such.

The complex's clade composition is resolved separately from the 13-clade SGB catalogue and reported under `copri_complex`. No clade has an established disease direction, so that section is composition only and contributes nothing to the score.

Replication of the species-level association is weak. Several cohorts found no group-wise difference, and in an at-risk cohort the strains moved in both directions at once. A high reading here is a weak signal from one Chinese cohort's accessory taxa, not evidence of arthritis risk.

Sources: Scher JU, et al. Expansion of intestinal Prevotella copri correlates with enhanced susceptibility to arthritis. eLife 2013;2:e01202; Zhang X, et al. The oral and gut microbiomes are perturbed in rheumatoid arthritis and partly normalized after treatment. Nat Med 2015;21:895-905; Pianta A, et al. Evidence of the immune relevance of Prevotella copri, a gut microbe, in patients with rheumatoid arthritis. Arthritis Rheumatol 2017;69:964.

## What the research says about this condition and this pattern

**Rheumatoid-arthritis pattern — small probiotic trials, positive and null** LANE C HUMAN TRIAL · LIMITED

Fired by: Rheumatoid arthritis pattern: notably high

Rheumatoid arthritis is diagnosed by rheumatology with serology and examination. Two small probiotic trials were positive on disease activity, one was null; many RA patients are immunosuppressed, which changes live-microbe safety.

**Studied:** Lacticaseibacillus casei 01 (Alipour 2014, rheumatoid arthritis) (probiotic) · **in:** people with clinically confirmed rheumatoid arthritis, ages 20–80, female, in Iran, n = 46 · **studied protocol (a source fact, not an instruction):** 10^8 CFU/day L. casei 01 for eight weeks

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: replicated. Strongest design: randomized trial.

- A 46-woman trial of L. casei 01 for eight weeks reported improved disease activity and cytokines in rheumatoid arthritis. [Alipour B et al. Int J Rheum Dis 2014;17:519] · lane C · randomized double-blind placebo-controlled · n = 46 · benefit
- A 60-person three-organism probiotic trial reported improved DAS28 and metabolic markers in rheumatoid arthritis. [Zamani B et al. Int J Rheum Dis 2016;19:869] · lane C · randomized double-blind placebo-controlled · n = 60 · benefit
- **Against or limiting —** A 29-person trial of L. rhamnosus GR-1 + L. reuteri RC-14 found no ACR20 benefit in rheumatoid arthritis. [Pineda M de L et al. Med Sci Monit 2011;17:CR347] · lane X · randomized double-blind placebo-controlled pilot · n = 29 · null
- **Against or limiting —** FMT in rheumatoid arthritis is case-report level and investigational. [FMT case report in rheumatoid arthritis (2021)] · lane X · case report

**Applicability:** The study required a clinical diagnosis of rheumatoid arthritis. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed. The study enrolled female participants only; sex is not on file here.

**Fecal microbiota transplantation:** investigational for this condition (IND-only; no treatment route from this report).

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “you have arthritis” · “probiotics treat RA”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance
what you can do about it ›

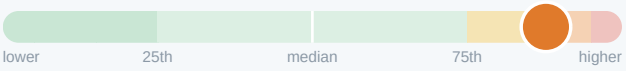
# Clinical Alzheimer's disease — stage-specific gut pattern

HIGH

Pattern version **1.0.0** (ad_clinical) · maturity **P2** (usable human shotgun cohort, or multiple compatible human association cohorts) · research beta · adult_chinese_cohort_derived

The only stage in the Alzheimer's continuum with a species-level panel that survives strict multiple-testing correction. Jia et al. profiled 476 participants across five stages of the continuum by faecal shotgun sequencing and found over 10% of microbial species and gene families significantly altered across progression, with cross-validated AUC around 0.80 and 0.75 in independent validation. At q < 0.05 no species qualified for the three pre-clinical stages. Only clinical AD has a strict species panel, which is why this profile scores and the earlier stages lean on evidence types this pipeline cannot yet measure. The directions below are findings in a Chinese staged cohort, not universal Alzheimer's biology.

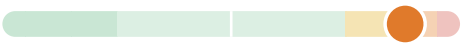
**COMBINED RESEMBLANCE**
percentile of the matched reference group




**88th**
likely range 46–97

## What is driving it

**Jia 2025 strict species panel (q < 0.05)**
species · study metaad_2025 · 6/6 features · 100% of combined



88th — HIGH

**Jia 2025 KEGG orthologue panel**
pathways · genomic capacity · study metaad_2025 · 0/8 features · not measurable — a HUMAnN 3 pathway/KO abundance stage is added



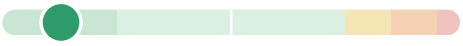

— NOT COMPUTABLE

**Alzheimer's-relevant metabolite capacity**
genes · genomic capacity · study mechanism_lanes · 5/5 features · not in combined



47th — TYPICAL

**Butyrate carriers (independent count)**
carriers · carrier proxy · study carrier_cross_check · 1/1 features · not in combined



13th — LOW

## How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

| | | |
|---|---|---|
| **Technical reliability** | 1.00 | sequencing depth, host fraction, reference match |
| **Independent evidence coverage** | 1.00 | share of the pattern's independent evidence clusters this sample could measure |
| **Evidence maturity** | P2 | usable human shotgun cohort, or multiple compatible human association cohorts |
| **Assay transportability** | native | whether the studies measured what this sample measured |
| **Population match** | 0.60 | how well the reference group matches you |
| **Specificity** | unknown | does this pattern separate its condition from general disturbance? measured where labelled data exist |
| **Direction conflict** | material | studies disagreeing on direction for one or more features |

**Competing patterns** scoring nearly as high: ra (95th), celiac (77th), t1d (64th). Margin over the next pattern: 8 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

## Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the

pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| **Jia 2025 strict species panel (q < 0.05)** · species · study metaad_2025 | | | | | | | | | |
| *Odoribacter splanchnicus* | 0.472% | 81 | ▲ | 1.00 | 0.50 | 0.50 | 1.00 | +0.29 | Jia 2025 Mol Psychiatry (simulator-supported) |
| *Barnesiella viscericola* | not detected | 50 | ▲ | 1.00 | 0.50 | 0.60 | 1.00 | -0.00 | Jia 2025 Mol Psychiatry (simulator-supported) |
| *Alistipes finegoldii* | 0.743% | 87 | ▲ | 1.00 | 0.50 | 0.40 | 0.80 | +0.37 | Jia 2025 Mol Psychiatry (simulator-supported) |
| *Alistipes timonensis* | not detected | 47 | ▲ | 0.70 | 0.50 | 0.40 | 0.80 | -0.03 | Jia 2025 Mol Psychiatry (lineage cluster) |
| *Prevotella bivia* | not detected | 48 | ▲ | 0.50 | 0.40 | 0.40 | 0.70 | -0.01 | Jia 2025 Mol Psychiatry |
| *Streptococcus infantis* | not detected | 43 | ▼ | 0.50 | 0.40 | 0.30 | 0.60 | +0.06 | Jia 2025 Mol Psychiatry |
| **Jia 2025 KEGG orthologue panel** · pathways · genomic capacity · study metaad_2025 | | | | | | | | | |
| K00857 | missing | — | ▲ | 1.00 | 1.00 | 1.00 | 1.00 | — | |
| K00876 | missing | — | ▲ | 1.00 | 1.00 | 1.00 | 1.00 | — | |
| K01031 | missing | — | ▲ | 1.00 | 1.00 | 1.00 | 1.00 | — | |
| K03179 | missing | — | ▲ | 1.00 | 1.00 | 1.00 | 1.00 | — | |
| K11782 | missing | — | ▲ | 1.00 | 1.00 | 1.00 | 1.00 | — | |
| K18285 | missing | — | ▲ | 1.00 | 1.00 | 1.00 | 1.00 | — | |
| K20036 | missing | — | ▲ | 1.00 | 1.00 | 1.00 | 1.00 | — | |
| K01754 | missing | — | ▼ | 1.00 | 1.00 | 1.00 | 1.00 | — | |
| **Alzheimer's-relevant metabolite capacity** · genes · genomic capacity · study mechanism_lanes | | | | | | | | | |
| bai | 3.5/100 | 72 | ▲ | 1.00 | 0.80 | 0.40 | 0.90 | +0.30 | MahmoudianDehkordi 2019 Alzheimers Dement |
| butyrate | 32.1/100 | 82 | ▼ | 1.00 | 0.80 | 0.30 | 0.90 | -0.26 | Zhang 2023 Mol Psychiatry |
| cutc | 3.3/100 | 66 | ▲ | 0.80 | 0.70 | 0.40 | 0.80 | +0.21 | Vogt 2018 Alzheimers Res Ther |
| ipa | 8.9/100 | 86 | ▼ | 0.70 | 0.60 | 0.40 | 0.80 | -0.61 | Pappolla 2021 Neurobiol Dis |
| propionate | 26.4/100 | 76 | ▲ | 0.50 | 0.50 | 0.30 | 0.40 | +0.36 | Mendelian randomisation, per Jia 2025 discussion |
| **Butyrate carriers (independent count)** · carriers · carrier proxy · study carrier_cross_check | | | | | | | | | |
| butyrate | 45.230% | 87 | ▼ | 1.00 | 0.80 | 0.30 | 0.90 | -0.43 | derived from the butyrate panel organism list |

FINE PRINT

**Confidence grade insufficient.** Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

**Measured discrimination: not available.** No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Not a diagnosis, a risk estimate, or a screening result. Alzheimer's disease is diagnosed clinically with cognitive assessment and, where indicated, amyloid and tau biomarkers. A gut community resembling a published Alzheimer's cohort says nothing about whether that is happening.

Derived from a Chinese staged cohort. Directions are source-cohort findings; conflicting directions in other cohorts are held as separate assertions rather than pooled into a consensus.

Four of the source panel's Alistipes species and both of its Streptococcus species have no entry in the reference catalogue this pattern is scored against. They are recorded as missing, which is why independent-cluster coverage here is well below the raw feature count.

Sources: Jia L, Ke Y, Zhao S, et al. Metagenomic analysis characterizes stage-specific gut microbiota in Alzheimer's disease. Mol Psychiatry 2025;30:3951-3962, doi:10.1038/s41380-025-02973-7; data PRJCA022804 / CRA014435 (NGDC), analysis github.com/JiaLonghao1997/metaAD_analysis.

## What the research says about this condition and this pattern

**Cognition — what has replicated evidence independent of the microbiome** LANE B HUMAN EVIDENCE · STRONG

Fired by: Clinical Alzheimer's disease pattern: high

Resemblance to an Alzheimer-cohort stool pattern does not indicate cognitive impairment. The multidomain FINGER lifestyle programme has randomized evidence for cognition in at-risk older adults; the MIND diet did not beat a control diet.

**Studied:** FINGER multidomain lifestyle intervention (diet, exercise, cognitive training, vascular risk monitoring) (lifestyle) · **in:** people with clinically confirmed elevated dementia risk (CAIDE score) with average or slightly lower cognition, ages 60–77, in Finland, n = 1260 · **studied protocol (a source fact, not an instruction):** two-year multidomain intervention (nutrition, exercise, cognitive training, vascular risk monitoring)

Evidence lane B: replicated human randomized clinical benefit for the exact intervention and population. Replication: single study. Strongest design: randomized trial.

- The FINGER trial showed that a two-year multidomain lifestyle programme (diet, exercise, cognitive training, vascular risk control) improved or maintained cognition in at-risk older adults — evidence independent of the microbiome. [Ngandu T et al. Lancet 2015;385:2255 (FINGER)] · lane B · randomized controlled, 2 years · n = 1260 · benefit
- **Against or limiting —** The MIND-diet trial did not improve cognition more than a control diet with the same mild caloric restriction over three years. [Barnes LL et al. N Engl J Med 2023;389:602 (MIND diet)] · lane X · randomized controlled, 3 years · n = 604 · null

**Conventional route:** Neurologic/cognitive evaluation only if memory or thinking symptoms are present.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “you have Alzheimer's” · “prevents dementia”

**Alzheimer disease — probiotic trials, positive and null** LANE C HUMAN TRIAL · LIMITED

Fired by: Clinical Alzheimer's disease pattern: high

One 60-person trial reported MMSE gains; a later trial in severe disease was null and pooled analyses are not robust. FMT in Alzheimer disease is animal and case-level only.

**Studied:** Probiotic milk with Lactobacillus acidophilus, L. casei, Bifidobacterium bifidum and L. fermentum (Akbari 2016) (probiotic) · **in:** people with clinically confirmed Alzheimer disease, ages 60–95, in Iran, n = 60 · **studied protocol (a source fact, not an instruction):** 200 mL/day probiotic milk (2 × 10^9 CFU/g of each organism) for 12 weeks

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: single study. Strongest design: randomized trial.

- One 60-person 12-week trial of a four-organism probiotic milk reported MMSE and metabolic-marker changes in people with Alzheimer disease. [Akbari E et al. Front Aging Neurosci 2016;8:256] · lane C · randomized double-blind controlled · n = 60 · benefit
- **Against or limiting —** A later trial in severe Alzheimer disease found no cognitive benefit from probiotics. [Agahi A et al. Front Neurol 2018 (severe AD, negative)] · lane X · randomized double-blind placebo-controlled · null
- **Against or limiting —** Conservative pooled analyses find the probiotic evidence in Alzheimer disease inconsistent and not robust. [meta-analysis of probiotics in Alzheimer disease/MCI (2021)] · lane X · systematic review / meta-analysis · mixed
- **Against or limiting —** Probiotic trials in mild cognitive impairment are small and inconsistent. [2024 evidence review — probiotics in mild cognitive impairment] · lane X · review · mixed

**Applicability:** The study required a clinical diagnosis of alzheimer disease. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed.

**Fecal microbiota transplantation:** investigational for this condition (IND-only; no treatment route from this report).

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “reverses dementia”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

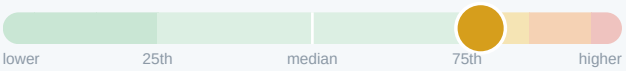
# Coeliac disease — gut pattern

ABOVE AVERAGE

Pattern version **1.0.0** (celiac) · maturity **P2** (usable human shotgun cohort, or multiple compatible human association cohorts) · experimental · mixed_european_cohorts

Coeliac disease has a confounder built into its treatment: the gluten-free diet removes most wheat-derived fermentable substrate and depletes Bifidobacterium and butyrate producers on its own (De Palma 2009, Bonder 2016). Untreated coeliac shows Proteobacteria and Bacteroides expansion with Bifidobacterium loss; treated coeliac looks like a gluten-free healthy person. The most specific finding is functional — the CELIAC Disease Genomic, Environmental, Microbiome and Metabolomic (CDGEMM) infant study and the Leonard 2021 prospective cohort found changes in carbohydrate-active enzyme and tryptophan metabolism before onset — which is why the CAZy module is declared even though this pipeline cannot compute it. Gluten-free diet status is the one thing needed to read this profile, and the confounder ledger will say whether it was supplied.

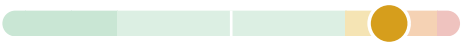
**COMBINED RESEMBLANCE**
percentile of the matched reference group




**77th**
likely range 13–100

## What is driving it

**Leonard 2021 CDGEMM prospective infants**
species · study cdgemm · 4/4 features · 50% of combined



85th — ABOVE AVERAGE

**Untreated coeliac consensus (16S, multiple cohorts)**
species · association only · 16S study · study untreated_consensus · 6/6 features · 50% of combined



54th — TYPICAL

**Carbohydrate-active enzyme shifts before onset**
CAZymes · genomic capacity · study cdgemm · 0/2 features · not measurable — assembly plus dbCAN3 carbohydrate-active enzyme annotation is added




— NOT COMPUTABLE

**Metabolite gene capacity**
genes · genomic capacity · study functional · 2/2 features · not in combined



11th — LOW

## How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

| | | |
|---|---|---|
| **Technical reliability** | 1.00 | sequencing depth, host fraction, reference match |
| **Independent evidence coverage** | 1.00 | share of the pattern's independent evidence clusters this sample could measure |
| **Evidence maturity** | P2 | usable human shotgun cohort, or multiple compatible human association cohorts |
| **Assay transportability** | labeled transport | whether the studies measured what this sample measured |
| **Population match** | 0.60 | how well the reference group matches you |
| **Specificity** | unknown | does this pattern separate its condition from general disturbance? measured where labelled data exist |
| **Direction conflict** | none | studies disagreeing on direction for one or more features |

**Competing patterns** scoring nearly as high: ad clinical (88th), alopecia areata (47th), acvd (39th). Margin over the next pattern: 8 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

## Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| **Leonard 2021 CDGEMM prospective infants** · species · study cdgemm | | | | | | | | | |
| *Dialister invisus* | not detected | 39 | ▲ | 0.60 | 0.50 | 0.50 | 0.60 | -0.09 | |
| *Parabacteroides distasonis* | 1.492% | 78 | ▲ | 0.50 | 0.50 | 0.40 | 0.50 | +0.24 | |
| *Bacteroides vulgatus* | not detected | 3 | ▼ | 0.50 | 0.50 | 0.30 | 0.50 | +1.00 | |
| *Faecalibacterium prausnitzii* | 9.840% | 81 | ▼ | 0.60 | 0.60 | 0.20 | 0.70 | -0.31 | |
| **Untreated coeliac consensus (16S, multiple cohorts)** · species · association only · 16S study · study untreated_consensus | | | | | | | | | |
| *Bifidobacterium longum* | not detected | 17 | ▼ | 0.80 | 0.60 | 0.30 | 0.70 | +0.32 | |
| *Bifidobacterium adolescentis* | not detected | 25 | ▼ | 0.60 | 0.60 | 0.30 | 0.70 | +0.23 | |
| *Escherichia coli* | not detected | 19 | ▲ | 0.70 | 0.60 | 0.30 | 0.70 | -0.29 | |
| *Bacteroides fragilis* | not detected | 24 | ▲ | 0.50 | 0.50 | 0.30 | 0.60 | -0.24 | |
| *Clostridium bolteae* | not detected | 29 | ▲ | 0.40 | 0.50 | 0.40 | 0.50 | -0.18 | |
| *Lactobacillus ruminis* | not detected | 44 | ▼ | 0.40 | 0.50 | 0.40 | 0.50 | +0.05 | |
| **Carbohydrate-active enzyme shifts before onset** · CAZymes · genomic capacity · study cdgemm | | | | | | | | | |
| GH13 subfamily shift | missing | — | ▲ | 1.00 | 1.00 | 1.00 | 1.00 | — | |
| GH43 shift | missing | — | ▼ | 1.00 | 1.00 | 1.00 | 1.00 | — | |
| **Metabolite gene capacity** · genes · genomic capacity · study functional | | | | | | | | | |
| ipa | 8.9/100 | 86 | ▼ | 0.80 | 0.60 | 0.50 | 0.70 | -0.61 | |
| butyrate | 32.1/100 | 82 | ▼ | 0.60 | 0.60 | 0.20 | 0.70 | -0.26 | |

FINE PRINT

**Confidence grade insufficient.** Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal; +1 more, all in results.json.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

**Measured discrimination: not available.** No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Coeliac disease is diagnosed by tissue transglutaminase serology and duodenal biopsy. This profile does not diagnose it, and a stool DNA pattern is irrelevant to whether someone should eat gluten.

A gluten-free diet produces this pattern in people without coeliac disease. A high score in someone avoiding gluten is expected.

The most specific pre-onset finding (CAZy families) is declared but not computable here.

Sources: Leonard MM, et al. Microbiome signatures of progression toward celiac disease onset in at-risk children in a longitudinal prospective cohort study. PNAS 2021;118:e2020322118; Lamas B, et al. Aryl hydrocarbon receptor ligand production by the gut microbiota is decreased in celiac disease leading to intestinal inflammation. Sci Transl Med 2020;12:eaba0624; Wacklin P, et al. Altered duodenal microbiota composition in celiac disease patients suffering from persistent symptoms on a long-term gluten-free diet. Am J Gastroenterol 2014;109:1933.

## What the research says about this condition and this pattern

**Celiac pattern — test before changing gluten** LANE A CLINICAL ROUTE

Fired by: Coeliac disease pattern: somewhat high

A gluten-free diet follows a proper celiac diagnosis. Removing gluten before serology and biopsy can make the tests falsely negative, so this pattern is a reason to test, not to change diet.

**Studied:** Coeliac serology and, when indicated, duodenal biopsy while still eating gluten (ACG 2023) (clinical route) · **in:** ages 0–99 · **studied protocol (a source fact, not an instruction):** tTG-IgA with total IgA while eating gluten; biopsy per ACG 2023

Evidence lane A: current authoritative guideline or regulatory label for an independently confirmed indication. Replication: single study. Strongest design: guideline. Guideline position: strong for.

- A gluten-free diet follows a proper clinical diagnosis (serology and biopsy while still eating gluten). Starting gluten avoidance from a microbiome pattern can impair the diagnostic tests. [Rubio-Tapia A et al. ACG Clinical Guidelines: Diagnosis and Management of Celiac Disease. Am J Gastroenterol 2023;118:59] · lane A · guideline · benefit

**Conventional route:** tTG-IgA with total IgA while still eating gluten; biopsy per ACG 2023.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “go gluten-free now” · “you have celiac”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

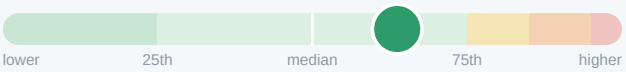
# Type 1 diabetes — gut pattern

TYPICAL

Pattern version **1.0.0** (t1d) · maturity **P2** (usable human shotgun cohort, or multiple compatible human association cohorts) · experimental · mixed_european_pediatric_adult

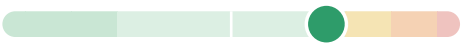
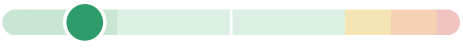
Most type 1 diabetes microbiome work is in infants at genetic risk, following them to seroconversion. Kostic et al. (DIABIMMUNE) found a drop in diversity and a rise in inflammation-associated taxa in the year before diagnosis; Vatanen et al. showed that Bacteroides-dominated Finnish and Estonian infants carried immunologically silent hexa-acylated-LPS-poor communities that failed to educate the immune system, unlike Russian Karelian infants. The TEDDY study found Bifidobacterium and short-chain fatty acid producers protective. Adult established-T1D cohorts show a weaker pattern of butyrate-producer depletion. An adult sample is far from the population these findings come from.

**COMBINED RESEMBLANCE**
percentile of the matched reference group




**64th**
likely range 13–95

## What is driving it

| | | | |
|---|---|---|---|
| **Vatanen 2018 TEDDY (10,913 metagenomes, 783 infants)**<br>species · study vatanen_2018 · 4/4 features · 67% of combined |  | 71st | TYPICAL |
| **Kostic 2015 DIABIMMUNE seroconverters**<br>species · study diabimmune · 5/5 features · 33% of combined |  | 40th | TYPICAL |
| **Diversity before diagnosis**<br>diversity · study diabimmune · 0/1 features · not in combined |  | — | NOT COMPUTABLE |
| **SCFA and LPS capacity**<br>genes · genomic capacity · study functional · 1/1 features · not in combined |  | 18th | BELOW AVERAGE |
| **LPS immunogenicity carriers**<br>carriers · carrier proxy · study carriers · 2/2 features · not in combined |  | 78th | ABOVE AVERAGE |

## How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

| | | |
|---|---|---|
| **Technical reliability** | 1.00 | sequencing depth, host fraction, reference match |
| **Independent evidence coverage** | 1.00 | share of the pattern's independent evidence clusters this sample could measure |
| **Evidence maturity** | P2 | usable human shotgun cohort, or multiple compatible human association cohorts |
| **Assay transportability** | native | whether the studies measured what this sample measured |
| **Population match** | 0.60 | how well the reference group matches you |
| **Specificity** | unknown | does this pattern separate its condition from general disturbance? measured where labelled data exist |
| **Direction conflict** | material | studies disagreeing on direction for one or more features |

**Competing patterns** scoring nearly as high: ad clinical (88th), alopecia areata (47th), acvd (39th). Margin over the next pattern: 8 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

## Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

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| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| **Vatanen 2018 TEDDY (10,913 metagenomes, 783 infants)** · species · study vatanen_2018 | | | | | | | | | |
| *Bifidobacterium longum* | not detected | 17 | ▼ | 0.80 | 0.80 | 0.30 | 0.80 | +0.32 | |
| *Bifidobacterium breve* | not detected | 48 | ▼ | 0.50 | 0.70 | 0.30 | 0.70 | +0.02 | |
| *Faecalibacterium prausnitzii* | 9.840% | 81 | ▼ | 0.70 | 0.80 | 0.20 | 0.80 | -0.31 | |
| *Streptococcus salivarius* | 0.186% | 82 | ▲ | 0.50 | 0.70 | 0.30 | 0.60 | +0.30 | |
| **Kostic 2015 DIABIMMUNE seroconverters** · species · study diabimmune | | | | | | | | | |
| *Blautia obeum* | 0.604% | 90 | ▲ | 0.50 | 0.50 | 0.40 | 0.50 | +0.37 | |
| *Ruminococcus gnavus* | not detected | 27 | ▲ | 0.60 | 0.60 | 0.30 | 0.60 | -0.21 | |
| *Streptococcus mitis* | not detected | 39 | ▲ | 0.40 | 0.50 | 0.30 | 0.50 | -0.09 | |
| *Roseburia intestinalis* | 0.350% | 68 | ▼ | 0.60 | 0.60 | 0.30 | 0.70 | -0.16 | |
| *Eubacterium rectale* | 1.082% | 52 | ▼ | 0.60 | 0.60 | 0.20 | 0.70 | -0.02 | |
| **Diversity before diagnosis** · diversity · study diabimmune | | | | | | | | | |
| shannon | not detected | — | ▼ | 0.70 | 0.60 | 0.10 | 0.70 | — | |
| **SCFA and LPS capacity** · genes · genomic capacity · study functional | | | | | | | | | |
| butyrate | 32.1/100 | 82 | ▼ | 0.80 | 0.70 | 0.20 | 0.80 | -0.26 | |
| **LPS immunogenicity carriers** · carriers · carrier proxy · study carriers | | | | | | | | | |
| hexa lps | 0.000% | 7 | ▼ | 0.60 | 0.60 | 0.50 | 0.60 | +0.84 | |
| butyrate | 45.230% | 87 | ▼ | 0.60 | 0.70 | 0.20 | 0.80 | -0.43 | |

FINE PRINT

**Confidence grade insufficient.** Why not higher: 67% of planned feature weight was measurable (below 75%); no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; +1 more, all in results.json.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

**Measured discrimination: not available.** No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Type 1 diabetes is diagnosed by glucose, autoantibodies and C-peptide. This profile does not diagnose or predict it in anyone, and its evidence base is infants, not adults.

The hexa-LPS direction here is reversed relative to every inflammatory disease profile. That is the finding, not an error, and it means a "healthy-looking" Bacteroides-dominant community scores toward T1D risk in this one profile.

Sources: Vatanen T, et al. The human gut microbiome in early-onset type 1 diabetes from the TEDDY study. Nature 2018;562:589-594; Kostic AD, et al. The dynamics of the human infant gut microbiome in development and in progression toward type 1 diabetes. Cell Host Microbe 2015;17:260-273; Vatanen T, et al. Variation in microbiome LPS immunogenicity contributes to autoimmunity in humans. Cell 2016;165:842-853.

‹ at a glance

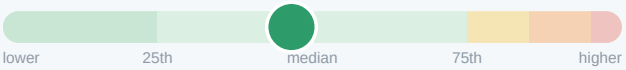
# Alopecia areata — gut pattern

TYPICAL

Pattern version **1.0.0** (alopecia_areata) · maturity **P1** (one small human cohort, amplicon-only, or inaccessible individual data) · experimental · pediatric_shotgun_derived

Alopecia areata is an autoimmune condition and biologically distinct from androgenetic alopecia; their evidence is never merged. The only shotgun cohort is pediatric — 41 children with AA against 41 unaffected siblings — and everything else is 16S, which binds here as a labelled transport rather than being discarded or quietly treated as equivalent. The published cohorts disagree with each other on direction. Bacteroides is case-enriched in the severe Moreno and Bain branches and lower in Lee. Dorea is higher at genus level in Lu while D. longicatena is control-associated in Moreno. Those conflicts are not pooled into a consensus: each study is its own module with its own vote, and the spread between them is the result worth reading.

**COMBINED RESEMBLANCE**
percentile of the matched reference group




**47th**
likely range 19–78

## What is driving it

**Rangu 2021 pediatric shotgun (41 AA / 41 siblings)**
species · study rangu_2021 · 1/1 features · 20% of combined
24th — BELOW AVERAGE

**Moreno-Arrones 2020 (15 alopecia universalis / 15 control, 16S)**
species · association only · 16S study · study moreno_2020 · 6/6 features · 20% of combined
16th — BELOW AVERAGE

**Lu 2021 (33 AA / 35 control, 16S)**
species · association only · 16S study · study lu_2021 · 4/4 features · 20% of combined
97th — NOTABLY HIGH

**Lee 2024 (19 AA / 20 control, 16S)**
species · association only · 16S study · study lee_2024 · 2/2 features · 20% of combined
39th — TYPICAL

**Nikoloudaki 2024 (24 AA / 18 control, 16S)**
species · association only · 16S study · study nikoloudaki_2024 · 3/3 features · 20% of combined
27th — TYPICAL

**Rangu 2021 KO set — figure encoding**
pathways · genomic capacity · study rangu_2021 · 0/4 features · not measurable — a HUMAnN 3 pathway/KO abundance stage is added
not measured
— — NOT COMPUTABLE

**Rangu 2021 KO set — prose encoding**
pathways · genomic capacity · study rangu_2021 · 0/4 features · not measurable — a HUMAnN 3 pathway/KO abundance stage is added
not measured
— — NOT COMPUTABLE

**Bain 2022 high-SALT subgroup (exploratory)**
species · association only · 16S study · study bain_2022 · 3/3 features · not in combined
78th — ABOVE AVERAGE

**Biotin biosynthesis capacity (mechanism, P0)**
genes · genomic capacity · study biotin_mechanism · 2/2 features · not in combined
32nd — TYPICAL

## How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

| | | |
|---|---|---|
| **Technical reliability** | 1.00 | sequencing depth, host fraction, reference match |
| **Independent evidence coverage** | 1.00 | share of the pattern's independent evidence clusters this sample could measure |
| **Evidence maturity** | P1 | one small human cohort, amplicon-only, or inaccessible individual data |
| **Assay transportability** | labeled transport | whether the studies measured what this sample measured |
| **Population match** | 0.60 | how well the reference group matches you |
| **Specificity** | unknown | does this pattern separate its condition from general disturbance? measured where labelled data exist |

| | | |
|---|---|---|
| **Direction conflict** | material | studies disagreeing on direction for one or more features |

**Competing patterns** scoring nearly as high: ra (95th), ad clinical (88th), celiac (77th). Margin over the next pattern: 8 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

## Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| **Rangu 2021 pediatric shotgun (41 AA / 41 siblings)** · species · study rangu_2021 | | | | | | | | | |
| *Ruminococcus bicirculans* | 0.406% | 76 | ▼ | 1.00 | 0.60 | 0.60 | 0.90 | **-0.24** | Rangu 2021 JID Innov (NCBI Taxonomy 1160721) |
| **Moreno-Arrones 2020 (15 alopecia universalis / 15 control, 16S)** · species · association only · 16S study · study moreno_2020 | | | | | | | | | |
| *Holdemania filiformis* | not detected 0.002% pooled | 30 | ▲ | 0.60 | 0.40 | 0.50 | 0.60 | **-0.17** | |
| *Parabacteroides johnsonii* | not detected | 43 | ▲ | 0.60 | 0.40 | 0.50 | 0.60 | **-0.06** | |
| *Bacteroides eggerthii* | not detected | 36 | ▲ | 0.50 | 0.40 | 0.40 | 0.50 | **-0.12** | |
| *Parabacteroides distasonis* | 1.492% | 78 | ▲ | 0.50 | 0.40 | 0.40 | 0.50 | **+0.24** | |
| *Phascolarctobacterium succinatutens* | 0.395% | 83 | ▼ | 0.60 | 0.40 | 0.50 | 0.60 | **-0.32** | |
| *Dorea longicatena* | 1.935% | 92 | ▼ | 0.50 | 0.40 | 0.40 | 0.40 | **-0.42** | |
| **Lu 2021 (33 AA / 35 control, 16S)** · species · association only · 16S study · study lu_2021 | | | | | | | | | |
| *Blautia (genus)* | 1.213% | 85 | ▲ | 0.50 | 0.40 | 0.30 | 0.40 | **+0.36** | |
| *Anaerostipes hadrus* | 11.119% | 98 | ▲ | 0.50 | 0.40 | 0.40 | 0.50 | **+0.66** | |
| *Collinsella aerofaciens* | 0.885% | 72 | ▲ | 0.50 | 0.40 | 0.40 | 0.50 | **+0.15** | |
| *Dorea (genus)* | 3.186% | 93 | ▲ | 0.40 | 0.30 | 0.30 | 0.30 | **+0.51** | |
| **Lee 2024 (19 AA / 20 control, 16S)** · species · association only · 16S study · study lee_2024 | | | | | | | | | |
| *Bacteroides (genus)* | 9.497% | 45 | ▼ | 0.50 | 0.40 | 0.30 | 0.30 | **+0.05** | |
| *Blautia wexlerae* | 0.014% | 31 | ▲ | 0.40 | 0.30 | 0.30 | 0.40 | **-0.16** | |
| **Nikoloudaki 2024 (24 AA / 18 control, 16S)** · species · association only · 16S study · study nikoloudaki_2024 | | | | | | | | | |
| *Blautia obeum* | 0.604% | 90 | ▲ | 0.40 | 0.30 | 0.30 | 0.40 | **+0.37** | |
| *Holdemania filiformis* | not detected 0.002% pooled | 30 | ▲ | 0.40 | 0.30 | 0.40 | 0.50 | **-0.17** | |
| *Coprococcus comes* | 1.842% | 96 | ▼ | 0.40 | 0.30 | 0.30 | 0.40 | **-0.58** | |
| **Rangu 2021 KO set — figure encoding** · pathways · genomic capacity · study rangu_2021 | | | | | | | | | |
| K06295 | missing | — | ▲ | 0.30 | 1.00 | 1.00 | 1.00 | — | |
| K06297 | missing | — | ▲ | 0.30 | 1.00 | 1.00 | 1.00 | — | |
| K02012 | missing | — | ▲ | 0.30 | 1.00 | 1.00 | 1.00 | — | |
| K07552 | missing | — | ▲ | 0.20 | 1.00 | 1.00 | 1.00 | — | |
| **Rangu 2021 KO set — prose encoding** · pathways · genomic capacity · study rangu_2021 | | | | | | | | | |
| K06295 | missing | — | ▼ | 0.30 | 1.00 | 1.00 | 1.00 | — | |
| K06297 | missing | — | ▼ | 0.30 | 1.00 | 1.00 | 1.00 | — | |
| K02012 | missing | — | ▼ | 0.30 | 1.00 | 1.00 | 1.00 | — | |
| K07552 | missing | — | ▼ | 0.20 | 1.00 | 1.00 | 1.00 | — | |
| **Bain 2022 high-SALT subgroup (exploratory)** · species · association only · 16S study · study bain_2022 | | | | | | | | | |
| *Alistipes (genus)* | 3.458% | 73 | ▲ | 0.30 | 0.20 | 0.30 | 0.30 | **+0.17** | |
| *Barnesiella intestinihominis* | 0.442% | 77 | ▲ | 0.30 | 0.20 | 0.30 | 0.30 | **+0.24** | |
| *Tyzzerella nexilis* | not detected | 45 | ▲ | 0.30 | 0.20 | 0.30 | 0.30 | **-0.05** | |
| **Biotin biosynthesis capacity (mechanism, P0)** · genes · genomic capacity · study biotin_mechanism | | | | | | | | | |
| biotin | 64.1/100 | 54 | ▼ | 0.60 | 0.30 | 0.30 | 0.60 | **-0.02** | Hayashi 2017 Cell Rep 20:1513 |

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| propionate | 26.4/100 | 76 | ▼ | 0.40 | 0.30 | 0.20 | 0.30 | -0.36 | Borde and Astrand 2018 |

**FINE PRINT**

**Confidence grade insufficient.** Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

**Measured discrimination: not available.** No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Alopecia areata is diagnosed by a clinician from the pattern of hair loss, not from stool. No microbiome test predicts it, and no result here should change treatment.

Six of the seven human cohorts here are 16S with fewer than 40 cases each, and their directions conflict. The score is an average of study votes and its spread is wide by construction — read the per-module scores, not just the fused number.

The pediatric shotgun cohort is the only module measured the way this sample was measured. An adult sample is outside its population.

Bacteroides and Dorea appear with opposite directions in different cohorts. Neither direction is endorsed here.

Sources: Rangu S, Lee J-J, Hu W, Bittinger K, Castelo-Soccio L. Understanding the gut microbiota in pediatric patients with alopecia areata and their siblings: a pilot study. JID Innov 2021;1:100051, doi:10.1016/j.xjidi.2021.100051; Moreno-Arrones et al. doi:10.1111/jdv.15885; Lu et al. doi:10.1016/j.jdermsci.2021.04.003; Bain et al. doi:10.1093/cei/uxac088; Lee JH, et al. IJMS 2024;25:4256; Nikoloudaki O, et al. Nutrients 2024;16:858; Hayashi A, et al. Cell Rep 2017;20:1513.

‹ at a glance

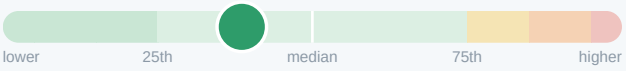
# Atherosclerotic cardiovascular disease — gut pattern

TYPICAL

Pattern version **1.0.0** (acvd) · maturity **P3** (independent human replication with coherent task and assay) · research beta · chinese_cohort_derived

Jie et al. sequenced 218 patients with atherosclerotic cardiovascular disease and 187 controls and found Enterobacteriaceae and oral-origin Streptococcus enriched, with Roseburia, Faecalibacterium and Bacteroides depleted — the same oral-invasion plus butyrate-loss pattern as cirrhosis, which is why the two are mutual challenge sets. The mechanistic layer is better than the taxonomy: trimethylamine production from choline and carnitine by cutC/cntA, converted to TMAO in the liver, is a replicated cardiovascular risk factor with a mouse causal chain. Statins, aspirin and PPIs are near-universal in ACVD cohorts and each shifts the community.

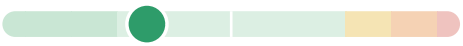
**COMBINED RESEMBLANCE**
percentile of the matched reference group




**39th**
likely range 18–64

## What is driving it

**Jie 2017 Chinese ACVD MGWAS**
species · study jie_2017 · 13/13 features · 50% of combined



32nd — TYPICAL

**Karlsson 2012 symptomatic atherosclerosis (Sweden)**
species · study karlsson_2012 · 3/3 features · 50% of combined



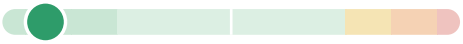
46th — TYPICAL

**TMA and metabolite gene capacity**
genes · genomic capacity · study functional · 3/3 features · not in combined



28th — TYPICAL

**Oral-origin and hexa-LPS carriers**
carriers · carrier proxy · study carriers · 3/3 features · not in combined



9th — LOW

## How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

| | | |
|---|---|---|
| **Technical reliability** | 1.00 | sequencing depth, host fraction, reference match |
| **Independent evidence coverage** | 1.00 | share of the pattern's independent evidence clusters this sample could measure |
| **Evidence maturity** | P3 | independent human replication with coherent task and assay |
| **Assay transportability** | native | whether the studies measured what this sample measured |
| **Population match** | 0.60 | how well the reference group matches you |
| **Specificity** | unknown | does this pattern separate its condition from general disturbance? measured where labelled data exist |
| **Direction conflict** | low | studies disagreeing on direction for one or more features |

**Competing patterns** scoring nearly as high: ra (95th), ad clinical (88th), celiac (77th). Margin over the next pattern: 8 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

## Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| **Jie 2017 Chinese ACVD MGWAS** · species · study jie_2017 | | | | | | | | | |

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| *Escherichia coli* | not detected | 19 | ▲ | 0.90 | 0.80 | 0.30 | 0.90 | -0.29 | |
| *Klebsiella pneumoniae* | not detected | 40 | ▲ | 0.70 | 0.70 | 0.40 | 0.70 | -0.09 | |
| *Enterobacter cloacae complex* | not detected | 46 | ▲ | 0.50 | 0.60 | 0.40 | 0.60 | -0.03 | |
| *Streptococcus salivarius* | 0.186% | 82 | ▲ | 0.70 | 0.70 | 0.40 | 0.80 | +0.30 | |
| *Streptococcus parasanguinis* | 0.048% | 80 | ▲ | 0.60 | 0.70 | 0.40 | 0.70 | +0.28 | |
| *Streptococcus anginosus group* | not detected | 45 | ▲ | 0.50 | 0.60 | 0.40 | 0.70 | -0.04 | |
| *Lactobacillus salivarius* | not detected | 48 | ▲ | 0.50 | 0.60 | 0.40 | 0.60 | -0.02 | |
| *Ruminococcus gnavus* | not detected | 27 | ▲ | 0.50 | 0.60 | 0.20 | 0.60 | -0.21 | |
| *Roseburia intestinalis* | 0.350% | 68 | ▼ | 0.90 | 0.80 | 0.30 | 0.80 | -0.16 | |
| *Faecalibacterium prausnitzii* | 9.840% | 81 | ▼ | 0.80 | 0.80 | 0.20 | 0.80 | -0.31 | |
| *Bacteroides uniformis* | 1.860% | 50 | ▼ | 0.50 | 0.60 | 0.40 | 0.70 | -0.00 | |
| *Prevotella copri* | not detected | 25 | ▼ | 0.50 | 0.60 | 0.30 | 0.50 | +0.23 | |
| *Alistipes shahii* | 0.012% | 39 | ▼ | 0.50 | 0.60 | 0.40 | 0.70 | +0.10 | |
| **Karlsson 2012 symptomatic atherosclerosis (Sweden)** · species · study karlsson_2012 | | | | | | | | | |
| *Collinsella aerofaciens* | 0.885% | 72 | ▲ | 0.60 | 0.50 | 0.40 | 0.60 | +0.15 | |
| *Roseburia intestinalis* | 0.350% | 68 | ▼ | 0.60 | 0.50 | 0.30 | 0.70 | -0.16 | |
| *Eubacterium rectale* | 1.082% | 52 | ▼ | 0.60 | 0.50 | 0.20 | 0.70 | -0.02 | |
| **TMA and metabolite gene capacity** · genes · genomic capacity · study functional | | | | | | | | | |
| cutc | 3.3/100 | 66 | ▲ | 1.00 | 0.90 | 0.60 | 0.90 | +0.21 | |
| butyrate | 32.1/100 | 82 | ▼ | 0.70 | 0.80 | 0.20 | 0.80 | -0.26 | |
| bsh | 123.0/100 | 96 | ▼ | 0.40 | 0.50 | 0.30 | 0.50 | -0.63 | |
| **Oral-origin and hexa-LPS carriers** · carriers · carrier proxy · study carriers | | | | | | | | | |
| oral origin | 0.234% | 70 | ▲ | 0.70 | 0.70 | 0.40 | 0.80 | +0.16 | |
| hexa lps | 0.000% | 7 | ▲ | 0.70 | 0.70 | 0.30 | 0.70 | -0.84 | |
| butyrate | 45.230% | 87 | ▼ | 0.60 | 0.80 | 0.20 | 0.80 | -0.43 | |

FINE PRINT

**Confidence grade insufficient.** Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

**Measured discrimination: not available.** No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Atherosclerosis is assessed by imaging, lipids and risk calculators. This profile is not a cardiovascular risk score.

The oral-invasion pattern is shared with cirrhosis and with PPI use; the butyrate-loss pattern with everything. Only the TMA lane is specific, and it is a capacity measurement, not a plasma TMAO level.

Sources: Jie Z, et al. The gut microbiome in atherosclerotic cardiovascular disease. Nat Commun 2017;8:845; Karlsson FH, et al. Symptomatic atherosclerosis is associated with an altered gut metagenome. Nat Commun 2012;3:1245; Tang WHW, et al. Intestinal microbial metabolism of phosphatidylcholine and cardiovascular risk. N Engl J Med 2013;368:1575.

‹ at a glance

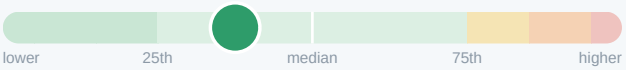
# Preclinical amyloid positivity — gut pattern

TYPICAL

Pattern version **1.0.0** (ad_preclinical_amyloid) · maturity **P2** (usable human shotgun cohort, or multiple compatible human association cohorts) · experimental · us_cognitively_unimpaired_adults

This is not an Alzheimer's disease label. Ferreiro et al. compared 49 amyloid-positive with 115 amyloid-negative participants who were all cognitively unimpaired — the question is whether the gut community differs in people carrying amyloid before any symptoms, which is a different question from whether it differs in people with the disease. Effect sizes at this stage are the smallest in the continuum, which is what you would expect and what the profile's low weights encode.

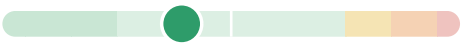
**COMBINED RESEMBLANCE**
percentile of the matched reference group




**38th**
likely range 4–94

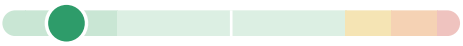
## What is driving it

**Ferreiro 2023 preclinical amyloid cohort**
species · study ferreiro_2023 · 4/4 features · 100% of combined



39th TYPICAL

**Amyloid-relevant metabolite capacity**
genes · genomic capacity · study amyloid_mechanism · 1/1 features · not in combined



14th LOW

## How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

| | | |
|---|---|---|
| **Technical reliability** | 1.00 | sequencing depth, host fraction, reference match |
| **Independent evidence coverage** | 1.00 | share of the pattern's independent evidence clusters this sample could measure |
| **Evidence maturity** | P2 | usable human shotgun cohort, or multiple compatible human association cohorts |
| **Assay transportability** | native | whether the studies measured what this sample measured |
| **Population match** | 0.60 | how well the reference group matches you |
| **Specificity** | unknown | does this pattern separate its condition from general disturbance? measured where labelled data exist |
| **Direction conflict** | none | studies disagreeing on direction for one or more features |

**Competing patterns** scoring nearly as high: ra (95th), celiac (77th), t1d (64th). Margin over the next pattern: 8 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

## Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| **Ferreiro 2023 preclinical amyloid cohort** · species · study ferreiro_2023 | | | | | | | | | |
| *Dorea formicigenerans* | 1.252% | 96 | ▲ | 0.60 | 0.40 | 0.40 | 0.60 | +0.54 | Ferreiro PMC10680783 |
| *Faecalibacterium prausnitzii* | 9.840% | 81 | ▼ | 0.60 | 0.40 | 0.20 | 0.60 | -0.31 | Ferreiro PMC10680783 |
| *Eubacterium eligens* | 10.449% | 97 | ▼ | 0.50 | 0.40 | 0.30 | 0.50 | -0.48 | Ferreiro PMC10680783 |
| *Bacteroides uniformis* | 1.860% | 50 | ▲ | 0.40 | 0.30 | 0.30 | 0.50 | +0.00 | Ferreiro PMC10680783 |
| **Amyloid-relevant metabolite capacity** · genes · genomic capacity · study amyloid_mechanism | | | | | | | | | |
| ipa | 8.9/100 | 86 | ▼ | 0.70 | 0.50 | 0.40 | 0.70 | -0.61 | Pappolla 2021 Neurobiol Dis 156:105403 |

FINE PRINT

**Confidence grade moderate.** Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

**Measured discrimination: not available.** No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Amyloid positivity is established by PET imaging or cerebrospinal fluid assay. Many amyloid-positive people never develop dementia. This profile is not a substitute for either measurement and not a prediction of either outcome.

A separate profile from clinical Alzheimer's disease and from mild cognitive impairment by design. The three are different prediction tasks and their scores must not be averaged or read as one continuum reading.

Sources: Ferreiro AL, et al. Gut microbiome composition may be an indicator of preclinical Alzheimer's disease. PMC10680783; data PRJNA798058.

‹ at a glance

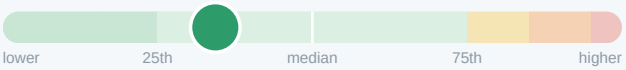
# IBS mixed type — gut pattern

TYPICAL

Pattern version **1.0.0** (ibs_m) · maturity **P1** (one small human cohort, amplicon-only, or inaccessible individual data) · experimental · mixed_adult_cohorts

Mixed-type IBS has no subtype-specific microbial marker of its own: the methanogen direction that separates IBS-C from IBS-D is undefined here, so it is left out entirely rather than scored in either direction. What remains is the shared IBS depletion pattern. This profile exists so the library is complete and so the subtype is not silently absorbed into the parent; it should score close to the parent and rarely above it.

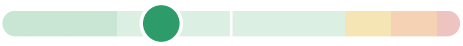
**COMBINED RESEMBLANCE**
percentile of the matched reference group




**34th**
likely range 5–83

## What is driving it

**Shared IBS depletion pattern**
species · association only · 16S study · study pittayanon_2019 · 5/5 features · 100% of combined



35th TYPICAL

**Butyrate capacity**
genes · genomic capacity · study mechanism · 1/1 features · not in combined



18th BELOW AVERAGE

## How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

| | | |
|---|---|---|
| **Technical reliability** | 1.00 | sequencing depth, host fraction, reference match |
| **Independent evidence coverage** | 1.00 | share of the pattern's independent evidence clusters this sample could measure |
| **Evidence maturity** | P1 | one small human cohort, amplicon-only, or inaccessible individual data |
| **Assay transportability** | labeled transport | whether the studies measured what this sample measured |
| **Population match** | 0.60 | how well the reference group matches you |
| **Specificity** | unknown | does this pattern separate its condition from general disturbance? measured where labelled data exist |
| **Direction conflict** | none | studies disagreeing on direction for one or more features |

**Competing patterns** scoring nearly as high: ra (95th), ad clinical (88th), celiac (77th). Margin over the next pattern: 8 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

## Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| **Shared IBS depletion pattern** · species · association only · 16S study · study pittayanon_2019 | | | | | | | | | |
| *Faecalibacterium prausnitzii* | 9.840% | 81 | ▼ | 0.70 | 0.60 | 0.20 | 0.70 | **-0.31** | |
| *Bifidobacterium adolescentis* | not detected | 25 | ▼ | 0.70 | 0.60 | 0.30 | 0.70 | **+0.23** | |
| *Bifidobacterium longum* | not detected | 17 | ▼ | 0.50 | 0.60 | 0.30 | 0.70 | **+0.32** | |
| *Ruminococcus gnavus* | not detected | 27 | ▲ | 0.60 | 0.50 | 0.20 | 0.60 | **-0.21** | |
| *Escherichia coli* | not detected | 19 | ▲ | 0.50 | 0.50 | 0.20 | 0.60 | **-0.29** | |
| **Butyrate capacity** · genes · genomic capacity · study mechanism | | | | | | | | | |
| butyrate | 32.1/100 | 82 | ▼ | 0.80 | 0.70 | 0.20 | 0.80 | **-0.26** | |

FINE PRINT

**Confidence grade insufficient.** Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

**Measured discrimination: not available.** No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

No microbial marker distinguishes mixed-type IBS from the other subtypes or from general dysbiosis. This profile is the shared IBS pattern with the subtype-specific methanogen direction deliberately omitted.

Sources: Pittayanon R, et al. Gut microbiota in patients with irritable bowel syndrome — a systematic review. Gastroenterology 2019;157:97-108.

‹ at a glance

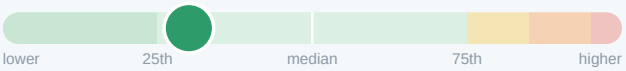
# IBS with constipation — gut pattern

TYPICAL

Pattern version **1.0.0** (ibs_c) · maturity **P1** (one small human cohort, amplicon-only, or inaccessible individual data) · experimental · mixed_adult_cohorts

Constipation-predominant IBS has one microbial feature the other subtypes lack: methanogen enrichment. Methane slows intestinal transit in animal models and breath methane tracks constipation in humans, so Methanobrevibacter smithii and the mcrA gene panel are the subtype-specific evidence here. The rest is inherited from the parent IBS pattern.

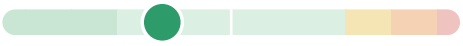
**COMBINED RESEMBLANCE**
percentile of the matched reference group




**30th**
likely range 8–65

## What is driving it

**Methanogen enrichment in constipation**
species · study methane_constipation · 1/1 features · 50% of combined



35th — TYPICAL

**Shared IBS depletion pattern**
species · association only · 16S study · study pittayanon_2019 · 2/2 features · 50% of combined



42nd — TYPICAL

**Methanogenesis gene capacity**
genes · genomic capacity · study methane_capacity · 1/1 features · not in combined



0 — NOTABLY LOW

## How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

| | | |
|---|---|---|
| **Technical reliability** | 1.00 | sequencing depth, host fraction, reference match |
| **Independent evidence coverage** | 1.00 | share of the pattern's independent evidence clusters this sample could measure |
| **Evidence maturity** | P1 | one small human cohort, amplicon-only, or inaccessible individual data |
| **Assay transportability** | labeled transport | whether the studies measured what this sample measured |
| **Population match** | 0.60 | how well the reference group matches you |
| **Specificity** | unknown | does this pattern separate its condition from general disturbance? measured where labelled data exist |
| **Direction conflict** | none | studies disagreeing on direction for one or more features |

**Competing patterns** scoring nearly as high: ra (95th), ad clinical (88th), celiac (77th). Margin over the next pattern: 8 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

## Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| **Methanogen enrichment in constipation** · species · study methane_constipation | | | | | | | | | |
| *Methanobrevibacter smithii* | not detected | 35 | ▲ | 1.00 | 0.70 | 0.60 | 0.80 | **-0.13** | Kim 2012 Dig Dis Sci; Ghoshal 2016 |
| **Shared IBS depletion pattern** · species · association only · 16S study · study pittayanon_2019 | | | | | | | | | |
| *Faecalibacterium prausnitzii* | 9.840% | 81 | ▼ | 0.60 | 0.60 | 0.20 | 0.70 | **-0.31** | |
| *Bifidobacterium adolescentis* | not detected | 25 | ▼ | 0.60 | 0.60 | 0.30 | 0.70 | **+0.23** | |
| **Methanogenesis gene capacity** · genes · genomic capacity · study methane_capacity | | | | | | | | | |
| methane | 0.0/100 | 31 | ▲ | 1.00 | 0.70 | 0.60 | 0.80 | **-0.17** | |

FINE PRINT

**Confidence grade insufficient.** Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

**Measured discrimination: not available.** No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

IBS subtypes are defined by stool form on the Bristol scale, not by stool DNA. This profile describes a community pattern associated with constipation, which many people without IBS also carry.

Methanogen carriage is common and partly heritable. High methanogen abundance is a normal state for a large minority of healthy adults.

Sources: Pimentel M, et al. Methane, a gas produced by enteric bacteria, slows intestinal transit and augments small intestinal contractile activity. Am J Physiol 2006;290:G1089; Kim G, et al. Methanobrevibacter smithii is the predominant methanogen in patients with constipation-predominant IBS and methane on breath. Dig Dis Sci 2012;57:3213.

‹ at a glance

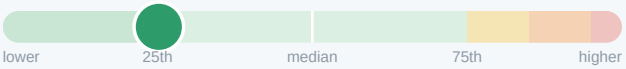
# Androgenetic alopecia — gut pattern (sex-stratified)

TYPICAL

Pattern version **1.0.0** (androgenetic_alopecia) · maturity **P1** (one small human cohort, amplicon-only, or inaccessible individual data) · experimental · mixed_adult_cohorts

Androgenetic alopecia is hormonal and genetic, biologically distinct from autoimmune alopecia areata, and its evidence is never merged with that profile. It is stratified on sex because the two sexes have different evidence: an all-male 16S cohort from China, a mixed-sex Korean cohort, and a female-only Italian cohort reporting sex-specific direction differences. Female pattern hair loss has different pathophysiology and cannot borrow male evidence. If sex is not supplied, the score is reported without stratification and this is recorded in the confounder ledger.

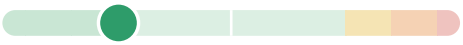
**COMBINED RESEMBLANCE**
percentile of the matched reference group




**25th**
likely range 6–68

## What is driving it

**Seo 2024 Korean cohort (mixed sex)**
species · association only · 16S study · study seo_2024 · 3/3 features · 100% of combined



25th — TYPICAL

**Huang 2024 all-male 16S (China)**
species · association only · 16S study · study huang_2024 · 0/0 features · not in combined




— NOT COMPUTABLE

**Savo Sardaro 2025 female-only (Italy)**
species · association only · 16S study · study savo_sardaro_2025 · 0/0 features · not in combined




— NOT COMPUTABLE

**Biotin biosynthesis capacity (mechanism, P0)**
genes · genomic capacity · study biotin_lane · 1/1 features · not in combined



46th — TYPICAL

## How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

| | | |
|---|---|---|
| **Technical reliability** | 1.00 | sequencing depth, host fraction, reference match |
| **Independent evidence coverage** | 1.00 | share of the pattern's independent evidence clusters this sample could measure |
| **Evidence maturity** | P1 | one small human cohort, amplicon-only, or inaccessible individual data |
| **Assay transportability** | labeled transport | whether the studies measured what this sample measured |
| **Population match** | 0.60 | how well the reference group matches you |
| **Specificity** | unknown | does this pattern separate its condition from general disturbance? measured where labelled data exist |
| **Direction conflict** | none | studies disagreeing on direction for one or more features |

**Competing patterns** scoring nearly as high: ra (95th), ad clinical (88th), celiac (77th). Margin over the next pattern: 8 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

## Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| **Seo 2024 Korean cohort (mixed sex)** · species · association only · 16S study · study seo_2024 | | | | | | | | | |
| *Faecalibacterium prausnitzii* | 9.840% | 81 | ▼ | 0.50 | 0.40 | 0.20 | 0.50 | **-0.31** | |
| *Ruminococcus gnavus* | not detected | 27 | ▲ | 0.50 | 0.40 | 0.30 | 0.50 | **-0.21** | |
| *Akkermansia muciniphila* | not detected<br>0.024% pooled | 24 | ▼ | 0.40 | 0.30 | 0.30 | 0.40 | **+0.23** | |
| **Huang 2024 all-male 16S (China)** · species · association only · 16S study · study huang_2024 | | | | | | | | | |
| **Savo Sardaro 2025 female-only (Italy)** · species · association only · 16S study · study savo_sardaro_2025 | | | | | | | | | |
| **Biotin biosynthesis capacity (mechanism, P0)** · genes · genomic capacity · study biotin_lane | | | | | | | | | |
| biotin | 64.1/100 | 54 | ▼ | 0.50 | 0.30 | 0.20 | 0.50 | **-0.02** | Hayashi 2017 Cell Rep 20:1513 |

FINE PRINT

**Confidence grade insufficient.** Why not higher: 33% of planned feature weight was measurable (below 75%); no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; +1 more, all in results.json.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

**Measured discrimination: not available.** No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Androgenetic alopecia is diagnosed from the pattern of hair loss and family history. Nothing here predicts it, and nothing here bears on treatment.

All three human cohorts are 16S with genus-level findings transported onto shotgun species. The transport penalty is shown in the confidence vector.

Without a supplied sex, the sex-specific modules are withheld and the score rests on the Korean mixed-sex module alone.

Sources: Huang J, et al. Skin Res Technol 2024;30:e13578; Seo Y, et al. Biomedicines 2024;12:2225; Savo Sardaro ML, et al. Int J Mol Sci 2025;26:5459; Hayashi A, et al. Cell Rep 2017;20:1513.

‹ at a glance

# Liver cirrhosis — gut pattern

BELOW AVERAGE

Pattern version **1.0.0** (cirrhosis) · maturity **P3** (independent human replication with coherent task and assay) · research beta · chinese_cohort_derived

One of the largest effect sizes in the gut microbiome literature. Qin et al. found that cirrhosis drives oral bacteria into the gut — Veillonella, Streptococcus, Haemophilus — because reduced bile acid output and gastric acid suppression remove the barriers that normally keep them out. The Bajaj cirrhosis dysbiosis ratio captures the same thing: oral and Enterobacteriaceae taxa up, Lachnospiraceae and Ruminococcaceae down. Cross-validated AUC for cirrhosis versus healthy exceeds 0.9 in the discovery cohort. Proton pump inhibitors reproduce the oral-invasion part of this pattern on their own, and most cirrhotic patients are on them.

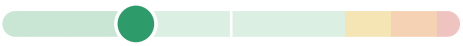
**COMBINED RESEMBLANCE**
percentile of the matched reference group




**22nd**
likely range 3–69

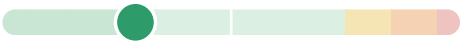
## What is driving it

**Bajaj cirrhosis dysbiosis ratio (16S, US)**
species · association only · 16S study · study bajaj_2014 · 4/4 features · 67% of combined



29th — TYPICAL

**Qin 2014 Chinese cirrhosis MGWAS**
species · study qin_2014 · 12/12 features · 33% of combined



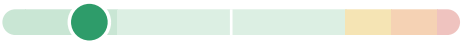
29th — TYPICAL

**Community ecology**
diversity · study qin_2014 · 0/1 features · not in combined



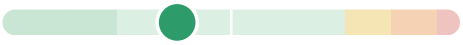

— NOT COMPUTABLE

**Metabolite gene capacity**
genes · genomic capacity · study functional · 4/4 features · not in combined



19th — BELOW AVERAGE

**Oral-origin carrier cross-check**
carriers · carrier proxy · study carriers · 2/2 features · not in combined



38th — TYPICAL

## How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

| | | |
|---|---|---|
| **Technical reliability** | 1.00 | sequencing depth, host fraction, reference match |
| **Independent evidence coverage** | 1.00 | share of the pattern's independent evidence clusters this sample could measure |
| **Evidence maturity** | P3 | independent human replication with coherent task and assay |
| **Assay transportability** | labeled transport | whether the studies measured what this sample measured |
| **Population match** | 0.60 | how well the reference group matches you |
| **Specificity** | unknown | does this pattern separate its condition from general disturbance? measured where labelled data exist |
| **Direction conflict** | none | studies disagreeing on direction for one or more features |

**Competing patterns** scoring nearly as high: ra (95th), ad clinical (88th), celiac (77th). Margin over the next pattern: 8 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

## Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| **Bajaj cirrhosis dysbiosis ratio (16S, US)** · species · association only · 16S study · study bajaj_2014 | | | | | | | | | |
| *Veillonella parvula* | not detected | 31 | ▲ | 0.70 | 0.70 | 0.50 | 0.80 | -0.17 | |
| *Escherichia coli* | not detected | 19 | ▲ | 0.60 | 0.70 | 0.30 | 0.70 | -0.29 | |
| *Roseburia intestinalis* | 0.350% | 68 | ▼ | 0.60 | 0.70 | 0.30 | 0.70 | -0.16 | |
| *Ruminococcus bromii* | not detected 3.40% pooled | 19 | ▼ | 0.60 | 0.70 | 0.40 | 0.70 | +0.29 | |
| **Qin 2014 Chinese cirrhosis MGWAS** · species · study qin_2014 | | | | | | | | | |
| *Veillonella parvula* | not detected | 31 | ▲ | 1.00 | 0.90 | 0.50 | 0.90 | -0.17 | |
| *Veillonella atypica* | not detected | 35 | ▲ | 0.80 | 0.80 | 0.50 | 0.80 | -0.13 | |
| *Streptococcus salivarius* | 0.186% | 82 | ▲ | 0.90 | 0.90 | 0.40 | 0.90 | +0.30 | |
| *Streptococcus parasanguinis* | 0.048% | 80 | ▲ | 0.80 | 0.80 | 0.40 | 0.80 | +0.28 | |
| *Streptococcus anginosus group* | not detected | 45 | ▲ | 0.60 | 0.70 | 0.40 | 0.70 | -0.04 | |
| *Haemophilus parainfluenzae* | not detected | 29 | ▲ | 0.80 | 0.80 | 0.40 | 0.80 | -0.18 | |
| *Lactobacillus salivarius* | not detected | 48 | ▲ | 0.50 | 0.60 | 0.40 | 0.60 | -0.02 | |
| *Eubacterium rectale* | 1.082% | 52 | ▼ | 0.70 | 0.80 | 0.20 | 0.80 | -0.02 | |
| *Faecalibacterium prausnitzii* | 9.840% | 81 | ▼ | 0.70 | 0.80 | 0.20 | 0.80 | -0.31 | |
| *Alistipes putredinis* | 1.648% | 63 | ▼ | 0.60 | 0.70 | 0.40 | 0.70 | -0.11 | |
| *Bacteroides uniformis* | 1.860% | 50 | ▼ | 0.50 | 0.70 | 0.40 | 0.70 | -0.00 | |
| *Coprococcus comes* | 1.842% | 96 | ▼ | 0.50 | 0.70 | 0.30 | 0.70 | -0.58 | |
| **Community ecology** · diversity · study qin_2014 | | | | | | | | | |
| shannon | not detected | — | ▼ | 0.60 | 0.80 | 0.10 | 0.80 | — | |
| **Metabolite gene capacity** · genes · genomic capacity · study functional | | | | | | | | | |
| bai | 3.5/100 | 72 | ▼ | 0.90 | 0.80 | 0.50 | 0.80 | -0.30 | |
| urease | 16.3/100 | 70 | ▲ | 0.70 | 0.70 | 0.40 | 0.70 | +0.25 | |
| butyrate | 32.1/100 | 82 | ▼ | 0.70 | 0.80 | 0.20 | 0.80 | -0.26 | |
| bsh | 123.0/100 | 96 | ▼ | 0.50 | 0.60 | 0.30 | 0.60 | -0.63 | |
| **Oral-origin carrier cross-check** · carriers · carrier proxy · study carriers | | | | | | | | | |
| oral origin | 0.234% | 70 | ▲ | 1.00 | 0.80 | 0.50 | 0.90 | +0.16 | |
| butyrate | 45.230% | 87 | ▼ | 0.60 | 0.80 | 0.20 | 0.80 | -0.43 | |

FINE PRINT

**Confidence grade insufficient.** Why not higher: 67% of planned feature weight was measurable (below 75%); no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; +1 more, all in results.json.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

**Measured discrimination: not available.** No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Cirrhosis is diagnosed by imaging, elastography, liver tests and, where needed, biopsy. This profile does not diagnose it.

Proton pump inhibitor use reproduces the oral-invasion pattern that carries this profile. A PPI user without liver disease can score high here. The confounder ledger shows whether PPI use was recorded.

Sources: Qin N, et al. Alterations of the human gut microbiome in liver cirrhosis. Nature 2014;513:59-64; Bajaj JS, et al. Altered profile of human gut microbiome is associated with cirrhosis and its complications. J Hepatol 2014;60:940-947; Kakiyama G, et al. Modulation of the fecal bile acid profile by gut microbiota in cirrhosis. J Hepatol 2013;58:949.

‹ at a glance

# Systemic lupus erythematosus — shared inflammatory gut pattern (not lupus-specific)

BELOW AVERAGE

Pattern version **0.1.0** (sle) · maturity **P1** (one small human cohort, amplicon-only, or inaccessible individual data) · research only · mixed_geography_treated_cohorts

Systemic lupus erythematosus is an autoimmune disease in which antibodies attack the body's own tissues, most seriously the kidneys, and it is diagnosed from clinical features together with blood autoantibody tests. A twenty-five-study audit of the gut-microbiome literature was carried out for this profile, and the honest result of that audit is that it does not support a lupus-specific stool score. What is scored here is deliberately narrower and deliberately named for what it is: the shared inflammatory pattern that lupus cohorts have in common with several other conditions. Two species-level directions have cross-geography, cross-platform support and are the only taxonomic features scored. Streptococcus anginosus and the closely related S. intermedius are raised — found by a Japanese metagenome-wide association study of 47 patients against 203 controls and corroborated in a Chinese amplicon cohort. Faecalibacterium is depleted, reported at genus level in a Chinese amplicon cohort and in a ninety-two-study systematic review. Both are pan-rheumatic: the same review that found them concluded there is little evidence of disease specificity for gut microbes in rheumatic disease, and a head-to-head comparison could not separate lupus from primary Sjögren's syndrome from stool at all, while oral samples could. An index built on these two measures inflammation, not lupus, and it is labelled that way here. Ruminococcus gnavus is the finding this profile is best known for, and it is deliberately not scored. Its abundance is reported in the body of the report with two caveats attached. The first is that the association is strain-level: lupus-derived and healthy-derived strains colonise mice equally well, and only the lupus strains cause the gut-permeability and autoimmunity phenotype, with only nephritis-derived strains expressing the lipoglycan that drives it. This pipeline measures the species accurately and the species is not the question. The second is that it is a transient bloom rather than a level, fluctuating greatly within the same patient between samples, so a single stool specimen is one draw from a volatile process: a low value means very little and a high one cannot be told apart from ordinary carriage. R. gnavus elevation is also already scored as a feature in fifteen other profiles in this library, which is the plainest possible demonstration of how non-specific it is. Two further limits are stated rather than worked around. The activity signal failed where it had the most power: in the largest single cohort, 157 Korean patients against 50 controls, there was no significant microbial difference by lupus-nephritis status and none by disease activity. And medication moves these biomarkers as much as the disease does. Proton pump inhibitors erased the diversity deficit entirely and were the only significant explanatory variable for composition among patients; hydroxychloroquine reduces Enterobacteriaceae; and in the one large treatment-naive cohort the lupus-enriched species were reduced after treatment. Nearly every real patient is taking at least one of these drugs, so a treated person's reading is partly a readout of their prescription. A high score is a research observation about community composition. It does not diagnose lupus, and a low score in someone with lupus does not argue against the diagnosis.

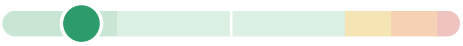
**COMBINED RESEMBLANCE**
percentile of the matched reference group




**16th**
likely range 4–42

## What is driving it

**Cross-platform species directions (inflammation-associated, not lupus-specific)**
species · association only · 16S study · study sle_cross_platform · 2/2 features · 100% of combined



17th BELOW AVERAGE

**Ruminococcus gnavus lipoglycan strain typing (not computable here)**
strain · study silverman_azzouz_strain · 0/1 features · not measurable — a validated reference panel for this specific strain-level target. Strain resolution itself is available and reported separately: the marker lane fingerprints the dominant population of every organism with enough coverage. What is missing here is not the engine but a published, discriminating marker set for this named target




— NOT COMPUTABLE

## How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

| | | |
|---|---|---|
| **Technical reliability** | 1.00 | sequencing depth, host fraction, reference match |

| | | |
|---|---|---|
| **Independent evidence coverage** | 1.00 | share of the pattern's independent evidence clusters this sample could measure |
| **Evidence maturity** | P1 | one small human cohort, amplicon-only, or inaccessible individual data |
| **Assay transportability** | labeled transport | whether the studies measured what this sample measured |
| **Population match** | 0.60 | how well the reference group matches you |
| **Specificity** | unknown | does this pattern separate its condition from general disturbance? measured where labelled data exist |
| **Direction conflict** | material | studies disagreeing on direction for one or more features |

**Competing patterns** scoring nearly as high: ra (95th), ad clinical (88th), celiac (77th). Margin over the next pattern: 8 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

## Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| **Cross-platform species directions (inflammation-associated, not lupus-specific)** · species · association only · 16S study · study sle_cross_platform | | | | | | | | | |
| *Streptococcus anginosus, measured as Streptococcus anginosus group* | not detected | 45 | ▲ | 0.60 | 0.40 | 0.20 | 0.60 | **-0.04** | |
| *Faecalibacterium (genus)* | 10.148% | 81 | ▼ | 0.50 | 0.30 | 0.20 | 0.70 | **-0.31** | |
| **Ruminococcus gnavus lipoglycan strain typing (not computable here)** · strain · study silverman_azzouz_strain | | | | | | | | | |
| Ruminococcus gnavus lipoglycan strains | missing | — | ▲ | 1.00 | 0.60 | 0.70 | 0.80 | — | |

## FINE PRINT

**Confidence grade insufficient.** Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

**Measured discrimination: not available.** No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Lupus is diagnosed from clinical features together with blood autoantibody tests, assessed by a clinician. Nothing in stool DNA makes or excludes that diagnosis, and this score is not a probability of having lupus. If you are worried about lupus, the tests that answer the question are blood tests.

This profile is named for a shared inflammatory pattern rather than for lupus, and that name is the finding. The systematic review that supplies both scored directions concluded there is little evidence of disease specificity for gut microbes in rheumatic disease. A head-to-head study could not tell lupus from primary Sjögren's syndrome using stool, though it could using oral samples. A high score here is equally consistent with several other inflammatory conditions, which is why the competing patterns are scored beside it.

Ruminococcus gnavus is the organism most associated with lupus in the press and it is deliberately excluded from this score. The association is strain-level: only lupus-derived strains cause the disease phenotype in mice, only nephritis-derived strains make the lipoglycan that drives it, and healthy-derived strains colonise just as well without causing it. This test measures the species, not the strain. R. gnavus is also a transient bloom rather than a stable level, and its elevation is already a scored feature in fifteen other profiles in this library.

Disease activity cannot be read from this score. In the largest cohort audited — 157 patients against 50 controls — there was no significant microbial difference between patients with and without lupus nephritis, and none between high and low disease activity. The activity associations reported elsewhere come from cohorts of sixteen to sixty-one people with largely uncorrected statistics.

In anyone already taking medication, this reading is partly a readout of the prescription. Proton pump inhibitors erased the diversity deficit altogether and explained community composition better than any clinical variable. Nothing in this profile is a reason to start, stop or change any lupus treatment, and no microbiome score authorises substituting a supplement for immunosuppression.

No microbiome-directed intervention in lupus has human evidence strong enough to recommend. Lactobacillus deserves specific mention because it cuts both ways: a Lactobacillus mixture improved kidney function in one mouse model of lupus, and L. reuteri worsened interferon-driven disease in another while escaping the gut. Sex hormones, strain and disease driver set which way it goes, and a stool report determines none of them. Live-microbe products also carry a documented bacteraemia risk in immunosuppressed people, which many lupus patients are.

Sources: Tomofuji Y, Maeda Y, Oguro-Igashira E, et al. Metagenome-wide association study revealed disease-specific landscape of the gut microbiome of systemic lupus erythematosus in Japanese. Ann Rheum Dis 2021;80:1575-1583. Wang Y, Wei J, Zhang W, et al. Gut dysbiosis in rheumatic diseases: a systematic review and meta-analysis of 92 observational studies. eBioMedicine 2022. Xiang S, Qu Y, Qian S, et al. Association between systemic lupus erythematosus and disruption of gut microbiota: a meta-analysis. Lupus Sci Med 2022. Azzouz DF, Omarbekova A, Heguy A, et al. Lupus nephritis is linked to disease-activity associated expansions and immunity to a gut commensal. Ann Rheum Dis 2019;78:947-956. Azzouz DF, Chen Z, Izmirly PM, et al. Longitudinal gut microbiome analyses and blooms of pathogenic strains during lupus disease flares. Ann Rheum Dis 2023;82:1315. Amarnani A, et al. A pathogenic gut lipoglycan drives systemic thromboinflammation in lupus nephritis. Ann Rheum Dis 2026, doi:10.1016/j.ard.2026.03.002. Silverman GJ, et al. Front Immunol 2022. Park Y, Yang J, Son H, et al. Distinct gut microbiome profiles in Korean systemic lupus erythematosus patients. J Transl Med 2025. Li XB, Chu XJ, Cao NW, et al. Proton pump inhibitors induce changes in the gut microbiome of systemic lupus erythematosus patients. BMC Microbiol 2022. Chen BD, Jia XM, Xu JY, et al. An autoimmunogenic and proinflammatory profile defined by the gut microbiota of patients with untreated systemic lupus erythematosus. Arthritis Rheumatol 2021. van der Meulen TA, Harmsen HJM, Vila AV, et al. Shared gut, but distinct oral microbiota composition in primary Sjögren's syndrome and systemic lupus erythematosus. J Autoimmun 2019. Ling Z, et al. Front Immunol 2023. Hevia A, et al. mBio 2014. Full twenty-five-study audit with per-finding directions, cohorts and verification notes: specs/research/sle_evidence.yaml.

‹ at a glance

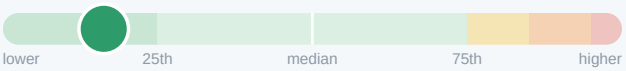
# Colorectal adenoma — gut pattern

BELOW AVERAGE

Pattern version **1.0.0** (adenoma) · maturity **P3** (independent human replication with coherent task and assay) · research beta · mixed_international_cohorts

Adenoma is the precursor lesion for most colorectal cancer, and the interesting finding is a negative one: the strong CRC stool signature — Fusobacterium nucleatum, Parvimonas micra, Peptostreptococcus stomatis, Gemella — is weak or absent in adenoma. Both large cross-cohort meta-analyses (Wirbel 2019, Thomas 2019) found adenoma barely separable from controls by stool metagenomics, with AUCs around 0.5-0.6. That makes adenoma the right challenge set for the CRC profile: a CRC score that is also high in adenoma is reading general dysbiosis rather than cancer. The modest adenoma-associated directions that do replicate are here, with low weights, because that is what the evidence supports.

**COMBINED RESEMBLANCE**
percentile of the matched reference group




**16th**
likely range 2–46

## What is driving it

| | | | |
|---|---|---|---|
| **Feng 2015 Austrian adenoma / carcinoma shotgun**<br>species · study feng_2015 · 5/5 features · 50% of combined | | 73rd | TYPICAL |
| **Thomas 2019 cross-cohort adenoma (7 cohorts)**<br>species · study thomas_2019 · 3/3 features · 50% of combined | | 4th | NOTABLY LOW |
| **Carcinogenic metabolite capacity**<br>genes · genomic capacity · study functional · 3/3 features · not in combined | | 69th | TYPICAL |

## How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

| | | |
|---|---|---|
| **Technical reliability** | 1.00 | sequencing depth, host fraction, reference match |
| **Independent evidence coverage** | 1.00 | share of the pattern's independent evidence clusters this sample could measure |
| **Evidence maturity** | P3 | independent human replication with coherent task and assay |
| **Assay transportability** | native | whether the studies measured what this sample measured |
| **Population match** | 0.60 | how well the reference group matches you |
| **Specificity** | unknown | does this pattern separate its condition from general disturbance? measured where labelled data exist |
| **Direction conflict** | low | studies disagreeing on direction for one or more features |

**Competing patterns** scoring nearly as high: ra (95th), ad clinical (88th), celiac (77th). Margin over the next pattern: 8 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

## Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| **Feng 2015 Austrian adenoma / carcinoma shotgun** · species · study feng_2015 | | | | | | | | | |
| *Bacteroides dorei* | 4.169% | 85 | ▲ | 0.50 | 0.60 | 0.40 | 0.60 | +0.35 | |
| *Bacteroides massiliensis* | 0.117% | 78 | ▲ | 0.50 | 0.60 | 0.40 | 0.60 | +0.26 | |
| *Bilophila wadsworthia* | not detected | 20 | ▲ | 0.50 | 0.60 | 0.40 | 0.60 | -0.28 | |
| *Alistipes putredinis* | 1.648% | 63 | ▲ | 0.40 | 0.60 | 0.40 | 0.50 | +0.11 | |
| *Fusobacterium nucleatum* | not detected | 49 | ▲ | 0.40 | 0.60 | 0.40 | 0.50 | -0.01 | |

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| **Thomas 2019 cross-cohort adenoma (7 cohorts)** · species · study thomas_2019 | | | | | | | | | |
| *Ruminococcus gnavus* | not detected | 27 | ▲ | 0.40 | 0.60 | 0.20 | 0.50 | -0.21 | |
| *Eubacterium eligens* | 10.449% | 97 | ▼ | 0.40 | 0.60 | 0.30 | 0.60 | -0.48 | |
| *Roseburia intestinalis* | 0.350% | 68 | ▼ | 0.40 | 0.60 | 0.30 | 0.60 | -0.16 | |
| **Carcinogenic metabolite capacity** · genes · genomic capacity · study functional | | | | | | | | | |
| bai | 3.5/100 | 72 | ▲ | 0.60 | 0.60 | 0.40 | 0.60 | +0.30 | |
| h2s | 96.5/100 | 80 | ▲ | 0.50 | 0.50 | 0.40 | 0.50 | +0.35 | |
| butyrate | 32.1/100 | 82 | ▼ | 0.60 | 0.60 | 0.20 | 0.70 | -0.26 | |

FINE PRINT

**Confidence grade moderate.** Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

**Measured discrimination: not available.** No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Adenomas are found and removed at colonoscopy. No stool DNA test replaces screening colonoscopy or FIT.

Adenoma is nearly inseparable from control by stool metagenomics. A low score here means nothing; a high score means dysbiosis. The profile is in the library chiefly as the challenge set for colorectal_cancer.

Sources: Feng Q, et al. Gut microbiome development along the colorectal adenoma- carcinoma sequence. Nat Commun 2015;6:6528; Thomas AM, et al. Metagenomic analysis of colorectal cancer datasets identifies cross-cohort microbial diagnostic signatures and a link with choline degradation. Nat Med 2019;25:667-678; Wirbel J, et al. Nat Med 2019;25:679-689.

‹ at a glance

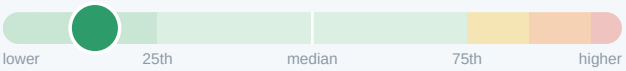
# Colorectal cancer-associated community pattern

LOW

Pattern version **1.0.0** (crc) · maturity **P3** (independent human replication with coherent task and assay) · validation control · mixed_international_cohorts

Colorectal cancer is the pattern on which this pipeline's scoring engine is validated. It is the only microbiome disease association with genuinely cross-cohort-validated performance, and it is the only one for which public cohorts carry disease labels — which makes it the single pattern on which the whole engine can be checked against known answers. If the engine cannot recover a known-good signal here, no number it produces for any other pattern means anything either. Two independent meta-analyses of eight shotgun cohorts across three continents agree on a core of oral-origin anaerobes enriched in CRC stool. The virulence-gene lane (fadA, bft, clbB) is the mechanistic layer and is reported as presence-grade only, for reasons recorded in the panel.

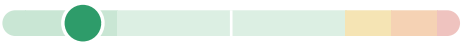
**COMBINED RESEMBLANCE**
percentile of the matched reference group




**15th**
likely range 0–37

## What is driving it

**Wirbel 2019 meta-analysis (5 cohorts, 768 metagenomes)**
species · study wirbel_2019 · 8/8 features · 50% of combined



18th — BELOW AVERAGE

**Thomas 2019 meta-analysis (7 cohorts) + choline**
species · study thomas_2019 · 7/7 features · 50% of combined



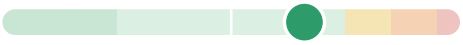
43rd — TYPICAL

**CRC virulence genes — presence grade**
genes · association only · study virulence · 1/1 features · not in combined



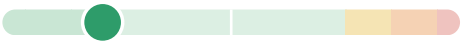
38th — TYPICAL

**Carcinogenic metabolite capacity**
genes · genomic capacity · study functional · 4/4 features · not in combined



66th — TYPICAL

**Oral-origin carriers**
carriers · carrier proxy · study carriers · 2/2 features · not in combined



22nd — BELOW AVERAGE

## How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

| | | |
|---|---|---|
| **Technical reliability** | 1.00 | sequencing depth, host fraction, reference match |
| **Independent evidence coverage** | 1.00 | share of the pattern's independent evidence clusters this sample could measure |
| **Evidence maturity** | P3 | independent human replication with coherent task and assay |
| **Assay transportability** | native | whether the studies measured what this sample measured |
| **Population match** | 0.60 | how well the reference group matches you |
| **Specificity** | unknown | does this pattern separate its condition from general disturbance? measured where labelled data exist |
| **Direction conflict** | none | studies disagreeing on direction for one or more features |

**Competing patterns** scoring nearly as high: ra (95th), ad clinical (88th), celiac (77th). Margin over the next pattern: 8 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

## Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| **Wirbel 2019 meta-analysis (5 cohorts, 768 metagenomes)** · species · study wirbel_2019 | | | | | | | | | |
| *Fusobacterium nucleatum* | not detected | 49 | ▲ | 1.00 | 1.00 | 0.80 | 1.00 | -0.01 | |
| *Parvimonas micra* | not detected | 48 | ▲ | 1.00 | 1.00 | 0.80 | 1.00 | -0.02 | |
| *Peptostreptococcus stomatis* | not detected | 49 | ▲ | 1.00 | 1.00 | 0.80 | 1.00 | -0.01 | |
| *Gemella morbillorum* | not detected | 49 | ▲ | 0.90 | 0.90 | 0.70 | 1.00 | -0.01 | |
| *Solobacterium moorei* | not detected | 48 | ▲ | 0.90 | 0.90 | 0.70 | 1.00 | -0.01 | |
| *Porphyromonas asaccharolytica* | not detected | 49 | ▲ | 0.80 | 0.90 | 0.70 | 0.90 | -0.01 | |
| *Clostridium symbiosum* | not detected | 36 | ▲ | 0.70 | 0.80 | 0.50 | 0.80 | -0.12 | |
| *Bacteroides fragilis* | not detected | 24 | ▲ | 0.60 | 0.80 | 0.40 | 0.80 | -0.24 | |
| **Thomas 2019 meta-analysis (7 cohorts) + choline** · species · study thomas_2019 | | | | | | | | | |
| *Fusobacterium nucleatum* | not detected | 49 | ▲ | 1.00 | 1.00 | 0.80 | 1.00 | -0.01 | |
| *Parvimonas micra* | not detected | 48 | ▲ | 1.00 | 1.00 | 0.80 | 1.00 | -0.02 | |
| *Peptostreptococcus stomatis* | not detected | 49 | ▲ | 0.90 | 1.00 | 0.80 | 1.00 | -0.01 | |
| *Gemella morbillorum* | not detected | 49 | ▲ | 0.80 | 0.90 | 0.70 | 0.90 | -0.01 | |
| *Porphyromonas asaccharolytica* | not detected | 49 | ▲ | 0.70 | 0.90 | 0.70 | 0.90 | -0.01 | |
| *Prevotella intermedia* | not detected | 50 | ▲ | 0.60 | 0.80 | 0.60 | 0.80 | -0.00 | |
| *Streptococcus anginosus group* | not detected | 45 | ▲ | 0.50 | 0.70 | 0.40 | 0.70 | -0.04 | |
| **CRC virulence genes — presence grade** · genes · association only · study virulence | | | | | | | | | |
| crc virulence | 0.0/100 | 38 | ▲ | 1.00 | 0.40 | 0.80 | 0.50 | -0.10 | |
| **Carcinogenic metabolite capacity** · genes · genomic capacity · study functional | | | | | | | | | |
| bai | 3.5/100 | 72 | ▲ | 0.70 | 0.70 | 0.40 | 0.70 | +0.30 | |
| h2s | 96.5/100 | 80 | ▲ | 0.60 | 0.60 | 0.40 | 0.60 | +0.35 | |
| cutc | 3.3/100 | 66 | ▲ | 0.60 | 0.70 | 0.40 | 0.70 | +0.21 | |
| butyrate | 32.1/100 | 82 | ▼ | 0.60 | 0.70 | 0.20 | 0.70 | -0.26 | |
| **Oral-origin carriers** · carriers · carrier proxy · study carriers | | | | | | | | | |
| oral origin | 0.234% | 70 | ▲ | 0.80 | 0.80 | 0.40 | 0.80 | +0.16 | |
| opportunistic pathogens | 0.000% | 3 | ▲ | 0.50 | 0.60 | 0.30 | 0.60 | -0.96 | |

FINE PRINT

**Validation control.** Included to check that the scoring engine recovers a known signal on labelled data. Not a cancer screen and not a risk estimate; validated clinical tests exist for this condition.

**Confidence grade moderate.** Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

**Measured discrimination: not available.** No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

This is not a cancer screen and not a risk estimate. It exists to test the scoring engine against labelled data. Colorectal cancer screening is done with faecal immunochemical testing, multi-target stool DNA testing, or colonoscopy, on a schedule set by a clinician.

The taxonomic features are oral-origin anaerobes. Their enrichment in faecal metagenomes is reproducible in cancer cohorts, but oral-gut translocation also occurs in periodontal disease and with reduced gastric acid, so the pattern is not unique to malignancy.

The virulence lane is presence-grade with unmeasurable sensitivity. Not detected does not mean absent.

Sources: Wirbel J, et al. Meta-analysis of fecal metagenomes reveals global microbial signatures that are specific for colorectal cancer. Nat Med 2019;25:679-689; Thomas AM, et al. Metagenomic analysis of colorectal cancer datasets identifies cross-cohort microbial diagnostic signatures and a link with choline degradation. Nat Med 2019;25:667-678.

‹ at a glance

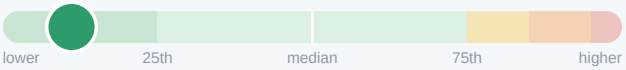
# Chronic kidney disease — gut pattern

LOW

Pattern version **1.0.0** (ckd) · maturity **P2** (usable human shotgun cohort, or multiple compatible human association cohorts) · research beta · mixed_chinese_european_cohorts

Kidney failure changes the gut chemically before it changes it ecologically: urea diffuses into the colon and is hydrolysed to ammonia by bacterial urease, raising luminal pH and selecting for urease-, uricase- and indole- and p-cresol-forming organisms (Wong 2014). Those uraemic toxins — indoxyl sulfate and p-cresyl sulfate — are made by gut bacteria from tryptophan and tyrosine and accumulate as filtration falls. Ren et al. and Wang et al. shotgun cohorts show Enterobacteriaceae and oral-origin taxa enriched with butyrate producers depleted, and the pattern tracks eGFR stage. Phosphate binders, potassium restriction and the low-protein renal diet each reshape the gut.

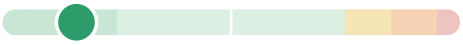
**COMBINED RESEMBLANCE**
percentile of the matched reference group




**11th**
likely range 1–47

## What is driving it

**Ren 2020 Chinese CKD shotgun**
species · study ren_2020 · 7/7 features · 50% of combined



16th — BELOW AVERAGE

**Wang 2020 end-stage renal disease shotgun + metabolomics**
species · study wang_2020 · 7/7 features · 50% of combined



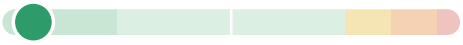
20th — BELOW AVERAGE

**Uraemic toxin and metabolite capacity**
genes · genomic capacity · study functional · 4/4 features · not in combined



34th — TYPICAL

**Oral-origin and hexa-LPS carriers**
carriers · carrier proxy · study carriers · 3/3 features · not in combined



7th — LOW

## How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

| | | |
|---|---|---|
| **Technical reliability** | 1.00 | sequencing depth, host fraction, reference match |
| **Independent evidence coverage** | 1.00 | share of the pattern's independent evidence clusters this sample could measure |
| **Evidence maturity** | P2 | usable human shotgun cohort, or multiple compatible human association cohorts |
| **Assay transportability** | native | whether the studies measured what this sample measured |
| **Population match** | 0.60 | how well the reference group matches you |
| **Specificity** | unknown | does this pattern separate its condition from general disturbance? measured where labelled data exist |
| **Direction conflict** | low | studies disagreeing on direction for one or more features |

**Competing patterns** scoring nearly as high: ra (95th), ad clinical (88th), celiac (77th). Margin over the next pattern: 8 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

## Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| **Ren 2020 Chinese CKD shotgun** · species · study ren_2020 | | | | | | | | | |
| *Escherichia coli* | not detected | 19 | ▲ | 0.80 | 0.70 | 0.30 | 0.80 | **-0.29** | |

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| *Klebsiella pneumoniae* | not detected | 40 | ▲ | 0.70 | 0.60 | 0.40 | 0.70 | -0.09 | |
| *Enterococcus faecalis* | not detected | 47 | ▲ | 0.60 | 0.60 | 0.40 | 0.60 | -0.02 | |
| *Streptococcus salivarius* | 0.186% | 82 | ▲ | 0.50 | 0.60 | 0.30 | 0.60 | +0.30 | |
| *Faecalibacterium prausnitzii* | 9.840% | 81 | ▼ | 0.80 | 0.70 | 0.20 | 0.80 | -0.31 | |
| *Roseburia intestinalis* | 0.350% | 68 | ▼ | 0.70 | 0.70 | 0.30 | 0.70 | -0.16 | |
| *Eubacterium rectale* | 1.082% | 52 | ▼ | 0.60 | 0.70 | 0.20 | 0.70 | -0.02 | |
| **Wang 2020 end-stage renal disease shotgun + metabolomics** · species · study wang_2020 | | | | | | | | | |
| *Eggerthella lenta* | 0.060% | 82 | ▲ | 0.70 | 0.70 | 0.50 | 0.70 | +0.30 | |
| *Flavonifractor plautii* | not detected<br>0.046% pooled | 12 | ▲ | 0.60 | 0.60 | 0.50 | 0.60 | -0.39 | |
| *Alistipes shahii* | 0.012% | 39 | ▲ | 0.40 | 0.60 | 0.40 | 0.50 | -0.10 | |
| *Fusobacterium nucleatum* | not detected | 49 | ▲ | 0.40 | 0.60 | 0.30 | 0.50 | -0.01 | |
| *Prevotella copri* | not detected | 25 | ▼ | 0.50 | 0.60 | 0.30 | 0.60 | +0.23 | |
| *Roseburia hominis* | 1.234% | 93 | ▼ | 0.60 | 0.70 | 0.30 | 0.70 | -0.41 | |
| *Faecalibacterium prausnitzii* | 9.840% | 81 | ▼ | 0.70 | 0.70 | 0.20 | 0.80 | -0.31 | |
| **Uraemic toxin and metabolite capacity** · genes · genomic capacity · study functional | | | | | | | | | |
| urease | 16.3/100 | 70 | ▲ | 1.00 | 0.70 | 0.50 | 0.80 | +0.25 | |
| butyrate | 32.1/100 | 82 | ▼ | 0.70 | 0.70 | 0.20 | 0.70 | -0.26 | |
| cutc | 3.3/100 | 66 | ▲ | 0.50 | 0.60 | 0.40 | 0.60 | +0.21 | |
| ipa | 8.9/100 | 86 | ▼ | 0.40 | 0.50 | 0.30 | 0.50 | -0.61 | |
| **Oral-origin and hexa-LPS carriers** · carriers · carrier proxy · study carriers | | | | | | | | | |
| hexa lps | 0.000% | 7 | ▲ | 0.60 | 0.60 | 0.30 | 0.60 | -0.84 | |
| oral origin | 0.234% | 70 | ▲ | 0.50 | 0.60 | 0.30 | 0.60 | +0.16 | |
| butyrate | 45.230% | 87 | ▼ | 0.60 | 0.70 | 0.20 | 0.70 | -0.43 | |

FINE PRINT

**Confidence grade insufficient.** Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

**Measured discrimination: not available.** No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Kidney function is measured by eGFR and urine albumin. Both are cheap blood and urine tests. This profile adds nothing to them.

The renal diet produces most of this pattern on its own.

Sources: Wang X, et al. Aberrant gut microbiota alters host metabolome and impacts renal failure in humans and rodents. Gut 2020;69:2131-2142; Ren Z, et al. Alterations of the human gut microbiome in chronic kidney disease. Front Cell Infect Microbiol 2020;10:5; Wong J, et al. Expansion of urease- and uricase-containing, indole- and p-cresol-forming and contraction of short-chain fatty acid-producing intestinal microbiota in ESRD. Am J Nephrol 2014;39:230.

‹ at a glance

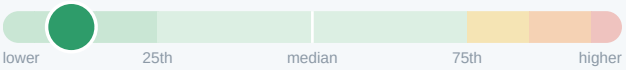
# Mild cognitive impairment — gut pattern

LOW

Pattern version **1.0.0** (ad_mci) · maturity **P2** (usable human shotgun cohort, or multiple compatible human association cohorts) · experimental · east_asian_cohort_derived

Mild cognitive impairment is a different prediction task from clinical Alzheimer's disease and is scored separately rather than folded into it. In the staged Chinese cohort no species reached q < 0.05 at this stage, so the strict panel that carries clinical AD does not exist here. What does exist is an independent Taiwanese cohort of 119 MCI against 320 normal controls, which is what this profile scores, plus the source cohort's exploratory q < 0.25 set reported as a sensitivity result and never called validated. The MCI stage is where the microbiome signal is smallest across the Alzheimer's continuum, so this score is read as supporting context for the clinical-stage pattern rather than on its own.

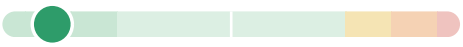
**COMBINED RESEMBLANCE**
percentile of the matched reference group




**11th**
likely range 6–22

## What is driving it

**Fan 2025 Taiwan MCI cohort**
species · study fan_2025 · 5/5 features · 100% of combined



11th — LOW

**Jia 2025 exploratory set (q < 0.25) — sensitivity only**
species · association only · study metaad_2025 · 2/2 features · not in combined



90th — HIGH

**Jia 2025 KEGG orthologue panel, MCI stage**
pathways · genomic capacity · study metaad_2025 · 0/3 features · not measurable — a HUMAnN 3 pathway/KO abundance stage is added




— NOT COMPUTABLE

## How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

| | | |
|---|---|---|
| **Technical reliability** | 1.00 | sequencing depth, host fraction, reference match |
| **Independent evidence coverage** | 1.00 | share of the pattern's independent evidence clusters this sample could measure |
| **Evidence maturity** | P2 | usable human shotgun cohort, or multiple compatible human association cohorts |
| **Assay transportability** | native | whether the studies measured what this sample measured |
| **Population match** | 0.60 | how well the reference group matches you |
| **Specificity** | unknown | does this pattern separate its condition from general disturbance? measured where labelled data exist |
| **Direction conflict** | none | studies disagreeing on direction for one or more features |

**Competing patterns** scoring nearly as high: ra (95th), celiac (77th), t1d (64th). Margin over the next pattern: 8 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

## Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| **Fan 2025 Taiwan MCI cohort** · species · study fan_2025 | | | | | | | | | |
| *Bacteroides fragilis* | not detected | 24 | ▲ | 0.70 | 0.60 | 0.30 | 0.70 | **-0.24** | Fan 2025 PMC12123878 |
| *Ruminococcus gnavus* | not detected | 27 | ▲ | 0.70 | 0.60 | 0.30 | 0.70 | **-0.21** | Fan 2025 PMC12123878 |

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| *Faecalibacterium prausnitzii* | 9.840% | 81 | ▼ | 0.80 | 0.60 | 0.20 | 0.80 | **-0.31** | Fan 2025 PMC12123878 |
| *Roseburia intestinalis* | 0.350% | 68 | ▼ | 0.60 | 0.50 | 0.20 | 0.70 | **-0.16** | Fan 2025 PMC12123878 |
| *Eubacterium rectale* | 1.082% | 52 | ▼ | 0.60 | 0.50 | 0.20 | 0.70 | **-0.02** | Fan 2025 PMC12123878 |
| **Jia 2025 exploratory set (q < 0.25) — sensitivity only** · species · association only · study metaad_2025 | | | | | | | | | |
| *Odoribacter splanchnicus* | 0.472% | 81 | ▲ | 0.40 | 0.30 | 0.40 | 0.50 | **+0.29** | |
| *Alistipes finegoldii* | 0.743% | 87 | ▲ | 0.40 | 0.30 | 0.40 | 0.50 | **+0.37** | |
| **Jia 2025 KEGG orthologue panel, MCI stage** · pathways · genomic capacity · study metaad_2025 | | | | | | | | | |
| K00857 | missing | — | ▲ | 1.00 | 1.00 | 1.00 | 1.00 | — | |
| K01031 | missing | — | ▲ | 1.00 | 1.00 | 1.00 | 1.00 | — | |
| K01754 | missing | — | ▼ | 1.00 | 1.00 | 1.00 | 1.00 | — | |

FINE PRINT

**Confidence grade insufficient.** Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

**Measured discrimination: not available.** No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Not a diagnosis or a risk estimate. Mild cognitive impairment is a clinical determination made with cognitive testing.

At strict multiple-testing correction, no species distinguished this stage in the largest staged cohort. This profile scores because an independent cohort supports a direction, not because the signal is strong.

Three of the five scored features are butyrate-producer depletion or R. gnavus enrichment, which appear in most gut disease cohorts. Read this score beside the general dysbiosis index and the shape analysis.

Sources: Fan et al. 2025, PMC12123878, data PRJNA1258384; Jia L, Ke Y, Zhao S, et al. Metagenomic analysis characterizes stage-specific gut microbiota in Alzheimer's disease. Mol Psychiatry 2025;30:3951-3962, doi:10.1038/s41380-025-02973-7.

‹ at a glance

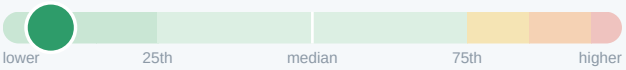
# Obesity — ecological pattern

LOW

Pattern version **1.0.0** (obesity) · maturity **P3** (independent human replication with coherent task and assay) · research beta · mixed_european_cohorts

In the obesity microbiome literature, species-level directions do not replicate across cohorts. The Firmicutes-to-Bacteroidetes ratio, once the headline finding, failed in meta-analyses (Sze and Schloss 2016; Walters 2014). What does replicate is ecological: low gene richness and low diversity are associated with adiposity, insulin resistance and dyslipidaemia (Le Chatelier 2013, Cotillard 2013), and the low-richness group gains more weight over time. So this profile scores ecology, not taxa. Akkermansia muciniphila is the one species with consistent direction and interventional support and is kept at moderate weight.

**COMBINED RESEMBLANCE**
percentile of the matched reference group




**8th**
likely range 8–8

## What is driving it

| | | | |
|---|---|---|---|
| **Le Chatelier 2013 MetaHIT gene richness**<br>diversity · study metahit_2013 · 1/2 features · 33% of combined | 7th | LOW |
| **Cotillard 2013 MicroObes dietary intervention**<br>diversity · study microobes_2013 · 1/1 features · 33% of combined | 7th | LOW |
| **Akkermansia muciniphila (replicated species direction)**<br>species · study dao_2016 · 1/1 features · 33% of combined | >99th | NOTABLY HIGH |
| **Firmicutes/Bacteroidetes ratio — sensitivity only**<br>species · association only · 16S study · study ley_2006 · 2/2 features · not in combined | 88th | HIGH |
| **Metabolite gene capacity**<br>genes · genomic capacity · study functional · 2/2 features · not in combined | 22nd | BELOW AVERAGE |
| **Mucin degraders**<br>carriers · carrier proxy · study carriers · 1/1 features · not in combined | 67th | TYPICAL |

## The Prevotella copri complex, clade by clade

No member of the Segatella (Prevotella) copri complex was detected. All 13 clades were assessed, including the 9 found in humans, so this is a measured absence across the whole complex rather than an unresolved species-level reading.

No clade of this complex has an established association with any disease, including rheumatoid arthritis. Manghi et al. tested all 13 clades against 1,635 disease cases and 1,854 controls across 22 studies and 11 conditions, a set that included a rheumatoid arthritis cohort, and found none of the complex's species associated with any health condition. Earlier rheumatoid-arthritis reports named the species as a whole, not a clade, and those reports have replicated poorly: in an anti-CCP-positive at-risk cohort three Prevotellaceae strains were enriched and five depleted in the people who progressed. So this section reports what is present and makes no disease claim from it.

Tett 2019 Cell Host Microbe doi:10.1016/j.chom.2019.08.018; Manghi 2023 Cell Host Microbe doi:10.1016/j.chom.2023.09.013; Hitch 2022 Syst Appl Microbiol doi:10.1016/j.syapm.2022.126354.

## How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

| | | |
|---|---|---|
| **Technical reliability** | 1.00 | sequencing depth, host fraction, reference match |
| **Independent evidence coverage** | 0.85 | share of the pattern's independent evidence clusters this sample could measure |
| **Evidence maturity** | P3 | independent human replication with coherent task and assay |
| **Assay transportability** | native | whether the studies measured what this sample measured |
| **Population match** | 0.60 | how well the reference group matches you |
| **Specificity** | unknown | does this pattern separate its condition from general disturbance? measured where labelled data exist |
| **Direction conflict** | low | studies disagreeing on direction for one or more features |

**Competing patterns** scoring nearly as high: ra (95th), ad clinical (88th), celiac (77th). Margin over the next pattern: 8 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

## Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| **Le Chatelier 2013 MetaHIT gene richness** · diversity · study metahit_2013 | | | | | | | | | |
| richness | 123.00 | 93 | ▼ | 1.00 | 0.90 | 0.20 | 0.90 | -0.55 | |
| shannon | not detected | — | ▼ | 0.80 | 0.90 | 0.20 | 0.90 | — | |
| **Cotillard 2013 MicroObes dietary intervention** · diversity · study microobes_2013 | | | | | | | | | |
| richness | 123.00 | 93 | ▼ | 0.80 | 0.80 | 0.20 | 0.80 | -0.55 | |
| **Akkermansia muciniphila (replicated species direction)** · species · study dao_2016 | | | | | | | | | |
| *Akkermansia muciniphila* | not detected<br>0.024% pooled | 24 | ▼ | 1.00 | 0.80 | 0.40 | 0.80 | +0.23 | |
| **Firmicutes/Bacteroidetes ratio — sensitivity only** · species · association only · 16S study · study ley_2006 | | | | | | | | | |
| *Bacteroides vulgatus* | not detected | 3 | ▼ | 0.30 | 0.20 | 0.10 | 0.20 | +1.00 | |
| *Bacteroides uniformis* | 1.860% | 50 | ▼ | 0.30 | 0.20 | 0.10 | 0.20 | -0.00 | |
| **Metabolite gene capacity** · genes · genomic capacity · study functional | | | | | | | | | |
| bcaa | 180.5/100 | 34 | ▲ | 0.70 | 0.70 | 0.40 | 0.70 | -0.15 | |
| butyrate | 32.1/100 | 82 | ▼ | 0.60 | 0.70 | 0.20 | 0.70 | -0.26 | |
| **Mucin degraders** · carriers · carrier proxy · study carriers | | | | | | | | | |
| mucin degraders | 1.464% | 33 | ▼ | 0.50 | 0.60 | 0.30 | 0.60 | +0.14 | |

FINE PRINT

**Confidence grade insufficient.** Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

**Measured discrimination: not available.** No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Adiposity is measured with a scale and a tape measure. This profile describes an ecological state associated with it, not a cause of it.

Low richness is the least specific finding in the microbiome. It marks antibiotics, illness, low-fibre diet and age as readily as adiposity.

The Firmicutes/Bacteroidetes ratio is included only to show that it does not work.

Sources: Le Chatelier E, et al. Richness of human gut microbiome correlates with metabolic markers. Nature 2013;500:541-546; Cotillard A, et al. Dietary intervention impact on gut microbial gene richness. Nature 2013;500:585-588; Sze MA, Schloss PD. Looking for a signal in the noise: revisiting obesity and the microbiome. mBio 2016;7:e01018; Dao MC, et al. Akkermansia muciniphila and improved metabolic health during a dietary intervention in obesity. Gut 2016;65:426.

‹ at a glance

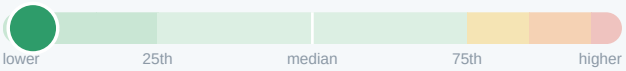
# IBS with diarrhoea — gut pattern

NOTABLY LOW

Pattern version **1.0.0** (ibs_d) · maturity **P1** (one small human cohort, amplicon-only, or inaccessible individual data) · experimental · mixed_adult_cohorts

Diarrhoea-predominant IBS overlaps with bile acid malabsorption in roughly a quarter to a third of patients, and the community pattern follows: higher secondary bile acid conversion, lower methanogens, and a larger dysbiotic subgroup than IBS-C. The parent IBS pattern is inherited; the bile acid and methanogen directions are the subtype-specific part.

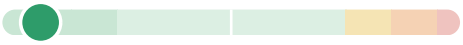
**COMBINED RESEMBLANCE**
percentile of the matched reference group




**5th**
likely range 0–26

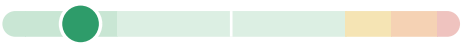
## What is driving it

**Jeffery 2020 IBS-D / bile acid malabsorption shotgun**
species · study jeffery_2020 · 5/5 features · 50% of combined



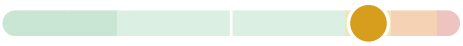
8th — LOW

**Mars 2020 longitudinal IBS-D shotgun**
species · study mars_2020 · 2/2 features · 50% of combined



17th — BELOW AVERAGE

**Secondary bile acid conversion capacity**
genes · genomic capacity · study bile_acid_capacity · 4/4 features · not in combined



80th — ABOVE AVERAGE

## How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

| | | |
|---|---|---|
| ● **Technical reliability** | 1.00 | sequencing depth, host fraction, reference match |
| ● **Independent evidence coverage** | 1.00 | share of the pattern's independent evidence clusters this sample could measure |
| ● **Evidence maturity** | P1 | one small human cohort, amplicon-only, or inaccessible individual data |
| ● **Assay transportability** | native | whether the studies measured what this sample measured |
| ● **Population match** | 0.60 | how well the reference group matches you |
| ● **Specificity** | unknown | does this pattern separate its condition from general disturbance? measured where labelled data exist |
| ● **Direction conflict** | none | studies disagreeing on direction for one or more features |

**Competing patterns** scoring nearly as high: ra (95th), ad clinical (88th), celiac (77th). Margin over the next pattern: 8 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

## Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| **Jeffery 2020 IBS-D / bile acid malabsorption shotgun** · species · study jeffery_2020 | | | | | | | | | |
| *Methanobrevibacter smithii* | not detected | 35 | ▼ | 0.80 | 0.70 | 0.50 | 0.80 | +0.13 | |
| *Bacteroides vulgatus* | not detected | 3 | ▲ | 0.60 | 0.60 | 0.30 | 0.60 | -1.00 | |
| *Ruminococcus gnavus* | not detected | 27 | ▲ | 0.60 | 0.60 | 0.20 | 0.60 | -0.21 | |
| *Faecalibacterium prausnitzii* | 9.840% | 81 | ▼ | 0.60 | 0.60 | 0.20 | 0.70 | -0.31 | |
| *Eubacterium rectale* | 1.082% | 52 | ▼ | 0.50 | 0.60 | 0.20 | 0.70 | -0.02 | |
| **Mars 2020 longitudinal IBS-D shotgun** · species · study mars_2020 | | | | | | | | | |
| *Bacteroides fragilis* | not detected | 24 | ▲ | 0.50 | 0.60 | 0.30 | 0.60 | -0.24 | |
| *Roseburia intestinalis* | 0.350% | 68 | ▼ | 0.50 | 0.60 | 0.30 | 0.70 | -0.16 | |

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| **Secondary bile acid conversion capacity** · genes · genomic capacity · study bile_acid_capacity | | | | | | | | | |
| bai | 3.5/100 | 72 | ▲ | 1.00 | 0.70 | 0.40 | 0.70 | **+0.30** | |
| bsh | 123.0/100 | 96 | ▲ | 0.50 | 0.60 | 0.20 | 0.50 | **+0.63** | |
| tryptamine | 0.1/100 | 44 | ▲ | 0.50 | 0.50 | 0.40 | 0.50 | **-0.01** | |
| methane | 0.0/100 | 31 | ▼ | 0.60 | 0.60 | 0.50 | 0.70 | **+0.17** | |

FINE PRINT

**Confidence grade insufficient.** Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

**Measured discrimination: not available.** No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

IBS subtypes are defined by stool form, not stool DNA. Bile acid malabsorption is diagnosed with SeHCAT retention or serum C4, not with a gene count.

Recent gastroenteritis, antibiotics, and any cause of diarrhoea reproduce this pattern. It is not specific to IBS-D.

Sources: Jeffery IB, et al. Gastroenterology 2020;158:1016-1028; Mars RAT, et al. Cell 2020;182:1460-1473; Bhattarai Y, et al. Cell Host Microbe 2018;23:775.

‹ at a glance

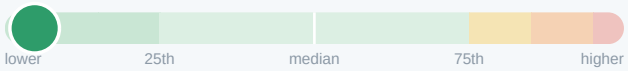
# Long COVID chronic-state community similarity

NOTABLY LOW

Pattern version **0.2.0** (longcovid) · maturity **P1** (one small human cohort, amplicon-only, or inaccessible individual data) · research only · mixed_hong_kong_us_cohorts

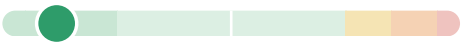
This scores resemblance to the chronic-state pattern: stool collected at least three months after infection with ongoing symptoms. Hong Kong and US cohorts agree on the core direction — depletion of the butyrate producers Faecalibacterium prausnitzii, Eubacterium rectale and Roseburia, with Ruminococcus gnavus and Bacteroides vulgatus enriched — and on reduced short-chain fatty acid and isoleucine synthesis capacity. Predicting future risk from stool collected in the first two months after infection is a separate task with different features, and those features are not used here. In the largest independent study (799 outpatients), species-level features reached AUROC 0.62 against 0.72 for clinical variables alone, so the pattern is read as a supporting signal alongside the clinical picture rather than in place of it.

**COMBINED RESEMBLANCE**
percentile of the matched reference group

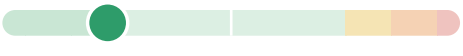



**5th**
likely range 1–23

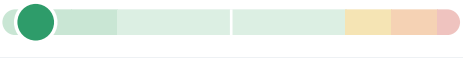
## What is driving it



**Taxonomic** — 12th — LOW
species · study taxonomic · 10/10 features · 50% of combined



**Functional** — 23rd — BELOW AVERAGE
genes · genomic capacity · study functional · 2/2 features · 35% of combined



**Ecological** — 7th — LOW
diversity · study ecological · 1/1 features · 15% of combined




**Phenotype** — — — NOT COMPUTABLE
diversity · study phenotype · 0/2 features · not in combined

## How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

| | | |
|---|---|---|
| **Technical reliability** | 1.00 | sequencing depth, host fraction, reference match |
| **Independent evidence coverage** | 1.00 | share of the pattern's independent evidence clusters this sample could measure |
| **Evidence maturity** | P1 | one small human cohort, amplicon-only, or inaccessible individual data |
| **Assay transportability** | native | whether the studies measured what this sample measured |
| **Population match** | 0.60 | how well the reference group matches you |
| **Specificity** | 0.40 | does this pattern separate its condition from general disturbance? measured where labelled data exist |
| **Direction conflict** | none | studies disagreeing on direction for one or more features |

**Competing patterns** scoring nearly as high: ra (95th), ad clinical (88th), celiac (77th). Margin over the next pattern: 8 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

## Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| **Taxonomic** · species · study taxonomic | | | | | | | | | |
| *Ruminococcus gnavus* | not detected | 27 | ▲ | 1.00 | 0.40 | 0.50 | 0.90 | **-0.21** | Liu 2022 Gut, Liu 2022 Nat Commun, Su 2024 Cell Host Microbe |

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| *Clostridium bolteae* | not detected | 29 | ▲ | 1.00 | 0.40 | 0.50 | 0.90 | **-0.18** | Liu 2022 Gut, Su 2024 Cell Host Microbe |
| *Clostridium innocuum* | not detected | 35 | ▲ | 1.00 | 0.40 | 0.50 | 0.90 | **-0.12** | Liu 2022 Gut, Su 2024 Cell Host Microbe |
| *Faecalibacterium prausnitzii* | 9.840% | 81 | ▼ | 1.00 | 0.50 | 0.30 | 0.90 | **-0.31** | Liu 2022 Gut, Zhang 2022 Gastroenterology |
| *Bifidobacterium adolescentis* | not detected | 25 | ▼ | 1.00 | 0.40 | 0.50 | 0.90 | **+0.23** | Liu 2022 Gut, Liu 2022 Nat Commun |
| *Bifidobacterium pseudocatenulatum* | not detected | 32 | ▼ | 1.00 | 0.40 | 0.50 | 0.90 | **+0.16** | Liu 2022 Gut, Liu 2022 Nat Commun |
| *Erysipelatoclostridium ramosum* | not detected | 39 | ▲ | 0.50 | 0.40 | 0.50 | 0.80 | **-0.10** | Su 2024 Cell Host Microbe |
| *Bacteroides vulgatus* | not detected | 3 | ▲ | 0.50 | 0.40 | 0.30 | 0.70 | **-1.00** | Liu 2022 Nat Commun |
| *Flavonifractor plautii* | not detected 0.046% pooled | 12 | ▲ | 0.50 | 0.40 | 0.50 | 0.80 | **-0.39** | Su 2024 Cell Host Microbe |
| *Gemmiger formicilis* | 0.273% | 88 | ▼ | 0.50 | 0.40 | 0.50 | 0.80 | **-0.39** | Liu 2022 Gut |
| **Functional** · genes · genomic capacity · study functional | | | | | | | | | |
| butyrate | 32.1/100 | 82 | ▼ | 1.00 | 0.60 | 0.40 | 0.90 | **-0.26** | Zhang 2022 Gastroenterology, Liu 2022 Gut |
| indole | 10.0/100 | 33 | ▼ | 0.50 | 0.50 | 0.50 | 0.80 | **+0.12** | Zhang 2022 Gastroenterology |
| **Ecological** · diversity · study ecological | | | | | | | | | |
| richness | 123.00 | 93 | ▼ | 0.50 | 0.60 | 0.20 | 0.60 | **-0.55** | Hamrefors 2024 Sci Rep (lower richness, no species survived correction) |
| **Phenotype** · diversity · study phenotype | | | | | | | | | |
| Actinomyces naeslundii | not detected | — | ▲ | 0.25 | 0.40 | 0.40 | 0.70 | — | Su 2024 Cell Host Microbe (fatigue / neuropsychiatric) |
| Streptococcus anginosus group | not detected | — | ▲ | 0.25 | 0.40 | 0.30 | 0.50 | — | Liu 2022 Gut (respiratory-associated) |

FINE PRINT

**Confidence grade insufficient.** Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

The 50/35/15 module split is an engineering starting point, not an evidence-derived weighting.

**Measured on labelled cohorts** (AUC, 0.5 = chance, 1.0 = perfect): IBD 0.80 (0.76–0.83); T2D 0.59 (0.53–0.64); CRC 0.53 (0.48–0.59). No labelled cohort exists for this condition, so its own discrimination is UNMEASURED (it scores 0.53-0.80 against other conditions).

Chronic-state resemblance only. Features from acute-phase (0-2 month) risk-prediction studies are not included and must not be read into a chronic sample.

Most taxonomic features originate in one institutional recruitment family; their cohort-independence factor is set to reflect roughly one independent vote rather than the number of papers.

E. coli, Klebsiella and antimicrobial-resistance genes track ICU exposure and antibiotics rather than the syndrome, and are deliberately excluded. Flagellin potential is shared with rheumatoid arthritis, ankylosing spondylitis and IBD, and is excluded for the same reason.

The two-year Hainan cohort (Zhang et al. 2026) is n=11 per arm with uninfected rather than recovered controls, and reports no deposited data, so its strain-level calls are neither reusable nor portable across database versions. It is treated as a hypothesis source only and none of its features carry weight here.

Sources: Liu Q, Mak JWY, Su Q, et al. Gut 2022;71:544-552, doi:10.1136/gutjnl-2021-325989 (PRJNA714459). Liu Q, Su Q, Zhang F, et al. Nat Commun 2022;13:6806, doi:10.1038/s41467-022-34535-8 (PRJNA876804). Su Q, et al. Cell Host Microbe 2024;32:651-660 (PRJNA1004982). Zhang F, et al. Gastroenterology 2022;162:548-561 (PRJNA689961). Comba IY, et al. Gut Microbes 2026 (PRJNA1167328).

‹ at a glance

# Multiple sclerosis — gut pattern

NOTABLY LOW

Pattern version **1.0.0** (ms) · maturity **P3** (independent human replication with coherent task and assay) · research beta · mixed_international_cohorts

The iMSMS consortium is the model of how to do this: 576 MS patients each paired with a genetically unrelated household control who shares their diet and environment, shotgun sequenced. That design removes most of the confounding that plagues every other disease here. The replicated MS pattern is modest: Akkermansia muciniphila higher, Ruminococcus torques and Eisenbergiella higher, and Faecalibacterium prausnitzii, Blautia and Eubacterium rectale lower. Effect sizes are small — the household design shows that MS shifts the gut less than diet and cohabitation do. Disease-modifying therapies have their own effects, some larger than the disease's, and are recorded in the ledger.

**COMBINED RESEMBLANCE**
percentile of the matched reference group




**5th**
likely range 1–22

## What is driving it

| | | | |
|---|---|---|---|
| **iMSMS 2022 household-paired shotgun (576 pairs)**<br>species · study imsms_2022 · 6/6 features · 33% of combined | 12th | LOW |
| **Jangi 2016 US cohort (16S + shotgun subset)**<br>species · association only · 16S study · study jangi_2016 · 3/3 features · 33% of combined | 21st | BELOW AVERAGE |
| **Cekanaviciute 2017 UCSF (gnotobiotic-validated)**<br>species · association only · 16S study · study cekanaviciute_2017 · 2/2 features · 33% of combined | 12th | LOW |
| **Metabolite gene capacity**<br>genes · genomic capacity · study functional · 3/3 features · not in combined | 12th | LOW |
| **Mucin degraders and butyrate carriers**<br>carriers · carrier proxy · study carriers · 2/2 features · not in combined | 8th | LOW |

## How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

| | | |
|---|---|---|
| **Technical reliability** | 1.00 | sequencing depth, host fraction, reference match |
| **Independent evidence coverage** | 1.00 | share of the pattern's independent evidence clusters this sample could measure |
| **Evidence maturity** | P3 | independent human replication with coherent task and assay |
| **Assay transportability** | labeled transport | whether the studies measured what this sample measured |
| **Population match** | 0.60 | how well the reference group matches you |
| **Specificity** | unknown | does this pattern separate its condition from general disturbance? measured where labelled data exist |
| **Direction conflict** | none | studies disagreeing on direction for one or more features |

**Competing patterns** scoring nearly as high: ad clinical (88th), alopecia areata (47th), acvd (39th). Margin over the next pattern: 8 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

## Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| **iMSMS 2022 household-paired shotgun (576 pairs)** · species · study imsms_2022 | | | | | | | | | |
| *Akkermansia muciniphila* | not detected 0.024% pooled | 24 | ▲ | 1.00 | 0.90 | 0.40 | 0.90 | **-0.23** | |
| *Ruminococcus torques* | 0.743% | 82 | ▲ | 0.70 | 0.80 | 0.40 | 0.80 | **+0.30** | |
| *Eisenbergiella tayi* | not detected | 37 | ▲ | 0.60 | 0.70 | 0.50 | 0.70 | **-0.11** | |
| *Faecalibacterium prausnitzii* | 9.840% | 81 | ▼ | 0.80 | 0.90 | 0.20 | 0.90 | **-0.31** | |
| *Eubacterium rectale* | 1.082% | 52 | ▼ | 0.60 | 0.80 | 0.20 | 0.80 | **-0.02** | |
| *Blautia obeum* | 0.604% | 90 | ▼ | 0.50 | 0.70 | 0.40 | 0.70 | **-0.37** | |
| **Jangi 2016 US cohort (16S + shotgun subset)** · species · association only · 16S study · study jangi_2016 | | | | | | | | | |
| *Akkermansia muciniphila* | not detected 0.024% pooled | 24 | ▲ | 0.80 | 0.70 | 0.40 | 0.80 | **-0.23** | |
| *Methanobrevibacter smithii* | not detected | 35 | ▲ | 0.60 | 0.70 | 0.40 | 0.70 | **-0.13** | |
| *Butyricimonas virosa* | not detected | 24 | ▼ | 0.50 | 0.60 | 0.50 | 0.60 | **+0.24** | |
| **Cekanaviciute 2017 UCSF (gnotobiotic-validated)** · species · association only · 16S study · study cekanaviciute_2017 | | | | | | | | | |
| *Akkermansia muciniphila* | not detected 0.024% pooled | 24 | ▲ | 0.70 | 0.70 | 0.40 | 0.80 | **-0.23** | |
| *Parabacteroides distasonis* | 1.492% | 78 | ▼ | 0.70 | 0.70 | 0.50 | 0.70 | **-0.24** | |
| **Metabolite gene capacity** · genes · genomic capacity · study functional | | | | | | | | | |
| butyrate | 32.1/100 | 82 | ▼ | 0.80 | 0.80 | 0.20 | 0.80 | **-0.26** | |
| propionate | 26.4/100 | 76 | ▼ | 0.70 | 0.70 | 0.30 | 0.70 | **-0.36** | |
| ipa | 8.9/100 | 86 | ▼ | 0.60 | 0.60 | 0.40 | 0.60 | **-0.61** | |
| **Mucin degraders and butyrate carriers** · carriers · carrier proxy · study carriers | | | | | | | | | |
| mucin degraders | 1.464% | 33 | ▲ | 0.70 | 0.70 | 0.40 | 0.70 | **-0.14** | |
| butyrate | 45.230% | 87 | ▼ | 0.70 | 0.80 | 0.20 | 0.80 | **-0.43** | |

FINE PRINT

**Confidence grade insufficient.** Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

**Measured discrimination: not available.** No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Multiple sclerosis is diagnosed by MRI, clinical course and cerebrospinal fluid. This profile does not diagnose or predict it.

The household-controlled design shows the MS gut effect is small. A modest score here is the expected result in MS, and a high score is more likely dysbiosis of another cause.

Akkermansia is higher in MS and in Parkinson's, and lower in most metabolic disease. Its direction here is disease-specific and the reason this profile has some specificity.

Sources: iMSMS Consortium. Gut microbiome of multiple sclerosis patients and paired household healthy controls reveal associations with disease risk and course. Cell 2022;185:3467-3486; Jangi S, et al. Alterations of the human gut microbiome in multiple sclerosis. Nat Commun 2016;7:12015; Cekanaviciute E, et al. Gut bacteria from multiple sclerosis patients modulate human T cells and exacerbate symptoms in mouse models. PNAS 2017;114:10713-10718.

‹ at a glance

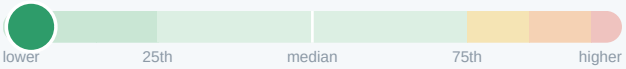
# Hypertension — gut pattern

NOTABLY LOW

Pattern version **1.0.0** (hypertension) · maturity **P2** (usable human shotgun cohort, or multiple compatible human association cohorts) · experimental · chinese_cohort_derived

Li et al. shotgun-sequenced 196 Chinese participants across normotension, pre-hypertension and hypertension and found reduced gene richness with Prevotella and Klebsiella enrichment, and faecal transplant from hypertensive humans raised blood pressure in germ-free mice. The mechanistic story is short-chain fatty acids acting on the Olfr78 and GPR41 receptors in the kidney and vasculature (Pluznick 2013), which makes butyrate and propionate capacity the lanes worth reading. Salt intake, adiposity and antihypertensive drugs all shape the gut, and the pattern is not separable from the metabolic-syndrome pattern.

**COMBINED RESEMBLANCE**
percentile of the matched reference group




**5th**
likely range 4–7

## What is driving it

| | | | |
|---|---|---|---|
| **Li 2017 Chinese hypertension shotgun (196)**<br>species · study li_2017 · 8/8 features · 50% of combined | 14th | LOW |
| **Gene richness**<br>diversity · study li_2017 · 1/2 features · 50% of combined | 7th | LOW |
| **SCFA and TMA capacity**<br>genes · genomic capacity · study functional · 3/3 features · not in combined | 20th | BELOW AVERAGE |

## How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

| | | |
|---|---|---|
| **Technical reliability** | 1.00 | sequencing depth, host fraction, reference match |
| **Independent evidence coverage** | 0.79 | share of the pattern's independent evidence clusters this sample could measure |
| **Evidence maturity** | P2 | usable human shotgun cohort, or multiple compatible human association cohorts |
| **Assay transportability** | native | whether the studies measured what this sample measured |
| **Population match** | 0.60 | how well the reference group matches you |
| **Specificity** | unknown | does this pattern separate its condition from general disturbance? measured where labelled data exist |
| **Direction conflict** | none | studies disagreeing on direction for one or more features |

**Competing patterns** scoring nearly as high: ra (95th), ad clinical (88th), celiac (77th). Margin over the next pattern: 8 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

## Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| **Li 2017 Chinese hypertension shotgun (196)** · species · study li_2017 | | | | | | | | | |
| *Prevotella copri* | not detected | 25 | ▲ | 0.70 | 0.60 | 0.30 | 0.60 | **-0.23** | |
| *Klebsiella pneumoniae* | not detected | 40 | ▲ | 0.70 | 0.60 | 0.40 | 0.70 | **-0.09** | |
| *Enterobacter cloacae complex* | not detected | 46 | ▲ | 0.40 | 0.50 | 0.40 | 0.50 | **-0.03** | |
| *Faecalibacterium prausnitzii* | 9.840% | 81 | ▼ | 0.70 | 0.60 | 0.20 | 0.70 | **-0.31** | |
| *Roseburia intestinalis* | 0.350% | 68 | ▼ | 0.60 | 0.60 | 0.30 | 0.70 | **-0.16** | |

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| *Bacteroides uniformis* | 1.860% | 50 | ▼ | 0.50 | 0.60 | 0.40 | 0.60 | **-0.00** | |
| *Oscillibacter sp 57 20* | 0.091% | 44 | ▼ | 0.40 | 0.50 | 0.40 | 0.50 | **+0.05** | |
| *Butyrivibrio crossotus* | not detected | 41 | ▼ | 0.40 | 0.50 | 0.50 | 0.50 | **+0.08** | |
| **Gene richness** · diversity · study li_2017 | | | | | | | | | |
| richness | 123.00 | 93 | ▼ | 0.70 | 0.60 | 0.10 | 0.70 | **-0.55** | |
| shannon | not detected | — | ▼ | 0.50 | 0.60 | 0.10 | 0.70 | — | |
| **SCFA and TMA capacity** · genes · genomic capacity · study functional | | | | | | | | | |
| butyrate | 32.1/100 | 82 | ▼ | 0.80 | 0.60 | 0.20 | 0.70 | **-0.26** | |
| propionate | 26.4/100 | 76 | ▼ | 0.60 | 0.60 | 0.30 | 0.60 | **-0.36** | |
| cutc | 3.3/100 | 66 | ▲ | 0.50 | 0.60 | 0.40 | 0.60 | **+0.21** | |

FINE PRINT

**Confidence grade insufficient.** Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

**Measured discrimination: not available.** No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Blood pressure is measured with a cuff. It is cheap and definitive. This profile adds nothing to that measurement.

The pattern is the metabolic-syndrome pattern plus a Prevotella enterotype signal that is diet- and geography-dependent.

Sources: Li J, et al. Gut microbiota dysbiosis contributes to the development of hypertension. Microbiome 2017;5:14; Pluznick JL, et al. Olfactory receptor responding to gut microbiota-derived signals plays a role in renin secretion and blood pressure regulation. PNAS 2013;110:4410.

‹ at a glance

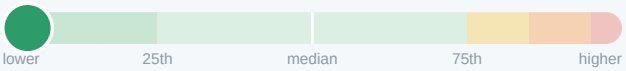
# Type 2 diabetes — gut pattern (metformin-gated)

NOTABLY LOW

Pattern version **1.0.0** (t2d) · maturity **P3** (independent human replication with coherent task and assay) · research beta · mixed_european_chinese_cohorts

Type 2 diabetes has large shotgun cohorts in China and Europe, and the first replicated lesson from them was a confounding one: much of the "diabetes signature" in early studies was metformin. Forslund et al. showed that metformin-treated patients carry more Escherichia and less Intestinibacter, and that the untreated-diabetes signal is a different, weaker pattern of butyrate-producer depletion. This profile therefore needs metformin status. In participant mode it abstains without it, because a score that cannot separate the drug from the disease measures the drug. In research and clinician mode an unsupplied medication list is read as "no metformin", the untreated-diabetes modules are scored on that stated assumption, and the card says how the reading would change if metformin were in fact being taken.

**COMBINED RESEMBLANCE**
percentile of the matched reference group




**2nd**
likely range 0–7

**Scored on an assumption.** This pattern depends on a fact about you that was not supplied, so the report assumed the ordinary case rather than withhold the score:

- **Metformin: assumed not taken.** If that is wrong: Metformin raises Escherichia, lowers Intestinibacter and shifts bile-salt hydrolase carriage (Forslund 2015). The type 2 diabetes pattern would then be partly measuring the drug, and its resemblance percentile would read higher than the disease alone warrants; it would need to be re-read against the metformin-treated stratum.

Supplying the fact (for example `--medications`) replaces the assumption and re-scores this pattern.

## What is driving it

| | | | |
|---|---|---|---|
| **Forlsund 2015 metformin-disentangled T2D signature**<br>species · study forslund_2015 · 7/7 features · 33% of combined | | 9th | LOW |
| **Qin 2012 Chinese MGWAS**<br>species · study qin_2012 · 6/6 features · 33% of combined | | 11th | LOW |
| **Karlsson 2013 Swedish postmenopausal women**<br>species · study karlsson_2013 · 5/5 features · 33% of combined | | 17th | BELOW AVERAGE |
| **Metabolite gene capacity**<br>genes · genomic capacity · study functional · 4/4 features · not in combined | | 47th | TYPICAL |
| **Butyrate carriers**<br>carriers · carrier proxy · study carriers · 1/1 features · not in combined | | 13th | LOW |

## The Prevotella copri complex, clade by clade

No member of the Segatella (Prevotella) copri complex was detected. All 13 clades were assessed, including the 9 found in humans, so this is a measured absence across the whole complex rather than an unresolved species-level reading.

No clade of this complex has an established association with any disease, including rheumatoid arthritis. Manghi et al. tested all 13 clades against 1,635 disease cases and 1,854 controls across 22 studies and 11 conditions, a set that included a rheumatoid arthritis cohort, and found none of the complex's species associated with any health condition. Earlier rheumatoid-arthritis reports named the species as a whole, not a clade, and those reports have replicated poorly: in an anti-CCP-positive at-risk cohort three Prevotellaceae strains were enriched and five depleted in the people who progressed. So this section reports what is present and makes no disease claim from it.

Tett 2019 Cell Host Microbe doi:10.1016/j.chom.2019.08.018; Manghi 2023 Cell Host Microbe doi:10.1016/j.chom.2023.09.013; Hitch 2022 Syst Appl Microbiol doi:10.1016/j.syapm.2022.126354.

## How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

| | | |
|---|---|---|
| **Technical reliability** | 1.00 | sequencing depth, host fraction, reference match |
| **Independent evidence coverage** | 1.00 | share of the pattern's independent evidence clusters this sample could measure |
| **Evidence maturity** | P3 | independent human replication with coherent task and assay |
| **Assay transportability** | native | whether the studies measured what this sample measured |

| | | |
|---|---|---|
| **Population match** | 0.60 | how well the reference group matches you |
| **Specificity** | unknown | does this pattern separate its condition from general disturbance? measured where labelled data exist |
| **Direction conflict** | material | studies disagreeing on direction for one or more features |

**Competing patterns** scoring nearly as high: ra (95th), ad clinical (88th), celiac (77th). Margin over the next pattern: 8 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

## Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| **Forslund 2015 metformin-disentangled T2D signature** · species · study forslund_2015 | | | | | | | | | |
| *Roseburia intestinalis* | 0.350% | 68 | ▼ | 0.90 | 0.80 | 0.30 | 0.80 | **-0.16** | |
| *Roseburia hominis* | 1.234% | 93 | ▼ | 0.70 | 0.80 | 0.30 | 0.80 | **-0.41** | |
| *Eubacterium rectale* | 1.082% | 52 | ▼ | 0.80 | 0.80 | 0.20 | 0.80 | **-0.02** | |
| *Faecalibacterium prausnitzii* | 9.840% | 81 | ▼ | 0.70 | 0.80 | 0.20 | 0.80 | **-0.31** | |
| *Clostridium bolteae* | not detected | 29 | ▲ | 0.60 | 0.70 | 0.40 | 0.70 | **-0.18** | |
| *Lactobacillus gasseri* | not detected | 49 | ▲ | 0.50 | 0.60 | 0.40 | 0.60 | **-0.01** | |
| *Streptococcus mutans* | not detected | 47 | ▲ | 0.50 | 0.60 | 0.40 | 0.60 | **-0.02** | |
| **Qin 2012 Chinese MGWAS** · species · study qin_2012 | | | | | | | | | |
| *Roseburia intestinalis* | 0.350% | 68 | ▼ | 0.70 | 0.70 | 0.30 | 0.70 | **-0.16** | |
| *Faecalibacterium prausnitzii* | 9.840% | 81 | ▼ | 0.70 | 0.70 | 0.20 | 0.70 | **-0.31** | |
| *Eubacterium rectale* | 1.082% | 52 | ▼ | 0.70 | 0.70 | 0.20 | 0.70 | **-0.02** | |
| *Clostridium bolteae* | not detected | 29 | ▲ | 0.50 | 0.60 | 0.40 | 0.60 | **-0.18** | |
| *Bacteroides caccae* | not detected | 20 | ▲ | 0.50 | 0.60 | 0.40 | 0.60 | **-0.28** | |
| *Akkermansia muciniphila* | not detected<br>0.024% pooled | 24 | ▲ | 0.40 | 0.50 | 0.30 | 0.40 | **-0.23** | |
| **Karlsson 2013 Swedish postmenopausal women** · species · study karlsson_2013 | | | | | | | | | |
| *Roseburia intestinalis* | 0.350% | 68 | ▼ | 0.70 | 0.70 | 0.30 | 0.70 | **-0.16** | |
| *Faecalibacterium prausnitzii* | 9.840% | 81 | ▼ | 0.60 | 0.70 | 0.20 | 0.70 | **-0.31** | |
| *Lactobacillus gasseri* | not detected | 49 | ▲ | 0.50 | 0.60 | 0.40 | 0.60 | **-0.01** | |
| *Streptococcus mutans* | not detected | 47 | ▲ | 0.50 | 0.60 | 0.40 | 0.60 | **-0.02** | |
| *Clostridium clostridioforme* | not detected | 42 | ▲ | 0.50 | 0.60 | 0.40 | 0.60 | **-0.07** | |
| **Metabolite gene capacity** · genes · genomic capacity · study functional | | | | | | | | | |
| butyrate | 32.1/100 | 82 | ▼ | 1.00 | 0.80 | 0.20 | 0.80 | **-0.26** | |
| urda | 53.5/100 | 87 | ▲ | 0.80 | 0.70 | 0.50 | 0.70 | **+0.32** | |
| bcaa | 180.5/100 | 34 | ▲ | 0.60 | 0.60 | 0.40 | 0.60 | **-0.15** | |
| cutc | 3.3/100 | 66 | ▲ | 0.40 | 0.50 | 0.30 | 0.50 | **+0.21** | |
| **Butyrate carriers** · carriers · carrier proxy · study carriers | | | | | | | | | |
| butyrate | 45.230% | 87 | ▼ | 0.80 | 0.80 | 0.20 | 0.80 | **-0.43** | |

FINE PRINT

**Confidence grade insufficient.** Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

**Measured discrimination: not available.** No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Type 2 diabetes is diagnosed with fasting glucose, HbA1c or a glucose tolerance test. Those are cheap, direct and definitive. This profile is none of those things.

Metformin status decides how to read this pattern. With metformin, the Escherichia expansion is the drug; in an untreated subject it may not be. The two cannot be told apart from stool alone, so when no medication list is supplied the score is printed on the stated assumption of no metformin, and would need re-reading if that assumption is wrong.

Butyrate-producer depletion is shared with every other dysbiosis pattern; see the specificity column in the confidence vector.

Sources: Forslund K, et al. Disentangling type 2 diabetes and metformin treatment signatures in the human gut microbiota. Nature 2015;528:262-266; Qin J, et al. A metagenome-wide association study of gut microbiota in type 2 diabetes. Nature 2012;490:55-60; Karlsson FH, et al. Gut metagenome in European women with normal, impaired and diabetic glucose control. Nature 2013;498:99-103; Koh A, et al. Microbially produced imidazole propionate impairs insulin signaling through mTORC1. Cell 2018;175:947.

‹ at a glance

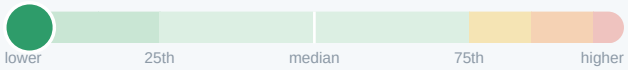
# ME/CFS-like gut microbiome similarity

NOTABLY LOW

Pattern version **0.2.0** (mecfs) · maturity **P2** (usable human shotgun cohort, or multiple compatible human association cohorts) · research only · us_adult_cohorts

The best-evidenced profile in this set. Two cohorts from different research groups, published simultaneously in the same journal, found the same core pattern: reduced abundance of the major butyrate-producing species Faecalibacterium prausnitzii and Eubacterium rectale, reduced microbial butyrate-synthesis capacity, and — crucially — confirmation by measured short-chain fatty acids and qPCR rather than DNA inference alone. Low F. prausnitzii tracked more severe fatigue. The pattern depends on how long someone has been ill. Xiong et al. found short-term patients (under 4 years) had marked microbial dysbiosis while long-term patients (over 10 years) had largely resolved microbial dysbiosis yet retained metabolic and clinical abnormalities. A low taxonomic score in someone ill for more than a decade is therefore consistent with the published pattern, not evidence against it.

**COMBINED RESEMBLANCE**
percentile of the matched reference group




**2nd**
likely range 0–4

## What is driving it

**Taxonomic**
species · study taxonomic · 4/4 features · 50% of combined
13th — LOW

**Functional**
genes · genomic capacity · study functional · 2/2 features · 35% of combined
18th — BELOW AVERAGE

**Ecological**
diversity · study ecological · 1/1 features · 15% of combined
<1st — NOTABLY LOW

## How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

| | | |
|---|---|---|
| **Technical reliability** | 1.00 | sequencing depth, host fraction, reference match |
| **Independent evidence coverage** | 1.00 | share of the pattern's independent evidence clusters this sample could measure |
| **Evidence maturity** | P2 | usable human shotgun cohort, or multiple compatible human association cohorts |
| **Assay transportability** | native | whether the studies measured what this sample measured |
| **Population match** | 0.60 | how well the reference group matches you |
| **Specificity** | 0.80 | does this pattern separate its condition from general disturbance? measured where labelled data exist |
| **Direction conflict** | none | studies disagreeing on direction for one or more features |

**Competing patterns** scoring nearly as high: ra (95th), ad clinical (88th), celiac (77th). Margin over the next pattern: 8 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

## Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| **Taxonomic** · species · study taxonomic | | | | | | | | | |
| *Faecalibacterium prausnitzii* | 9.840% | 81 | ▼ | 1.00 | 1.00 | 0.50 | 1.00 | **-0.31** | Guo 2023 Cell Host Microbe, Xiong 2023 Cell Host Microbe |
| *Eubacterium rectale* | 1.082% | 52 | ▼ | 1.00 | 0.70 | 0.60 | 1.00 | **-0.02** | Guo 2023 Cell Host Microbe |

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| *Roseburia intestinalis* | 0.350% | 68 | ▼ | 0.60 | 0.60 | 0.50 | 0.90 | **-0.16** | Guo 2023 Cell Host Microbe (network analysis) |
| *Coprococcus comes* | 1.842% | 96 | ▼ | 0.50 | 0.60 | 0.50 | 0.90 | **-0.58** | Guo 2023 Cell Host Microbe (network analysis) |
| **Functional** · genes · genomic capacity · study functional | | | | | | | | | |
| butyrate | 32.1/100 | 82 | ▼ | 1.00 | 1.00 | 0.50 | 1.00 | **-0.26** | Guo 2023 Cell Host Microbe, Xiong 2023 Cell Host Microbe |
| propionate | 26.4/100 | 76 | ▼ | 0.40 | 0.60 | 0.40 | 0.80 | **-0.36** | Xiong 2023 Cell Host Microbe (SCFA panel) |
| **Ecological** · diversity · study ecological | | | | | | | | | |
| shannon diversity | 3.72 | 100 | ▼ | 0.50 | 0.70 | 0.20 | 0.60 | **-0.60** | Guo 2023 Cell Host Microbe |

FINE PRINT

**Confidence grade insufficient.** Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

The 50/35/15 module split is an engineering starting point, not an evidence-derived weighting.

**Measured on labelled cohorts** (AUC, 0.5 = chance, 1.0 = perfect): T2D 0.60 (0.55–0.66); CRC 0.55 (0.50–0.60); IBD 0.50 (0.46–0.55). No labelled cohort exists for this condition, so its own discrimination is UNMEASURED (it scores 0.50-0.60 against other conditions).

Do not read the 90% accuracy figure from the 2025 BioMapAI paper alongside this profile. That figure came from integrating metagenomics with plasma metabolomics, immune-cell profiling and blood tests; microbiome data alone best predicted gastrointestinal, emotional and sleep symptoms specifically. A stool-only tool cannot reproduce it.

Butyrate-producer depletion is one of the least specific findings in the microbiome literature. It appears in inflammatory bowel disease, after antibiotics, with low-fibre diets, and in general ill health. A high score here should be read beside the dysbiosis anchor.

If illness duration is unknown, both strata are reported and confidence is reduced, because the expected pattern differs between them and they must not be averaged.

Sources: Guo C, Che X, Briese T, et al. Deficient butyrate-producing capacity in the gut microbiome is associated with bacterial network disturbances and fatigue symptoms in ME/CFS. Cell Host Microbe 2023;31:288-304, doi:10.1016/j.chom.2023.01.004. Xiong R, Gunter C, Fleming E, et al. Multi-omics of gut microbiome-host interactions in short- and long-term ME/CFS patients. Cell Host Microbe 2023;31:273-287, doi:10.1016/j.chom.2023.01.001; data PRJNA878603, code github.com/ohlab/MECFS_2021.

‹ at a glance

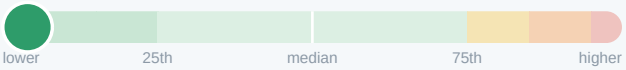
# Symptomatic dermographism (chronic inducible hives from skin stroking) — translated 16S pattern

NOTABLY LOW

Pattern version **0.1.0** (symptomatic_dermographism) · maturity **P1** (one small human cohort, amplicon-only, or inaccessible individual data) · research only · chinese_adult_cohort

Symptomatic dermographism is the commonest chronic inducible urticaria: hives that appear along the line where the skin has been stroked, scratched or rubbed, itching for minutes to an hour. One 16S amplicon study supplies the pattern here — Liu et al. compared stool from 22 adults with symptomatic dermographism against 22 healthy controls and described a healthy-control core community that the patient group lacked. Only two candidates from that core are imported: the genus Subdoligranulum and the species Ruminococcus bromii, both depletion-oriented. They are scored, but every number on this card carries a permanent transport warning, because a 16S study names organisms by amplicon and this report measures them by shotgun sequencing. The two are not the same measurement, and no shotgun cohort has tested this pattern. Two features is also a very small panel: coverage is printed beside the score for that reason, and one missing feature halves it. This profile is deliberately separate from the chronic spontaneous urticaria profile. The two conditions coexist often, and when they do both eligible cards are shown side by side — there is no combined score, and the two indices are never multiplied or averaged into a third number.

**COMBINED RESEMBLANCE**
percentile of the matched reference group




**2nd**
likely range 0–100

### The same species on the source study's own scale

**Not computed on this scale** (no usable features). No number is shown rather than a zero, because a zero here would be a measurement that was never made. Panel features and why: *Subdoligranulum* — reference ineligible; *Ruminococcus bromii* — below reporting limit.

The percentile above still has a value because the two calculations treat a nondetection differently, and the difference is the point of showing both. The percentile places an undetected organism at a fixed floor and ranks the sample from there, which is how every reading in this report is built and what makes them comparable with each other. This scale declines to convert an undetected organism into a number at all unless a validated detection limit exists for it, and none does at this sequencing depth. Read the percentile as the position of this community, and read the blank here as the reason not to put much weight on it.

## What is driving it

**Liu 2021 symptomatic-dermographism core-community candidates (16S, translated)**
species · association only · 16S study · study liu_sd_2021 · 2/2 features · 100% of combined



2nd NOTABLY LOW

## How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

| | | |
|---|---|---|
| **Technical reliability** | 1.00 | sequencing depth, host fraction, reference match |
| **Independent evidence coverage** | 1.00 | share of the pattern's independent evidence clusters this sample could measure |
| **Evidence maturity** | P1 | one small human cohort, amplicon-only, or inaccessible individual data |
| **Assay transportability** | labeled transport | whether the studies measured what this sample measured |
| **Population match** | 0.60 | how well the reference group matches you |
| **Specificity** | unknown | does this pattern separate its condition from general disturbance? measured where labelled data exist |
| **Direction conflict** | low | studies disagreeing on direction for one or more features |

**Competing patterns** scoring nearly as high: ra (95th), ad clinical (88th), celiac (77th). Margin over the next pattern: 8 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

## Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| **Liu 2021 symptomatic-dermographism core-community candidates (16S, translated)** · species · association only · 16S study · study liu_sd_2021 | | | | | | | | | |
| *Subdoligranulum (genus)* | 0.758% | 100 | ▼ | 1.00 | 0.30 | 0.20 | 0.50 | **-1.00** | |
| *Ruminococcus bromii* | not detected<br>3.40% pooled | 19 | ▼ | 1.00 | 0.30 | 0.20 | 0.50 | **+0.29** | |

FINE PRINT

**Confidence grade insufficient.** Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

**Measured discrimination: not available.** No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Symptomatic dermographism is diagnosed by stroking the skin and watching what happens, usually with a calibrated dermographometer. A stool profile cannot make that observation, and this figure is a research pattern index, not a probability of having the condition.

Every number here crosses assay types. Until an independent shotgun cohort tests these two directions, the transport warning stays on the card regardless of how the score comes out.

The other chronic inducible urticarias — the other forms of hives brought on by a trigger: cold, delayed pressure, solar, heat, vibratory, cholinergic, contact and aquagenic — have no stool signature this audit could locate. They are reported as evidence not established rather than scored with this profile or with the chronic spontaneous urticaria (spontaneous chronic hives) profile standing in for them. Skin and sweat Malassezia mechanisms proposed for cholinergic urticaria — hives brought on by heat, exercise or stress — are not gut findings.

Having both chronic spontaneous urticaria and dermographism at once — spontaneous hives as well as hives raised by stroking the skin — is common. When both profiles are eligible, both scores are shown; there is no third combined index, and neither score modifies the other.

Sources: Liu R, Peng C, Jing D, et al. Biomarkers of gut microbiota in chronic spontaneous urticaria and symptomatic dermographism. Exp Dermatol 2021;30:1657-1665, doi:10.1111/exd.14326; raw data OEP001229 (BioSino). Coexisting-phenotype cohort: Liu R, et al. Front Cell Infect Microbiol 2021;11:703126, doi:10.3389/fcimb.2021.703126, raw data OEP001891. Clinical classification of the inducible urticarias: Zuberbier T, et al., International guideline for the definition, classification, diagnosis and management of urticaria, Allergy 2026, doi:10.1111/all.70210.

‹ at a glance

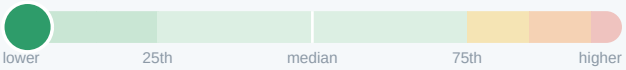
# Irritable bowel syndrome — gut pattern

NOTABLY LOW

Pattern version **1.0.0** (ibs) · maturity **P2** (usable human shotgun cohort, or multiple compatible human association cohorts) · research beta · mixed_adult_cohorts

A symptom-based diagnosis, so the microbiome literature is large, mostly 16S, and unusually inconsistent between cohorts. Systematic reviews agree on little beyond depleted butyrate producers and Bifidobacterium, and a subgroup of patients with a distinctly dysbiotic community — most others look like healthy controls. That heterogeneity is why the shape analysis matters more than the score here: an IBS-like community is also a post-antibiotic community, a low-fibre community, and a post-infectious community. Subtypes are scored separately in ibs_c, ibs_d and ibs_m, which share this parent's modules and add subtype-specific ones.

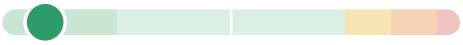
**COMBINED RESEMBLANCE**
percentile of the matched reference group




**1st**
likely range 0–17

## What is driving it

**Mars 2020 longitudinal shotgun (Mayo)**
species · study mars_2020 · 4/4 features · 40% of combined



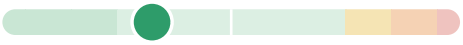
9th — LOW

**Jeffery 2020 (Irish IBS shotgun)**
species · study jeffery_2020 · 4/4 features · 40% of combined



5th — LOW

**Pittayanon 2019 systematic review consensus**
species · association only · 16S study · study pittayanon_2019 · 6/6 features · 20% of combined



33rd — TYPICAL

**Community ecology**
diversity · study pittayanon_2019 · 0/1 features · not in combined




— NOT COMPUTABLE

**Gas, bile acid and neuroactive capacity**
genes · genomic capacity · study mechanism · 4/4 features · not in combined



53rd — TYPICAL

## How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

| | | |
|---|---|---|
| **Technical reliability** | 1.00 | sequencing depth, host fraction, reference match |
| **Independent evidence coverage** | 1.00 | share of the pattern's independent evidence clusters this sample could measure |
| **Evidence maturity** | P2 | usable human shotgun cohort, or multiple compatible human association cohorts |
| **Assay transportability** | labeled transport | whether the studies measured what this sample measured |
| **Population match** | 0.60 | how well the reference group matches you |
| **Specificity** | unknown | does this pattern separate its condition from general disturbance? measured where labelled data exist |
| **Direction conflict** | material | studies disagreeing on direction for one or more features |

**Competing patterns** scoring nearly as high: ra (95th), ad clinical (88th), celiac (77th). Margin over the next pattern: 8 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

## Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| **Mars 2020 longitudinal shotgun (Mayo)** · species · study mars_2020 | | | | | | | | | |
| *Faecalibacterium prausnitzii* | 9.840% | 81 | ▼ | 0.70 | 0.70 | 0.20 | 0.70 | -0.31 | |
| *Roseburia intestinalis* | 0.350% | 68 | ▼ | 0.60 | 0.60 | 0.30 | 0.70 | -0.16 | |
| *Bacteroides fragilis* | not detected | 24 | ▲ | 0.50 | 0.60 | 0.30 | 0.60 | -0.24 | |
| *Ruminococcus gnavus* | not detected | 27 | ▲ | 0.60 | 0.60 | 0.20 | 0.60 | -0.21 | |
| **Jeffery 2020 (Irish IBS shotgun)** · species · study jeffery_2020 | | | | | | | | | |
| *Bacteroides vulgatus* | not detected | 3 | ▲ | 0.50 | 0.60 | 0.30 | 0.60 | -1.00 | |
| *Eubacterium rectale* | 1.082% | 52 | ▼ | 0.60 | 0.60 | 0.20 | 0.70 | -0.02 | |
| *Coprococcus comes* | 1.842% | 96 | ▼ | 0.50 | 0.50 | 0.30 | 0.60 | -0.58 | |
| *Methanobrevibacter smithii* | not detected | 35 | ▼ | 0.50 | 0.50 | 0.40 | 0.50 | +0.13 | |
| **Pittayanon 2019 systematic review consensus** · species · association only · 16S study · study pittayanon_2019 | | | | | | | | | |
| *Faecalibacterium prausnitzii* | 9.840% | 81 | ▼ | 0.80 | 0.60 | 0.20 | 0.70 | -0.31 | |
| *Bifidobacterium adolescentis* | not detected | 25 | ▼ | 0.80 | 0.60 | 0.30 | 0.70 | +0.23 | |
| *Bifidobacterium longum* | not detected | 17 | ▼ | 0.60 | 0.60 | 0.30 | 0.70 | +0.32 | |
| *Lactobacillus ruminis* | not detected | 44 | ▲ | 0.50 | 0.50 | 0.30 | 0.50 | -0.05 | |
| *Escherichia coli* | not detected | 19 | ▲ | 0.60 | 0.50 | 0.20 | 0.60 | -0.29 | |
| *Ruminococcus gnavus* | not detected | 27 | ▲ | 0.60 | 0.50 | 0.20 | 0.60 | -0.21 | |
| **Community ecology** · diversity · study pittayanon_2019 | | | | | | | | | |
| shannon | not detected | — | ▼ | 0.50 | 0.60 | 0.10 | 0.70 | — | |
| **Gas, bile acid and neuroactive capacity** · genes · genomic capacity · study mechanism | | | | | | | | | |
| butyrate | 32.1/100 | 82 | ▼ | 0.80 | 0.70 | 0.20 | 0.80 | -0.26 | |
| bai | 3.5/100 | 72 | ▲ | 0.60 | 0.60 | 0.30 | 0.60 | +0.30 | |
| h2s | 96.5/100 | 80 | ▲ | 0.50 | 0.50 | 0.30 | 0.50 | +0.35 | |
| tryptamine | 0.1/100 | 44 | ▲ | 0.40 | 0.40 | 0.40 | 0.40 | -0.01 | |

FINE PRINT

**Confidence grade insufficient.** Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

**Measured discrimination: not available.** No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

IBS is a symptom-based diagnosis made by a clinician after excluding organic disease. No stool test diagnoses it or excludes it.

The IBS-like community is the general dysbiosis community. Every scored direction here also appears in IBD, post-antibiotic and low-fibre states. Read this beside the shape analysis: if many profiles are high together, that is dysbiosis, not IBS.

Most IBS patients in the shotgun cohorts had communities indistinguishable from controls, so a low score is common in IBS.

Sources: Pittayanon R, et al. Gut microbiota in patients with irritable bowel syndrome — a systematic review. Gastroenterology 2019;157:97-108; Mars RAT, et al. Longitudinal multi-omics reveals subset-specific mechanisms underlying irritable bowel syndrome. Cell 2020;182:1460-1473; Jeffery IB, et al. Differences in fecal microbiomes and metabolomes of people with vs without irritable bowel syndrome and bile acid malabsorption. Gastroenterology 2020;158:1016-1028.

‹ at a glance

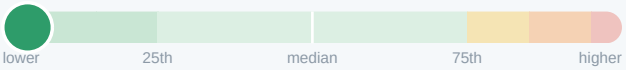
# Chronic spontaneous urticaria (chronic hives) gut-microbiome similarity

NOTABLY LOW

Pattern version **0.1.0** (csu) · maturity **P2** (usable human shotgun cohort, or multiple compatible human association cohorts) · research only · chinese_adult_cohort

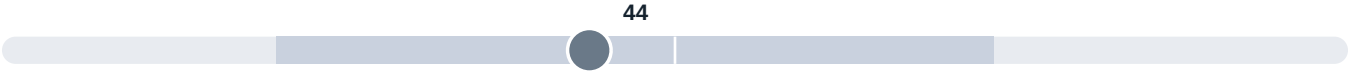
Chronic spontaneous urticaria — chronic hives, historically called chronic idiopathic urticaria — is itchy weals and often deep swelling that keep recurring for more than six weeks without an external trigger anyone can point to. One human shotgun-metagenomic cohort supplies the pattern scored here: Zhu et al. sequenced stool from 26 adults with chronic hives and 26 healthy controls and reported fifteen species-level directions. The depletions sit in the fibre-fermenting, short-chain-fatty-acid-producing end of the community — three Alistipes species, Coprococcus catus, Roseburia hominis, Blautia (Ruminococcus) obeum, two Bacteroides — and the enrichments in the Gram-negative and mucin-degrading end, led by Ruminococcus gnavus, with Escherichia coli, Klebsiella pneumoniae, Veillonella parvula and Bacteroides stercoris. Eight of the fifteen survive Benjamini-Hochberg correction in the source table; the other seven are exploratory and labelled as such. Both are scored, the strict eight separately as a sensitivity panel, because a fifteen-species panel selected in fifty-two people is a discovery result and the sensitivity panel is how the report says so out loud. Every feature carries the same scoring weight: the published association coefficients are recorded as provenance and are deliberately not used as model weights, because a MaAsLin2 coefficient fitted on a discovery cohort is not a deployable classifier. Amplicon cohorts from other centres agree on some of this and disagree on the rest. Diversity has been reported lower, unchanged and higher in chronic hives; genus Lactobacillus has been reported both depleted and enriched. That is why this profile contains no diversity term and no genus-level term, and why the conflicting cohorts are listed rather than averaged away. A high score is a research observation about community composition. It does not diagnose hives, and a low score in someone with hives does not argue against the diagnosis.

**COMBINED RESEMBLANCE**
percentile of the matched reference group




**1st**
likely range 0–14

## The same species on the source study's own scale




**44 out of 100** — with no percent sign, because this is not a percentage, a probability or a match score. Coverage 7 of 15 panel features; the pale band, 20 to 74, is where the figure could sit once the unmeasured features are accounted for. Fifty is neutral signed deviation under this transformation, not a definition of health, and higher means stronger alignment with the directions the source study reported.

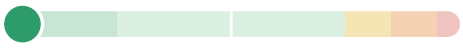
Strict subset, the 8 species that survive multiple-testing correction in the source table: **48** at coverage 4 of 8. This is a sensitivity analysis on the same measurements, not a second independent test — the two panels share every one of these features, so a gap between them means the wider figure is being carried by its exploratory members.

These are coverage and censoring bounds, not a 95% confidence interval: they widen when a feature cannot be measured and say nothing about sampling error, about how well a single-centre discovery cohort transports to this population, or about how specific the pattern is to this condition. The percentile above and the figure here read the same measurements two ways, and are not two opinions about one quantity: the percentile is a rank, so it says how this sample compares with a matched reference group and it moves when that group's spread changes; this figure is a fixed transformation, so it says how far each organism sits from its reference centre in the direction the study reported, and it stays put. A high rank can therefore sit beside a middling figure, or the reverse, depending on how tightly the reference group clusters and on which features could be measured at all.

Features outside the figure: *Alistipes onderdonkii* (below reporting limit); *Bacteroides intestinalis* (below reporting limit); *Veillonella parvula* (below reporting limit); *Bacteroides stercoris* (below reporting limit); *Bilophila wadsworthia* (below reporting limit); *Escherichia coli* (below reporting limit); *Klebsiella pneumoniae* (below reporting limit); *Ruminococcus gnavus* (below reporting limit). None of them was scored as zero.

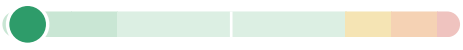
## What is driving it

**Zhu 2024 shotgun species panel (all 15 reported directions)**
species · study zhu_2024 · 15/15 features · 100% of combined



1st NOTABLY LOW

**Zhu 2024 strict subset (BH q < 0.05) — sensitivity only**
species · study zhu_2024 · 8/8 features · not in combined



6th LOW

**Gram-negative envelope and endotoxin capacity (mechanism context)**
carriers · carrier proxy · study zhu_2024_function · 2/2 features · not in combined



3rd NOTABLY LOW

## How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

| | | |
|---|---|---|
| **Technical reliability** | 1.00 | sequencing depth, host fraction, reference match |
| **Independent evidence coverage** | 1.00 | share of the pattern's independent evidence clusters this sample could measure |
| **Evidence maturity** | P2 | usable human shotgun cohort, or multiple compatible human association cohorts |
| **Assay transportability** | native | whether the studies measured what this sample measured |
| **Population match** | 0.60 | how well the reference group matches you |
| **Specificity** | unknown | does this pattern separate its condition from general disturbance? measured where labelled data exist |
| **Direction conflict** | none | studies disagreeing on direction for one or more features |

**Competing patterns** scoring nearly as high: ra (95th), ad clinical (88th), celiac (77th). Margin over the next pattern: 8 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

## Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| **Zhu 2024 shotgun species panel (all 15 reported directions)** · species · study zhu_2024 | | | | | | | | | |
| *Alistipes onderdonkii* | not detected 0.304% pooled | 43 | ▼ | 1.00 | 0.40 | 0.30 | 0.70 | +0.06 | |
| *Alistipes putredinis* | 1.648% | 63 | ▼ | 1.00 | 0.40 | 0.30 | 0.70 | -0.11 | |
| *Alistipes shahii* | 0.012% | 39 | ▼ | 1.00 | 0.40 | 0.30 | 0.70 | +0.10 | |
| *Coprococcus catus* | 1.281% | 99 | ▼ | 1.00 | 0.40 | 0.20 | 0.80 | -0.76 | |
| *Bacteroides intestinalis* | not detected | 36 | ▼ | 1.00 | 0.40 | 0.30 | 0.70 | +0.12 | |
| *Ruminococcus obeum, measured as Blautia obeum* | 0.604% | 90 | ▼ | 1.00 | 0.40 | 0.30 | 0.60 | -0.37 | |
| *Veillonella parvula* | not detected | 31 | ▲ | 1.00 | 0.40 | 0.20 | 0.60 | -0.17 | |
| *Ruminococcus gnavus* | not detected | 27 | ▲ | 1.00 | 0.40 | 0.30 | 0.80 | -0.21 | |
| *Bacteroides cellulosilyticus* | 0.875% | 88 | ▼ | 1.00 | 0.40 | 0.30 | 0.50 | -0.39 | |
| *Odoribacter splanchnicus* | 0.472% | 81 | ▼ | 1.00 | 0.40 | 0.20 | 0.40 | -0.29 | |
| *Roseburia hominis* | 1.234% | 93 | ▼ | 1.00 | 0.40 | 0.20 | 0.60 | -0.41 | |
| *Bacteroides stercoris* | not detected | 24 | ▲ | 1.00 | 0.40 | 0.30 | 0.50 | -0.24 | |
| *Bilophila wadsworthia* | not detected | 20 | ▼ | 1.00 | 0.40 | 0.30 | 0.40 | +0.28 | |
| *Escherichia coli* | not detected | 19 | ▲ | 1.00 | 0.40 | 0.20 | 0.60 | -0.29 | |
| *Klebsiella pneumoniae* | not detected | 40 | ▲ | 1.00 | 0.40 | 0.20 | 0.60 | -0.09 | |
| **Zhu 2024 strict subset (BH q < 0.05) — sensitivity only** · species · study zhu_2024 | | | | | | | | | |
| *Alistipes onderdonkii* | not detected 0.304% pooled | 43 | ▼ | 1.00 | 0.40 | 0.30 | 0.70 | +0.06 | |
| *Alistipes putredinis* | 1.648% | 63 | ▼ | 1.00 | 0.40 | 0.30 | 0.70 | -0.11 | |
| *Alistipes shahii* | 0.012% | 39 | ▼ | 1.00 | 0.40 | 0.30 | 0.70 | +0.10 | |
| *Bacteroides intestinalis* | not detected | 36 | ▼ | 1.00 | 0.40 | 0.30 | 0.70 | +0.12 | |
| *Coprococcus catus* | 1.281% | 99 | ▼ | 1.00 | 0.40 | 0.20 | 0.80 | -0.76 | |
| *Ruminococcus obeum, measured as Blautia obeum* | 0.604% | 90 | ▼ | 1.00 | 0.40 | 0.30 | 0.60 | -0.37 | |

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| *Veillonella parvula* | not detected | 31 | ▲ | 1.00 | 0.40 | 0.20 | 0.60 | **-0.17** | |
| *Ruminococcus gnavus* | not detected | 27 | ▲ | 1.00 | 0.40 | 0.30 | 0.80 | **-0.21** | |
| **Gram-negative envelope and endotoxin capacity (mechanism context)** · carriers · carrier proxy · study zhu_2024_function | | | | | | | | | |
| hexa lps | 0.000% | 7 | ▲ | 0.60 | 0.40 | 0.20 | 0.50 | **-0.84** | |
| opportunistic pathogens | 0.000% | 3 | ▲ | 0.50 | 0.40 | 0.20 | 0.50 | **-0.96** | |

FINE PRINT

**Confidence grade insufficient.** Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

**Measured discrimination: not available.** No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Chronic hives is a clinical diagnosis made from weals, itch and their duration. Nothing in stool DNA makes or excludes it, and this score is not a probability of having it. A diagnosis of chronic spontaneous urticaria needs weals or angioedema recurring for more than six weeks without an identifiable external trigger, assessed by a clinician.

The fifteen species were selected in the same fifty-two people whose data produced them. Any evaluation on that cohort is reconstruction, not external validation, and no independent shotgun cohort has yet tested this panel. The strict-eight sensitivity figure is the same measurements re-summarised, not a second opinion.

Diversity is deliberately absent from this profile. It has been reported lower (Lu 2019), unchanged (Wang 2021, Ciftci 2025) and higher (Podder 2025) in chronic hives, and the cohorts that disagree differ in treatment exposure more than in phenotype. A Shannon or evenness figure elsewhere in this report says nothing about this pattern.

Detection of Escherichia coli, Klebsiella pneumoniae or Ruminococcus gnavus in this panel is a relative abundance measurement. It is not an infection, not an overgrowth diagnosis and not an indication for antimicrobials, herbal or prescription. No microbiome score authorises eradication, a transplant, or stopping any hives treatment already prescribed.

Only chronic spontaneous urticaria — spontaneous chronic hives — is scored here. Symptomatic dermographism, the hives raised by stroking the skin, has its own separate and explicitly translated profile. The other inducible forms of hives (cold, delayed pressure, solar, heat, vibratory, cholinergic, contact and aquagenic urticaria) have no stool-signature cohort that this audit could locate: evidence not established, so they carry no score at all rather than borrowing this one. Short-lived acute hives, urticarial vasculitis, mastocytosis and bradykinin-mediated angioedema are different conditions from chronic hives and are not scored here either.

Sources: Zhu L, Jian X, Zhou B, Liu R, Muñoz M, Sun W, Xie L, Chen X, Peng C, Maurer M, Li J. Gut microbiota facilitate chronic spontaneous urticaria. Nat Commun 2024;15:112, doi:10.1038/s41467-023-44373-x; raw data OEP002960 (BioSino), metabolomics CNP0005021, mouse 16S OEP002997. Conflicting and supporting amplicon cohorts: Lu 2019 PMC6881578; Wang 2020 doi:10.3389/fcimb.2020.00024; Wang 2021 doi:10.3389/fimmu.2021.691304; Luo 2023 doi:10.3389/fcimb.2022.1094737; Cesic 2023 doi:10.3390/life13061280; Podder 2025 doi:10.1002/clt2.70027; Park 2025 doi:10.1111/all.16601; Ciftci 2025 Pathogens 14(11):1140; Cho 2026 doi:10.1111/exd.70208; Wang 2026 doi:10.1111/1751-7915.70316. Clinical classification: Zuberbier T, et al. International guideline for the definition, classification, diagnosis and management of urticaria, Allergy 2026, doi:10.1111/all.70210.

‹ at a glance

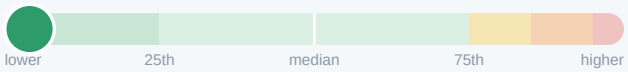
# Ulcerative colitis — gut pattern

NOTABLY LOW

Pattern version **1.0.0** (uc) · maturity **P3** (independent human replication with coherent task and assay) · research beta · mixed_north_american_european_cohorts

Ulcerative colitis shares most of its gut pattern with Crohn's disease — butyrate-producer depletion, Ruminococcus gnavus and Escherichia coli expansion — with a milder dysbiosis on average and without the oral-origin taxa that mark Crohn's. Roseburia hominis and Faecalibacterium prausnitzii depletion were shown in UC specifically by Machiels, and in both HMP2 and PRISM the UC dysbiotic state was less frequent and less extreme than in CD. Because the two IBD patterns overlap so heavily, the difference between the uc and crohns scores on one sample is rarely meaningful. The ibd composite profile is the one to read when the question is "IBD-like or not"; these two exist so the library records what each disease's own literature says.

**COMBINED RESEMBLANCE**
percentile of the matched reference group




**1st**
likely range 0–2

## What is driving it

| | | | |
|---|---|---|---|
| **Machiels 2014 Belgian UC cohort**<br>species · study machiels_2014 · 2/2 features · 40% of combined | | 7th | LOW |
| **Franzosa 2019 PRISM — UC directions**<br>species · study franzosa_2019 · 5/5 features · 40% of combined | | 2nd | NOTABLY LOW |
| **HMP2 IBDMDB — UC directions**<br>species · study hmp2_2019 · 6/6 features · 20% of combined | | 13th | LOW |
| **Community ecology**<br>diversity · study hmp2_2019 · 0/1 features · not in combined | not measured | — | NOT COMPUTABLE |
| **Metabolite gene capacity**<br>genes · genomic capacity · study functional · 3/3 features · not in combined | | 21st | BELOW AVERAGE |
| **Butyrate carriers and sulfate reducers**<br>carriers · carrier proxy · study carriers · 2/2 features · not in combined | | 7th | LOW |

## How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

| | | |
|---|---|---|
| **Technical reliability** | 1.00 | sequencing depth, host fraction, reference match |
| **Independent evidence coverage** | 1.00 | share of the pattern's independent evidence clusters this sample could measure |
| **Evidence maturity** | P3 | independent human replication with coherent task and assay |
| **Assay transportability** | native | whether the studies measured what this sample measured |
| **Population match** | 0.60 | how well the reference group matches you |
| **Specificity** | unknown | does this pattern separate its condition from general disturbance? measured where labelled data exist |
| **Direction conflict** | low | studies disagreeing on direction for one or more features |

**Competing patterns** scoring nearly as high: ra (95th), ad clinical (88th), celiac (77th). Margin over the next pattern: 8 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

## Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection

methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| **Machiels 2014 Belgian UC cohort** · species · study machiels_2014 | | | | | | | | | |
| *Roseburia hominis* | 1.234% | 93 | ▼ | 1.00 | 0.90 | 0.30 | 0.90 | **-0.41** | |
| *Faecalibacterium prausnitzii* | 9.840% | 81 | ▼ | 1.00 | 0.90 | 0.20 | 0.90 | **-0.31** | |
| **Franzosa 2019 PRISM — UC directions** · species · study franzosa_2019 | | | | | | | | | |
| *Ruminococcus gnavus* | not detected | 27 | ▲ | 0.80 | 0.80 | 0.30 | 0.90 | **-0.21** | |
| *Escherichia coli* | not detected | 19 | ▲ | 0.60 | 0.70 | 0.30 | 0.80 | **-0.29** | |
| *Roseburia intestinalis* | 0.350% | 68 | ▼ | 0.60 | 0.80 | 0.20 | 0.80 | **-0.16** | |
| *Faecalibacterium prausnitzii* | 9.840% | 81 | ▼ | 0.80 | 0.90 | 0.20 | 0.90 | **-0.31** | |
| *Bacteroides vulgatus* | not detected | 3 | ▲ | 0.50 | 0.60 | 0.30 | 0.60 | **-1.00** | |
| **HMP2 IBDMDB — UC directions** · species · study hmp2_2019 | | | | | | | | | |
| *Ruminococcus gnavus* | not detected | 27 | ▲ | 0.90 | 0.80 | 0.30 | 0.90 | **-0.21** | |
| *Escherichia coli* | not detected | 19 | ▲ | 0.70 | 0.80 | 0.30 | 0.80 | **-0.29** | |
| *Faecalibacterium prausnitzii* | 9.840% | 81 | ▼ | 0.90 | 0.90 | 0.20 | 0.90 | **-0.31** | |
| *Roseburia intestinalis* | 0.350% | 68 | ▼ | 0.70 | 0.80 | 0.20 | 0.80 | **-0.16** | |
| *Eubacterium rectale* | 1.082% | 52 | ▼ | 0.70 | 0.80 | 0.20 | 0.80 | **-0.02** | |
| *Akkermansia muciniphila* | not detected<br>0.024% pooled | 24 | ▼ | 0.60 | 0.60 | 0.40 | 0.60 | **+0.23** | |
| **Community ecology** · diversity · study hmp2_2019 | | | | | | | | | |
| shannon | not detected | — | ▼ | 0.60 | 0.80 | 0.10 | 0.80 | — | |
| **Metabolite gene capacity** · genes · genomic capacity · study functional | | | | | | | | | |
| butyrate | 32.1/100 | 82 | ▼ | 1.00 | 0.90 | 0.20 | 0.90 | **-0.26** | |
| h2s | 96.5/100 | 80 | ▲ | 0.70 | 0.60 | 0.40 | 0.70 | **+0.35** | |
| bsh | 123.0/100 | 96 | ▼ | 0.40 | 0.60 | 0.30 | 0.60 | **-0.63** | |
| **Butyrate carriers and sulfate reducers** · carriers · carrier proxy · study carriers | | | | | | | | | |
| butyrate | 45.230% | 87 | ▼ | 0.80 | 0.80 | 0.20 | 0.80 | **-0.43** | |
| sulfate reducers | 0.000% | 16 | ▲ | 0.60 | 0.60 | 0.40 | 0.60 | **-0.34** | |

FINE PRINT

**Confidence grade moderate.** Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

**Measured discrimination: not available.** No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Ulcerative colitis is diagnosed by colonoscopy and histology. This profile cannot diagnose or exclude it, and it cannot reliably separate UC from Crohn's disease.

Every direction here is shared with the general dysbiosis pattern. Read it beside the shape analysis and the crohns and ibd scores.

Sources: Machiels K, et al. A decrease of the butyrate-producing species Roseburia hominis and Faecalibacterium prausnitzii defines dysbiosis in patients with ulcerative colitis. Gut 2014;63:1275-1283; Lloyd-Price J, et al. Nature 2019;569:655-662; Franzosa EA, et al. Nat Microbiol 2019;4:293-305.

‹ at a glance

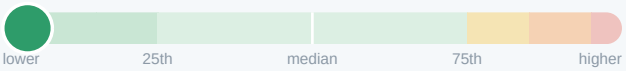
# Ankylosing spondylitis — gut pattern

NOTABLY LOW

Pattern version **1.0.0** (ankylosing_spondylitis) · maturity **P2** (usable human shotgun cohort, or multiple compatible human association cohorts) · research beta · chinese_cohort_derived

Up to 60% of ankylosing spondylitis patients have subclinical gut inflammation, and about 10% develop overt IBD, so the AS gut pattern overlaps with Crohn's disease. Wen et al. shotgun-sequenced 97 AS patients and 114 controls in China and found Prevotella species, Bacteroides coprophilus and Bifidobacterium enriched with Roseburia, Faecalibacterium and Eubacterium depleted; Zhou et al. replicated in a second Chinese cohort with Bifidobacterium and Klebsiella higher, and Costello's Australian ileal biopsy study found Lachnospiraceae, Ruminococcaceae, Bacteroidaceae and Porphyromonadaceae differences. The HLA-B27 host genotype itself shapes the gut community. TNF inhibitors, the standard AS treatment, partly normalise the pattern.

**COMBINED RESEMBLANCE**
percentile of the matched reference group




**1st**
likely range 0–3

## What is driving it

| | | | |
|---|---|---|---|
| **Wen 2017 Chinese AS shotgun (97 / 114)**<br>species · study wen_2017 · 8/8 features · 50% of combined | 5th | NOTABLY LOW |
| **Zhou 2020 Chinese AS shotgun (replication)**<br>species · study zhou_2020 · 5/5 features · 50% of combined | 3rd | NOTABLY LOW |
| **Metabolite gene capacity**<br>genes · genomic capacity · study functional · 1/1 features · not in combined | 18th | BELOW AVERAGE |
| **Hexa-LPS and butyrate carriers**<br>carriers · carrier proxy · study carriers · 2/2 features · not in combined | 2nd | NOTABLY LOW |

## How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

| | | |
|---|---|---|
| **Technical reliability** | 1.00 | sequencing depth, host fraction, reference match |
| **Independent evidence coverage** | 1.00 | share of the pattern's independent evidence clusters this sample could measure |
| **Evidence maturity** | P2 | usable human shotgun cohort, or multiple compatible human association cohorts |
| **Assay transportability** | native | whether the studies measured what this sample measured |
| **Population match** | 0.60 | how well the reference group matches you |
| **Specificity** | unknown | does this pattern separate its condition from general disturbance? measured where labelled data exist |
| **Direction conflict** | low | studies disagreeing on direction for one or more features |

**Competing patterns** scoring nearly as high: ad clinical (88th), alopecia areata (47th), acvd (39th). Margin over the next pattern: 8 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

## Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| **Wen 2017 Chinese AS shotgun (97 / 114)** · species · study wen_2017 | | | | | | | | | |

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| *Prevotella copri* | not detected | 25 | ▲ | 0.80 | 0.70 | 0.30 | 0.70 | -0.23 | |
| *Prevotella stercorea* | not detected | 40 | ▲ | 0.50 | 0.60 | 0.40 | 0.60 | -0.08 | |
| *Bacteroides coprophilus* | not detected | 47 | ▲ | 0.60 | 0.60 | 0.50 | 0.60 | -0.02 | |
| *Bifidobacterium adolescentis* | not detected | 25 | ▲ | 0.60 | 0.60 | 0.40 | 0.60 | -0.23 | |
| *Roseburia intestinalis* | 0.350% | 68 | ▼ | 0.70 | 0.70 | 0.30 | 0.70 | -0.16 | |
| *Faecalibacterium prausnitzii* | 9.840% | 81 | ▼ | 0.70 | 0.70 | 0.20 | 0.70 | -0.31 | |
| *Eubacterium rectale* | 1.082% | 52 | ▼ | 0.60 | 0.70 | 0.20 | 0.70 | -0.02 | |
| *Eubacterium eligens* | 10.449% | 97 | ▼ | 0.50 | 0.60 | 0.40 | 0.70 | -0.48 | |
| **Zhou 2020 Chinese AS shotgun (replication)** · species · study zhou_2020 | | | | | | | | | |
| *Bifidobacterium adolescentis* | not detected | 25 | ▲ | 0.60 | 0.60 | 0.40 | 0.60 | -0.23 | |
| *Bifidobacterium longum* | not detected | 17 | ▲ | 0.50 | 0.60 | 0.30 | 0.60 | -0.32 | |
| *Klebsiella pneumoniae* | not detected | 40 | ▲ | 0.60 | 0.60 | 0.40 | 0.60 | -0.09 | |
| *Faecalibacterium prausnitzii* | 9.840% | 81 | ▼ | 0.60 | 0.70 | 0.20 | 0.70 | -0.31 | |
| *Roseburia hominis* | 1.234% | 93 | ▼ | 0.50 | 0.60 | 0.30 | 0.70 | -0.41 | |
| **Metabolite gene capacity** · genes · genomic capacity · study functional | | | | | | | | | |
| butyrate | 32.1/100 | 82 | ▼ | 0.80 | 0.70 | 0.20 | 0.70 | -0.26 | |
| **Hexa-LPS and butyrate carriers** · carriers · carrier proxy · study carriers | | | | | | | | | |
| hexa lps | 0.000% | 7 | ▲ | 0.60 | 0.60 | 0.30 | 0.60 | -0.84 | |
| butyrate | 45.230% | 87 | ▼ | 0.60 | 0.70 | 0.20 | 0.70 | -0.43 | |

FINE PRINT

**Confidence grade insufficient.** Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

**Measured discrimination: not available.** No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Ankylosing spondylitis is diagnosed by imaging of the sacroiliac joints, HLA-B27 and clinical criteria. This profile does not diagnose it.

The Prevotella direction is confounded with diet and geography; a high P. copri sample from a high-fibre diet will score toward AS here for reasons that have nothing to do with joints.

Sources: Wen C, et al. Quantitative metagenomics reveals unique gut microbiome biomarkers in ankylosing spondylitis. Genome Biol 2017;18:142; Zhou C, et al. Metagenomic profiling of the pro-inflammatory gut microbiota in ankylosing spondylitis. J Autoimmun 2020;107:102360; Costello ME, et al. Brief report: intestinal dysbiosis in ankylosing spondylitis. Arthritis Rheumatol 2015;67:686.

‹ at a glance

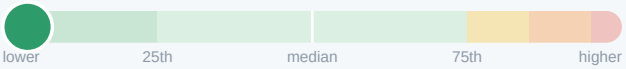
# Parkinson's disease — gut pattern

NOTABLY LOW

Pattern version **1.0.0** (parkinsons) · maturity **P3** (independent human replication with coherent task and assay) · research beta · mixed_us_european_cohorts

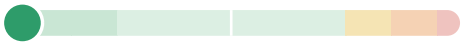
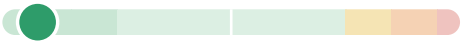
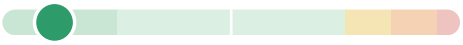
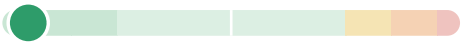
Parkinson's has the most consistently replicated gut pattern of any neurological disease. Across more than twenty cohorts and a 2022 meta- analysis of shotgun and 16S data, Akkermansia muciniphila and Bifidobacterium are elevated, and the Lachnospiraceae butyrate producers — Roseburia, Faecalibacterium, Blautia — are depleted. Wallen et al. 2022 confirmed this at species level in 490 PD cases with 234 controls by shotgun. Constipation precedes motor symptoms by years and is the most obvious confounder, since slow transit alone raises Akkermansia and methanogens.

**COMBINED RESEMBLANCE**
percentile of the matched reference group




**1st**
likely range 0–2

## What is driving it

| | | | |
|---|---|---|---|
| **Wallen 2022 US shotgun (490 PD / 234 controls)**<br>species · study wallen_2022 · 15/15 features · 33% of combined |  | 1st | NOTABLY LOW |
| **Romano 2021 meta-analysis of 10 cohorts**<br>species · association only · 16S study · study romano_2021 · 5/5 features · 33% of combined |  | 8th | LOW |
| **Boktor 2023 integrated shotgun (Wallen + Bedarf + Boktor)**<br>species · study boktor_2023 · 4/4 features · 33% of combined |  | 11th | LOW |
| **Metabolite gene capacity**<br>genes · genomic capacity · study functional · 3/3 features · not in combined |  | 14th | LOW |
| **Mucin degraders and butyrate carriers**<br>carriers · carrier proxy · study carriers · 3/3 features · not in combined |  | 6th | LOW |

## How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

| | | |
|---|---|---|
| **Technical reliability** | 1.00 | sequencing depth, host fraction, reference match |
| **Independent evidence coverage** | 1.00 | share of the pattern's independent evidence clusters this sample could measure |
| **Evidence maturity** | P3 | independent human replication with coherent task and assay |
| **Assay transportability** | labeled transport | whether the studies measured what this sample measured |
| **Population match** | 0.60 | how well the reference group matches you |
| **Specificity** | unknown | does this pattern separate its condition from general disturbance? measured where labelled data exist |
| **Direction conflict** | none | studies disagreeing on direction for one or more features |

**Competing patterns** scoring nearly as high: ra (95th), ad clinical (88th), celiac (77th). Margin over the next pattern: 8 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

## Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| **Wallen 2022 US shotgun (490 PD / 234 controls)** · species · study wallen_2022 | | | | | | | | | |
| *Akkermansia muciniphila* | not detected 0.024% pooled | 24 | ▲ | 1.00 | 0.90 | 0.50 | 0.90 | -0.23 | |
| *Bifidobacterium dentium* | not detected | 44 | ▲ | 0.80 | 0.80 | 0.50 | 0.80 | -0.05 | |
| *Bifidobacterium longum* | not detected | 17 | ▲ | 0.60 | 0.80 | 0.30 | 0.70 | -0.32 | |
| *Lactobacillus gasseri* | not detected | 49 | ▲ | 0.60 | 0.70 | 0.40 | 0.70 | -0.01 | |
| *Methanobrevibacter smithii* | not detected | 35 | ▲ | 0.70 | 0.80 | 0.50 | 0.80 | -0.13 | |
| *Escherichia coli* | not detected | 19 | ▲ | 0.60 | 0.70 | 0.20 | 0.70 | -0.29 | |
| *Klebsiella pneumoniae* | not detected | 40 | ▲ | 0.50 | 0.60 | 0.30 | 0.60 | -0.09 | |
| *Roseburia intestinalis* | 0.350% | 68 | ▼ | 0.90 | 0.90 | 0.30 | 0.90 | -0.16 | |
| *Roseburia hominis* | 1.234% | 93 | ▼ | 0.70 | 0.80 | 0.30 | 0.80 | -0.41 | |
| *Faecalibacterium prausnitzii* | 9.840% | 81 | ▼ | 0.90 | 0.90 | 0.20 | 0.90 | -0.31 | |
| *Eubacterium rectale* | 1.082% | 52 | ▼ | 0.70 | 0.80 | 0.20 | 0.80 | -0.02 | |
| *Blautia obeum* | 0.604% | 90 | ▼ | 0.60 | 0.70 | 0.40 | 0.70 | -0.37 | |
| *Blautia wexlerae* | 0.014% | 31 | ▼ | 0.50 | 0.70 | 0.40 | 0.70 | +0.16 | |
| *Coprococcus comes* | 1.842% | 96 | ▼ | 0.60 | 0.70 | 0.30 | 0.70 | -0.58 | |
| *Prevotella copri* | not detected | 25 | ▼ | 0.50 | 0.60 | 0.30 | 0.50 | +0.23 | |
| **Romano 2021 meta-analysis of 10 cohorts** · species · association only · 16S study · study romano_2021 | | | | | | | | | |
| *Akkermansia muciniphila* | not detected 0.024% pooled | 24 | ▲ | 0.90 | 0.80 | 0.50 | 0.90 | -0.23 | |
| *Bifidobacterium longum* | not detected | 17 | ▲ | 0.60 | 0.70 | 0.30 | 0.70 | -0.32 | |
| *Lactobacillus gasseri* | not detected | 49 | ▲ | 0.50 | 0.60 | 0.40 | 0.60 | -0.01 | |
| *Roseburia intestinalis* | 0.350% | 68 | ▼ | 0.80 | 0.80 | 0.30 | 0.80 | -0.16 | |
| *Faecalibacterium prausnitzii* | 9.840% | 81 | ▼ | 0.80 | 0.80 | 0.20 | 0.80 | -0.31 | |
| **Boktor 2023 integrated shotgun (Wallen + Bedarf + Boktor)** · species · study boktor_2023 | | | | | | | | | |
| *Akkermansia muciniphila* | not detected 0.024% pooled | 24 | ▲ | 0.80 | 0.80 | 0.50 | 0.80 | -0.23 | |
| *Bifidobacterium dentium* | not detected | 44 | ▲ | 0.60 | 0.70 | 0.50 | 0.70 | -0.05 | |
| *Roseburia intestinalis* | 0.350% | 68 | ▼ | 0.70 | 0.80 | 0.30 | 0.80 | -0.16 | |
| *Faecalibacterium prausnitzii* | 9.840% | 81 | ▼ | 0.70 | 0.80 | 0.20 | 0.80 | -0.31 | |
| **Metabolite gene capacity** · genes · genomic capacity · study functional | | | | | | | | | |
| butyrate | 32.1/100 | 82 | ▼ | 1.00 | 0.80 | 0.20 | 0.80 | -0.26 | |
| dopamine | 0.8/100 | 34 | ▲ | 0.60 | 0.60 | 0.60 | 0.70 | -0.11 | |
| methane | 0.0/100 | 31 | ▲ | 0.60 | 0.70 | 0.40 | 0.70 | -0.17 | |
| **Mucin degraders and butyrate carriers** · carriers · carrier proxy · study carriers | | | | | | | | | |
| mucin degraders | 1.464% | 33 | ▲ | 0.70 | 0.70 | 0.40 | 0.70 | -0.14 | |
| butyrate | 45.230% | 87 | ▼ | 0.80 | 0.80 | 0.20 | 0.80 | -0.43 | |
| methanogens | 0.000% | 35 | ▲ | 0.60 | 0.70 | 0.40 | 0.70 | -0.13 | |

FINE PRINT

**Confidence grade insufficient.** Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

**Measured discrimination: not available.** No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Parkinson's disease is a clinical diagnosis. No stool test predicts it, and this profile is not a prodromal screen.

Chronic constipation of any cause produces most of this pattern. A high score in someone with slow transit is expected and says nothing about Parkinson's.

Bifidobacterium and Akkermansia are higher here — the opposite of most gut disease patterns — which is what gives this profile its unusually good specificity across the library.

Sources: Wallen ZD, et al. Metagenomics of Parkinson's disease implicates the gut microbiome in multiple disease mechanisms. Nat Commun 2022;13:6958; Romano S, et al. Meta-analysis of the Parkinson's disease gut microbiome suggests alterations linked to intestinal inflammation. NPJ Parkinsons Dis 2021;7:27; Boktor JC, et al. Integrated multi-cohort analysis of the Parkinson's disease gut metagenome. Mov Disord 2023;38:399-409; van Kessel SP, et al. Gut bacterial tyrosine decarboxylases restrict levels of levodopa in the treatment of Parkinson's disease. Nat Commun 2019;10:310.

‹ at a glance

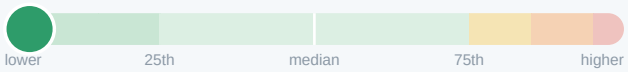
# Inflammatory bowel disease (composite) — gut pattern

NOTABLY LOW

Pattern version **1.0.0** (ibd) · maturity **P3** (independent human replication with coherent task and assay) · production candidate · mixed_north_american_european_cohorts

The composite IBD pattern: what Crohn's disease and ulcerative colitis share. This is the profile validated against labelled cohorts in the profile_validation file, because the public cohorts label IBD far more often than they separate CD from UC. It scores the core replicated directions — butyrate-producer depletion, Ruminococcus gnavus and Escherichia coli expansion, diversity loss — and leaves the CD-specific oral-origin taxa and the UC-specific sulfide story to the two disease-specific profiles. When this profile is high and the shape analysis reads specific, the sample resembles the IBD dysbiotic state. When many profiles are high together it is dysbiosis of some cause, and IBD is one of many.

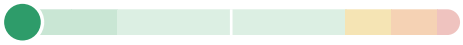
**COMBINED RESEMBLANCE**
percentile of the matched reference group




**<1st**
likely range 0–1

## What is driving it

**Franzosa 2019 PRISM + Dutch validation**
species · study franzosa_2019 · 7/7 features · 33% of combined



4th — NOTABLY LOW

**Hall 2017 R. gnavus clade expansion**
species · study hall_2017 · 2/2 features · 33% of combined



12th — LOW

**HMP2 IBDMDB core dysbiosis directions**
species · study hmp2_2019 · 10/10 features · 17% of combined



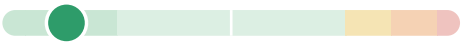
8th — LOW

**Community ecology**
diversity · study hmp2_2019 · 1/2 features · 17% of combined



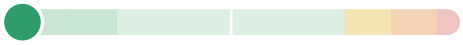

— SCORED LOW COVERAGE

**Metabolite gene capacity**
genes · genomic capacity · study functional · 4/4 features · not in combined



14th — LOW

**Carrier cross-check**
carriers · carrier proxy · study carriers · 2/2 features · not in combined



2nd — NOTABLY LOW

## How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

| | | |
|---|---|---|
| **Technical reliability** | 1.00 | sequencing depth, host fraction, reference match |
| **Independent evidence coverage** | 0.90 | share of the pattern's independent evidence clusters this sample could measure |
| **Evidence maturity** | P3 | independent human replication with coherent task and assay |
| **Assay transportability** | native | whether the studies measured what this sample measured |
| **Population match** | 0.60 | how well the reference group matches you |
| **Specificity** | unknown | does this pattern separate its condition from general disturbance? measured where labelled data exist |
| **Direction conflict** | none | studies disagreeing on direction for one or more features |

**Competing patterns** scoring nearly as high: ra (95th), ad clinical (88th), celiac (77th). Margin over the next pattern: 8 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

## Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| **Franzosa 2019 PRISM + Dutch validation** · species · study franzosa_2019 | | | | | | | | | |
| *Ruminococcus gnavus* | not detected | 27 | ▲ | 0.90 | 0.80 | 0.30 | 0.90 | -0.21 | |
| *Escherichia coli* | not detected | 19 | ▲ | 0.80 | 0.80 | 0.30 | 0.90 | -0.29 | |
| *Clostridium bolteae* | not detected | 29 | ▲ | 0.60 | 0.70 | 0.40 | 0.70 | -0.18 | |
| *Faecalibacterium prausnitzii* | 9.840% | 81 | ▼ | 0.90 | 0.90 | 0.20 | 0.90 | -0.31 | |
| *Roseburia intestinalis* | 0.350% | 68 | ▼ | 0.70 | 0.80 | 0.20 | 0.80 | -0.16 | |
| *Alistipes shahii* | 0.012% | 39 | ▼ | 0.50 | 0.60 | 0.40 | 0.70 | +0.10 | |
| *Eubacterium eligens* | 10.449% | 97 | ▼ | 0.60 | 0.70 | 0.30 | 0.70 | -0.48 | |
| **Hall 2017 R. gnavus clade expansion** · species · study hall_2017 | | | | | | | | | |
| *Ruminococcus gnavus* | not detected | 27 | ▲ | 1.00 | 0.80 | 0.30 | 0.90 | -0.21 | |
| *Faecalibacterium prausnitzii* | 9.840% | 81 | ▼ | 0.70 | 0.80 | 0.20 | 0.80 | -0.31 | |
| **HMP2 IBDMDB core dysbiosis directions** · species · study hmp2_2019 | | | | | | | | | |
| *Ruminococcus gnavus* | not detected | 27 | ▲ | 1.00 | 0.90 | 0.30 | 0.90 | -0.21 | |
| *Escherichia coli* | not detected | 19 | ▲ | 0.90 | 0.90 | 0.30 | 0.90 | -0.29 | |
| *Ruminococcus torques* | 0.743% | 82 | ▲ | 0.60 | 0.70 | 0.40 | 0.70 | +0.30 | |
| *Clostridium bolteae* | not detected | 29 | ▲ | 0.60 | 0.70 | 0.40 | 0.70 | -0.18 | |
| *Faecalibacterium prausnitzii* | 9.840% | 81 | ▼ | 1.00 | 0.90 | 0.20 | 0.90 | -0.31 | |
| *Roseburia hominis* | 1.234% | 93 | ▼ | 0.80 | 0.90 | 0.30 | 0.90 | -0.41 | |
| *Roseburia intestinalis* | 0.350% | 68 | ▼ | 0.80 | 0.80 | 0.20 | 0.80 | -0.16 | |
| *Eubacterium rectale* | 1.082% | 52 | ▼ | 0.80 | 0.80 | 0.20 | 0.80 | -0.02 | |
| *Alistipes putredinis* | 1.648% | 63 | ▼ | 0.60 | 0.70 | 0.40 | 0.70 | -0.11 | |
| *Bifidobacterium adolescentis* | not detected | 25 | ▼ | 0.60 | 0.70 | 0.30 | 0.70 | +0.23 | |
| **Community ecology** · diversity · study hmp2_2019 | | | | | | | | | |
| shannon | not detected | — | ▼ | 0.80 | 0.90 | 0.10 | 0.90 | — | |
| richness | 123.00 | 93 | ▼ | 0.60 | 0.90 | 0.10 | 0.90 | -0.55 | |
| **Metabolite gene capacity** · genes · genomic capacity · study functional | | | | | | | | | |
| butyrate | 32.1/100 | 82 | ▼ | 1.00 | 0.90 | 0.20 | 0.90 | -0.26 | |
| bsh | 123.0/100 | 96 | ▼ | 0.50 | 0.70 | 0.30 | 0.70 | -0.63 | |
| bai | 3.5/100 | 72 | ▼ | 0.50 | 0.70 | 0.30 | 0.70 | -0.30 | |
| h2s | 96.5/100 | 80 | ▲ | 0.50 | 0.60 | 0.30 | 0.60 | +0.35 | |
| **Carrier cross-check** · carriers · carrier proxy · study carriers | | | | | | | | | |
| butyrate | 45.230% | 87 | ▼ | 0.80 | 0.80 | 0.20 | 0.80 | -0.43 | |
| hexa lps | 0.000% | 7 | ▲ | 0.60 | 0.70 | 0.30 | 0.70 | -0.84 | |

FINE PRINT

**Confidence grade moderate.** Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

**Measured discrimination: not available.** No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

IBD is diagnosed by endoscopy, histology and imaging. This profile does not diagnose or exclude it. Calprotectin is the stool test that matters clinically, and it is a protein, not DNA.

This is the general dysbiosis pattern. Antibiotics, bowel preparation, gastroenteritis, hospitalisation and low-fibre diets all produce it.

Sources: Lloyd-Price J, et al. Nature 2019;569:655-662; Franzosa EA, et al. Nat Microbiol 2019;4:293-305; Hall AB, et al. A novel Ruminococcus gnavus clade enriched in inflammatory bowel disease patients. Genome Med 2017;9:103.

‹ at a glance

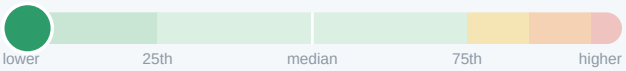
# Crohn's disease — gut pattern

NOTABLY LOW

Pattern version **1.0.0** (crohns) · maturity **P3** (independent human replication with coherent task and assay) · research beta · mixed_north_american_european_cohorts

The best-replicated disease pattern in the gut microbiome literature after colorectal cancer, with independent shotgun cohorts in North America and Europe agreeing on direction for a core set of species: depletion of Faecalibacterium prausnitzii and the Roseburia and Eubacterium butyrate producers, and expansion of Escherichia coli, Ruminococcus gnavus and the oral-origin Veillonella and Haemophilus. Treatment-naive paediatric cohorts show the same pattern, so it is not a treatment artefact. The pattern is strongest in a dysbiotic subgroup. The HMP2 study found IBD patients moved in and out of a dysbiotic state over time, and at any one sampling many Crohn's patients look like controls. A high score here is also the general dysbiosis pattern, so read it with the shape analysis.

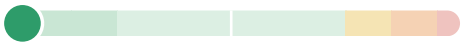
**COMBINED RESEMBLANCE**
percentile of the matched reference group




**<1st**
likely range 0–1

## What is driving it

**Gevers 2014 RISK treatment-naive paediatric cohort**
species · study gevers_2014 · 10/10 features · 25% of combined



4th NOTABLY LOW

**Franzosa 2019 PRISM + validation cohort**
species · study franzosa_2019 · 7/7 features · 25% of combined



8th LOW

**Pascal 2017 Spanish / Belgian / German cohorts**
species · association only · 16S study · study pascal_2017 · 7/7 features · 25% of combined



4th NOTABLY LOW

**HMP2 IBDMDB longitudinal multi-omics**
species · study hmp2_2019 · 8/8 features · 12% of combined



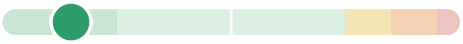
5th LOW

**Community ecology**
diversity · study hmp2_2019 · 1/2 features · 12% of combined



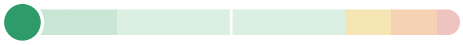

— SCORED LOW COVERAGE

**Metabolite gene capacity**
genes · genomic capacity · study functional · 5/5 features · not in combined



15th LOW

**Butyrate carriers and hexa-LPS carriers**
carriers · carrier proxy · study carriers · 2/2 features · not in combined



2nd NOTABLY LOW

## How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

| | | |
|---|---|---|
| **Technical reliability** | 1.00 | sequencing depth, host fraction, reference match |
| **Independent evidence coverage** | 0.93 | share of the pattern's independent evidence clusters this sample could measure |
| **Evidence maturity** | P3 | independent human replication with coherent task and assay |
| **Assay transportability** | labeled transport | whether the studies measured what this sample measured |
| **Population match** | 0.60 | how well the reference group matches you |
| **Specificity** | unknown | does this pattern separate its condition from general disturbance? measured where labelled data exist |
| **Direction conflict** | none | studies disagreeing on direction for one or more features |

**Competing patterns** scoring nearly as high: ra (95th), ad clinical (88th), celiac (77th). Margin over the next pattern: 8 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

## Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| **Gevers 2014 RISK treatment-naive paediatric cohort** · species · study gevers_2014 | | | | | | | | | |
| *Escherichia coli* | not detected | 19 | ▲ | 1.00 | 0.80 | 0.30 | 0.90 | -0.29 | |
| *Veillonella parvula* | not detected | 31 | ▲ | 0.80 | 0.70 | 0.40 | 0.80 | -0.17 | |
| *Haemophilus parainfluenzae* | not detected | 29 | ▲ | 0.80 | 0.70 | 0.40 | 0.80 | -0.18 | |
| *Fusobacterium nucleatum* | not detected | 49 | ▲ | 0.60 | 0.60 | 0.30 | 0.70 | -0.01 | |
| *Faecalibacterium prausnitzii* | 9.840% | 81 | ▼ | 1.00 | 0.90 | 0.20 | 0.90 | -0.31 | |
| *Bifidobacterium adolescentis* | not detected | 25 | ▼ | 0.70 | 0.70 | 0.30 | 0.80 | +0.23 | |
| *Eubacterium rectale* | 1.082% | 52 | ▼ | 0.80 | 0.80 | 0.20 | 0.80 | -0.02 | |
| *Roseburia intestinalis* | 0.350% | 68 | ▼ | 0.80 | 0.80 | 0.20 | 0.80 | -0.16 | |
| *Coprococcus comes* | 1.842% | 96 | ▼ | 0.60 | 0.70 | 0.30 | 0.70 | -0.58 | |
| *Dorea longicatena* | 1.935% | 92 | ▼ | 0.50 | 0.60 | 0.30 | 0.70 | -0.42 | |
| **Franzosa 2019 PRISM + validation cohort** · species · study franzosa_2019 | | | | | | | | | |
| *Ruminococcus gnavus* | not detected | 27 | ▲ | 0.90 | 0.80 | 0.30 | 0.90 | -0.21 | |
| *Escherichia coli* | not detected | 19 | ▲ | 0.80 | 0.80 | 0.30 | 0.90 | -0.29 | |
| *Clostridium bolteae* | not detected | 29 | ▲ | 0.60 | 0.70 | 0.40 | 0.70 | -0.18 | |
| *Roseburia intestinalis* | 0.350% | 68 | ▼ | 0.70 | 0.80 | 0.20 | 0.80 | -0.16 | |
| *Faecalibacterium prausnitzii* | 9.840% | 81 | ▼ | 0.90 | 0.90 | 0.20 | 0.90 | -0.31 | |
| *Alistipes putredinis* | 1.648% | 63 | ▼ | 0.60 | 0.70 | 0.40 | 0.70 | -0.11 | |
| *Alistipes shahii* | 0.012% | 39 | ▼ | 0.50 | 0.60 | 0.40 | 0.70 | +0.10 | |
| **Pascal 2017 Spanish / Belgian / German cohorts** · species · association only · 16S study · study pascal_2017 | | | | | | | | | |
| *Fusobacterium nucleatum* | not detected | 49 | ▲ | 0.60 | 0.60 | 0.30 | 0.70 | -0.01 | |
| *Escherichia coli* | not detected | 19 | ▲ | 0.70 | 0.70 | 0.30 | 0.80 | -0.29 | |
| *Faecalibacterium prausnitzii* | 9.840% | 81 | ▼ | 0.80 | 0.70 | 0.20 | 0.80 | -0.31 | |
| *Methanobrevibacter smithii* | not detected | 35 | ▼ | 0.70 | 0.70 | 0.50 | 0.80 | +0.13 | |
| *Christensenella minuta* | not detected | 46 | ▼ | 0.60 | 0.60 | 0.50 | 0.70 | +0.03 | |
| *Collinsella aerofaciens* | 0.885% | 72 | ▼ | 0.50 | 0.60 | 0.40 | 0.70 | -0.15 | |
| *Anaerostipes hadrus* | 11.119% | 98 | ▼ | 0.60 | 0.60 | 0.30 | 0.70 | -0.66 | |
| **HMP2 IBDMDB longitudinal multi-omics** · species · study hmp2_2019 | | | | | | | | | |
| *Ruminococcus gnavus* | not detected | 27 | ▲ | 1.00 | 0.90 | 0.30 | 0.90 | -0.21 | |
| *Escherichia coli* | not detected | 19 | ▲ | 0.90 | 0.80 | 0.30 | 0.90 | -0.29 | |
| *Ruminococcus torques* | 0.743% | 82 | ▲ | 0.60 | 0.70 | 0.40 | 0.70 | +0.30 | |
| *Clostridium bolteae* | not detected | 29 | ▲ | 0.60 | 0.70 | 0.40 | 0.70 | -0.18 | |
| *Faecalibacterium prausnitzii* | 9.840% | 81 | ▼ | 1.00 | 0.90 | 0.20 | 0.90 | -0.31 | |
| *Roseburia hominis* | 1.234% | 93 | ▼ | 0.70 | 0.80 | 0.30 | 0.80 | -0.41 | |
| *Alistipes putredinis* | 1.648% | 63 | ▼ | 0.60 | 0.70 | 0.40 | 0.70 | -0.11 | |
| *Eubacterium siraeum* | 1.213% | 85 | ▼ | 0.50 | 0.60 | 0.40 | 0.70 | -0.34 | |
| **Community ecology** · diversity · study hmp2_2019 | | | | | | | | | |
| shannon | not detected | — | ▼ | 0.80 | 0.90 | 0.10 | 0.90 | — | |
| richness | 123.00 | 93 | ▼ | 0.60 | 0.90 | 0.10 | 0.90 | -0.55 | |
| **Metabolite gene capacity** · genes · genomic capacity · study functional | | | | | | | | | |
| butyrate | 32.1/100 | 82 | ▼ | 1.00 | 0.90 | 0.20 | 0.90 | -0.26 | |
| bsh | 123.0/100 | 96 | ▼ | 0.60 | 0.70 | 0.30 | 0.70 | -0.63 | |
| bai | 3.5/100 | 72 | ▼ | 0.60 | 0.70 | 0.30 | 0.70 | -0.30 | |
| h2s | 96.5/100 | 80 | ▲ | 0.50 | 0.60 | 0.30 | 0.60 | +0.35 | |
| urease | 16.3/100 | 70 | ▲ | 0.50 | 0.60 | 0.30 | 0.60 | +0.25 | Ni J, et al. Sci Transl Med 2017;9:eaah6888 (urease and CD dysbiosis) |
| **Butyrate carriers and hexa-LPS carriers** · carriers · carrier proxy · study carriers | | | | | | | | | |
| butyrate | 45.230% | 87 | ▼ | 0.80 | 0.80 | 0.20 | 0.80 | -0.43 | |

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| hexa lps | 0.000% | 7 | ▲ | 0.60 | 0.70 | 0.30 | 0.70 | -0.84 | Vatanen T, et al. Cell 2016;165:842 (hexa- vs penta-acylated LPS immunogenicity) |

**FINE PRINT**

**Confidence grade moderate.** Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

**Measured discrimination: not available.** No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Crohn's disease is diagnosed by endoscopy, histology and imaging. This profile cannot diagnose it, exclude it, or distinguish it from ulcerative colitis with any reliability.

The Crohn's pattern is the general dysbiosis pattern with oral-origin taxa and E. coli on top. Antibiotics, bowel prep, gastroenteritis and hospitalisation produce it. If the shape analysis reads diffuse, this score is dysbiosis, not Crohn's.

Many people with Crohn's disease have a community indistinguishable from controls at any one sampling.

Sources: Gevers D, et al. The treatment-naive microbiome in new-onset Crohn's disease. Cell Host Microbe 2014;15:382-392; Lloyd-Price J, et al. Multi-omics of the gut microbial ecosystem in inflammatory bowel diseases. Nature 2019;569:655-662; Franzosa EA, et al. Gut microbiome structure and metabolic activity in inflammatory bowel disease. Nat Microbiol 2019;4:293-305; Pascal V, et al. A microbial signature for Crohn's disease. Gut 2017;66:813-822.

‹ at a glance

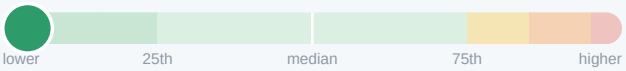
# Metabolic dysfunction-associated steatotic liver disease — gut pattern

NOTABLY LOW

Pattern version **1.0.0** (masld) · maturity **P2** (usable human shotgun cohort, or multiple compatible human association cohorts) · research beta · mixed_us_european_cohorts

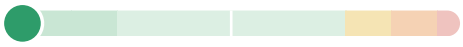
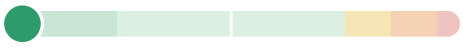
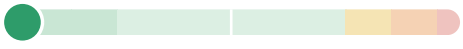
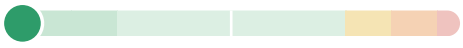
Fatty liver disease (formerly NAFLD) is common enough that its gut pattern is well sampled, and confounded enough by adiposity and diabetes that the pattern is hard to attribute. Loomba et al. built a shotgun signature for advanced fibrosis in biopsy-proven patients — Escherichia coli and Bacteroides vulgatus higher, Eubacterium rectale lower — and Hoyles et al. linked hepatic steatosis in non-diabetic obese women to low gene richness and higher branched-chain amino acid and aromatic amino acid biosynthesis potential. Endogenous ethanol production by Klebsiella pneumoniae strains is a reproducible mechanism in a subset of patients (Yuan 2019). Obesity and T2D are mutual challenge sets because most MASLD patients have one or both.

**COMBINED RESEMBLANCE**
percentile of the matched reference group




**<1st**
likely range 0–2

## What is driving it

| | | | |
|---|---|---|---|
| **Loomba 2017 biopsy-proven NAFLD (fibrosis signature)**<br>species · study loomba_2017 · 4/4 features · 33% of combined |  | 2nd | NOTABLY LOW |
| **Yuan 2019 high-alcohol-producing Klebsiella**<br>species · study yuan_2019 · 1/1 features · 33% of combined |  | 0 | NOTABLY LOW |
| **Hoyles 2018 FLORINASH non-diabetic obese women**<br>species · study hoyles_2018 · 3/3 features · 17% of combined |  | 2nd | NOTABLY LOW |
| **Gene richness**<br>diversity · study hoyles_2018 · 1/2 features · 17% of combined |  | 7th | LOW |
| **Metabolite gene capacity**<br>genes · genomic capacity · study functional · 4/4 features · not in combined |  | 46th | TYPICAL |
| **Hexa-LPS carriers**<br>carriers · carrier proxy · study carriers · 2/2 features · not in combined |  | 2nd | NOTABLY LOW |

## How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

| | | |
|---|---|---|
| **Technical reliability** | 1.00 | sequencing depth, host fraction, reference match |
| **Independent evidence coverage** | 0.94 | share of the pattern's independent evidence clusters this sample could measure |
| **Evidence maturity** | P2 | usable human shotgun cohort, or multiple compatible human association cohorts |
| **Assay transportability** | native | whether the studies measured what this sample measured |
| **Population match** | 0.60 | how well the reference group matches you |
| **Specificity** | unknown | does this pattern separate its condition from general disturbance? measured where labelled data exist |
| **Direction conflict** | none | studies disagreeing on direction for one or more features |

**Competing patterns** scoring nearly as high: ra (95th), ad clinical (88th), celiac (77th). Margin over the next pattern: 8 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

## Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| **Loomba 2017 biopsy-proven NAFLD (fibrosis signature)** · species · study loomba_2017 | | | | | | | | | |
| *Escherichia coli* | not detected | 19 | ▲ | 0.90 | 0.70 | 0.30 | 0.80 | -0.29 | |
| *Bacteroides vulgatus* | not detected | 3 | ▲ | 0.80 | 0.70 | 0.30 | 0.70 | -1.00 | |
| *Eubacterium rectale* | 1.082% | 52 | ▼ | 0.80 | 0.70 | 0.20 | 0.80 | -0.02 | |
| *Blautia obeum* | 0.604% | 90 | ▼ | 0.50 | 0.60 | 0.40 | 0.60 | -0.37 | |
| **Yuan 2019 high-alcohol-producing Klebsiella** · species · study yuan_2019 | | | | | | | | | |
| *Klebsiella pneumoniae* | not detected | 40 | ▲ | 0.60 | 0.60 | 0.40 | 0.50 | -0.09 | |
| **Hoyles 2018 FLORINASH non-diabetic obese women** · species · study hoyles_2018 | | | | | | | | | |
| *Escherichia coli* | not detected | 19 | ▲ | 0.60 | 0.60 | 0.30 | 0.60 | -0.29 | |
| *Eubacterium eligens* | 10.449% | 97 | ▼ | 0.60 | 0.60 | 0.40 | 0.70 | -0.48 | |
| *Faecalibacterium prausnitzii* | 9.840% | 81 | ▼ | 0.60 | 0.60 | 0.20 | 0.70 | -0.31 | |
| **Gene richness** · diversity · study hoyles_2018 | | | | | | | | | |
| richness | 123.00 | 93 | ▼ | 0.80 | 0.70 | 0.10 | 0.80 | -0.55 | |
| shannon | not detected | — | ▼ | 0.50 | 0.70 | 0.10 | 0.80 | — | |
| **Metabolite gene capacity** · genes · genomic capacity · study functional | | | | | | | | | |
| bcaa | 180.5/100 | 34 | ▲ | 0.80 | 0.70 | 0.40 | 0.70 | -0.15 | |
| bai | 3.5/100 | 72 | ▲ | 0.60 | 0.60 | 0.30 | 0.60 | +0.30 | |
| butyrate | 32.1/100 | 82 | ▼ | 0.70 | 0.70 | 0.20 | 0.80 | -0.26 | |
| urda | 53.5/100 | 87 | ▲ | 0.40 | 0.50 | 0.40 | 0.50 | +0.32 | |
| **Hexa-LPS carriers** · carriers · carrier proxy · study carriers | | | | | | | | | |
| hexa lps | 0.000% | 7 | ▲ | 0.70 | 0.70 | 0.30 | 0.70 | -0.84 | |
| butyrate | 45.230% | 87 | ▼ | 0.60 | 0.70 | 0.20 | 0.80 | -0.43 | |

FINE PRINT

**Confidence grade insufficient.** Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

**Measured discrimination: not available.** No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Steatotic liver disease is diagnosed by imaging or elastography and staged by biopsy or fibrosis scores. This profile does none of that.

The pattern is largely the obesity and metabolic-syndrome pattern. A high score in someone with obesity is expected and not informative about the liver.

Sources: Loomba R, et al. Gut microbiome-based metagenomic signature for non-invasive detection of advanced fibrosis in human nonalcoholic fatty liver disease. Cell Metab 2017;25:1054-1062; Hoyles L, et al. Molecular phenomics and metagenomics of hepatic steatosis in non-diabetic obese women. Nat Med 2018;24:1070-1080; Yuan J, et al. Fatty liver disease caused by high-alcohol-producing Klebsiella pneumoniae. Cell Metab 2019;30:675-688.

‹ at a glance

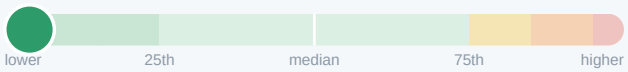
# Major depressive disorder — gut pattern

NOTABLY LOW

Pattern version **1.0.0** (mdd) · maturity **P2** (usable human shotgun cohort, or multiple compatible human association cohorts) · research beta · mixed_european_chinese_cohorts

Depression is scored and reported like every other pattern here, because the gut evidence for it is published and specific enough to measure. It remains a resemblance score, not a mood measurement: depression is diagnosed by a clinician, and a stool pattern cannot make or exclude that diagnosis. Transfer experiments are why this pattern is worth measuring rather than merely noting. Stool from people with major depression induced depression-like behaviour in germ-free mice, and the recipients took on the donors' carbohydrate and amino-acid metabolic shifts (Zheng 2016). The result has been reproduced in microbiota-depleted rats, which developed anhedonia, anxiety-like behaviour and altered tryptophan metabolism (Kelly 2016); in germ-free mice given communities from women with prenatal depression, which added hippocampal neuroinflammation and raised plasma LPS (Cao 2025); and in adolescent rats given pooled paediatric MDD stool, which showed reduced sucrose preference and fourteen amino-acid deficiencies in the prefrontal cortex (Teng 2025). One human-donor experiment cuts the other way, and it is why the wording here stays careful: in Flinders rats, healthy-donor recipients did better than depression-donor recipients, but the depression-donor group was no worse than untreated controls — so that gap may be benefit from healthy microbes rather than harm from patient microbes (Knudsen 2021). The case-control evidence is real but modest. Valles-Colomer et al. found in two large population cohorts that Coprococcus and Dialister were depleted in depression, and that butyrate-producing Faecalibacterium and Coprococcus tracked higher quality-of-life scores; they also showed the microbial capacity to synthesise the dopamine metabolite DOPAC was associated with quality of life. Zheng et al. found Bacteroides higher and Blautia, Eubacterium lower in a Chinese MDD cohort with faecal transfer inducing depression-like behaviour in germ-free mice. Antidepressants themselves have antimicrobial activity and reshape the community.

**COMBINED RESEMBLANCE**
percentile of the matched reference group




**<1st**
likely range 0–4

## What is driving it

**Valles-Colomer 2019 Flemish Gut Flora Project + LifeLines DEEP**
species · association only · 16S study · study valles_colomer_2019 · 4/4 features · 33% of combined



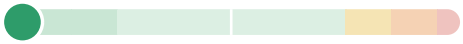
2nd — NOTABLY LOW

**Zheng 2016 Chinese MDD (16S, GF-mouse transfer)**
species · association only · 16S study · study zheng_2016 · 3/3 features · 33% of combined



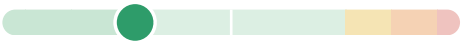
4th — NOTABLY LOW

**Yang 2020 Chinese MDD shotgun + metabolomics**
species · study yang_2020 · 5/5 features · 33% of combined



4th — NOTABLY LOW

**Neuroactive metabolite capacity**
genes · genomic capacity · study functional · 4/4 features · not in combined



29th — TYPICAL

## How much to trust it

Confidence is reported beside the score, never multiplied into it. Each row answers a different question; a weak row means 'less certain', not 'lower'.

| | | |
|---|---|---|
| **Technical reliability** | 1.00 | sequencing depth, host fraction, reference match |
| **Independent evidence coverage** | 1.00 | share of the pattern's independent evidence clusters this sample could measure |
| **Evidence maturity** | P2 | usable human shotgun cohort, or multiple compatible human association cohorts |
| **Assay transportability** | labeled transport | whether the studies measured what this sample measured |
| **Population match** | 0.60 | how well the reference group matches you |
| **Specificity** | unknown | does this pattern separate its condition from general disturbance? measured where labelled data exist |
| **Direction conflict** | material | studies disagreeing on direction for one or more features |

**Competing patterns** scoring nearly as high: ra (95th), ad clinical (88th), celiac (77th). Margin over the next pattern: 8 points. Where several patterns share features, a high score on one is not evidence for it over the others.

no matched reference stratum; full cohort used

## Every feature that produced this score

Direction is what the research reports for the condition (▲ higher, ▼ lower). **v** is how far this sample moves in that direction, from –1 (opposite) to +1 (strongly concordant). **w** is the feature's weight in its module; q, s, c (cohort independence, specificity, cross-study agreement) are recorded for audit and not multiplied into the score.

Every species value here is the reference catalogue's own measurement (MetaPhlAn 3, the catalogue the reference cohort was profiled with), because that is what makes a percentile against that cohort mean anything. It is not the organism's share of the community in section 6, which pools nine detection methods and the current catalogues: a species that catalogue has no genome for reads as *absent* here and still has a share there; where that is so, the pooled share is printed beneath. Section 6 is the account of what is present; this table is the account of what produced this score.

| FEATURE | MEASURED | REF PCT | DIR | w | q | s | c | v | SOURCES |
|---|---|---|---|---|---|---|---|---|---|
| **Valles-Colomer 2019 Flemish Gut Flora Project + LifeLines DEEP** · species · association only · 16S study · study valles_colomer_2019 | | | | | | | | | |
| *Coprococcus comes* | 1.842% | 96 | ▼ | 0.90 | 0.80 | 0.40 | 0.80 | -0.58 | |
| *Coprococcus catus* | 1.281% | 99 | ▼ | 0.60 | 0.70 | 0.40 | 0.70 | -0.76 | |
| *Dialister invisus* | not detected | 39 | ▼ | 0.80 | 0.80 | 0.50 | 0.70 | +0.09 | |
| *Faecalibacterium prausnitzii* | 9.840% | 81 | ▼ | 0.60 | 0.70 | 0.20 | 0.70 | -0.31 | |
| **Zheng 2016 Chinese MDD (16S, GF-mouse transfer)** · species · association only · 16S study · study zheng_2016 | | | | | | | | | |
| *Bacteroides vulgatus* | not detected | 3 | ▲ | 0.50 | 0.60 | 0.30 | 0.50 | -1.00 | |
| *Eubacterium rectale* | 1.082% | 52 | ▼ | 0.50 | 0.60 | 0.20 | 0.60 | -0.02 | |
| *Blautia obeum* | 0.604% | 90 | ▼ | 0.40 | 0.50 | 0.40 | 0.50 | -0.37 | |
| **Yang 2020 Chinese MDD shotgun + metabolomics** · species · study yang_2020 | | | | | | | | | |
| *Bacteroides fragilis* | not detected | 24 | ▲ | 0.60 | 0.70 | 0.30 | 0.60 | -0.24 | |
| *Bacteroides vulgatus* | not detected | 3 | ▲ | 0.50 | 0.70 | 0.30 | 0.60 | -1.00 | |
| *Eubacterium rectale* | 1.082% | 52 | ▼ | 0.60 | 0.70 | 0.20 | 0.70 | -0.02 | |
| *Blautia wexlerae* | 0.014% | 31 | ▼ | 0.50 | 0.60 | 0.40 | 0.60 | +0.16 | |
| *Eubacterium eligens* | 10.449% | 97 | ▼ | 0.50 | 0.60 | 0.40 | 0.70 | -0.48 | |
| **Neuroactive metabolite capacity** · genes · genomic capacity · study functional | | | | | | | | | |
| butyrate | 32.1/100 | 82 | ▼ | 0.80 | 0.70 | 0.20 | 0.70 | -0.26 | |
| gaba | 10.5/100 | 21 | ▼ | 0.60 | 0.50 | 0.40 | 0.50 | +0.29 | |
| ipa | 8.9/100 | 86 | ▼ | 0.50 | 0.50 | 0.30 | 0.50 | -0.61 | |
| dopamine | 0.8/100 | 34 | ▼ | 0.40 | 0.40 | 0.40 | 0.40 | +0.11 | |

FINE PRINT

**Confidence grade insufficient.** Why not higher: no matching criteria requested; missing metadata: antibiotics, stool_form, onset_date, sampling_date, medications — confounders cannot be excluded; no exposure-matched contrast exists for this profile, so syndrome-specific signal cannot be separated from post-exposure signal.

Independent study modules are averaged with equal weight. Modules that share participants share one vote, so a study contributing twenty correlated taxa cannot outweigh a study contributing one replicated feature.

**Measured discrimination: not available.** No labelled cohort exists for this pattern, so its ability to separate cases from controls is unmeasured.

Depression is diagnosed by a clinician. A stool pattern is not a mood disorder, and this profile is not a mental-health screen. A high score is not evidence that you are depressed, and a low score is not evidence that you are not.

Butyrate-producer depletion is shared with everything. The Coprococcus and Dialister findings are the more specific part and are weighted accordingly.

If this score is high, the appropriate response is nothing at all without a conversation with a clinician.

Transfer of depression-like features between animals does not establish that a community causes depression in people, and one human-donor transfer experiment found no worsening from patient microbiota at all.

Sources: Zheng P, et al. Gut microbiome remodeling induces depressive-like behaviors through a pathway mediated by the host's metabolism. Mol Psychiatry 2016;21:786-796, doi:10.1038/mp.2016.44; Kelly JR, et al. Transferring the blues: depression-associated gut microbiota induces neurobehavioural changes in the rat. J Psychiatr Res 2016;82:109-118, doi:10.1016/j.jpsychires.2016.07.019; Cao Y, et al. Prenatal depression-associated gut microbiota induces depressive-like behaviors and hippocampal neuroinflammation in germ-free mice. Transl Psychiatry 2025;15, doi:10.1038/s41398-025-03606-x; Teng T, et al. Microbiota alterations leading to amino acid deficiency contribute to depression in children and adolescents. Microbiome 2025;13, doi:10.1186/s40168-025-02122-w; Knudsen JK, et al. Faecal microbiota transplantation from patients with depression or healthy individuals into rats modulates mood-related behaviour. Sci Rep 2021;11, doi:10.1038/s41598-021-01248-9; Valles-Colomer M, et al. The neuroactive potential of the human gut microbiota in quality of life and depression. Nat Microbiol 2019;4:623-632; Zheng P, et al. Gut microbiome remodeling induces depressive-like behaviors through a pathway mediated by the host's metabolism. Mol Psychiatry 2016;21:786-796; Yang J, et al. Landscapes of bacterial and metabolic signatures and their interaction in major depressive disorders. Sci Adv 2020;6:eaba8555; Strandwitz P, et al. GABA-modulating bacteria of the human gut microbiota. Nat Microbiol 2019;4:396.

# Topics people ask about, and where the answers are

Fifteen topics people ask about, each pointing at what this report actually measured and at what it did not. Where a pattern score for a condition already exists it is linked exactly as calculated; no score here is new, and no topic without a score has been given one.

**This is a way of navigating what was measured, not a test for the condition. Nothing in this card raises or lowers the chance that you have it.**

## Anxiety and depression

**What this keeps separate.** Human microbiome associations with depression are observational and mostly cross-sectional; the behavioural findings are from animals. Neither establishes cause, and no stool measurement diagnoses anxiety or depression.

Several gut bacteria decarboxylate glutamate to GABA, and GABA-modulating strains were isolated from human stool. The blood-brain barrier is not permeable to GABA, so any effect would have to run through the vagus nerve or the enteric nervous system rather than through GABA reaching the brain.

Tryptophan is the shared precursor of serotonin, indoles and kynurenines. Bacteria competing for it is a plausible mechanism and is not a measurement of anybody's serotonin.

| IN YOUR REPORT | WHY IT IS HERE | WHAT IT SHOWS |
|---|---|---|
| **mdd** | the existing depression gut-pattern similarity, linked as it stands | linked as calculated |
| **gaba** | GABA production capacity, one of the routes this literature discusses | 11 per 100 genomes |
| **gababreak** | GABA degradation, shown beside production and never subtracted from it | 8 per 100 genomes |
| **indole** | indole formation from tryptophan, the other much-discussed route | 10 per 100 genomes |
| **tryptamine** | tryptamine formation, which competes for the same tryptophan | 0 per 100 genomes |
| **neuro.gaba_synthesis** | the specific decarboxylation step, with its own limits | supported |

6 readings in this report are relevant to this topic. That is a count of things to read, not a measure of how likely the condition is.

**What this cannot tell you:** Whether any of these genes are being expressed, which sequencing DNA cannot show. Your symptoms, which are not in this data at all and matter more than any of it. Whether you take medication affecting these systems, which changes the picture entirely.

**If you want to take this further:** If you are looking for help with mood, that is a conversation with a clinician, not a microbiome result. A symptom diary alongside a repeat sample would at least make a within-person comparison possible, which a single timepoint cannot support.

## Autism research

**What this keeps separate.** The published classifiers were built in paediatric cohorts, and diet differs systematically between the groups they compared. This report has no autism classifier and produces no score; these are the pathways the research names.

A 2024 multikingdom study reported a panel spanning bacteria, fungi, archaea and viruses, and highlighted B1 and quinone pathways among its features.

Whether these differences are a cause, a consequence of diet and behaviour, or an artefact of cohort design is unresolved in the source literature itself.

| IN YOUR REPORT | WHY IT IS HERE | WHAT IT SHOWS |
|---|---|---|
| **thiamine** | B1 synthesis, one of the pathways the multikingdom work highlighted | 231 per 100 genomes |
| **k2** | quinone synthesis, highlighted in the same work | 12 per 100 genomes |
| **mycobiome** | the fungal and multikingdom component those studies used | linked as calculated |
| **ecology** | community composition and type, which those cohorts stratified on | linked as calculated |

4 readings in this report are relevant to this topic. That is a count of things to read, not a measure of how likely the condition is.

**What this cannot tell you:** Diet, which differs between the compared groups and is not recorded here. Age: these cohorts were children, and the reference set behind this report is adult. Everything a diagnosis actually rests on, none of which is microbial.

**If you want to take this further:** There is no action to take from this card. It exists so the pathways named in the research can be found in your own results.

## Serotonin, melatonin and sleep

**What this keeps separate.** Serotonin, melatonin and tryptamine are three different molecules made by three different reactions. Gut serotonin and brain serotonin are separate pools that do not exchange. No number here is a concentration of any of them.

Most of the body's serotonin is made in the gut by host enterochromaffin cells, and specific bacteria can increase that host production in gnotobiotic mice. That is host synthesis being modulated, not bacterial synthesis.

A traceable candidate decarboxylase produces tryptamine from tryptophan. Tryptamine is not serotonin, and an enzyme that makes one is not a detector for the other.

| IN YOUR REPORT | WHY IT IS HERE | WHAT IT SHOWS |
|---|---|---|
| **tryptamine** | tryptamine, a distinct molecule from serotonin, made from the same precursor | 0 per 100 genomes |
| **indole** | the bacterial route that competes for that precursor | 10 per 100 genomes |
| **dopamine** | the other monoamine chemistry measured here, for contrast | 1 per 100 genomes |

| IN YOUR REPORT | WHY IT IS HERE | WHAT IT SHOWS |
|---|---|---|
| **histamine** | another amine from the same decarboxylase chemistry | 0 per 100 genomes |

4 readings in this report are relevant to this topic. That is a count of things to read, not a measure of how likely the condition is.

**What this cannot tell you:** Your sleep, which is not measurable from stool. Whether the candidate enzymes here act on the substrates the papers used, which is unresolved for some of them in the source.

**If you want to take this further:** A sleep or symptom diary is the only thing that would make this comparable over time, and it is worth more than the gene counts.

## Atherosclerosis and cardiovascular context

**What this keeps separate.** Microbes make TMA. Your liver makes TMAO from it. These are different steps in different organs, and the existing cardiovascular score is linked exactly as it is rather than recomputed from these genes.

Gut bacteria cleave choline and carnitine to trimethylamine, which the liver oxidises to TMAO. Higher circulating TMAO is associated with cardiovascular events in cohort studies, and the causal question remains open.

Host flavin-containing monooxygenase 3 activity, which varies genetically and by sex, sits between the bacterial step and the circulating molecule.

| IN YOUR REPORT | WHY IT IS HERE | WHAT IT SHOWS |
|---|---|---|
| **acvd** | the existing cardiovascular gut-pattern similarity, linked unchanged | linked as calculated |
| **cutc** | choline TMA-lyase, the most studied microbial link | 3 per 100 genomes |
| **bcaa** | amino-acid fermentation, the other route to circulating microbial metabolites | 181 per 100 genomes |
| **pcresol** | p-cresol, another microbial metabolite the liver conjugates before it circulates | 11 per 100 genomes |

4 readings in this report are relevant to this topic. That is a count of things to read, not a measure of how likely the condition is.

**What this cannot tell you:** Your blood TMAO, which this cannot estimate and which a laboratory can measure. Your lipids, blood pressure and family history, which dominate cardiovascular risk.

**If you want to take this further:** Cardiovascular risk is assessed with blood pressure, lipids and history. If that is the question, those are the measurements. A measured TMAO, if you have one, can be entered in the lab results section and will be shown beside this rather than replaced by it.

## Blood pressure context

**What this keeps separate.** The existing profile is the result. This card adds navigation and the functional context, and deliberately does not produce a second number for the same thing.

Short-chain fatty acids act on host receptors including GPR41 and Olfr78 that have opposing effects on vascular tone in rodent experiments, which is why no single direction can be assigned to a butyrate reading.

| IN YOUR REPORT | WHY IT IS HERE | WHAT IT SHOWS |
|---|---|---|
| **hypertension** | the existing gut-pattern similarity, reused rather than duplicated | linked as calculated |
| **butyrate** | short-chain fatty acid production, the mechanism most often proposed | 32 per 100 genomes |
| **propionate** | propionate specifically, which has its own receptor literature | 26 per 100 genomes |

3 readings in this report are relevant to this topic. That is a count of things to read, not a measure of how likely the condition is.

**What this cannot tell you:** Your blood pressure, which takes thirty seconds to measure properly and is not inferable from stool. Salt intake, weight and medication, which are the things that actually move it.

**If you want to take this further:** If blood pressure is the question, measure blood pressure. Repeatedly, at home, is better than once in a clinic.

## Endogenous ethanol and auto-brewery evaluation

**What this keeps separate.** Carrying adhE is ordinary; the liver findings concern particular high-alcohol-producing strains. Genetic potential is not measured ethanol, and auto-brewery syndrome is diagnosed with a carbohydrate challenge and serial blood alcohol measurements, not with a stool gene count.

Specific high-alcohol-producing Klebsiella pneumoniae strains isolated from people with fatty liver disease produced liver changes when colonised into mice. Strains of the same species differ enormously in how much they produce.

Acetaldehyde, the intermediate, is the genotoxic part of this chemistry, and is handled by an enzyme family too broad to measure specifically here.

| IN YOUR REPORT | WHY IT IS HERE | WHAT IT SHOWS |
|---|---|---|
| **capacity.ethanol** | the ethanol-forming route, newly measured | complete |
| **ethanol** | the underlying gene readings, including the ethanolamine route | 70 per 100 genomes |
| **masld** | the existing fatty-liver gut-pattern similarity, linked unchanged | linked as calculated |
| **mycobiome** | yeasts, which ferment ethanol and are measured separately | linked as calculated |
| **strain_resolution** | strain-level evidence, since this is a strain-level question | linked as calculated |

5 readings in this report are relevant to this topic. That is a count of things to read, not a measure of how likely the condition is.

**What this cannot tell you:** Whether the organisms carrying these genes here are high producers, which requires culture and measurement rather than sequence. Any actual blood alcohol measurement, which is what the diagnosis rests on.

**If you want to take this further:** If episodes of unexplained intoxication are the reason for asking, that is a medical evaluation with a supervised carbohydrate challenge, and this card is not a substitute for it.

## Chronic constipation

**What this keeps separate.** Methanogen genes in stool are not a breath test. The association between methane and slow transit comes from breath measurements, and a gene count cannot substitute for one or diagnose intestinal methanogen overgrowth.

Methane slows intestinal transit in animal models, and breath methane is associated with constipation in humans. Methanogens are archaea and are measured here by their own marker genes.

Fermentable substrate reaching the colon affects stool water and bulk, which is why the substrate readings are linked here.

| IN YOUR REPORT | WHY IT IS HERE | WHAT IT SHOWS |
|---|---|---|
| **ibs_c** | the existing constipation-predominant IBS similarity, linked unchanged | linked as calculated |
| **methane** | methanogen capacity, associated with slower transit | 0 per 100 genomes |
| **substrates** | which fibres this community can actually open, newly measured | linked as calculated |
| **butyrate** | the fermentation end-product most linked to colonic function | 32 per 100 genomes |

4 readings in this report are relevant to this topic. That is a count of things to read, not a measure of how likely the condition is.

**What this cannot tell you:** Your stool form and frequency, which can be recorded and are more informative than any of this. Transit time, which is directly measurable and is not inferable from stool DNA.

**If you want to take this further:** Recording stool form on the Bristol scale for two weeks costs nothing and gives a better picture than a single sequencing run. A breath test is the measurement that corresponds to the methane literature.

## Chronic diarrhoea

**What this keeps separate.** Bile-acid diarrhoea is one cause among many, and it is diagnosed with a SeHCAT scan or a measured faecal bile acid, not with these genes. The pathogen screen is linked here because it answers a separate question that matters more when diarrhoea is acute.

Bacteria deconjugate and then dehydroxylate bile acids. Reduced conversion leaves more primary bile acid reaching the colon, which draws in water. The bacterial step is real; the amount arriving is set by the liver and the ileum.

| IN YOUR REPORT | WHY IT IS HERE | WHAT IT SHOWS |
|---|---|---|
| **ibs_d** | the existing diarrhoea-predominant IBS similarity, linked unchanged | linked as calculated |
| **bsh** | bile salt hydrolase, the first step of bile-acid transformation | 123 per 100 genomes |
| **bai** | the 7-alpha-dehydroxylation route to secondary bile acids | 3 per 100 genomes |
| **pathogens** | the pathogen screen, which is a different question with a different answer | linked as calculated |

4 readings in this report are relevant to this topic. That is a count of things to read, not a measure of how likely the condition is.

**What this cannot tell you:** Whether your ileum is reabsorbing bile acids normally, which is the actual question in bile-acid diarrhoea and needs a scan or a stool measurement. Recent antibiotics and travel, which change the interpretation completely.

**If you want to take this further:** Persistent diarrhoea, and particularly diarrhoea with weight loss, blood or fever, needs a clinician rather than a microbiome report.

## Recurring abdominal pain

**What this keeps separate.** The histamine mechanism is strain-dependent. A species that contains histamine-producing strains is not evidence that yours produces histamine, and species-level abundance cannot establish it.

A specific Klebsiella aerogenes strain producing histamine was linked to visceral hypersensitivity in a mouse model and to a subgroup of patients. The work is explicitly strain-level.

Gas production from fermentation distends the colon, and distension is painful at lower volumes in people with visceral hypersensitivity.

| IN YOUR REPORT | WHY IT IS HERE | WHAT IT SHOWS |
|---|---|---|
| **ibs** | the existing IBS similarity, linked as it stands | linked as calculated |
| **histamine** | bacterial histamine production, a strain-specific mechanism | 0 per 100 genomes |
| **fermentation** | gas-forming fermentation routes, newly measured | linked as calculated |
| **h2s** | hydrogen sulfide, which has its own visceral sensitivity literature | 97 per 100 genomes |

4 readings in this report are relevant to this topic. That is a count of things to read, not a measure of how likely the condition is.

**What this cannot tell you:** Where the pain is, when it happens and what changes it, none of which is here and all of which is more diagnostic than this report. Whether the strains present are the producing ones, which needs strain-level resolution this depth cannot always reach.

**If you want to take this further:** Pain with weight loss, bleeding, fever or waking you at night is a reason to see a clinician promptly, whatever a microbiome report says.

## Wheat, gluten and fructan sensitivity

**What this keeps separate.** Coeliac disease, wheat allergy and non-coeliac wheat sensitivity are three different conditions with three different mechanisms. Coeliac disease is diagnosed serologically and by biopsy, while still eating gluten. Nothing here tests for any of them.

Trials that separated gluten from fructans found that fructans, not gluten, reproduced symptoms in many people who believed they were gluten-sensitive. Wheat carries both, which is why the substrate reading is linked here.

| IN YOUR REPORT | WHY IT IS HERE | WHAT IT SHOWS |
|---|---|---|
| **celiac** | the existing coeliac gut-pattern similarity, linked unchanged | linked as calculated |
| **substrates** | fructan utilisation specifically, newly measured | linked as calculated |
| **carbohydrates** | the wider carbohydrate-active enzyme readings | 549 per 100 genomes |

3 readings in this report are relevant to this topic. That is a count of things to read, not a measure of how likely the condition is.

**What this cannot tell you:** Whether you are currently eating gluten, which determines whether coeliac testing would even be valid. Your symptoms and their timing relative to meals.

**If you want to take this further:** If coeliac disease has never been excluded, that test is done before removing gluten, not after: stopping first makes the test unreliable. A supervised fructan reduction is the way to separate the two, and it is temporary rather than permanent.

## Irritable bowel syndrome

**What this keeps separate.** All four IBS scores already exist in this report and are shown as they are. This card adds navigation and substrate context only, and produces no new score. IBS is a clinical diagnosis made on symptom criteria.

The microbiome literature in IBS is large, mostly 16S, and inconsistent between cohorts, which is what a symptom-defined diagnosis tends to produce.

Fermentable substrate reaching the colon is the mechanism behind the low-FODMAP response, and substrate capacity is the part of that this report can measure.

| IN YOUR REPORT | WHY IT IS HERE | WHAT IT SHOWS |
|---|---|---|
| **ibs** | the existing overall IBS similarity | linked as calculated |
| **ibs_c** | the constipation-predominant variant | linked as calculated |
| **ibs_d** | the diarrhoea-predominant variant | linked as calculated |
| **ibs_m** | the mixed variant | linked as calculated |
| **substrates** | which fermentable substrates this community can open | linked as calculated |
| **fermentation** | the gas-forming routes downstream of them | linked as calculated |

6 readings in this report are relevant to this topic. That is a count of things to read, not a measure of how likely the condition is.

**What this cannot tell you:** Your symptom pattern against the Rome criteria, which is what the diagnosis is. What you eat, which determines how much fermentable substrate arrives.

**If you want to take this further:** A low-FODMAP trial is a short diagnostic exercise with a structured reintroduction, not a long-term diet, and is best done with a dietitian.

## Thyroid context

**What this keeps separate.** Hypothyroidism, Hashimoto's thyroiditis and Graves' disease are separate conditions. Host blood markers and host gene expression cannot be inferred from stool DNA, and this report has no thyroid score.

Observational studies report compositional differences in thyroid conditions. They are small, cross-sectional and confounded by treatment, and no mechanism has been established in humans.

Selenium and iodine handling are frequently discussed, but the bacterial contribution to either has not been quantified in people.

| IN YOUR REPORT | WHY IT IS HERE | WHAT IT SHOWS |
|---|---|---|
| **bglucuronidase** | beta-glucuronidase, often invoked in this context | 83 per 100 genomes |
| **ecology** | overall community composition, which the observational work compared | linked as calculated |

2 readings in this report are relevant to this topic. That is a count of things to read, not a measure of how likely the condition is.

Not produced in this run, and so not part of the card: input_register.

**What this cannot tell you:** Your TSH, free T4 and antibody status, which are blood tests and are the actual measurements for this. Whether you take thyroid hormone, which changes everything about interpretation.

**If you want to take this further:** If thyroid function is the question, a blood test answers it directly. Results can be entered in the lab section and will be shown alongside, not merged in.

## Food allergy and immune tolerance

**What this keeps separate.** Mouse transfer and consortium experiments show that particular bacteria affect tolerance in mice. They do not establish that any food is safe or unsafe for you, and nothing here identifies an allergen.

Consortia from healthy infants protected mice against cow's milk allergy, and consortia from allergic infants did not. The effect travelled with specific organisms rather than with diversity.

Butyrate induces regulatory T cells in the colon in mouse work, which is the usual proposed bridge between fibre and tolerance.

| IN YOUR REPORT | WHY IT IS HERE | WHAT IT SHOWS |
|---|---|---|
| **csu** | the existing chronic urticaria pattern, linked unchanged | linked as calculated |
| **ATD_ADULT_WANG2023_TRANSLATED_V1** | the existing atopic dermatitis pattern index | linked as calculated |
| **butyrate** | butyrate, the metabolite in the regulatory T cell literature | 32 per 100 genomes |
| **substrates** | the fibre substrates upstream of that butyrate | linked as calculated |

4 readings in this report are relevant to this topic. That is a count of things to read, not a measure of how likely the condition is.

**What this cannot tell you:** Which foods, if any, cause you problems, which is established by history and by supervised challenge rather than by sequencing. Your IgE status, which is a blood or skin test.

**If you want to take this further:** Never use a microbiome result to decide whether a food is safe. Suspected allergy is assessed by an allergist, and a genuine anaphylaxis risk is not something to experiment with.

## GLP-1-related microbial mechanisms

**What this keeps separate.** Microbial genetic potential is not a hormone concentration and is not comparable to a GLP-1 medication. Host receptors are host genes and are not targets in stool sequencing. Indole does not have one universal direction: prolonged exposure reduced secretion in the source experiments.

Short-chain fatty acids trigger GLP-1 secretion through FFAR2 in experimental systems, and a targeted colonic propionate delivery study in humans increased secretion.

Bile acids act through TGR5 on the same cells, which is a separate mechanism that happens to converge on the same hormone.

A specific Akkermansia muciniphila protein has experimental support, and detecting it requires the tested sequence rather than the species.

| IN YOUR REPORT | WHY IT IS HERE | WHAT IT SHOWS |
|---|---|---|
| **biotransformation** | the GLP-1 channels, kept as separate mechanisms | linked as calculated |
| **butyrate** | the short-chain fatty acid channel | 32 per 100 genomes |
| **propionate** | the colonic propionate experiments specifically | 26 per 100 genomes |
| **bsh** | the bile-acid channel, which acts through a different receptor | 123 per 100 genomes |
| **indole** | the indole channel, whose direction depends on exposure duration | 10 per 100 genomes |

5 readings in this report are relevant to this topic. That is a count of things to read, not a measure of how likely the condition is.

**What this cannot tell you:** Your actual GLP-1, which requires a blood assay with the active-versus-total distinction, fasting or postprandial timing, and correct sample handling. Whether any of these genes are expressed or their products secreted.

**If you want to take this further:** A measured GLP-1 can be entered in the lab section with its assay and timing, and will be shown next to this rather than combined with it.

## Eczema

**What this keeps separate.** The existing skin pattern index is preserved exactly and is not recomputed here. Gut composition is one factor among genetics, skin barrier function and the skin's own microbiome, which a stool sample does not sample.

Filaggrin variants and skin barrier function dominate atopic dermatitis risk. Gut microbial associations are real in cohort studies and much weaker than those factors.

| IN YOUR REPORT | WHY IT IS HERE | WHAT IT SHOWS |
|---|---|---|
| **ATD_ADULT_WANG2023_TRANSLATED_V1** | the existing atopic dermatitis pattern index, preserved | linked as calculated |
| **butyrate** | the short-chain fatty acid arm of the tolerance literature | 32 per 100 genomes |
| **substrates** | the fibre substrates upstream of it | linked as calculated |

3 readings in this report are relevant to this topic. That is a count of things to read, not a measure of how likely the condition is.

**What this cannot tell you:** Your skin, which is not what was sampled. The skin microbiome is a different community and would need a different swab. Whether the flares follow anything identifiable, which a diary answers.

**If you want to take this further:** Emollients and the treatments a dermatologist recommends have far stronger evidence than anything derived from a stool sample.

PART B — BIOFILM FINDINGS IN FULL

# 21 Biofilm findings in full

This section lists every biofilm measurement, every module in the registry with its actual status, and every matched piece of evidence — including the evidence that points the other way. Nothing here is truncated to a top-N list.

## The reference population

Percentiles are ranks against **1,127 independent adults** from 14 public control cohorts, aged 19 to 107 (median 52), profiled on the same taxonomy as this sample. 3,027 samples were collapsed to 1,127 people: several of the source studies sampled the same person repeatedly, and counting those as separate people would have made the reference look larger and steadier than it is.

Control status in these studies means no recorded disease for the study's purpose. It does not mean microscopy-confirmed absence of mucosal biofilm, which was never assessed in any of them.

### H-C · Biofilm-associated community pattern

**Percentile 5.0** (lower reference range) against 1,127 adults. Calibration 2ce8f21198799625.

**Escherichia coli and its close relatives** = 0%, rank 10 — *not detected above this assay's limit*. Species summed: none.

**Mediterraneibacter gnavus** = 0%, rank 22 — *not detected above this assay's limit*. Species summed: none.

**Why this direction:** Higher means more of this two-feature association pattern, which BF-S01 reported enriched in biofilm-positive participants.

**What it is not:** Not measured harmful biofilm, not a stool biofilm probability, and not a pathotype. BF-S01's 16S Escherichia/Shigella association cannot be resolved identically by a WGS species-complex measurement, and not every Escherichia/Shigella read belongs to pathogenic E. coli.

Recorded omissions: ecoli_complex: not detected above this assay's limit; m_gnavus: not detected above this assay's limit

### P-E · Protective ecological support (partial panel)

**Percentile 96.9** (upper reference range) against 1,127 adults. Calibration 73125c04fea5c96c.

**Faecalibacterium prausnitzii** = 9.84%, rank 85. Species summed: Faecalibacterium_prausnitzii 9.84%.

**Coprococcus species** = 8.897%, rank 98. Species summed: Coprococcus_catus 1.28%, Coprococcus_comes 1.84%, Coprococcus_eutactus 5.77%.

**Blautia species** = 0.6182%, rank 70. Species summed: Blautia_obeum 0.604%, Blautia_wexlerae 0.0138%.

**Why this direction:** Higher means more of this narrow community pattern, which BF-S01 reported relatively depleted in biofilm-positive biopsies.

**What it is not:** Not a measurement of beneficial biofilm amount. Low values do not diagnose a missing protective biofilm, a species deficiency or a need for transplant, and this proxy does not instruct supplementation with these organisms. This is a partial panel: subdoligranulum could not be measured in this reference. It is calibrated separately and is not the named 4-feature index.

Recorded omissions: subdoligranulum: dropped from the panel - detected in 2 of 1127 reference participants (0.2%), below the 5% floor needed to rank anyone. The organism is not rare; this taxonomy release simply has no named species leaf that captures it, so the feature is not assayed here rather than absent.

## Axes with no measurement

• **Harm-associated mechanisms** — No mechanism in this axis has a validated sequence panel yet. This is a gap in the reference data, not a finding of absence.

• **Protective-associated mechanisms** — No mechanism in this axis has a validated sequence panel yet. This is a gap in the reference data, not a finding of absence.

## Do the different kinds of evidence agree?

Each row is a different way of looking at the same named concern: biofilm-associated community pattern (BF-S01 association). There is deliberately no combined confidence number, because two scores computed from the same reads are not independent confirmations of each other.

| | | |
|---|---|---|
| Molecular mechanism | not measured | No mechanism module has a validated sequence panel, so no molecular mechanism measurement exists to agree or disagree. |
| Community proxy | opposes named concern | The two-feature H-C pattern, ranked against the adult control reference. |
| Expression or activity | not measured | No RNA or protein assay was supplied. |
| Directly measured architecture | not measured | No endoscopy or imaging was supplied. Generic inflammation cannot fill this row. |
| Measured host consequence | not measured | No host protein, permeability or injury assay was supplied. |

## Every mechanism in the registry

All 24 modules appear here with their real status. A module that cannot be measured says so; it is not left out, because a silently missing mechanism is indistinguishable from one that was checked and found absent.

| ID | MECHANISM | STATUS | AXIS | EVIDENCE |
|---|---|---|---|---|
| BF-M01 | Curli-family bacterial amyloid genes | candidate, not validated | context dependent | mixed |
| BF-M02 | Biofilm-associated protein (BAP) family targets | candidate, not validated | context dependent | mixed |
| BF-M03 | PNAG/PGA matrix synthesis and export locus | candidate, not validated | context dependent | human association |
| BF-M04 | Bile-acid-diarrhoea-associated biofilm gene pattern | candidate, not validated | context dependent | human association |
| BF-M05 | Klebsiella type-1 and type-3 adhesion loci | candidate, not validated | context dependent | animal mechanism |
| BF-M06 | Extracellular polysaccharide and matrix-associated proteins | context only | context dependent | resource |
| BF-M07 | eDNA, quorum sensing and cyclic-di-GMP regulatory context | context only | context dependent | in vitro mechanism |
| BF-M08 | C. difficile adhesion, matrix and spore context | context only | context dependent | review |
| BF-M09 | Candida adhesion, matrix and regulatory context | not measurable from stool DNA | context dependent | review |
| BF-M10 | L. reuteri GtfW protective mechanism | candidate, not validated | protective associated | in vitro mechanism |
| BF-M11 | F. prausnitzii host-glycan protective mechanism | no reference panel yet | context dependent | animal mechanism |
| BF-M12 | Fibre-associated mixed-community mechanisms | context only | context dependent | in vitro mechanism |
| BF-M13 | Other verified probiotic-strain adhesion and EPS mechanisms | candidate, not validated | context dependent | animal mechanism |
| BF-M14 | S. aureus host fibrin-interaction capability | context only | context dependent | in vitro mechanism |
| BF-M15 | Co-occurring pathogen, toxin and AMR calls | implemented | context dependent | human association |
| BF-M16 | Measured host thrombin, protein activity and tissue architecture | not measurable from stool DNA | context dependent | animal mechanism |
| BF-M17 | Longitudinal shared strain with biofilm-associated loci | implemented | context dependent | animal mechanism |
| BF-M18 | Two-feature biofilm-associated community proxy (H-C) | implemented | harmful associated | human association |
| BF-M19 | Four-genus biofilm-negative-associated ecological proxy (P-E) | implemented | protective associated | human association |
| BF-M20 | Linked matrix/adhesion plus independently supported damaging capability | implemented | context dependent | human association |
| BF-M21 | Faeces-derived community behaviour | not measurable from stool DNA | context dependent | human derived ex vivo |
| BF-M22 | Tissue architecture and host-interface observations | not measurable from stool DNA | context dependent | human association |
| BF-M23 | Matched expression and activity support | not measurable from stool DNA | context dependent | in vitro mechanism |
| BF-M24 | Additional source-specific protective-antagonism evidence | candidate, not validated | context dependent | animal mechanism |

## Matched research evidence

9 records matched the findings above. These are things to read, not things to take. Where a study points the other way it is listed beside the one that points favourably, at the same size. Laboratory concentrations are shown as laboratory concentrations and are never converted into a dose.

### L. plantarum CIRM653 opposing evidence - review before considering

Reduced K. pneumoniae biofilm in vitro, but treated mice MAINTAINED intestinal carriage while untreated controls declined.

Measured endpoint: **viable organism burden**. Model: animal gut. Strain: CIRM653. Laboratory exposure: live strain, in vitro and mouse. Delivery to the colon: plausible unverified.

**Caution:** The essential regression case: a favourable in-vitro biomass result with an unfavourable in-vivo carriage result. It must never be ranked as a selected decolonisation treatment.

BF-I45 · 10.3920/bm2017.0002

### Host-glycan support of F. prausnitzii matched research evidence

Protective mechanism in experimental colitis with correlations in human tissue.

Measured endpoint: **protective function**. Model: animal gut. Laboratory exposure: glycan intervention, mouse. Delivery to the colon: plausible unverified.

**Caution:** Not a validated oral treatment and not a unique DNA biomarker.

BF-I36 · 10.1080/19490976.2026.2665870

### L. johnsonii studied isolate matched research evidence

Direct EPEC biofilm experiments plus improved pathogen and inflammation outcomes in a C. rodentium mouse model.

Measured endpoint: **protective function**. Model: animal gut. Strain: the studied isolate; PV739486 is a 16S sequence only. Laboratory exposure: live isolate, in vitro and mouse. Delivery to the colon: plausible unverified.

**Caution:** Do not substitute a commercial L. johnsonii strain without identity evidence. Public 16S data do not provide an exact strain genome, and putatively annotated metabolites are not proven active ingredients.

BF-I44 · 10.3389/fimmu.2026.1749001

### Biofilm-state L. plantarum Y42 and its matrix products matched research evidence

Barrier and immune outcomes improved in a mouse challenge model.

Measured endpoint: **protective function**. Model: animal gut. Strain: Y42. Laboratory exposure: biofilm-state cells and components, mouse. Delivery to the colon: plausible unverified.

**Caution:** Exact strain, state and components matter. Ordinary planktonic products and generic EPS genes are not equivalent.

BF-I47 · 10.1021/acs.jafc.4c00460

### Biofilm-state Lactiplantibacillus LR-1 matched research evidence

Biofilm-state cells outperformed planktonic cells on selected colitis outcomes.

Measured endpoint: **protective function**. Model: animal gut. Strain: LR-1. Laboratory exposure: biofilm-state cells, mouse. Delivery to the colon: plausible unverified.

**Caution:** Experimental protective support, with no claimed human treatment effect.

BF-I48 · 10.1039/d3fo02733c

### Multi-strain restorative consortium matched research evidence

Mouse recovery after antibiotics.

Measured endpoint: **protective function**. Model: animal gut. Strain: the administered consortium. Laboratory exposure: administered consortium, mouse. Delivery to the colon: plausible unverified.

**Caution:** Full tested composition and methods must be imported before any exact product matching. Co-detection of these taxa in stool or donor material does not recreate the consortium.

BF-I49 · 10.1007/s11274-026-05071-0

### L. reuteri ATCC 23272 in microsphere formulation matched research evidence

Improved probiotic properties in the biofilm state, with GtfW implicated.

Measured endpoint: **protective function**. Model: gut relevant in vitro. Strain: ATCC 23272. Laboratory exposure: microsphere formulation, in vitro. Delivery to the colon: plausible unverified.

**Caution:** The ATCC 23272 and GtfW context is not a generic probiotic-species effect, and the microsphere formulation is not naturally present in a donor sample.

BF-I34 · 10.3389/fmicb.2017.00489

### Resveratrol with L. paracasei ATCC334 matched research evidence

Strain-dependent promotion of adhesion and biofilm formation in vitro.

Measured endpoint: **adhesion**. Model: gut relevant in vitro. Strain: ATCC334. Laboratory exposure: resveratrol in culture. Delivery to the colon: plausible unverified.

**Caution:** A supportive formulation hypothesis. Not proven human engraftment, not a beneficial-biofilm quantity, and not a reason to take concentrated resveratrol.

BF-I46 · 10.3390/ijms21155423

### SB-121 (L. reuteri with dextran microparticles and maltose) matched research evidence

Small randomised placebo-controlled crossover study in 15 participants with autism, reporting safety, tolerability and exploratory outcomes.

Measured endpoint: **clinical outcome, no biofilm measured**. Model: human gut controlled. Strain: SB-121 formulation. Laboratory exposure: oral SB-121, clinical. Delivery to the colon: target exposure measured.

An oral formulation studied in humans.

**Caution:** Relevant evidence that biofilm-oriented supportive formulations have reached human research. Not validation of a stool protective score and not proof of broad clinical efficacy.

BF-I35 · 10.1038/s41598-023-30909-0

## Reading this section

### What are biofilms?

Biofilms are organised communities of microbes held together by a shared matrix. They are a common microbial way of living, not automatically a disease. In the gut, some communities can support stable colonisation and a protective relationship with the intestinal lining. Others can help pathogens persist, grow too close to tissue, or participate in inflammation. Their effect depends on the organisms, their activity, their location and the host.

### What did this test measure?

This test ranks selected biofilm-related genetic capabilities and community patterns. We show concerning mechanisms and protective support separately, so one does not cancel the other. The community-pattern scores are experimental proxies. DNA alone does not directly measure an attached biofilm, its thickness, location or current activity.

### Why can harmful biofilms matter?

Some biofilms help organisms survive immune defences or antimicrobial exposure. That can contribute to persistent or recurrent infection in the right setting. Biofilm-related tolerance is different from carrying an antibiotic-resistance gene, although both may occur together. A genetic finding alone cannot tell us which treatment will work.

### Can harmful biofilms be reduced?

Research includes targeted treatment of specific infections, dietary and community-supporting approaches, probiotics, compounds and enzymes that affect matrix or adhesion, and emerging precision therapies. Some have encouraging laboratory or animal evidence; others have human evidence in particular conditions. Each option above states exactly what was tested. Breaking up a matrix does not necessarily kill the organisms, and removing protective communities or mucus can be counterproductive.

### What about fibrin and other organs?

Fibrin is a human clotting protein that some microbes can exploit in blood or tissue infections. It is not the same as bacterial amyloid or every biofilm matrix. Biofilm-associated problems can occur on teeth, wounds, airways, urinary devices, implants and heart valves. Stool DNA does not locate or diagnose those problems or measure circulating clots.

### What can I do with this result?

Use the supported findings to identify which mechanisms deserve attention, which evidence-linked options are relevant, and what additional information would make a decision more reliable. Changes over time can show changes in the measured genetic signals; they do not by themselves prove that a biofilm has been removed.

## What this section cannot tell you

• Age was not supplied. This percentile assumes the subject is an adult, because the reference is adults aged 18 and over. If the subject is a child the percentile does not apply and should be disregarded.

• DNA shows genetic potential, not activity, attachment or location.

• Both computable cards are experimental research proxies, not measured biofilm quantity.

• No validated stool-DNA classifier of protective versus harmful biofilm exists; this module does not claim one.

• A negative or low result never means biofilm-free, infection-free or cleared for donation.

# 22 Your gut mycobiome, in detail

Every quality-passed read pair (5,660,139) was aligned competitively against 85 fungal assemblies chosen from a reference of 1,696 (961 species), their near neighbours, and 33 bacterial and control decoys; a separate single-copy-marker search (EukDetect2 8d69014) ran on the same reads. A fungus is supported when its fragments are spread across separated regions of its genome at high identity and no decoy competes for them. Below: each fungus found, then the score trace.

Ledger: 1410 fungal · 89 cross-kingdom ambiguous (excluded) · 5541 fungal hits below 97% identity (excluded) · 1,434,214 decoy · 4,218,885 unassigned. Each pair counted once.

‹ at a glance

## *Geotrichum candidum* food-associated

SUPPORTED

### What it is

The white rind yeast of soft cheeses; a stool hit is a cheese hit.

### How it was found

| | |
|---|---|
| WHOLE GENOME | **1053** fragments over **728** separated regions on 728 contigs · identity 99.1% · backbone GCA_023629795.1 |
| ORIGIN | compatible with food or supplement — a single stool sample cannot prove residence |

### Strain species only · complete

| | |
|---|---|
| COVERAGE | mean depth 0.0131x · 4,918 bases at ≥3x of 26,474,063 · expected 0.0119x from the fungal fraction |
| NEXT | No discriminating marker panel exists for this species yet; intraspecies references from CGF/NCBI would need to be assembled into one before a strain claim is possible. |

### What the evidence says 0 claims

No health evidence is held for this organism. It is listed because it was found.

**Not inferred from this finding:** Residence or infection.

‹ at a glance

## *Penicillium* environmental or food mould

SUPPORTED

### What it is

Cheese, fruit and indoor-air moulds; P. camemberti and P. roqueforti make soft and blue cheeses. Stool DNA is usually dietary passage.

### How it was found

| | |
|---|---|
| WHOLE GENOME | **21** fragments over **6** separated regions on 6 contigs · identity 99.6% · backbone GCA_037042535.1 |
| ORIGIN | compatible with food or supplement — a single stool sample cannot prove residence |

### Strain species only · complete

| | |
|---|---|
| COVERAGE | mean depth 0.0011x · 2,733 bases at ≥3x of 28,184,015 · expected 0.0002x from the fungal fraction |
| NEXT | No discriminating marker panel exists for this species yet; intraspecies references from CGF/NCBI would need to be assembled into one before a strain claim is possible. |

### What the evidence says 1 claim

| DIRECTION | ENDPOINT | EVIDENCE | LIMIT |
|---|---|---|---|
| context | recurrent culture from stool over time<br>human · stool · longitudinal culture study | **human association**<br>https://doi.org/10.3389/fmicb.2019.01575 | Recurrence can still reflect repeated ingestion. |

**Not inferred from this finding:** Mycotoxin exposure or infection.

‹ at a glance

### *Geotrichum bryndzae* no health evidence

SUPPORTED

#### What it is

No curated description; listed because it was found.

#### How it was found

| | |
|---|---|
| MARKERS | EukDetect2: 14 markers, 32 reads, RPKS 0.107 (marker-length-normalised support, not load) |
| ORIGIN | unknown — a single stool sample cannot prove residence |

#### Strain not assessed

Strain analysis was not attempted for this finding.

#### What the evidence says 0 claims

No health evidence is held for this organism. It is listed because it was found.

‹ at a glance

### *Rhizomucor* environmental or food mould

PROVISIONAL

#### What it is

Thermophilic moulds of compost and rennet production (R. miehei makes microbial rennet for cheese); dietary or environmental passage.

#### How it was found

| | |
|---|---|
| WHOLE GENOME | **45** fragments over **4** separated regions on 4 contigs · identity 97.9% · backbone GCA_040807125.1 |
| NOTE | locus concentrated possible rdna mito or contaminant, mostly on short contigs repeat or organelle likely, identity 0.979 below species confidence nearest Rhizomucor miehei |
| ORIGIN | compatible with food or supplement — a single stool sample cannot prove residence |

#### Strain not assessed

Strain analysis was not attempted for this finding.

#### What the evidence says 0 claims

No health evidence is held for this organism. It is listed because it was found.

**Not inferred from this finding:** Infection from DNA alone.

## How the Myco-Score was computed myco-score-v2

Inputs: opportunistic fungi are 0% of all read pairs and 0.0% of the fungal DNA; 0 opportunist(s) confirmed at species level; 0 alert-route pathogen(s); Saccharomyces not supported; evidence rules B = 0, C = 0.

| COMPONENT | POINTS | WHAT IT MEANS |
|---|---|---|
| **Neutral start** | **50** | Every completed analysis starts at 50. |
| **Opportunists absent or very little** | **+20.0** | Under a tenth of the fungal DNA and under 0.001% of all read pairs, none confirmed at species level. |
| **Myco-Score** | **70**/100 | Sum, held within 0–100, then the alert ceiling if any. Higher is healthier. |

Basis: Huang et al. 2024, Gut Microbes 16:2440111 (review); policy weights are OpenBiota's. No component rewards fungal load or diversity: the review describes fungi at 0.01–0.1% of the microbiome with no lower bound, and diversity moves in opposite directions across diseases. Experimental; not validated against outcomes.

## The evidence rules behind it MHS-E1 · policy mhs-e1-policy-f6de290eb9d36b3c

B = 0, C = 0; score = 100 × 0.5 × (1 – C) × (1 + B) = **50.00**. B and C are each the single strongest supported contribution, not sums: more studies of the same organism do not add, and a benefit cannot erase the dominant concern. Caps are policy, not measured effects.

| RULE | DIR | CAP | ACTIVATION | STATUS | WHY |
|---|---|---|---|---|---|
| **P4013B_B** | B | 0.5 | 0 | inactive | Clavispora lusitaniae not supported in this sample |
| **CL_LUSITANIAE_B** | B | 0.25 | 0 | inactive | Clavispora lusitaniae not supported in this sample |
| **CA_HD_C** | C | 0.5 | 0 | inactive | Candida albicans not supported in this sample |
| **CA_LD_CONTEXT** | B | 0 | 0 | inactive | Candida albicans not supported in this sample |
| **CPAR_GUT_C** | C | 0.25 | 0 | inactive | Candida parapsilosis not supported in this sample |

| | | | | | |
|---|---|---|---|---|---|
| **DH_REPAIR_C** | C | 0.25 | 0 | inactive | Debaryomyces hansenii not supported in this sample |
| **MR_COLITIS_C** | C | 0.25 | 0 | inactive | Malassezia restricta not supported in this sample |
| **CT_EC_SM_INTERACTION_C** | C | 0.125 | 0 | inactive | Candida tropicalis not supported in this sample |
| **SB_STUDIED_B** | B | 0.5 | 0 | inactive | unbound rule: Unbound: no verified genotype-to-trial join exists for the studied strain. A boulardii-like lineage, product label or supplement history is not that join. The finding stays visible; it scores nothing. |
| **FUNGAL_ABUNDANCE_ASSOCIATION** | C | 0.25 | 0 | inactive | unbound rule: Generic adapter for a reviewed human stool abundance-outcome rule with a qualified same-pipeline reference. No such reference is qualified; unbound. |

Sensitivity: resolution alternatives 50.0–50.0; policy caps ×0.5/×1.5 50.0–50.0; combined (displayed) 50.0–50.0. Scenario ranges, not confidence intervals. Interpretation confidence is experimental limited at any depth.

## Limits

• Fungal DNA is not living biomass: genome size, cell-wall lysis, ploidy and dietary DNA all shape the signal.

• Within-fungi percentages and the total-sample fungal signal use different denominators and are shown apart.

• Reference: 1,696 fungal assemblies covering 961 species; a fungus outside them is found only by the marker lane, and one outside both catalogues is not found at all.

• The support rule (20 fragments over 3 separated regions) is a locked starting value from the specification's calibration grid; qualification on mock communities is pending and is recorded as such.

• Stool does not fully represent the intestinal mucosa; a mucosal fungus can be absent from stool.

PART B — SKIN-CONDITION RESEARCH PATTERNS

# 23 Gut-skin axis, in detail

Each panel below gets its own page: what the source study compared, every organism or gene family it declared, what this sample measured for each, and the evidence that argues against reading too much into it. Slots that could not be measured are listed as such, because leaving them out would make a partly-measured panel look complete.

‹ at a glance

## 23.1 Plaque psoriasis — Deng 2026 single-marker SGB4348

**Plaque psoriasis (raised scaly skin plaques).** Plaque psoriasis is a chronic immune-mediated skin disease with raised, scaly plaques, commonly on elbows, knees, scalp and trunk. Four separate stool-shotgun studies are carried here as separate readings rather than one score, because they disagree: Deng 2026 nominates a single SGB, Chang 2022 and Sá 2026 report largely non-overlapping species, and Bacteroides coprocola is enriched in cases in one study and in controls in another.

**No index — no usable features.** This is not a negative result and not clearance. It means the number could not be produced, for the reasons below.

Reasons recorded: no usable features; age unknown scored in research mode without a population guarantee; antibiotic exposure unknown and unknown is not none.

### What was measured, and against whom

Source comparison: **psoriasis vs controls**. Measurement: shotgun taxonomic fraction. The source reported a single-feature ROC AUC of 0.84 (95% CI 0.74-0.94) on a same-centre test group. That figure belongs to the source's own threshold on its own cohort. It is not the accuracy of this index and is never attached to an individual result.

No percentile is offered on this scale: that would need a held-out reference score distribution on this same set of features, and none exists.

### Every declared feature, including the ones that could not be measured

| ORGANISM OR GENE FAMILY | MEASURED | STUDY DIRECTION | CONTRIBUTION | NOTE |
|---|---|---|---|---|
| *GGB51647 SGB4348* | not in database | higher in cases | — | Requires the same biological SGB definition as its sources. A database without that SGB returns unavailable: the same number in another namespace is a different organism, and no phylum, family or unclassified bucket substitutes for it. |

### Evidence pointing the other way

**This marker is not specific to psoriasis.** The same SGB4348 was reported as increased in pemphigus foliaceus, a different blistering skin disease, in a separate cohort. A high reading on a single shared marker therefore does not point at psoriasis in particular. A further study of 53 people with psoriasis and 47 controls found no broad diversity difference at all, with inflammation confounding the apparent profiles.

Sources recorded as counterevidence for this panel: E03, E05.

## What the human research has actually tested for this condition

**Plaque psoriasis — what human trials have tested** LANE C HUMAN TRIAL · LIMITED

Fired by: plaque psoriasis stool pattern: somewhat high

Resembling a published psoriasis stool pattern is a research observation, not a diagnosis and not an indication for treatment. The randomized evidence here is for one exact three-strain product added to topical therapy in people who already had a confirmed diagnosis. It does not transfer to other probiotics, it does not follow from this index, and no dose is issued from this report.

**Studied:** Bifidobacterium longum CECT 7347 + Bifidobacterium lactis CECT 8145 + Lacticaseibacillus rhamnosus CECT 8361, 1:1:1 (probiotic) · **in:** people with clinically confirmed plaque psoriasis, ages 18–99, in Spain, n = 90 · **studied protocol (a source fact, not an instruction):** 1:1:1 combination, total 10^9 CFU/day for 12 weeks, adjunct to topical treatment received by both arms

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: single study. Strongest design: randomized trial.

- In 90 adults with plaque psoriasis, adding one exact three-strain probiotic to topical treatment for 12 weeks raised the proportion reaching at least 75% improvement from 41.9% to 66.7%. The endpoint is that 75% threshold, despite looser wording in the abstract. This is evidence about that formulation as an add-on to topical therapy; it is not evidence that a stool profile predicts who responds, that other probiotic blends work, or that treatment normalises any organism measured in this report. [Navarro-Lopez V et al. Acta Derm Venereol 2019;99:1078] · lane C · 12-week randomized placebo-controlled adjunct trial, both arms on topical treatment · n = 90 · benefit

- **Against or limiting —** The relapse follow-up reported after this trial was not part of the original protocol. It is kept as a caution about how far the result extends, not as an additional benefit claim, and trial adverse-event reporting does not establish universal safety. [Navarro-Lopez V et al. Acta Derm Venereol 2019;99:1078, post-trial follow-up] · lane X · post-trial relapse follow-up outside the original protocol

**Applicability:** The study required a clinical diagnosis of plaque psoriasis. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed.

**Fecal microbiota transplantation:** not indicated.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “treats psoriasis” · “clears plaques” · “normalises SGB4348” · “antimicrobials for psoriasis” · “eradicate the raised organisms”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

# 23.2 Plaque psoriasis — Chang 2022 seven-taxon panel

**Plaque psoriasis (raised scaly skin plaques).** Plaque psoriasis is a chronic immune-mediated skin disease with raised, scaly plaques, commonly on elbows, knees, scalp and trunk. Four separate stool-shotgun studies are carried here as separate readings rather than one score, because they disagree: Deng 2026 nominates a single SGB, Chang 2022 and Sá 2026 report largely non-overlapping species, and Bacteroides coprocola is enriched in cases in one study and in controls in another.




**Measured-feature index 75**, over the 1 of 7 declared features that could be measured. **Possible full-panel range 11–96**, which is where the index could sit once the 6 unmeasured features are accounted for. These two describe different things: the range is not a margin of error around the index, and the index can fall outside it without any arithmetic being wrong.

## What was measured, and against whom

Source comparison: **psoriasis vs healthy**. Measurement: shotgun taxonomic fraction.

No percentile is offered on this scale: that would need a held-out reference score distribution on this same set of features, and none exists.

## Every declared feature, including the ones that could not be measured

| ORGANISM OR GENE FAMILY | MEASURED | STUDY DIRECTION | CONTRIBUTION | NOTE |
|---|---|---|---|---|
| *Megasphaera unclassified* | not in database | higher in cases | — | The source's unclassified Megasphaera bucket, not the genus total and not M. elsdenii. Without that exact bucket in the production catalogue the slot is unavailable. |
| *Dialister invisus* | below reporting limit | higher in cases | — | the profiler reports only the organisms it detects, and this one is not in this sample's table. No validated reporting limit exists for it at this sequencing depth, so its point is omitted and only its bounds are kept: a nondetection, not a measured absence. It was detected in 22% of the reference participants. |
| *Bacteroides vulgatus* | below reporting limit | higher in cases | — | the profiler reports only the organisms it detects, and this one is not in this sample's table. No validated reporting limit exists for it at this sequencing depth, so its point is omitted and only its bounds are kept: a nondetection, not a measured absence. It was detected in 89% of the reference participants. |
| *Phascolarctobacterium succinatutens* | 0.395% | lower in cases | 75 | |
| *Bifidobacterium pseudocatenulatum* | below reporting limit | lower in cases | — | the profiler reports only the organisms it detects, and this one is not in this sample's table. No validated reporting limit exists for it at this sequencing depth, so its point is omitted and only its bounds are kept: a nondetection, not a measured absence. It was detected in 36% of the reference participants. |
| *Coprococcus sp ART55 1* | not in database | lower in cases | — | An exact source concept, not the genus Coprococcus and not any named Coprococcus species. Unavailable unless the catalogue carries this specific entry. |
| *Bacteroides coprocola* | below reporting limit | higher in cases | — | the profiler reports only the organisms it detects, and this one is not in this sample's table. No validated reporting limit exists for it at this sequencing depth, so its point is omitted and only its bounds are kept: a nondetection, not a measured absence. It was detected in 13% of the reference participants. |

## Evidence pointing the other way

**Two of these studies disagree on direction.** *Bacteroides coprocola* is recorded here as increased in psoriasis, following its source, but a separate psoriasis cohort reported that organism as enriched in *controls*. The slot is kept as its own source described it and the disagreement is not resolved by preferring one paper.

Sources recorded as counterevidence for this panel: E04.

## What the human research has actually tested for this condition

**Plaque psoriasis — what human trials have tested** LANE C HUMAN TRIAL · LIMITED

Fired by: plaque psoriasis stool pattern: somewhat high

Resembling a published psoriasis stool pattern is a research observation, not a diagnosis and not an indication for treatment. The randomized evidence here is for one exact three-strain product added to topical therapy in people who already had a confirmed diagnosis. It does not transfer to other probiotics, it does not follow from this index, and no dose is issued from this report.

**Studied:** Bifidobacterium longum CECT 7347 + Bifidobacterium lactis CECT 8145 + Lacticaseibacillus rhamnosus CECT 8361, 1:1:1 (probiotic) · **in:** people with clinically confirmed plaque psoriasis, ages 18–99, in Spain, n = 90 · **studied protocol (a source fact, not an instruction):** 1:1:1 combination, total 10^9 CFU/day for 12 weeks, adjunct to topical treatment received by both arms

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: single study. Strongest design: randomized trial.

- In 90 adults with plaque psoriasis, adding one exact three-strain probiotic to topical treatment for 12 weeks raised the proportion reaching at least 75% improvement from 41.9% to 66.7%. The endpoint is that 75% threshold, despite looser wording in the abstract. This is evidence about that formulation as an add-on to topical therapy; it is not evidence that a stool profile predicts who responds, that other probiotic blends work, or that treatment normalises any organism measured in this report. [Navarro-Lopez V et al. Acta Derm Venereol 2019;99:1078] · lane C · 12-week randomized placebo-controlled adjunct trial, both arms on topical treatment · n = 90 · benefit

- **Against or limiting —** The relapse follow-up reported after this trial was not part of the original protocol. It is kept as a caution about how far the result extends, not as an additional benefit claim, and trial adverse-event reporting does not establish universal safety. [Navarro-Lopez V et al. Acta Derm Venereol 2019;99:1078, post-trial follow-up] · lane X · post-trial relapse follow-up outside the original protocol

**Applicability:** The study required a clinical diagnosis of plaque psoriasis. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed.

**Fecal microbiota transplantation:** not indicated.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

This report will not say: “treats psoriasis” · “clears plaques” · “normalises SGB4348” · “antimicrobials for psoriasis” · “eradicate the raised organisms”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

# 23.3 Severe plaque psoriasis — Sá 2026 three-taxon panel

**Plaque psoriasis (raised scaly skin plaques).** Plaque psoriasis is a chronic immune-mediated skin disease with raised, scaly plaques, commonly on elbows, knees, scalp and trunk. Four separate stool-shotgun studies are carried here as separate readings rather than one score, because they disagree: Deng 2026 nominates a single SGB, Chang 2022 and Sá 2026 report largely non-overlapping species, and Bacteroides coprocola is enriched in cases in one study and in controls in another.




**Measured-feature index 41**, over the 1 of 3 declared features that could be measured. **Possible full-panel range 14–80**, which is where the index could sit once the 2 unmeasured features are accounted for. These two describe different things: the range is not a margin of error around the index, and the index can fall outside it without any arithmetic being wrong.

## What was measured, and against whom

Source comparison: **severe psoriasis vs healthy**. Measurement: shotgun taxonomic fraction. Twenty-four psoriasis participants, of whom 18 were receiving immunosuppression and 5 had parasitic infection, against 10 controls. Treatment and infection are not separable from disease in this contrast.

No percentile is offered on this scale: that would need a held-out reference score distribution on this same set of features, and none exists.

## Every declared feature, including the ones that could not be measured

| ORGANISM OR GENE FAMILY | MEASURED | STUDY DIRECTION | CONTRIBUTION | NOTE |
|---|---|---|---|---|
| *Enterocloster* | below reporting limit | lower in cases | — | the profiler reports only the organisms it detects, and this one is not in this sample's table. No validated reporting limit exists for it at this sequencing depth, so its point is omitted and only its bounds are kept: a nondetection, not a measured absence. It was detected in 54% of the reference participants. |
| *Phascolarctobacterium faecium* | below reporting limit | lower in cases | — | the profiler reports only the organisms it detects, and this one is not in this sample's table. No validated reporting limit exists for it at this sequencing depth, so its point is omitted and only its bounds are kept: a nondetection, not a measured absence. It was detected in 44% of the reference participants. |
| *Bacteroides finegoldii* | 0.531% | lower in cases | 41 | Bacteroides finegoldii, which is a different organism from Alistipes finegoldii in the pooled psoriatic-disease panel. Same epithet, different genus; no fuzzy match. |

## Evidence pointing the other way

**Treatment is not separable from disease here.** Eighteen of the twenty-four psoriasis participants were on immunosuppression and five had parasitic infection, against ten controls. Differences attributed to psoriasis may belong to its treatment.

## What the human research has actually tested for this condition

**Plaque psoriasis — what human trials have tested** LANE C HUMAN TRIAL · LIMITED

Fired by: plaque psoriasis stool pattern: somewhat high

Resembling a published psoriasis stool pattern is a research observation, not a diagnosis and not an indication for treatment. The randomized evidence here is for one exact three-strain product added to topical therapy in people who already had a confirmed diagnosis. It does not transfer to other probiotics, it does not follow from this index, and no dose is issued from this report.

**Studied:** Bifidobacterium longum CECT 7347 + Bifidobacterium lactis CECT 8145 + Lacticaseibacillus rhamnosus CECT 8361, 1:1:1 (probiotic) · **in:** people with clinically confirmed plaque psoriasis, ages 18–99, in Spain, n = 90 · **studied protocol (a source fact, not an instruction):** 1:1:1 combination, total 10^9 CFU/day for 12 weeks, adjunct to topical treatment received by both arms

Evidence lane C: single, pilot or conflicting human randomized evidence (clinical or surrogate endpoint). Replication: single study. Strongest design: randomized trial.

- In 90 adults with plaque psoriasis, adding one exact three-strain probiotic to topical treatment for 12 weeks raised the proportion reaching at least 75% improvement from 41.9% to 66.7%. The endpoint is that 75% threshold, despite looser wording in the abstract. This is evidence about that formulation as an add-on to topical therapy; it is not evidence that a stool profile predicts who responds, that other probiotic blends work, or that treatment normalises any organism measured in this report. [Navarro-Lopez V et al. Acta Derm Venereol 2019;99:1078] · lane C · 12-week randomized placebo-controlled adjunct trial, both arms on topical treatment · n = 90 · benefit

- **Against or limiting —** The relapse follow-up reported after this trial was not part of the original protocol. It is kept as a caution about how far the result extends, not as an additional benefit claim, and trial adverse-event reporting does not establish universal safety. [Navarro-Lopez V et al. Acta Derm Venereol 2019;99:1078, post-trial follow-up] · lane X · post-trial relapse follow-up outside the original protocol

**Applicability:** The study required a clinical diagnosis of plaque psoriasis. This report provides microbiome resemblance or capacity only. Clinical eligibility is therefore unconfirmed.

**Fecal microbiota transplantation:** not indicated.

Safety: No known safety signal is on record for this context; that is not a statement that it is safe for you.

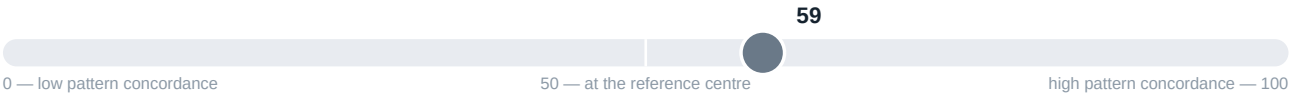
This report will not say: “treats psoriasis” · “clears plaques” · “normalises SGB4348” · “antimicrobials for psoriasis” · “eradicate the raised organisms”

Not a treatment score, a probability of benefit or a personalised prescription. Discuss any change with a clinician or registered dietitian.

‹ at a glance

# 23.4 Psoriatic disease, skin and joint pooled — Xiao 2024 three species

**Psoriatic disease, skin and joint forms pooled.** Xiao 2024 compared 44 people with skin-only plaque psoriasis and 26 with psoriatic arthritis, pooled, against 25 controls. The three species below come from that pooled comparison. No species difference separated psoriatic arthritis from skin-only psoriasis in that cohort, so this reading cannot tell the two apart.




**Index 59 of 100** over all 3 declared features, each measured as a point, so the coverage range collapses onto the index itself. That says every declared organism was measured; it says nothing about sampling error, which is not estimated here.

## What was measured, and against whom

Source comparison: **pooled plaque psoriasis and psoriatic arthritis vs healthy**. Measurement: shotgun taxonomic fraction. The q-values here belong to the pooled psoriatic-disease-versus-healthy contrast and to nothing else. The source reported roughly 0.74-0.77 AUC for Eubacterium rectale abundance alone; that is one feature's figure, not this three-species index's, and subgroup-specific estimates for the two Alistipes species are unknown.

No percentile is offered on this scale: that would need a held-out reference score distribution on this same set of features, and none exists.

## Every declared feature, including the ones that could not be measured

| ORGANISM OR GENE FAMILY | MEASURED | STUDY DIRECTION | CONTRIBUTION | NOTE |
|---|---|---|---|---|
| *Eubacterium rectale* | 1.08% | lower in cases | 60 | Eubacterium rectale was transferred to Agathobacter rectalis (Rosero 2016): one organism, two names. The pinned catalogue is checked for both and the one it carries is used; a catalogue carrying both would withhold the slot rather than sum them. |
| *Alistipes finegoldii* | 0.743% | lower in cases | 36 | |
| *Alistipes shahii* | 0.0123% | lower in cases | 81 | |

## Evidence pointing the other way

**This is a pooled contrast.** Skin-only psoriasis and psoriatic arthritis were combined against healthy controls, and no species difference separated the two from each other in that cohort. A high reading here cannot be read as psoriatic arthritis rather than psoriasis, and the source's q-values belong to the pooled comparison only.

## Matched intervention evidence

No supported signature-matched intervention evidence was identified for this pattern. Resembling a research pattern is not a diagnosis and does not establish treatment eligibility, predicted benefit, or any need to remove organisms. Nothing on this page triggers antibiotics, antifungals, botanical antimicrobials, a prescription change, or a stool transplant.

‹ at a glance

# 23.5 Psoriatic arthritis — Liu 2024 nine gene families

**Psoriatic arthritis (psoriasis with joint inflammation).** Psoriatic arthritis is inflammatory arthritis occurring with psoriasis. The panel here is nine microbial gene families (KEGG orthologues) reported by Liu 2024 in 20 treatment-naive patients against 10 controls. Ten controls do not meet this build's reference minimum, so the panel scores only against an independently compatible functional reference, or not at all.

**No index — none of the organisms could be measured.** This is not a negative result and not clearance. It means the number could not be produced, for the reasons below.

Reasons recorded: age unknown scored in research mode without a population guarantee; antibiotic exposure unknown and unknown is not none; no pinned gene family annotation was supplied; no taxa to function imputation is permitted.

This panel needs directly annotated gene-family abundances from a pinned annotation pipeline, normalised across every measured microbial gene family in that same space. The taxonomic table cannot stand in for it.

## What was measured, and against whom

Source comparison: **treatment naive psoriatic arthritis vs healthy**. Measurement: shotgun ko fraction. Selected by LEfSe at p<0.05 with LDA>2 in 20 patients and 10 controls, with no external validation and no recovered public accession. Ten discovery controls do not meet this build's reference minimum of 20 per feature.

No percentile is offered on this scale: that would need a held-out reference score distribution on this same set of features, and none exists.

## Every declared feature, including the ones that could not be measured

No feature-level measurements were produced for this panel, so there is nothing to break down. The declared organisms are listed in the panel definition and all of them remain in the coverage denominator.

## Matched intervention evidence

No supported signature-matched intervention evidence was identified for this pattern. Resembling a research pattern is not a diagnosis and does not establish treatment eligibility, predicted benefit, or any need to remove organisms. Nothing on this page triggers antibiotics, antifungals, botanical antimicrobials, a prescription change, or a stool transplant.

‹ at a glance

# 23.6 Psoriatic arthritis versus skin-only psoriasis — KEGG 00072 capacity

**Psoriatic arthritis (psoriasis with joint inflammation).** Psoriatic arthritis is inflammatory arthritis occurring with psoriasis. The panel here is nine microbial gene families (KEGG orthologues) reported by Liu 2024 in 20 treatment-naive patients against 10 controls. Ten controls do not meet this build's reference minimum, so the panel scores only against an independently compatible functional reference, or not at all.

**No index — none of the organisms could be measured.** This is not a negative result and not clearance. It means the number could not be produced, for the reasons below.

Reasons recorded: age unknown scored in research mode without a population guarantee; antibiotic exposure unknown and unknown is not none; no pinned gene family annotation was supplied; no taxa to function imputation is permitted.

This panel needs directly annotated gene-family abundances from a pinned annotation pipeline, normalised across every measured microbial gene family in that same space. The taxonomic table cannot stand in for it.

## What was measured, and against whom

Source comparison: **psoriatic arthritis vs skin only psoriasis**. Measurement: shotgun ko fraction. This contrast compares two groups who both have psoriatic disease. It answers nothing about a healthy person and is never inferred from the three taxa in the pooled panel from the same participants. Butanoate metabolism in the same source was not significant (q=0.157).

No percentile is offered on this scale: that would need a held-out reference score distribution on this same set of features, and none exists.

## Every declared feature, including the ones that could not be measured

No feature-level measurements were produced for this panel, so there is nothing to break down. The declared organisms are listed in the panel definition and all of them remain in the coverage denominator.

## Matched intervention evidence

No supported signature-matched intervention evidence was identified for this pattern. Resembling a research pattern is not a diagnosis and does not establish treatment eligibility, predicted benefit, or any need to remove organisms. Nothing on this page triggers antibiotics, antifungals, botanical antimicrobials, a prescription change, or a stool transplant.

‹ at a glance

# 23.7 Adult atopic dermatitis (eczema) — Wang 2023 thirteen slots, translated from 16S

**Atopic dermatitis in adults (eczema).** Atopic dermatitis, commonly called eczema, is a chronic itchy inflammatory skin disease. Wang 2023 studied 104 adult cases and 130 controls by 16S amplicon sequencing, which names organisms differently from the shotgun sequencing used here. Thirteen slots are declared; five are source groups from a 16S reference database with no exact shotgun equivalent, and those stay in the coverage denominator as unavailable rather than being approximated.




**Measured-feature index 41**, over the 2 of 13 declared features that could be measured. **Possible full-panel range 6–91**, which is where the index could sit once the 11 unmeasured features are accounted for. These two describe different things: the range is not a margin of error around the index, and the index can fall outside it without any arithmetic being wrong.

## What was measured, and against whom

Source comparison: **adult atopic dermatitis vs healthy**. Measurement: shotgun genus fraction translated from 16s. **This is a 16S-derived pattern applied to shotgun measurements.** The source study named organisms with a different method from the one used here, so the translation is unvalidated and inherits none of the source's sensitivity or specificity. Selected by LEfSe at p<0.05 with LDA>2. Only Clostridium sensu stricto 1 was also significant under ANCOM (W=218). Neither Shannon diversity nor the Firmicutes-to-Bacteroidetes ratio differed, so neither appears here.

No percentile is offered on this scale: that would need a held-out reference score distribution on this same set of features, and none exists.

## Every declared feature, including the ones that could not be measured

| ORGANISM OR GENE FAMILY | MEASURED | STUDY DIRECTION | CONTRIBUTION | NOTE |
|---|---|---|---|---|
| *Blautia* | 0.618% | higher in cases | 59 | |
| *Butyricicoccus* | below reporting limit | higher in cases | — | the profiler reports only the organisms it detects, and this one is not in this sample's table. No validated reporting limit exists for it at this sequencing depth, so its point is omitted and only its bounds are kept: a nondetection, not a measured absence. It was detected in 3% of the reference participants. |
| *Lachnoclostridium* | below reporting limit | higher in cases | — | the profiler reports only the organisms it detects, and this one is not in this sample's table. No validated reporting limit exists for it at this sequencing depth, so its point is omitted and only its bounds are kept: a nondetection, not a measured absence. It was detected in 10% of the reference participants. |
| *Eubacterium hallii group* | not in database | higher in cases | — | A SILVA138 group, not a genus and not a species. Its membership must be resolved to a frozen non-overlapping set of catalogue entries before it can be summed; until then the slot is unavailable and stays in the denominator. |
| *Erysipelatoclostridium* | below reporting limit | higher in cases | — | the profiler reports only the organisms it detects, and this one is not in this sample's table. No validated reporting limit exists for it at this sequencing depth, so its point is omitted and only its bounds are kept: a nondetection, not a measured absence. It was detected in 23% of the reference participants. |
| *Megasphaera* | below reporting limit | higher in cases | — | the profiler reports only the organisms it detects, and this one is not in this sample's table. No validated reporting limit exists for it at this sequencing depth, so its point is omitted and only its bounds are kept: a nondetection, not a measured absence. It was detected in 16% of the reference participants. |
| *Oscillibacter* | 0.135% | higher in cases | 22 | |

| ORGANISM OR GENE FAMILY | MEASURED | STUDY DIRECTION | CONTRIBUTION | NOTE |
|---|---|---|---|---|
| *Flavonifractor* | below reporting limit | higher in cases | — | the profiler reports only the organisms it detects, and this one is not in this sample's table. No validated reporting limit exists for it at this sequencing depth, so its point is omitted and only its bounds are kept: a nondetection, not a measured absence. It was detected in 77% of the reference participants. |
| *unclassified Oscillospiraceae* | not in database | higher in cases | — | The source's unclassified bucket within this family. It is not the family total: summing the whole family would count organisms the source resolved separately. |
| *Romboutsia* | below reporting limit | lower in cases | — | the profiler reports only the organisms it detects, and this one is not in this sample's table. No validated reporting limit exists for it at this sequencing depth, so its point is omitted and only its bounds are kept: a nondetection, not a measured absence. It was detected in 11% of the reference participants. |
| *Clostridium sensu stricto 1* | not in database | lower in cases | — | A SILVA138 group, and the one finding here that also survived ANCOM. It is never substituted with the genus Clostridium, which is far broader. |
| *unclassified Butyricicoccaceae* | not in database | lower in cases | — | The source's unclassified bucket within this family, not the family total. |
| *unclassified Erysipelotrichaceae* | not in database | lower in cases | — | The source's unclassified bucket within this family, not the family total. |

## Evidence pointing the other way

Sources recorded as counterevidence for this panel: E07.

## Matched intervention evidence

No supported signature-matched intervention evidence was identified for this pattern. Resembling a research pattern is not a diagnosis and does not establish treatment eligibility, predicted benefit, or any need to remove organisms. Nothing on this page triggers antibiotics, antifungals, botanical antimicrobials, a prescription change, or a stool transplant.

‹ at a glance

# 23.8 Advanced non-segmental vitiligo — Luan 2023 eleven species

**Non-segmental vitiligo (patches of skin losing pigment).** Vitiligo is loss of skin pigment in patches; the non-segmental form is the commoner, usually symmetrical type. Two panels are carried separately. Luan 2023 is a single matched cohort of 25 cases and 25 controls with native shotgun sequencing. Ju 2025 compared 10 actively spreading cases against 20 controls selected out of a different study, so its cases and controls were never processed together.




**Measured-feature index 24**, over the 2 of 11 declared features that could be measured. **Possible full-panel range 4–86**, which is where the index could sit once the 9 unmeasured features are accounted for. These two describe different things: the range is not a margin of error around the index, and the index can fall outside it without any arithmetic being wrong.

## What was measured, and against whom

Source comparison: **advanced nonsegmental vitiligo vs healthy**. Measurement: shotgun taxonomic fraction. The source's published AUC of 0.786 belongs to a 28-feature random forest under nested cross-validation, which is a different estimator on a different feature set from the eleven-slot index computed here. Beta diversity did not differ (p=0.495). Individual q-values were not published per feature and are not invented.

No percentile is offered on this scale: that would need a held-out reference score distribution on this same set of features, and none exists.

## Every declared feature, including the ones that could not be measured

| ORGANISM OR GENE FAMILY | MEASURED | STUDY DIRECTION | CONTRIBUTION | NOTE |
|---|---|---|---|---|
| *Bacteroides fragilis* | below reporting limit | higher in cases | — | the profiler reports only the organisms it detects, and this one is not in this sample's table. No validated reporting limit exists for it at this sequencing depth, so its point is omitted and only its bounds are kept: a nondetection, not a measured absence. It was detected in 52% of the reference participants. |
| *Massilioclostridium coli* | not in database | higher in cases | — | this organism has no slot in the pinned reference catalogue used here, so its absence has not been measured; the slot lowers coverage instead of scoring zero |
| *Lachnospiraceae bacterium BX3* | not in database | higher in cases | — | An exact source concept, not the family Lachnospiraceae. |
| *TM7 phylum sp oral taxon 348* | not in database | higher in cases | — | An exact source concept, not the TM7 phylum as a whole. |
| *Prevotella copri clade C* | not in database | lower in cases | — | One clade of Prevotella copri, not all P. copri and not an unverified Segatella relabelling. Without clade resolution in the catalogue the slot is unavailable. |
| *Dorea longicatena* | 1.93% | lower in cases | 27 | |
| *Allisonella histaminiformans* | below reporting limit | lower in cases | — | the profiler reports only the organisms it detects, and this one is not in this sample's table. No validated reporting limit exists for it at this sequencing depth, so its point is omitted and only its bounds are kept: a nondetection, not a measured absence. It was detected in 15% of the reference participants. |
| *Coprobacter secundus* | below reporting limit | lower in cases | — | the profiler reports only the organisms it detects, and this one is not in this sample's table. No validated reporting limit exists for it at this sequencing depth, so its point is omitted and only its bounds are kept: a nondetection, not a measured absence. It was detected in 13% of the reference participants. |
| *Coprococcus comes* | 1.84% | lower in cases | 21 | |

| ORGANISM OR GENE FAMILY | MEASURED | STUDY DIRECTION | CONTRIBUTION | NOTE |
|---|---|---|---|---|
| *Sellimonas intestinalis* | below reporting limit | lower in cases | — | the profiler reports only the organisms it detects, and this one is not in this sample's table. No validated reporting limit exists for it at this sequencing depth, so its point is omitted and only its bounds are kept: a nondetection, not a measured absence. It was detected in 15% of the reference participants. |
| *Bacteroides bouchesdurhonensis* | not in database | lower in cases | — | this organism has no slot in the pinned reference catalogue used here, so its absence has not been measured; the slot lowers coverage instead of scoring zero |

## Matched intervention evidence

No supported signature-matched intervention evidence was identified for this pattern. Resembling a research pattern is not a diagnosis and does not establish treatment eligibility, predicted benefit, or any need to remove organisms. Nothing on this page triggers antibiotics, antifungals, botanical antimicrobials, a prescription change, or a stool transplant.

‹ at a glance

# 23.9 Actively spreading vitiligo — Ju 2025 four species, exploratory

**Non-segmental vitiligo (patches of skin losing pigment).** Vitiligo is loss of skin pigment in patches; the non-segmental form is the commoner, usually symmetrical type. Two panels are carried separately. Luan 2023 is a single matched cohort of 25 cases and 25 controls with native shotgun sequencing. Ju 2025 compared 10 actively spreading cases against 20 controls selected out of a different study, so its cases and controls were never processed together.

**No index — no usable features.** This is not a negative result and not clearance. It means the number could not be produced, for the reasons below.

Reasons recorded: no usable features; age unknown scored in research mode without a population guarantee; antibiotic exposure unknown and unknown is not none.

## What was measured, and against whom

Source comparison: **active spreading vitiligo vs external controls**. Measurement: shotgun taxonomic fraction. Ten cases from one project and twenty controls selected out of a different project of 37 people. Cases and controls were never processed together, so study batch and disease are not separable. Alpha diversity did not differ significantly.

No percentile is offered on this scale: that would need a held-out reference score distribution on this same set of features, and none exists.

## Every declared feature, including the ones that could not be measured

| ORGANISM OR GENE FAMILY | MEASURED | STUDY DIRECTION | CONTRIBUTION | NOTE |
|---|---|---|---|---|
| *Faecalibacterium prausnitzii* | below reporting limit | lower in cases | — | the profiler reports only the organisms it detects, and this one is not in this sample's table. No validated reporting limit exists for it at this sequencing depth, so its point is omitted and only its bounds are kept: a nondetection, not a measured absence. It was detected in 98% of the reference participants. |
| *Faecalibacterium duncaniae* | not in database | lower in cases | — | Split out of historical F. prausnitzii. Quarantined together with it unless the catalogue proves the two inputs disjoint. |
| *Megamonas funiformis* | below reporting limit | lower in cases | — | the profiler reports only the organisms it detects, and this one is not in this sample's table. No validated reporting limit exists for it at this sequencing depth, so its point is omitted and only its bounds are kept: a nondetection, not a measured absence. It was detected in 10% of the reference participants. |
| *Bifidobacterium bifidum* | below reporting limit | higher in cases | — | the profiler reports only the organisms it detects, and this one is not in this sample's table. No validated reporting limit exists for it at this sequencing depth, so its point is omitted and only its bounds are kept: a nondetection, not a measured absence. It was detected in 22% of the reference participants. |

## Evidence pointing the other way

**Cases and controls came from different studies.** The ten cases and twenty controls were deposited by separate projects, so sequencing batch and disease cannot be separated. This panel is also not a replication of the other vitiligo panel: they share almost no features.

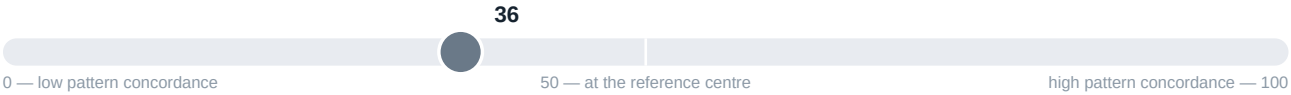
## Matched intervention evidence

No supported signature-matched intervention evidence was identified for this pattern. Resembling a research pattern is not a diagnosis and does not establish treatment eligibility, predicted benefit, or any need to remove organisms. Nothing on this page triggers antibiotics, antifungals, botanical antimicrobials, a prescription change, or a stool transplant.

‹ at a glance

# 23.10 Hidradenitis suppurativa — Ogut 2022 single genus, translated from 16S

**Hidradenitis suppurativa (recurrent painful skin abscesses in folds).** Hidradenitis suppurativa causes recurrent painful nodules and abscesses in skin folds. The stool evidence is thin: one 16S study of 15 cases and 15 controls supplies a single genus at p=0.046, and a separate shotgun study found no significant difference at all between hidradenitis cases and controls. Both are shown.




**Index 36 of 100 — a single-marker result.** This panel declares one feature, so the figure is one organism's position against the reference group, not a signature. It cannot be more than that however far from 50 it sits.

## What was measured, and against whom

Source comparison: **hidradenitis suppurativa vs healthy**. Measurement: shotgun genus fraction translated from 16s. **This is a 16S-derived pattern applied to shotgun measurements.** The source study named organisms with a different method from the one used here, so the translation is unvalidated and inherits none of the source's sensitivity or specificity. One genus at p=0.046 in 15 cases and 15 controls, by 16S amplicon, with no verified raw accession. A separate shotgun study (Sá 2026) found no significant taxonomic difference between hidradenitis cases and controls at all.

No percentile is offered on this scale: that would need a held-out reference score distribution on this same set of features, and none exists.

## Every declared feature, including the ones that could not be measured

| ORGANISM OR GENE FAMILY | MEASURED | STUDY DIRECTION | CONTRIBUTION | NOTE |
|---|---|---|---|---|
| *Fusicatenibacter* | 1.91% | lower in cases | 36 | |

## Evidence pointing the other way

**A separate shotgun study found nothing.** Comparing hidradenitis suppurativa against controls, it reported no significant taxonomic difference and no significant diversity difference. With one translated feature here against an explicit negative comparison there, the honest summary is that stool evidence for this condition is very weak.

Sources recorded as counterevidence for this panel: E03.

## Matched intervention evidence

No supported signature-matched intervention evidence was identified for this pattern. Resembling a research pattern is not a diagnosis and does not establish treatment eligibility, predicted benefit, or any need to remove organisms. Nothing on this page triggers antibiotics, antifungals, botanical antimicrobials, a prescription change, or a stool transplant.

# 23.11 Seborrheic dermatitis — the evidence, and why there is no score

**Gut involvement has been reported, but no adequately specified stool-DNA signature was identified for this implementation. This result cannot assess whether you have seborrheic dermatitis.**

Seborrheic dermatitis is a scaly, greasy rash of the scalp, face and upper trunk; dandruff is its mildest form. Gut involvement has been reported, and the three blocks below are what that reporting actually consists of. None of them defines a stool-DNA signature, and publishing a number anyway would be inventing one.

## Stool bacterial culture

A 2019 series enrolled 67 people aged 18 to 57 and describes 30 controls, using stool **bacteriology** rather than sequencing. It reported how often counts were low: bifidobacteria in 59.4%, enterococci 62.5%, *E. coli* 65.7%, lactobacilli 90.6%, bacteroides 71.9%. Those percentages imply a denominator near 32, which is not explained against the 67 people enrolled, and the denominator discrepancy is preserved here rather than tidied away. Per-subject values, control distributions and usable thresholds are absent. Counts of living organisms grown on a plate are a different quantity from the share of DNA in a sequencing run, so these cannot be converted into abundance coefficients — and a study reporting *low* E. coli does not become a report of high E. coli because a generic dysbiosis template expects it.

## Genetic hypotheses

A 2024 study used genetic instruments and national disease records rather than measuring stool from people with the condition. Its nominal associations point in various directions across broad, overlapping taxonomic ranks. These concern genetically instrumented exposures, not an observed diagnostic stool profile, so no organism here receives a weight. The printed significance threshold in that paper is also internally inconsistent. Human genetic risk is not recovered from stool sequencing in this report.

## Scalp and skin findings

Studies of *Malassezia*, of Staphylococcus-to-Cutibacterium ratios, of scalp lipases and of a topical probiotic all measure **skin**, not stool. They cannot seed a gut classifier. Fungal organisms seen in stool remain exactly that: measured stool fungal observations, which neither demonstrate scalp overgrowth nor prove contamination.

## What would have to exist for this to become a score

A measured seborrheic-dermatitis case-control stool dataset with exact phenotype, body site, feature definitions, assay, clinical metadata and a reproducible analysis, tested against psoriasis and sebopsoriasis, atopic dermatitis, rosacea and healthy controls. A culture-based module would additionally need quantitative culture inputs and reference distributions; it is not built by converting published percentages into colony counts.

A reported figure such as "60% had dysbiosis" in a clinical series is not a screening prevalence, a disease prior, or a cutoff, and is not used as one.

# 23.12 Infant atopic dermatitis (eczema) — Seong 2025 strain-level finding, not scored

The infant finding here is a distinction between two subclades inside one bacterial subspecies, which this pipeline cannot yet resolve. No species or genus measurement substitutes for it, and the finding does not transfer to an adult or to a child aged 8 to 18.

**What would have to exist:** The exact assemblies and marker reference, source subclade labels, sequence quality and unique-marker alignment criteria, strain coverage and mixture behaviour, and independent age-matched validation. No read-depth threshold is asserted for strain reconstruction until such a validation bundle exists.

# 23.13 Plaque psoriasis — Deng 2026 original classifier, not reproduced

The source study's own classifier is not reproduced here. Its published text and supplement disagree with each other: a three-feature AUC appears as both 0.74 and 0.76, and two precision-recall figures, 0.83 and 0.96, are swapped between the two accounts. The methods mix relative abundance with a read-count adjustment, the cross-validation is not documented as grouped by household, and the declared code repository was unreachable. Picking whichever number is more favourable would be the wrong repair.

**What would have to exist:** A reproducible locked implementation with source labels, oversampling confined to inner training folds, original untouched test prevalence retained, and household grouping. The separate directional index above remains implementable and is what is shown.

## Measured alongside, with no disease-score weight

These were reported by the source studies and are shown because they were measured, not because they contribute to any score. DNA capacity for a pathway is not a measurement of a metabolite, and an annotation is not a validated gene call.

| SOURCE FINDING | WHAT IT MAY BE READ AS |
|---|---|
| **PWY-5005 (biotin biosynthesis)**<br>increased in discovery, failed to validate | Measured biotin-biosynthesis DNA potential. It failed validation in its own source, so it carries no disease-score weight and is not a replicated finding. |
| **P163-PWY**<br>decreased | One reported pathway-capacity observation. DNA capacity for a pathway is not a measurement of any metabolite in stool or blood. |

| SOURCE FINDING | WHAT IT MAY BE READ AS |
|---|---|
| **PWY-5304**<br>decreased | One reported pathway-capacity observation; no metabolite level is inferred from it. |
| **PWY-7200**<br>increased | One reported pathway-capacity observation; no metabolite level is inferred from it. |
| **KEGG 00500, 02030, 02040**<br>decreased in all psoriasis versus healthy controls | Source-associated functional potential. Not skin cytokines, and not a measurement of gut motility or any clinical feature. |
| **CAZy GH43 increased; CE4 and CBM50 decreased**<br>as stated | Source-specific gene-family potential from the same samples as the nine gene families scored above, so it is the same evidence group rather than an independent replication. |
| **vBin_422 and three published contig names**<br>annotation only | A lead with a name and no sequence. Without the actual assembly and a validated mapping and reference bundle there is nothing to score, and no reference sequence is generated to fill the gap. |
| **enzyme annotations labelled virulence or resistance**<br>reported by the source as annotation categories | Annotation categories, not validated resistance or virulence gene calls. The source's clinical framing of them is not imported. |

# 24 Your Information & Report Context

This report used 1 fact you supplied and 25 assumptions the report had to make because they were not. No external laboratory results were supplied, and none are needed for anything above.

**1** supplied **25** assumed **7** not supplied **0** laboratory results

## Everything this report used

| INFORMATION | VALUE USED | SUPPLIED OR ASSUMED | WHAT IT AFFECTS |
|---|---|---|---|
| **PARTICIPANT AND SPECIMEN** | | | |
| Sample identifier | **MM1_FXX745** | Supplied | **adds context** — Names this report and links it to your sequencing files. |
| Sample type | **stool** | Assumed — existing default<br>This pipeline analyses stool. No other specimen type is accepted, so the value is fixed rather than guessed. | **changes how a reading is read** — Every reading here describes stool. A measurement from another body site cannot be read off this report. |
| age 18 or over | **yes** | Assumed — existing default<br>Not supplied. In research mode the report assumes True so the reading can be produced at all. | **changes how a reading is read** — If the person is under 18, every supplement, probiotic, diet and medication card would be withheld: none of the underlying evidence was generated in children. |
| **DIET AND TOLERANCE** | | | |
| Diet used for the synbiotic scenarios | **Average European diet** | Assumed for this run<br>No record of what you actually eat was supplied, so the scenarios use the published reference diet the protocol declares, and are additionally run against every other declared diet so the range is visible. | **changes a number** — Sets how much of each nutrient the modelled community has. The sensitivity run shows how much the answer moves when this assumption changes - here it moves a great deal, which is why the range is printed. |
| **MEDICATIONS AND SUPPLEMENTS** | | | |
| metformin | **no** | Assumed — existing default<br>Not supplied. In research mode the report assumes False so the reading can be produced at all. | **changes how a reading is read** — Metformin raises Escherichia, lowers Intestinibacter and shifts bile-salt hydrolase carriage (Forslund 2015). The type 2 diabetes pattern would then be partly measuring the drug, and its resemblance percentile would read higher than the disease alone warrants; it would need to be re-read against the metformin-treated stratum. |
| proton-pump inhibitor | **no** | Assumed — existing default<br>Not supplied. In research mode the report assumes False so the reading can be produced at all. | **changes how a reading is read** — Proton-pump inhibitors are the medication with the largest known effect on stool composition: they let Streptococcus, Veillonella and other mouth bacteria reach the colon and lower diversity (Imhann 2016, Jackson 2016). The oral-origin group and any pattern that leans on oral taxa would read higher than the underlying gut state. |
| statin | **no** | Assumed — existing default<br>Not supplied. In research mode the report assumes False so the reading can be produced at all. | **changes how a reading is read** — Statin use is associated with a lower prevalence of the Bacteroides 2 enterotype and a less disturbed community in obese adults (Vieira-Silva 2020); the general index and the cardiometabolic patterns would need to be read with the drug in view. |
| GLP-1 receptor agonist | **no** | Assumed — existing default<br>Not supplied. In research mode the report assumes False so the reading can be produced at all. | **changes how a reading is read** — GLP-1 agonists slow gastric emptying and change bile-acid handling; the type 2 diabetes, obesity and MASLD patterns would be read against a treated rather than an untreated community. |
| immunosuppressant | **no** | Assumed — existing default<br>Not supplied. In research mode the report assumes False so the reading can be produced at all. | **changes how a reading is read** — Live-microbe interventions carry case reports of bacteraemia and fungaemia in immunocompromised adults; the safety rules would move every probiotic and fermented-food card to clinician review. |
| anticoagulant or antiplatelet | **no** | Assumed — existing default<br>Not supplied. In research mode the report assumes False so the reading can be produced at all. | **changes how a reading is read** — Vitamin K2 and several botanicals (garlic, ginkgo, high-dose fish oil) interact with anticoagulation; those cards would carry a caution and move to clinician review. |
| **CLINICAL HISTORY** | | | |

| INFORMATION | VALUE USED | SUPPLIED OR ASSUMED | WHAT IT AFFECTS |
|---|---|---|---|
| 13 safety questions | **assumed no** | Assumed — existing default<br>None were supplied. Each is assumed absent so the action cards can be produced; any one of them being true changes them. | **changes how a reading is read** — These are the safety rules on the intervention cards (pregnancy, breastfeeding, immune compromise, central venous catheter, critical illness or ICU care, short bowel or structural GI disease, active IBD flare, kidney disease, liver disease, kidney stone history, GI bleeding history, eating disorder or malnutrition, surgery in the next 14 days). If any is true, the affected probiotic, supplement, diet or drug card moves to clinician review or is withdrawn. |
| **COLLECTION AND PROCESSING** | | | |
| Sequencing quality gates | **warn** | Worked out from what you supplied | **changes how a reading is read** — Decides whether the readings are fit to interpret at all. The full gate table, with every threshold and what was measured against it, is in the sequencing-quality section. 5,660,139 usable non-host pairs. |
| **LONGITUDINAL EXPOSURES** | | | |
| antibiotics in the last 90 days | **no** | Assumed — existing default<br>Not supplied. In research mode the report assumes False so the reading can be produced at all. | **changes how a reading is read** — A recent antibiotic course depresses diversity, butyrate producers and most of the species the disease patterns are built on. Every pattern would abstain, and the low readings in the organism section would be attributed to the drug rather than to the person's baseline community. |
| laxative or bowel preparation | **no** | Assumed — existing default<br>Not supplied. In research mode the report assumes False so the reading can be produced at all. | **changes how a reading is read** — Osmotic laxatives and colonoscopy preparation transiently collapse diversity and enrich Proteobacteria and mucin degraders; readings taken within about two weeks would reflect the preparation rather than the baseline community. |
| **ANALYSIS CONFIGURATION** | | | |
| Reference cohort for the community type | **3,027 samples** | Assumed for this run<br>No tighter matching information was supplied, so the documented eligible reference population is used: 3027 samples from 22 studies, frozen as ext083.community-type/2-medoid/1.0. | **changes a number** — Every distance on the community-type card is measured against this frozen cohort, and never against the other samples analysed alongside yours. |

## Not supplied, and what each would add

every reading above was produced without them

| CATEGORY | NOT SUPPLIED | WHAT IT WOULD ADD |
|---|---|---|
| **Participant and specimen** | Collection date | Without it, a repeat sample cannot be placed on a timeline and an imported laboratory result cannot be checked for date compatibility. |
| **Diet and tolerance** | Your actual dietary pattern, Known fibre intolerances or allergies | With it, the scenarios could be run on the diet you actually eat instead of a reference diet, and the fibre rankings would apply to you directly. |
| **Clinical history** | sex, age in years, country | Sex-specific patterns could be scored and the reference cohort matched more tightly. |

## External laboratory results

optional; this report is complete without them

**Not supplied.** Every reading in this report is produced from your sequencing data. If you have results for short-chain fatty acids, inflammation and immune markers, digestion, other stool chemistry, targeted metabolomics, endocrine and nutrient context, absolute microbial load, molecular pathogen assays, they can be imported and will be shown beside the finding each one relates to — as a separate, differently-labelled measurement, never merged into a genetic reading or used to overrule it. Nothing here is missing without them.

· An assumption is recorded as an assumption. It never becomes a supplied fact.

· An absent laboratory result has no imputed normal value and produces no normal or abnormal classification.

· A stored value with no consumer is labelled 'Recorded; not used in this report' rather than implied to be integrated.

· Conflicting supplied records are shown as conflicting; neither is silently overwritten.

# 25 What this report cannot tell you

The four boundaries of what a stool metagenome can show.

- **It does not measure metabolite levels.** Carrying the gene means the bacteria *can* make the metabolite. How much they actually make depends on your diet, your gut transit time and much else. Measuring the metabolites themselves requires a blood or stool chemistry test.

- **It does not diagnose or predict disease.** No threshold in this report corresponds to a clinical diagnosis, because no such threshold has been published for any of these pathways.

- **It is a snapshot.** Gut composition shifts with diet over days to weeks. A single measurement is far less informative than the same test repeated over time, which is why the trend matters more than any one value.

- **Pattern resemblance is resemblance, not diagnosis.** A resemblance percentile says how closely this community resembles the *average* pattern published for a group of patients. It is not a probability of having the condition. The underlying associations largely do not distinguish one condition from general gut disturbance — the same species shift in inflammatory bowel disease, after antibiotics, and with a low-fibre diet — which is why every pattern score is shown beside the general gut-health index.

FINE PRINT

Research use only. Nothing in this report is a diagnosis, a risk estimate or a treatment recommendation. If you have symptoms or concerns, the right next step is a clinician and the validated tests they can order.

## Terms this report is produced under

FINE PRINT

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# 26 Sequencing quality and reference coverage

Whether the sample was sequenced well enough for the readings above to mean what they say. Each gate has a prespecified threshold; a warning means the reading is shown but should be read with that in mind, and a failure means the affected section says so.

**WARN**
OVERALL
across all gates

**5,660,139**
USABLE NON-HOST PAIRS
minimum 500,000

**94.8%**
BASES AT Q30
≤ 0.1% error probability

**0.99%**
DUPLICATE RATE
whole-file, fastp

**147 bp**
READ LENGTH
paired end (151 cycles + 1…

| GATE | VALUE | | NOTE |
|---|---|---|---|
| **Input read pairs** | 5,660,139 | pass | read pairs in the FASTQ files as delivered |
| **Pairs passing quality filters** | 5,660,139 (100.0%) | pass | fastp: length ≥ 60 after polyG and adapter trimming, ≤ 5 N, mean Q ≥ 15; assessment only — original reads went downstream |
| **Human DNA fraction** | not assessable | not_assessable | no human reads at all — the sequencing provider removed them before delivery (routine for consumer kits), so the host fraction of the original specimen is not measurable here (not_assessable_upstream_removed) |
| **Usable non-host pairs** | 5,660,139 | pass | prespecified minimum 500,000 |
| **Reads assigned to a known species** | 45.5% | warn | the remainder is organisms without a reference genome, not a quality problem — 30-55% unclassified is normal for stool |
| **Duplicate read rate** | 0.99% | pass | fastp, whole file; high duplication means less unique information than the read count suggests |
| **Bases at Q30 or better** | 94.8% | pass | fraction of bases with ≤ 0.1% error probability |
| **Reads with adapter sequence** | 0.03% | pass | detected by read-pair overlap; 45% of pairs overlap, peak insert 105 bp; reads average 147 bp against 151 cycles with almost no adapter found, so the provider had already trimmed adapters to the insert |
| **Mean read length** | 147 bp | pass | translated search needs ≥ 75 bp to place a read on a protein; ≥ 100 bp preferred |
| **GC content** | 45.2% | pass | stool metagenomes sit near 45-50%; far outside suggests contamination or a different sample type |
| **Contamination flags** | none raised | pass | composite of host fraction, GC and classified fraction |
| **rpoB fragments (genome denominator)** | 6,574 | pass | single-copy marker used to express gene counts per 100 genomes; ≥ 2,000 for a stable denominator |
| **Sequencing batch** | LH00469 run 646, flowcell 23VJTHLT4, lane 8 | pass | from read headers; samples from different flowcells carry a batch difference |

Sequenced on instrument LH00469, run 646, flowcell 23VJTHLT4, lane(s) 8. Samples from different flowcells carry a batch difference that can exceed the biology; compare across samples only with that in mind.

Read preprocessing: fastp 1.3.6 (assess mode). 45% of pairs overlap, peak insert 105 bp; reads average 147 bp against 151 cycles with almost no adapter found, so the provider had already trimmed adapters to the insert.

## 26.1 How much of your community has a name

Your reads were searched against every reference catalogue available, and the organism list in section 6 is the merged result. The largest of them holds about 26,000 organisms assembled from roughly a million genomes, including some 4,900 that have never been grown in a laboratory and so have no formal name. The figures below say how much of your community each search could account for. Percentiles are the one thing that cannot be pooled: a position is only meaningful against a reference population measured the same way, and that population exists for one catalogue.

**253**
GENOME BINS (SGBS)
240 species · 150 genera

**100**
WITHOUT A NAME
27.3% of classified reads

**0**
ARCHAEA
methanogens and relatives

**0**
EUKARYOTES
fungi and protists in the cat…

**10.9%**
UNCLASSIFIED
reads matching nothing on…

**How much the catalogues agree.** 66 organisms were found by all of them. 174 were found only by the larger catalogue — recently described organisms, renamed genera, and organisms with no name — and 57 only by the smaller one, mostly names that have since been split or merged. All of them appear in your organism list: pooling the catalogues is what makes that list complete, and a detection found by one search and not another is a difference in reference coverage, not a contradiction.

Largest found only by the wider search: *Ruminococcus bromii*, *GGB6612 SGB9346*, *Ruminococcus SGB4421*, *GGB3267 SGB4316*, *GGB51269 SGB5062*, *GGB9342 SGB14306*, *GGB3175 SGB4191*, *GGB1495 SGB2071*, *Faecalibacterium SGB15346*, *GGB6601 SGB9333*.

| IDENTIFIER | NEAREST NAMED LINEAGE | ABUNDANCE |
|---|---|---|
| SGB9346 | Bacteria · *GGB6612* | 2.603% |
| SGB4421 | Bacteria · *Ruminococcus* | 2.359% |
| SGB4316 | Bacteria · *GGB3267* | 2.109% |
| SGB5062 | Bacteria · *GGB51269* | 1.898% |
| SGB14306 | Bacteria · *GGB9342* | 1.367% |
| SGB4191 | Bacteria · *GGB3175* | 1.294% |
| SGB2071 | Bacteria · *GGB1495* | 1.119% |
| SGB15346 | Bacteria · *Faecalibacterium* | 0.949% |

Unnamed organisms are known only from assembled genomes. They are real and often abundant; what is missing is a cultured isolate and therefore any physiology to report. Their share is one measure of how much of your community current science has not yet characterised.

# 27 How this test performs

Every figure below was measured, not estimated. The method was tested against synthetic samples built from reference genomes whose gene content is known exactly, so the correct answer was known in advance and could be compared against what the pipeline reported.

**97.0%**
SENSITIVITY
genes present that were found

**100.0%**
SPECIFICITY
genes absent, correctly absent

**100.0%**
PRECISION
of detections that were real

**0.0%**
FALSE POSITIVES
spurious detection rate

Based on 24 reference genomes across 5 synthetic samples (32 correct detections, 62 correct non-detections, 0 false positives, 1 missed).

## Are the numbers themselves accurate?

Beyond simply detecting a gene, the reported quantity was compared against the known quantity in each synthetic sample. A slope of 1.0 would mean the reported figure matches truth exactly, and an $R^2$ of 1.0 would mean a perfect relationship. Measured: slope **1.13**, $R^2$ **0.72** across 37 comparisons.

## How rare an organism can this still find?

| Carrier abundance | True copies/100 | Result | Fragments |
|---|---|---|---|
| 5.00% | 5.0 | found | 224 |
| 1.00% | 1.0 | found | 86 |
| 0.20% | 0.2 | found | 64 |

## Performance by pathway

| Pathway | Sensitivity | Specificity | Precision | TP/TN/FP/FN |
|---|---|---|---|---|
| bai | 100% | 100% | 100% | 6/24/0/0 |
| bsh | 100% | — | 100% | 5/0/0/0 |
| butyrate | 100% | 100% | 100% | 6/4/0/0 |
| cutc | 100% | 100% | 100% | 2/8/0/0 |
| indole | 100% | — | 100% | 5/0/0/0 |
| ipa | 100% | 100% | 100% | 2/8/0/0 |
| pcresol | 100% | 100% | 100% | 1/4/0/0 |
| propionate | 100% | 100% | 100% | 2/13/0/0 |
| urda | 75% | 100% | 100% | 3/1/0/1 |

## The reference group

100 adult stool shotgun metagenomes from ENA study PRJNA1271016, a uniform random 2,000,000 read pairs of each (seed 20260909). Same body site, library strategy and NovaSeq X platform family as the screened sample. ENA describes the study as: MARS shotgun metagenomics — Gut microbiome and AD study in MARS cohort.

## What is still uncertain

The measured figures above come from synthetic samples, where the answer is known. They establish that the method finds what is there and does not invent what is not. They do not establish that any of these pathways predicts a health outcome — that is a question about the biology, not about the test, and for most of these metabolites it remains open. The research summaries in section 19 give the current state of the evidence, including how strong it is in each case.

## Part B — how the resemblance scoring was tested

Species abundances are compared against **3,027 healthy adult stool metagenomes** from 22 published studies (curatedMetagenomicData snapshot 2021-10-14), measured the same way as your sample so the two are directly comparable. Reference identifier 95c86bd4c759748f.

The scoring engine was run end to end on public cohorts that carry disease labels, and asked to separate cases from controls. AUC is the standard measure: 0.5 is a coin flip, 1.0 is perfect. **The same pattern was also scored against unrelated conditions** — a pattern that scores high on everything is measuring general gut disturbance, not its own condition.

| PATTERN | vs CRC | vs IBD | vs T2D |
|---|---|---|---|
| **longcovid** | 0.53<br>0.48–0.59 | 0.80<br>0.76–0.83 | 0.59<br>0.53–0.64 |
| **mecfs** | 0.55<br>0.50–0.60 | 0.50<br>0.46–0.55 | 0.60<br>0.55–0.66 |

**acvd:** not enough labelled conditions to judge specificity.

**ad_clinical:** not enough labelled conditions to judge specificity.

**ad_mci:** not enough labelled conditions to judge specificity.

**ad_preclinical_amyloid:** not enough labelled conditions to judge specificity.

**adenoma:** not enough labelled conditions to judge specificity.

**alopecia_areata:** not enough labelled conditions to judge specificity.

**androgenetic_alopecia:** not enough labelled conditions to judge specificity.

**ankylosing_spondylitis:** not enough labelled conditions to judge specificity.

**celiac:** not enough labelled conditions to judge specificity.

**cirrhosis:** not enough labelled conditions to judge specificity.

**ckd:** not enough labelled conditions to judge specificity.

**crc:** not enough labelled conditions to judge specificity.

**crohns:** not enough labelled conditions to judge specificity.

**csu:** not enough labelled conditions to judge specificity.

**hypertension:** not enough labelled conditions to judge specificity.

**ibd:** not enough labelled conditions to judge specificity.

**ibs:** not enough labelled conditions to judge specificity.

**ibs_c:** not enough labelled conditions to judge specificity.

**ibs_d:** not enough labelled conditions to judge specificity.

**ibs_m:** not enough labelled conditions to judge specificity.

**longcovid:** no labelled cohort exists for this condition, so its own discrimination is UNMEASURED (it scores 0.53-0.80 against other conditions).

**masld:** not enough labelled conditions to judge specificity.

**mdd:** not enough labelled conditions to judge specificity.

**mecfs:** no labelled cohort exists for this condition, so its own discrimination is UNMEASURED (it scores 0.50-0.60 against other conditions).

**ms:** not enough labelled conditions to judge specificity.

**obesity:** not enough labelled conditions to judge specificity.

**parkinsons:** not enough labelled conditions to judge specificity.

**ra:** not enough labelled conditions to judge specificity.

**sle:** not enough labelled conditions to judge specificity.

**symptomatic_dermographism:** not enough labelled conditions to judge specificity.

**t1d:** not enough labelled conditions to judge specificity.

**t2d:** not enough labelled conditions to judge specificity.

**uc:** not enough labelled conditions to judge specificity.

**Unmeasured:** acvd, ad_clinical, ad_mci, ad_preclinical_amyloid, adenoma, alopecia_areata, androgenetic_alopecia, ankylosing_spondylitis, celiac, cirrhosis, ckd, crc, crohns, csu, hypertension, ibd, ibs, ibs_c, ibs_d, ibs_m, masld, mdd, ms, obesity, parkinsons, ra, sle, symptomatic_dermographism, t1d, t2d, uc. No labelled cohort with outcome keys was available for these patterns, so their discrimination is unknown and no accuracy figure is printed for them.

Reading the table: a pattern should score highest against its own condition and near 0.5 against the others. Here the **longcovid** pattern scores *higher* against IBD (AUC 0.80) than the 0.5 a pattern with no cross-condition signal would give, which means part of what it detects is general disturbance rather than that one condition. That is why the general gut-health index accompanies every resemblance score.

# 28 Technical data

## Detection lanes and reference releases

Every organism in this report was found by one or more of these methods, each run on the same host-filtered reads against a fixed reference release. A lane recorded as not run says nothing about which organisms it would have found.

| LANE | TOOL | REFERENCE | TAXONOMY | RESULT |
|---|---|---|---|---|
| **Marker lane** | metaphlan 4.2.6 (tag c55b299; self-reports 4.2.5) | mpa_vJan26_CHOCOPhlAnSGB_202605 | SGB Jan26; GTDB R232 via refs/crosswalk (assembly-membership walk from the R226 bridge) | 256 species-level observations |
| **Broad discovery** | sylph 1.0.0 | GlobDB r232 | GlobDB r232 (GTDB R232 for GTDB members; GlobDB-native clusters otherwise) | 321 species-level observations |
| **Universal markers** | motus 4.1.0 | mOTUs DB 4.1 | mOTUs 4.1 taxonomy, GTDB R226 names (walked to R232 by the crosswalk) | 299 species-level observations |
| **Gut catalogue classifier** | kraken2+bracken kraken2 2.17.1; bracken Bracken v3.0.1 | UHGG v2.0.2 Kraken2 (publisher build) | UHGG v2.0.2 (GTDB r95-era names; MGYG representatives) | 3179 species-level observations |
| **Gut rescue panel** | kraken2+bracken kraken2 2.17.1; bracken Bracken v3.0.1 | gut rescue panel (local build: reconciliation targets, near neighbours, detected alternatives, decoys) built 2026-09-27T22:00:03-07:00 | custom; GlobDB / GTDB R232 lineages, taxid <-> source id in panel_manifest.tsv | 897 species-level observations |
| **Lineage rescue** | singlem 0.21.4 | GlobDB_r232.metapackage_v4 | GlobDB r232 metapackage | 130 species-level observations |

### Reference sources locked for this run

| SOURCE | RELEASE | COUNT | LICENCE | VERIFIED |
|---|---|---|---|---|
| elgg | ELGG (Zenodo 6969520) | 2,172 species representative | CC BY 4.0 (Zenodo record 6969520) | yes |
| globdb_genomes | globdb_r232 | 346,233 representative genome | CC BY-SA 4.0 | yes |
| globdb_r232 | globdb_r232 | 346,233 species representative | CC BY-SA 4.0 (GlobDB); index by sylph authors | yes |
| gtdb_r207 | R207 | — | CC BY-SA 4.0 | yes |
| gtdb_r220 | R220 | — | CC BY-SA 4.0 | yes |
| gtdb_r226 | R226 | — | CC BY-SA 4.0 | yes |
| gtdb_r232 | R232 | 901,341 genome | CC BY-SA 4.0 | yes |
| hrgm2 | HRGMv2 | 4,824 species representative | CC BY 4.0 (Zenodo record 19482781) | yes |
| hrom | HROM | — | see file | yes |
| humgut2 | HumGut2 | 31,225 genome (97.5% ANI dereplicated) | CC BY 4.0 (authors) | yes |
| motus_db | mOTUs DB 4.1 (Zenodo 20322482) | mOTU (species-level unit) | CC BY 4.0 (mOTUs data) | yes |
| mpa_jan26 | mpa_vJan26_CHOCOPhlAnSGB_202605 | 72,000 SGB | MIT (MetaPhlAn); database CC BY-NC-SA per bioBakery | yes |
| sgb_bridges | MetaPhlAn utils @ master 2026-09 | — | MIT (MetaPhlAn) | yes |
| singlem_globdb | GlobDB_r232.metapackage_v4 | — | CC BY-SA 4.0 | yes |
| uhgg_kraken2 | v2.0.2 | 4,744 species | EMBL-EBI terms of use | yes |
| uhgg_v2.0.2 | v2.0.2 | 289,232 genome | EMBL-EBI terms of use (open) | yes |

GlobDB propagates CC BY-SA 4.0; mOTUs data are CC BY 4.0; GTDB is CC BY-SA 4.0; BacDive is CC BY 4.0 and asks to be contacted for commercial use. Full URLs, checksums and retrieval times are in refs/expanded/locks and in results.json under detection.reference_releases.

### Measured capability

On 2 simulated communities of 600 species at 8,000,000 read pairs with known composition (benchmark *overnight_v4*, 2026-09-29), detection found 94.2% of the species present, and 92.6% of the species it reported as supported were present (95% CI 91.0–94.0%); the species reported in error (33.0 per community) were relatives of species that were present. On 2 organism-free host and food backgrounds, 0 microbial species were reported as supported. These are measurements on simulated data, not a statement about this specimen.

## Per-gene results

| Pathway | Gene | Fragments | Copies/100 | Pathway total |
|---|---|---|---|---|
| agmatine | adiA | 1,062 | 32.5 | 32.5 |

| Pathway | Gene | Fragments | Copies/100 | Pathway total |
|---|---|---|---|---|
| b12 | cbiA | 1,089 | 46.7 | 83.4 |
| b12 | cobS | 479 | 36.7 | 83.4 |
| b12 | btuB | 1,006 | 29.9 | 83.4 |
| b6 | pdxS | 653 | 43.0 | 59.8 |
| b6 | pdxT | 454 | 44.5 | 59.8 |
| b6 | pdxA | 286 | 16.8 | 59.8 |
| b6 | pdxJ | 203 | 16.2 | 59.8 |
| b6 | pdxH | 92 | 8.5 | 59.8 |
| b6 | pdxK | 964 | 67.1 | 59.8 |
| bai | baiCD | 41 | 1.3 | 3.5 |
| bai | baiH | 190 | 5.7 | 3.5 |
| bai | baiB | 7 | 0.3 | 3.5 |
| bai | baiF | 1 | 0.048 | 3.5 |
| bai | baiE | 0 | 0 | 3.5 |
| bai | baiA | 100 | 7.7 | 3.5 |
| bcaa | ilvC | 1,617 | 93.3 | 180.5 |
| bcaa | ilvD | 2,603 | 87.2 | 180.5 |
| bcaa | leuA | 2,461 | 88.8 | 180.5 |
| bglucuronidase | gus | 2,529 | 83.1 | 83.1 |
| biotin | bioB | 459 | 26.4 | 64.1 |
| biotin | bioF | 306 | 15.3 | 64.1 |
| biotin | bioD | 260 | 22.5 | 64.1 |
| bsh | bsh | 2,023 | 123.0 | 123.0 |
| butyrate | but | 720 | 32.1 | 32.1 |
| butyrate | buk | 17 | 0.9 | 32.1 |
| butyrate | lctA | 122 | 9.8 | 32.1 |
| butyrate | lctB | 163 | 6.7 | 32.1 |
| butyrate | lctC | 195 | 17.2 | 32.1 |
| carbohydrates | celA | 1,708 | 86.0 | 549.3 |
| carbohydrates | celB | 1,350 | 48.9 | 549.3 |
| carbohydrates | bglA | 3,042 | 129.1 | 549.3 |
| carbohydrates | amyA | 3,315 | 120.6 | 549.3 |
| carbohydrates | pulA | 3,732 | 101.9 | 549.3 |
| carbohydrates | gluA | 1,171 | 39.5 | 549.3 |
| carbohydrates | aglA | 5,685 | 141.7 | 549.3 |
| carbohydrates | malL | 2,589 | 91.1 | 549.3 |
| carbohydrates | chiA | 3,642 | 133.4 | 549.3 |
| carbohydrates | chiD | 2,005 | 138.5 | 549.3 |
| carbohydrates | nagZ | 54 | 3.1 | 549.3 |
| carbohydrates | pgl | 2,582 | 103.3 | 549.3 |
| carbohydrates | pelA | 1,698 | 76.8 | 549.3 |
| carbohydrates | pme | 804 | 45.3 | 549.3 |
| carbohydrates | rgl | 304 | 11.2 | 549.3 |
| carbohydrates | sacA | 2,356 | 93.2 | 549.3 |
| carbohydrates | levB | 415 | 15.2 | 549.3 |
| carbohydrates | lacZ | 5,806 | 177.6 | 549.3 |
| carbohydrates | melA | 4,390 | 119.7 | 549.3 |

| Pathway | Gene | Fragments | Copies/100 | Pathway total |
|---|---|---|---|---|
| carbohydrates | xylB | 2,279 | 85.3 | 549.3 |
| carbohydrates | xynA | 2,502 | 147.8 | 549.3 |
| carbohydrates | abfA | 907 | 51.0 | 549.3 |
| carbohydrates | axe | 973 | 53.6 | 549.3 |
| carbohydrates | fae | 119 | 4.4 | 549.3 |
| carbohydrates | licB | 135 | 6.6 | 549.3 |
| carbohydrates | glcA | 149 | 3.2 | 549.3 |
| carbohydrates | manB | 737 | 34.9 | 549.3 |
| carbohydrates | manA | 1,147 | 27.3 | 549.3 |
| crc_virulence | fadA | 0 | 0 | 0 |
| crc_virulence | bft | 0 | 0 | 0 |
| crc_virulence | clbB | 0 | 0 | 0 |
| cutc | cutC | 140 | 3.3 | 3.3 |
| cutc | cntA | 0 | 0 | 3.3 |
| dopamine | tyrDC | 17 | 0.8 | 0.8 |
| dopamine | dadh | 69 | 1.3 | 0.8 |
| equol | dznr | 4 | 0.1 | 0.1 |
| equol | ddr | 3 | 0.2 | 0.1 |
| equol | tdr | 5 | 0.2 | 0.1 |
| ethanol | adhE | 3,128 | 69.8 | 69.8 |
| ethanol | adhP | 94 | 5.4 | 69.8 |
| ethanol | pdc | 278 | 9.7 | 69.8 |
| ethanol | eutB | 24 | 0.6 | 69.8 |
| ethanol | eutC | 36 | 3.2 | 69.8 |
| fermentation | pta | 2,370 | 66.3 | 151.2 |
| fermentation | acsB | 763 | 21.1 | 151.2 |
| fermentation | cooS | 911 | 67.4 | 151.2 |
| fermentation | fhs | 2,915 | 102.6 | 151.2 |
| fermentation | ldhA | 1,995 | 84.9 | 151.2 |
| fermentation | mutB | 877 | 24.6 | 151.2 |
| fermentation | scpA | 337 | 13.6 | 151.2 |
| fermentation | pduC | 23 | 0.8 | 151.2 |
| fermentation | hydA | 4,366 | 167.0 | 151.2 |
| fermentation | ldhL | 1,077 | 65.3 | 151.2 |
| fermentation | xfp | 959 | 23.7 | 151.2 |
| fermentation | ldh | 1,048 | 62.8 | 151.2 |
| fermentation | mdh | 585 | 34.9 | 151.2 |
| folate | folP | 919 | 64.8 | 102.1 |
| folate | folK | 315 | 37.3 | 102.1 |
| folate | folC | 791 | 34.2 | 102.1 |
| gaba | gadB | 249 | 10.5 | 10.5 |
| gaba | gabT | 116 | 5.3 | 10.5 |
| gababreak | gabD | 212 | 8.2 | 8.2 |
| gababreak | abfH | 349 | 18.5 | 8.2 |
| glucosinolate | BT2158 | 36 | 2.4 | 2.4 |
| glutathione | gshA | 81 | 3.5 | 5.2 |
| glutathione | gshB | 2 | 0.1 | 5.2 |

| Pathway | Gene | Fragments | Copies/100 | Pathway total |
|---|---|---|---|---|
| glutathione | gshF | 61 | 1.6 | 5.2 |
| glutathione | gor | 123 | 5.3 | 5.2 |
| glutathione | ggt | 145 | 5.2 | 5.2 |
| h2s | dsrA | 7 | 0.6 | 96.5 |
| h2s | asrA | 1,508 | 95.9 | 96.5 |
| h2s | cgl | 382 | 20.6 | 96.5 |
| h2s | tpa | 588 | 25.3 | 96.5 |
| h2s | isl | 216 | 7.5 | 96.5 |
| histamine | hdcA | 9 | 0.5 | 0.5 |
| histamine | hdcP | 1,016 | 41.4 | 0.5 |
| indole | tnaA | 237 | 10.0 | 10.0 |
| ipa | fldBC | 181 | 8.9 | 8.9 |
| ipa | fldH | 15 | 0.9 | 8.9 |
| k2 | menA | 16 | 1.0 | 11.7 |
| k2 | menD | 235 | 8.1 | 11.7 |
| k2 | mqnE | 50 | 2.6 | 11.7 |
| methane | mcrA | 0 | 0 | 0 |
| methane | mcrB | 0 | 0 | 0 |
| methane | mtaB | 0 | 0 | 0 |
| mucinlps | nanH | 618 | 29.5 | 136.7 |
| mucinlps | fucA | 2,320 | 93.3 | 136.7 |
| mucinlps | sulf | 4,030 | 152.9 | 136.7 |
| mucinlps | lpxA | 186 | 13.9 | 136.7 |
| mucinlps | lpxC | 224 | 14.5 | 136.7 |
| mucinlps | lpxL | 288 | 18.6 | 136.7 |
| mucinlps | hdhA | 397 | 29.7 | 136.7 |
| niacin | nadA | 1,311 | 72.4 | 192.9 |
| niacin | nadB | 1,362 | 49.9 | 192.9 |
| niacin | nadC | 1,077 | 70.5 | 192.9 |
| niacin | nadD | 691 | 67.1 | 192.9 |
| niacin | nadE | 360 | 23.7 | 192.9 |
| niacin | pncA | 480 | 45.6 | 192.9 |
| niacin | pncB | 1,531 | 66.2 | 192.9 |
| niacin | pnuC | 144 | 13.5 | 192.9 |
| nitrogen | pepT | 1,630 | 78.0 | 298.5 |
| nitrogen | dppA | 181 | 6.7 | 298.5 |
| nitrogen | grdA | 43 | 7.9 | 298.5 |
| nitrogen | prdA | 181 | 15.4 | 298.5 |
| nitrogen | ilvE | 1,807 | 100.2 | 298.5 |
| nitrogen | bkdA | 17 | 0.9 | 298.5 |
| nitrogen | gdhA | 2,604 | 120.4 | 298.5 |
| nitrogen | aspA | 998 | 41.3 | 298.5 |
| nitrogen | ansA | 1,402 | 82.1 | 298.5 |
| nitrogen | porA | 23 | 0.4 | 298.5 |
| nitrogen | padI | 1,113 | 50.3 | 298.5 |
| oxalate | oxc | 2 | 0.068 | 0.4 |
| oxalate | frc | 7 | 0.3 | 0.4 |

| Pathway | Gene | Fragments | Copies/100 | Pathway total |
|---|---|---|---|---|
| p9 | Amuc_1631 | 2 | 0.053 | 0.053 |
| pantothenate | panC | 746 | 51.0 | 160.9 |
| pantothenate | panB | 641 | 45.7 | 160.9 |
| pantothenate | panD | 422 | 64.2 | 160.9 |
| pantothenate | panE | 1,207 | 75.2 | 160.9 |
| pantothenate | panF | 678 | 27.3 | 160.9 |
| pcresol | hpdB | 492 | 11.3 | 11.3 |
| peptides | oppA | 2,438 | 85.9 | 108.7 |
| peptides | oppB | 930 | 54.7 | 108.7 |
| peptides | oppD | 10,126 | 612.7 | 108.7 |
| peptides | dtpT | 575 | 22.8 | 108.7 |
| peptides | livK | 55 | 2.8 | 108.7 |
| peptides | brnQ | 79 | 3.5 | 108.7 |
| propionate | pct | 292 | 10.8 | 26.4 |
| propionate | pduP | 374 | 15.6 | 26.4 |
| propionate | lcdA | 299 | 19.8 | 26.4 |
| propionate | lcdB | 3,352 | 164.3 | 26.4 |
| propionate | mmdA | 803 | 61.4 | 26.4 |
| putrescine | speC | 2,565 | 125.2 | 206.2 |
| putrescine | speB | 392 | 24.7 | 206.2 |
| putrescine | aguA | 937 | 51.4 | 206.2 |
| putrescine | aguB | 833 | 56.3 | 206.2 |
| riboflavin | ribB | 1,191 | 61.9 | 200.1 |
| riboflavin | ribE | 731 | 69.0 | 200.1 |
| riboflavin | ribH | 557 | 69.1 | 200.1 |
| spermidine | speE | 786 | 54.0 | 118.1 |
| spermidine | speD | 95 | 13.7 | 118.1 |
| spermidine | casDH | 1,341 | 64.8 | 118.1 |
| spermidine | casDC | 1,236 | 64.1 | 118.1 |
| thiamine | thiE | 1,078 | 98.2 | 230.5 |
| thiamine | thiC | 1,727 | 58.8 | 230.5 |
| thiamine | thiD | 1,384 | 99.6 | 230.5 |
| thiamine | thiM | 880 | 64.4 | 230.5 |
| thiamine | thiG | 1,009 | 73.5 | 230.5 |
| thiamine | thiH | 3,412 | 174.2 | 230.5 |
| thiamine | thiS | 1,870 | 81.1 | 230.5 |
| thiamine | thiB | 2 | 0.1 | 230.5 |
| thiamine | thiP | 36 | 1.3 | 230.5 |
| thiamine | thiQ | 3,154 | 135.5 | 230.5 |
| tryptamine | tdc | 3 | 0.1 | 0.1 |
| urate | xdhA | 3,355 | 86.1 | 180.3 |
| urate | xdhB | 792 | 50.3 | 180.3 |
| urate | xdhC | 404 | 43.9 | 180.3 |
| urate | uox | 116 | 7.6 | 180.3 |
| urate | hpt | 1,042 | 112.7 | 180.3 |
| urda | urdA | 1,749 | 53.5 | 53.5 |
| urease | ureC | 475 | 16.3 | 16.3 |

| Pathway | Gene | Fragments | Copies/100 | Pathway total |
|---|---|---|---|---|
| urease | ureB | 100 | 18.4 | 16.3 |
| urolithin | ucdO | 0 | 0 | 0 |
| urolithin | ucdC | 0 | 0 | 0 |
| urolithin | ucdF | 0 | 0 | 0 |

## Sample and run details

| | |
|---|---|
| Sample | MM1_FXX745 |
| Read pairs | 5,660,139 |
| Total bases sequenced | 1.67 Gbp |
| Read length | 100–151 bp |
| GC content | 45.3% |
| Mean quality | Phred 40.1 |
| Calibration gene fragments | 6,574 rpoB |
| Reference database | 462,075 (209,105,182 aa) |
| Search tool | diamond version 2.2.6 |
| Analysis time | 1m45s |
| Pipeline | OpenBiota Gut Health Test (openbiota) v0.8.4 |
| Reference set fingerprint | 6d80b0d2b6ff5841 |
| Organism identification | MetaPhlAn 3 3.1.0 |
| Taxonomic database | mpa_v31_CHOCOPhlAn_201901 |
| Detection catalogues | mpa_vJun23_CHOCOPhlAnSGB_202403 (36,822 species-level bins); GTDB r232 via sylph 1.0.0 (199,923 species clusters, ANI ≥ 95%) |
| Host reads removed | not assessable — removed by the provider before delivery |
| Species reference cohort | 3,027 samples, snapshot 2021-10-14, id 95c86bd4c759748f |
| Reference matching | no matching criteria requested |
| Profiles scored | acvd v1.0.0, ad_clinical v1.0.0, ad_mci v1.0.0, ad_preclinical_amyloid v1.0.0, adenoma v1.0.0, alopecia_areata v1.0.0, androgenetic_alopecia v1.0.0, ankylosing_spondylitis v1.0.0, celiac v1.0.0, cirrhosis v1.0.0, ckd v1.0.0, crc v1.0.0, crohns v1.0.0, csu v0.1.0, hypertension v1.0.0, ibd v1.0.0, ibs v1.0.0, ibs_c v1.0.0, ibs_d v1.0.0, ibs_m v1.0.0, longcovid v0.2.0, masld v1.0.0, mdd v1.0.0, mecfs v0.2.0, ms v1.0.0, obesity v1.0.0, parkinsons v1.0.0, ra v1.0.0, sle v0.1.0, symptomatic_dermographism v0.1.0, t1d v1.0.0, t2d v1.0.0, uc v1.0.0 |
| Dysbiosis anchor | GMWI2 +2.00 |

## How the calculation works

Each pathway gene's fragment count is divided by the average length of that gene, then divided again by the same quantity for *rpoB* — a gene every bacterium carries exactly once. That ratio gives copies per bacterial genome, which is then multiplied by 100.

Two consequences are worth knowing. Sequencing depth cancels out, so a deeper sample does not produce bigger numbers. And human DNA in the sample cancels out too, because human cells carry neither the pathway genes nor *rpoB* — which is why Part A needs no human-DNA removal step.

Part B works differently. Species abundances are compositional — they sum to 100%, so one going up forces others down — and are therefore compared in centred log-ratio space, against the median and spread of the reference cohort, using a robust scale that a single outlier cannot dominate. Each feature's deviation is clipped, signed by the published direction, and weighted by four documented factors; module scores are weighted means; and percentiles come from scoring every reference sample the same way, never from rescaling. Human DNA does *not* cancel here, so it is removed first.

## Full method and source code

The complete method, the validation protocol and every gene query used are documented alongside the source code. The machine-readable form of this report, containing every number shown here plus a good deal more, is in **results.json**.

# Complete index of the additional measurements

**150 additional measurements** in this report, 132 of them quantified for your sample, across 13 capabilities. 6 could not be produced from this sample and 0 are not built yet; the two are listed separately because only one of them is a limit of the science. Each row below is one unique measurement — a reading shown in more than one place is counted once.

| METRIC ID | READING | VALUE | STATUS | WHERE IT APPEARS |
|---|---|---|---|---|
| **5.8 · 6 measurements** | | | | |
| ext083.capacity.agmatine | Agmatine formation | **32.54 copies per 100 bacterial genomes** | measured | — |
| ext083.capacity.ethanol | Microbial ethanol formation | **69.78 copies per 100 bacterial genomes** | measured | — |
| ext083.capacity.glutathione | Glutathione synthesis | **5.22 copies per 100 bacterial genomes** | measured | — |
| ext083.capacity.putrescine | Putrescine formation | **206.2 copies per 100 bacterial genomes** | measured | — |
| ext083.capacity.spermidine | Spermidine formation | **118.1 copies per 100 bacterial genomes** | measured | — |
| ext083.capacity.urate_anaerobic | Anaerobic urate degradation | **180.3 copies per 100 bacterial genomes** | measured | — |
| **A01 · 14 measurements** | | | | |
| ext083.carb.arabinoxylan | Arabinoxylan | **5,688 fragments** | measured | — |
| ext083.carb.beta_glucan | Beta-glucans | **3,326 fragments** | measured | — |
| ext083.carb.cellulose | Cellulose | **3,058 fragments** | partly measured | — |
| ext083.carb.chitin | Chitin | **3,696 fragments** | measured | — |
| ext083.carb.fos | Fructooligosaccharides | **2,771 fragments** | partly measured | — |
| ext083.carb.galactomannan | Galactomannan and guar | **1,884 fragments** | measured | — |
| ext083.carb.gos | Galactooligosaccharides | **6,953 fragments** | partly measured | — |
| ext083.carb.imo | Isomaltooligosaccharides | **1.532e+04 fragments** | measured | — |
| ext083.carb.inulin | Inulin | **2,771 fragments** | partly measured | — |
| ext083.carb.lactose | Lactose | **6,953 fragments** | partly measured | — |
| ext083.carb.pectin | Pectin | **5,388 fragments** | measured | — |
| ext083.carb.resistant_starch | Resistant starch | **9,636 fragments** | partly measured | — |
| ext083.carb.starch_general | General starch | **1.649e+04 fragments** | measured | — |
| ext083.carb.xos | Xylooligosaccharides | **2,279 fragments** | partly measured | — |
| **A02 · 13 measurements** | | | | |
| ext083.fermentation.acetate.pta_ack | Acetate production | **1 genes present** | measured | — |
| ext083.fermentation.acetate.wood_ljungdahl | Acetate made from hydrogen and CO2 | **2 genes present** | measured | — |
| ext083.fermentation.hydrogen.consumption_methanogenesis | Hydrogen used to make methane | **—** | searched, nothing above the floor | — |
| ext083.fermentation.hydrogen.consumption_sulfate | Hydrogen used to make sulfide | **1 genes present** | measured | — |
| ext083.fermentation.hydrogen.production_fefe | Hydrogen production | **1 genes present** | measured | — |
| ext083.fermentation.lactate.d_formation | D-lactate production | **1 genes present** | measured | — |
| ext083.fermentation.lactate.l_formation | L-lactate production | **1 genes present** | measured | — |
| ext083.fermentation.lactate.utilisation_butyrate | Turning lactate into butyrate | **4 genes present** | measured | — |
| ext083.fermentation.lactate.utilisation_propionate | Turning lactate into propionate | **2 genes present** | measured | — |
| ext083.fermentation.propionate.propanediol | Turning propanediol into propionate | **2 genes present** | measured | — |
| ext083.fermentation.succinate.formation | Succinate production | **1 genes present** | measured | — |
| ext083.fermentation.succinate.to_propionate | Turning succinate into propionate | **2 genes present** | measured | — |
| ext083.fermentation.sulfur.taurine | Turning taurine into sulfide | **2 genes present** | measured | — |
| **A03 · 4 measurements** | | | | |
| ext083.vitamin.b1 | Vitamin B1 | **230.5 copies per 100 bacterial genomes** | measured | — |

| METRIC ID | READING | VALUE | STATUS | WHERE IT APPEARS |
|---|---|---|---|---|
| ext083.vitamin.b3 | Vitamin B3 | **192.9 copies per 100 bacterial genomes** | measured | — |
| ext083.vitamin.b5 | Vitamin B5 | **160.9 copies per 100 bacterial genomes** | measured | — |
| ext083.vitamin.b6 | Vitamin B6 | **59.77 copies per 100 bacterial genomes** | measured | — |
| **A04 · 6 measurements** | | | | |
| ext083.nitrogen.amino_acid_fermentation | Amino-acid fermentation, including Stickland pairs | **3 informative genes present** | partly measured | — |
| ext083.nitrogen.ammonia | Ammonia generation and assimilation | **3 informative genes present** | partly measured | — |
| ext083.nitrogen.aromatic | Aromatic precursor routes | **3 informative genes present** | partly measured | — |
| ext083.nitrogen.bcfa | Branched-chain fatty acid formation | **2 informative genes present** | partly measured | — |
| ext083.nitrogen.peptide_transport | Peptide uptake and amino-acid use | **6 informative genes present** | measured | — |
| ext083.nitrogen.proteolysis | Protein and peptide breakdown | **2 informative genes present** | partly measured | — |
| **A05 · 3 measurements** | | | | |
| ext083.neuro.gaba_degradation | GABA degradation | **2 genes present** | measured | — |
| ext083.neuro.gaba_synthesis | GABA synthesis | **1 genes present** | measured | — |
| ext083.neuro.gaba_to_butyrate | 4-aminobutyrate to butyrate | **2 genes present** | partly measured | — |
| **A06 · 3 measurements** | | | | |
| ext083.diet.glucosinolate_isothiocyanate_conversion | Glucosinolate to isothiocyanate | **1 genes present** | partly measured | — |
| ext083.polyphenol.equol_daidzein_conversion | Daidzein to equol | **3 genes present** | measured | — |
| ext083.polyphenol.urolithin_9_dehydroxylation | Urolithin C to urolithin A (9-dehydroxylation) | **—** | this assay cannot answer | — |
| **A07 · 4 measurements** | | | | |
| ext083.glp1.bile_acid_signalling | Bile-acid transformation | **1 genes present** | measured | — |
| ext083.glp1.indole_signalling | Indole and tryptophan routes | **2 genes present** | measured | — |
| ext083.glp1.microbial_protein | Experimentally supported microbial proteins | **1 genes present** | measured | — |
| ext083.glp1.scfa_signalling | Short-chain fatty acid signalling | **3 genes present** | measured | — |
| **A08 · 5 measurements** | | | | |
| ext083.bile.deconjugation | Bile-salt deconjugation | **1 genes present** | measured | — |
| ext083.bile.dehydroxylation | 7-alpha-dehydroxylation (bai pathway) | **3 genes present** | partly measured | — |
| ext083.bile.hsdh_transformation | Hydroxysteroid dehydrogenase transformations | **1 genes present** | measured | — |
| ext083.lps.lipid_a_modification | Lipid A acylation and modification | **3 genes present** | measured | — |
| ext083.mucin.glycan_foraging | Mucin glycan foraging | **3 genes present** | measured | — |
| **A09 · 51 measurements** | | | | |
| A09.redundancy.panels.b12 | Vitamin B12 (cobalamin): organisms holding it | **5.736 effective organisms** | measured | Your gut community |
| A09.redundancy.panels.b6 | Vitamin B6 (pyridoxal phosphate): organisms holding it | **3.587 effective organisms** | measured | Your gut community |
| A09.redundancy.panels.bcaa | Branched-chain amino acids (valine, leucine, isoleucine): organisms holding it | **4.875 effective organisms** | measured | Your gut community |
| A09.redundancy.panels.bglucuronidase | Deconjugated glucuronides (oestrogens, drug metabolites, bilirubin): organisms holding it | **1.829 effective organisms** | measured | Your gut community |
| A09.redundancy.panels.biotin | Biotin (vitamin B7): organisms holding it | **4.055 effective organisms** | measured | Your gut community |
| A09.redundancy.panels.bsh | Unconjugated bile acids: organisms holding it | **2.858 effective organisms** | measured | Your gut community |

| METRIC ID | READING | VALUE | STATUS | WHERE IT APPEARS |
|---|---|---|---|---|
| A09.redundancy.panels.butyrate | Butyrate (SCFA): organisms holding it | **4.203 effective organisms** | measured | Your gut community |
| A09.redundancy.panels.carbohydrates | Dietary fibre and carbohydrate breakdown: organisms holding it | **3.679 effective organisms** | measured | Your gut community |
| A09.redundancy.panels.ethanol | Microbial ethanol formation: organisms holding it | **1.808 effective organisms** | measured | Your gut community |
| A09.redundancy.panels.fermentation | Fermentation routes and cross-feeding: organisms holding it | **2.571 effective organisms** | measured | Your gut community |
| A09.redundancy.panels.folate | Folate (vitamin B9): organisms holding it | **1.998 effective organisms** | measured | Your gut community |
| A09.redundancy.panels.gaba | gamma-aminobutyric acid (GABA): organisms holding it | **1 effective organisms** | measured | Your gut community |
| A09.redundancy.panels.glutathione | Glutathione synthesis: organisms holding it | **2.996 effective organisms** | measured | Your gut community |
| A09.redundancy.panels.h2s | Hydrogen sulfide (H2S): organisms holding it | **2.876 effective organisms** | measured | Your gut community |
| A09.redundancy.panels.histamine | Histamine: organisms holding it | **1 effective organisms** | measured | Your gut community |
| A09.redundancy.panels.indole | Indole / indoxyl sulfate: organisms holding it | **3.958 effective organisms** | measured | Your gut community |
| A09.redundancy.panels.ipa | Indole-3-propionate (IPA): organisms holding it | **1.649 effective organisms** | measured | Your gut community |
| A09.redundancy.panels.k2 | Vitamin K2 (menaquinone, MK-n): organisms holding it | **1.997 effective organisms** | measured | Your gut community |
| A09.redundancy.panels.mucinlps | Mucin glycans and lipid A: organisms holding it | **1.997 effective organisms** | measured | Your gut community |
| A09.redundancy.panels.niacin | Vitamin B3 (niacin and NAD precursors): organisms holding it | **1.946 effective organisms** | measured | Your gut community |
| A09.redundancy.panels.nitrogen | Protein and nitrogen metabolism: organisms holding it | **2 effective organisms** | measured | Your gut community |
| A09.redundancy.panels.pantothenate | Vitamin B5 (pantothenate): organisms holding it | **5.572 effective organisms** | measured | Your gut community |
| A09.redundancy.panels.pcresol | p-Cresol / p-cresyl sulfate: organisms holding it | **1 effective organisms** | measured | Your gut community |
| A09.redundancy.panels.peptides | Peptide uptake: organisms holding it | **1.335 effective organisms** | measured | Your gut community |
| A09.redundancy.panels.propionate | Propionate (SCFA): organisms holding it | **2.836 effective organisms** | measured | Your gut community |
| A09.redundancy.panels.putrescine | Putrescine formation: organisms holding it | **2.119 effective organisms** | measured | Your gut community |
| A09.redundancy.panels.riboflavin | Riboflavin (vitamin B2): organisms holding it | **1.996 effective organisms** | measured | Your gut community |
| A09.redundancy.panels.spermidine | Spermidine formation: organisms holding it | **7.367 effective organisms** | measured | Your gut community |
| A09.redundancy.panels.thiamine | Vitamin B1 (thiamine): organisms holding it | **3.6 effective organisms** | measured | Your gut community |
| A09.redundancy.panels.urate | Urate degradation: organisms holding it | **4.227 effective organisms** | measured | Your gut community |
| A09.redundancy.panels.urda | Imidazole propionate (ImP): organisms holding it | **4.304 effective organisms** | measured | Your gut community |
| A09.redundancy.panels.urease | Ammonia (from urea): organisms holding it | **1 effective organisms** | measured | Your gut community |
| ext083.ecology.aerotolerant_fraction | Aerotolerance balance | **0.01328 fraction** | measured | Your gut community |
| ext083.ecology.community_type.distance.Bacteroides-dominant | Distance to the Bacteroides-dominant reference group | **0.4859 sqrt(Jensen-Shannon divergence), nats^0.5** | measured | Your gut community |
| ext083.ecology.community_type.distance.Prevotella-dominant | Distance to the Prevotella-dominant reference group | **0.5856 sqrt(Jensen-Shannon divergence), nats^0.5** | measured | Your gut community |
| ext083.ecology.composition.anaerostipes | Anaerostipes share of the community | **0.04671 fraction of 1** | measured | Your gut community |
| ext083.ecology.composition.bacteroides | Bacteroides share of the community | **0.03848 fraction of 1** | measured | Your gut community |

| METRIC ID | READING | VALUE | STATUS | WHERE IT APPEARS |
|---|---|---|---|---|
| ext083.ecology.composition.blautia | Blautia share of the community | **0.1039 fraction of 1** | measured | Your gut community |
| ext083.ecology.composition.faecalibacterium | Faecalibacterium share of the community | **0.07784 fraction of 1** | measured | Your gut community |
| ext083.ecology.composition.lachnospira | Lachnospira share of the community | **0.05842 fraction of 1** | measured | Your gut community |
| ext083.ecology.composition.ruminococcus | Ruminococcus share of the community | **0.04284 fraction of 1** | measured | Your gut community |
| ext083.ecology.composition_unresolved_share | Share of the community outside the composition's genus universe | **0 fraction of 1** | measured | Your gut community |
| ext083.ecology.dominance_top1 | Share of the single most abundant organism | **0.07434 fraction** | measured | Your gut community |
| ext083.ecology.dominance_top5 | Share of the five most abundant organisms | **0.2699 fraction** | measured | Your gut community |
| ext083.ecology.effective_shannon_species | Effective number of species | **94.3 species_equivalents** | measured | Your gut community |
| ext083.ecology.gini_simpson | Gini-Simpson index | **0.9777 fraction** | measured | Your gut community |
| ext083.ecology.inverse_simpson | Inverse Simpson diversity | **44.86 species_equivalents** | measured | Your gut community |
| ext083.ecology.mapi | Microbial aerotolerance profile index (MAPI) | **-4.308 ln_ratio** | measured | Your gut community |
| ext083.ecology.ratio.fusobacterium_faecalibacterium | Fusobacterium : Faecalibacterium | **—** | searched, nothing above the floor | Your gut community |
| ext083.ecology.ratio.prevotella_bacteroides | Prevotella : Bacteroides | **0.02829 ratio** | measured | Your gut community |
| ext083.ecology.ratio.proteobacteria_actinobacteriota | Proteobacteria (Pseudomonadota) : Actinobacteriota | **2.576 ratio** | measured | Your gut community |
| **A10 · 2 measurements** | | | | |
| ext083.explorer.interpretation_coverage | Share of the community with an interpretation | **1 fraction** | measured | Every organism found |
| ext083.explorer.organisms_listed | Organisms in the complete explorer | **469 organisms** | measured | Every organism found |
| **A11 · 7 measurements** | | | | |
| ext083.resistance.class.aminoglycoside | Aminoglycosides: determinant richness | **4 distinct_gene_families** | measured | Pathogen screening |
| ext083.resistance.class.macrolide_lincosamide_streptogramin | Macrolides, lincosamides and streptogramin B: determinant richness | **3 distinct_gene_families** | measured | Pathogen screening |
| ext083.resistance.class.oxazolidinone_phenicol | Oxazolidinones and phenicols: determinant richness | **2 distinct_gene_families** | measured | Pathogen screening |
| ext083.resistance.class.sulfonamide_trimethoprim | Sulfonamides and trimethoprim: determinant richness | **1 distinct_gene_families** | measured | Pathogen screening |
| ext083.resistance.class.tetracycline | Tetracyclines: determinant richness | **3 distinct_gene_families** | measured | Pathogen screening |
| ext083.resistance.determinant_richness | Resistance determinant richness | **13 distinct_gene_families** | measured | Pathogen screening |
| ext083.resistance.fragment_abundance | Resistance determinant abundance | **486 fragments** | measured | Pathogen screening |
| **A16 · 32 measurements** | | | | |
| A16.arabinoxylan.both.butyrate_potential | Arabinoxylan — both: butyrate potential | **300.8 mmol/gDW/h** | measured | What you can do about it |
| A16.arabinoxylan.interaction | Arabinoxylan + consortium: interaction contrast | **71.44 mmol/gDW/h** | measured | What you can do about it |
| A16.arabinoxylan.neither.butyrate_potential | Arabinoxylan — neither: butyrate potential | **219.4 mmol/gDW/h** | measured | What you can do about it |
| A16.arabinoxylan.probiotic_only.butyrate_potential | Arabinoxylan — probiotic only: butyrate potential | **219 mmol/gDW/h** | measured | What you can do about it |
| A16.arabinoxylan.substrate_only.butyrate_potential | Arabinoxylan — substrate only: butyrate potential | **229.8 mmol/gDW/h** | measured | What you can do about it |
| A16.cellulose.both.butyrate_potential | Cellulose — both: butyrate potential | **219 mmol/gDW/h** | measured | What you can do about it |
| A16.cellulose.interaction | Cellulose + consortium: interaction contrast | **0 mmol/gDW/h** | measured | What you can do about it |
| A16.cellulose.neither.butyrate_potential | Cellulose — neither: butyrate potential | **219.4 mmol/gDW/h** | measured | What you can do about it |
| A16.cellulose.probiotic_only.butyrate_potential | Cellulose — probiotic only: butyrate potential | **219 mmol/gDW/h** | measured | What you can do about it |

| METRIC ID | READING | VALUE | STATUS | WHERE IT APPEARS |
|---|---|---|---|---|
| A16.cellulose.substrate_only.butyrate_potential | Cellulose — substrate only: butyrate potential | **219.4 mmol/gDW/h** | measured | What you can do about it |
| A16.goal_coverage | Goal coverage (C) | **83.3 % of weighted goals** | measured | What you can do about it |
| A16.inulin.both.butyrate_potential | Inulin — both: butyrate potential | **440.1 mmol/gDW/h** | measured | What you can do about it |
| A16.inulin.interaction | Inulin + consortium: interaction contrast | **21.86 mmol/gDW/h** | measured | What you can do about it |
| A16.inulin.neither.butyrate_potential | Inulin — neither: butyrate potential | **219.4 mmol/gDW/h** | measured | What you can do about it |
| A16.inulin.probiotic_only.butyrate_potential | Inulin — probiotic only: butyrate potential | **219 mmol/gDW/h** | measured | What you can do about it |
| A16.inulin.substrate_only.butyrate_potential | Inulin — substrate only: butyrate potential | **418.6 mmol/gDW/h** | measured | What you can do about it |
| A16.maltodextrin.both.butyrate_potential | Maltodextrin — both: butyrate potential | **437 mmol/gDW/h** | measured | What you can do about it |
| A16.maltodextrin.interaction | Maltodextrin + consortium: interaction contrast | **19.56 mmol/gDW/h** | measured | What you can do about it |
| A16.maltodextrin.neither.butyrate_potential | Maltodextrin — neither: butyrate potential | **219.4 mmol/gDW/h** | measured | What you can do about it |
| A16.maltodextrin.probiotic_only.butyrate_potential | Maltodextrin — probiotic only: butyrate potential | **219 mmol/gDW/h** | measured | What you can do about it |
| A16.maltodextrin.substrate_only.butyrate_potential | Maltodextrin — substrate only: butyrate potential | **417.9 mmol/gDW/h** | measured | What you can do about it |
| A16.pair_supported_goal_coverage | Pair-supported goal coverage (P) | **66.7 % of weighted goals** | measured | What you can do about it |
| A16.pectin.both.butyrate_potential | Pectin — both: butyrate potential | **375.2 mmol/gDW/h** | measured | What you can do about it |
| A16.pectin.interaction | Pectin + consortium: interaction contrast | **20.65 mmol/gDW/h** | measured | What you can do about it |
| A16.pectin.neither.butyrate_potential | Pectin — neither: butyrate potential | **219.4 mmol/gDW/h** | measured | What you can do about it |
| A16.pectin.probiotic_only.butyrate_potential | Pectin — probiotic only: butyrate potential | **219 mmol/gDW/h** | measured | What you can do about it |
| A16.pectin.substrate_only.butyrate_potential | Pectin — substrate only: butyrate potential | **355 mmol/gDW/h** | measured | What you can do about it |
| A16.starch.both.butyrate_potential | Starch — both: butyrate potential | **343.6 mmol/gDW/h** | measured | What you can do about it |
| A16.starch.interaction | Starch + consortium: interaction contrast | **-3.882e-11 mmol/gDW/h** | measured | What you can do about it |
| A16.starch.neither.butyrate_potential | Starch — neither: butyrate potential | **219.4 mmol/gDW/h** | measured | What you can do about it |
| A16.starch.probiotic_only.butyrate_potential | Starch — probiotic only: butyrate potential | **219 mmol/gDW/h** | measured | What you can do about it |
| A16.starch.substrate_only.butyrate_potential | Starch — substrate only: butyrate potential | **344 mmol/gDW/h** | measured | What you can do about it |

Every value here also appears in **results.json** and **all_metrics.tsv** beside this report, with its full precision, its method identifier and the fingerprint of the inputs that produced it. Rounding on this page is presentation only. An optional laboratory field nobody supplied is not counted as a reading.